Timepiece: A Verified Mathematical Tour

The Timepiece Project

This book is a pedagogical, machine-verified tour of the mathematical ideas behind the *Timepiece* programme: the observation that an ordinary probability measure can be **parametrized by a wave-function**, and that this single change of variables naturally reproduces the structural backbone of quantum mechanics — Born's rule, unitary time-evolution, gauge symmetry, quantization, and the classical limit — as theorems of probability theory rather than as physical postulates.

Every mathematical claim that can be made precise here is backed by a formal proof in the Lean 4 proof assistant, checked against the Mathlib library. The companion library `BookProof` contains these proofs; this book presents them.

Table of Contents

Preface: How to Read This Book

1. Introduction

2. A Decidable, Complete Foundation

3. Completeness without Peano Arithmetic

4. The Solovay–Kopperman Tensor Product

5. Statistical Model Theory, and Statements as Operators

6. Probability as Coherent Belief

7. The Dutch-book Theorem: Probability Is Coherence

8. Bayesian Updating Is Associative

9. Maximum Entropy Selects the Uniform Prior

10. The Law of Total Variance

11. Determinism, Complementarity, and Collapse

12. Symmetries, Determinism, and Complementarity

13. Collapse Keeps Quantum Mechanics a Probability Theory

14. The Double-Slit Experiment

15. The Bell Inequalities

16. EPR-Completeness, Causality, and a Deterministic Theory

17. Wave-functions, Euler's Formula, and the Born Rule

18. The Probability Clock and Euler's Formula

19. The Born Rule Reproduces Every Distribution

20. The Gauge Ambiguity of the Born Parametrization

21. Conditional Probability Is Parametrized by a Unitary

22. The Stern-Gerlach Experiment

23. Relativity, Gauge Theory, and Gravity

24. Gauge Symmetry and Dissipative Dynamics

25. Free Fields: the Gaussian and the Uniform Sphere

26. The Spin–Statistics Dichotomy

27. Real Representations, the CPT Theorem and the Relativistic Position Operator

28. Quantization Due to Time-Evolution: Yang–Mills and Classical Statistical Field Theory

29. Timepiece and the Gribov Ambiguity

30. The Physical Parity Transformation and Antiparticles

31. Diffeomorphisms and Gravity

32. The Fourier Elimination of the Derivative Variables

33. The Navier–Stokes One-Particle Hamiltonian and Its Fock Enclosure

34. The Comparison Operator of the Navier–Stokes Hamiltonian

35. Second Quantization of an Essentially Self-Adjoint One-Particle Operator

36. Eliminating the Derivative Variables in Quantum Gravity

37. The Starobinsky Scalaron: from Potential to Fock Space

38. The Standard Model Hamiltonian

39. Shift-Invert Selection for the Navier–Stokes Generator

40. Carleman Flux Criteria for Lattice Hops

41. Reliability of the Shift-Invert Rational Krylov Scheme

42. The Constants Suite: g-2, Lamb Shift, Positronium, Blackbody

43. NS Numerics: Laminar Decay, Advection Symmetry, Turbulence Laws

44. Consciousness, Deep Learning, and the Bayesian Prior

45. Consciousness as a Representation of a Bayesian Prior

46. Aligned Deep Learning as a Random Sampling Method

47. Dimension-Robust Bayesian Adaptation: Hyperspherical nGPT Geometry and SG-pCNL

48. The Coherent State of Attention

49. Entropy, Irreversibility, and the Arrow of Time

50. Irreversibility: Injective but Not Surjective

51. A Random Map Is Almost Surely Non-Invertible

52. Null-Measure Sets Need Not Be Small

53. Cosmological Amplification of the Matter/Radiation Ratio

54. Measurements Reproduce the Distribution

55. The ODE Singularity

56. Resolution of the Singularity of an ODE

Appendix: Proof Plans

Index

Preface: How to Read This Book

The book is written to be read in two layers at once.

The mathematical layer
: Each chapter develops a self-contained mathematical idea in ordinary prose and notation, with a complete **sketch proof**. These sketches are meant to be readable: they explain *why* a statement is true, not merely *that* it is true.

The verified layer
: Each section ends with one or more Lean 4 code blocks naming the theorem that formalizes the claim, identified by its exact Lean name (for example `ChapterTotalVariance.total_variance`). The full proof lives in the `BookProof` library. Where a claim is not yet proved, we say so and give a proof plan in the appendix.

Verifying everything for yourself. Two commands reproduce the whole verification. From the repository root:

`lake build BookProof`

compiles the entire library of formal proofs. It is `sorry`-free, and every theorem cited in this book relies only on Lean's standard `propext`, `Classical.choice`, and `Quot.sound`. The only `sorry`s in the repository are the two intentional, quarantined ones in `UnusedRoute/SchoenfeldPRA.lean` (a historical PRA spine this edition does not import); the claims they leave open are stated and planned in the appendix. And

`lake build book && lake exe book`

rebuilds this book and renders the HTML you are reading as a single page at `_out/html-single/index.html`. (The Lean statements are currently shown as plain code blocks; upgrading them to elaborated, hover-enabled blocks, and migrating to `verso-blueprint`, are planned — see the appendix.)

Scope of this edition. This is a curated edition. It follows the structure of the source manuscript `book.tex`, but it selects the threads whose mathematics is both self-contained and already formalized. Two chapters of the manuscript are deliberately replaced here:

- the chapter on the ordinary differential equation $x' = x^2$ is presented in its expanded, operator-theoretic form (Resolution of the singularity of an ODE);
- the chapters on *P versus NP* and on the *Riemann Hypothesis* are replaced by a single, self-contained metamathematical chapter (Completeness without Peano Arithmetic) that captures the precise point those chapters were making: a completed Hilbert space can be a complete, decidable extension that does not leak undecidable arithmetic, provided its infinite elements are kept *internally unselectable*.

A note on what remains open. Where a statement is mathematically relevant to the narrative but is not yet proved in `BookProof`, we say so explicitly and give a detailed proof plan in Appendix: Proof Plans. Nothing in the verified layer is asserted without a proof; the open items are quarantined there.

1. Introduction

1.1. What Probability Does: Relating the Complex to the Intuitive

We begin not with a definition of probability, but with a description of what a probability **does**. The point of view of this book is that a probability is a *map from complex random events to standard, intuitive random events*. It does not tell us what randomness *is*; it tells us how an event we do not understand lines up with one we do.

Here is the picture from the source manuscript. Suppose someone tells you that a certain event has probability 0.32. The number is a shorthand for the following sentence:

This event is as likely as finding a small object — say, a lost ring — that we know is buried somewhere in the sand of a one-kilometre beach, if we search for it with a metal detector over a 320-metre interval.

The treasure hunt is itself ambiguous: we do not know whether there are clues to the ring's location, how it was lost, or who else has walked the beach. But that ambiguity has been **moved** by the map. Whatever is mysterious about our original event has been transferred onto a standard random event — *a point chosen on an interval* — that everyone already understands intuitively. The map from the complex event to the treasure hunt is unambiguous, even though the treasure hunt is not.

This is the whole game. We do not need to settle what probability *means*. We need only a disciplined way of **relating** an arbitrary, possibly very complicated random event to a standard one — a point on a line, a draw from an urn, a spin of a wheel — whose behaviour is transparent. Kolmogorov's axioms are precisely the rules that make this relating consistent: a probability space is a sample space (the set of possible states), a Boolean algebra of events (subsets of states), and a measure assigning each event a number in $[0, 1]$, with the whole space carrying measure 1.

Most probability spaces that arise in applications are **standard** measure spaces: up to sets of measure zero they are the unit interval with Lebesgue measure, a finite or countable discrete space, or a mixture of the two. On a standard measure space one can always form regular conditional probabilities, and — this is the technical fact the

whole book rests on — one can always parametrize the resulting probability distributions by a wave-function on a sphere.

1.2. The Central Idea

A **probability distribution** on a finite set of outcomes $\{1, \dots, n\}$ is a list of non-negative numbers that sum to one:

$$p_1, \dots, p_n \geq 0, \qquad \sum_{k=1}^{n} p_k = 1.$$

The set of all such lists is the **probability simplex**. It is a convex body: a mixture $\lambda p + (1-\lambda) q$ of two distributions is again a distribution. This convexity is the mathematical expression of the fact that one can be *uncertain about which of two models is correct*.

Now observe a curious fact. Every point of the simplex can be written as the coordinate-wise square of a point on the unit sphere. If

$$\psi = (\psi_1, \dots, \psi_n), \qquad \sum_{k=1}^{n} \psi_k^2 = 1,$$

then setting $p_k = \psi_k^2$ automatically gives a probability distribution: each $p_k \geq 0$ and $\sum_k p_k = 1$. The vector ψ is a **wave-function**, and the rule $p_k = \psi_k^2$ (more generally $p_k = |\psi_k|^2$ over the complex numbers) is **Born's rule**.

This change of variables — from a distribution p on the simplex to a wave-function ψ on the sphere — is the seed from which the whole book grows. It is a *parametrization*: it does not change the probabilities, only the coordinates we use to describe them. The source manuscript is emphatic that this is all it is — a "mere (but very useful) parametrization" of probability. The wave-function is "nothing else than one possible parametrization of any probability distribution"; the "coherence," "interference," and "collapse" that appear in these coordinates are features of the parametrization, not new physical phenomena. The whole quantum formalism is read this way here, against the "exotic" view that quantum mechanics is a new physics that "shook our sense of reality."

1.3. One Answer Among Many: Coherent Belief

Because a probability is a *relation* between events rather than a substance, one can ask what constraints this relation must obey. Different answers give different foundations of the same calculus.

One influential answer treats probabilities as **degrees of belief** and asks: when is a system of beliefs internally consistent? The **Dutch-book** argument shows that a bettor whose degrees of belief violate the probability axioms can be offered a combination of bets that loses money no matter what happens — a *sure loss*. Avoiding such a sure loss forces the degrees of belief to obey exactly the probability axioms. We develop this in Part I. It is a beautiful and useful answer, and it is *one* example of what probability can be.

But it is not the answer this book is built around. The Dutch-book view explains what coherence *demands* of a single agent's beliefs. The view we take here is broader and, we think, closer to how probability is actually used in physics and engineering: probability is the **bridge** that lets us reason about a complex random event by relating it to an intuitive one. The wave-function parametrization is a construction *on that bridge*. Everything else in the book — Born's rule, unitary evolution, gauge symmetry, the classical limit — is read off from the geometry of the bridge, not from any single philosophical account of what probability *is*.

1.4. Why a Simple Solution Exists

The source manuscript opens with a diagnosis. Real-world engineering is impossible without managing uncertainty, yet the mathematical tools of engineering (differential geometry, the calculus of variations) are built on deterministic logic. The difficulty is sharpest in infinite-dimensional spaces: there is no Lebesgue measure on an infinite-dimensional Euclidean-like space, and once probabilities are placed on a space of functions one typically leaves the separable world in which computation is possible. The result is a zoo of *ad hoc* methods.

The manuscript's claim is that the obstruction is narrower than it looks. Infinite dimensions are not themselves the problem. A **uniform, Lebesgue-like measure on an infinite-dimensional sphere** *can* be defined, using the Gaussian measure and the **Fock space** (the separable Hilbert space used in the second quantization of free fields). Such a sphere parametrizes the probability distribution of another probability distribution — a *free-field parametrization*. We make the finite-dimensional version of this construction precise in the free-field chapter.

The real problem, then, is not infinity but **constraints**: how to impose an exact constraint inside a separable probability space *without* giving the constrained set measure zero. The manuscript's thesis is that **quantum constraints** — which, unlike classical ones, need not commute with the variables that define the sample space — solve precisely this. This is the thread that connects the foundations of probability to the foundations of quantum field theory, and it is the reason a "simple solution" is even plausible.

1.5. Why "Timepiece"

The name of the programme refers to a piece of structure that the manuscript insists on. A **dynamical system** has two parts: a *state*, which is a point of a state space and involves no time (it is the *present* state); and an *evolution rule*, which says how a future state is produced from the present one. The notion of "time" is determined by the evolution rule itself; it is not a pre-existing concept, and it may be measured by integers, by real or complex numbers, or by a more general algebraic object.

In a Hamiltonian formalism with **time-dependent** transformations, the symplectic form of conservative classical mechanics is not invariant. The cleanest way to make non-relativistic mechanics consistent with that fact is to formulate it as a field theory whose phase space is a fibred manifold over an extra "time" axis. The name *Timepiece* refers to this extra axis — defined by the phase space — which is complementary to the ordinary time defined by the time-evolution operator. The distinction between these two notions of time recurs throughout the book, and it is what makes the wave-function parametrization compatible with both classical Hamiltonian mechanics and quantum mechanics.

1.6. Aim, and What Quantization Is Not

The aim is a simple, mathematically meaningful account of **quantization** that applies uniformly to quantum mechanics and to (classical and quantum) statistical field theory, and that shows the "mystery" of quantum mechanics to be a feature of the parametrization rather than of nature. To clear the ground, the manuscript is explicit about what quantization is **not**:

- It is not **prequantization** — the mechanical replacement of Poisson brackets by commutators — which can always be done for analytic functions but does not by itself yield useful results.
- It is not **second quantization** (the Fock-space passage from one particle to many), which can only be applied to a theory that is already quantum — hence "second".
- It is not the **Feynman path integral**, which lacks the σ-additivity that would make it an integral in the first place.
- It is not a **perturbative expansion** or a **lattice regularization**: both are well-defined approximations, but they are complementary, so neither can serve as the *definition*.

The positive proposal is that quantization is a consequence of **time-evolution**: the wave-function is one parametrization of an arbitrary probability distribution, the parametrization is a surjective map from a hypersphere to the set of all probability distributions, and two wave-functions are always related by a rotation of that hypersphere. The non-commutativity of operators is then intrinsic to *any* statistical theory, not a deformation imposed on a classical algebra.

1.7. Why Parametrize a Probability by a Wave-function?

The parametrization $p_k = \psi_k^2$ is many-to-one. Both ψ and $-\psi$ give the same distribution, and over the complex numbers the whole **phase** $e^{i\theta}$ is invisible to $p_k = |\psi_k|^2$. The wave-function therefore carries *more* information than the distribution: it remembers a phase that the probabilities have forgotten.

This redundancy is not a defect; it is the origin of **interference**. On the simplex, probabilities only ever add. On the sphere, wave-functions add *first* and are squared *afterwards*, so two alternatives ψ and ϕ combine as

$$|\psi + \phi|^2 = |\psi|^2 + |\phi|^2 + 2\,\mathrm{Re}(\overline{\psi}\,\phi),$$

and the final **cross term** is precisely the interference that has no analogue on the simplex. The double-slit experiment, the Stern–Gerlach experiment, and the rest of the phenomenology of quantum mechanics are, in this view, consequences of doing linear algebra on ψ before applying Born's rule.

The programme of this book is to make this precise and to prove it. The slogan is: **quantum mechanics is what probability theory looks like when you parametrize the simplex by the sphere.**

The slogan is rhetoric, and it should be read with the manuscript's own caveat attached: quantum mechanics is a generalization of **classical statistical mechanics**, **not** of probability theory itself. The parametrization by a wave-function is available for any probability distribution and changes none of the Kolmogorov axioms; what quantum mechanics adds is the freedom to act on the parametrized distribution **non-deterministically**. That distinction is stated precisely in the chapter on deterministic transformations, and again in the chapter on collapse and the Kolmogorov axioms.

The unitary time-evolution of quantum mechanics becomes a rotation of the sphere; the quantization of energy becomes a statement about periodic rotations; gauge symmetry becomes the invisibility of the phase; and the **classical limit** becomes the question of when the interference cross terms become negligible.

1.8. Euler's Formula as the Parametrization

There is an even more explicit form of the same parametrization, and it is the one that gives the book its name. Any probability distribution on n outcomes can be written using **Euler angles** $\theta_1, \dots, \theta_{n-1}$ as a telescoping product of sines and cosines:

$$p_1 = \cos^2\theta_1, \quad p_2 = \sin^2\theta_1 \cos^2\theta_2, \quad p_3 = \sin^2\theta_1 \sin^2\theta_2 \cos^2\theta_3, \quad \dots$$

with the last outcome carrying the remaining product of sines. Because every number in $[0,1]$ is a $\cos^2$ of some angle, this parametrization reaches *every* distribution; and because the sines and cosines telescope, the probabilities sum to one **identically**, for every choice of angles. We prove this in The Born rule reproduces every distribution.

For a single two-outcome system the parametrization is the **probability clock** $\Psi(t) = (\cos t, \sin t)$. As t advances, the point travels around the unit circle and the probabilities $(\cos^2 t, \sin^2 t)$ oscillate. The infinitesimal generator of this motion is the matrix

$$J = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}, \qquad J^2 = -\mathbf{1},$$

which squares to minus the identity — it is a real incarnation of the imaginary unit, and **Euler's formula** $e^{tJ} = \cos t\,\mathbf{1} + \sin t\,J$ is the statement that the rotation of the probability clock is the exponential of this generator. This is the bridge from probability to the complex numbers, and we make it explicit in The probability clock and Euler's formula.

1.9. What Is Verified, and How

The formal counterpart of this book is the Lean 4 library `BookProof`. It currently contains well over one hundred modules, each formalizing one self-contained mathematical claim from the source manuscript. The whole library is **`sorry`-free** (no proof is omitted) and **`axiom`-free** in substance: it relies only on Lean's standard `propext`, `Classical.choice`, and `Quot.sound`. (The only `axiom` declarations are two `axiom … : True` placeholders in the P-versus-NP module, which this edition does not cite; since `True` is already provable, they add no logical strength.) The statements quoted in this book are drawn from that library; each is identified by its Lean name so that you can find its full proof.

As a first example, here is the verified statement that the Born-rule/Euler-angle parametrization really does sum to one for any choice of angles — the formal counterpart of the telescoping identity previewed above. It lives in the module `BookProof.ChapterEulerNState`:

```
#check @ChapterEulerNState.euler_sum_one
```

Reading the type: for any sequence of angles θ and any number of outcomes $n \geq 1$, the sum of the Born probabilities $\sum_{k<n} \mathrm{bornProb}\,\theta\,n\,k$ equals 1. The symbol `#check` asks Lean to print the precise statement of an existing theorem; the fact that this block elaborates at all is the kernel confirming that the theorem exists and has exactly this type.

1.10. Roadmap

The book is organized as a tour in eight parts, followed by the proof-plan appendix. The order is the reading order, and it is deliberate: the metamathematical foundation is laid before the probability it hosts, the probability toolkit before the wave-function parametrization that turns it into physics, and the applications — gravity, learning, the arrow of time — after the physics they reuse. Each part opens with its own **How to read this part** note; the list below is the map, in reading order.

A Decidable, Complete Foundation
: Where the sample spaces live before anything is measured on them: the completed Hilbert space is a complete, decidable, conservative extension that does not leak undecidable arithmetic, because its infinite elements are kept internally unselectable. The Solovay–Kopperman tensor product then builds the two-factor spaces used everywhere later — an arbitrary law on a finite factor, tensored with an infinite factor whose law is forced to be the Mehler measure precisely because the language cannot distinguish its elements — and the statistical-model-theory chapter reads statements as operators on these spaces.

Probability as Coherent Belief
: The probabilistic toolkit: the Dutch-book derivation of the probability axioms, the associativity of Bayesian updating, the maximum-entropy characterization of the uniform prior (and why it holds only relative to a chosen parametrization), and the law of total variance.

Determinism, Complementarity, and Collapse
: The conceptual core of the manuscript's quantum-foundations chapters: symmetries as unitary representations; deterministic versus non-deterministic transformations and the origin of complementarity; why wave-function collapse keeps quantum mechanics an ordinary (Kolmogorov) probability theory, and how this differs from Gleason's theorem; the theorem that time-translation is a stochastic process *if and only if* it is deterministic; the double-slit and Bell/CHSH experiments read through that lens; EPR-completeness, relativistic causality, and the classical limit.

Wave-functions, Euler's Formula, and the Born Rule
: The heart of the book: the probability clock and Euler's formula, the fact that the Born rule reproduces every distribution (finite, countable, complex, quaternionic), the gauge ambiguity (phase) of the parametrization, the parametrization of a joint probability by a unitary operator, and information erasure in the Stern–Gerlach experiment.

Relativity, Gauge Theory, and Gravity
: The longest part and the field-theoretic one: gauge symmetry as parametrization redundancy and dissipative dynamics, the free-field construction of a uniform measure on a sphere out of the Gaussian, the spin–statistics dichotomy, real representations and CPT, quantization due to time-evolution (Yang–Mills, the Weyl/CCR relations, the BRST charge), the Gribov ambiguity, physical parity and antiparticles, diffeomorphisms and gravity — and the thread on the **derivative variables** of a field (the Fourier elimination, the Navier–Stokes one-particle Hamiltonian and comparison operator, the quantum-gravity elimination) that closes with the Starobinsky scalaron, the Standard Model, and the SIRK numerics chapters.

Consciousness, Deep Learning, and the Bayesian Prior
: The Bayesian foundations of agency in three registers: no prior is best for all cases and a deterministic prior is still subjective; randomized (deep) learning as a random sampling method that induces a posterior over

models; and the coherent-state reading of Softmax attention, where the Born rule on coherent states is the same equation as the attention mechanism.

Entropy, Irreversibility, and the Arrow of Time
: How an irreversible, entropy-increasing dynamics coexists with deterministic evolution: injective-but-not-surjective maps, the vanishing probability that a random map is invertible, uncountable null-measure sets, cosmological amplification of the matter/radiation ratio, and the law of large numbers.

The ODE Singularity
: An operator-theoretic resolution of the blow-up of $x' = x^2$, via Koopman–von Neumann theory, Weyl quantization, Nelson's essential self-adjointness theorem, and the complexification argument (`ae_no_real_singular_time`) — together with the book's most careful accounting of which results are genuine theorems and which are algebraic certificates.

Appendix: Proof Plans
: Where the unproved claims are quarantined, each with a detailed plan. Nothing in the verified layer is asserted without a proof.

Two reading paths are worth knowing. For the **idea**, the shortest route through the tour is this chapter, the probability clock (the smallest instance of the whole construction), the Dutch-book chapter (what the construction must land in), and the free-field chapter (the same construction at field scale); the part introductions say where to go next. For the **verified machinery**, take the parts in order: the field-theoretic part assumes both the probability toolkit and the Born-rule parametrization, and every later part reuses something from it.

2. A Decidable, Complete Foundation

A metamathematical chapter replacing the manuscript's chapters on *P versus NP* and the *Riemann Hypothesis*: the completed Hilbert space is complete and decidable because its infinite elements are kept internally unselectable. The second chapter builds the Solovay–Kopperman tensor product on this foundation: a finite-dimensional factor carrying an arbitrary probability law, tensored with a separable infinite-dimensional factor whose law is forced to be the Mehler measure precisely because the language cannot distinguish its elements.

How to read this part. Read Completeness without Peano Arithmetic first: its one verified engine is the Riesz–Fischer characterization of completeness by absolutely convergent series, and its one discipline is the refusal to name infinite elements as constants. Everything else in the part — the head–tail decomposition, the Mehler measure on the tail, the decidability of tensor products — is the same discipline applied to two factors instead of one. When the later physics chapters speak of a "free field" or of the "infinite tail" of a state space, this is the precise object they mean; the free-field chapter in Part Relativity is the direct descendant of these two chapters and repays re-reading them.

3. Completeness without Peano Arithmetic

3.1. Orientation and Status

This is the first chapter of the tour, and its foundation: the only place where the **sample space** itself is built. Nothing from the rest of the book is assumed beyond the Introduction.

The chapter runs: Kopperman's warning (why one named infinite constant destroys decidability), the construction of the completed space, the verified core — the Riesz–Fischer characterization of completeness by absolutely convergent series — the **verifiable** unselectability facts, the metamathematical interpretation that is deliberately **not** claimed as a theorem, why this chapter replaces the manuscript's P-versus-NP and Riemann-hypothesis chapters, and the tensor products of decidable languages that the next chapter generalizes.

The verified core lives in `BookProof/ChapterRieszFischer` and `BookProof/ChapterEll`; everything else in the chapter is the analysis of those results and is labelled as such. The continuation is the Solovay–Kopperman tensor product, which needs precisely the tensor-product section at the end here.

This chapter replaces the manuscript's chapters on *P versus NP* and on the *Riemann Hypothesis*. Those chapters both turned on a single metamathematical point, isolated here in self-contained form: a completed Hilbert space can be **topologically complete** and yet **decidable**, provided its infinite elements are kept *internally unselectable*. The chapter is careful to separate what is a verified theorem of analysis from what is a metamathematical interpretation.

3.2. Kopperman's Warning

In Section V of his 1967 paper, Kopperman proved a striking negative result about the logic of Hilbert spaces:

"If even one constant c_0 is added to the language of Hilbert systems, the new theory is no longer compact. This results from the fact that it is possible to introduce a predicate into the language which is satisfied by only a single complex number."

The mechanism is important. A **named constant vector** c_0 lets you write a first-order formula that isolates a specific complex number. From one named complex number you can reconstruct the integers; from the integers you get **Peano Arithmetic (PA)**; and PA is undecidable and non-compact (by Gödel's theorem). So the moment you add a single infinite constant to the language, the clean decidable geometry is contaminated by undecidable arithmetic.

The goal is therefore to have a **complete** Hilbert space — one that contains its limits — **without** adding any infinite element to the language as a primitive constant. The space should be complete *ontologically* (its limits exist) but *epistemologically unselectable* (no infinite element can be named).

The verified formalization of Kopperman's setting confirms that the substrate Hilbert space is **separable** (has a countable dense subset), which is the topological precondition for his model-theoretic analysis:

```
#check @kopperman_substrate_separable
#check @all_formalism_models_separable
```

3.3. The Construction

The construction has three layers.

The dense, decidable core
: Start with the **finitely-supported** vectors $\mathcal{H}_0 = P \to_0 \mathbb{R}_{\text{alg}}$ (a `Finsupp`). Every vector in $\mathcal{H}_0$ has **finite support** by definition, so its representation is strictly finite and decidable. There is no infinite object here at all.

The metric completion
: Define the completed Hilbert space $\mathcal{H}$ as the **metric completion** of $\mathcal{H}_0$: Cauchy sequences of finite vectors, modulo the equivalence relation of converging to the same limit. This is a standard, purely topological construction; it adds limit points without naming any of them.

Existence of elements = absolute convergence
: By the Banach-space characterization below, every element of the completion corresponds to an **absolutely convergent series** of finite vectors. So the infinite elements exist (as equivalence classes of Cauchy sequences / convergent series) but are never introduced as primitive constants.

3.4. The Verified Core: Riesz–Fischer / Banach by Summable Series

The mathematical engine of the whole argument is a genuine, verified theorem of functional analysis, present in Mathlib as `completeSpace_iff_summable_norm`:

A normed space E is complete (a Banach space) if and only if every absolutely convergent series in E converges.

This is the standard tool for proving that L^p spaces are complete, and it is the fact that lets the completion be characterized by convergent series of finite vectors. The Mathlib statement:

```
#check @completeSpace_iff_summable_norm
```

and the completeness of the metric completion itself:

```
#check @UniformSpace.Completion.completeSpace
```

3.5. Sketch Proof of the Characterization

Because this theorem is the verified core, it is worth seeing the proof in full.

(⇒) Complete ⇒ absolutely convergent series converge. Let $\sum_n u_n$ be absolutely convergent, $\sum_n \|u_n\| < \infty$, and let $s_N = \sum_{n<N} u_n$ be the partial sums. For $M > N$, the triangle inequality gives

$$\|s_M - s_N\| = \Big\| \sum_{N \le n < M} u_n \Big\| \le \sum_{N \le n < M} \|u_n\|.$$

Since the real series $\sum \|u_n\|$ converges, its partial sums are Cauchy, so the right-hand side is $< \varepsilon$ for all large N. Hence (s_N) is Cauchy in E, and completeness makes it converge.

(⇐) Absolutely convergent ⇒ complete. Let (x_n) be a Cauchy sequence. Choose a subsequence $x_{k(j)}$ with $\|x_p - x_q\| < 2^{-(j+2)}$ for all $p, q \ge k(j)$, and set

$$u_0 = x_{k(0)}, \qquad u_j = x_{k(j)} - x_{k(j-1)} \ \ (j \ge 1).$$

The partial sums **telescope** to the subsequence: $u_0 + \sum_{j=1}^{M} u_j = x_{k(M)}$. And the series is absolutely convergent, because

$$\sum_j \|u_j\| \le \|x_{k(0)}\| + \sum_{j \ge 1} 2^{-j} = \|x_{k(0)}\| + 1 < \infty.$$

By hypothesis the series converges to some x, so the subsequence $x_{k(M)} \to x$; and a Cauchy sequence with a convergent subsequence converges to the same limit. Hence E is complete. ■

3.6. The Verifiable "Unselectability" Facts

Two pieces of the safety argument are themselves ordinary, verifiable mathematics.

Every writable vector has finite support. This is immediate from the definition of `Finsupp`: a term of type $P \to_0 \mathbb{R}$ carries an explicit finite support, so anything you can actually write down is finitely supported and decidable.

```
#check @Finsupp.finite_support
```

The completion adds exactly the limit points, and nothing else. This is the Riesz–Fischer half of the construction, now proved rather than asserted: the sequence space $\ell^2(\mathbb{N})$ is complete; every one of its vectors is the norm-limit of its finitely-supported truncations; the image of the writable core $\mathbb{N} \to_0 \mathbb{R}$ is **exactly** the set of finitely-supported vectors; and that image is a **proper** subset, so the completion is not a vacuous operation.

```
#check @riesz_fischer
#check @range_ofCore
#check @denseCore_dense
#check @denseCore_proper
#check @completion_conservative_over_core
```

The Riesz–Fischer facts themselves live in `BookProof.ChapterRieszFischer`: the completeness of $\ell^2(\mathbb{N})$ (`ell2_completeSpace`), the unconditional sum representation of every vector (`riesz_fischer_hasSum`), and the properness witness that the completion genuinely adds elements — the geometric vector $n \mapsto 2^{-n}$ is square-summable but has infinite support:

```
#check @BookProof.ChapterRieszFischer.ell2_completeSpace
#check @BookProof.ChapterRieszFischer.riesz_fischer_hasSum
#check @BookProof.ChapterRieszFischer.finSupport_dense
```

```
#check @BookProof.ChapterRieszFischer.finSupport_ne_univ
#check @BookProof.ChapterRieszFischer.geomVec_not_mem_finSupport
#check @BookProof.ChapterRieszFischer.core_proper_and_dense
```

Infinite elements are null events under a diffuse prior. Since the infinite elements cannot be named as constants, the manuscript represents them as measurable sets under a diffuse probability prior. Under any **diffuse** measure (every singleton has measure zero), any particular infinite vector — being a single point of the completed space — has probability zero:

```
#check @MeasureTheory.Measure.noAtoms
```

This is the measure-theoretic form of "the infinite elements are unselectable": no single one of them can be picked out with positive probability.

Only countably many elements are definable. A countable formal language can name only countably many reals, so **most** reals — and hence most elements of the completed space — are not definable in the base language:

```
#check @definable_reals_countable
#check @exists_nondefinable_real
```

This is the definability-theoretic form of unselectability: the elements that Kopperman's trap would need to isolate simply cannot be named.

Moreover the completed space $\ell^2(\mathbb{N})$ is **separable** — a countable language suffices to name a **dense** fragment of it. The finitely-supported **rational** vectors form a countable set (`ratVec_range_countable`) which is still dense (`ratVec_dense`), so the whole completion is the closure of a countable set of nameable vectors:

```
#check @BookProof.ChapterEll2Separable.ratVec_range_countable
#check @BookProof.ChapterEll2Separable.ratVec_dense
#check @BookProof.ChapterEll2Separable.ell2_separable
```

3.7. The Metamathematical Interpretation (Not a Theorem)

We must now be precise about what has and has not been proved.

Proved (verified mathematics)

> The completion $\mathcal{H}$ is a complete space; its elements are absolutely convergent series of finite vectors (Riesz–Fischer). Every *writable* vector is finitely supported and decidable. Under a diffuse prior, individual infinite elements are null events.
>
> Additionally, a **bounded-arithmetic prior** on any finite type is a genuine probability measure, and its "certain extension" to known propositions is verified:
>
> ```
> #check @prior_is_probability
> #check @certainExtension_known
> ```

Interpretation (metamathematics, not a Lean theorem)

> The claim that "the completed space **does not leak Peano Arithmetic**" / "is a **conservative, decidable extension** of the base theories" is a statement about **formal languages and definability**, not an internal statement of analysis. It says: because we never add an infinite element as a primitive constant, no first-order formula of the base language can isolate one, so Kopperman's reconstruction of the integers never gets started.

This interpretation is mathematically reasonable and is exactly the point the manuscript's *P vs NP* and *Riemann Hypothesis* chapters were making, but it is **not** a single theorem that Lean can state and prove inside the theory. Making it fully rigorous would require formalizing a notion of "definable constant in the base language" and proving a conservativity / non-definability result about the completion — a model-theoretic task well beyond the verified analysis above. We record what *is* formalizable in the proof-plan appendix and flag the rest as interpretation.

3.8. Why This Replaces the Two Chapters

Both replaced chapters relied on the same maneuver: an apparently undecidable or intractable object (the complexity of a decision problem; the zeros of the zeta function) is represented not as a **named infinite constant** but as a **limit / expectation over a measure space** — an element of a completion that exists ontologically but is kept unselectable, so that Kopperman's trap never springs. The verified content is always the same analytic fact, the Riesz–Fischer characterization of completeness by absolutely convergent series; the rest is the disciplined refusal to add the infinite object to the language as a constant.

3.9. Tensor Products of Decidable Languages

The construction above gives a single decidable language. The manuscript's quantum chapters require **composing** two such languages — the tensor product of two Hilbert spaces — and the question is whether decidability survives composition.

3.9.1. Head ⊗ Tail Decomposition

Every state in the Solovay–Kopperman space decomposes as a **finite head** (the first N coordinates, carrying the observable content) tensored with an **infinite tail** (the remaining coordinates, carrying the measure-theoretic substrate). The verified equivalence:

```
#check @headSumEquiv
```

splits $\mathbb{R}^{N_1+N_2} \times \mathrm{Tail}$ into $\mathbb{R}^{N_1} \times (\mathbb{R}^{N_2} \times \mathrm{Tail})$, showing that tensoring two finite heads simply concatenates their coordinates while the tail is shared. The state measure respects this:

```
#check @coordinateStateMeasure
```

takes an **arbitrary** probability distribution on the finite head and pairs it with the fixed Mehler (Gaussian) measure on the tail.

3.9.2. Decidability is Preserved

The central theorem: if two observables each depend only on their respective finite heads, then their tensor product depends only on the concatenated head, and the expectation over the product state is decidable.

```
#check @tensor_language_decidable
```

This is the formal content of "the tensor product of two decidable languages is decidable": the infinite tail never needs to be inspected to compute any observable expectation.

3.9.3. The Tail Admits the Mehler Measure

While the head carries an arbitrary probability law, the tail is rigid. The verified theorem:

```
#check @only_mehler_on_tail
```

states that the Mehler (standard Gaussian product) measure is an admissible prior on the tail: it is a probability measure, atomless (no singleton is distinguishable), and invariant under every measure-preserving transformation of the tail. These three properties are the admissibility criteria; the theorem confirms the Mehler prior satisfies all of them. (Full uniqueness — that no other measure can satisfy them — is not asserted; the formal docstring explicitly disclaims it.) The contrast with the head is made explicit:

```
#check @head_vs_tail_admissibility
```

Any probability distribution whatsoever is admissible on the finite head; the Mehler measure is the canonical admissible prior on the infinite tail.

3.9.4. Cross-Dimensional Inner Products

A subtle point: the Solovay inner product of two states with **different** finite-part dimensions is well-defined, because the uniform Mehler measure on the tail **splits** so that the finite coordinates match. The coordinate-level realization:

```
#check @tailSplitEquiv
```

is a measure-preserving equivalence that peels off the first k coordinates from the tail, and:

```
#check @tailTensorEquiv
```

identifies the product of two tails with a single tail (via a bijection $\mathrm{Fin}\,2 \times \mathbb{N} \simeq \mathbb{N}$). Together they show that padding a state with extra zero coordinates does not change its measure class:

```
#check @enlargeEquiv
```

3.9.5. Invariance Under Finite Orthogonal Symmetries

The Mehler measure is invariant under any measure-preserving transformation of the tail coordinates:

```
#check @mehler_invariant_under_finite_orthogonal
```

At the current certificate layer, "finite-orthogonal symmetry" is defined as measure-preservation of the tail prior, so this theorem confirms the interface is consistent (the conclusion is exactly the hypothesis). A coordinate-level finite-rank orthogonal group on the substrate is future work; the physical content — no finite rotation of the unselectable tail can be detected by any observable, because observables depend only on the finite head — is captured by the head-only decidability predicate.

3.9.6. Summary of the Tensor-Product Layer

Verified (Lean theorems)

Head–tail decomposition (`headSumEquiv`); tensor closure and decidability (`tensor_language_decidable`); admissibility of the Mehler tail prior (`only_mehler_on_tail`); head/tail asymmetry (`head_vs_tail_admissibility`); coordinate splitting (`tailSplitEquiv`, `tailTensorEquiv`); enlargement invariance (`enlargeEquiv`); measure-preserving invariance (`mehler_invariant_under_finite_orthogonal`).

Structural consequence (metamathematical)

Because every observable in the tensor-product language still depends only on a finite head, Kopperman's trap never springs: no infinite constant is ever named, and the completed tensor-product space remains a conservative, decidable extension of the base theories.

4. The Solovay–Kopperman Tensor Product

4.1. Orientation and Status

The second chapter of the foundation, and the construction the physics parts run on: a single sample space that is infinite-dimensional yet separable, built as a tensor product of two factors. It answers the open problem the Introduction posed — a separable probability space with an arbitrary measure over real functions — and it needs only the completed-space discipline of the previous chapter: never name an infinite element.

The route: the three-instant ABL reconstruction and its joint and marginal laws (the finite-dimensional model), post-selection and the reconstruction formula, the tensor product itself and why decidable languages close under it, the two factors and their laws (arbitrary on the finite part, the Mehler measure forced on the infinite part), the cross-dimensional inner product, and a closing account of what is proved versus planned.

The proofs are spread across `BookProof/ChapterSolovayHilbertTensor`, `ChapterSolovaySeparableExistence`, `ChapterSolovayCrossDim`, `ChapterSolovayTailDimension`, `ChapterTensorCompleteness`, and the Mehler uniqueness and orthogonal-invariance chapters; the planned items are stated in the chapter and collected in the appendix. The chapter below on statistical model theory closes this part, and the free-field chapter later reuses the Mehler measure as its Gaussian.

4.2. The Question

The Introduction posed a concrete open problem. Engineering needs a probability space over real functions of real variables, but:

"no one has found yet a systematic way to define a separable probability space with an arbitrary probability measure over real functions of real variables (and infinite-dimensional spaces in general)."

The difficulty is the infinite-dimensional part. A separable space is one whose elements can be approximated by a countable set; a non-separable space has elements that "cannot be approximated by a finite set of elements, up to an arbitrarily small error" — and then they cannot be distinguished, named, or computed. So we want a space that is infinite-dimensional (to carry a field, or a trajectory, or a wave-function) yet still separable (so its elements remain approximable), and on which we may still choose a probability measure.

The manuscript also repeatedly forms **tensor products of sample spaces**. A joint probability density lives on a product $X \times Y$; the prior wave-function can be "redefined as a tensor product of two Fock-spaces"; and often "we are interested in a tensor product of sample spaces, some of which have finite degrees of freedom," for instance $\mathbb{Z}_2^n \times \mathbb{R}^m$. The construction below answers the Introduction's problem precisely by building such a tensor product: a **finite-dimensional** factor that carries an **arbitrary** probability law, tensored with a **separable infinite-dimensional** factor that carries a **forced** law.

A simpler finite-dimensional instance of the same problem arises in the double-slit chapter: given only the final screen position, can one reconstruct the intermediate "which slit" trajectory? The Aharonov–Bergmann–Lebowitz (ABL) two-state reconstruction answers this affirmatively. The reconstruction is a finite-phase-space version of the same idea: a quantum trajectory can be measured directly only at its final time, but the intermediate instants can be recovered from the statistics of runs that ended in a chosen final outcome — using probabilities conditional on the final state and the same quantum time-evolution.

4.3. The Three-Instant ABL Reconstruction

Model three instants — initial, intermediate, final — on a finite phase space. A unit initial wave-function Ψ is evolved by a unitary U to the intermediate time, where a measurement in the standard basis yields outcome a with the Born probability and **collapses** the state to e_a:

```
#check @midProb
```

The collapsed state is then evolved by a unitary V to the final time, where outcome f occurs with probability

```
#check @transProb
```

Both are genuine probabilities — non-negative, and each sums to one over its outcomes:

```
#check @midProb_nonneg
#check @midProb_sum
#check @transProb_sum
```

4.4. The Joint and Marginal Laws

The joint law of the intermediate outcome a and the final outcome f is the product of the two Born factors:

```
#check @jointProb
```

Summing over the intermediate outcome gives the marginal law of the final outcome:

```
#check @finalProb
```

The collapsed three-instant process is a genuine probability law: the marginal final probabilities sum to one. This uses only the unitarity of U and V and the normalization $\|\Psi\| = 1$:

```
#check @finalProb_total
```

4.5. Post-Selection: the Reconstruction Formula

Now condition on the final outcome f. The **post-selected** law of the intermediate outcome — the probability that the trajectory passed through a, given that it ended at f — is the conditional probability

$$\mathrm{condProb}(f, a) = \frac{\mathrm{jointProb}(f, a)}{\mathrm{finalProb}(f)}.$$

```
#check @condProb
#check @condProb_nonneg
```

For each fixed final outcome f, this is again a probability distribution over the intermediate outcomes:

```
#check @condProb_sum
```

This is the reconstruction: although a was not recorded, its conditional distribution is determined by the final outcome and the same unitary evolution, and it is a honest probability law.

4.6. Consistency of the Reconstruction

The reconstruction does not depend on which final outcome one post-selects. Summing the post-selected joint law over **all** final outcomes recovers the original intermediate Born distribution:

```
#check @jointProb_sum_final_eq_midProb
```

So the intermediate statistics are stable: marginalizing the reconstructed joint law back over the final outcomes returns exactly the Born probabilities one would have measured directly at the intermediate time. The post-selection adds information about individual runs without distorting the ensemble.

4.7. The Solovay–Kopperman Tensor Product

The ABL reconstruction is a finite-phase-space instance of the general problem. The general construction — the Solovay–Kopperman tensor product — answers the Introduction's question directly: it is a systematic separable probability space carrying an arbitrary law on a finite part and a controlled (forced) law on an infinite part.

4.8. The Two Factors

The total sample space splits as a finite **head** times an infinite **tail**:

```
#check @InnerSpace
#check @InnerHead
#check @InnerTail
```

The head $\mathrm{InnerHead}\, N = \mathrm{Fin}\, N \to \mathbb{R} \cong \mathbb{R}^N$ is an ordinary finite-dimensional Euclidean Hilbert space. The tail $\mathrm{InnerTail}$ is the Kopperman **substrate**, the standard separable real Hilbert space $L^2([0,1])$. Its separability is the topological precondition for everything that follows, and it is verified directly:

```
#check @kopperman_substrate_separable
#check @substrate_decidable_skeleton
```

Separable, but not small: the tail carries a **countable** orthonormal family — the normalized indicators of the disjoint intervals $(1/(n+2),\, 1/(n+1)]$ — and therefore has infinitely many independent directions. The head–tail split is not a finite-dimensional artefact:

```
#check @BookProof.ChapterSolovayTailDimension.substrateBasisVector_orthonormal
#check @BookProof.ChapterSolovayTailDimension.substrate_infinite_dimensional
#check @BookProof.ChapterSolovayTailDimension.tail_infinite_dimensional
```

Thus the tail is infinite-dimensional but **separable**: it possesses a countable dense skeleton of computable approximants. This is exactly the property the Introduction asked for. The finite head is the part we *know*; the

separable infinite tail is the part we are *uncertain* about.

4.9. Decidable Languages Close Under Tensor Product

A **language** here is an explicit Boolean decision procedure: a classifier $\mathrm{decide} : \alpha \to \mathrm{Bool}$. The tensor product of two languages is conjunction on pairs — a point of the product carrier is accepted exactly when both components accept it:

```
#check @tensor_decide_apply
```

Because a conjunction of two decidable propositions is decidable, membership in the tensor language is decidable for every point of the product carrier:

```
#check @tensor_language_membership_decidable
```

This is the first half of the author's goal: **the tensor product of two decidable languages is a decidable language.** The same closure holds for the Hilbert-space languages used in this book. The "cylindrical" observables — those that depend only on the finite head — form a decidable language, because the head is finite-dimensional and hence Tarski-decidable. A tensor product of two such finite-head observables is again a finite-head observable on the combined head:

```
#check @tensorHeadObservable_dependsOnlyOnHead
#check @tensor_language_decidable
```

The combined head is split into its two finite blocks by an explicit equivalence, so "tensor product of languages" and "concatenation of finite heads" are the same operation:

```
#check @headSumEquiv
```

4.10. The Tensor-Product Hilbert Space

The tensor product of the two factors is the total space $\mathrm{InnerSpace}\, N = \mathrm{InnerHead}\, N \times \mathrm{InnerTail}$, and its Hilbert completion is the **Solovay–Hilbert space**:

```
#check @SolovayHilbertSpace
```

The infinite tail itself tensors with a copy of itself back into a single tail: two infinite Gaussian coordinate sequences interleave into one, and the product Mehler law is carried exactly to the Mehler law. This is the measure-theoretic content of $\Gamma \otimes \Gamma \cong \Gamma$ for the infinite factor:

```
#check @tailTensorEquiv
#check @tailTensorEquiv_map
```

The embedding of a wave-function into the completed space preserves the inner product, so the completion is a genuine Hilbert-space tensor product rather than a mere product of sets:

```
#check @toSolovay_inner
```

In the coordinate model the tensor identification is available in full. Two Solovay systems combine into one — the finite heads concatenate and the two Mehler tails interleave — and the combination carries the **product** of the two state laws to the state law of the combined system. That measure-preserving isomorphism induces a unitary of the corresponding L^2 spaces, which is the Hilbert-space form of $H(N_1) \otimes H(N_2) \cong H(N_1 + N_2)$:

```
#check @BookProof.ChapterSolovayHilbertTensor.solovayTensorEquiv
#check @BookProof.ChapterSolovayHilbertTensor.solovayTensorEquiv_map
#check @BookProof.ChapterSolovayHilbertTensor.solovayTensorUnitary
```

And the elementary tensors behave as elementary tensors must: a product $f(x)g(y)$ of two square-integrable factors is square-integrable for the product law, and the inner product of two such pure tensors is the product of the two inner products.

```
#check @BookProof.ChapterSolovayHilbertTensor.tensorLp
#check @BookProof.ChapterSolovayHilbertTensor.inner_tensorLp
```

The completeness half holds too, for any two finite laws: the closed linear span of the pure tensors is **all** of $L^2(\mu \otimes \nu)$. The proof is the classical π–λ argument — the indicator of a measurable rectangle $s \times t$ is the pure tensor $1_s \otimes 1_t$; the measurable sets whose indicator lies in the closed span contain the rectangles and are stable under complements and countable disjoint unions, hence are all of them; and indicators generate L^2. Written as an approximation statement, this is **separation of variables**: every square-integrable function of two variables is an L^2-limit of finite sums $\sum_k f_k(x) g_k(y)$. Together with the multiplicativity of the inner product on pure tensors, it says that $L^2(\mu \otimes \nu)$ **is** the Hilbert tensor product of $L^2(\mu)$ and $L^2(\nu)$ — a statement one can make here even though the library has no Hilbert tensor product to name:

```
#check @BookProof.ChapterTensorCompleteness.pureTensors
#check @BookProof.ChapterTensorCompleteness.indicator_mem_tensorSpan
#check @BookProof.ChapterTensorCompleteness.tensorSpan_eq_top
#check @BookProof.ChapterTensorCompleteness.pureTensors_dense
#check @BookProof.ChapterTensorCompleteness.exists_tensor_approx
```

The practical corollary is the **product basis**. Writing $u \otimes v$ for the pure tensor of two L^2 elements, the inner product multiplies and the norm multiplies, the operation is bilinear and continuous in each slot; so the products $e_i \otimes f_j$ of two orthonormal families are orthonormal, and if both families are total then so is the family of products. A basis of the two-factor state space is therefore obtained by multiplying a basis of each factor — the familiar bookkeeping of a composite quantum system:

```
#check @BookProof.ChapterTensorCompleteness.tensorOf
#check @BookProof.ChapterTensorCompleteness.inner_tensorOf
#check @BookProof.ChapterTensorCompleteness.norm_tensorOf
#check @BookProof.ChapterTensorCompleteness.orthonormal_tensorOf
#check @BookProof.ChapterTensorCompleteness.tensorFamily_span_eq_top
```

Orthonormality and totality can be packaged into a single object. Given a Hilbert basis (b_i) of $L^2(\mu)$ and a Hilbert basis (c_j) of $L^2(\nu)$, the products $b_i \otimes c_j$, indexed by pairs, **are** a Hilbert basis of $L^2(\mu \otimes \nu)$. Its coefficients are the two-variable Fourier coefficients $\langle b_i \otimes c_j, F \rangle$, every square-integrable function of two variables is the unconditional sum of its components, and Parseval holds in the product form $\sum_{i,j} |\langle b_i \otimes c_j, F \rangle|^2 = \|F\|^2$. This is the bookkeeping a composite quantum system actually uses:

```
#check @BookProof.ChapterTensorCompleteness.tensorHilbertBasis
#check @BookProof.ChapterTensorCompleteness.coe_tensorHilbertBasis
#check @BookProof.ChapterTensorCompleteness.tensorHilbertBasis_repr_apply
#check @BookProof.ChapterTensorCompleteness.hasSum_tensorHilbertBasis
#check @BookProof.ChapterTensorCompleteness.hasSum_sq_norm_inner_tensorHilbertBas
```

The decisive simplification is the **decoupling theorem**: because the cylindrical wave-functions depend only on the finite head, and the tail measure is an independent probability measure, the L^2 inner product over the infinite-dimensional total space collapses exactly to a finite-dimensional integral over the head $\mathbb{R}^N$:

```
#check @inner_reduces_to_head
```

This is the dimensional reduction that makes the whole construction *decidable*: an a-priori infinite-dimensional computation reduces to a finite Tarski-decidable integral. It is the formal version of the manuscript's claim that probabilities let us relate arbitrarily complex random events to standard, intuitive ones.

4.11. An Arbitrary Law on the Finite Part

On the finite head we may choose **any** probability law. Given any probability measure $\mathrm{headDist}$ on $\mathbb{R}^N$, the product with the tail law is a probability measure on the total space, and its finite marginal is exactly the law we chose:

```
#check @stateMeasure_isProbability
#check @stateMeasure_finite_marginal
```

Put together, that is precisely the Introduction's question answered: a separable probability space, over an infinite-dimensional carrier, on which the law of the finite part may be chosen arbitrarily while the infinite part carries the Mehler law — in the coordinate model and in the abstract substrate model alike:

```
#check @BookProof.ChapterSolovaySeparableExistence.exists_separable_prob_with_arb
#check @BookProof.ChapterSolovaySeparableExistence.exists_separable_prob_with_arb
#check @BookProof.ChapterSolovaySeparableExistence.separable_carrier_tail_infinit
```

On the finite part the wave-function parametrization is always available: any finite joint law is the squared modulus of a wave-function, and taking the marginal of the Born probabilities gives the marginal of the law. And any joint law whose second factor is standard Borel factors as its own marginal composed with a Markov kernel — the disintegration used to condition without rewriting history:

```
#check @BookProof.ChapterSolovaySeparableExistence.joint_prob_has_wavefunction
#check @BookProof.ChapterSolovaySeparableExistence.joint_prob_has_wavefunction_pr
#check @BookProof.ChapterSolovaySeparableExistence.prod_disintegration
```

The same statement in the explicit coordinate model, where the tail is a countable product of standard Gaussians:

```
#check @coordinateStateMeasure
#check @coordinateStateMeasure_isProbability
```

This is the finite-dimensional freedom the author wants: **our knowledge of the finite-dimensional part is an arbitrary probability distribution.** Nothing constrains it; it is the part we are free to choose.

4.12. Only the Mehler Measure on the Infinite Part

The infinite tail is different. The Kopperman language is **cylindrical**: it can query only finitely many coordinates at a time. Consequently it cannot distinguish two tail points that agree on every finite set of coordinates — no individual element of the infinite-dimensional Hilbert space can be named or singled out. (This is the same mechanism as Completeness without Peano Arithmetic: adding a named infinite element to the language would destroy decidability.)

A probability law on the tail that respects this blindness must be **atomless** (no single state carries mass) and **invariant** under the symmetries the language can express. The selected Mehler/Kopperman prior has precisely these admissibility properties:

```
#check @TailPriorAdmissible
#check @only_mehler_on_tail
```

Concretely, the tail prior is atomless, and it is invariant under every admitted finite orthogonal symmetry:

```
#check @tailMeasure_singleton
#check @mehler_invariant_under_finite_orthogonal
```

And it really lives on the infinite-dimensional sphere: the normalized empirical squared norm of the first k coordinates converges to 1 almost surely (Poincaré–Borel / the strong law), which is Mehler's 1866 limit of the uniform measure on the high-dimensional sphere:

```
#check @mehler_concentrates_on_unit_sphere
```

Mehler's 1866 statement is not a slogan; it is a genuine limit, and it is the single most important structural fact of this part. On the surface of a k-sphere the rotation-invariant (uniform) area element factorizes into single-coordinate weights $(1-x^2)^{(\alpha-1)/2}$ — exactly the **Gegenbauer** weight of order α. The uniform hyperspherical measure is therefore the natural **prior**: it assigns equal weight to every direction, expressing that we have no information. Rescale the coordinate $x \mapsto \sqrt{2/\alpha}\,x$ and let $\alpha \to \infty$. The single-coordinate hyperspherical weight tends to the Gaussian weight e^{-x^2}:

$$\lim_{\alpha\to\infty}\left(1-\tfrac{x^2}{\alpha}\right)^{\alpha-1/2}=e^{-x^2}.$$

At the same time the Gegenbauer polynomials $C_n^{(\alpha/2)}$ (orthogonal with respect to the uniform measure, and defining the hyperspherical harmonics) become the Hermite polynomials $H_n/n!$ (with the physicists' H_n from the recurrence $H_0=1, H_1=2x, H_{n+2}=2xH_{n+1}-2(n+1)H_n$), in the scaled limit

$$\lim_{\alpha\to\infty}\left(\tfrac{\alpha}{2}\right)^{-n/2}C_n^{(\alpha/2)}\left(\sqrt{\tfrac{2}{\alpha}}\,x\right)=\frac{H_n(x)}{n!}.$$

And the normalizations converge consistently: the (squared) Gegenbauer normalization integral against the hyperspherical weight tends to the Hermite normalization integral against the Gaussian weight, $\int_{\mathbb{R}}[H_n(x)/n!]^2e^{-x^2}dx=\sqrt{\pi}\,2^n/n!$. The verified statements live in `BookProof.PhysHSGaussian`:

```
#check @PhysHSGaussian.physHermite
#check @PhysHSGaussian.gegenbauer
#check @PhysHSGaussian.gegenbauerScaled
#check @PhysHSGaussian.gegenbauerScaled_tendsto_hermite
#check @PhysHSGaussian.weight_tendsto_gaussian
#check @PhysHSGaussian.hermite_normalization
#check @PhysHSGaussian.normalization_tendsto
#check @PhysHSGaussian.gaussian_concentration_sphere
```

This is the precise sense in which the uniform (hyperspherical) prior and the Gaussian (Fock) vacuum coincide, and in which the orthogonal excitations of the Fock space are the images of the hyperspherical harmonics. The dictionary is complete:

Finite hypersphere ($\alpha<\infty$)	Infinite-dimensional limit (Fock)
uniform surface measure $d\sigma$	Gaussian measure $e^{-x^2}dx$
Gegenbauer weight $(1-x^2)^{(\alpha-1)/2}$	Gaussian weight e^{-x^2}
Gegenbauer polynomials $C_n^{(\alpha/2)}$	Hermite polynomials $H_n/n!$
hyperspherical harmonics	number states $\lvert n\rangle$
uniform "no-information" prior	Gaussian source qsample $\Psi_0=\lvert 0\rangle$

So the tail prior that the blind language is forced to adopt is not merely **a** rotation-invariant measure: it is the infinite-dimensional **limit** of the family of uniform hyperspherical priors, with the Fock number states as the limits of the hyperspherical harmonics. This is why the free-field prior of the free-field chapter is the same object, and why the Gaussian source qsample $\sqrt{p_0}$ of a flow model coincides with the Fock vacuum. The finite-hypersphere-to-Fock dictionary is the structural backbone of this part and of the whole book.

In the coordinate model the forcing is explicit. The language sees only finite coordinate sets, and restriction to any finite set has the corresponding finite product Gaussian law:

```
#check @finiteCoordinateMarginal
```

So the only data the language can observe is the family of finite-dimensional marginals, and a probability measure on $\mathbb{N}\to\mathbb{R}$ is determined by its finite marginals. The intended conclusion — that the Mehler measure is the *unique* probability law whose every finite marginal is the standard Gaussian product, and hence the *only* law the blind language can use — is now proved: cylinder sets form a generating π-system, so a projective limit of finite measures is unique.

```
#check @BookProof.ChapterMehlerUniqueness.mehler_unique_by_finite_marginals
#check @BookProof.ChapterMehlerUniqueness.mehler_characterization
#check @BookProof.ChapterMehlerUniqueness.language_blind_implies_mehler
#check @BookProof.ChapterMehlerUniqueness.solovay_kopperman_probability_clas
```

The other half is admissibility: the Mehler law is a probability measure, atomless, and invariant. The two halves together give the author's classification:

```
#check @head_vs_tail_admissibility
```

The invariance half is now available in **concrete coordinates**, not only at the abstract measure-preserving interface. The characteristic function of the standard k-dimensional Gaussian is $\exp(-\|t\|^2/2)$, a function of the norm alone, so the finite Gaussian head is invariant under the whole orthogonal group $O(k)$; and rotating the first k coordinates of the infinite tail therefore leaves the Mehler law unchanged:

```
#check @BookProof.ChapterMehlerOrthogonalInvariance.charFun_stdGaussianEucli
#check @BookProof.ChapterMehlerOrthogonalInvariance.stdGaussianEuclidean_map_
#check @BookProof.ChapterMehlerOrthogonalInvariance.gaussianHead_map_orthogo
#check @BookProof.ChapterMehlerOrthogonalInvariance.coordinateTailMeasure_ma
```

Heads admit an arbitrary law; the tail admits the Mehler law. That single statement is the probabilistic content of the Solovay–Kopperman tensor product.

4.13. The Cross-Dimensional Inner Product

Because the Mehler tail **splits** — the first k Gaussian coordinates separate off as an independent finite Gaussian block, leaving a fresh copy of the tail — we may enlarge the finite head by adjoining k tail coordinates:

```
#check @tailSplitEquiv
#check @enlargeEquiv
```

This makes the inner product well-defined **across dimensions**: an element with N head-coordinates and an element with $N + k$ head-coordinates can be compared after padding the smaller one with k fresh Gaussian coordinates, and the result does not depend on the padding, because the added coordinates integrate to 1. The finite dimensions "match" precisely because the Mehler measure is a product. The measure-preservation of the coordinate tail-split — the theorem that splitting off the first k coordinates carries the Mehler law to the product of a finite Gaussian head with a fresh copy of the tail — is proved:

```
#check @tailSplitEquiv_map
```

The cross-dimensional statement is now a theorem, not just an assertion: the enlargement map $N \mapsto N + k$ is a **measure-preserving isomorphism** of the corresponding state spaces, so it is a genuine embedding of the N-dimensional coordinate model into the $(N + k)$-dimensional one. Consequently every expectation, and in particular the L^2 inner product of two wave-functions, is unchanged when it is computed after enlarging the head — the value does not depend on which head dimension the computation is performed in. This is the manuscript's "the inner product is well defined across head dimensions," made precise:

```
#check @BookProof.ChapterSolovayCrossDim.cross_dim_embedding
#check @BookProof.ChapterSolovayCrossDim.integral_cross_dim_well_defined
#check @BookProof.ChapterSolovayCrossDim.inner_cross_dim_well_defined
#check @BookProof.ChapterSolovayCrossDim.lintegral_cross_dim_well_defined
#check @BookProof.ChapterSolovayCrossDim.cross_dim_isProbability
```

4.14. What Is Proved and What Is Planned

This edition separates the verified layer from the open one.

Verified. The ABL three-instant reconstruction (midProb, transProb, jointProb, condProb, consistency); the two factors and their separability; closure of decidable languages under tensor product; the tensor-product Hilbert space and the decoupling theorem; the arbitrary finite-head law and its marginal; the admissibility of the Mehler tail (probability, atomlessness, invariance, concentration on the sphere); and the self-interleaving of the infinite tail. Every theorem cited above is `sorry`-free.

Newly verified. The two statements previously quarantined as planned are now theorems: the **uniqueness** of the Mehler tail law from its finite marginals (`mehler_unique_by_finite_marginals`, the forcing half of "only the Mehler measure") and the **measure-preservation** of the coordinate tail-split

(`tailSplitEquiv_map`), from which the cross-dimensional inner product follows — together with the coordinate-level orthogonal invariance of the Mehler law cited above.

4.15. Why This Matters Here

The ABL reconstruction shows that even when only the final outcome is known, the intermediate trajectory can be recovered as a probability law — a finite-phase-space instance of the general principle. The Solovay–Kopperman tensor product extends this to the infinite-dimensional case: it is a systematic separable probability space carrying an arbitrary law on a finite part and a controlled (forced) law on an infinite part. Together they explain the recurring tensor products of the manuscript — a joint density on $X \times Y$, a prior as a tensor product of two Fock-spaces, a sample space $\mathbb{Z}_2^n \times \mathbb{R}^m$ — as instances of a single principle: tensor a finite, freely-chosen factor with a separable infinite factor whose law is forced by the very fact that the language cannot distinguish its elements. Probability on the infinite part is not a choice; it is the Mehler measure.

This also identifies the construction as the **single exception** to the slogan "there are no non-informative priors." On a general continuous space a uniform prior stops being uniform under a change of coordinates (non-informativeness is coordinate-dependent); here the cylindrical language admits no such change — arbitrary unitary transforms are not defined on the non-metrically-complete restricted space — so the Mehler prior on the infinite-dimensional hypersphere stays uniform under every symmetry the language can express. It is uniform not by a parametrization-dependent trick (the finite case) but intrinsically, because the offending reparametrizations are absent from the language.

5. Statistical Model Theory, and Statements as Operators

5.1. Orientation and Status

The last chapter of the foundation part, read best after the Solovay–Kopperman tensor product: it uses the same completed-space discipline (no named infinite elements) but turns from **spaces** to **language** — what a theory can pin down, and what a statement is once its models form a Hilbert space. No field theory and no probability beyond the Introduction is needed.

The chapter has two halves and says so: first-order axioms never select a single uncountable structure over a countable language (while second-order axioms can), then a statement becomes an orthogonal projection whose truth value is a number rather than a bit. The probabilistic toolkit that the next part builds — the Dutch-book chapter and after — is exactly what turns those numbers into degrees of belief.

This chapter formalizes the model-theoretic content of the manuscript's closing chapter, **Statistical Model Theory and Bayesian priors where the Riemann Hypothesis is true** — everything of it except the Riemann-hypothesis prior, which stays out of scope by construction. It has two halves that answer one question from opposite sides. The first asks what a **theory** can pin down: over a countable language, first-order axioms never select a single uncountable structure, while second-order axioms can. The second asks what a **statement** is when the models it talks about form a Hilbert space: an orthogonal projection, with a truth value that is a number rather than a bit.

Both halves are `sorry`-free and `axiom`-free, in `BookProof/ChapterStatisticalModelTheory.lean` and `BookProof/Probability/ChapterStatementOperator.lean`. They are the only chapters of the manuscript's closing section with a counterpart in the proof library, and they deliberately stop at the boundary the manuscript itself draws: no infinitary $L_{\omega_1\omega_1}$ logic, no Abstraction Logic, no completeness claim for either, and no Bayesian prior on the Möbius function.

5.2. What First-Order Logic Cannot Do

The chapter opens with the upward Löwenheim–Skolem theorem, and it is worth seeing why that theorem is the engine of everything below. If a first-order theory has an infinite model, it has models of **every** infinite cardinality large enough for the language (`upward_lowenheim_skolem`); in particular a theory with an infinite model has an

uncountable one (`exists_uncountable_model`), and over a countable language it has one of cardinality exactly the continuum (`exists_uncountable_model_of_countable_language`) — the "monster" model (`monster_model_continuum`) that model theorists reason with.

The negative consequence is the one the manuscript quotes as a slogan: **you cannot filter out the uncountable models using first-order axioms** (`countability_not_first_order_axiomatizable`). Whatever countable first-order axioms you write, if they have an infinite model at all they have uncountable ones, and pairwise non-isomorphic ones (`exists_nonisomorphic_models`) — which is exactly why Presburger arithmetic, for instance, has non-standard models (`exists_nonstandard_model`) rather than being pinned down to $\mathbb{N}$. Every countable model of a countable theory is embeddable (`exists_countable_model`), and no uncountable model is bijective with a countable one under those hypotheses (`exists_model_not_bijective_of_uncountable`).

The contrast that gives the section its point is the second-order case: `secondOrder_categoricity_real` states that every conditionally complete linear ordered field is order-ring-isomorphic to $\mathbb{R}$, uniquely — the second-order completeness axiom **is** categorical where the first-order scheme is not. The two results together are the manuscript's contrast between the two logics made precise: first-order axioms over a countable language never pin down an uncountable structure, while a second-order axiom can.

```
#check @BookProof.ChapterStatisticalModelTheory.upward_lowenheim_skolem
#check @BookProof.ChapterStatisticalModelTheory.exists_uncountable_model
#check @BookProof.ChapterStatisticalModelTheory.monster_model_continuum
#check @BookProof.ChapterStatisticalModelTheory.exists_uncountable_model_of_countable
#check @BookProof.ChapterStatisticalModelTheory.countability_not_first_order_axiomati
#check @BookProof.ChapterStatisticalModelTheory.exists_nonisomorphic_models
#check @BookProof.ChapterStatisticalModelTheory.exists_nonstandard_model
#check @BookProof.ChapterStatisticalModelTheory.exists_countable_model
#check @BookProof.ChapterStatisticalModelTheory.exists_model_not_bijective_of_uncount
#check @BookProof.ChapterStatisticalModelTheory.secondOrder_categoricity_real
```

5.3. Statements as Operators in a Hilbert Space

The second half implements the chapter's other image: **statements become operators in a Hilbert space**. A `ModelStatement` is an orthogonal projection — a bounded, self-adjoint, idempotent operator `S.op` on the Hilbert space of models — and its **truth value** on a model vector ψ is $\mathrm{Re}\langle\psi, S\psi\rangle$, a real number in $[0, \|\psi\rangle^2]$ (`truthValue_nonneg`, `truthValue_le_norm_sq`, `truthValue_mem_unitInterval`). The extreme cases are identified exactly: the value vanishes precisely on the models in the kernel (`truthValue_eq_zero_iff`) and reaches the norm squared precisely on the range (`truthValue_eq_norm_sq_iff`). So "true", "false" and "undecidable" are no longer three colors of a bit but positions of a spectrum — which is what the manuscript means by statements becoming "more than true/false/undecidable".

The propositional calculus is then the operator calculus: negation is $1 - S$ with `truthValue_not` making the values complementary, and conjunction of two **commuting** statements is the composition of their projections, with `truthValue_and_le_left` and `truthValue_and_le_right` the monotonicity the reader expects. An **undecidable** statement is precisely a projection with a proper nonzero range (`Undecidable`, `undecidable_iff`). The chapter then steps outside projections: `UncertainStatement` is the book's "operators but not projections", with `halfUncertain_not_idempotent` as the witness that a half-true statement is not idempotent, and `approximate_proof_error` bounds how well an operator within ε of a statement evaluates every model: $|\langle A\rangle - \langle P\rangle| \le \|A - P\| \cdot \|\psi\|^2$, the "approximated proof" of the manuscript.

```
#check @BookProof.ChapterStatementOperator.ModelStatement
#check @BookProof.ChapterStatementOperator.ModelStatement.truthValue
#check @BookProof.ChapterStatementOperator.ModelStatement.truthValue_nonneg
#check @BookProof.ChapterStatementOperator.ModelStatement.truthValue_mem_unitInterval
#check @BookProof.ChapterStatementOperator.ModelStatement.truthValue_eq_zero_iff
#check @BookProof.ChapterStatementOperator.ModelStatement.truthValue_eq_norm_sq_iff
#check @BookProof.ChapterStatementOperator.ModelStatement.not
#check @BookProof.ChapterStatementOperator.ModelStatement.truthValue_not
#check @BookProof.ChapterStatementOperator.ModelStatement.and
#check @BookProof.ChapterStatementOperator.ModelStatement.truthValue_and_le_left
#check @BookProof.ChapterStatementOperator.ModelStatement.Undecidable
#check @BookProof.ChapterStatementOperator.ModelStatement.undecidable_iff
```

```
#check @BookProof.ChapterStatementOperator.UncertainStatement
#check @BookProof.ChapterStatementOperator.halfUncertain_not_idempotent
#check @BookProof.ChapterStatementOperator.approximate_proof_error
```

Honest boundary. Nothing above is claimed about infinitary $L_{\omega_1\omega_1}$ logic, about Abstraction Logic, or about the completeness of either; and nothing is claimed about the Bayesian prior on the Möbius function, which is the part of the manuscript's closing chapter this chapter does not touch. The projection model of a statement is a **model** — it formalizes the chapter's image, not a thesis about the foundations of mathematics.

5.4. Summary

- `upward_lowenheim_skolem`, `monster_model_continuum`, `countability_not_first_order_axiomatizable` — first-order theories with an infinite model have models of every sufficiently large cardinality, so countability is not first-order axiomatizable.
- `exists_nonisomorphic_models`, `exists_nonstandard_model` — the non-uniqueness that Presburger arithmetic's non-standard models exemplify.
- `secondOrder_categoricity_real` — the second-order completeness axiom is categorical: $\mathbb{R}$ is unique up to order-ring isomorphism.
- `ModelStatement.truthValue`, `truthValue_not`, `truthValue_and_le_left` — statements as projections, with truth values in $[0, \|\psi\|^2]$ and the propositional calculus as the operator calculus.
- `undecidable_iff`, `halfUncertain_not_idempotent`, `approximate_proof_error` — undecidability as a proper nonzero range, uncertainty as non-idempotence, and the error bound of an approximated proof.
- open: infinitary and Abstraction Logic, their completeness, and the Bayesian prior on the Möbius function — out of scope by construction.

6. Probability as Coherent Belief

The second part builds the probabilistic foundation on which the rest of the book rests. We begin from the most primitive question — *what does it mean for a system of degrees of belief to be consistent?* — and recover the probability axioms as the answer (the Dutch-book theorem). We then establish the three working tools used throughout: the associativity of Bayesian updating, the maximum-entropy justification of the uniform prior, and the decomposition of variance into *within-group* and *between-group* parts.

How to read this part. The four chapters are a single toolkit, and they are used constantly in the rest of the book. The Dutch-book theorem fixes the **target** of every later construction: whatever the wave-function parametrization produces, it must land inside the coherent price systems, i.e. inside the simplex. Bayesian updating is associative is the coherence property that makes the posterior a genuine state of knowledge, and maximum entropy and the law of total variance are the two quantitative instruments — the former a principle for **choosing** priors within a parametrization, the latter a bookkeeping identity for **combining** sources of uncertainty. A reader who wants the conceptual core can read just the Dutch-book chapter and the law of total variance; the other two are needed by the consciousness and deep-learning parts later.

7. The Dutch-book Theorem: Probability Is Coherence

7.1. Orientation and Status

The Dutch-book result is a finite sample-space theorem. It establishes the precise coherence/probability equivalence used by the manuscript, while continuous-space and decision-theoretic extensions require separate measure and utility hypotheses.

This chapter opens the probability part of the tour and fixes its **target**: every later construction — above all the wave-function parametrization — must land inside the coherent price systems, i.e. inside the simplex. It is self-contained; the route is the question (why bookies force a probability), the formal setup, the sketch proof, the

verified statement, and two closing readings: why the result matters here, and why the **rules** are objective while the **priors** are not.

The proof is `BookProof/ChapterDutchBook`. A reader in a hurry can read this chapter and the law of total variance alone for the conceptual core of the part.

7.2. The Question

Before we can parametrize probability distributions by wave-functions, we should ask what a probability distribution *is*, and why its rules are the rules they are. The cleanest answer is due to de Finetti and Ramsey, and it is operational: probabilities are **coherent betting prices**.

Imagine a bookmaker who posts a price $\Pr(A)$ for each event A in a finite sample space Ω. A bettor may buy, for a stake s, a ticket on A that pays s if A happens and returns the price $s \cdot \Pr(A)$ regardless. The bookmaker's **net payoff** in a state ω from a finite family of such bets is

$$\mathrm{payoff}(\omega) = \sum_i s_i \big(\mathbf{1}_{A_i}(\omega) - \Pr(A_i)\big).$$

The posted prices are **incoherent** if the bettor can choose stakes so that the bookmaker loses money in *every* state — that is, $\mathrm{payoff}(\omega) < 0$ for all ω. Such a guaranteed-loss portfolio is called a **Dutch book**. The prices are **coherent** if no Dutch book exists.

The theorem, made precise below, is the striking fact:

The coherent price systems are exactly the probability distributions.

In other words, the Kolmogorov axioms — non-negativity, normalization, and finite additivity — are not arbitrary conventions. They are precisely the condition under which a bookmaker cannot be forced into a sure loss. Probability is *consistency of partial belief*.

7.3. The Formal Setup

We work on a finite sample space Ω. A price system is a function $\Pr : \mathrm{Finset}\,\Omega \to \mathbb{R}$ assigning a real number to each event. The two central definitions are:

```
#check @ChapterDutchBook.HasDutchBook
#check @ChapterDutchBook.Coherent
```

Thus `HasDutchBook Pr` says there is a finite family of bets whose payoff is strictly negative in every state, and `Coherent Pr` is its negation. A genuine probability distribution is recorded by:

```
#check @ChapterDutchBook.IsProb
#check @ChapterDutchBook.Represents
```

Here `Represents Pr p` means that the price of every event is the p-measure of that event, $\Pr(A) = \sum_{\omega \in A} p(\omega)$.

7.4. Sketch Proof

Easy direction — a probability is coherent. Suppose $\Pr(A) = \sum_{\omega \in A} p(\omega)$ for a genuine distribution p. Take the p-expectation of the payoff:

$$\sum_\omega p(\omega)\,\mathrm{payoff}(\omega) = \sum_i s_i \Big(\sum_\omega p(\omega) \mathbf{1}_{A_i}(\omega) - \Pr(A_i) \Big) = \sum_i s_i \big(\Pr(A_i) - \Pr(A_i)\big) = 0.$$

A quantity whose p-average is 0 cannot be strictly negative at every state (since p is a probability and at least one state has positive weight). So no Dutch book exists.

Hard direction — coherence forces the axioms. Assume $\Pr$ is coherent. Each axiom is forced by an explicit Dutch book that would otherwise be available to the bettor:

- $\Pr(\varnothing) = 0$: if the price of the empty event were positive, buy a ticket on it — it never pays, so the bookmaker loses the price in every state.
- $\Pr(\Omega) = 1$: if the price of the certain event were not 1, a single ticket on Ω (which always pays its stake) yields a sure gain or loss.
- $0 \le \Pr(A) \le 1$: a negative price, or a price above 1, is exploited by a single ticket on A.
- **Finite additivity on disjoint events**: if $A \cap B = \varnothing$ but $\Pr(A \cup B) \neq \Pr(A) + \Pr(B)$, a three-ticket portfolio — long $A \cup B$ and short A and B (or the reverse) — locks in the discrepancy as a sure profit.

Once additivity holds, the price of any event is the sum of the prices of its singletons, $\Pr(A) = \sum_{\omega \in A} \Pr(\{\omega\})$. So the distribution $p(\omega) := \Pr(\{\omega\})$ **represents** $\Pr$, and the axioms above make p a genuine probability. This is the converse direction.

7.5. The Verified Statement

The full equivalence is the headline theorem of `BookProof.ChapterDutchBook`:

```
#check @ChapterDutchBook.coherent_iff_exists_prob
```

The two directions are also available separately. That a represented probability is coherent:

```
#check @ChapterDutchBook.represents_isProb_coherent
```

and that coherence yields a representing distribution, built from the singleton prices:

```
#check @ChapterDutchBook.coherent_exists_prob
#check @ChapterDutchBook.Coherent.additive
```

The last of these is the finite-additivity step, the only part of the argument that needs a genuinely new idea (the three-ticket book); the rest are one-ticket constructions.

7.6. Why This Matters Here

The Dutch-book theorem fixes the stage on which the rest of the book is played. Whatever parametrization we choose for a probability distribution — and we will choose a wave-function — it must land inside the coherent price systems, i.e. inside the simplex. The parametrizations of Part II will be maps *into* this simplex, and the theorem above tells us exactly what it means for the image to be a legitimate probability.

7.7. The Rules Are Objective; The Priors Are Not

There is a misreading the theorem invites, and the source manuscript is explicit in resisting it. The Dutch-book argument makes the **rules** of probability objective: coherence forces non-negativity, normalization, and finite additivity, and there is no choice in the matter. It does **not** make the **priors** objective. Coherence constrains the *form* a probability must take — it must be a point of the simplex — but it is silent on *which* point of the simplex is the right one. The prior is a free, unavoidable act of theoretical prejudice.

The manuscript leans on this repeatedly, citing the result of Eaton and Freedman (*Dutch book against some 'objective' priors*, Bernoulli 10(5), 2004, 861–872): "in Bayesian inference there is always a prior probability distribution, and there is no prior which is better for all cases." Priors that are marketed as "objective" — symmetry priors, reference priors, and the uniform prior among them — can themselves be Dutch-booked; no single prior is good for every problem. The same source draws the sharper conclusion that "there are no non-informative priors in Bayesian inference, therefore theoretical prejudice is unavoidable."

This is the betting-theoretic complement to the coordinate argument of the previous chapter. There the point was that the uniform prior's distinguishing properties — relabeling-invariance, maximum entropy — hold only *within a fixed parametrization*, and that any non-null finite prior can be reparametrized into the uniform one. Here the point is that coherence cannot certify any prior as the correct one, because against any proposed "objective" prior an

adversary can construct a Dutch book in some problem. The uniform prior is appropriate "in many cases, not in all cases": a useful default, not an objective truth.

So the Dutch-book theorem fixes the **objective** part of probability — the calculus, the simplex, the rules any coherent belief must obey — and leaves the **subjective** part exactly where it belongs, in the choice of prior. The parametrizations of Part II map into this objectively fixed simplex; the point they land on is never forced by coherence alone. The one place a prior *is* forced is the infinite-dimensional tail of the Solovay–Kopperman construction, and there it is forced not by Dutch-book objectivity but by the blindness of the decidable language: the Mehler measure is the only law the language can express, not the only law a rational agent may hold.

8. Bayesian Updating Is Associative

8.1. Orientation and Status

The second instrument of the probability part: the coherence property that makes a posterior a genuine state of knowledge — updating in any order gives the same answer. It follows the Dutch-book chapter (which fixed the simplex as the target) and precedes the maximum-entropy chapter, which builds on the uniform prior characterized here.

The route: one-piece-at-a-time updating and its associativity, the sketch proof, the verified statement, the fact that every prior is a posterior from the uniform prior, the precise sense in which non-informativeness is coordinate-dependent, why the uniform prior is special only within a parametrization — and the exception that the later physics leans on: infinite-dimensional spaces need the Mehler formalism of the Solovay–Kopperman chapter.

The proof is `BookProof/ChapterSequentialBayes`.

8.2. Updating One Piece of Evidence at a Time

Bayesian inference updates a **prior** distribution into a **posterior** by multiplying by a likelihood and renormalizing. On a finite hypothesis space X, with prior $\pi : X \to \mathbb{R}$ and likelihood weight $\ell : X \to \mathbb{R}$, the update is

$$\mathrm{bayesUpdate}(\pi, \ell)(x) = \frac{\pi(x)\,\ell(x)}{\sum_{x'} \pi(x')\,\ell(x')}.$$

The denominator is the **evidence** — the total probability of the observation — and it is what makes the posterior sum to one.

Now suppose two observations arrive. There are two ways to use them:

- **Sequentially.** Update on the first observation, take the resulting posterior as the new prior, and update again on the second.
- **In a batch.** Multiply the prior by the product likelihood $\ell_1(x)\,\ell_2(x)$ and renormalize once.

The fundamental coherence property of Bayesian inference is that **these two procedures give exactly the same posterior**. The order in which evidence is processed, and whether it is processed one item at a time or all at once, is irrelevant. This is the associativity of Bayesian updating.

8.3. Sketch Proof

Write the evidence of the first update as $Z_1 = \sum_x \pi(x)\ell_1(x)$. After the first update the prior is $\pi_1(x) = \pi(x)\ell_1(x)/Z_1$. Updating π_1 on ℓ_2 gives

$$\pi_2(x) = \frac{\pi_1(x)\,\ell_2(x)}{\sum_{x'} \pi_1(x')\,\ell_2(x')} = \frac{\dfrac{\pi(x)\ell_1(x)}{Z_1}\,\ell_2(x)}{\sum_{x'} \dfrac{\pi(x')\ell_1(x')}{Z_1}\,\ell_2(x')}.$$

The factor $1/Z_1$ appears in every term of both numerator and denominator, so it **cancels**:

$$\pi_2(x) = \frac{\pi(x)\,\ell_1(x)\,\ell_2(x)}{\sum_{x'} \pi(x')\,\ell_1(x')\,\ell_2(x')} = \mathrm{bayesUpdate}(\pi,\, \ell_1\ell_2)(x).$$

The intermediate normalization is exactly what disappears. This is why a Bayesian agent never needs to remember *how* it accumulated its evidence — only the current posterior matters, and it is the same as if all the evidence had been weighed at once.

8.4. The Verified Statement

The associativity, and the bookkeeping that the second-step evidence factorizes correctly, are proved in `BookProof.ChapterSequentialBayes`:

```
#check @ChapterSequentialBayes.sequential_eq_batch
#check @ChapterSequentialBayes.bayesUpdate_evidence
```

The same module also confirms the basic sanity of the update — the posterior is non-negative and, given positive evidence, sums to one:

```
#check @ChapterSequentialBayes.bayesUpdate_nonneg
#check @ChapterSequentialBayes.bayesUpdate_sum_one
```

8.5. Every Prior Is a Posterior From the Uniform Prior

A fact the source manuscript uses repeatedly is that **no finite prior is primitive**. Any finite prior can be obtained by starting from the uniform prior and conditioning on suitable data.

Given a target distribution q on X, define a binary likelihood

$$L(x, \mathsf{true}) = q(x), \qquad L(x, \mathsf{false}) = 1 - q(x).$$

Starting from any positive constant prior and conditioning on the outcome `true`, the posterior is exactly q. So the choice of a prior can always be re-expressed as the choice of a likelihood acting on the uniform prior.

```
#check @ChapterUniformPriorPosterior.exists_likelihood_uniform_prior_posterior
```

This is sometimes misread as saying that the uniform prior is the natural, non-informative starting point. It says no such thing. The information has not vanished; it has only **moved from the prior into the likelihood**. The uniform prior is not privileged here — it is merely a convenient origin from which every other prior is reachable. The next sections make precise both why the uniform prior is *not* intrinsically special, and what is actually true in the slogan "there are no non-informative priors."

8.6. The Real Sense: Non-Informativeness Is Coordinate-Dependent

The slogan refers to a different and deeper fact: **non-informativeness is not invariant under a change of coordinates.** A density that is flat — and therefore apparently non-informative — in one parametrization is generally *not* flat in another. Under a smooth reparametrization $y = f(x)$ the density picks up the Jacobian factor $|dx/dy|$, so a constant p_x becomes a non-constant p_y. The prior that "assumed nothing" in the x-coordinates quietly assumes something in the y-coordinates.

The converse is just as important, and it is what removes any special status the uniform prior might seem to have: **any prior that is non-null on a finite set of events can be made uniform on that same set by a reparametrization.** Given a distribution p on $\{0, \dots, n-1\}$ with $p_k > 0$, form its cumulative sum $\mathrm{cdf}(k) = \sum_{i<k} p_i$ and partition the unit interval into the seed intervals $[\mathrm{cdf}(k), \mathrm{cdf}(k+1))$, whose lengths are exactly the p_k. A *uniform* seed on $[0, 1)$, decoded by which interval it falls in, reproduces p; read the other way, the cumulative-sum map sends p back to the uniform measure. The two distributions are the *same* prior written in different coordinates:

```
#check @InverseTransform.seedSet_measure
#check @InverseTransform.seedSet_cover
#check @InverseTransform.seedSet_total_measure
```

So uniformity is not an intrinsic property of a prior at all. It is a property of a prior *relative to a chosen parametrization*. On a finite set every non-null prior is "uniform in disguise," and which disguise counts as the neutral one is itself an informative choice. *This* is the precise sense of "there are no non-informative priors": an apparently non-informative prior becomes apparently informative the moment the coordinates are changed — and an apparently informative one can be made to look non-informative just as easily. The measure-theoretic obstructions behind the same slogan are taken up in null-measure sets need not be small and the free-field chapter.

8.7. The Uniform Prior Is Special Only Within a Parametrization

None of this means the standard characterizations of the uniform prior are false. They are true — and worth proving — but each of them is a statement *relative to a fixed parametrization*, and that caveat is essential.

Within a fixed labeling of the hypotheses, the uniform prior is the **unique** prior that treats all labels symmetrically: a prior unchanged under every permutation of the labels must be constant, and a constant probability distribution on a finite non-empty space is the uniform one, $1/|X|$.

```
#check @ChapterUniformPrior.isRelabelingInvariant_iff_constant
#check @ChapterUniformPrior.normalized_isRelabelingInvariant_eq_uniform
```

But "relabeling-invariant" privileges one particular labeling of the hypotheses. The previous section showed that any non-null prior can be reparametrized into the uniform one; in those new coordinates it is the reparametrized prior, not the old uniform one, that is "the invariant" prior. Symmetry under relabeling is a property of the *coordinate system*, not an intrinsic property that singles out one prior across all parametrizations. The same caveat attaches to the maximum-entropy characterization (the maximum-entropy chapter): entropy is computed with respect to a chosen base measure, and changing coordinates changes which distribution maximizes it.

A useful corollary connects two estimators that are often confused. The **maximum a posteriori** (MAP) estimate maximizes the posterior $\pi(x)\,\ell(x)$; the **maximum-likelihood** (MLE) estimate maximizes $\ell(x)$ alone. Under a positive uniform prior the factor $\pi(x)$ is constant, so the two coincide:

```
#check @ChapterUniformPrior.uniform_prior_isMAP_iff_isMLE
```

This is the rigorous content of the common advice that "maximum likelihood is Bayesian inference with a flat prior" — always understood as "flat *in the chosen coordinates*."

8.8. The Exception: Infinite-Dimensional Spaces Need the Mehler Formalism

On a **finite** set of events, then, uniform priors are cheap: any non-null prior can be reparametrized into one. But that uniformity is parametrization-dependent, hence not genuinely non-informative. The genuinely new case is not a merely *countably* infinite set of events but an **infinite-dimensional space** of events — infinitely many degrees of freedom. There is no Lebesgue probability measure on such a space (no translation-invariant finite measure), and the reparametrization trick that worked in finite dimension is exactly what destroys uniformity there. To have a uniform prior on an infinite-dimensional space at all, one needs the **Mehler formalism**.

A countably infinite set of events is a different, and worse, case still: there the uniform prior is not merely parametrization-dependent but **impossible outright**. There is no uniform probability measure on a countable space — assigning every outcome the same positive mass gives an infinite total, while mass zero gives total zero:

```
#check @ChapterNoUniformCountable.no_uniform_countable_measure
```

This is the measure-theoretic reason the construction below lives on a genuine continuum (the rationals are not enough) rather than on a countable set of events.

This is the one setting in the manuscript where the slogan "there are no non-informative priors" does not bite, and it is the setting this book is built around. In the Solovay–Kopperman inner-product space the infinite tail is not a completed Hilbert space on which arbitrary unitary transforms act. The admissible language is **cylindrical**: it may query only finitely many coordinates, so it cannot name or distinguish individual elements of the infinite-

dimensional space. Because of that restriction the space is **not necessarily metrically complete**, and arbitrary unitary transforms are simply **not defined** within the language. The reparametrizations that would turn a uniform prior into an informative one are not available.

The symmetries that survive are precisely the **admissible finite-orthogonal symmetries** of the tail — the measure-preserving maps the language can express — and the Mehler prior is invariant under every one of them. It is a probability measure, it is atomless, and it concentrates on the infinite-dimensional unit sphere:

```
#check @only_mehler_on_tail
#check @mehler_invariant_under_finite_orthogonal
#check @mehler_concentrates_on_unit_sphere
```

So within the restricted decidable language the Mehler uniform prior on the infinite-dimensional hypersphere **is** genuinely uniform: every symmetry the language can express leaves it unchanged, and there is no admissible change of coordinates that could make it informative. Finite sets admit uniform priors only by a parametrization-dependent trick; a countable infinity of events admits none at all; the infinite-dimensional space admits one only through the Mehler formalism, and there it is intrinsic. Probability on the infinite-dimensional part is not a choice; it is the forced Mehler measure (The Solovay–Kopperman Tensor Product).

9. Maximum Entropy Selects the Uniform Prior

9.1. Orientation and Status

The second, independent route to the uniform prior: not relabeling invariance (the previous chapter) but Jaynes' maximum-entropy principle. The two routes agree — and both are relative to a fixed parametrization, a point the closing section drives home. The chapter is self-contained once the Bayesian-updating chapter is read.

The route: measuring ignorance (the entropy functional), the Gibbs-inequality sketch proof, the verified statement (the uniform distribution is the unique entropy maximizer, with maximum $\log n$), the thermal law where a fixed mean replaces a fixed count, and the reconciliation of the two roads to one prior. The maximum-entropy instrument is reused by the consciousness and learning chapters at the end of the book.

The proofs are `BookProof/ChapterMaxEntropy` and `BookProof/ChapterThermalMaxEntropy`.

9.2. Measuring Ignorance

The previous chapter singled out the uniform prior as the unique *relabeling-invariant* one. There is a second, independent route to the same destination, due to Jaynes: the uniform prior is the distribution that **maximizes the Shannon entropy**. Within a fixed parametrization this is the prior that assumes the least — but, as we stress at the end, "least" is measured relative to the chosen coordinates, and the conclusion is no more objective than the parametrization it rests on.

For a probability distribution p on a finite non-empty space α with $n = |\alpha|$ outcomes, the **entropy** is

$$H(p) = \sum_i -p_i \log p_i,$$

with the convention $0 \log 0 = 0$. Entropy measures spread: it is small when the distribution is concentrated on a few outcomes and large when the mass is spread out. The claim we prove is:

Among all distributions on n outcomes, the uniform distribution $u_i = 1/n$ uniquely maximizes the entropy, and the maximum value is $\log n$.

This is the **maximum-entropy principle**: *relative to a fixed set of labeled outcomes*, if the only thing you know is the set of possibilities, the prior with the largest entropy is the uniform one, and any other prior assumes extra information beyond the labeling. The italicized clause is essential. Entropy is computed against the counting measure on the chosen labels; change the parametrization and a different prior becomes the maximum-entropy one. The principle selects the uniform prior within a coordinate system — it does not certify the uniform prior as objectively non-informative.

9.3. Sketch Proof (Gibbs' Inequality)

The whole argument rests on the elementary bound $\log t \le t - 1$ for $t > 0$, with equality only at $t = 1$. Apply it with $t = 1/(n\,p_i)$:

$$-\log(n\,p_i) \le \frac{1}{n\,p_i} - 1.$$

Multiplying by $p_i \ge 0$ and rearranging gives the **pointwise Gibbs bound**

$$-p_i \log p_i \;-\; p_i \log n \;\le\; \frac{1}{n} - p_i.$$

Now sum over all i. The left-hand side becomes $H(p) - \log n \sum_i p_i = H(p) - \log n$. The right-hand side becomes $\sum_i \frac{1}{n} - \sum_i p_i = 1 - 1 = 0$, because there are n terms each equal to $1/n$ and $\sum_i p_i = 1$. Therefore

$$H(p) - \log n \le 0, \qquad \text{i.e.} \qquad H(p) \le \log n.$$

For the uniform distribution $u_i = 1/n$ one computes directly $H(u) = \sum_i -\frac{1}{n} \log \frac{1}{n} = \log n$, so the bound is attained.

Finally, equality in $H(p) \le \log n$ requires equality in $\log t \le t - 1$ for **every** i, which forces $1/(np_i) = 1$, i.e. $p_i = 1/n$, for every i. So the uniform distribution is the **unique** maximizer.

9.4. The Verified Statement

The maximum-entropy bound and its sharpness are proved in `BookProof.ChapterMaxEntropy`. The headline is that every distribution has entropy at most that of the uniform one:

```
#check @ChapterMaxEntropy.entropy_le_entropy_uniform
```

The stronger statement, that equality holds **only** for the uniform distribution:

```
#check @ChapterMaxEntropy.entropy_eq_log_card_iff
```

and the supporting facts — the bound $H(p) \le \log n$, the value $H(u) = \log n$, and the non-negativity of entropy:

```
#check @ChapterMaxEntropy.entropy_le_log_card
#check @ChapterMaxEntropy.entropy_uniform
#check @ChapterMaxEntropy.entropy_nonneg
```

The pointwise Gibbs bound that drives the proof is itself verified:

```
#check @ChapterMaxEntropy.negMulLog_sub_le
```

9.5. A Fixed Mean Instead of a Fixed Count — The Thermal Law

The uniform distribution maximizes entropy among distributions on a **fixed number of outcomes**. The same principle, applied with a **fixed mean value** rather than a fixed count, selects the thermal law: among finitely supported occupation distributions with a prescribed mean occupation, the Gibbs law is the unique maximizer of the Shannon entropy. This variational characterization of the thermal state is proved in `BookProof.ChapterThermalMaxEntropy`:

```
#check @BookProof.ChapterThermalMaxEntropy.thermalEntropy
#check @BookProof.ChapterThermalMaxEntropy.thermalEntropy_eq
#check @BookProof.ChapterThermalMaxEntropy.gibbs_pointwise
#check @BookProof.ChapterThermalMaxEntropy.shannonEntropy_le_thermalEntropy
```

9.6. Two Roads, One Prior — Both Within a Parametrization

We now have two independent characterizations of the uniform prior: it is the unique *relabeling-invariant* distribution (previous chapter) and the unique *maximum-entropy* distribution. That a symmetry principle and an optimization principle agree makes the uniform prior a natural default **within a fixed parametrization**. It is not evidence that the uniform prior is objectively "the" non-informative prior. Both characterizations are parametrization-dependent: relabeling-invariance privileges a labeling, and entropy privileges the counting measure on that labeling, and the previous chapter showed that any non-null finite prior can be reparametrized into the uniform one — which then becomes the maximum-entropy prior in the new coordinates.

The source manuscript leans on the uniform prior as a convenient default while insisting it is never objectively forced. Citing Eaton and Freedman (Dutch book against some 'objective' priors), it holds that "there is no prior which is better for all cases" and that "there are no non-informative priors in Bayesian inference." On a *continuous* space the point is sharper still: there is in general **no** non-informative prior at all (Null-measure sets need not be small). The maximum-entropy principle is a rule for choosing a prior once coordinates are fixed — not a proof that any prior, uniform or otherwise, is written into the problem.

10. The Law of Total Variance

10.1. Orientation and Status

The closing chapter of the probability part, and its bookkeeping identity: how uncertainty splits into a within-group and a between-group part. It is self-contained (the conditional expectation is set up from scratch), and it is one of the two chapters — with the Dutch-book theorem — that give the conceptual core of the part.

The route: the two kinds of uncertainty (aleatoric and epistemic), the setup, the "Pythagoras for conditional expectation" sketch proof, the verified statement, and a reading of what the decomposition tells you when sources of uncertainty are combined. The identity is reused wherever the book later combines uncertainties — notably in the learning chapters.

The proof is `BookProof/ChapterTotalVariance`.

10.2. Two Kinds of Uncertainty

When a model makes a prediction, its error has two conceptually distinct sources:

- **Aleatoric uncertainty** — the noise *inside* the data-generating process, which no amount of extra information about the model would remove.
- **Epistemic uncertainty** — the ignorance about *which* model (or which group, or which ensemble member) generated the data.

The source manuscript frames this distinction in the language of **ensemble forecasting**: a forecast is a *probability distribution over probability distributions*, and its total spread combines the variability *within* each member with the variability *between* members.

The exact mathematical statement underlying this picture is the **law of total variance**: the total variance of a quantity splits, with no remainder, into a within-group part and a between-group part.

10.3. The Setup

Take a finite sample space Ω with non-negative weights $w : \Omega \to \mathbb{R}$ (think of a probability distribution, though the algebraic identity needs only non-negativity), a real-valued quantity $Y : \Omega \to \mathbb{R}$, and a **grouping** $X : \Omega \to \kappa$ that assigns each outcome to one of finitely many groups. Define:

- the **mean** $\mathbb{E}[Y] = \sum_\omega w_\omega \, Y_\omega$;
- the **total variance** $\mathrm{Var}[Y] = \sum_\omega w_\omega \, (Y_\omega - \mathbb{E}[Y])^2$;
- the **conditional mean** in group g, $\mu_g = \mathbb{E}[Y \mid X = g]$;

- the **within-group variance** $\mathrm{within} = \mathbb{E}[\mathrm{Var}(Y \mid X)]$, the average of the variances inside each group;
- the **between-group variance** $\mathrm{between} = \mathrm{Var}(\mathbb{E}[Y \mid X])$, the variance of the group means.

The claim is the clean identity

$$\mathrm{Var}[Y] \;=\; \underbrace{\mathbb{E}[\mathrm{Var}(Y \mid X)]}_{\text{within (aleatoric)}} \;+\; \underbrace{\mathrm{Var}(\mathbb{E}[Y \mid X])}_{\text{between (epistemic)}}.$$

10.4. Sketch Proof (Pythagoras for Conditional Expectation)

The proof is a single orthogonal decomposition. Write each value as

$$Y_\omega - \mathbb{E}[Y] = \underbrace{\left(Y_\omega - \mu_{X(\omega)}\right)}_{\text{residual within its group}} + \underbrace{\left(\mu_{X(\omega)} - \mathbb{E}[Y]\right)}_{\text{group mean vs. overall mean}}.$$

Square and sum with weights w_ω. The two squared terms give exactly within and $\mathrm{between}$. The only thing to check is that the **cross term vanishes**:

$$2\sum_\omega w_\omega \left(Y_\omega - \mu_{X(\omega)}\right)\left(\mu_{X(\omega)} - \mathbb{E}[Y]\right) = 0.$$

Factor out the group-level quantity $\left(\mu_g - \mathbb{E}[Y]\right)$, which is constant within group g. What remains in each group is $\sum_{\omega : X(\omega) = g} w_\omega \left(Y_\omega - \mu_g\right)$, which is **zero by the definition of the conditional mean** μ_g as the w-weighted average of Y over the group. This is the **balance** (orthogonality) property of conditional expectation: the residuals sum to zero within every group. With the cross term gone, the identity follows.

Geometrically, conditional expectation is an orthogonal projection, and the law of total variance is the Pythagorean theorem for that projection.

10.5. The Verified Statement

The decomposition and its ingredients are proved in `BookProof.ChapterTotalVariance`. The headline identity:

```
#check @ChapterTotalVariance.total_variance
```

The two facts that make the cross term disappear — the within-group balance of the residuals, and the resulting vanishing of the cross term:

```
#check @ChapterTotalVariance.groupBalance
#check @ChapterTotalVariance.crossTerm_zero
```

Because both summands are non-negative, each is bounded by the total variance — a fact the manuscript uses to control epistemic error:

```
#check @ChapterTotalVariance.within_nonneg
#check @ChapterTotalVariance.between_nonneg
#check @ChapterTotalVariance.within_le_variance
#check @ChapterTotalVariance.between_le_variance
```

10.6. Reading the Decomposition

The identity is an *equality*, not an inequality: there is no leftover term. If a prediction is uncertain, the law tells you precisely how much of that uncertainty would disappear if you learned the group X (the between-group part) and how much is irreducible noise (the within-group part). This is the quantitative backbone of the manuscript's discussion of **systematic uncertainties as Bayesian priors** and of **ensemble forecasting**, and it reappears in the law of large numbers when we ask how repeated measurement reduces the epistemic part.

11. Determinism, Complementarity, and Collapse

The third part is the conceptual core of the manuscript's quantum-foundations chapters. We begin with the group-theoretic fact that a symmetry group of canonical transformations is a unitary representation, taking the one-parameter time-translation group as the prototype; characterize deterministic versus non-deterministic symmetry transformations and locate the origin of complementarity; show why wave-function collapse keeps quantum mechanics an ordinary Kolmogorov probability theory, and how this differs from Gleason's theorem; prove that time-translation is a stochastic process *if and only if* it is deterministic; reconstruct a quantum trajectory at intermediate times by post-selection; work the double-slit and Bell/CHSH experiments; and close with EPR-completeness, relativistic causality, a concrete deterministic theory, and the classical limit.

How to read this part. Symmetries, determinism, and complementarity is the load-bearing chapter: everything else in the part is an application of its one theorem — time-translation is a stochastic process iff it is deterministic. Collapse keeps quantum mechanics a probability theory answers the standard objection, and the double-slit experiment and the Bell inequalities are the two laboratory situations the manuscript reads through that lens; EPR-completeness closes the logical circle. These four form a single argument and are best read in sequence; the double-slit chapter additionally introduces the post-selection and weak-value machinery that the Solovay–Kopperman tensor product chapter later generalizes.

12. Symmetries, Determinism, and Complementarity

12.1. Orientation and Status

The load-bearing chapter of this part: every chapter after it is an application of its one theorem — **time-translation is a stochastic process if and only if it is deterministic**. It sits between the probability toolkit of the previous part (the coherence target and the bookkeeping of total variance) and the laboratory chapters that follow; no field theory is needed.

The route: symmetries are unitary representations (canonical transformations, the one-parameter group law), symmetry transformations act on events, deterministic versus non-deterministic transformations, complementarity as randomness, the reading of quantum mechanics as a generalization of statistical mechanics rather than of probability, the stochastic-process theorem itself, and conservative transformations.

The proofs are `BookProof/ChapterDeterministic` and `BookProof/ChapterTimeTranslation` (the main theorem), `BookProof/ChapterConservative` with `BookProof/ChapterConservativeDiagonal` (the conservative case), and `BookProof/ChapterReconstruct` (trajectory reconstruction).

12.2. Symmetries Are Unitary Representations

A conditional probability measure that depends on a one-dimensional parameter — which we may call time — "can be parametrized by a quantum process: a time-ordered product integral of unitary operators, defined by a time-dependent Hamiltonian operator." The manuscript's conclusion is that such parametrizations are "certainly natural and common," and the reason is group-theoretic:

"When a subset of canonical transformations forms a group, such subset is always a linear unitary representation of a group which is then called symmetry group and the canonical transformations are then called symmetry transformations."

This chapter makes that sentence precise in the cleanest possible case: the one-parameter group of time translations. We show that the time-evolution $U(t) = e^{iHt}$ generated by a self-adjoint Hamiltonian is, literally, a group homomorphism from the additive real line into the unitary group — that is, a **unitary representation** of the symmetry group $\mathbb{R}$.

12.3. Canonical Transformations Are Unitary

For a Hermitian matrix H, the time-evolution is the matrix exponential $U(t) = e^{iHt}$:

```
#check @timeEvo
```

Because H is self-adjoint, $U(t)$ is unitary, and hence lives in the unitary group:

```
#check @timeEvo_unitary
#check @timeEvo_mem_unitaryGroup
```

The measure-preserving (conservative) content of this — that conjugating a density operator by $U(t)$ preserves its trace, so total probability is conserved — is taken up in the section on conservative transformations below. Here we first record the *group* structure.

12.4. The One-Parameter Group Law

The family $U(t)$ is a group action of $(\mathbb{R}, +)$. It sends the identity 0 to the identity operator, it respects addition, and every element is invertible with the time-reversed evolution as its inverse:

$$U(0) = 1, \qquad U(s)\,U(t) = U(s+t), \qquad U(t)^{-1} = U(-t).$$

```
#check @timeEvoU_one
#check @timeEvoU_mul
#check @timeEvoU_inv
```

This is precisely the structure the manuscript contrasts with Markov processes. A Markov process is a **semigroup**: it has no inverses, and "one has not only to deal with linear algebra but also with convex geometry at the infinitesimal level." A unitary time-evolution is a **group**, and Lie theory "uses the group structure to translate global geometric and analytic problems into local and infinitesimal ones," which are "essentially linear algebra." That is why quantum processes are not intrinsically exotic — they are simply *more general* than Markov processes, and far easier to reduce to an infinitesimal generator (the Hamiltonian).

12.5. A Symmetry Group Is a Unitary Representation

We now package the group law as a single object. Bundling each $U(t)$ with the proof that it is unitary gives an element of the unitary group:

```
#check @timeEvoU
```

And the whole family is a **monoid homomorphism** from the additive reals (viewed multiplicatively) into the unitary group — i.e. a unitary representation of the one-parameter symmetry group $\mathbb{R}$:

```
#check @timeEvoRep
#check @timeEvoRep_apply
```

This is the verified form of the manuscript's claim. A subset of canonical transformations that forms a group — here the time translations — is a linear unitary representation of that group, and the group is then called the symmetry group. The representation is `sorry`-free: the homomorphism property is exactly the group law $U(s)U(t) = U(s+t)$ proved above. Symmetry, in this book, is not an extra physical postulate; it is the statement that the canonical transformations preserving a conditional probability measure form a group, and that any such group acts by unitaries. The spin-dependent representations that select bosonic versus fermionic statistics (the spin–statistics dichotomy) are further instances of the same principle.

12.6. Symmetry Transformations Act on Events

A symmetry transformation is, at bottom, a unitary operator U on the wave-function. But the physical content of the transformation is not what it does to ψ; it is what it does to **events**. An event A is represented by a projection P_A, and the transformed event is

$$P_A \;\longmapsto\; U^{\dagger} P_A\, U.$$

In the measurement basis $\mathrm{Fin}\, n$, the algebra $L^{\infty}(X, \mu)$ of events is just the algebra of **diagonal** matrices, and a single outcome a is the rank-one projection $P_a = |e_a\rangle\langle e_a|$. The transformed projection $U^{\dagger} P_a U$ has matrix entries

$$(U^\dagger P_a U)_{ij} = \overline{U_{ai}}\, U_{aj}.$$

The question that drives the rest of this chapter is: *when is the transformed event again a diagonal event?*

12.7. Deterministic Transformations

A transformation is **deterministic** when it sends every diagonal element of the event algebra to another diagonal element — equivalently, when the conditional probability it defines is a genuine function (the Radon–Nikodym derivative of one measure with respect to another) rather than a genuinely random kernel. In the finite-dimensional basis this means every column of U has at most one non-zero entry: each input outcome is sent to a single output outcome.

The manuscript's headline result is a clean commutation criterion:

An automorphism U is deterministic **if and only if** P_A and $U P_B U^\dagger$ commute, for all events A, B.

The verified statement (module `BookProof.ChapterDeterministic`):

```
#check @ChapterDeterministic.commute_proj_measOp_iff_isDeterministic
#check @ChapterDeterministic.commute_projSet_measOpSet_iff_isDeterministic
```

The second of these promotes the statement from single outcomes a, b to arbitrary events $A, B \subseteq \mathrm{Fin}\, n$, by summing the rank-one projections into the projection onto a subset. The same criterion appears, in the language of off-diagonal parts, in module `BookProof.ChapterReconstruct`:

```
#check @ChapterReconstruct.offDiag_eq_zero_iff_isDeterministic
```

Reading it: the off-diagonal part of a column of U vanishes for every column exactly when U is deterministic. The off-diagonal part is precisely the interference that a deterministic map must destroy.

12.8. Complementarity Is Randomness

Here is the conceptual payoff. Take two observables, say position and momentum. Their projections P_X and P_p are diagonal in *different* bases, related by a Fourier transform U; so P_X and $U P_p U^\dagger$ do **not** commute. By the criterion above, the symmetry transformation U relating the two observables is **non-deterministic**.

The manuscript reads this as the *definition* of complementarity: two observables are complementary precisely because the symmetry transformation relating them is random. There is no mystery, and no failure of probability theory — probability theory has no trouble with non-commuting observables, **as long as the wave-function collapses** (the collapse chapter makes this precise). What the criterion shows is that the non-commutativity is not an exotic addition to probability; it is the signature of a non-deterministic symmetry.

12.9. Quantum Mechanics Generalizes Statistical Mechanics, Not Probability

The deterministic transformations form the special case in which the symmetry transformation is a mere change of variables. Classical statistical mechanics lives entirely inside this case: its conditional probability measures are deterministic, transforming a deterministic ensemble into a deterministic ensemble. Quantum mechanics generalizes classical statistical mechanics by allowing the symmetry transformations — in particular the time-evolution — to be **non-deterministic**.

This is the precise sense in which quantum mechanics is a generalization of *classical statistical mechanics* but **not** of probability theory itself. The parametrization by a wave-function is available for any probability distribution; what quantum mechanics adds is the freedom to act on it non-deterministically.

12.10. Time-Translation Is a Stochastic Process If and Only If It Is Deterministic

A **stochastic process** is a family of random events indexed by time: there is a probability distribution for *each* time, and the times are glued together by conditional probabilities. Does the time-evolution of quantum mechanics define such a process? The manuscript's answer is a precise and surprising one:

There is a group action of a symmetry group on the **probability distribution** if and only if the symmetry transforms **deterministic** distributions into deterministic distributions. In particular, **time-translation in quantum mechanics is a stochastic process if and only if it is deterministic.**

This single fact is, the manuscript argues, overlooked by the assumptions of both Bell's theorem and the EPR paradox.

Start with a probability distribution $\mathrm{diag}(\rho_1)$. Acting on the wave-function with a group element g and then collapsing gives a new distribution $\mathrm{diag}(\rho_g)$. Wigner's theorem says the action of a symmetry on the wave-function is linear and unitary. For the symmetry to act on the **probability distribution** (and hence for a stochastic process to exist), the collapsed trace $\mathrm{tr}\big(\mathrm{diag}(\rho_g)\, U_h P_A U_h^\dagger\big)$ and the uncollapsed trace $\mathrm{tr}\big(\rho_g\, U_h P_A U_h^\dagger\big)$ must agree for every pure ρ_g, every event A, and every group element h.

The two traces differ exactly by the off-diagonal part of ρ_g. The verified accounting (module `BookProof.ChapterTimeTranslation`):

```
#check @ChapterTimeTranslation.trace_rho_measOp
#check @ChapterTimeTranslation.trace_diagPart_measOp
#check @ChapterTimeTranslation.trace_diff
```

The first two compute the two traces; the third isolates their difference. The headline equivalence then falls out:

```
#check @ChapterTimeTranslation.trace_eq_iff_isDeterministic
#check @ChapterTimeTranslation.trace_eq_iff_isDeterministic_pure
```

Reading them: the two traces agree for every state and every event **if and only if** U is deterministic (the second states the same for pure states only). The mechanism is concrete: if U is deterministic, then $\overline{U_{ja}}\, U_{jl} = 0$ whenever $a \neq l$, so the off-diagonal contribution vanishes; if U is **not** deterministic, some $\overline{U_{ma}}\, U_{la} \neq 0$ with $l \neq m$, and the superposition $\Psi = (\delta_m + \delta_l)/\sqrt{2}$ isolates it. The same content, phrased through the off-diagonal part (module `BookProof.ChapterReconstruct`):

```
#check @ChapterReconstruct.offDiag_eq_zero_iff_isDeterministic
```

If the time-evolution is non-deterministic, then there is **no** probability distribution attached to each intermediate time — the symmetry does not descend to the distributions, so no stochastic process is defined. This is the mathematical core behind several of the manuscript's most provocative claims:

- **EPR.** After the entangled particles separate and before the measurement transformation, there is no probability distribution for the state — because the (non-deterministic) time-evolution is not a stochastic process.
- **Bell.** The Bell assumptions implicitly treat the time-evolution as a stochastic process; the theorem above shows that this is legitimate only in the deterministic case.
- **The double-slit.** The product of two non-deterministic transformations is not the same as a stochastic process in which they are applied in sequence — which is exactly why the interference pattern appears (see the double-slit chapter).

12.11. Conservative Transformations

A transformation is **conservative** when it preserves the measure μ of the event algebra. A continuous conservative transformation is a unitary time-evolution $U(t) = e^{iHt}$ generated by a self-adjoint operator H. The verified construction (module `BookProof.ChapterConservative`):

```
#check @ChapterConservative.timeEvo_unitary
#check @ChapterConservative.timeEvo_add
```

The first says e^{iHt} is unitary whenever H is Hermitian (so total probability is conserved); the second says these unitaries form a one-parameter group, $U(s)U(t) = U(s+t)$.

Which Hamiltonians generate **deterministic** conservative evolutions? Exactly the diagonal ones. A Hamiltonian commutes with every event projection if and only if it is itself diagonal in the measurement basis (module `BookProof.ChapterConservativeDiagonal`):

```
#check @ConservativeDiagonal.commutes_all_events_iff_isDiag
#check @ConservativeDiagonal.conservative_iff_isDiag
```

So a conservative evolution is deterministic exactly when its Hamiltonian is diagonal — when it merely re-phases the outcomes without mixing them. The moment H has off-diagonal entries, the evolution becomes non-deterministic, and we have left classical statistical mechanics for quantum mechanics.

12.12. Summary

- A symmetry group of canonical transformations is a **unitary representation**; the one-parameter time-translation group $U(t) = e^{iHt}$ satisfies the group law $U(s)U(t) = U(s+t)$.
- A symmetry acts on events by $P_A \mapsto U^\dagger P_A U$. It is **deterministic** iff every transformed event stays diagonal, iff P_A and $UP_BU^\dagger$ commute for all events A, B, iff every column of U has a single non-zero entry.
- **Complementarity** is the non-commutativity that arises when the symmetry relating two observables is non-deterministic.
- Time-translation is a **stochastic process** iff it is deterministic — the symmetry descends to the probability distributions exactly in the deterministic case.
- A **conservative** evolution e^{iHt} is deterministic iff H is diagonal.
- Quantum mechanics = classical statistical mechanics + non-deterministic symmetry transformations. It generalizes statistical mechanics, not probability theory.

13. Collapse Keeps Quantum Mechanics a Probability Theory

13.1. Orientation and Status

The second chapter of the part answers the standard objection to the ensemble reading — "quantum mechanics is a non-commutative generalization of probability theory" — and shows instead that collapse keeps quantum mechanics an ordinary (Kolmogorov) probability theory. Read it after the deterministic-transformations chapter; the contrast with Gleason's theorem at the end marks exactly where this book diverges from the mainstream foundations.

The route: the apparent non-commutativity of position and momentum projections, why the state of an ensemble is diagonal, why measuring momentum still changes the ensemble, the collapse as a recursion of two-dimensional maps, and the contrast with Gleason's theorem.

The proofs are `BookProof/ChapterCollapseDiagonal`, `BookProof/ChapterEulerDensityMatrix`, `BookProof/ChapterEulerGenericDensity`, and the Gleason comparison `BookProof/ChapterGleason` / `BookProof/ChapterGleasonPureMixed`.

13.2. The Apparent Non-Commutativity

It is often said that quantum mechanics is a **non-commutative generalization of probability theory**. The evidence seems clear: the projection P_X onto a region of space and the projection $UP_pU^\dagger$ onto a region of momentum are diagonal in different bases (related by a Fourier transform U), so they do not commute. Since we may choose to measure position *or* momentum, the algebra of "events" looks non-commutative.

The manuscript's claim is that this appearance is created by **forgetting the collapse**. Once the collapse is taken seriously, quantum mechanics remains an ordinary Kolmogorov probability theory, even when complementary observables are considered.

13.3. The State of an Ensemble Is Diagonal

The state of a statistical ensemble is a linear functional $E(A) = \mathrm{tr}(\rho A)$ on operators, where ρ is a self-adjoint operator with $\mathrm{tr}\,\rho = 1$. After a measurement, the wave-function has collapsed, so ρ is **diagonal** in the measurement basis. The defining property of a diagonal state is:

$$E(O) = \mathrm{tr}(\rho\, O) = 0 \qquad \text{for every operator } O \text{ with null diagonal.}$$

The verified statement (module `BookProof.ChapterCollapseDiagonal`):

```
#check @ChapterCollapseDiagonal.trace_diag_nullDiag_zero
#check @ChapterCollapseDiagonal.trace_diagonal_mul_diag
```

The first is exactly the property above: a diagonal ρ pairs to zero with any null-diagonal operator. The second says that, for diagonal ρ, the expectation of any operator depends only on that operator's diagonal part. **The off-diagonal (coherence) part is invisible to a collapsed ensemble.**

13.4. Measuring Momentum Changes the Ensemble

Now suppose we want the probability that the system lies in a region of momentum p. We cannot read this off the same ensemble E; we must first apply the physical transformation U that relates the position and momentum bases, and then collapse. The result is a **new** ensemble E_U, diagonal in the momentum basis, defined on diagonal operators D by

$$E_U(D) = \mathrm{tr}(\rho_U D) = \mathrm{tr}(\rho\, U D\, U^\dagger) = E(U D\, U^\dagger).$$

For a null-diagonal operator O, the collapse gives $E_U(O) = 0$. Hence

$$E_U(P_p) = E(U P_p U^\dagger)$$

is a perfectly good probability — but it is a probability for the *new* ensemble E_U, not for E. The ensembles E and E_U are different; a physical transformation relates them.

Without the collapse we would instead have $E_U(O) = E(UOU^\dagger) \neq 0$ for null-diagonal O, and we could speak of a single common state assigning probabilities to a non-commutative algebra. **The collapse is what prevents this.** It keeps quantum mechanics a Kolmogorov probability theory: every actual measurement sees a single commutative (diagonal) algebra.

13.5. The Collapse Is a Recursion of Two-Dimensional Maps

The collapse that makes the ensemble diagonal has a description in wave-function coordinates that is worth making explicit, because it links the abstract argument above to the concrete two-dimensional maps of the manuscript. In the probability clock the collapse of a two-state wave-function $\Psi(t) = (\cos t, \sin t)$ is exactly the map that takes the rank-one projector

$$\rho(t) = \Psi(t)\Psi(t)^\top = \tfrac{1}{2}\,\mathbf{1} + Z\,(\cos 2t\,\mathbf{1} + \sin 2t\, J)$$

to its diagonal by setting to zero the term proportional to J — the imaginary unit of the plane, $J^2 = -\mathbf{1}$. This is "taking the real part" of the "complex number" $\cos(2t) + J\sin(2t)$. The verified statements (module `BookProof.ChapterEulerDensityMatrix`):

```
#check @ChapterEulerDensityMatrix.density_euler
#check @ChapterEulerDensityMatrix.density_collapse
#check @ChapterEulerDensityMatrix.Jdens_sq
```

The manuscript lifts this from two states to n and to a countable partition by constructing the wave-function **recursively** from two-dimensional angles — **Euler angles** $v_k = \cos\theta_k\, l_k + \sin\theta_k\, v_{k+1}$. The recursion step is formalized by `eulerVec`; the expanded density matrix is given by `outer_eulerVec`. At each level the same two-dimensional identity holds, with a generator J_k in the plane spanned by $\{l_k, v_{k+1}\}$:

$$v_k v_k^\dagger = \tfrac{1}{2}\,\mathbf{1} + \tfrac{1}{2}\left(l_k l_k^\dagger - v_{k+1} v_{k+1}^\dagger\right)\left(\cos 2\theta_k + J_k \sin 2\theta_k\right).$$

Collapse at level k deletes the J_k-proportional term, leaving the diagonal operator whose entries are the **conditional** probabilities $P(k \mid k \text{ or above}) = \cos^2(\theta_k)$. Iterating over all levels gives a probability on a single commutative algebra:

$$P(n) = P((n \text{ or above}))\, P(n \mid n \text{ or above}) = \Big(\prod_{k=1}^{n-1} P\big((k{+}1 \text{ or above}) \mid (k \text{ or above})\big)\Big) P(n \mid n \text{ or above}).$$

This is the manuscript's statement that **the collapse of the wave-function for a generic phase space is a recursion of collapses of two-dimensional real wave-functions** — the probability clock, nested. The verified statements (module `BookProof.ChapterEulerGenericDensity`):

```
#check @ChapterEulerGenericDensity.eulerVec
#check @ChapterEulerGenericDensity.outer_eulerVec
#check @ChapterEulerGenericDensity.Jgen_sq
#check @ChapterEulerGenericDensity.density_euler_generic
#check @ChapterEulerGenericDensity.density_collapse_generic
#check @ChapterEulerGenericDensity.density_idempotent
#check @ChapterEulerGenericDensity.conditional_probability_at
#check @ChapterEulerGenericDensity.product_probability
```

Here $J_k^2 = -\mathbf{1}$, so each recursion step is again the two-dimensional map of the probability clock. This doubly confirms the message of the previous sections. First, the collapse seen ensemble by ensemble ($E_U(O) = 0$ for null-diagonal O) is the same operation as this level-by-level deletion of the J_k-coherence. Second, and more important, what a measurement observes is only the sequence of **conditional** probabilities — a family of real numbers in $[0, 1]$ — that is, an ordinary commutative algebra. Every non-commutativity lives in the invisible phases J_k that the collapse discards, exactly as in the probability clock. The recursion shows that no dimensional escape from this conclusion exists: quantum mechanics is Kolmogorov at every level, and the "non-commutative" structure is the same two-dimensional bookkeeping of the parametrization at each stage.

13.6. Contrast with Gleason's Theorem

This result resembles Gleason's theorem but differs from it in exactly the place that matters. Gleason's theorem says that any probability measure on **all the non-commuting projections** of a Hilbert space of dimension ≥ 3 is given by a **density matrix** — which includes mixed states. Our result parametrizes **commuting** projections by a **wave-function** — a pure state.

The difference is visible in the smallest non-trivial real example, two dimensions. Take the two non-commuting projections

$$P_1 = \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix}, \qquad Q = \frac{1}{2}\begin{pmatrix} 1 & 1 \\ 1 & 1 \end{pmatrix}.$$

The verified facts (module `BookProof.ChapterGleason2D`):

```
#check @ChapterGleason2D.P1_Q_noncomm
#check @ChapterGleason2D.pure_realizes_P1
#check @ChapterGleason2D.pure_realizes_Q
#check @ChapterGleason2D.no_pure_state_both
#check @ChapterGleason2D.mixed_state_both
```

Reading them: P_1 and Q do not commute; there is a pure state realizing $\mathrm{tr}(\rho P_1) = \frac{1}{2}$, and a (different) pure state realizing $\mathrm{tr}(\rho Q) = \frac{1}{2}$; but there is **no** pure state realizing *both* simultaneously. By contrast, the **mixed** state $\rho =$

$\frac{1}{2}\mathbf{1}$ realizes both at once. The same contrast, stated as a single theorem (module `BookProof.ChapterGleasonPureMixed`):

```
#check @ChapterGleasonPureMixed.pure_vs_mixed_gleason_contrast
#check @ChapterGleasonPureMixed.exists_mixed_state_both
```

The lesson: a single **wave-function** (pure state) can reproduce the probabilities of any *one* commuting family of events — which is all a single measurement ever sees, because of collapse. A **density matrix** (mixed state) is needed only if one insists on attaching probabilities to several non-commuting families at once — which is precisely what the collapse forbids. Gleason's theorem is relevant only when the collapse is neglected.

13.7. Summary

- A collapsed ensemble is diagonal: it assigns zero to every null-diagonal operator, so it sees only one commutative algebra of events.
- Measuring a complementary observable applies a physical transformation **and** a collapse, producing a *new* diagonal ensemble — not a single non-commutative state.
- Therefore quantum mechanics is **not** a non-commutative generalization of probability theory; the collapse keeps it Kolmogorov.
- In wave-function coordinates the same collapse is a **recursion of collapses of two-dimensional real wave-functions** — the probability clock nested — deleting the J_k-coherence at every level.
- The wave-function parametrizes **commuting** projections (pure states); Gleason's density matrix parametrizes **non-commuting** projections (mixed states). The two differ exactly where the collapse is or is not used.

In the manuscript's phrasing, quantum mechanics is "a generalization of classical statistical mechanics (but not of probability theory)." The non-commutativity and the off-diagonal coherence are artifacts of the wave-function parametrization; once the collapse is taken seriously they dissolve, and what remains is an ordinary Kolmogorov probability theory.

14. The Double-Slit Experiment

14.1. Orientation and Status

The first laboratory situation of the part, and the chapter that introduces the post-selection and weak-value machinery that the Solovay–Kopperman tensor product later generalizes. It follows the previous chapter and uses its theorem directly: time-evolution is a stochastic process only when it is deterministic.

The route: the puzzle of self-interference under the ensemble reading, the two transformations (source to slits, slits to screen), a two-by-two model — one slit closed gives 50/50, both open gives 100/0 — where the "mystery" goes, and weak measurements reconstructing the trajectory at intermediate times.

The proofs are `BookProof/ChapterDoubleSlit`, `BookProof/ChapterTrajectory`, and `BookProof/ChapterWeakValue`.

14.2. The Puzzle

The ensemble interpretation, by itself, does not explain why an electron's wave-function appears to **interfere with itself** in Young's double-slit experiment — that would seem to require the wave-function to describe an *individual* system. This chapter fills the gap using the result of the previous chapter: the time-evolution is a stochastic process only when it is deterministic.

14.3. The Two Transformations

After the electron is fired (call this instant S_1), its evolution to the detector (F) is the product of **two** non-deterministic symmetry transformations:

- $S_1 \to S_2$: passing through one slit or the other, with probability $50/50$;
- $S_2 \to F$: a non-deterministic propagation from the slits to the screen.

Because **both** transformations are non-deterministic, the theorem of the previous chapter says that no stochastic process can be defined through the intermediate instant S_2. The only stochastic process that exists connects S_1 directly to F; the transformations through S_2 "never occurred" as a sequence of random events. This is why one cannot say which slit the electron went through.

14.4. A Two-by-Two Model

Restrict to the electrons that reach the detector along one of two angles. The wave-function at S_1 is

$$\Psi = \begin{pmatrix} 1 \\ 0 \end{pmatrix}.$$

The propagation from the slits to the screen is the **Hadamard** matrix, which adds the two slit-amplitudes for the first angle and subtracts them for the second:

$$H = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix}.$$

The verified properties of H (module `BookProof.ChapterDoubleSlit`):

```
#check @ChapterDoubleSlit.H_unitary
#check @ChapterDoubleSlit.H_involutive
#check @ChapterDoubleSlit.Hpsi0
```

The matrix is unitary (probability-conserving) and **involutive**: $H^2 = \mathbf{1}$. Acting once on Ψ it produces the balanced superposition $H\Psi = \frac{1}{\sqrt{2}}(1, 1)$.

14.5. One Slit Closed: 50/50

With the second slit **closed**, only the propagation $S_2 \to F$ acts, so the state at the screen is $H\Psi = \frac{1}{\sqrt{2}}(1, 1)$. Born's rule gives equal probability for the two angles:

```
#check @ChapterDoubleSlit.slit_closed_born
```

14.6. Both Slits Open: 100/0

With the second slit **open**, both transformations act: first $S_1 \to S_2$ (which is again H, putting the electron into an equal superposition of the two slits), then $S_2 \to F$ (another H). The state at the screen is

$$H\,(H\Psi) = H^2\Psi = \Psi = \begin{pmatrix} 1 \\ 0 \end{pmatrix},$$

because H is involutive. Born's rule now gives probability 1 for the first angle and 0 for the second:

```
#check @ChapterDoubleSlit.slit_open_state
#check @ChapterDoubleSlit.slit_open_born
```

Opening the second slit has turned a $50/50$ distribution into a $100/0$ distribution. The first angle is **constructive** interference (the two slit-amplitudes add), the second **destructive** (they cancel).

14.7. Where the Mystery Goes

The "mystery" is the same as for the probability clock: how can $50/50$ become $100/0$? The answer is that $H \cdot H$ is **not** a stochastic process in which H is applied and then H is applied again with a collapse in between. If one *collapses* after the first H (i.e. actually measures which slit), the interference is destroyed and one recovers $50/50$. The coherent product H^2 and the incoherent sequence give different answers, and there is no reason they should agree — because time plays no fundamental role, and there is no probability distribution at the intermediate instant S_2.

And just as for the probability clock, the "interference" is not a physical wave interfering with itself. The wave-function is one parametrization of a probability distribution; in these coordinates the two slit-amplitudes add (the H^2 product) and a cross term appears. Call that cross term "interference" if you like, but it is a feature of the parametrization, not a phenomenon in the world — the manuscript's point that the wave-function is "nothing else than one possible parametrization of any probability distribution." The exotic reading, on which the electron literally passes through both slits as a wave, is precisely what the parametrization dissolves.

The verified contrast between the coherent and the incoherent composition (module `BookProof.ChapterTrajectory`):

```
#check @ChapterTrajectory.jointProb_sum_final_eq_midProb
#check @ChapterTrajectory.dslit_finalProb
#check @ChapterTrajectory.dslit_coherentFinal
```

The first says that if one *does* record the intermediate outcome (summing the joint probabilities over the final index), one recovers the middle distribution. The second, `dslit_finalProb`, is the **collapsed** (incoherent) final law: once the intermediate measurement is made, the outcome is uniform $50/50$. The third, `dslit_coherentFinal`, is the **coherent** double-slit result: without intermediate measurement, the state is $(1, 0)$. The difference between them is exactly the interference cross term — present in the coherent product, erased by the intermediate measurement.

14.8. Weak Measurements and the Reconstruction of the Trajectory

The double-slit reconstruction is the book's instance of the **weak measurement** idea (the manuscript's "reconstruction of the trajectory", citing Tamir and Cohen on weak measurements and weak values). The point is subtle: one cannot directly measure the state of a system at an **intermediate** time without disturbing it. But one can **post-select** — run the experiment many times, keep only the trials that end at a chosen final outcome f, and inspect the conditional distribution of the intermediate outcome a given that final state. Because the intermediate distribution is recovered **consistently** (summing the post-selected joint law over all final outcomes returns the un-post-selected middle law, `jointProb_sum_final_eq_midProb`), the reconstructed picture is a genuine probability theory, not an artifact.

The weak-value reading sharpens this. For a **pre-selected** state $\langle i|$ and a **post-selected** state $|f\rangle$, the **weak value** of an observable A is the ratio

$$\langle A \rangle_w = \frac{\langle f|A|i\rangle}{\langle f|i\rangle}.$$

This is the natural continuation of the post-selected conditional probability of the reconstruction above: where the ABL / two-state formula (`condProb = jointProb / finalProb`) reconstructs a **probability** distribution for the intermediate outcome, the weak value reconstructs the **expectation** of an observable under the same pre- and post-selection — and, unlike a strong measurement, it does so without collapsing the intermediate state into an eigenstate. The trajectory is reconstructed by stitching together such weak measurements at finitely many intermediate times, the manuscript's procedure of "repeating the experiment in the same conditions."

The **conditional** (post-selected) probability — not the total probability — is what one divides by here, and the manuscript notes this may "mimic super-determinism" and so evade some assumptions of relevant no-go theorems. The three-instant post-selected core (`midProb`, `transProb`, `jointProb`, `finalProb`, `condProb`) is formalized in `BookProof.ChapterTrajectory`:

```
#check @ChapterTrajectory.finalProb_total
#check @ChapterTrajectory.jointProb_sum_final_eq_midProb
```

```
#check @ChapterTrajectory.condProb_sum
#check @ChapterTrajectory.dslit_condProb
```

The weak-value ratio itself is formalized in `BookProof.ChapterWeakValue`, on the finite complex Hilbert space $\mathrm{Fin}\, n \to \mathbb{C}$ with the standard inner product:

```
#check @ChapterWeakValue.weakValue_wellDefined
#check @ChapterWeakValue.weakValue_unique
#check @ChapterWeakValue.weakValue_diag
#check @ChapterWeakValue.weakValue_diag_isReal
#check @ChapterWeakValue.weakValue_linear
#check @ChapterWeakValue.weakValue_proj_sum
```

`weakValue_wellDefined` (with `weakValue_unique`) says the weak value is the unique number w with $w\,\langle f|i\rangle = \langle f|A|i\rangle$, so the ratio is well-defined exactly when the pre- and post-selected states are not orthogonal. `weakValue_diag` is the collapse to the ordinary expectation $\langle i|A|i\rangle$ when the post-selection is the pre-selection, and `weakValue_diag_isReal` records that this expectation is real for a Hermitian observable — the weak value only leaves the real line once the post-selection differs from the pre-selection. `weakValue_linear` is the linearity in the observable, and `weakValue_proj_sum` is the weak-value counterpart of `condProb_sum`: the weak values of a complete family of projectors sum to 1.

The tie back to the post-selected probabilities is explicit — the ABL joint law is the squared modulus of the weak-value numerator for the post-selection covector $b \mapsto \overline{V_{fb}}$, and the conditional law is its normalization:

```
#check @ChapterWeakValue.jointProb_eq_normSq_weakNumerator
#check @ChapterWeakValue.condProb_eq_weakNumerator_ratio
#check @ChapterWeakValue.dslit_weakValue
```

The last is the double-slit capstone: pre-selecting the both-slits superposition $H\Psi$ and post-selecting the definite state $\Psi = (1, 0)$, the two which-slit projectors have weak values 1 and 0.

14.9. Summary

- The electron's evolution is a product of two non-deterministic transformations, so no stochastic process exists through the slits; "which slit?" has no answer.
- In a two-angle model the propagation is the involutive Hadamard H.
- One slit closed: state $H\Psi$, giving $50/50$ at the two angles.
- Both slits open: state $H^2\Psi = \Psi$, giving $100/0$ — constructive and destructive interference.
- The coherent product H^2 differs from an incoherent, measured sequence; the difference is the interference cross term.
- Post-selection reconstructs the intermediate distribution consistently (`condProb` / ABL), and the weak value $\langle f|A|i\rangle / \langle f|i\rangle$ extends this to expectations of observables without collapsing the intermediate state; it is well-defined off orthogonality, linear in the observable, and reduces to the ordinary expectation on the diagonal (`BookProof.ChapterWeakValue`).

15. The Bell Inequalities

15.1. Orientation and Status

The second laboratory situation of the part: the CHSH inequality, its quantum violation, the Tsirelson bound, and the manuscript's reading of what the violation does and does not show. Read it after the double-slit chapter — the two experiments are read through the same lens and are best taken together.

The route: what the CHSH combination bounds, how quantum mechanics violates it, where the Tsirelson bound sits, and the manuscript's reading.

The proofs are `BookProof/ChapterBell` and `BookProof/ChapterTsirelson`.

15.2. What the Inequality Bounds

The CHSH inequality is a constraint that any **local** theory must satisfy. Two parties, Alice and Bob, each choose one of two measurements, labelled a_0, a_1 and b_0, b_1, each returning a value ± 1. The CHSH combination is

$$S = a_0 b_0 + a_0 b_1 + a_1 b_0 - a_1 b_1.$$

If the outcomes are determined by a shared local hidden variable — that is, if each of a_0, a_1, b_0, b_1 is a definite number ± 1 before measurement — then a pointwise algebraic bound gives $|S| \leq 2$. The verified pointwise bound and its expectation-value form (module `BookProof.ChapterBell`):

```
#check @ChapterBell.chsh_pointwise
#check @ChapterBell.chsh_local
```

The first is the elementary fact that for any four numbers ± 1 the combination S lies in $[-2, 2]$; the second lifts it to the **local bound** on the expectation value, $\langle S \rangle \leq 2$, for any local (hidden-variable) model.

15.3. Quantum Mechanics Violates It

Quantum mechanics does not assign definite values to all four measurements at once. Take two qubits in the **Bell state** and the Pauli observables

$$A_0 = \sigma_z, \quad A_1 = \sigma_x, \quad B_0 = \frac{\sigma_z + \sigma_x}{\sqrt{2}}, \quad B_1 = \frac{\sigma_z - \sigma_x}{\sqrt{2}}.$$

The expectation of the CHSH operator in the Bell state is then $2\sqrt{2}$, which exceeds the local bound 2. The verified statements (module `BookProof.ChapterBell`):

```
#check @ChapterBell.chsh_quantum_value
#check @ChapterBell.chsh_quantum_violates_local_bound
```

The first computes the quantum value $\langle S \rangle = 2\sqrt{2}$ exactly; the second is the bare inequality $2 < 2\sqrt{2}$ that records the violation.

15.4. The Tsirelson Bound

The value $2\sqrt{2}$ is not arbitrary: it is the **maximum** quantum violation, the **Tsirelson bound**. The verified statements (module `BookProof.ChapterTsirelson`):

```
#check @ChapterTsirelson.chshTuple_isCHSHTuple
#check @ChapterTsirelson.tsirelson_value_eq
#check @ChapterTsirelson.tsirelson_bound_tight
```

The first checks that the four observables above form a valid CHSH tuple (self-adjoint involutions with Alice's observables commuting with Bob's); the second records the identity $2\sqrt{2} = (\sqrt{2})^3$; the third proves the bound is **tight**: the Bell state is an eigenvector of the CHSH operator with eigenvalue $2\sqrt{2}$, so this concrete tuple saturates the Tsirelson bound. (The universal upper bound — no quantum state can exceed $2\sqrt{2}$ — is Mathlib's `tsirelson_inequality`.) So quantum mechanics sits strictly between the local bound 2 and the algebraic maximum 4, at exactly $2\sqrt{2}$.

15.5. The Manuscript's Reading

The standard reading is that the Bell violation rules out **local hidden-variable** theories. The manuscript offers a more specific reading, tied to the stochastic-process theorem: the Bell assumptions implicitly treat the time-evolution as a **stochastic process** — a probability distribution at each time, glued by conditionals. But that theorem shows the time-evolution is a stochastic process **if and only if it is deterministic**.

On this reading, what the Bell inequalities actually distinguish is quantum mechanics from a theory whose time-evolution *is* a stochastic process — not from "any complete statistical theory." A complete statistical theory whose time-evolution is non-deterministic does not define a probability distribution at each intermediate time, so it need

not satisfy the Bell assumptions. The inequalities are mathematically valid; the question is whether their assumptions are physically mandatory. The manuscript's claim is that they are not — that they encode the extra, optional assumption that the evolution is a stochastic process.

This is a **interpretive** claim about the physical meaning of the Bell assumptions; the machine-verified content is the mathematics above: the local bound 2, the quantum value $2\sqrt{2}$, and the tight Tsirelson bound.

15.6. Summary

- Any local hidden-variable model satisfies the CHSH bound $\langle S \rangle \leq 2$ (a pointwise algebraic fact).
- Quantum mechanics, with the Bell state and Pauli observables, achieves $2\sqrt{2} > 2$.
- $2\sqrt{2}$ is the **Tsirelson bound**: the tight quantum maximum.
- The manuscript reads the violation as distinguishing quantum mechanics from theories whose time-evolution is a stochastic process — an interpretation built on the theorem that such an evolution is stochastic iff deterministic.

16. EPR-Completeness, Causality, and a Deterministic Theory

16.1. Orientation and Status

The chapter that closes the logical circle of the part: EPR-completeness, relativistic causality, an explicit deterministic theory, and the classical limit. It opens with the negative fact that shapes everything below — there is no uniform probability measure on a countable space — and needs only the chapters before it in this part.

The route: no uniform measure on a countable space, the two kinds of incompleteness (quantum mechanics has only the first), ensemble forecasting as many models in one, when probabilities become calculable (the classical limit), the monotonicity of entropy for Markov processes, relativistic causality, the existing deterministic theory (inverse-transform sampling), and why the uniform countable measure cannot exist.

The proofs are `BookProof/ChapterNoUniformCountable`, `BookProof/ChapterClassicalLimit`, `BookProof/ChapterMarkovEntropy`, `BookProof/ChapterCausality`, and `BookProof/ChapterInverseTransform`.

16.2. No Uniform Measure on a Countable Space

We begin with a negative fact that shapes everything below. There is **no uniform probability measure on a countable space**: the rationals are not enough for probability theory. The verified statements (module `BookProof.ChapterNoUniformCountable`):

```
#check @ChapterNoUniformCountable.no_uniform_countable_measure
#check @ChapterNoUniformCountable.no_uniform_measure_nat
```

A standard probability space — which mixes a countable part and a continuous part — is therefore **irreducible**: it cannot be replaced by a countable model carrying a uniform measure. This is the measure-theoretic reason the wave-function parametrization lives on a genuine continuum.

16.3. Two Kinds of Incompleteness

The manuscript distinguishes **two** kinds of incompleteness that a stochastic process can have, and argues that conflating them is the source of the EPR paradox.

Stochastic incompleteness

> The root of probability is **absence of information**. A deterministic process can be turned into a stochastic one unambiguously (using trivial, delta-function distributions), but a stochastic process cannot be turned back into a deterministic one without **new information**. This kind of incompleteness is intrinsic to any statistical theory.

Non-Markov incompleteness

Any **non-Markov** stochastic process can be described as a **Markov** process in which some variables defining the state are hidden (unknown). Conversely, hiding variables from an irreducible Markov process produces a non-Markov process. Brownian motion is the textbook example: the underlying molecular dynamics is deterministic and Markov, but the observed motion is non-Markov because the full state cannot be measured.

16.4. Quantum Mechanics Has Only the First Kind

In quantum mechanics, any sequence of measurements **is** a Markov stochastic process: the outcome of each measurement, together with the state update, depends only on the present state. So quantum mechanics has the **stochastic** kind of incompleteness — as every statistical theory must. But it does **not** have the non-Markov kind: there are no hidden variables whose revelation would restore a deterministic Markov description.

The EPR argument tries to infer the non-Markov kind from the stochastic kind — to argue that because position and momentum cannot be simultaneously measured, they must be simultaneous "elements of reality" in some deeper description. The manuscript's response is that this inference has no mathematical basis: the stochastic kind of incompleteness is **harmless**, and the wave-function parametrization shows that no deterministic model is more fundamental than quantum mechanics. In this precise sense, **quantum mechanics is a complete statistical theory** as EPR defined completeness.

The structural reason, again, is the stochastic-process theorem: the EPR setup assumes a probability distribution for the state of the separated system *between* preparation and measurement, but a non-deterministic time-evolution defines no such distribution.

16.5. Ensemble Forecasting: Many Models as One

The rules for updating a probability measure are not obvious, because one can always consider a **probability space of probability spaces** — an ensemble of statistical models, which the manuscript calls **ensemble forecasting**. Assuming a quantum (linear) time-evolution for the ensemble imposes essentially no restriction on the time-evolution of the individual models, which may still be non-linear.

Using the free-field parametrization, one can **diagonalize** the quantum time-evolution, decomposing a non-linear infinite-dimensional model into a direct integral of **linear models with a single Boolean variable**. A direct integral can still be uncomputable, so the next step is to decompose a continuous spectrum into a direct **sum** of small energy intervals, each described by **few variables**. The manuscript presents this as a mathematical definition of **renormalization**: for reasonable initial conditions (we never have access to an infinite range of energies in a real experiment) one can predict the time-evolution using a model with few variables, using different unrelated models for different energy ranges.

The key conceptual point: the wave-function defines an **ensemble of deterministic transformations** (functions). When few variables are relevant, each such function can be studied and computed individually. This is why deterministic logic and deterministic mathematics find application in a world full of uncomputable functions — not because the functions are computable, but because the *relevant* ones, in a given energy range, are.

16.6. The Classical Limit: When Probabilities Become Calculable

When can one discard probabilities in coordinate space and recover classical mechanics? The manuscript's answer is deliberately non-rigorous — it often works, but not always, so **classical mechanics is never enough at any energy scale; quantum mechanics is always needed**. The mechanism is the density of calculable functions.

Step functions, polynomials, and smooth functions are **dense** in the L^2 measure: any bounded function on a compact domain can be approximated, in the average over that domain, by calculable functions. The verified density theorems (module `BookProof.ChapterClassicalLimit`):

```
#check @ChapterClassicalLimit.simpleFunc_dense_L2
#check @ChapterClassicalLimit.continuousMap_denseRange_L2
```

```
#check @ChapterClassicalLimit.polynomial_denseRange_L2
#check @ChapterClassicalLimit.exists_polynomial_approx_L2
```

Reading them: simple (step) functions are dense in L^2; continuous functions are dense; polynomial functions are dense; and concretely, every L^2 function admits a polynomial approximation within any prescribed $\varepsilon > 0$.

The crucial caveat: approximating a function **in the average over a compact domain** does not make the function itself calculable. In macroscopic phenomena one usually *does* average over a large domain (partitioning it for a numerical approximation), so calculable functions suffice. In **microscopic** phenomena — the double-slit experiment is the example — opening one slit or two makes a macroscopic difference, and averaging destroys exactly the effect one wants. There, only the **probabilities** admit calculable L^2 approximations, not the underlying function. Chaos and finite-time singularities of ODEs are phenomena of *computable* functions (computable for a small finite time); genuinely uncomputable functions are uncomputable for *any* finite time.

16.7. Markov Processes Are Monotone in Entropy

Finally, the structural reason a Markov process cannot produce an arbitrary function of time: there is an **ordering**, related to entropy, with respect to which all continuous-time Markov processes are monotone. A doubly stochastic (Markov) map **increases** entropy, while a mere permutation of outcomes preserves it. The verified statements (module `BookProof.ChapterMarkovEntropy`):

```
#check @ChapterMarkovEntropy.entropy_applyMarkov_ge
#check @ChapterMarkovEntropy.permMatrix_doublyStochastic
#check @ChapterMarkovEntropy.entropy_applyMarkov_permMatrix
```

Reading them: applying a doubly stochastic matrix to a distribution does not decrease its entropy; a permutation matrix is doubly stochastic; and a permutation leaves the entropy **unchanged**. This is the precise sense in which Bayesian inference (a Markov process) is **irreversible**, and why the reversible unitary models of this book are a genuine generalization of it: a unitary evolution can be non-deterministic without being entropy-increasing in the Markov sense.

16.8. Relativistic Causality

Now restrict to a **free** system, where relativistic quantum mechanics is well-defined (the free Dirac equation). Relativistic causality holds there: the propagator vanishes for space-like separation, so the probability that the system moves faster than light is **null**.

A deterministic theory **compatible** with relativistic quantum mechanics is one that, applied to an ensemble of free systems, reproduces quantum mechanics' statistical predictions. Since in relativistic quantum mechanics the probability of superluminal motion is null, **no** system in the ensemble moves faster than light; hence any such deterministic theory necessarily respects relativistic causality. The verified measure-theoretic statement (module `BookProof.ChapterCausality`):

```
#check @Causality.seedSet_biUnion_measure
#check @Causality.causality
#check @Causality.causality_ae
```

Reading them: the seed sets (the regions of the random seed that produce each outcome) have a controlled union measure; `causality` bounds the measure of the "superluminal" set; and `causality_ae` states that this set is **almost everywhere empty** — the probability of violating causality is zero.

16.9. A Deterministic Theory Exists: Inverse-Transform Sampling

Does such a deterministic theory actually exist? Yes, and the construction is the classical **inverse-transform sampling** method. An experiment always yields a discrete set of outcomes, so quantum mechanics predicts a cumulative distribution function. Partition the unit interval into **seed sets** whose lengths are exactly the predicted probabilities:

$$\mathrm{seedSet}(k) = [\,\mathrm{cdf}(k),\ \mathrm{cdf}(k+1)\,).$$

The verified partition (module `BookProof.ChapterInverseTransform`):

```
#check @InverseTransform.cdf_monotone
#check @InverseTransform.seedSet_measure
#check @InverseTransform.seedSet_disjoint
#check @InverseTransform.seedSet_cover
#check @InverseTransform.seedSet_total_measure
```

Reading them: the CDF is monotone; the k-th seed set has measure exactly p_k; the seed sets are pairwise disjoint; they cover the whole unit interval; and their total measure is 1. The deterministic theory is then: pick a seed (a point of $[0,1]$), find which seed set it lies in, and set the outcome accordingly. Each run uses a different seed, so the procedure reproduces the quantum probabilities exactly — and is therefore **experimentally indistinguishable** from quantum mechanics.

The manuscript is candid that this theory is metaphysically unappealing (it relies on pseudo-random number generation — one would have to "program" each particle), and that it is **not** super-deterministic: the experimenter remains free to choose which measurements to perform. Its purpose is logical, not persuasive: it exhibits a complete, deterministic, causality-respecting theory compatible with quantum mechanics, which is enough to show that the EPR and Bell arguments rest on an extra assumption rather than on a theorem.

16.10. Why No Uniform Countable Measure

One subtlety underlies the whole construction: there is **no** uniform probability measure on a countable space. The verified statement (module `BookProof.ChapterNoUniformCountable`):

```
#check @ChapterNoUniformCountable.no_uniform_countable_measure
```

This is why the seed space must be the **continuum** $[0,1]$ (a standard measure space) rather than a countable set of seeds, and it is the measure-theoretic reason the deterministic theory uses a continuous random seed.

16.11. Summary

- There is no uniform measure on a countable space; the standard (countable + continuous) probability space is irreducible.
- Ensemble forecasting + the free-field parametrization decompose a non-linear infinite-dimensional model into linear models with few variables — a mathematical definition of renormalization, and the reason deterministic mathematics applies to a world of uncomputable functions.
- The classical limit rests on the L^2-density of step, continuous, and polynomial functions: calculable on average over a compact domain, but not pointwise — which is why microscopic phenomena still need probabilities.
- Markov processes are monotone in entropy (doubly stochastic maps increase it, permutations preserve it), so they are irreversible; unitary evolution generalizes them.
- A stochastic process can be incomplete in two ways: **stochastic** (missing information) and **non-Markov** (hidden variables).
- Quantum mechanics has only the stochastic kind, so it is a **complete** statistical theory in EPR's sense; the EPR inference to the non-Markov kind has no mathematical basis.
- Any deterministic theory compatible with relativistic quantum mechanics respects causality, because the superluminal set has measure zero.
- Such a theory exists, by inverse-transform sampling on $[0,1]$; it reproduces the quantum probabilities exactly and is not super-deterministic.
- The construction needs a continuous seed because there is no uniform measure on a countable space.

17. Wave-functions, Euler's Formula, and the Born Rule

The heart of the book. We make precise the parametrization of a probability distribution by a wave-function: the probability clock and Euler's formula in two dimensions, the fact that the Born rule reproduces *every* distribution in any finite (or countable) dimension, the gauge ambiguity (the invisible phase) of the parametrization, the parametrization of any joint or conditional probability by a unitary operator (the finite-dimensional core of the book's commutative Wigner theorem), and information erasure in the Stern–Gerlach experiment.

How to read this part. The probability clock is the smallest possible instance of the whole book — read it first even if you read nothing else in this part, since it introduces the generator $J^2 = -1$, Euler's formula, and the singular collapse matrix in two dimensions. The Born rule reproduces every distribution lifts the construction to n outcomes, countable chains, and the complex and quaternionic cases; the gauge ambiguity explains what is **invisible** in the parametrization. The remaining two chapters are the bridge to physics: a joint probability is a wave-function is the finite-dimensional core of the commutative Wigner theorem, and the Stern–Gerlach experiment is the first laboratory application. This part is the prerequisite for the field-theory and relativity part that follows.

18. The Probability Clock and Euler's Formula

18.1. Orientation and Status

This chapter develops the two-state model that the manuscript uses as its smallest example of wave-function parametrization. The finite matrix identities below are fully formalized; the interpretive claims about collapse and the physical meaning of quantum mechanics remain explanatory, not additional Lean theorems. Later chapters reuse this clock as a two-dimensional building block.

This chapter opens the heart of the book — the wave-function parametrization — in its smallest possible instance: two outcomes, one angle. Read it first even if you read nothing else in this part; it introduces the generator $J^2 = -1$, Euler's formula as a matrix identity, and the singular collapse matrix that later chapters reuse as a two-dimensional building block.

The route: the two-state system, the generator that squares to minus one, Euler's formula as a matrix identity, why the parametrization acts on the wave-function and not on the probability, the density matrix and collapse, and why coherence is a feature of the parametrization rather than a phenomenon.

The finite matrix identities are `BookProof/ChapterProbabilityClockStochastic` and `BookProof/ChapterEulerStochastic` / `ChapterEulerDensityMatrix`; the interpretive claims stay explanatory, as the status paragraph above says. The next chapter lifts the construction to n outcomes.

18.2. A Two-State System

The smallest non-trivial probability space has two outcomes. A distribution on it is a pair $(p, 1-p)$, a single number $p \in [0,1]$. The source manuscript calls the wave-function parametrization of this space the **probability clock**:

$$\Psi(t) = (\cos t,\ \sin t), \qquad p = \cos^2 t, \quad 1 - p = \sin^2 t.$$

As the parameter t advances, the point $\Psi(t)$ travels around the unit circle and the probabilities oscillate between the two outcomes. This is the real, two-dimensional incarnation of a quantum two-level system (a *qubit*).

18.3. The Generator Squares to Minus One

What moves the clock? The infinitesimal generator of the rotation $\Psi(t) \mapsto \Psi(t+a)$ is the matrix

$$J = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}.$$

A direct multiplication shows the single most important fact about this matrix:

$$J^2 = \begin{pmatrix} -1 & 0 \\ 0 & -1 \end{pmatrix} = -\mathbf{1}.$$

The generator squares to **minus the identity**. This is exactly the defining property of the imaginary unit i, and it is the reason the complex numbers appear in quantum mechanics: the real rotation generator behaves algebraically like i.

18.4. Euler's Formula, as a Matrix Identity

Because $J^2 = -\mathbf{1}$, the exponential of aJ collapses, by the power series of the exponential, into sines and cosines:

$$e^{aJ} = \cos a\, \mathbf{1} + \sin a\, J = \begin{pmatrix} \cos a & -\sin a \\ \sin a & \cos a \end{pmatrix} =: R(a).$$

This is **Euler's formula**, lifted from complex numbers to the rotation matrix. The matrix $R(a)$ is a genuine rotation: its determinant is 1, so it is invertible, and it acts on the clock by advancing the parameter:

$$\Psi(t + a) = R(a)\, \Psi(t).$$

The verified statements (module `BookProof.ChapterProbabilityClockStochastic`):

```
#check @ProbabilityClockStochastic.Jgen_sq
#check @ProbabilityClockStochastic.rotMat_eq_exp
#check @ProbabilityClockStochastic.rotMat_det
#check @ProbabilityClockStochastic.rotMat_mulVec_clockPsi
#check @ProbabilityClockStochastic.clockPsi_eq_exp
```

18.5. Why Act on the Wave-function and Not on the Probability?

Here is the crux of the whole book, in the smallest possible setting. We could try to act **directly** on the probability vector $(p, 1-p)$ instead of on the wave-function. Which linear maps preserve the space of probability vectors?

A linear map preserves probability vectors **exactly when it is column-stochastic**: each column is itself a probability vector. The most general such 2×2 map is parametrized by two angles:

$$M(a, b) = \begin{pmatrix} \cos^2 a & \cos^2 b \\ \sin^2 a & \sin^2 b \end{pmatrix}.$$

The verified classification (module `BookProof.ChapterEulerStochastic`):

```
#check @ChapterEulerStochastic.preservesProb_iff_exists_angles
#check @ProbabilityClockStochastic.isColumnStochastic_eq_Mab
```

Now ask: is there such a map that sends the **uniform** distribution $\frac{1}{2}(1,1)$ to a **vertex** $(1,0)$ (a deterministic outcome)? There is — but any such map has determinant **zero**:

```
#check @ChapterEulerStochastic.uniform_to_vertex_singular
#check @ProbabilityClockStochastic.stochastic_uniform_to_deterministic_singular
```

A singular matrix is **not invertible**, so it cannot represent a symmetry (a symmetry must be reversible). This is the precise obstruction the manuscript identifies: on the probability simplex, the only linear "symmetries" are stochastic matrices, and the operation that turns a superposition into a definite outcome — **collapse** — is necessarily irreversible.

By contrast, the rotation $R(a)$ acting on the wave-function has determinant 1: it is invertible, hence a genuine symmetry. **Reversible dynamics lives on the wave-function; irreversible collapse lives on the probabilities.** But this contrast is a feature of the parametrization, not a physical dichotomy: it says only that the wave-function coordinates admit invertible maps that the probability coordinates do not. The manuscript reads it exactly this way, as a property of the chosen coordinates rather than a new fact about the world.

18.6. The Density Matrix and Collapse

It is illuminating to write the **density matrix** of the clock, the rank-one projector $\rho(t) = \Psi(t)\,\Psi(t)^{\top}$:

$$\rho(t) = \begin{pmatrix} \cos^2 t & \cos t \sin t \\ \cos t \sin t & \sin^2 t \end{pmatrix}.$$

Using J and the diagonal generator $Z = \frac{1}{2}\,\mathrm{diag}(1, -1)$, this projector can be rewritten in Euler form:

$$\rho(t) = \tfrac{1}{2}\,\mathbf{1} + Z\,(\cos 2t\,\mathbf{1} + \sin 2t\,J).$$

The term proportional to J is the **off-diagonal** (coherence) part. The manuscript's model of **wave-function collapse** is simply: set the J-proportional part to zero. What remains is the diagonal, classical distribution:

$$\rho(t) \;\longrightarrow\; \begin{pmatrix} \cos^2 t & 0 \\ 0 & \sin^2 t \end{pmatrix}.$$

The verified statements (module `BookProof.ChapterEulerDensityMatrix`):

```
#check @ChapterEulerDensityMatrix.density_euler
#check @ChapterEulerDensityMatrix.density_collapse
#check @ChapterEulerDensityMatrix.Jdens_sq
#check @ChapterEulerDensityMatrix.densityMatrix_trace
#check @ChapterEulerDensityMatrix.densityMatrix_idempotent
```

The last of these, $\rho^2 = \rho$, says the clock is in a **pure** state; the trace $\mathrm{tr}\,\rho = 1$ says it is normalized. Collapse destroys the idempotent purity by deleting the coherence term, leaving a genuine classical probability on the diagonal.

18.7. Coherence Is a Feature of the Parametrization, Not a Phenomenon

A word of interpretation the source manuscript insists on. The off-diagonal "coherence" term above is **not a physical phenomenon**. It is an artifact of describing a probability distribution in wave-function coordinates: the same distribution, written on the simplex, carries no such term. The wave-function is "nothing else than one possible parametrization of any probability distribution," and the coherence is bookkeeping the parametrization carries, not a substance in the world.

Read this way, **collapse** is not a physical process either. It is exactly the operation the manuscript calls "a generalization of taking the real part of Euler's formula": delete the J-proportional (off-diagonal) part and keep the diagonal probabilities. And because *every* probability distribution admits a wave-function parametrization, the same "collapse" can be written down for any random phenomenon whatsoever. This is the manuscript's conclusion that "the wave-function collapse is a feature of all random phenomena" — not a mysterious quantum event, but a consequence of choosing these coordinates.

18.8. Summary

In two dimensions we have already seen the entire mechanism of the book:

- parametrizing the probability by a wave-function $\Psi = (\cos t, \sin t)$;
- a rotation generator J with $J^2 = -1$, i.e. an emergent imaginary unit;
- Euler's formula $e^{aJ} = R(a)$ giving reversible, unitary dynamics;
- the obstruction that direct linear action on probabilities is stochastic and singular at collapse, hence irreversible;
- collapse as deletion of the J-proportional coherence in the density matrix.

The rest of Part II lifts each of these facts from two dimensions to arbitrary finite dimension, and then to the countable and continuous settings.

19. The Born Rule Reproduces Every Distribution

19.1. Orientation and Status

The lifting chapter: the probability clock's one angle becomes $n - 1$ Euler angles, and the Born rule is shown to reproduce **every** distribution. Read it after the probability clock; it is the second of the three parametrization chapters, and the next chapter asks what the map forgets.

The route: from two states to n states (the telescoping stick-breaking process), the construction always sums to one, it reaches every distribution, the countable case as an infinite stick-breaking chain, the complex and quaternionic wave-functions, and the average-versus-maximal error comparison.

The proofs are `BookProof/ChapterEulerNState` (the parametrization) and `BookProof/ChapterErrorNorms` (the error comparison).

19.2. From Two States to n States

The probability clock parametrized a two-outcome distribution by one angle. The same idea extends to any finite number n of outcomes using $n - 1$ **Euler angles** $\theta_0, \ldots, \theta_{n-2}$. The construction is a telescoping **stick-breaking** process: at step k we break off a fraction $\cos^2 \theta_k$ of the probability mass that remains, and pass the rest $\sin^2 \theta_k$ onward.

Concretely, define the **tail product** (the mass still unassigned before outcome m)

$$T(\theta, m) = \prod_{i<m} \sin^2 \theta_i,$$

and the **Born probability** of outcome k

$$\mathrm{bornProb}(\theta, n, k) = T(\theta, k)\, \cos^2 \theta_k = \Big(\prod_{i<k} \sin^2 \theta_i\Big) \cos^2 \theta_k,$$

with the last outcome $k = n - 1$ carrying the remaining product of sines (no cosine factor).

19.3. It Always Sums to One

The first thing to check is that this really is a probability distribution, for *every* choice of angles. The proof is a telescoping sum: since $\cos^2 \theta_k = 1 - \sin^2 \theta_k$, the partial sums collapse:

$$\sum_{k<n} \mathrm{bornProb}(\theta, n, k) = 1 - T(\theta, n) + T(\theta, n) \cdot (\text{last term}) = 1.$$

The verified statement (module `BookProof.ChapterEulerNState`):

```
#check @ChapterEulerNState.euler_sum_one
#check @ChapterEulerNState.euler_wave_unit
```

The second of these says the underlying **Euler wave-function** $\varphi_k = (\prod_{i<k} \sin \theta_i) \cos \theta_k$ is a **unit vector**, $\sum_k \varphi_k^2 = 1$; the Born probabilities are its squared coordinates $\mathrm{bornProb} = \varphi_k^2$.

19.4. It Reaches Every Distribution

The converse is the substantive claim: **every** probability distribution on n outcomes arises this way. Given $p_0, \ldots, p_{n-1} \geq 0$ summing to one, choose the angles backwards from the tail. The key elementary fact is that every number in $[0, 1]$ is a $\cos^2$ (and a $\sin^2$) of some angle:

```
#check @ChapterEulerNState.exists_cos_sq
```

Setting $\cos^2\theta_k = p_k / T(\theta, k)$ — the fraction of the *remaining* mass that outcome k should carry — reproduces p exactly:

```
#check @ChapterEulerNState.euler_reproduces
```

This is the finite-dimensional heart of the book's thesis: the Born-rule parametrization is **surjective** onto the simplex. No distribution is left out.

19.5. The Countable Case: an Infinite Stick-Breaking Chain

The manuscript emphasizes that "the recursion does not need to stop." For a **countably infinite** partition, write the conditional probabilities $c_n = P(n \mid n \text{ or above}) \in [0,1]$ and define

$$T(c, N) = \prod_{k<N} (1 - c_k), \qquad P(n) = T(c, n)\, c_n.$$

The same telescoping gives the exact finite normalization

$$\sum_{n<N} P(n) = 1 - T(c, N),$$

and if the tail $T(c, N) \to 0$ (the stick is eventually fully broken), the point masses sum to exactly one:

```
#check @ChapterEulerCountableChain.partial_sum
#check @ChapterEulerCountableChain.stick_tsum_one
```

Writing $c_n = \cos^2\theta_n$ recovers the Euler-angle form $P(n) = (\prod_{k<n} \sin^2\theta_k)\cos^2\theta_n$, the countable analogue of the finite formula — and `stickProb_euler` identifies each stick-breaking probability with a $\cos^2$ of an angle:

```
#check @ChapterEulerCountableChain.stickProb_euler
#check @ChapterEulerCountableChain.euler_tsum_one
```

19.6. Complex and Quaternionic Wave-functions

So far the wave-function has been real. Over the **complex** numbers, the Born probability of coordinate k is $|v_k|^2 = (\mathrm{Re}\, v_k)^2 + (\mathrm{Im}\, v_k)^2$: a complex wave-function is just a real one on a state space with **twice** as many outcomes, and its Born probability is the sum of the two real Born probabilities. The manuscript's quaternionic case is the same with **four** real coordinates, $P(n) = \sum_{m=1}^{4} P(n, m)$:

```
#check @ChapterEulerComplexQuat.complex_born_split
#check @ChapterEulerComplexQuat.quat_born_split
```

And the surjectivity survives: every distribution is still reproduced by a complex (resp. quaternionic) unit wave-function:

```
#check @ChapterEulerComplexQuat.complex_reproduces
#check @ChapterEulerComplexQuat.quat_reproduces
```

This justifies the book's use of complex (and quaternionic) Hilbert spaces while keeping the real Euler-angle parametrization as the underlying construction: passing to $\mathbb{C}$ or $\mathbb{H}$ does not change which distributions are reachable, only how many real coordinates each outcome hides.

The manuscript's argument that these are the **only** fields worth considering runs through the **real Schur's lemma**: if a set of normal operators (the projections together with a unitary representation of a symmetry group) leaves no non-trivial closed subspace invariant, then the algebra of operators commuting with all of them is a **real associative division algebra** — and such an algebra is isomorphic to exactly one of the real numbers, the complex numbers, or the quaternions. The field over which the wave-function lives is therefore **not** an extra physical postulate: it is read off from the commuting algebra of the symmetry representation, and the parametrization works uniformly over all three. The verified norm fact underlying the complex case is that the norm of a complex wave-function equals the norm of its **realification** (a complex Hilbert space is a real one of twice the dimension, with the same norm):

```
#check @ChapterEulerComplexQuat.cbornProb_nonneg
#check @ChapterEulerComplexQuat.complex_realification_norm
```

19.7. Average versus Maximal Error

The parametrization has a pleasant analytic consequence that the manuscript relies on when comparing models. When a probability density ρ is written as the square of a wave-function, $\rho = \psi^2$ with $\psi = \sqrt{\rho}$, the wave-function is automatically **normalized** whenever the density is: its squared L^2 norm is the total mass of ρ, namely 1:

```
#check @BookProof.ChapterErrorNorms.wavefunction_l2_norm_sq_eq_integral
#check @BookProof.ChapterErrorNorms.wavefunction_lintegral_sq_eq_one
```

The manuscript contrasts two ways of measuring the error of a model: the **average** error (the L^2 norm of the wave-function) and the **maximal** error (the L^∞ essential supremum). On a probability space the average error is always dominated by the maximal error — the two live in different L^p spaces and the order of the L^p norms on a probability space is $\|f\|_2 \le \|f\|_\infty$:

```
#check @BookProof.ChapterErrorNorms.l2_le_linfty_of_finite
#check @BookProof.ChapterErrorNorms.l2_le_of_ae_bound
```

So bounding the pointwise (maximal) error automatically bounds the average error, and the wave-function normalization is not an additional constraint but a direct consequence of the density's being a probability density.

20. The Gauge Ambiguity of the Born Parametrization

20.1. Orientation and Status

The chapter of the part that asks what the parametrization **forgets**: the Born map is surjective (the previous chapter) but far from injective, and its fibers are exactly the coordinate-wise phases (or signs). That redundancy is the book's first **gauge freedom**, and the chapter closes with the structural consequence — no continuous symmetry survives on the simplex.

The route: the many-to-one question and its precise answer, the complex fiber (coordinate-wise phases), the real fiber (coordinate-wise signs), why this is "gauge", and the consequence for continuous symmetries on the simplex.

The fiber theorems are `ChapterBornPhaseFiber.born_fiber_complex` and `ChapterBornPhaseFiber.born_fiber_real`. The geometric version of the same statement — the simplex as the sphere modulo the gauge — is developed in the free-field chapter.

20.2. The Parametrization Is Many-to-One

We have seen that the Born map — sending a unit wave-function ψ to the distribution $p_k = |\psi_k|^2$ — is **surjective**: every distribution is reached. It is emphatically **not injective**: many different wave-functions give the same probabilities. The most familiar example is the global phase: over the complex numbers, ψ and $e^{i\theta}\psi$ give identical Born probabilities.

The natural question is: *exactly* how many wave-functions map to a given distribution? The answer, made precise below, is that two wave-functions give the same Born probabilities **if and only if** they differ by an invisible phase (or sign, in the real case) at each coordinate. This invisible redundancy is the **gauge ambiguity** of the parametrization, and it is the seed of gauge symmetry.

20.3. The Complex Fiber: Coordinate-wise Phases

For complex wave-functions $u, v : \{0, \dots, n-1\} \to \mathbb{C}$, the Born probabilities agree, $|u_k|^2 = |v_k|^2$ for all k, **if and only if** each coordinate differs by a phase $e^{i\theta_k}$:

$$|u_k| = |v_k| \iff \exists \theta_k,\ v_k = e^{i\theta_k} u_k.$$

The single-coordinate fact is just the polar form of a complex number; the multi-coordinate statement is its pointwise application:

```
#check @ChapterBornPhaseFiber.Complex.normSq_eq_iff_exists_phase
#check @ChapterBornPhaseFiber.born_fiber_complex
```

So the **fiber** of the Born map over a distribution with all-positive entries is an n-torus $(S^1)^n$: one independent phase per coordinate. A wave-function is a probability distribution together with a choice of these invisible phases.

20.4. The Real Fiber: Coordinate-wise Signs

Over the **real** numbers the only numbers with a given squared value differ by a **sign** ± 1, so the fiber is the finite group $\{\pm 1\}^n$:

```
#check @ChapterBornPhaseFiber.sq_eq_iff_exists_sign
#check @ChapterBornPhaseFiber.born_fiber_real
```

The real parametrization of the previous chapter thus double-covers the interior of the simplex: each strictly-positive distribution is the Born image of 2^n real wave-functions, related by coordinate-wise sign flips.

20.5. Why This Is "Gauge"

A redundancy in a parametrization that has no observable effect is precisely what physicists call a **gauge freedom**. Here the observables are the probabilities $p_k = |\psi_k|^2$, and the gauge transformations are the coordinate-wise phases $\psi_k \mapsto e^{i\theta_k}\psi_k$ (or signs, over $\mathbb{R}$). Two wave-functions related by such a transformation describe the **same** physical probability distribution.

The manuscript's refrain — "two wave-functions are always related by a rotation of the hypersphere" — is the geometric version of this: the Born fibers are the orbits of a group of invisible rotations, and the simplex is the **quotient** of the sphere by that group. Making that quotient precise (the sphere modulo the phase/sign gauge) is the subject of the free-field chapters; the fiber computation above is its algebraic core.

20.6. A Consequence: No Continuous Symmetry on the Simplex

Recall from the probability clock that the only linear maps preserving the probability simplex are stochastic matrices, and the collapse-to-a-vertex one is singular. The gauge picture explains *why* the genuine (invertible) symmetries live one level up, on the wave-function: a symmetry of the probabilities must lift to a symmetry of the sphere that respects the fibers, i.e. a **unitary** transformation. The phases we just identified are exactly the kernel of the projection from unitary dynamics on the sphere to stochastic dynamics on the simplex.

21. Conditional Probability Is Parametrized by a Unitary

21.1. Orientation and Status

This chapter separates two levels that are easy to conflate. The finite-dimensional Gram--Schmidt/SVD construction is proved and is the main formal result here. The manuscript's statement for arbitrary standard measure spaces requires additional measurability, tensor-product, and disintegration infrastructure; the cited finite results should therefore be read as a verified core, not as that full infinite- dimensional theorem.

In the tour this chapter is the bridge from probability to physics: the finite-dimensional core of the book's commutative Wigner theorem, where a **joint** probability is parametrized by a unitary operator rather than by angles. It follows the three parametrization chapters (the clock, n outcomes, the fiber) and precedes the Stern–Gerlach application that closes the part.

The route: the question, a joint probability is a wave-function, the converse (an operator defines a joint probability), the kernel operator is bounded, the singular-value expansion, marginal and conditional probability, the concrete

Pauli–Grover rotation, what is verified versus what is infinite-dimensional, the less arbitrary unitary built from the dynamics, and the generalization to conditional probability on a standard Hilbert space.

The verified core includes `BookProof/ChapterBornMeasure`, `BookProof/ChapterPauliGrover`, `BookProof/ChapterUnitaryTransport`, and the selecting-events chapter `BookProof/ChapterSelectingEvents`; the infinite-dimensional bridge is `BookProof/ChapterContinuityUnitary` (with its infinite variants and the Stone/unbounded-spectral chapters).

21.2. The Question

The Born rule (the Born rule reproduces every distribution) parametrizes a single probability distribution by a wave-function: $p(x) = |\Psi(x)|^2$. The manuscript goes further. **Any joint probability density** $p(x, y)$ between two standard measure spaces X, Y can be written as

$$p(x, y) = |\mathcal{U}(y, x, 0)|^2,$$

where $\mathcal{U} : L^2(\mathbb{Z}) \to L^2(X \times Y)$ is a **unitary** operator. The book calls this "a commutative version of Wigner's theorem." Its consequence is that any conditional probability measure in a standard measure space is parametrized by a unitary operator — and therefore quantum processes are not exotic, they are simply *more general* than Markov processes. This chapter verifies the finite-dimensional core of that statement, following the book's own proof: write the joint density as a wave-function, build the unitary by Gram–Schmidt, bound the wave-function as an operator, take its singular-value expansion, and read off the marginal and conditional probabilities.

21.3. A Joint Probability Is a Wave-function

Since $p(x, y) \geq 0$ and $\sum_{x,y} p(x, y) = 1$, there is a normalized wave-function $\Psi \in L^2(X \times Y)$ with $|\Psi(x, y)|^2 = p(x, y)$: take $\Psi(x, y) = \sqrt{p(x, y)}$. The book then builds a unitary $\mathcal{U}$ whose chosen column *is* Ψ, "through the Gram–Schmidt process." In finite dimension this is the statement that every unit vector is a column of some unitary matrix — extend it to an orthonormal basis:

```
#check @exists_unitary_column
```

Applying this to $v(i) = \sqrt{p(i)}$, every finite probability distribution is the squared modulus of a column of a unitary matrix:

```
#check @exists_unitary_of_prob
```

And in the book's exact two-space form: **every joint probability distribution** $p(x, y)$ on a finite $X \times Y$ is $|U|^2$ on a column of a unitary matrix indexed by $X \times Y$:

```
#check @exists_unitary_joint
```

21.4. The Converse: an Operator Defines a Joint Probability

The converse also holds, and it is the direction the book uses to define a probability from an operator. Given any bounded operator B with $\mathrm{tr}(BB^\dagger) = 1$, the squared modulus $p(x, y) = |B(y, x)|^2$ is a genuine joint probability distribution:

```
#check @sqAbs_isProb_of_frobenius_one
#check @frobenius_eq_trace
```

So probability distributions and (trace-normalized) operators are two descriptions of the same object. This is the algebraic backbone of the wave-function parametrization.

21.5. The Kernel Operator Is Bounded

The book now views Ψ as an **integral operator** $\Psi : L^2(X) \to L^2(Y)$, acting by $\Psi\{\Phi\}(y) = \int dx\, \Psi(x, y)\, \Phi(x)$. By Cauchy–Schwarz, this operator is bounded, with operator norm at most the Hilbert–

Schmidt norm of the kernel. In the finite (discretized) model the kernel operator and its bound are explicit:

```
#check @kernelOp
#check @kernel_row_bound
#check @kernel_l2_bound
```

In particular, when the kernel has L^2 norm at most 1 — which holds for a normalized wave-function — the operator is a **contraction**:

```
#check @kernel_contraction
```

Boundedness is the prerequisite for the singular-value expansion: only a bounded operator admits a polar decomposition.

21.6. The Singular-Value Expansion

Because Ψ is bounded, it admits a singular-value expansion (through the polar decomposition) $\Psi = WDU^\dagger$, with $D \geq 0$ diagonal and W, U unitary. In finite dimension this is the ordinary SVD, discharged by diagonalizing the positive-semidefinite Gram matrix $A^\dagger A$ with the spectral theorem and taking square roots of its eigenvalues:

```
#check @gram_svd
#check @svd_completion
#check @denseCore_svd
```

The factor V that appears before completion to the unitary W is a **partial isometry**: $V^\dagger V$ and $VV^\dagger$ are orthogonal projections, and $VV^\dagger V = V$:

```
#check @IsPartialIsometry
```

The book then changes the marginal from a fixed reference $p_0 > 0$ to an arbitrary p, using $TpT^\dagger = WDU^\dagger(p/p_0)UDW^\dagger$. This change-of-marginal identity is verified at the operator level:

```
#check @conditional_operator_identity
```

21.7. Marginal and Conditional Probability

Finally the book reads the probabilities back off the operator. For a bounded operator B, the joint distribution is $p(x,y) = |B(y,x)|^2$, the **marginal** is the diagonal of the Gram matrix $B^\dagger B$, and the normalization is the trace:

```
#check @pJoint
#check @pMarg
#check @pMarg_eq_diagBHB
#check @trace_gram_eq_one
```

When the marginal is positive, $p(x) = \{B^\dagger B\}(x,x) > 0$, the **regular conditional probability** $p(y|x) = p(x,y)/p(x)$ is defined, is non-negative, and sums to one over y:

```
#check @pCond
#check @pCond_sum_one
```

And the joint distribution factors as marginal times conditional, exactly as the book parametrizes it, $p(x,y) = p(y|x)\,p(x)$:

```
#check @pJoint_eq_cond_mul_marg
```

The same marginal/conditional split is available for a **density matrix**. The spectral decomposition $\rho = U\,\mathrm{diag}(d)\,U^\dagger$ has a diagonal d which is a probability distribution — the marginal of the initial state — while the Born matrix $\|U_{ij}\|^2$ of the unitary is a **doubly stochastic** conditional probability, and the final-state marginal is obtained from d by that conditional, $\rho_{ii} = \sum_k \|U_{ik}\|^2 d_k$:

```
#check @BookProof.DensitySpectral.bornKernel
#check @BookProof.DensitySpectral.bornKernel_row_sum
#check @BookProof.DensitySpectral.bornKernel_col_sum
#check @BookProof.DensitySpectral.density_diag_eq_kernel_apply
```

```
#check @BookProof.DensitySpectral.density_diag_isProbability
#check @BookProof.DensitySpectral.density_marginal_conditional
```

21.8. A Concrete Parametrization: the Pauli–Grover Rotation

The abstract theorem of this chapter — every conditional probability measure is parametrized by a unitary operator — has a concrete finite-dimensional instance: the Pauli–Grover rotation of `QFM.tex` (**Alternate Hamiltonian: the Pauli--Grover construction**). For each training pair `(i, f_i)` the generator acts as a Pauli-X rotation in the two-dimensional subspace `{ |i,0⟩, |i,f_i⟩ }` of input `i`'s output space. At the ideal coupling `a = 1` it is the exact swap, so evolving `|0⟩` for time `τ = π/2` yields `|f⟩` up to an unobservable global phase. The readout probability $p(f|i) = |\langle i, f|\, e^{-iH_{\mathrm{PG}}\pi/2}\, |i, 0\rangle|^2 = 1$ is therefore a **deterministic** regular conditional probability — a perfect classifier on the training pair.

The finite case $X = Y = \mathrm{Fin}\, 2$ is formalized in `BookProof.ChapterPauliGrover`. The swap `pauliX` is a unitary, so it is a valid instance of the parametrization above; its column 0 is the delta on output 1, so the joint concentrates on `(0,1)`, the marginal at input 0 is 1, and the regular conditional probability $p(f|0)$ is 1:

```
#check @BookProof.ChapterPauliGrover.pauliX_unitary
#check @BookProof.ChapterPauliGrover.pauliX_rotates
#check @BookProof.ChapterPauliGrover.pauliX_parametrizes_delta
#check @BookProof.ChapterPauliGrover.pauliGrover_cond_one
#check @exists_unitary_joint
#check @pCond
#check @pJoint_eq_cond_mul_marg
```

Two parts of the construction are book prose rather than formalized statements: the imperfect rotation `a < 1`, whose generator retains a small `|0⟩` component and which the config field `pauli_grover_a` tunes (default `1.0`); and the two global ingredients — the sum of the per-input generators over the training set, together with a $\sqrt{N}$-weighted vacuum projector on the unseen inputs, and the Krylov starting vector v_0 that the SIRK iteration needs. The chapter formalizes the ideal `a = 1` swap, which is the case `QFM.tex` reports as achieving 100% training accuracy.

21.9. What Is Verified and What Is Infinite-Dimensional

Everything cited above is verified `sorry`-free, over **finite** index sets (the discretized models used throughout `BookProof`). This is the concrete content of the book's Gram–Schmidt, Cauchy–Schwarz, and singular-value arguments. The manuscript states the result for arbitrary **standard** measure spaces, possibly with continuous parts; the abstract measure-theoretic layer — the classification of standard measure spaces, the identification of commutative von Neumann algebras with $L^\infty(X, \mu)$, and regular conditional probabilities via disintegration on a standard Borel space — is the infinite-dimensional extension. Parts of that layer are proved in `BookProof.ChapterSelectingEvents`: regular conditional probabilities exist for any finite measure on a standard Borel space (`exists_regular_conditional_probability`), singletons and finite sets are null in a continuous space (`singleton_null_in_continuous`, `finite_set_null_in_continuous`), every probability measure splits into a continuous and an atomic part (`exists_continuous_atomic_decomposition`), and the finite type-I_n case of the von Neumann classification is a theorem (`vonNeumann_abelian_classification_typeI`):

```
#check @BookProof.ChapterSelectingEvents.exists_regular_conditional_probability
#check @BookProof.ChapterSelectingEvents.singleton_null_in_continuous
#check @BookProof.ChapterSelectingEvents.finite_set_null_in_continuous
#check @BookProof.ChapterSelectingEvents.exists_continuous_atomic_decomposition
#check @BookProof.ChapterSelectingEvents.selecting_events_not_rewriting_history
```

The finite-dimensional algebraic core proved here is what makes the parametrization work, and it is the part the book actually computes.

21.10. A Less Arbitrary Construction: the Unitary from the Dynamics

The construction above builds the unitary $\mathcal{U}$ by completing the wave-function $\Psi = \sqrt{p}$ to an orthonormal basis (Gram–Schmidt). That is **correct** but **arbitrary**: the columns after the first are chosen by the completion, not fixed by the probability data. The manuscript's own field-theoretic thread (QFM.tex) supplies a construction that is **less arbitrary**, because the unitary is **pinned down by the dynamics** rather than by a basis choice.

The point is the passage from a **function** to a **unitary**. A velocity field $v_t(x)$ — a function on configuration space — determines a Hermitian generator by the continuity (Weyl-symmetrized) prescription

$$\mathbf{H}_t = \tfrac{1}{2}\big[\hat{p}\cdot v_t(\hat{x}) + v_t(\hat{x})\cdot\hat{p}\big],$$

and hence a unitary $\mathbf{U} = e^{i\mathbf{H}t}$. The unitary is **determined by the function** v_t; there is no free choice of extra columns. This is the "function $\leftrightarrow$ unitary" translation of QFM.tex, and it is the natural replacement for the Gram–Schmidt completion: instead of arbitrarily extending a wave-function to a unitary, one declares the **transition law** (a conditional probability) and lets the dynamics build the unitary.

21.11. The Generalization: Conditional Probability and a Standard Hilbert Space

The same construction generalizes to a **conditional probability** (a regular conditional probability / Markov kernel) on continuous inputs, and it does so without any Bochner-space machinery. Let $p(y|x)$ be a transition law from X to Z, and let $e_0 \in L^2(Z,\nu)$ be a fixed **background** wave-function — for concreteness the standard Gaussian state $e_0(z) = \pi^{-1/4}e^{-z^2/2}$. Form the joint space by the canonical tensor–product identification

$$L^2(X,\mu)\otimes L^2(Z,\nu) \;\cong\; L^2(X\times Z,\ \mu\times\nu).$$

Then $\mathbf{H}$ is a **standard** Hermitian operator on the scalar space $L^2(X\times Z)$ — for instance a Schrödinger-type Hamiltonian

$$\mathbf{H} = -\tfrac{1}{2}\,\Delta_z + V(x,z),$$

with x acting as an external coordinate that modifies the potential felt by the z variable. Starting from the product wave-function $\Psi_0(x,z) = f(x)\,e_0(z)$ and evolving by the unitary gives $\Psi_1 = e^{i\mathbf{H}}\Psi_0$, and the conditional probability is recovered by the **ordinary Born rule**:

$$P(x,B) = \int_B \big|\Psi_1(x,z)\big|^2\, d\nu(z).$$

This is the key simplification: no operator-valued inner products, no module algebra. Any classical stochastic transition (regular conditional probability) is simulated by an ordinary quantum system — a standard Hermitian $\mathbf{H}$ on a standard $L^2(X\times Z)$, a fixed background wave-function, and the Born rule. The conditional probability $\leftrightarrow$ unitary map of this chapter is thereby re-grounded in standard quantum mechanics, and the arbitrary Gram–Schmidt completion is replaced by a unitary that the dynamics determines.

The shift in viewpoint is the same one that makes the finite core of this chapter honest: the Gram–Schmidt and SVD results record what **any** unitary must do with the wave-function data; the dynamics-based construction records **which** unitary the physics picks. Both are needed — the former is the algebraic backbone, the latter the less-arbitrary physical selection.

The finite (discretized) form of this construction is formalized in `BookProof.ChapterContinuityUnitary`, on the cyclic lattice with the symmetric-difference momentum $(\hat{p}\,\psi)_k = -\frac{i}{2}(\psi_{k+1} - \psi_{k-1})$:

```
#check @BookProof.ChapterContinuityUnitary.continuityHamiltonian
#check @BookProof.ChapterContinuityUnitary.continuityHamiltonian_hermitian
#check @BookProof.ChapterContinuityUnitary.momentum_mul_velocityOp_not_hermitian
#check @BookProof.ChapterContinuityUnitary.continuityUnitary
#check @BookProof.ChapterContinuityUnitary.continuityUnitary_unitary
#check @BookProof.ChapterContinuityUnitary.continuityUnitary_add
```

The generator $\mathbf{H} = \frac{1}{2}(\hat{p}\,v + v\,\hat{p})$ is Hermitian, and the unsymmetrized product $\hat{p}\,v$ is **not** — the Weyl symmetrization is exactly what makes the generator an observable. The exponential $\mathbf{U}_t = e^{it\mathbf{H}}$ is then unitary

and forms a one-parameter group, with no free choice of columns anywhere: the function v determines it.

The Born-rule recovery of the conditional probability, and the tensor–product identification that keeps everything on the scalar space, are formalized as well:

```
#check @BookProof.ChapterContinuityUnitary.bornRecover
#check @BookProof.ChapterContinuityUnitary.bornRecover_union
#check @BookProof.ChapterContinuityUnitary.bornRecover_univ
#check @BookProof.ChapterContinuityUnitary.condProb_of_continuity
#check @BookProof.ChapterContinuityUnitary.tensorIsom
#check @BookProof.ChapterContinuityUnitary.bornRecover_product_state
```

`bornRecover` is $P(x, B) = \sum_{z \in B} |\Psi_t(x, z)|^2$; it is nonnegative, finitely additive, and of total mass 1 (`bornRecover_univ`, a consequence of unitarity alone). The capstone `condProb_of_continuity` packages this as a genuine probability distribution on the lattice for **every** input x — a Markov kernel built from the dynamics rather than from a basis choice. `tensorIsom` is the finite index-level identification $L^2(X) \otimes L^2(Z) \cong L^2(X \times Z)$, and `bornRecover_product_state` runs the recovery on a product initial state $\Psi_0(x, z) = f(x) e_0(z)$.

The same construction runs on the **infinite** lattice, with bounded operators on the genuine Hilbert space $\ell^2(\mathbb{Z})$ in place of matrices (`BookProof.ChapterContinuityUnitaryInfinite`). The lattice translations are unitaries, the symmetric-difference momentum and a bounded velocity field $v \in \ell^\infty(\mathbb{Z})$ are bounded self-adjoint operators, the Weyl-symmetrized generator is again self-adjoint, and $\mathbf{U}_t = e^{itH}$ is the Banach-algebra exponential of $\ell^2(\mathbb{Z}) \to \ell^2(\mathbb{Z})$:

```
#check @BookProof.ChapterContinuityUnitaryInfinite.momentum_isSelfAdjoint
#check @BookProof.ChapterContinuityUnitaryInfinite.velocityOp_isSelfAdjoint
#check @BookProof.ChapterContinuityUnitaryInfinite.continuityHamiltonian_isSelfAdjoir
#check @BookProof.ChapterContinuityUnitaryInfinite.continuityUnitary_unitary
#check @BookProof.ChapterContinuityUnitaryInfinite.continuityUnitary_add
#check @BookProof.ChapterContinuityUnitaryInfinite.bornRecover_tsum_univ
#check @BookProof.ChapterContinuityUnitaryInfinite.condProb_of_continuity_infinite
```

On the infinite lattice the Born recovery is **countably** additive: the total mass $\sum_{z \in \mathbb{Z}} |\Psi_t(z)|^2 = 1$ is Parseval plus unitarity (`bornRecover_tsum_univ`), and `condProb_of_continuity_infinite` packages it as a probability distribution on $\mathbb{Z}$ for every input. What stays outside the statement is unboundedness — the position and momentum operators of the continuum — not infinite-dimensionality.

The discretization can be dropped entirely on the **probabilistic** side. On any measure space (α, μ) and for a state $\Psi \in L^2(\mu)$, the Born prescription $P(B) = \int_B |\Psi|^2 \, d\mu$ is defined in `BookProof.ChapterBornMeasure` as a **measure**, so countable additivity holds by construction; it is a probability measure exactly when Ψ is normalized, and it is absolutely continuous with respect to μ:

```
#check @BookProof.ChapterBornMeasure.bornMeasure
#check @BookProof.ChapterBornMeasure.lintegral_bornDensity
#check @BookProof.ChapterBornMeasure.isProbabilityMeasure_bornMeasure
#check @BookProof.ChapterBornMeasure.bornMeasure_absolutelyContinuous
#check @BookProof.ChapterBornMeasure.condProb_of_bounded_dynamics
```

The capstone `condProb_of_bounded_dynamics` runs the whole construction on the continuum: for a bounded self-adjoint generator H on $L^2(\mu)$ and the unitary group $\mathbf{U}_t = e^{itH}$, the evolved state carries a Born law that is a countably additive probability measure at every time, and it charges no μ-null set. The integrability that the proof-plan appendix deferred is thus settled; what remains open is the **unbounded** generator of the continuum.

That last layer is at least made precise. `BookProof.ChapterUnboundedPosition` builds the lattice position operator $\hat{x}\,\psi_k = k\,\psi_k$ on its natural domain $D = \{\psi \in \ell^2(\mathbb{Z}) : \hat{x}\psi \in \ell^2(\mathbb{Z})\}$ and proves that the domain is dense, that the operator is symmetric on it, and that it is genuinely unbounded — not the restriction of any bounded operator:

```
#check @BookProof.ChapterUnboundedPosition.mulDomain
#check @BookProof.ChapterUnboundedPosition.mulOp_symmetric
#check @BookProof.ChapterUnboundedPosition.mulDomain_dense
```

```
#check @BookProof.ChapterUnboundedPosition.position_unbounded
#check @BookProof.ChapterUnboundedPosition.position_not_boundedOperator
```

That operator is not merely symmetric. Its adjoint domain is **exactly** the natural domain and the adjoint acts by the same multiplication, so the lattice position operator is a bona fide self-adjoint observable:

```
#check @BookProof.ChapterUnboundedPosition.adjointDomain_eq_mulDomain
#check @BookProof.ChapterUnboundedPosition.adjoint_eq_mulOp
```

And it generates its unitary group. The pointwise phase $(\mathbf{U}_t\psi)_k = e^{itf_k}\psi_k$ is a surjective linear isometry of $\ell^2(\mathbb{Z})$ satisfying $\mathbf{U}_0 = 1$ and $\mathbf{U}_{s+t} = \mathbf{U}_s\mathbf{U}_t$; it is strongly continuous at 0 for **every** state, with no domain hypothesis; and on the natural domain its difference quotient converges in $\ell^2(\mathbb{Z})$ to $i\hat{x}\psi$, which is Stone's relation $\frac{d}{dt}\mathbf{U}_t\big|_{t=0} = iA$ for an unbounded self-adjoint A:

```
#check @BookProof.ChapterUnboundedPosition.phaseUnitary
#check @BookProof.ChapterUnboundedPosition.phaseUnitary_add
#check @BookProof.ChapterUnboundedPosition.tendsto_phaseUnitary
#check @BookProof.ChapterUnboundedPosition.tendsto_slope_phaseUnitary
```

None of that is special to the lattice. `BookProof.ChapterUnitaryTransport` proves that the whole package is invariant under a unitary change of Hilbert space: for any unitary $W : H \to K$ and any densely defined `A` on `D \subseteq H`, the transported operator WAW^{-1} on $W(D)$ has a dense domain, is symmetric when `A` is, has adjoint domain $W(\mathrm{dom}\, A^\dagger)$ — so **self-adjointness** transports — and the transported group WU_tW^{-1} is strongly continuous with WAW^{-1} as its generator:

```
#check @BookProof.ChapterUnitaryTransport.transport_isSelfAdjointOn
#check @BookProof.ChapterUnitaryTransport.tendsto_transportUnitary
#check @BookProof.ChapterUnitaryTransport.tendsto_slope_transportUnitary
```

Combining the two halves: **every** operator unitarily equivalent to a lattice multiplication operator — on any complex Hilbert space — is self-adjoint on a dense domain and generates a strongly continuous unitary group satisfying Stone's relation.

```
#check @BookProof.ChapterUnitaryTransport.transported_position_domain_dense
#check @BookProof.ChapterUnitaryTransport.transported_position_isSelfAdjointOn
#check @BookProof.ChapterUnitaryTransport.transported_position_group
#check @BookProof.ChapterUnitaryTransport.tendsto_slope_transported_position
```

So the boundary of the formalized theory is stated inside the theory, and it now lies two layers further out than the proof plan first drew it. Bounded generators give the unitary group and the Born law outright; the unbounded position observable is densely defined, symmetric, self-adjoint, and does generate a strongly continuous unitary group with itself as generator; and that conclusion is inherited by anything unitarily equivalent to it. And the general Stone theorem is itself proved in this development — the **existence** half first: an arbitrary unbounded self-adjoint operator on a (complete, separable) Hilbert space generates a weakly measurable one-parameter unitary group `e^{-itA}` satisfying the Schrödinger equation on its domain, and conversely every such group arises from its self-adjoint generator (`ChapterStoneResolvent` through `ChapterStoneSeparable`: `stoneU`, `stoneU_mem_domain`, `hasDerivAt_stoneU`, `stone_bijection`, `stoneEquiv`, and the concrete `ℓ²(ℤ)` instance `stoneU_mulSA`). The bridge from a **selected** self-adjoint extension to the flow is then packaged by `ChapterStoneBridge` (`unboundedSelfAdjointOf`, `IsStoneFlow`, `isStoneFlow_stoneU`, `exists_stone_flow_of_selfAdjointExtension` / `of_positive` / `of_esa`), so the passage "essentially self-adjoint on a core ⇒ complete unitary flow" is a theorem rather than a promise. What a continuum Laplacian would still need is the **spectral theorem** — the existence of the diagonalizing unitary — and that step is now supplied as well; see the paragraph below.

```
#check @BookProof.ChapterStoneSeparable.stoneEquiv
#check @BookProof.ChapterStoneSeparable.stoneU_mulSA
#check @BookProof.StoneBridge.unboundedSelfAdjointOf
#check @BookProof.StoneBridge.IsStoneFlow
#check @BookProof.StoneBridge.isStoneFlow_stoneU
#check @BookProof.StoneBridge.exists_stone_flow_of_esa
#check @BookProof.StoneFlows.ym_fock_stone_flow
```

The diagonalizing unitary itself is constructed in `BookProof.ChapterUnboundedSpectralModel` by the classical resolvent (Cayley) route. For a densely defined self-adjoint A the resolvent $R = (A - i)^{-1}$ is a **bounded** operator: it is injective, its range is exactly $\mathrm{dom}\, A$, its adjoint is the resolvent at the conjugate point, and resolvents commute — so R is normal and the bounded multiplication model of `ChapterSpectralDirectSum` applies to it. Reading $A = R^{-1} + i$ back through that model, the representing measure gives no mass to $z = 0$ and is carried by the Cayley circle $|z|^2 = \mathrm{Im}\, z$ — the image of the real line under $t \mapsto 1/(t - i)$ — so the multiplier $1/z + i$ coincides almost everywhere with the **real** function $\mathrm{Re}\, z/|z|^2$. The conclusion: **every** densely defined self-adjoint operator on a complex Hilbert space is multiplication by a real function on a Hilbert sum of $L^2(\mu_x)$ spaces, with no cyclic vector and no separability assumed; on a separable space the family of summands is countable.

```
#check @BookProof.UnboundedSpectralModel.resOp_injective
#check @BookProof.UnboundedSpectralModel.exists_resOp_eq
#check @BookProof.UnboundedSpectralModel.isStarNormal_resOp
#check @BookProof.UnboundedSpectralModel.model_apply
#check @BookProof.UnboundedSpectralModel.model_ae_circle
#check @BookProof.UnboundedSpectralModel.model_ae_real_multiplier
#check @BookProof.UnboundedSpectralModel.unbounded_multiplication_model_cyclic
#check @BookProof.UnboundedSpectralModel.unbounded_multiplication_model_general
#check @BookProof.UnboundedSpectralModel.unbounded_multiplication_model_separable
```

21.12. Why This Matters Here

This is the theorem that turns probability theory into quantum mechanics. A conditional probability measure is parametrized by a unitary operator; a joint probability is the squared modulus of a wave-function; the wave-function is a bounded operator with a singular-value expansion; and the marginal and conditional probabilities are read off its Gram matrix. None of this is a physical postulate — it is the structure of probability itself, once we parametrize the simplex by the sphere. The unitary time-evolution of the symmetry chapter and the collapse of the collapse chapter are both instances of this single parametrization.

22. The Stern-Gerlach Experiment

22.1. Orientation and Status

The chapter that closes the Born-rule part with its first laboratory application: Stern–Gerlach, where a symmetry of the wave-function is non-deterministic on the phase space and information is erased by a unitary. Read it after the probability clock — the two-state phase space is the same one — and note the part's closing point: this part is the prerequisite for the field-theory and relativity part that follows.

The route: the stronger case for non-deterministic symmetry, the experiment, the two-state phase space, why sequential measurements give 50/50, and information erasure by a unitary.

The proofs are `BookProof/ChapterSternGerlach` and `BookProof/ChapterEulerDensityMatrix` (the density-matrix language reused from the clock).

22.2. A Stronger Case for Non-Deterministic Symmetry

In the probability clock the symmetry transformations used so far are **deterministic**: they send a point of the phase space to a point of the phase space. The manuscript argues that the Stern-Gerlach experiment makes a strong case for **generalizing** the symmetry transformations to be **non-deterministic**, and that its theoretical predictions only require a phase space with two states like the one already discussed in the previous chapter. It is a concrete setting in which the wave-function parametrization — and the fact that a symmetry of the wave-function can be non-deterministic on the probabilities — is put to work.

22.3. The Experiment

A beam of silver atoms is sent through a magnetic field with a gradient along the z-axis or the x-axis and the deflection is observed. The result is that the silver atoms possess an **intrinsic angular momentum** (spin) that takes only one of two possible values, represented by the symbols $+$ and $-$. Moreover, in **sequential** Stern-Gerlach experiments, the measurement of the spin along the z-axis destroys the information about the spin of an atom along the x-axis, and vice versa.

22.4. A Two-State Phase Space

We take as the phase space not only the spin of an atom of the beam but also the **angle of orientation of a macroscopic reference object**, for pedagogical purposes. A rotation of the phase space is then a **non-deterministic** transformation of the spin of the atom and a **deterministic** transformation of the macroscopic object. To keep track of the part of the wave-function corresponding to the orientation angle we only need its **central value**, which we call simply "the angle," and we then consider only the part of the wave-function corresponding to the spin.

With $\cos^2(t)$ the probability that the spin is in the state $+$, and $\sin^2(t)$ the probability that it is in the state $-$, the non-deterministic symmetry transformation given by a rotation of the spin in the $x-z$ plane is parametrized by the parameter t, and its linear representation on the wave-function is exactly the probability clock of that chapter. The Euler identity for the 2-state density matrix (module `BookProof.ChapterEulerDensityMatrix`):

```
#check @ChapterEulerDensityMatrix.density_euler
#check @ChapterEulerDensityMatrix.density_collapse
```

We only make measurements along the z and x-axis; if we also measured along the y-axis, the phase space would require **four** states, or equivalently, a parametrization with a complex wave-function.

22.5. Sequential Measurements Give 50/50

In the first measurement the reference angle is 0 with respect to the z-axis, and we know for sure that the spin is in the state $+$ because we measure the spin along the z-axis of atoms previously filtered to be in the state $+$.

A second, sequential measurement along the x-axis means that we rotate the reference object by 90 degrees, so the new angle is 90 degrees; and we rotate the spin of the atom by 45 degrees in the $x-z$ plane, that is $t = \pi/4$, because the spin group is a double cover of the orthogonal group, for which the angle would be 90 degrees. We then determine whether the spin is in the $+$ or $-$ state — the wave-function collapses. The probability of each outcome is now $50\%/50\%$:

```
#check @ChapterSternGerlach.sg_cos_sq_quarter
#check @ChapterSternGerlach.sg_sin_sq_quarter
```

since $\cos^2(\pi/4) = \sin^2(\pi/4) = 1/2$.

A third, sequential measurement along the z-axis means that we rotate the reference object by -90 degrees (so the new angle is again 0) and apply a $-\pi/4$ rotation to the atoms with spin $+$; we then determine the spin one more time, and the wave-function collapses again. Although in the first measurement the spin was in the state $+$, the probability is still $50\%/50\%$ — because the same non-deterministic rotation was applied in the second and third steps to switch from the z to the x-axis and back.

Generalizing the symmetry transformations to be non-deterministic therefore suffices to account for all the experimental results described by quantum mechanics, with the Stern-Gerlach experiment being one example. The question remaining is whether the Euler's formula applies for phase-spaces with more than two states — which would imply that the collapse of the wave-function is merely a mathematical artifact of the wave-function parametrization. That is the subject of the Euler chapters.

22.6. Information Erasure by a Unitary

The manuscript draws a second, sharper conclusion from this experiment, now about the **reversibility** of the erasure. Consider a quantum theory whose phase space is an orthogonal basis of a Hilbert space of finite

dimension n. There is **always a unitary transformation** U such that the corresponding probability distribution is necessarily the **constant** distribution, for all initial states in the same orthogonal basis. Under Born's rule, feeding any basis state e_j into such a unitary and measuring in the same basis gives the uniform output $i \mapsto |U_{ij}|^2 = 1/n$. A unitary time evolution can therefore turn any incoming basis state into a maximally mixed, **information-erased** distribution while remaining perfectly reversible — erasure is not in tension with unitary.

One such unitary for every n is the **normalized discrete Fourier transform** (a complex Hadamard matrix):

$$U_{ij} = \frac{1}{\sqrt{n}}\, e^{2\pi i\, ij/n}.$$

Every entry has the same squared modulus $|U_{ij}|^2 = 1/n$, and the matrix is unitary because the columns are orthonormal, which follows from the vanishing of the geometric sum over one full period. For $n = 2$ this is the Hadamard gate, the case of the two-state Stern-Gerlach model. The construction and its properties are in `BookProof.ChapterSternGerlach`:

```
#check @ChapterSternGerlach.dft_normSq
#check @ChapterSternGerlach.dft_unitary
#check @ChapterSternGerlach.exists_uniform_unitary
```

The headline `exists_uniform_unitary` is exactly the manuscript's assertion: for every $n \geq 1$ there is a unitary sending every basis state to the constant law $i \mapsto 1/n$. The subtlety — and the resolution of the apparent paradox with the singular collapse matrix — is that this erasure is **unitary**, hence reversible at the level of the wave-function: the information is not destroyed but delocalized into the relative phases between the $1/\sqrt{n}$ amplitudes. What is irreversible is only the subsequent restriction to the probabilities $|U_{ij}|^2 = 1/n$, which discards those phases. The unitary mixes the state perfectly; Born's rule then forgets how. This is the cleanest finite illustration of the manuscript's distinction between reversible wave-function dynamics and the irreversible appearance of measurement statistics.

22.7. Summary

- The Stern-Gerlach experiment requires as a phase space only two states, and motivates generalizing the symmetry transformations to be **non-deterministic**.
- Sequential measurements along the z and x-axis then give only $50/50$ outcomes, the spin being rotated by $\pi/4$ on a double-cover of the orthogonal group; each measurement collapses the wave-function and destroys the orthogonal information. Yet the same non-deterministic rotation is why the third measurement still yields $50/50$.
- A unitary transformation always exists sending every basis state of any finite dimension to the uniform distribution, so this erasure is reversible at the level of the wave-function.

23. Relativity, Gauge Theory, and Gravity

The manuscript's relativistic and field-theoretic programme. We develop the Lorentz group and its real representations, the CPT theorem and the relativistic position operator; quantization arising from time-evolution (Yang–Mills, the Weyl/CCR relations, the nilpotent BRST charge); the Gribov ambiguity and the abelian (electromagnetic) case; the physical parity transformation and antiparticles; and the diffeomorphism/3+1 structure of gravity. We also develop the field-theoretic foundations: gauge symmetry and dissipative dynamics, the free-field construction of a uniform measure on a sphere out of the Gaussian, and the spin–statistics dichotomy that distinguishes bosonic commutation from fermionic anticommutation.

How to read this part. This is the longest part and the one with the most independent threads. The two foundation chapters — Gauge symmetry and dissipative dynamics and the free-field construction — set the vocabulary (gauge as parametrization redundancy, the Gaussian as the rotation-invariant prior) that the rest assume. The middle block (Real representations, Quantization due to time-evolution, the Gribov ambiguity, the physical parity transformation) is the Standard-Model thread, and can be read largely independently of the free-field thread. The final chapters — Diffeomorphisms and gravity, the Starobinsky scalaron, and the Navier–Stokes and Carleman threads — apply the same parametrization to the systems whose numerical validation is carried out in the

companion solver work: the 3D gauge-fixed gravity Hamiltonian, the Yang–Mills Hamiltonian, and the Eulerian Navier–Stokes fiber. The spin– statistics chapter belongs with the free-field thread (it is the finite-degree instance of the tensor products of sample spaces).

Three of these chapters form a single thread on the **derivative variables** of a field — the places where a first-order system is written with the derivatives of the fields as fields. The Fourier elimination of the derivative variables states the strategy of record: **eliminate** them by a spatial Fourier transform inside the squares that make up the Hamiltonian, rather than constrain them by a gauge symmetry, so that no ghost sector and no cubic derivative symbol is needed for the definition. It also settles the Lagrangian face of the same question, where the eliminated kinematics turns out to be rank-one — the cofactor of the deformation gradient vanishes, so the volume constraint collapses to a constant and no deformation-gradient square carries field content. The Navier–Stokes one-particle Hamiltonian and its Fock enclosure then does the operator work for the Eulerian route: it exhibits the reduced one-parcel family, proves the one-body generator $H_{\mathrm{sp}} = H_{\mathrm{visc}} + H_{\mathrm{advect}}$ symmetric and positive for the exact nonlinearity, and identifies the final Hamiltonian with its second quantization $d\Gamma(H_{\mathrm{sp}})$ — the one-particle Hamiltonian enclosed in creation on the left and annihilation on the right, so every particle-number sector is preserved. The comparison operator of the Navier–Stokes Hamiltonian supplies the other half of that operator story: the Faris–Lavine comparison operator N — the Leray energy on the mainstream leg and the lifted auxiliary Friedrichs extension on the full nonlinear leg — with the theorems that establish each, and the reasons the Hamiltonian itself (or its square) cannot serve as N. Eliminating the derivative variables in quantum gravity is the same service for the vielbein sector of the R^2 Starobinsky model, deleting twenty-seven derivative components per momentum while reproducing the vielbein self-interaction and the scalaron coupling verbatim. Those three chapters are the structural companion of the Starobinsky scalaron and of the Navier–Stokes and numerics chapters that follow, and they are where the final-Hamiltonian convention of the whole programme is explained for these two systems.

24. Gauge Symmetry and Dissipative Dynamics

24.1. Orientation and Status

The opening chapter of the field-theory part, and one of its two vocabulary chapters: **gauge** is the redundancy of a many-to-one parametrization — exactly the freedom the fiber chapter isolated algebraically — and the BRST machinery is how that redundancy is carried into dynamics. The other vocabulary chapter is the free-field construction; the rest of the part assumes both.

The route: gauge as parametrization redundancy (and its Born-parametrization instance), first-class constraints as gauge generators and their conservation, the BRST charge and the ghost field, reduced dynamics on BRST cohomology, the abelian case (no Gribov ambiguity — the non-abelian one is its own chapter), dissipative dynamics as irreversibility without a classical Hamiltonian, complete and comprehensive gauge fixing, parametrizations as gauge symmetries, and Casimir constraints, gauge averaging, and the absence of anomalies.

The proofs are spread over `BookProof/BRSTNilpotent`, `BookProof/GhostField`, `BookProof/BrstReducedTransfer`, `BookProof/FreeFieldConstraint`, `BookProof/IrreversibleDynamics`, and the gauge-fixing and Casimir-averaging chapters; the same BRST pattern is re-used, system by system, in the Navier–Stokes and quantum-gravity chapters below.

24.2. The Idea: Gauge Is a Redundancy of the Parametrization

The wave-function parametrization of a probability measure is **many-to-one**. The same distribution $p(x) = |\Psi(x)|^2$ is reproduced by many different wave-functions, and the freedom between them is a **gauge symmetry**: a group of transformations of the wave-function that leave the probabilities unchanged. This chapter studies that redundancy and its consequences. The manuscript's thesis is that the wave-function parametrization lets us implement gauge symmetries as **exact constraints** in a standard probability space **without** giving the constrained space null measure, and that the same parametrization extends to classical dissipative dynamics. The verified content here is the finite, algebraic core of that story: the gauge redundancy of the Born map, the identity showing constraints that commute with the Hamiltonian are conserved, the nilpotency of the BRST charge (which makes the gauge-invariant algebra well-defined), and the abelian case where the Gribov ambiguity does not arise.

24.3. The Gauge Redundancy of the Born Parametrization

Over $\mathbb{R}^n$, the Born map sends a (normalized) wave-function to its squared components, $x \mapsto (x_k)^2$ under the constraint $\sum_k x_k^2 = 1$. The manuscript stresses that **"two wave-functions are always related by a rotation of the hypersphere"** — the parametrization is a surjection of the sphere onto the simplex, and it is never a bijection. The first, minimal source of redundancy is the **antipodal** map: replacing x by $-x$ leaves every Born probability unchanged:

```
#check @BookProof.ChapterFreeFieldBornGauge.bornMap_neg
#check @BookProof.ChapterFreeFieldBornGauge.bornMap_not_injOn_sphere
```

The second states the redundancy as a **group**: the full diagonal sign group $\{-1,+1\}^n$ acts by coordinate-wise reflections $x_k \mapsto s_k x_k$, each an isometry of the sphere, and the born image is invariant under the whole group:

```
#check @BookProof.ChapterFreeFieldBornSignGauge.signFlip
#check @BookProof.ChapterFreeFieldBornSignGauge.bornMap_signFlip
```

So the gauge group of the real Born parametrization contains at least the finite abelian group $\{-1,+1\}^n$; the antipodal map is the special case $s \equiv -1$. This is the concrete, verified seed of the manuscript's claim that the parametrization is genuinely redundant — a surjection with fibers, never a bijection.

24.4. First-Class Constraints as Gauge Generators

The manuscript's "gauge transformations, constrained systems and conditioned probability" section develops the general picture: a **first-class constraint** is the generator of a unitary gauge group, all of whose operators are constrained to be the identity. The dynamical content is that an observable must commute with the gauge generator, and therefore the **gauge-invariant algebra** is what is constrained — not the Hilbert space, and not the individual wave-function (the manuscript quotes Dirac on the "standard ket" that cannot be gauge invariant).

Formalizing constraints requires the notion of a **conditional probability**: given a marginal $p(x) > 0$, the regular conditional probability $p(y|x) = p(x,y)/p(x)$ is a genuine probability distribution on Y, and the joint distribution factors as marginal times conditional. These are the verified facts that make "conditioning on a constraint" a well-defined operation:

```
#check @BookProof.ChapterConditional.pCond
#check @BookProof.ChapterConditional.pCond_sum_one
#check @BookProof.ChapterConditional.pJoint_eq_cond_mul_marg
```

Because it is always possible to define such regular conditional probabilities in a standard measure space, the manuscript argues that exact constraints can in principle be implemented without giving the constrained space null measure. The verified finite-dimensional content is the conditional-probability core above.

A device used twice in this book puts that picture to work on a constraint that is a **spatial derivative** rather than an algebraic relation. The relations that define the derivative modes of a field — $u_{i,j} = \partial_j u_i$ for Navier–Stokes, the derivatives of the vielbein for gravity — have no explicit solution, so they are the ones a gauge generator has to impose. The manoeuvre is to adjoin each derivative as an **independent canonical variable** and let the gauge condition set it back equal to the derivative of the field. What is bought is that the constraint becomes algebraic and abelian, hence first class, so the BRST machinery of the following sections applies, whereas the differential relation it replaces generates no finitely generated gauge algebra. The Navier–Stokes instantiation is the generator $G_j = \partial/\partial y_j - u_{i,j}\,\partial/\partial u_i$ (`BookProof.NavierStokesGaugeY.genY`), which translates a second coordinate y while shifting each velocity mode by its own first derivative; the gravitational one adjoins one derivative mode per spatial derivative of every vielbein component, and in those variables the torsion becomes linear in the auxiliaries with no derivative left (`BookProof.QgBrstDerivativeGauge.extTorsionCoef`). The BRST completion of each is `BookProof.NsBrstDerivativeGauge` in the free-field chapter and `BookProof.QgBrstDerivativeGauge`, with its substitution operator `gaugeReduce`, in the diffeomorphism chapter. Both chapters keep this gauge fixing separate from the algebraic constraints above precisely because the derivative variable, not the field, is what the gauge condition acts on: the field $u_i(y)$ is invariant under G_j (`BookProof.NavierStokesGaugeY.genY_uField`), and it is the derivative mode that moves.

24.5. Constraints That Commute with the Hamiltonian Are Conserved

The key dynamical identity comes from the free-field section of the manuscript (the momentum constraint `i D_x = 0`). If a constraint D commutes with the Hamiltonian H — $\{D, H\} = 0$ — then the constrained quantity is **conserved** by the Hamiltonian flow: for every operator A,

$$\{\{D, A\}, H\} = -\{D, \{H, A\}\}.$$

This is a direct consequence of the Jacobi identity together with $\{D, H\} = 0$. Its meaning is that the set of D-invariant observables is closed under evolution by H — a constraint that commutes with the Hamiltonian is preserved, which is exactly what makes it a **gauge** constraint rather than a merely initial-condition-imposed one. The verified statement is the general algebraic fact, with the book's two literal instances (for the field $\varphi^{(0)}$ and the momentum $p_{(1)}$):

```
#check @BookProof.FreeFieldConstraint.bracket
#check @BookProof.FreeFieldConstraint.bracket_jacobi
#check @BookProof.FreeFieldConstraint.constraint_commutation_identity
#check @BookProof.FreeFieldConstraint.constraint_preserved_under_bracket
```

24.6. The BRST Charge and the Ghost Field

The manuscript's gauge-mechanics example quantizes a classical gauge system on the Hilbert space $L^2(\mathbb{R}^2 \times \mathbb{Z}_2)$, where the $\mathbb{Z}_2$ factor is the ghost degree of freedom. The gauge generator is the charge $Q = \pi\phi + \pi^*\phi^*$, and the BRST charge is $\Omega = (\pi\phi + \pi^*\phi^*)\,\psi^\dagger$ with the ghost field ψ satisfying the canonical anticommutation relation $\{\psi, \psi^\dagger\} = 1$. The wave-function itself need not be gauge invariant — only the observables must commute with Q. The manuscript separates the gauge generator from the gauge-invariant algebra by requiring the latter to be a subalgebra of the commutative von Neumann algebra generated by ϕ, ϕ^*, k.

The verified content is the ghost-field algebra and the single property that makes the whole BRST construction well-defined: the nilpotency of the BRST charge, $\Omega^2 = 0$. Concretely, on the two-dimensional $\mathbb{Z}_2$ Fock factor the ghost operators are the matrices

$$\psi = \begin{pmatrix} 0 & 0 \\ 1 & 0 \end{pmatrix}, \qquad \psi^\dagger = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix},$$

with $\psi^2 = 0$, $\psi^{\dagger 2} = 0$ (Pauli exclusion), and the canonical anticommutation relation $\{\psi, \psi^\dagger\} = 1$; the number operator $N = \psi^\dagger\psi$ is an orthogonal projection, so the ghost occupation is 0 or 1:

```
#check @BookProof.GhostField.psi
#check @BookProof.GhostField.psiDag
#check @BookProof.GhostField.psi_sq
#check @BookProof.GhostField.car
#check @BookProof.GhostField.numberOp_idempotent
#check @BookProof.GhostField.numberOp_selfAdjoint
```

The nilpotency itself holds abstractly in any ring: if the ghost factor f is square-zero and the bosonic field factor b commutes with it, then the composite $\Omega = b \cdot f$ satisfies $\Omega^2 = 0$. With $f = \psi^\dagger$ and b the divergence field factor this is the book's $\Omega^2 = 0$. For the non-abelian case, the cubic ghost term $Q = \sum_{a,b,e} f_{abe}\,(\psi_a^\dagger \psi_b^\dagger \psi_e)$ of the BRST charge also squares to zero, given the canonical anticommutation relations and structure constants that are antisymmetric in their first two indices and satisfy the Jacobi identity:

```
#check @BookProof.GhostField.brst_charge_nilpotent
#check @BookProof.BRSTNilpotent.GhostCAR
#check @BookProof.BRSTNilpotent.Q
#check @BookProof.BRSTNilpotent.brst_charge_nilpotent
```

This nilpotency is what makes the BRST cohomology — and hence the physical, gauge-invariant algebra that the manuscript proposes as the definition of the constrained theory — well defined. It is the concrete counterpart of the manuscript's claim that the gauge generators of the group "cannot be interpreted literally": the physical content lives in the gauge-invariant (cohomological) algebra, not in the individual wave-function.

24.7. The Reduced Dynamics on BRST Cohomology

Nilpotency makes the cohomology well defined; it does not yet say that the **dynamics** respects it. That is a separate requirement, and it is the one the manuscript's transfer operator has to meet: the evolution must map the physical (BRST-closed) subspace to itself, and it must not mix physical states with pure-gauge (BRST-exact) ones, so that it descends to the quotient. Writing $\ker \Omega$ for the closed states and $\overline{\operatorname{ran} \Omega}$ for the exact ones — the closure is what makes the quotient a topological object — the BRST cohomology is $\ker \Omega / \overline{\operatorname{ran} \Omega}$:

```
#check @BookProof.BrstReducedTransfer.physicalStates
#check @BookProof.BrstReducedTransfer.exactStates
#check @BookProof.BrstReducedTransfer.exactStates_le_physicalStates
#check @BookProof.BrstReducedTransfer.Cohomology
```

Any bounded operator commuting with Ω preserves both subspaces — for the exact states this needs a continuity argument, since the range is only dense in its closure — and therefore induces a linear map on the cohomology. Applied to a one-parameter family this gives the **reduced transfer**: a one-parameter group of linear automorphisms of the BRST cohomology, which moreover preserves the quotient (BRST) norm when the family is isometric.

```
#check @BookProof.BrstReducedTransfer.physicalStates_invariant
#check @BookProof.BrstReducedTransfer.exactStates_invariant
#check @BookProof.BrstReducedTransfer.reducedMap
#check @BookProof.BrstReducedTransfer.transfer
#check @BookProof.BrstReducedTransfer.transfer_comp
#check @BookProof.BrstReducedTransfer.transfer_bijective
#check @BookProof.BrstReducedTransfer.infDist_exactStates_eq
```

The case of interest is the unitary group e^{-itT} of an unbounded self-adjoint Hamiltonian commuting with the BRST charge — the half-density evolution of the quantization chapters. It maps the physical subspace to itself, it moves two states that differ by a gauge vector to two states that still differ by a gauge vector, and the induced group on cohomology is norm-preserving:

```
#check @BookProof.BrstReducedTransfer.stoneU_mem_physicalStates
#check @BookProof.BrstReducedTransfer.stoneU_sub_mem_exactStates
#check @BookProof.BrstReducedTransfer.stoneTransfer
#check @BookProof.BrstReducedTransfer.stoneTransfer_comp
#check @BookProof.BrstReducedTransfer.stoneTransfer_bijective
#check @BookProof.BrstReducedTransfer.infDist_exactStates_stoneU_eq
```

What is **not** claimed here is a construction of the concrete gauge-fixed field-space Hamiltonian and its ghost-sector BRST charge; the reduction statement above is what such a construction has to feed. Nor does the exactness of the reduced dynamics say anything about the **truncated** dynamics, which does leak out of the physical subspace — that leakage is bounded separately in the reliability chapter.

24.8. The Abelian Case: No Gribov Ambiguity

The manuscript contrasts gauge symmetry with anomalies and discusses the Gribov ambiguity — the fact that the Dirac bracket requires a gauge-fixing that is both **complete** and **unconstrained**, which is not always possible. For the abelian (electromagnetic) case the situation is clean: reducing the non-abelian structure constants to zero, the field strength becomes the linear abelian curl, which is **gauge invariant** — invariant under the shift $A_j \mapsto A_j + \partial_j \theta$, the curl of a gradient vanishing by commutativity of the partial derivatives. This is the manuscript's "no Gribov ambiguity" for the abelian theory:

```
#check @BookProof.FreeEMField.emFieldStrength
#check @BookProof.FreeEMField.emFieldStrength_gauge_invariant
#check @BookProof.FreeEMField.fieldStrengthMul_eq_emFieldStrength_of_commute
```

The last statement is the reduction itself: when the connection components commute, the non-abelian field strength loses its quadratic correction and becomes the linear abelian one — which is why the Hamiltonian is quadratic in the fields in this case.

24.9. Dissipative Dynamics: Irreversibility Without a Classical Hamiltonian

The manuscript also uses the wave-function parametrization to describe **classical dissipative dynamics**. Its example is two classical coupled oscillators with different frequencies and different damping constants,

$$\ddot{x}_1 + \lambda_1 \dot{x}_1 + \omega_1^2 x_1 - c_2 x_2 = 0, \qquad \ddot{x}_2 + \lambda_2 \dot{x}_2 + \omega_2^2 x_2 - c_1 x_1 = 0.$$

This motion cannot be derived from a Lagrangian, and so has no classical Hamiltonian without enlarging the system; yet because a probability measure exists for the system, we can still define a **quantum** Hamiltonian and quantum constraints — which are more general than the classical ones because they do not need to commute with the variables defining the sample space. The system is dissipative (the energy is not conserved) but the degrees of freedom do not disappear, so the **probability** is conserved. The manuscript's slogan is that "the quantum formalism is the most general formalism whenever there is a conserved probability."

The verified content here is the **dissipative** (irreversible, non-conservative) character of such dynamics, which is what obliges one to work with a probability measure rather than a conserved energy. In a **finite** world a deterministic self-map is injective iff surjective iff bijective, so there is no injective-but-not-surjective (i.e. irreversible, dissipative) map at all — the discrete counterpart of the manuscript's remark that "the rationals are not enough":

```
#check @BookProof.IrreversibleDynamics.finite_injective_iff_surjective
#check @BookProof.IrreversibleDynamics.finite_no_irreversible
```

On an infinite state space such a map exists. The concrete dissipative example is the halving map $x \mapsto x/2$ on the unit interval: it is **injective** (no two points collapse together) but **not surjective** onto its domain (the point 1 is not hit), hence irreversible — the halving of an interval is genuinely "dissipative", shrinking its length by a factor of two — yet it is **non-singular**: it maps every interval of positive length to a set of positive measure, so no probability mass is lost. This is exactly the manuscript's **"a process with a dissipative time-evolution is irreversible: the deterministic time-evolution is not an invertible function (it is injective but not surjective). Then there is time asymmetry"**:

```
#check @BookProof.IrreversibleDynamics.dissipative_injective
#check @BookProof.IrreversibleDynamics.dissipative_not_surjective_unitInterval
#check @BookProof.IrreversibleDynamics.dissipative_volume_Icc
#check @BookProof.IrreversibleDynamics.dissipative_nonsingular_Icc
```

This is the precise sense in which dissipative dynamics escapes the classical Hamiltonian framework while remaining within the wave-function parametrization: the dynamics is irreversible (injective, not surjective) and loses energy, but since it is non-singular it conserves probability, and that conserved probability is all the quantum formalism needs.

The manuscript's **specific** dissipative system is the damped coupled oscillator, and its energy balance is proved rather than read off. For a solution of $\ddot{x} + \lambda \dot{x} + \omega^2 x = 0$ the energy $E = \frac{1}{2}\dot{x}^2 + \frac{1}{2}\omega^2 x^2$ obeys $\dot{E} = -\lambda \dot{x}^2$, so it is non-increasing for non-negative damping, and it is **not** constant at any instant at which the damping is positive and the oscillator moves. The coupled system of the manuscript,

$$\ddot{x}_1 + \lambda_1 \dot{x}_1 + \omega_1^2 x_1 - c_2 x_2 = 0, \qquad \ddot{x}_2 + \lambda_2 \dot{x}_2 + \omega_2^2 x_2 - c_1 x_1 = 0,$$

satisfies the same identity, $\dot{E} = -\lambda_1 \dot{x}_1^2 - \lambda_2 \dot{x}_2^2$, with monotonicity and non-conservation inherited. An explicit critically damped solution shows the statement is not vacuous: its energy is strictly decreasing. And the total probability is conserved by **any** measurable deterministic evolution, so energy loss and probability conservation are not in tension.

```
#check @BookProof.DampedOscillatorEnergy.hasDerivAt_dampedEnergy
#check @BookProof.DampedOscillatorEnergy.dampedEnergy_antitone
#check @BookProof.DampedOscillatorEnergy.dampedEnergy_not_constant_of_damped
#check @BookProof.DampedOscillatorEnergy.hasDerivAt_coupledEnergy
#check @BookProof.DampedOscillatorEnergy.coupledEnergy_antitone
#check @BookProof.DampedOscillatorEnergy.coupledEnergy_not_constant_of_damped
#check @BookProof.DampedOscillatorEnergy.criticallyDamped_energy
```

```
#check @BookProof.DampedOscillatorEnergy.criticallyDamped_energy_strictAnti
#check @BookProof.DampedOscillatorEnergy.total_probability_conserved
```

24.10. Complete and Comprehensive Gauge Fixing

The manuscript distinguishes two properties a gauge fixing may have. It is **comprehensive** if it meets every gauge orbit at least once (nothing physical is dropped), and **complete** if it meets every orbit **at most** once (no remnant symmetry is left constraining the surface). `IsComprehensiveGaugeFixing` and `IsCompleteGaugeFixing'` are those two predicates, and the chapter proves that the pair is not a wish but a construction: the set of orbit representatives is at once comprehensive and complete, so a faithful gauge fixing always exists.

```
#check @BookProof.ChapterGaugeComprehensiveFixing.orbitRepresentatives_isComprehensiv
#check @BookProof.ChapterGaugeComprehensiveFixing.orbitRepresentatives_isCompleteGaug
#check @BookProof.ChapterGaugeComprehensiveFixing.exists_comprehensive_complete_gauge
```

The pay-off is that on such a gauge fixing every function **extends** to a gauge-invariant observable, and extends **uniquely** — the gauge fixing is a faithful parametrization of the physical algebra, not merely a choice of coordinates. The two axes are then characterized exactly, as the two directions of the restriction map on the physical observables: comprehensiveness is its injectivity, completeness its surjectivity. Both converses are proved with explicit counter-observables, not asserted.

```
#check @BookProof.ChapterGaugeComprehensiveFixing.existsUnique_physical_extension_of_
#check @BookProof.ChapterGaugeComprehensiveFixing.physical_ext_iff_comprehensive
#check @BookProof.ChapterGaugeComprehensiveFixing.physical_extension_iff_complete
#check @BookProof.ChapterGaugeComprehensiveFixing.exists_physical_ne_agreeing_of_not_
#check @BookProof.ChapterGaugeComprehensiveFixing.exists_not_extendable_of_not_comple
```

The manuscript's **unconstrained** condition is what makes this non-trivial, and its consequence is formalized as a Gribov-type obstruction: on a connected spectrum, with the gauge group acting freely and non-trivially, a fixing that is at once complete and comprehensive is **never** clopen — a complete choice of representatives cannot be a continuous one. The manuscript's spurious-field device is shown to produce a complete **and** comprehensive fixing for an arbitrary gauge group, and for the concrete translation gauge symmetry of the line all four combinations of the two axes are realized by explicit witnesses.

```
#check @BookProof.ChapterGaugeComprehensiveFixing.not_isClopen_of_complete_comprehens
#check @BookProof.ChapterGaugeComprehensiveFixing.not_isPhysicalObservable_indicator
#check @BookProof.ChapterGaugeComprehensiveFixing.spuriousSection_isComprehensiveGaug
#check @BookProof.ChapterGaugeComprehensiveFixing.spuriousSection_isCompleteGaugeFixi
#check @BookProof.ChapterGaugeComprehensiveFixing.unitCell_isComprehensiveGaugeFixing
#check @BookProof.ChapterGaugeComprehensiveFixing.unitCell_isCompleteGaugeFixing'
#check @BookProof.ChapterGaugeComprehensiveFixing.unitCell_not_isClopen
#check @BookProof.ChapterGaugeComprehensiveFixing.shift_gaugeFixing_classification
```

24.11. Parametrizations Are Gauge Symmetries

A more structural reading of the same phenomenon: **every** parametrization carries a gauge group. For a map $\pi : X \to Y$ the group is the set of permutations of the parameter space that fix the parametrized point, `fiberGauge π`; its orbits are exactly the fibers, and its invariant observables are exactly the functions of the parametrized point. The group is trivial precisely when the parametrization is redundant-free, and the two axes above reappear as injectivity of the parametrization on the gauge fixing (completeness) and surjectivity onto the fibers (comprehensiveness).

```
#check @BookProof.ChapterGaugeParametrization.fiberGauge
#check @BookProof.ChapterGaugeParametrization.orbit_eq_fiber
#check @BookProof.ChapterGaugeParametrization.isPhysicalObservable_iff_factors_throug
#check @BookProof.ChapterGaugeParametrization.fiberGauge_eq_bot_iff
#check @BookProof.ChapterGaugeParametrization.fiberGauge_ne_bot_iff
#check @BookProof.ChapterGaugeParametrization.isCompleteGaugeFixing'_iff_injOn
#check @BookProof.ChapterGaugeParametrization.isComprehensiveGaugeFixing_iff_surjOn
```

24.12. Casimir Constraints, Gauge Averaging and the Absence of Anomalies

The manuscript's device for imposing **all** constraints at once is the quadratic Casimir of a family of Hermitian generators. That it works is a theorem here: the expectation of the Casimir in a state is the sum of the squared norms of the constraint expectations, so its kernel is **exactly** the intersection of the kernels of the individual constraints — setting the single Casimir to zero really does impose every constraint.

```
#check @BookProof.ChapterGaugeCasimirAverage.casimir_apply
#check @BookProof.ChapterGaugeCasimirAverage.inner_casimir
#check @BookProof.ChapterGaugeCasimirAverage.casimir_apply_eq_zero_iff
#check @BookProof.ChapterGaugeCasimirAverage.ker_casimir
```

Averaging is the other half: the Haar average (for a right-invariant measure) and the finite normalized average of any observable are gauge-invariant, normalized on the constants, and fix the observables that were already invariant — averaging is a projection onto the physical algebra, not merely a smoothing. Gauge invariance is preserved by limits, which is the manuscript's statement that a gauge symmetry can never exhibit an anomaly; and pushing a probability measure forward along a measurable projection into the constrained set gives that set probability one, with the expectations of observables that do not see the projection left unchanged — the constraint imposed exactly, without a null set.

```
#check @BookProof.ChapterGaugeCasimirAverage.gaugeAverage_isPhysicalObservable
#check @BookProof.ChapterGaugeCasimirAverage.gaugeAverage_of_isPhysicalObservable
#check @BookProof.ChapterGaugeCasimirAverage.finiteGaugeAverage_isPhysicalObservable
#check @BookProof.ChapterGaugeCasimirAverage.finiteGaugeAverage_of_isPhysicalObservab
#check @BookProof.ChapterGaugeCasimirAverage.isPhysicalObservable_of_tendsto
#check @BookProof.ChapterGaugeCasimirAverage.map_measure_constrainedSet
#check @BookProof.ChapterGaugeCasimirAverage.physical_invariant_along_gaugeProjectior
```

24.13. Summary

The gauge symmetry of the wave-function parametrization is a redundancy of the coordinates, not a new physical phenomenon. Concretely:

- the Born map is a surjection of the sphere onto the simplex that is never a bijection, with at least the diagonal sign group $\{-1,+1\}^n$ in its fiber;
- constraints that commute with the Hamiltonian are conserved, by the constraint-commutation identity (Jacobi);
- the BRST charge is nilpotent, $\Omega^2 = 0$, so the gauge-invariant algebra is well-defined — one never needs the wave-function itself to be gauge invariant;
- in the abelian limit the field strength is gauge invariant and the Gribov ambiguity does not arise;
- a fixing that is at once comprehensive and complete always exists (the orbit representatives) and makes every observable extend **uniquely** to a physical one, but on a connected spectrum with a free non-trivial action it can never be clopen — completeness forces a discontinuous choice;
- the quadratic-Casimir constraint imposes every Hermitian constraint at once, and gauge averaging is a projection onto the physical algebra that survives limits (no anomaly);
- the dissipative coupled oscillators lose energy at the rate $\dot{E} = -\lambda_1 \dot{x}_1^2 - \lambda_2 \dot{x}_2^2$ while conserving total probability — irreversibility without a classical Hamiltonian.

25. Free Fields: the Gaussian and the Uniform Sphere

25.1. Orientation: From the Manuscript to the Formal Development

This chapter follows the free-field thread of `book.tex`, but the formal development has two distinct layers. The probability and Gaussian constructions describe the infinite-dimensional parametrization; the later

Hamiltonian/SIRK material describes finite truncations and their limits. A theorem citation below certifies only the exact statement of its `BookProof` declaration. In particular, the QYM certificate currently certifies a truncated value; the remaining specialist target is the lowest positive one-particle edge and its lift through the free nested-Fock `dGamma` Hamiltonian.

Object	Formal status
Gaussian/Mehler and Born parametrization	proved model theorems
finite Krylov/SIRK approximation	proved/conditional error theorems
emitted QYM certificate	proved truncated enclosure
QYM Friedrichs/Hashimoto-selected operator	proved construction/selection
one-particle gap, lifted to Fock space	proved for the concrete gauge-fixed QYM one-particle operator, conditional on its one-particle **form** gap (`ChapterYangMillsFockGapChain`)
unbounded field perturbations of the gap	proved for the linear coupling `Φ(f)` (`ChapterFockFieldPerturbation`, `sorry`-free) and for the quadratic pair coupling `P(f,g)` (`ChapterFockPairPerturbation`); proved **impossible** for a bare cubic term (`ChapterFockCubicUnbounded`); the cubic–quartic pair is bounded below on all finite states (`ChapterFockCubicQuarticStability`)
gap for sectors with constant or diagonal one-particle energy	proved unconditionally: the R² scalaron at `m = 1/√(12α)` (`ChapterScalaronFockGapChain`), any diagonal dispersion `ω_k = √(p_k² + m²)` (`ChapterFockDiagonalGapChain`), and the full-exponential scalaron fiber's strict edge (`ChapterScalaronEdge`) — no certificate, no form-gap hypothesis
QED photon sector	proved: positivity with **no gap** for the massless photon (`ChapterQedFockGapChain`), the no-gap obstruction proved for every `m > 0` under infrared momenta accumulating at zero, and gapped statements attached only to the regulated/massive instantiations (`irPhoton_fock_mass_gap`, `proca_fock_mass_gap`)
real Fock mass gap of the continuum operator	still conditional on the one-particle form gap (the certificate supplies a truncated bound); no continuum claim

In the tour this is the second vocabulary chapter of the part (the first is gauge symmetry) and the direct descendant of the Solovay–Kopperman tensor product — the Mehler measure reappears here as the rotation-invariant Gaussian, so re-reading that chapter first repays itself. From the Born-rule chapters it takes the push from the sphere to the simplex.

The route: the problem (there is no infinite-dimensional Lebesgue measure), what masslessness costs in QED, the rotation-invariant Gaussian as substitute, pushing the Gaussian to the sphere, the Mehler limit (why the Gaussian **is** the uniform measure on the infinite sphere), Born pushing the sphere to the simplex, differentiability in the Fourier-transformed space, the countability sanity check, and the dimensional-reduction/Krylov material that lands on the Navier–Stokes Hamiltonian as a complete flow on the truncation.

The Gaussian layers are `BookProof/ChapterNoLebesgue`, `BookProof/ChapterFreeFieldGaussian`, `BookProof/PhysHSGaussian` and the Mehler chapters; the Hamiltonian layers are `BookProof/ChapterF`, `BookProof/ChapterH`, the Navier–Stokes chapters, and the ESA toolbox (`ComparisonCoreEsa`, `SelfAdjointCoreEsa`, `KatoRellich`, and the truncation-gap chapters).

25.2. The Problem: There Is No Infinite-Dimensional Lebesgue Measure

The new gap modules make the operator-level boundary explicit. `BookProof/Toolbox/ChapterFockOneParticleGap.lean` proves the positive one-particle edge statement under its spectral hypotheses; `ChapterFriedrichsFormGap.lean` packages the corresponding closed-form/Friedrichs route; and `ChapterBandEnclosure.lean` shows how nested finite bands can provide the remaining enclosure input. These are conditional analytic bridges, not a claim that numerical data alone proves the continuum gap.

Two of those bridges have since been made **theorems about recorded data**, which is what a certificate can actually supply. In the final object of the programme — the nested-Fock Hamiltonian $H = \sum_{i,j} h_{ij}\, C^\dagger(e_i) A(e_j)$ built from the gauge-fixed one-particle operator h — the gap of $\sum_i h_{ii}\, C^\dagger(e_i) A(e_i)$ of the **free** n-particle sector is what `ChapterFockDiagonalGapChain` and its instantiation for the massless photon certify. The newly landed QYM-1 wave closes the same gap for the genuinely coupled gauge-fixed Yang–Mills one-particle operator. `BookProof/Sirk/ChapterTruncationGapLift.lean` splits each finite-mode core vector $v = x + w$ into its order-m Galerkin part and its tail, and proves that a gap certified on the truncation lifts to the core once the tail is coercive and the coupling across the split is bounded (`gap_of_level_gap_and_tail`, `gap_of_uniform_truncated_gap`), and further via the dGamma operator applied to the outer-enclosed H (the

ym_fock_* compositions). BookProof/Sirk/ChapterSchurGershgorinGap.lean then proves those two analytic inputs **from the matrix elements** the certificate records: diagonal dominance on the tail gives coercivity (tail_coercive_of_gershgorin) and a Schur test on the coupling block gives the cross bound (coupling_bound_of_schur), composed into the strict positivity $\lambda_1(H_1 \mid \text{core}) > 0$ of QYM-1 task 3 (strict_pos_of_matrix_bounds) — the single remaining hypothesis of the gap chain is now a family of checkable inequalities on the recorded entries, not an unanalysed spectral assumption.

```
#check @BookProof.TruncationGapLift.gap_of_uniform_truncated_gap
#check @BookProof.TruncationGapLift.gap_of_level_gap_and_tail
#check @BookProof.SchurGershgorin.tail_coercive_of_gershgorin
#check @BookProof.SchurGershgorin.coupling_bound_of_schur
#check @BookProof.SchurGershgorin.strict_pos_of_matrix_bounds
```

25.3. QED: What Masslessness Costs, Formally

The QED sector is the cleanest illustration of why a **gap** is a separate, stronger statement than **positivity**. BookProof/Qed/ChapterQedFockGapChain.lean (namespace BookProof.QedFockGapChain) runs the diagonal gap chain (ChapterFockDiagonalGapChain — the instrument for one-particle energies $e_k \mapsto \omega_k e_k$) on the photon dispersion. For the massless photon $\omega_k = |p_k|$ the sector is positive (photon_fock_positivity) but **gapless**; the obstruction is proved, not asserted: photon_no_one_particle_gap shows the one-particle form gap fails for **every** $m > 0$ whenever the momenta accumulate at zero (the infrared). The mechanism is exactly the diagonal-chain boundary: the form gap needs a uniform positive lower bound $\omega_k \geq m$ across the modes, and the massless dispersion violates it precisely at the infrared modes — a statement that survives any relabelling of the momentum sequence.

```
#check @BookProof.QedFockGapChain.photon_fock_positivity
#check @BookProof.QedFockGapChain.photon_no_one_particle_gap
#check @BookProof.QedFockGapChain.irPhoton_fock_mass_gap
#check @BookProof.QedFockGapChain.proca_fock_mass_gap
```

Where a gap **is** claimed for QED, it is claimed honestly — attached to the regulated or massive instantiation, not to the free photon. The infrared- regulated dispersion (a mass cutoff $\mu > 0$ away from zero) recovers $\omega_k \geq \mu$ and therefore the Fock mass gap irPhoton_fock_mass_gap; the Proca (massive photon) dispersion $\omega_k = \sqrt{p_k^2 + m^2}$ gives proca_fock_mass_gap. Both are labelled exactly as what they are: gapped statements for **regulated/massive** QED, with the masslessness of the physical photon kept as the explicit reason the free sector has no gap. This is the same honesty discipline as the rest of the programme: the gap chain attaches to the one-particle energy that the model actually specifies, and every gapped claim names its sector.

The finite-dimensional Born parametrization puts a uniform-looking measure on the sphere and pushes it to the simplex. To pass to **field theory** — infinitely many degrees of freedom — we would like a uniform (Lebesgue-like) measure on an infinite-dimensional sphere. There is an immediate obstruction:

There is no non-zero, translation-invariant measure that is finite on bounded sets in an infinite-dimensional normed space.

The proof is a clean volume argument. In a translation-invariant measure, all balls of a given radius have the same measure. In infinite dimension a ball of radius r contains **infinitely many** disjoint balls of radius $r/2$ (because there is an infinite orthonormal set, and the small balls around well-separated points are disjoint). If the small balls have measure $m > 0$, the big ball has infinite measure; if $m = 0$, countable additivity is not even needed to see the big ball has measure 0 once it is covered by countably many null balls in the finite-on-bounded case. Either way, no measure can be both translation-invariant and finite and non-zero on bounded sets.

The verified statements (module BookProof.ChapterNoLebesgue):

```
#check @ChapterNoLebesgue.measure_ball_eq_measure_ball_zero
#check @ChapterNoLebesgue.not_exists_nonzero_isAddLeftInvariant_finite_on_bounded
```

This is the rigorous content of the manuscript's claim that "there is no non-informative prior" on a continuous infinite-dimensional space, and it forces the free-field construction to build its uniform measure **out of the Gaussian** rather than out of a nonexistent Lebesgue measure.

25.4. The Substitute: the Rotation-Invariant Gaussian

The standard Gaussian on $\mathbb{R}^n$,

$$\gamma_n(dx) = (2\pi)^{-n/2} e^{-\|x\|^2/2}\, dx,$$

depends only on the radius $\|x\|$. Consequently it is **rotation-invariant**: pushing it forward by any orthogonal map L leaves it unchanged. Its characteristic function $t \mapsto e^{-\|t\|^2/2}$ likewise depends only on $\|t\|$, which pins down the measure and makes the invariance manifest.

The verified statements (module `BookProof.ChapterFreeFieldGaussian`):

```
#check @ChapterFreeFieldGaussian.instIsProbabilityMeasure_stdGaussian
#check @ChapterFreeFieldGaussian.charFun_stdGaussian
#check @ChapterFreeFieldGaussian.stdGaussian_map_linearIsometryEquiv
```

The last is the headline: $(\gamma_n).\mathrm{map}(L) = \gamma_n$ for every orthogonal L. The Gaussian is the canonical **rotation-invariant probability prior**, the finite-dimensional stand-in for the uniform measure that does not exist in infinite dimension.

25.5. Pushing the Gaussian to the Sphere

Because the Gaussian has no atom at the origin (in dimension $n \geq 1$), radial normalization $x \mapsto x/\|x\|$ is defined almost everywhere and pushes γ_n to a probability measure **on the unit sphere**. By rotation-invariance of the Gaussian, this induced sphere measure is itself rotation-invariant — it is the **uniform measure on the sphere**, constructed without ever invoking a Lebesgue measure on the infinite-dimensional ambient space. This is the manuscript's free-field prior: "a uniform measure of an infinite-dimensional sphere defined using the Gaussian measure and the Fock space."

This uniform sphere measure is the **Mehler measure**, and it is the exception to the slogan this chapter opened with. There is no *Lebesgue* uniform probability measure on an infinite-dimensional space, but the Gaussian-built sphere measure is a genuine uniform prior on the infinite-dimensional hypersphere. It is genuinely non-informative — invariant under every symmetry the language can express — in the Solovay–Kopperman space, where the restricted decidable language forbids the reparametrizations that would otherwise turn a uniform prior into an informative one (non-informativeness is coordinate-dependent). The free-field construction is the finite-dimensional shadow of that infinite-dimensional exception.

25.6. Why the Gaussian Is the Uniform Measure on the Infinite Sphere: the Mehler Limit

This probabilistic construction is also the domain in which the later Fock proofs operate. The unbounded-perturbation ladder is now complete at degrees one and two. `ChapterFockFieldPerturbation.lean` is `sorry`-free and controls the **linear** field coupling $\Phi(f) = a^\dagger(f) + a(f)$ — genuinely unbounded and number-changing — leaving the gap `(μ - 2‖f‖)‖u‖²` on vacuum-orthogonal states. `ChapterFockPairPerturbation.lean` carries the same package to the **quadratic**, pair-creating coupling $P(f,g) = a^\dagger(f)a^\dagger(g) + a(g)a(f)$ (particle number changes by two), leaving the gap `(μ - 2√2‖f‖‖g‖)‖u‖²` under the smallness condition `2√2‖f‖‖g‖ < μ`. And `ChapterFockCubicUnbounded.lean` proves that the route **stops** there: a bare cubic term has no relative form bound at all, so `dΓ(N) + λC` is unbounded below at every coupling strength — the divergence is cured only by the (bounded-below) quartic term. `ChapterFockInteractionStability.lean` records the quantitative loss of a gap under a bounded or relatively form-bounded interaction, and `ChapterYangMillsFockGapChain.lean` instantiates the whole chain for the concrete gauge-fixed QYM one-particle operator, conditionally on its one-particle form gap.

The claim that the Gaussian is the uniform measure "on an infinite-dimensional sphere" is not a slogan; it is a genuine limit, due in spirit to Mehler (1866) and made quantitative by López and Temme. On the surface of a k-sphere the rotation-invariant (uniform) area element factorizes into single-coordinate weights $(1-x^2)^{(\alpha-1)/2}$ — exactly the **Gegenbauer** weight of order α. The uniform hyperspherical measure is therefore the natural **prior**: it

assigns equal weight to every direction, expressing that we have no information. Its orthogonal polynomials are the Gegenbauer polynomials $C_n^{(\alpha/2)}$.

The infinite-dimensional limit is where the identification becomes the Gaussian. Rescale the coordinate $x \mapsto \sqrt{2/\alpha}\,x$ and let $\alpha \to \infty$. The single-coordinate hyperspherical weight tends to the Gaussian weight e^{-x^2}:

$$\lim_{\alpha\to\infty}\left(1-\tfrac{x^2}{\alpha}\right)^{\alpha-1/2} = e^{-x^2}.$$

The verified statement (module `BookProof.PhysHSGaussian`):

```
#check @PhysHSGaussian.weight_tendsto_gaussian
```

At the same time the Gegenbauer polynomials become the Hermite polynomials $H_n/n!$ (with the physicists' H_n defined by the recurrence $H_0 = 1, H_1 = 2x, H_{n+2} = 2xH_{n+1} - 2(n+1)H_n$), in the scaled limit

$$\lim_{\alpha\to\infty}\left(\tfrac{\alpha}{2}\right)^{-n/2} C_n^{(\alpha/2)}\left(\sqrt{\tfrac{2}{\alpha}}\,x\right) = \frac{H_n(x)}{n!}.$$

The verified statement (module `BookProof.PhysHSGaussian`):

```
#check @PhysHSGaussian.physHermite
#check @PhysHSGaussian.gegenbauer
#check @PhysHSGaussian.gegenbauerScaled
#check @PhysHSGaussian.gegenbauerScaled_tendsto_hermite
```

The normalizations converge consistently as well: the (squared) Gegenbauer normalization integral against the hyperspherical weight tends to the Hermite normalization integral against the Gaussian weight,

$$\int_{\mathbb{R}}\Big[\frac{H_n(x)}{n!}\Big]^2 e^{-x^2}\,dx = \frac{\sqrt{\pi}\,2^n}{n!}.$$

The verified statements (module `BookProof.PhysHSGaussian`):

```
#check @PhysHSGaussian.hermite_normalization
#check @PhysHSGaussian.normalization_tendsto
```

This is the precise sense in which the uniform (hyperspherical) prior and the Gaussian (Fock) vacuum coincide, and in which the orthogonal excitations of the Fock space are the images of the hyperspherical harmonics. The dictionary is complete:

Finite hypersphere ($\alpha < \infty$)	Infinite-dimensional limit (Fock)
uniform surface measure $d\sigma$	Gaussian measure $e^{-x^2}dx$
Gegenbauer weight $(1-x^2)^{(\alpha-1)/2}$	Gaussian weight e^{-x^2}
Gegenbauer polynomials $C_n^{(\alpha/2)}$	Hermite polynomials $H_n/n!$
hyperspherical harmonics	number states $\lvert n\rangle$
uniform "no-information" prior	Gaussian source qsample $\Psi_0 = \lvert 0\rangle$

That the Gaussian pushes forward to the **uniform** sphere measure, and is itself rotation-invariant, is proved in the same module (and is the mechanism used above to build the sphere measure in the first place):

```
#check @PhysHSGaussian.gaussianE_rotation_invariant
#check @PhysHSGaussian.sphereUniform_rotation_invariant
```

And the concentration is quantitative: for an almost-every sequence of standard Gaussian coordinates, the empirical squared norm $\sum_{i<k}\omega_i^2$ grows like k, i.e. the Gaussian sample sits on the sphere of radius $\sqrt{k}$ in the limit — the strong-law form of "the Gaussian concentrates on the sphere":

```
#check @PhysHSGaussian.gaussian_concentration_sphere
```

So the free-field prior is not merely **a** rotation-invariant measure: it is the infinite-dimensional limit of the family of uniform hyperspherical priors, with the Fock excitations as the limits of the hyperspherical harmonics. This is what makes the Gaussian source qsample $\sqrt{p_0}$ of a flow model coincide with the Fock vacuum that the companion algebra manipulates natively.

25.7. Born Pushes the Sphere to the Simplex

Composing the two maps — normalize the Gaussian to the sphere, then apply the Born rule $\psi \mapsto |\psi|^2$ — parametrizes the probability simplex by the Gaussian-built sphere measure. The finite-dimensional Born map is continuous and surjective (The Born rule reproduces every distribution), and its fibers are the phase/sign gauge of the gauge ambiguity. The simplex is thus realized as a **quotient** of the Gaussian-built sphere by the gauge group, with a canonical rotation-invariant measure on top.

25.8. Differentiability in the Fourier-Transformed Space

The manuscript's free-field section is careful about a cost of the parametrization: because the uniform prior attributes null measure to **deterministic** functions, one **sacrifices point-evaluation** of a classical field, and so "continuity and differentiability notions must be redefined for quantum fields". Its justification that this is not a problem is that **the smooth wave-functions are dense in the Hilbert space** — an approximation statement.

That justification is correct but incomplete. The stronger fact is that, in the Fourier-transformed space, the field is already smooth almost everywhere, not merely approximable by smooth functions. The momentum constraint of the free-field section is $iD_x = 0$ with p the eigenvalue of the derivative operator that the Fourier transform diagonalizes: in the Fourier picture, differentiation is multiplication by a real frequency, and the (Fourier-transformed) wave-function that defines the creation operator is the analytic continuation of that frequency function. Analytic functions are smooth everywhere, so the represented field is smooth on the transformed space — the null-measure caveat of the manuscript's point-evaluation discussion is discharged in the Fourier domain **before** any density argument is needed.

The same phenomenon appears in number-theoretic series: the series used in Bagchi's universality theorem (and the Dirichlet series it is built from) is convergent and analytic on a half-plane, and its analytic continuation is smooth almost everywhere in the transformed (frequency) variable. There too the smoothness is not an approximation imposed by a choice of prior; it is a property the analytic continuation already has. The manuscript's density argument describes what is true in the **original** (position) space; the sharper statement is that the **Fourier** or frequency representation of the same object is already smooth, so the differentiability notions the manuscript says must be "redefined" are in fact satisfied pointwise (almost everywhere) on the transformed side.

The density assertion the manuscript relies on is itself verified (`polynomial_dense_L2` in `BookProof.PhysFunctionalAnalysis`: polynomials — hence smooth functions — are dense in L^2 with respect to the uniform (unit-interval) measure):

```
#check @PhysFunctionalAnalysis.polynomial_dense_L2
```

This is the approximation statement: it says smooth functions are **close** to every L^2 field. The Fourier-domain observation above is the strictly stronger one: in the transformed space the field is already smooth a.e., so no approximation step is needed there. Both facts coexist — the former concerns the original space, the latter concerns its Fourier representation.

This is consistent with, and sharpens, the rest of the free-field thread: the Gaussian-built uniform measure is rotation-invariant, its Fourier transform is the same Gaussian (self-dual), and the frequency representation of a field drawn from that prior is a smooth, analytically-continuable object rather than a merely approximable one.

25.9. A Countability Sanity Check

One last measure-theoretic fact keeps the countable Euler-angle parametrization of the Born-rule chapter honest. The manuscript asserts that "any partition of the phase-space (where each part has a non-null Lebesgue measure) is countable." This is a general fact about s-finite measures: a pairwise-disjoint family of measurable sets, **every** one of positive measure, must be countable — because in each measure band $[1/(k+1), 1/k]$ there can be only finitely many disjoint sets, and a countable union of finite sets is countable.

The verified statements (module `BookProof.ChapterCountablePartition`):

```
#check @ChapterCountablePartition.countable_of_partition_pos
#check @ChapterCountablePartition.prob_partition_countable
```

So when the book indexes the parts of a phase-space partition by a countable orthonormal basis (the setting of the countable stick-breaking chain), it is not making an extra assumption: positivity of the part measures **forces** the index set to be countable.

25.10. Dimensional Reduction: the Krylov Shortcut and the Compiled Sketch

The same Gaussian/Fock substrate is what makes the **computational** side of the free-field thread tractable. Two reductions do the work. The first is the inversion-free Krylov shortcut: the Krylov subspace $\mathrm{Kry}_m(\bar{H}, v_0) = \mathrm{span}\{v_0, \bar{H}v_0, \ldots, \bar{H}^{m-1}v_0\}$ is unchanged when the generator is shifted by a multiple of the identity, because the shift only adds lower-order terms. So the sequence $w_k = (\bar{H} - \gamma I)w_{k-1}$, which never solves a linear system, spans exactly the same subspace as the rational construction that would; and pre-conditioning the seed makes the resolvent merely lower the polynomial degree. The generator itself is bounded — the Mehler projector is rank-one, the number operators diagonal.

```
#check @BookProof.ChapterH5.krylovSpan
#check @BookProof.ChapterH5.krylov_subspace_span
#check @BookProof.ChapterH5.shift_pow_sub_pow_mem
#check @BookProof.ChapterH5.krylovSpan_shift_eq
#check @BookProof.ChapterH5.inversion_free_seed
#check @BookProof.ChapterH5.generator_bounded_of_rankOneProjector
#check @BookProof.ChapterH5.krylov_no_inversion_eq_standard
```

The second reduction reads the Krylov projection as a **spectral low-pass filter**. The compression $\bar{H}_{\mathrm{reduced}} = V^{\mathsf{H}}\bar{H}V$ has the restricted quadratic form, so each of its eigenvalues is a Rayleigh quotient of the full generator and can never exceed its norm: the projection keeps frequencies, it does not invent them. What it discards costs only the e^{-hm} term already carried by the SIRK error bound, which decays to zero as the retained dimension grows. And generation is then a **single** matrix exponential of an explicitly $m \times m$ object.

```
#check @BookProof.ChapterH6.sirk_error_decay_exponential
#check @BookProof.ChapterH6.sirk_error_bound_antitone
#check @BookProof.ChapterH6.krylov_rayleigh_transfer
#check @BookProof.ChapterH6.krylovRetainsDominantSpectrum
#check @BookProof.ChapterH6.reduce_generator_mul_m
#check @BookProof.ChapterH6.generation_single_exponential
```

The reductions form a **nested tower of approximation orders**, and this nesting is itself a decidable, finite-dimensional statement — the part of the Hashimoto convergence guarantee that needs no analytic input. Write Kry_n for the order-n Krylov subspace $\mathrm{Kry}_n(\bar{H}, v_0)$. Two facts are purely algebraic. First, the subspaces nest: $\mathrm{Kry}_n \subseteq \mathrm{Kry}_{n+1}$ ($\mathrm{krylovSpan_mono}$): the order-n basis is the first n vectors of the order-(n+1) basis. Second, the order-n reduced generator is the top-left $n \times n$ block of the order-(n+1) reduced generator, because the compression is upper-Hessenberg — so the order-n approximant is the **projection** of the order-(n+1) approximant onto Kry_n: the finer band, restricted to the coarser information, reproduces the coarser band exactly. And the error bound is monotone in the same direction: the band $[0, \mathrm{sirkBound}(n+1)]$ is contained in $[0, \mathrm{sirkBound}(n)]$ ($\mathrm{sirk_error_bound_antitone}$), and the bands collapse to $\{0\}$ as n grows ($\mathrm{sirk_error_decay_exponential}$).

```
#check @BookProof.ChapterH5.krylovSpan_mono
#check @BookProof.ChapterH5.krylovSpan_map_le
#check @BookProof.ChapterH6.sirk_error_bound_antitone
#check @BookProof.ChapterH6.sirk_error_decay_exponential
#check @BookProof.ChapterH6.sirk_error_tendsto_zero
#check @BookProof.ChapterH7.compress_isSelfAdjoint
#check @BookProof.ChapterH7.compression_eigenvalue_mem_numericalRange
#check @BookProof.ChapterH8.sirk_krylov_tower
#check @BookProof.ChapterH8.sirk_compression_block
#check @BookProof.ChapterH8.sirk_compression_submatrix
#check @BookProof.ChapterH8.sirk_band_refinement
#check @BookProof.ChapterH8.sirk_band_refinement_proj
```

```
#check @BookProof.ChapterH8.sirk_approx_projection
#check @BookProof.ChapterH8.sirk_band_refinement_poly
#check @BookProof.ChapterH8.sirk_band_refinement_rational
#check @BookProof.ChapterH8.sirk_approx_projection_poly
#check @BookProof.ChapterH8.sirk_approx_projection_rational
#check @BookProof.ChapterH8.sirk_band_refinement_of_orthonormal
#check @BookProof.ChapterH8.sirk_compression_submatrix_of_orthonormal
#check @BookProof.ChapterH8.krylovOrthonormal_span
#check @BookProof.ChapterH8.krylovEmbedding_range
#check @BookProof.ChapterH8.sirk_band_refinement_krylov
#check @BookProof.ChapterH8.sirk_band_contained
#check @BookProof.ChapterH8.sirk_band_contained_le
#check @BookProof.ChapterH8.sirk_compression_submatrix_le
#check @BookProof.ChapterH8.sirk_nested_orders
#check @BookProof.ChapterH8.sirk_nested_orders_le
```

The honest boundary: the **nesting** above is finite-dimensional linear algebra over the decidable skeleton (Solovay–Mehler–Kopperman: every inner product over the infinite substrate collapses to a finite head integral), so it is provable without any analytic hypothesis. What the nesting does **not** supply is the numerical **width** of the bands: that the true error lies inside $\mathrm{sirkBound}(n)$ is conditional on Crouzeix's inequality ($\mathrm{sirk_error_bound_decay}$, `ChapterH4`), which is recorded in `BookProof/` as a named hypothesis rather than an axiom. The band-containment statement itself — order-(n+1) refines order-n, iterated — is proved in `ChapterH8`: the block identity $\mathrm{sirk_compression_block}$, the projection identities $\mathrm{sirk_band_refinement}$ and $\mathrm{sirk_approx_projection}$, and the tower $\mathrm{sirk_nested_orders}$ — which holds not only one step at a time but between any two orders $m \le n$ ($\mathrm{sirk_nested_orders_le}$, with the leading-block identity $\mathrm{sirk_compression_submatrix_le}$). The refinement is not restricted to powers of the generator: it holds for an arbitrary polynomial of the reduced generator ($\mathrm{sirk_band_refinement_poly}$) and for the rational functions the SIRK step actually evaluates — numerator polynomial over an invertible denominator whose compression is invertible ($\mathrm{sirk_band_refinement_rational}$), with the whole-space projection forms $\mathrm{sirk_approx_projection_poly}$ and $\mathrm{sirk_approx_projection_rational}$. The hypotheses are not vacuous: any pair of nested orthonormal Krylov bases realizes them, through the embedding of an orthonormal family and the coordinate inclusion along castLE, which gives the hypothesis-free instances $\mathrm{sirk_band_refinement_of_orthonormal}$ and $\mathrm{sirk_compression_submatrix_of_orthonormal}$. Those bases exist: Gram–Schmidt on the Krylov sequence $k \mapsto H^k v$ produces one orthonormal sequence whose prefixes are orthonormal, nested by construction and span the Krylov subspaces ($\mathrm{krylovOrthonormal_span}$, $\mathrm{krylovEmbedding_range}$), which gives the refinement statement for the Krylov flag itself ($\mathrm{sirk_band_refinement_krylov}$, under the hypothesis that the Krylov sequence has not broken down before the finer order). Two provisos are recorded there. The refinement of the **approximants** needs the order-n subspace to be invariant under the compressed operator — the block-triangularity that makes the leading block a generator in its own right; and the whole-space form of the projection identity needs that subspace to **reduce** the operator (invariance under the adjoint as well), since without it the leading block controls only the coarse data.

The nesting has a spectral face as well: not only do the approximants refine, the **frequencies** the reduced generators can see nest. Writing $W(\cdot)$ for the numerical range — the set of Rayleigh quotients over unit vectors — the reduced generators $B_k = V_k^* X V_k$ satisfy $W(B_m) \subseteq W(B_n) \subseteq W(X)$ for $m \le n$ ($\mathrm{sirk_numRange_nested_orders}$, and $\mathrm{sirk_numRange_krylov}$ for the Krylov flag the method actually builds), with the matching norm chain $\|B_m\| \le \|B_n\| \le \|X\|$ and the uniform envelope $W(X) \subseteq \{|z| \le \|X\|\}$. Every Ritz value of a coarse order is therefore a Rayleigh quotient of every finer order, and of the full generator ($\mathrm{ritz_mem_numRange}$, $\mathrm{ritz_mem_numRange_compress}$): refining the order can only **add** frequencies, and never one the physics does not already have. Since a finite-dimensional reduced generator has only eigenvalues in its spectrum, the whole Ritz **spectrum** nests the same way ($\mathrm{spectrum_compress_subset_numRange_orthonormal}$). Positivity and any real window $[a, b]$ of the quadratic form survive compression at every order ($\mathrm{compress_nonneg}$, $\mathrm{compress_re_inner_mem_Icc}$). The direction matters for honesty: since the numerical ranges **grow** with the order, a Crouzeix-type bound $C \sup_{W(B)} |f|$ is non-decreasing in the order — the band decay of $\mathrm{sirkBound}$ comes from the approximation quality, not from shrinking numerical ranges. Convexity of the numerical range (Toeplitz–Hausdorff) is neither used nor claimed. The approximation quality itself does improve, unconditionally: the Krylov subspaces are finite-dimensional, so the orthogonal projection onto them exists, and since they nest, the best-approximation error is

antitone in the order — for $m \le n$ the order-n subspace approximates any target at least as well as the order-m one ($\mathrm{krylov_bestApprox_antitone}$), and when the Krylov flag is dense — a cyclic seed — the error tends to 0 ($\mathrm{krylov_bestApprox_tendsto_zero}$). Those statements need no Crouzeix constant; what they do **not** give is a rate.

```
#check @BookProof.ChapterH9.numRange_compress_subset
#check @BookProof.ChapterH9.numRange_compress_mono
#check @BookProof.ChapterH9.numRange_compress_chain
#check @BookProof.ChapterH9.numRange_subset_closedBall
#check @BookProof.ChapterH9.convexHull_numRange_compress_mono
#check @BookProof.ChapterH9.norm_compress_le
#check @BookProof.ChapterH9.norm_compress_mono
#check @BookProof.ChapterH9.ritz_mem_numRange
#check @BookProof.ChapterH9.ritz_mem_numRange_compress
#check @BookProof.ChapterH9.ritz_re_mem_Icc_of_fine
#check @BookProof.ChapterH9.compress_nonneg
#check @BookProof.ChapterH9.compress_re_inner_mem_Icc
#check @BookProof.ChapterH9.numRange_compress_orthonormal_mono
#check @BookProof.ChapterH9.spectrum_compress_subset_numRange
#check @BookProof.ChapterH9.spectrum_compress_subset_numRange_compress
#check @BookProof.ChapterH9.spectrum_compress_subset_numRange_orthonormal
#check @BookProof.ChapterH9.sirk_numRange_nested_orders
#check @BookProof.ChapterH9.sirk_numRange_krylov
#check @BookProof.ChapterH9.norm_sub_starProjection_antitone
#check @BookProof.ChapterH9.krylov_bestApprox_antitone
#check @BookProof.ChapterH9.krylov_bestApprox_tendsto_zero
```

Finally, the offline stage: a two-level sketch hashes raw coordinates into k features and places the features on distinct modes of a K_2-mode Fock space, producing a genuine single-excitation state from which the features can be read back; and all observables of the reduced space are pre-projected into a complete basis of m^2 matrix units. The corpus size enters only there — the online cost carries no such term.

```
#check @BookProof.ChapterF8.singleExcitation
#check @BookProof.ChapterF8.twoLevelHash_total
#check @BookProof.ChapterF8.featureHash_decodes
#check @BookProof.ChapterF8.offline_operatorBasis
#check @BookProof.ChapterF8.online_cost_independent_of_M
#check @BookProof.ChapterF8.tsr_offline_compiles
```

Two properties make the reduction safe to **run**. First, when the full generator is self-adjoint the compression is self-adjoint too, so every eigenvalue of the reduced generator is real and is pinned between any bounds satisfied by the quadratic form of the full generator: the reduced dynamics can neither go complex nor leave the frequency band of the physics it came from. Second, the reduced propagator $e^{-\mathrm{i}t\bar{H}}$ of a Hermitian generator is unitary, hence generation conserves the ℓ^2 mass exactly — the generated state is a state, at every step and for every horizon t.

```
#check @BookProof.ChapterH7.compress_isSelfAdjoint
#check @BookProof.ChapterH7.compression_rayleigh_real
#check @BookProof.ChapterH7.compression_eigenvalue_mem_numericalRange
#check @BookProof.ChapterH7.reduceGenerator_isHermitian
#check @BookProof.ChapterH7.generationOperator_mem_unitaryGroup
#check @BookProof.ChapterH7.generation_preserves_l2
```

25.11. The Navier–Stokes Hamiltonian: a Complete Flow on the Truncation

The same free-field parametrization is what the manuscript applies to the Navier–Stokes equations: the velocity field, **its derivatives** $u_{k,j}$, $u_{k,jj}$, and the conjugate momenta π^i are all treated as independent canonical degrees of freedom, and the Hamiltonian is the Weyl-symmetrized $H = \sum_i (\pi_i A_i + A_i \pi_i)$ with $A_i = \sum_j u_j u_{i,j} - \nu\, u_{i,jj}$. One scoping sentence, because the word "Hamiltonian" is doing two jobs in this chapter: this Weyl-symmetrized operator is the **one-particle** operator — the manuscript's $H(\vec{x})$ that sits **inside** the enclosure — and the final Hamiltonian of record is its creation-left/annihilation-right second quantization $d\Gamma(h) = \sum_{ij} h_{ij} C^\dagger(e_i) A(e_j)$, as the Navier–Stokes one-particle Hamiltonian chapter states and as Second Quantization and Essential Self-Adjointness makes rigorous. Everything in this section — the truncation's completeness, the deficiency computation, the Faris–Lavine lift below — is about that operator and its lift, not about a different final

Hamiltonian. That the derivatives may be treated as fields is an operator statement: an operator-valued field $\varphi(X) = \varphi + \varphi_i (X_i - x_i)$ collapses to its point value on the position eigenstates, so the first-order Taylor coefficients are free modes with commutation relations of their own.

```
#check @BookProof.NavierStokesFlow.fieldTaylor
#check @BookProof.NavierStokesFlow.field_evaluates_to_value
#check @BookProof.NavierStokesFlow.field_evaluates_to_value_diagonal
#check @BookProof.NavierStokesFlow.ccr_field
#check @BookProof.NavierStokesFlow.derivativeField_momentum
#check @BookProof.NavierStokesFlow.secondDerivativeField_momentum
#check @BookProof.NavierStokesFlow.momentumConstraint_preserved
```

The Eulerian side of that construction carries its own constraints, and the manuscript's own taxonomy splits them in two. The relations that **define** the derivative modes — $u_{i,j} = \partial_j u_i$ and $u_{i,jk} = \partial_k u_{i,j}$ — have no explicit solution, so they are the ones a gauge generator has to impose; their consequence is Clairaut's condition $u_{i,jk} = u_{i,kj}$, which makes the second-derivative modes symmetric. The momenta conjugate to the derivative modes pair with them by a Kronecker delta, which is what makes each derivative an independent canonical variable. Incompressibility $\partial_j u_j = 0$, by contrast, **does** have an explicit solution — the substitution $u_{3,3} = -(u_{1,1} + u_{2,2})$ — so it is imposed by initial data rather than by a gauge symmetry, and it is satisfied by genuinely non-constant fields such as the cyclic shear $u_i(x) = x_{i+1}$.

```
#check @BookProof.NavierStokesEulerian.u_evaluates_to_value
#check @BookProof.NavierStokesEulerian.eulerian_momentum_constraint
#check @BookProof.NavierStokesEulerian.eulerian_momentum_dual
#check @BookProof.NavierStokesEulerian.derivativeField_relates_to_field
#check @BookProof.NavierStokesEulerian.derivativeField_second
#check @BookProof.NavierStokesEulerian.derivativeField_consistency
#check @BookProof.NavierStokesEulerian.eulerian_divergence_constraint
#check @BookProof.NavierStokesEulerian.cyclicShear_divergence_free
```

The constraint is stated properly only once a **second coordinate** y is adjoined to the space coordinate x: the field that enters the Hamiltonian is the expansion $u_i(y) = u_i + u_{i,j} y_j$ in that second coordinate. Each coordinate then comes with a gauge generator. For x it is the standard momentum $\pi^j = \partial/\partial x_j$; for y it is the generator built from the **derivatives of** u_i, $G_j = \partial/\partial y_j - u_{i,j}\, \partial/\partial u_i$, which translates the second coordinate while shifting each velocity mode by its own first derivative. Both generators annihilate $u_i(y)$ and the Hamiltonian symbol $A_i = u_j(y) u_{i,j} - \nu u_{i,jj}$, and they commute among themselves, so the constraints are first class; and the coefficient $u_{i,j}$ is the only one for which the invariance holds, which is exactly the content of $u_{i,j} = \partial u_i / \partial y_j$. In the initial state the second coordinate evaluates to $y = 0$, so the field collapses to its point value u_i there and the Hamiltonian acts as the ordinary Navier–Stokes one: the second coordinate carries the constraint without changing the dynamics of the initial data.

```
#check @BookProof.NavierStokesGaugeY.uField
#check @BookProof.NavierStokesGaugeY.genX
#check @BookProof.NavierStokesGaugeY.genY
#check @BookProof.NavierStokesGaugeY.uField_pderiv_y
#check @BookProof.NavierStokesGaugeY.genX_ccr_x
#check @BookProof.NavierStokesGaugeY.genY_ccr_y
#check @BookProof.NavierStokesGaugeY.genY_shifts_velocity
#check @BookProof.NavierStokesGaugeY.genY_uField
#check @BookProof.NavierStokesGaugeY.genX_uField
#check @BookProof.NavierStokesGaugeY.genY_uField_perturbed_ne_zero
#check @BookProof.NavierStokesGaugeY.genY_genY_commute
#check @BookProof.NavierStokesGaugeY.genX_genY_commute
#check @BookProof.NavierStokesGaugeY.genY_nsSymbol
#check @BookProof.NavierStokesGaugeY.setYZero_uField
#check @BookProof.NavierStokesGaugeY.setYZero_nsSymbol
#check @BookProof.NavierStokesGaugeY.uFieldOp_apply_of_y_zero
#check @BookProof.NavierStokesGaugeY.hamiltonianOp_apply_of_y_zero
```

That generator compensates only the **first** derivatives, so the Laplacian modes $u_{i,jj}$ — the ones the viscous term is built from — are still spectators of the expansion. Carrying the Taylor expansion in the second coordinate one order further repairs this. The field is $u_i(y) = u_i + u_{i,j} y_j + \tfrac{1}{2} u_{i,jj} y_j^2$, its y-derivative is the **derivative field** $u_{i,j}(y) = u_{i,j} + u_{i,jj} y_j$, and the gauge generator becomes $G_j^2 = \partial/\partial y_j - u_{i,j}\, \partial/\partial u_i - u_{i,jj}\, \partial/\partial u_{i,j}$: it

shifts the velocity modes by their first derivatives **and** the first-derivative modes by the Laplacian modes. G_j^2 annihilates both fields, hence the symbol $A_i(y) = u_j(y)u_{i,j}(y) - \nu u_{i,jj}$ built from them, and the second-order generators still commute with each other and with the momenta, so the constraints remain first class. The two orders are genuinely different: the first-order generator leaves $u_{i,jj}\,y_j$ behind on the second-order field, the second-order generator does not annihilate the first-order field, and the Taylor coefficient $\frac{1}{2}$ is the only one that works. On the initial state $y = 0$ the second-order symbol collapses, once more, to the ordinary Navier–Stokes symbol $u_j u_{i,j} - \nu u_{i,jj}$.

```
#check @BookProof.NavierStokesGaugeY2.uField2
#check @BookProof.NavierStokesGaugeY2.uDField
#check @BookProof.NavierStokesGaugeY2.genY2
#check @BookProof.NavierStokesGaugeY2.uField2_pderiv_y
#check @BookProof.NavierStokesGaugeY2.uField2_pderiv_y_twice
#check @BookProof.NavierStokesGaugeY2.genY2_leibniz
#check @BookProof.NavierStokesGaugeY2.genY2_uField2
#check @BookProof.NavierStokesGaugeY2.genY2_uDField
#check @BookProof.NavierStokesGaugeY2.genY_uField2_ne_zero
#check @BookProof.NavierStokesGaugeY2.genY2_uField_ne_zero
#check @BookProof.NavierStokesGaugeY2.genY2_uField2_perturbed_ne_zero
#check @BookProof.NavierStokesGaugeY2.genY2_genY2_commute
#check @BookProof.NavierStokesGaugeY2.genX_genY2_commute
#check @BookProof.NavierStokesGaugeY2.genY_genY2_not_commute
#check @BookProof.NavierStokesGaugeY2.genY2_nsSymbol2
#check @BookProof.NavierStokesGaugeY2.genX_nsSymbol2
#check @BookProof.NavierStokesGaugeY2.setYZero_nsSymbol2
```

Both families of generators are abelian, which is the situation BRST was made for: one ghost χ_j per generator, carried on $\Lambda(\mathbb{C}^3)$ (`nsGhost_car`), no cubic ghost term, and the charge $\Omega = \sum_j G_j\chi_j$ (`nsDerivBrstCharge`), which squares to zero (`nsDerivBrstCharge_nilpotent`) without being the zero operator (`nsDerivBrstCharge_ne_zero`), so the statements below are not vacuous. The Hamiltonian must be built from the momentum conjugate to the **field**, not from $\partial/\partial x_i$: with $\pi^i = \partial/\partial u_i$ the momentum fails to commute with the symbol $A_i = u_j(y)u_{i,j} - \nu u_{i,jj}$ (`genU_nsSymbol`), so the symmetrised ordering $\pi^i A_i + A_i\pi^i$ carries real content and differs from the superseded version (`nsHamAlg_ne_nsHamAlgX`) — already on the constant state. The charge is odd and the Hamiltonian even, so the compatibility of the gauge fixing with the dynamics is the **commutator** $[\Omega, H] = 0$, and it holds (`nsDerivBrstCharge_comm_hamiltonian`): the Hamiltonian maps BRST-closed states to BRST-closed states and BRST-exact states to BRST-exact states (`nsHamiltonian_mapsTo_ker`, `nsHamiltonian_mapsTo_range`), hence descends to the cohomology (`nsBrstCohomologyMap`). The order of the generator genuinely matters — the first-order generator is not a symmetry of the second-order field (`genY_uField2_ne_zero`) — so the second-order charge is a separate statement, proved separately (`nsDerivBrstCharge2_comm_hamiltonian2`).

```
#check @BookProof.NsBrstDerivativeGauge.nsGaugeConstraint
#check @BookProof.NsBrstDerivativeGauge.nsGhost_car
#check @BookProof.NsBrstDerivativeGauge.nsDerivBrstCharge
#check @BookProof.NsBrstDerivativeGauge.nsDerivBrstCharge_nilpotent
#check @BookProof.NsBrstDerivativeGauge.nsDerivBrstCharge_ne_zero
#check @BookProof.NsBrstDerivativeGauge.nsGaugeConstraint_comm
#check @BookProof.NsBrstDerivativeGauge.genU_ccr_u
#check @BookProof.NsBrstDerivativeGauge.genU_nsSymbol
#check @BookProof.NsBrstDerivativeGauge.nsHamAlg_ne_nsHamAlgX
#check @BookProof.NsBrstDerivativeGauge.nsDerivBrstCharge_comm_hamiltonian
#check @BookProof.NsBrstDerivativeGauge.nsDerivBrstCharge2_comm_hamiltonian2
#check @BookProof.NsBrstDerivativeGauge.nsHamiltonian_mapsTo_ker
#check @BookProof.NsBrstDerivativeGauge.nsHamiltonian_mapsTo_range
#check @BookProof.NsBrstDerivativeGauge.nsBrstCohomologyMap
#check @BookProof.NavierStokesGaugeY2.genY_uField2_ne_zero
```

On a **finite truncation** — finitely many modes, each a Hermitian matrix, the field modes commuting as multiplication operators do — the whole claim is provable. The Hamiltonian is Hermitian (the anticommutator of two Hermitian factors is Hermitian), it is a polynomial of degree at most three in the generators, and therefore $U(t) = e^{\mathrm{i}tH_N}$ is a one-parameter unitary group defined for **every** real time: the flow is complete, it preserves the ℓ^2 mass, and every coefficient of the evolved state stays bounded by the initial mass — no finite-time singularity on the

truncation. The truncated generator also has vanishing deficiency, which on a finite-dimensional space is exactly essential self-adjointness.

```
#check @BookProof.NavierStokesFlow.NSTruncation
#check @BookProof.NavierStokesFlow.nsHamiltonian
#check @BookProof.NavierStokesFlow.nsHamiltonian_hermitian
#check @BookProof.NavierStokesFlow.nsHamiltonian_isPolynomial
#check @BookProof.NavierStokesFlow.nsWord_length_le_three
#check @BookProof.NavierStokesFlow.nsHamiltonian_ne_zero_example
#check @BookProof.NavierStokesFlow.nsFlow_zero
#check @BookProof.NavierStokesFlow.nsFlow_group
#check @BookProof.NavierStokesFlow.nsFlow_unitary
#check @BookProof.NavierStokesFlow.nsFlow_norm_preserving
#check @BookProof.NavierStokesFlow.nsFlow_noBlowup
#check @BookProof.NavierStokesFlow.nsFlow_groupOnEvolved
#check @BookProof.NavierStokesFlow.nsHamiltonian_hasZeroDeficiency
```

Completeness of the flow has a differential counterpart, which is the form in which the manuscript states existence and uniqueness. On the truncation the curve $\psi(t) = U(t)\psi$ is differentiable and solves $\dot{\psi}(t) = \mathrm{i}H_N\psi(t)$, and it is the **only** solution: any differentiable curve satisfying the same equation with the same initial value coincides with it at every time. So the Cauchy problem has exactly one global solution, defined for all $t \in \mathbb{R}$ and confined to the sphere of the initial mass. The proof of uniqueness is the classical one — the derivative of $t \mapsto U(-t)y(t)$ vanishes, so that curve is constant. Along the way the energy is conserved too: the expectation of H_N in the evolved state does not depend on time.

```
#check @BookProof.NavierStokesFlow.nsFlow_hasDerivAt
#check @BookProof.NavierStokesFlow.nsFlow_solves_schrodinger
#check @BookProof.NavierStokesFlow.nsFlow_unique_solution
#check @BookProof.NavierStokesFlow.nsCauchy_existsUnique
#check @BookProof.NavierStokesFlow.nsFlow_energy_conserved
#check @BookProof.NavierStokesFlow.nsFlow_continuous
```

The divergence constraint enters à la BRST. Its resolution is the book's own substitution $u_{3,3} = -(u_{1,1} + u_{2,2})$, and the truncated BRST charge $\Omega = u_{j,j} \otimes \psi^\dagger$ is nilpotent because the ghost creation operator is — the charge itself is not Hermitian, which is why the physical space is a cohomology and not an eigenspace. In parcel (Lagrangian) variables the constraint becomes volume preservation: a unit Jacobian determinant preserves the volume of every set, and its infinitesimal form is exactly $\nabla \cdot u = 0$, since the derivative of $t \mapsto \det(1 + tA)$ at $t = 0$ is $\operatorname{tr} A$. In those variables the advection term is a **positive** second-order operator (a sum of squares of the parcel momenta), the viscosity likewise, the force a first-order drift and the pressure the 0-order constraint.

```
#check @BookProof.NavierStokesFlow.nsDivergenceConstraint_resolution
#check @BookProof.NavierStokesFlow.nsBrstCharge
#check @BookProof.NavierStokesFlow.nsBrst_nilpotent
#check @BookProof.NavierStokesFlow.nsBrst_adjoint
#check @BookProof.NavierStokesEulerian.nsBrst_not_hermitian
#check @BookProof.NavierStokesEulerian.nsBrst_symmetrization_hermitian
#check @BookProof.NavierStokesFlow.lagrangian_velocity
#check @BookProof.NavierStokesFlow.volume_preservation_constraint
#check @BookProof.NavierStokesFlow.det_one_add_smul_hasDerivAt
#check @BookProof.NavierStokesFlow.LagrangianNS.transformed_hamiltonian_decompositior
#check @BookProof.NavierStokesFlow.LagrangianNS.kinetic_posSemidef
#check @BookProof.NavierStokesFlow.LagrangianNS.viscous_posSemidef
#check @BookProof.NavierStokesFlow.LagrangianNS.transformed_hamiltonian_hermitian
#check @BookProof.NavierStokesFlow.LagrangianNS.flowUnitary_unitary
#check @BookProof.NavierStokesFlow.LagrangianNS.flowUnitary_group
#check @BookProof.NavierStokesFlow.LagrangianNS.cauchy_existsUnique
```

The gauge-fixing machinery behind such charges has an abstract skeleton worth isolating on its own. Grade the fields by (form degree, ghost number): the physical scalar φ in $(0,0)$, the gauge field v in $(1,0)$, the ghost c in $(1,1)$, the anti-ghost $\bar{c}$ in $(1,-1)$ and the Nakanishi–Lautrup field B in $(1,0)$. The BRST operator s raises the ghost number and is nilpotent, and $sv = c$, $s\bar{c} = B$, $s\varphi = 0$ make (v, c) and $(\bar{c}, B)$ **contractible pairs**, which is the formal content of "the gauge field and the ghosts decouple". Since a product is now involved, s acts as an odd derivation, picking up a sign across the anti-ghost. The Gauge-Fixing Fermion $\Psi = \bar{c}\,(v - d\varphi)$, of bidegree

$(2,-1)$, then has BRST variation $s\Psi = B\,(v - d\varphi) - \bar{c}\,c$: BRST exactness **generates** the Lagrange multiplier that enforces $v = d\varphi$ together with a ghost term carrying no momentum. Nilpotency gives $s(s\Psi) = 0$, so the gauge-fixing Lagrangian is BRST-invariant, and under any evaluation that annihilates exact terms it contributes nothing to observables. The axioms are not vacuous: an explicit 2×2 matrix model realizes them with non-zero ghost, auxiliary field, Fermion and Lagrangian.

```
#check @BookProof.GaugeFixing.GaugeFixingSystem
#check @BookProof.GaugeFixing.Psi
#check @BookProof.GaugeFixing.s_c_eq_zero
#check @BookProof.GaugeFixing.s_B_eq_zero
#check @BookProof.GaugeFixing.s_gaugeField
#check @BookProof.GaugeFixing.L_gf_evaluation
#check @BookProof.GaugeFixing.L_gf_invariant
#check @BookProof.GaugeFixing.int_L_gf_eq_zero
#check @BookProof.GaugeFixing.matrixModel
#check @BookProof.GaugeFixing.matrixModel_s_Psi_ne_zero
```

Three ways to relate a field to its derivatives. The relation "v is the spatial derivative of φ" can be imposed in (at least) three distinct ways, and the difference between them is exactly the difference between a **physical constraint** and a **gauge-fixing condition** — the choice the skeleton above has to make. The three are:

- Method A — the topological (gauge-invariant) constraint. One introduces a symmetry that shifts both fields at once, $\delta\varphi = \varepsilon$ and $\delta v = d\varepsilon$, so that the combination $v - d\varphi$ is **gauge-invariant** ($\delta(v - d\varphi) = 0$) and hence a candidate physical observable. Promoting the parameter to a ghost, $s\varphi = c$, $sv = dc$, $sc = 0$ is nilpotent, and the constraint is BRST-closed but not BRST-exact: it sits in the BRST cohomology. That is the correct structure when $v = d\varphi$ is a genuinely topological relation — a Bogomol'nyi equation, a field-strength relation $F = dA$ — a piece of physics one wants to keep. It is the wrong structure for the Navier–Stokes thread, where the derivative modes are redundant canonical coordinates one wants to eliminate, not promote to observables.
- Method B — the gauge-fixing condition (the one chosen here). Here v starts as an arbitrary unconstrained field with its own shift symmetry, $\delta v = \varepsilon$, $\delta\varphi = 0$, and the relation $v = d\varphi$ is a condition one imposes by gauge fixing. The Gauge-Fixing Fermion $\Psi = \bar{c}\,(v - d\varphi)$ is chosen so that its BRST variation $s\Psi = B(v - d\varphi) - \bar{c}\,c$ is a BRST-exact term: the Lagrange multiplier B enforces $v = d\varphi$ as a delta function in the path integral, and the ghost term carries no derivatives, so the ghosts decouple. Being BRST-exact, the whole term integrates to zero against any evaluation functional that annihilates exact terms (`int_L_gf_eq_zero`) — it contributes nothing to physical observables, which is precisely what makes it a pure gauge fixing rather than new physics. This is the method formalized in `BookProof/Qym/ChapterGaugeFixing.lean`.
- A complex gauge field (a flat-connection reading of the same content). One might instead replace the derivative $\partial_\mu u$ in the Hamiltonian by $A_\mu u$ for a complex gauge field A_μ (a complexified vector potential), fix the magnetic field to zero as an initial condition, and impose covariant constancy $D_\mu u = \partial_\mu u + A_\mu u = 0$ to fix the field derivatives. This is geometrically elegant — the derivative field becomes the connection coefficient of a flat line bundle — but it carries no new content: wherever $u \neq 0$ the constraint forces $A_\mu = -\partial_\mu \log u$, a pure gauge with curvature automatically zero, so A is the derivative field in disguise and the "magnetic field is zero" is the flatness consistency, not a dynamical condition. It also costs the structure the proofs here depend on: a gauge field is not a constant of the motion (it is coupled to u through the covariant constraint), so the block decomposition over the derivative spectrum that drives the ESA chain collapses; and a complex coefficient breaks the real Hermitian form of the Navier–Stokes Hamiltonian, re-opening the symmetrization question. It is kept here only as a geometric interpretation of what the formalized gauge fixing does, not as an alternative implementation.

Why Method B is chosen. The task in the Navier–Stokes thread is to **remove** the derivative coordinates, not to make them physical: the Eulerian derivatives-as-fields construction promotes $u_{i,j}$ and $u_{i,jj}$ to independent canonical coordinates purely so the Hamiltonian carries momenta only for the velocity modes — which is what makes the derivative field a constant of the motion and the block decomposition possible. Method B is the instrument that completes that construction: it eliminates the redundant coordinates by a BRST-exact gauge fixing whose ghosts decouple and whose Lagrange multiplier enforces the constraint as a strict delta function, leaving the physical cohomology untouched. Method A would invert the logic (it would make $v - d\varphi$ a physical

observable), and the flat-connection reading, while insightful, is the same logic repackaged at the cost of a new gauge redundancy and a complexified Hermitian structure. Method B is also the formalization-friendly choice: one graded Leibniz rule, no analysis, a purely algebraic skeleton (plan item E.6, `BookProof/Qym/ChapterGaugeFixing.lean`), and it is the gauge fixing the A.6/A.7 genY/genY2 generators perform when they tie the derivative fields to the velocity field — the step on which the Hashimoto shift-invert selection for the Navier–Stokes Hamiltonian (§9 item 8 of `CONSOLIDATED_PLAN.md`) is to be built.

Because that transformed operator is Hermitian on the truncation, everything said about the flow of H_N holds for it verbatim: $e^{\mathrm{i}t\hat{h}}$ is a one-parameter unitary group and the corresponding Cauchy problem has exactly one global solution. That is the truncated form of the Lagrangian route — the continuum statement remains unclaimed.

The honest boundary. Nothing above claims essential self-adjointness of the **untruncated continuum** operator — and with it, nothing claims global existence/uniqueness for the **classical** Navier–Stokes equations (the Clay regularity problem, deliberately out of scope). The distinction matters: once ESA **is** proved, global existence of the operator flow follows automatically — Stone's theorem (a self-adjoint operator generates a complete unitary group for every real time) is itself proved in this project (2026-08-20e, `ChapterStoneResolvent`–`ChapterStoneSeparable`) — and that is exactly what `book.tex` §4210-4216 means by "the solution ... exists and it is unique", and the truncation already proves it as `nsCauchy_existsUnique`. The open step is ESA itself for the continuum operator, not a separate global-existence claim. The ODE chapter is the standing warning: for $\dot{x} = x^2$ the Hamiltonian $x^2\hat{p} - \mathrm{i}\hat{x}$ is a polynomial of degree three whose classical flow is incomplete, so "low degree in the fields" cannot by itself give self-adjointness — it gives symmetry. The continuum route that remains is the Faris–Lavine commutator criterion applied to the Lagrangian-transformed operator, whose second-quantized form on the Fock-of-a-Fock space is at most **quadratic** in the outer ladder operators, and whose comparison operator is the (positive) outer number operator. That criterion is recorded as a named hypothesis, never an axiom, exactly as Crouzeix's inequality is in `ChapterH4`; its two analytic inequalities for the continuum operator are a research target, not a result.

```
#check @BookProof.NavierStokesFlow.nsFockOfFock
#check @BookProof.NavierStokesFlow.nsSecondQuant
#check @BookProof.NavierStokesFlow.ns_outer_degree_le_two
#check @BookProof.NavierStokesFlow.nsNumberOp
#check @BookProof.NavierStokesFlow.nsNumberOp_eq_secondQuant
#check @BookProof.NavierStokesFlow.nsNumberOp_posSemidef
#check @BookProof.NavierStokesFlow.HasZeroDeficiency
#check @BookProof.NavierStokesFlow.symmetric_hasZeroDeficiency
#check @BookProof.NavierStokesFlow.ns_esa_of_farisLavine
```

The named hypothesis has to be the **honest** criterion. Stated without the symmetry of the operator it is contradictory — $H = \mathrm{i}\cdot 1$ with comparison operator the identity satisfies both inequalities while having a full deficiency space — so a conditional theorem resting on it would be vacuous. With symmetry the criterion is satisfiable, and for operators defined on the whole space it is automatic: symmetry alone already forces the deficiency to vanish. The analytic content of Faris–Lavine therefore lives entirely in the densely defined case, where the deficiency spaces are those of the **adjoint**; that is the predicate under which the conditional theorem is restated, and where the truncation — whose domain is the whole space — is the only case proved here.

```
#check @BookProof.NavierStokesFlow.farisLavine_without_symmetry_forces_trivial
#check @BookProof.NavierStokesFlow.farisLavine_holds_of_everywhereDefined
#check @BookProof.NavierStokesFlow.HasZeroDeficiencyOn
#check @BookProof.NavierStokesFlow.hasZeroDeficiencyOn_top_of_symmetric
#check @BookProof.NavierStokesFlow.nsHamiltonian_hasZeroDeficiencyOn
#check @BookProof.NavierStokesFlow.ns_esa_of_farisLavine_dense
```

There is one criterion under which the passage from the truncation to an infinite-dimensional operator is not a change of kind: **a complete flow**. If a symmetric operator on a dense domain generates a norm-preserving flow that is defined for **every** real time and leaves the domain invariant, then the deficiency spaces of its adjoint vanish — the operator is essentially self-adjoint. The proof is the classical orbit argument: a deficiency vector w with $H^*w = \pm\mathrm{i}w$ makes the orbit function $g(t) = \langle w, U(t)v\rangle$ satisfy $g' = \pm g$, so $g(t) = g(0)e^{\pm t}$, while unitarity of the flow keeps $|g|$ bounded on the whole line; hence $g(0) = \langle w, v\rangle = 0$ for every v in the dense domain, and $w = 0$. Completeness is exactly the hypothesis that a finite-time blow-up would destroy, which is why the ODE chapter's $\dot{x} = x^2$ example is the standing warning for the continuum case.

```
#check @BookProof.NavierStokesFlow.eq_zero_of_hasDerivAt_smul_of_bounded
#check @BookProof.NavierStokesFlow.hasZeroDeficiencyOn_of_completeUnitaryFlow
#check @BookProof.NavierStokesFlow.nsHamiltonian_hasZeroDeficiencyOn_of_flow
```

The criterion is applied where its hypotheses are actually available: to the truncation, whose flow is complete by construction. Nothing is claimed for the untruncated generator, whose flow completeness is the open problem.

A second, genuinely infinite-dimensional case is available on the bounded $\ell^2(\mathbb{Z})$ layer: a bounded symmetric operator is essentially self-adjoint on **every** dense invariant domain, and the Weyl-symmetrized continuity generator $H = \frac{1}{2}(pv + vp)$ leaves the domain of finitely supported modes invariant. That domain is dense but is **not** the whole space, so this is the first instance in the development of vanishing adjoint deficiency on a proper dense domain — the setting in which the notion has analytic content.

```
#check @BookProof.NavierStokesFlow.hasZeroDeficiencyOn_of_bounded_symmetric
#check @BookProof.NavierStokesFlow.finiteModes
#check @BookProof.NavierStokesFlow.finiteModes_dense
#check @BookProof.NavierStokesFlow.finiteModes_ne_top
#check @BookProof.NavierStokesFlow.continuityHamiltonian_hasZeroDeficiencyOn_finiteMc
```

Both criteria are needed, because symmetry on a dense domain is by itself never enough. On $\ell^2(\mathbb{N})$, with the finitely supported states as domain, the tridiagonal operator $(Hf)(0) = a_0 f(1)$, $(Hf)(n+1) = a_n f(n) + a_{n+1} f(n+2)$ with real weights $a_0 = 2$, $a_{n+1} = 4a_n + 2$ is symmetric on that dense domain, yet the square-summable geometric sequence $w(n) = (\mathrm{i}/2)^n$ satisfies $Hw = \mathrm{i}w$ coefficientwise and is therefore a deficiency vector of the adjoint: essential self-adjointness **fails**. The one computation behind both statements is the discrete Green identity — the truncated sum of Wronskian increments telescopes to a boundary term, which vanishes for finitely supported states.

So the "polynomial of low degree in the fields" hypothesis, which gives symmetry on the finite-particle domain, cannot by itself yield essential self-adjointness; an analytic input — a complete flow, boundedness, or the Faris–Lavine inequalities — is required.

```
#check @BookProof.NavierStokesFlow.JacobiDeficiency.jacobi_wronskian
#check @BookProof.NavierStokesFlow.JacobiDeficiency.jacobiOp_symmetric
#check @BookProof.NavierStokesFlow.JacobiDeficiency.defState_deficiency
#check @BookProof.NavierStokesFlow.JacobiDeficiency.jacobiOp_not_hasZeroDeficiencyOn
#check @BookProof.NavierStokesFlow.JacobiDeficiency.jacobi_symmetric_dense_not_esa
```

Unboundedness by itself is not the obstruction either. A symmetric operator that carries a **total** family of eigenvectors inside its domain — only the zero vector is orthogonal to all of them — is essentially self-adjoint, because testing the deficiency identity against an eigenvector e with real eigenvalue λ gives $(\lambda \mp \mathrm{i})\langle e, w\rangle = 0$. A diagonal operator on $\ell^2(\mathbb{N})$ with an arbitrary, possibly unbounded, real sequence of entries satisfies this on the same finitely supported domain on which the tridiagonal example fails. What separates the two is whether the domain carries enough eigenvectors. That diagonal operator is genuinely unbounded whenever its sequence of entries is: no constant dominates it on the finite-mode domain, so essential self-adjointness there is not a boundedness phenomenon in disguise.

```
#check @BookProof.NavierStokesFlow.hasZeroDeficiencyOn_of_total_eigenvectors
#check @BookProof.NavierStokesFlow.DiagonalEsa.diagOp
#check @BookProof.NavierStokesFlow.DiagonalEsa.diagOp_basis
#check @BookProof.NavierStokesFlow.DiagonalEsa.diagOp_hasZeroDeficiencyOn
#check @BookProof.NavierStokesFlow.DiagonalEsa.diagOp_not_bounded
```

The Lagrangian change of variables is what turns these criteria into a statement about the Navier–Stokes generator itself. Writing the flow in parcel coordinates — the trajectory operators $X(\xi)$ with conjugate momenta $P(\xi) = \dot{X}(\xi) = u(X(\xi))$ in place of the Eulerian velocity — the operator becomes $\hat{h}_{\mathrm{full}} = \frac{1}{2}\sum_i P_i^2 + \nu \sum_i Q_i^2 + \sum_i f_i D_i + C$: two second-order terms, a first-order force drift and the zeroth-order volume-preservation constraint. Untruncated — all of these are (possibly unbounded) operators on a dense domain of an arbitrary complex inner-product space — the transformed operator is symmetric, and the quadratic forms of its two second-order terms are $\frac{1}{2}\sum_i \|P_i v\|^2$ and $\nu \sum_i \|Q_i v\|^2$: the advection term, which in Eulerian variables is the troublesome $-u_j \partial_j u_i$, has become **positive**.

```
#check @BookProof.NavierStokesFlow.LagrangianEsa.LagrangianFullData
#check @BookProof.NavierStokesFlow.LagrangianEsa.LagrangianFullData.hFull
#check @BookProof.NavierStokesFlow.LagrangianEsa.LagrangianFullData.hFull_isSymmetric
#check @BookProof.NavierStokesFlow.LagrangianEsa.LagrangianFullData.kinetic_inner
#check @BookProof.NavierStokesFlow.LagrangianEsa.LagrangianFullData.kinetic_nonneg
#check @BookProof.NavierStokesFlow.LagrangianEsa.LagrangianFullData.viscous_nonneg
```

Essential self-adjointness then follows in the transformed variables. A total family of common eigenvectors of the constituents — the Lagrangian momentum representation — makes the full transformed Hamiltonian essentially self-adjoint, with eigenvalue $\frac{1}{2}\sum_i p_i^2 + \nu\sum_i q_i^2 + \sum_i f_i d_i + c$; and because vanishing adjoint deficiency is invariant under a unitary change of variables, what is proved after the change of variables holds for the operator it came from. Two untruncated realizations make this unconditional: on $\ell^2(\mathbb{Z})$, with the parcel momenta the symmetric-difference lattice momentum — so $\frac{1}{2}\sum_i P_i^2$ is a discrete Laplacian — the transformed Hamiltonian is essentially self-adjoint on the **proper** dense domain of finitely supported modes and is not the zero operator; on $\ell^2(\mathbb{N})$, with diagonal constituents of arbitrary real symbols, it is essentially self-adjoint while being genuinely unbounded. The limits are equally explicit: an unbounded first-order drift term alone already destroys the property, so no criterion-free statement about the transformed data is available, and essential self-adjointness of the **continuum** transformed generator — the operator-flow global existence would follow from ESA by the Stone theorem of 2026-08-20e, but the classical global existence for Navier–Stokes would not — is not claimed here.

```
#check @BookProof.NavierStokesFlow.LagrangianEsa.LagrangianFullData.hasZeroDeficiency
#check @BookProof.NavierStokesFlow.LagrangianEsa.hasZeroDeficiencyOn_iff_of_linearIsc
#check @BookProof.NavierStokesFlow.LagrangianEsa.NSFullData.hasZeroDeficiencyOn_of_la
#check @BookProof.NavierStokesFlow.LagrangianEsa.latticeLag_hasZeroDeficiencyOn
#check @BookProof.NavierStokesFlow.LagrangianEsa.latticeLag_hFull_ne_zero
#check @BookProof.NavierStokesFlow.LagrangianEsa.diagLag_hasZeroDeficiencyOn
#check @BookProof.NavierStokesFlow.LagrangianEsa.diagLag_not_bounded
#check @BookProof.NavierStokesFlow.LagrangianEsa.exists_lagrangianFullData_not_hasZer
```

The last step of the chain is the passage from one particle to the Fock space, and it is where the comparison operator of the Faris–Lavine criterion has to be rebuilt. Fock space here is the ℓ^2 direct sum of the particle sectors, and the finite-particle domain is the span of states supported in finitely many sectors with each component in that sector's core. That domain is dense as soon as every sector core is, and it is a **proper** subspace of the Fock space, so nothing below is a statement about a boundedly extended operator. The second quantization $d\Gamma(A)$ acts sector by sector, and vanishing adjoint deficiency lifts: if every sector operator has it on its own core, so does $d\Gamma(A)$ on the finite-particle domain. Applied to a one-particle comparison operator this is the "cage" the criterion needs.

```
#check @BookProof.NavierStokesFlow.SecondQuant.fockCore
#check @BookProof.NavierStokesFlow.SecondQuant.fockCore_dense
#check @BookProof.NavierStokesFlow.SecondQuant.fockOp
#check @BookProof.NavierStokesFlow.SecondQuant.fockOp_hasZeroDeficiencyOn
#check @BookProof.NavierStokesFlow.SecondQuant.fockCore_ne_top
```

The one-particle comparison operator is $n = \sum_i \pi_i^2 + \sum_i V_i^2 + I$, with the π_i the fiber momenta and the V_i the fiber realization of the advection field, all symmetric on a common dense domain. Because both families enter squared, the quadratic form of n is $\sum_i \|\pi_i v\|^2 + \sum_i \|V_i v\|^2 + \|v\|^2$, so $n \geq I$ — the strict positivity that a comparison operator must have. In the momentum representation, where the constituents are diagonal with arbitrary real symbols, n is essentially self-adjoint on the finite-mode domain and is genuinely unbounded whenever its symbols are; second-quantizing that realization gives a Fock-space comparison operator that is essentially self-adjoint on the finite-particle domain, and it inherits $\hat{\mathrm{N}} \geq \mathrm{I}$ from the sectors.

```
#check @BookProof.NavierStokesFlow.FarisLavineLift.ComparisonData
#check @BookProof.NavierStokesFlow.FarisLavineLift.ComparisonData.comparison
#check @BookProof.NavierStokesFlow.FarisLavineLift.ComparisonData.comparison_ge_norm_
#check @BookProof.NavierStokesFlow.FarisLavineLift.diagComparison_hasZeroDeficiencyOr
#check @BookProof.NavierStokesFlow.FarisLavineLift.diagComparison_not_bounded
#check @BookProof.NavierStokesFlow.SecondQuant.fockComparison_hasZeroDeficiencyOn
#check @BookProof.NavierStokesFlow.SecondQuant.fockOp_ge_norm_sq
#check @BookProof.NavierStokesFlow.SecondQuant.fockComparison_ge_norm_sq
#check @BookProof.NavierStokesFlow.SecondQuant.fockComparison_domain_ne_top
```

The two Faris–Lavine bounds then have to survive the lift, and here the two halves behave differently. The form-commutator bound does lift as expected: the commutator of the sums is the sum of the commutators once distinct particles commute, and summing $|\langle v, [h_k, n_k]v\rangle| \le c_2\langle v, n_k v\rangle$ over the particles gives the same bound for the totals — stated with a modulus on the left, since for symmetric constituents the commutator expectation is purely imaginary and its real part carries no information. The operator bound does **not** lift by the triangle inequality alone: the informal chain $\|\sum_k h_k v\| \le \sum_k \|h_k v\| \le c_1 \sum_k \|n_k v\| \le c_1 \|\sum_k n_k v\|$ uses a last step that is false, and two two-by-two operator pairs with $\|h_k x\| \le \|n_k x\|$ for every x but $\|(n_0 + n_1)v\| < \|(h_0 + h_1)v\|$ witness the failure. What does lift is the pairwise-dominated form of the hypothesis, and with the two bounds in that shape the Faris–Lavine criterion — taken as a named hypothesis, not an axiom — yields vanishing adjoint deficiency for the second-quantized Hamiltonian on the finite-particle domain. Essential self-adjointness of the **continuum** Navier–Stokes Hamiltonian is not claimed; the operator-flow global existence would follow from it by the Stone theorem of 2026-08-20e, while classical global existence is a separate, deliberately out-of-scope statement.

```
#check @BookProof.NavierStokesFlow.FarisLavineLift.norm_sum_le_of_pairwise
#check @BookProof.NavierStokesFlow.FarisLavineLift.not_forall_norm_sum_le_of_pointwis
#check @BookProof.NavierStokesFlow.FarisLavineLift.norm_inner_commutator_sum_le
#check @BookProof.NavierStokesFlow.SecondQuant.fockOp_norm_le_of_sectors
#check @BookProof.NavierStokesFlow.SecondQuant.fockOp_norm_inner_le_of_sectors
#check @BookProof.NavierStokesFlow.SecondQuant.fockOp_hasZeroDeficiencyOn_of_farisLav
```

That last statement still carries the criterion as a hypothesis, and in the shape it is stated there — a relative bound plus a commutator bound, with no positivity of N and no surjectivity of $N + 1$ — the criterion is in fact refutable, by the limit-circle Jacobi operator. The remedy is to supply exactly the input the Faris–Lavine proof uses, and this the momentum representation does provide. Take the comparison operator on its **maximal** domain in ℓ^2: multiplication by the symbol σ, defined on every state whose product with σ is again square summable. For a non-negative symbol that operator is symmetric, positive, $N + 1$ maps the maximal domain onto the whole space because $(\sigma + 1)^{-1}$ is a contraction, and the finite-mode states — the momentum-space stand-in for C_c^∞ — are an operator core, since the truncations of a state converge to it in the graph norm. That package is the Ikebe–Kato-type input, and it is proved, not assumed: for a non-negative symbol, multiplication by σ is essentially self-adjoint already on the finite-mode core, and self-adjoint on its maximal domain.

```
#check @BookProof.NavierStokesFlow.IkebeKato.maxDom
#check @BookProof.NavierStokesFlow.IkebeKato.diagMax
#check @BookProof.NavierStokesFlow.IkebeKato.diagMax_quadForm_ge_norm_sq
#check @BookProof.NavierStokesFlow.IkebeKato.diagMax_add_one_surjective
#check @BookProof.NavierStokesFlow.IkebeKato.exists_finiteModes_graph_approx
#check @BookProof.NavierStokesFlow.IkebeKato.diagMax_essentiallySelfAdjointOn
#check @BookProof.NavierStokesFlow.IkebeKato.ikebeKato_momentum
```

With that input the Faris–Lavine theorem of the criterion chapter applies with no hypothesis left over: any symmetric Hamiltonian on the maximal domain which is relatively bounded by N and whose form commutator with N is dominated by N is essentially self-adjoint on the finite-mode core. Specialised to the Navier–Stokes symbol $\sigma(k) = \sum_i p_i(k)^2 + \sum_i q_i(k)^2 + 1$ this is the one-particle statement, and the restriction of the maximal-domain operator to the core is literally the comparison operator $n = \sum_i \pi_i^2 + \sum_i V_i^2 + I$ of the fiber. In the occupation-number representation the bosonic Fock space over the fiber is ℓ^2 over the configurations $\alpha : \mathbb{N} \to_f \mathbb{N}$ and $\hat{N} = d\Gamma(n) + I$ is again a multiplication operator, by the total energy $\Sigma(\alpha) = \sum_k \alpha(k)\, n(k) + 1$; so the same theorem gives essential self-adjointness of the second-quantized Navier–Stokes Hamiltonian on the finite-particle, finite-mode core. What remains hypothetical are exactly the two Faris–Lavine inequalities for the Hamiltonian itself; they are not vacuous — an unbounded Hamiltonian $N + B$ with a rank-two perturbation satisfies both while its commutator with N is non-zero. Essential self-adjointness of the **continuum** generator, and global existence, are still not claimed.

```
#check @BookProof.NavierStokesFlow.IkebeKato.essentiallySelfAdjointOn_finiteModes_of_
#check @BookProof.NavierStokesFlow.MomentumEsa.nsComparison_restrict_eq
#check @BookProof.NavierStokesFlow.MomentumEsa.nsComparison_ikebeKato
#check @BookProof.NavierStokesFlow.MomentumEsa.ns_hamiltonian_essentiallySelfAdjointO
#check @BookProof.NavierStokesFlow.MomentumEsa.fockSymbol
#check @BookProof.NavierStokesFlow.MomentumEsa.fockSymbol_add
#check @BookProof.NavierStokesFlow.MomentumEsa.fockComparison_ikebeKato
#check @BookProof.NavierStokesFlow.MomentumEsa.navierStokes_fock_hamiltonian_essentia
```

```
#check @BookProof.NavierStokesFlow.MomentumPerturbation.pertHam_essentiallySelfAdjoir
#check @BookProof.NavierStokesFlow.MomentumPerturbation.pertHam_not_bounded
#check @BookProof.NavierStokesFlow.MomentumPerturbation.exists_commForm_ne_zero
```

The quadratic Navier–Stokes symbol is a **direct sum** of linear ones. The Hamiltonian carries momenta only for the velocity modes, so the derivative modes $u_{i,j}$ commute with it: they are constants of the motion. Diagonalising the derivative field splits the Hilbert space as $\ell^2(\mathbb{N} \times J)$ — the Hermite levels of the velocity fiber times the spectrum J of the derivative field — and in the block j the bilinear symbol $A = u_{,1}\, u$ becomes the **linear** advection field $V(u) = \kappa_j u$, whose fiber Hamiltonian is exactly the one for which the two Faris–Lavine inequalities are proved. The strain rates κ_j range over that spectrum and are in general unbounded, so no single pair of Faris–Lavine constants can serve the whole operator; the deficiency problem, however, decomposes over the blocks, because a deficiency vector restricts to one in each block. Hence the bilinear (quadratic-symbol) Navier–Stokes Hamiltonian is symmetric and essentially self-adjoint on the finite-mode core for an arbitrary family of strain rates, and it is genuinely unbounded whenever they are. The viscous term and the cross terms add a constant to the fiber field — an affine $V(u) = \kappa_j u + c_j$, a ± 1 shift on top of the ± 2 shift.

```
#check @BookProof.NavierStokesFlow.BilinearEsa.bilH
#check @BookProof.NavierStokesFlow.BilinearEsa.bilFun_embFun
#check @BookProof.NavierStokesFlow.BilinearEsa.hasSum_inner_blocks
#check @BookProof.NavierStokesFlow.BilinearEsa.blockVec_bilH
#check @BookProof.NavierStokesFlow.BilinearEsa.bilH_symmetricOn
#check @BookProof.NavierStokesFlow.BilinearEsa.deficiencyTrivialAt_bilH
#check @BookProof.NavierStokesFlow.BilinearEsa.bilH_essentiallySelfAdjointOn_core
#check @BookProof.NavierStokesFlow.BilinearEsa.bilH_ne_zero
#check @BookProof.NavierStokesFlow.BilinearEsa.bilH_domain_dense
#check @BookProof.NavierStokesFlow.BilinearEsa.bilH_not_bounded
```

That last shift is supplied by summing two hopping operators against one comparison operator. The Faris–Lavine hypotheses used here — symmetry, a relative bound $\|Hx\|^2 \le a\|Nx\|^2 + b\|x\|^2$ with no smallness required of a, and a commutator bound $|\langle x, i[H,N]x\rangle| \le c\,\langle x, Nx\rangle$ — are all stable under sums, because $\|(H_1 + H_2)x\|^2 \le 2\|H_1 x\|^2 + 2\|H_2 x\|^2$ and the commutator form is additive in H. Taking the number operator $N = \mu(2n+1) + 1$ with $\mu = \kappa + c + 1$ as the common comparison operator, the ± 2 hopping $(\kappa/2)\sqrt{(n+1)(n+2)}$ of $\kappa\,\frac{1}{2}(\pi u + u\pi)$ and the ± 1 hopping $(c/\sqrt{2})\sqrt{n+1}$ of $c\,\pi$ are both dominated by it, so the affine fiber Hamiltonian is symmetric and essentially self-adjoint on the finite-mode core, and both hoppings are genuinely present. Running the block decomposition again with the affine fiber Hamiltonian in place of the linear one covers the viscous term and the cross terms. Assumed throughout: $\kappa_j \ge 0$ and $c_j \ge 0$, the latter only because a hopping amplitude is required to be non-negative. Only one velocity component is carried, and nothing here claims global regularity for the classical Navier–Stokes equation.

```
#check @BookProof.NavierStokesFlow.AffineFiber.PairShift
#check @BookProof.NavierStokesFlow.AffineFiber.PairShift.pairH
#check @BookProof.NavierStokesFlow.AffineFiber.PairShift.pairH_relative_bound
#check @BookProof.NavierStokesFlow.AffineFiber.PairShift.pairH_commForm_bound
#check @BookProof.NavierStokesFlow.AffineFiber.PairShift.pairH_essentiallySelfAdjoint
#check @BookProof.NavierStokesFlow.AffineFiber.affH
#check @BookProof.NavierStokesFlow.AffineFiber.affH_symmetricOn
#check @BookProof.NavierStokesFlow.AffineFiber.affH_essentiallySelfAdjointOn_core
#check @BookProof.NavierStokesFlow.AffineFiber.affH_coord_succ
#check @BookProof.NavierStokesFlow.AffineFiber.affH_coord_succ_succ
#check @BookProof.NavierStokesFlow.AffineFiber.affH_not_bounded
#check @BookProof.NavierStokesFlow.AffineBlock.affBlockH
#check @BookProof.NavierStokesFlow.AffineBlock.blockVec_affBlockH
#check @BookProof.NavierStokesFlow.AffineBlock.affBlockH_symmetricOn
#check @BookProof.NavierStokesFlow.AffineBlock.affBlockH_essentiallySelfAdjointOn_cor
#check @BookProof.NavierStokesFlow.AffineBlock.affBlockH_not_bounded
```

Both restrictions just recorded are artefacts of the bookkeeping, and both are now removed. The sign condition goes first: essential self-adjointness is a **unitary invariant**, so if a unitary U of the ambient space preserves the core and intertwines H with H', one is essentially self-adjoint exactly when the other is. The sign-flip $(Ux)_n = (-1)^n x_n$ is such a unitary; it reverses the sign of a ± 1 hopping and preserves a ± 2 hopping, so it conjugates the

affine fiber Hamiltonian with constant c into the one with constant $|c|$. Hence the fiber and the block Hamiltonians are essentially self-adjoint for a fiber constant of **arbitrary sign**.

```
#check @BookProof.NavierStokesFlow.SignFlip.flipU
#check @BookProof.NavierStokesFlow.SignFlip.essentiallySelfAdjointOn_of_intertwine
#check @BookProof.NavierStokesFlow.SignFlip.shiftH_flip
#check @BookProof.NavierStokesFlow.SignFlip.saffH
#check @BookProof.NavierStokesFlow.SignFlip.saffH_symmetricOn
#check @BookProof.NavierStokesFlow.SignFlip.saffH_essentiallySelfAdjointOn_core
#check @BookProof.NavierStokesFlow.SignFlip.sblockH_essentiallySelfAdjointOn_core
```

The one-component restriction goes next. At one fiber the velocity is the vector $u = (u_1, u_2, u_3)$, the Hermite basis is indexed by $\beta \in \mathbb{N}^3$, and the fiber fields are the affine $V_i(u) = \sum_k A_{ik} u_k + c_i$ for an **arbitrary** real 3×3 matrix A — the velocity gradient at the fiber, with no symmetry, positivity or sign assumption — and an arbitrary real vector c. The Weyl ordering $H = \sum_i \frac{1}{2}(\pi_i V_i + V_i \pi_i)$ becomes twenty-four hopping terms: a ± 2 hopping per component, a ± 1 hopping per fiber constant, and, for each pair of distinct components, a double-raising **strain** hopping of amplitude $\frac{1}{2}(A_{ik} + A_{ki})\sqrt{(\beta_i + 1)(\beta_k + 1)}$ and a number-conserving **vorticity** hopping of amplitude $\frac{1}{2}(A_{ik} - A_{ki})\sqrt{(\beta_i + 1)\beta_k}$. The vorticity amplitude increases in one coordinate and decreases in the other, so it is not monotone along its shift, and the strain rates and the constants have arbitrary signs. The instrument that copes with all of this is a hopping term whose amplitude is an arbitrary real function dominated by $\frac{1}{4}\sigma + K$: the estimates never need the amplitude itself, only a majorant that is non-negative, monotone along the shift and dominated by the comparison symbol. A finite family of such terms sharing one comparison symbol again sums to an operator that is symmetric and essentially self-adjoint on the finite-mode core. Applied to the twenty-four terms above with $N = \mu(2|\beta| + 3) + 1$, this gives the coupled three-component fiber Hamiltonian, essentially self-adjoint on the finite-mode core of $\ell^2(\mathbb{N}^3)$ for every real A and every real c; the strain and vorticity matrix entries are computed, so the coupling is genuinely there, and the operator is unbounded. The statement is proved in the Hermite (sequence-space) realization of the fiber; the next paragraph reads the same operator back in canonical form. Nothing here claims global regularity for the classical Navier–Stokes equation.

```
#check @BookProof.NavierStokesFlow.SignedShift.SignedHop
#check @BookProof.NavierStokesFlow.SignedShift.SignedHop.hopH_relative_bound
#check @BookProof.NavierStokesFlow.SignedShift.SignedHop.hopH_commForm_bound
#check @BookProof.NavierStokesFlow.SignedShift.listH_essentiallySelfAdjointOn_core
#check @BookProof.NavierStokesFlow.SignedShift.gaffH
#check @BookProof.NavierStokesFlow.SignedShift.gaffH_essentiallySelfAdjointOn_core
#check @BookProof.NavierStokesFlow.ThreeComponent.velH
#check @BookProof.NavierStokesFlow.ThreeComponent.velH_symmetricOn
#check @BookProof.NavierStokesFlow.ThreeComponent.velH_essentiallySelfAdjointOn_core
#check @BookProof.NavierStokesFlow.ThreeComponent.velH_coord_pair
#check @BookProof.NavierStokesFlow.ThreeComponent.velH_coord_rot
#check @BookProof.NavierStokesFlow.ThreeComponent.velH_not_bounded
```

The hopping description is a matrix, and a matrix is not obviously a Hamiltonian. What closes the gap is to build the canonical pairs **inside** the sequence space and check that the matrix is the Weyl-ordered expression one started from. On the finite-mode core of $\ell^2(\mathbb{N}^3)$ the three ladder pairs are the coordinate shifts $(a_i X)(\beta) = \sqrt{\beta_i + 1}\, X(\beta + e_i)$ and $(a_i^\dagger X)(\beta) = \sqrt{\beta_i}\, X(\beta - e_i)$; they satisfy the full canonical commutation relations, $[a_i, a_k] = [a_i^\dagger, a_k^\dagger] = 0$ and $[a_i, a_k^\dagger] = \delta_{ik}$, and the resulting $u_i = (a_i + a_i^\dagger)/\sqrt{2}$, $\pi_i = i(a_i^\dagger - a_i)/\sqrt{2}$ are three commuting canonical pairs with $[\pi_i, u_k] = -i\delta_{ik}$. Writing $H_{\mathrm{can}} = \sum_i \frac{1}{2}(\pi_i V_i + V_i \pi_i)$ with $V_i(u) = \sum_k A_{ik} u_k + c_i$ literally in those operators and expanding by the commutation relations produces, term by term, exactly the twenty-four hopping amplitudes of the previous paragraph: the two $\sqrt{2}$'s of a Weyl-ordered product combine into the $\frac{1}{4}$ of the Hermite amplitudes, the double raisings give the strain, the number-conserving terms the vorticity. So $H_{\mathrm{can}} = H$ as operators on the core, and the essential self-adjointness proved for the matrix is essential self-adjointness of the canonically written quadratic symbol. Spelling the coefficients out the way the Navier–Stokes symbol supplies them — linear part the velocity gradient, constant part $-\nu$ times the velocity Laplacian at the fiber — gives the statement in its intended form: for every viscosity, every velocity gradient and every velocity Laplacian, the quantized full quadratic Navier–Stokes symbol $\sum_i \frac{1}{2}(\pi_i A_i + A_i \pi_i)$, $A_i(u) = \sum_j u_{i,j} u_j - \nu u_{i,jj}$, is essentially self-adjoint on the Hermite core of the three velocity components. The canonical pairs used here are abstract, characterized by their commutation relations; the next paragraph transports them to the concrete realization on $L^2(du_1 du_2 du_3)$.

```
#check @BookProof.NavierStokesFlow.CanonicalVector.comm_ann_cre
#check @BookProof.NavierStokesFlow.CanonicalVector.comm_mom_pos
#check @BookProof.NavierStokesFlow.CanonicalVector.comm_mom_pos_of_ne
#check @BookProof.NavierStokesFlow.CanonicalVector.canH
#check @BookProof.NavierStokesFlow.CanonicalVector.canH_eq_velH
#check @BookProof.NavierStokesFlow.CanonicalVector.canH_essentiallySelfAdjointOn_core
#check @BookProof.NavierStokesFlow.CanonicalVector.canH_not_bounded
#check @BookProof.NavierStokesFlow.CanonicalVector.nsQuadraticH
#check @BookProof.NavierStokesFlow.CanonicalVector.nsQuadraticH_essentiallySelfAdjoir
```

The abstract picture is now carried to the concrete one. The product Hermite functions $\psi_\alpha(u) = \prod_i He_{\alpha_i}(u_i) e^{-\|u\|^2/4}/\|\cdot\|$, indexed by the three-mode multi-indices, are an orthonormal basis of $L^2(du_1du_2du_3)$ (`hermiteMvBasis`, `velBasis`), so the sequence space of the previous paragraph **is** $L^2(\mathbb{R}^3)$ under the unitary `velUnitary`. On the Gauss–polynomial core of that space the two physical operators are honest: `posOp` is multiplication by the coordinate u_i and `momOp` is $-i\partial/\partial u_i$ — `momOp_apply_eq_differential` states that its value at $p(u)e^{-\|u\|^2/4}$ is, pointwise, $-i$ times Mathlib's `deriv` of that function along the i-th coordinate — and they satisfy the canonical commutation relation $[\pi_i, u_k] = -i\delta_{ik}$ (`comm_momOp_posOp`). The unitary carries the finite-mode core onto the Gauss–polynomial core and the ladder operators onto the differential ones (`intertwine_ann`, `intertwine_cre`), hence the canonical Hamiltonian onto the differentially written one (`intertwined_canH`). The conclusion is the statement in its intended realization: for every real velocity gradient and every constant part, the Weyl-ordered $\sum_i \frac{1}{2}(\pi_i V_i + V_i \pi_i)$ with $\pi_i = -i\partial/\partial u_i$ and V_i multiplication by $\sum_k A_{ik}u_k + c_i$ is essentially self-adjoint on the Hermite core of $L^2(du_1du_2du_3)$, the operator is unbounded there, and the domain is dense. Nothing here claims global regularity of the classical Navier–Stokes PDE: the theorem is about the Hilbert-space operator at one Eulerian fiber, where the derivative fields are independent canonical coordinates.

```
#check @BookProof.HermiteProductBasis.hermiteMvBasis
#check @BookProof.NavierStokesFlow.DifferentialL2.momOp_apply_eq_differential
#check @BookProof.NavierStokesFlow.DifferentialL2.comm_momOp_posOp
#check @BookProof.NavierStokesFlow.DifferentialL2.velUnitary
#check @BookProof.NavierStokesFlow.DifferentialL2.intertwined_canH
#check @BookProof.NavierStokesFlow.DifferentialL2.nsDiffH
#check @BookProof.NavierStokesFlow.DifferentialL2.nsDiffH_essentiallySelfAdjointOn_cc
#check @BookProof.NavierStokesFlow.DifferentialL2.nsDiffH_not_bounded
#check @BookProof.NavierStokesFlow.DifferentialL2.nsQuadraticDiffH_essentiallySelfAdj
```

That realization inherited its essential self-adjointness by transport: the estimate was proved in the sequence picture and carried across `velUnitary`. The Faris–Lavine argument can instead be run in $L^2(du_1du_2du_3)$ itself, against a comparison operator that is also a differential operator. Take the harmonic oscillator $N_\mu = 2\mu\sum_i(\pi_i^2 + u_i^2/4) + 1$ (`nsDiffN`). The identity that makes the two pictures one is `oscOp_eq_number`: on the Gauss–polynomial core, $\pi_i^2 + u_i^2/4 = a_i^\dagger a_i + \frac{1}{2}$, which is the polynomial identity `oscPoly_eq` obtained from the Leibniz rule $\partial_i(u_i p) = p + u_i\partial_i p$. So N_μ is the transport of multiplication by the comparison symbol $\mu(2|\beta| + 3) + 1$ (`intertwined_nsDiffN`), and `embedCore_surjective` says the Gauss–polynomial core **is** the transported finite-mode core. With that, the two Faris–Lavine inequalities hold for the differential operator itself: the relative bound $\|Hf\|^2 \le a\|Nf\|^2 + b\|f\|^2$ (`nsDiffH_relative_bound`) and the form-commutator bound $|\langle f, i[H,N]f\rangle| \le c\langle f, Nf\rangle$ (`nsDiffH_commForm_bound`), with $\langle f, Nf\rangle \ge \|f\|^2$ (`nsDiffN_quadForm_ge_norm_sq`). Carried to the maximal domain of N inside $L^2(\mathbb{R}^3)$ (`diffMaxH_relative_bound`, `diffMaxH_commForm_bound`, `diffMaxN_add_one_surjective`, `diffMaxN_core_approx`, `diffMaxH_restrict`), the Faris–Lavine criterion then yields `nsDiffH_esa_of_farisLavine` — the same conclusion as above, but obtained from an estimate on the differential operator rather than from a transported theorem.

```
#check @BookProof.NavierStokesFlow.DiffFarisLavine.nsDiffN
#check @BookProof.NavierStokesFlow.DiffFarisLavine.oscPoly_eq
#check @BookProof.NavierStokesFlow.DiffFarisLavine.oscOp_eq_number
#check @BookProof.NavierStokesFlow.DiffFarisLavine.intertwined_nsDiffN
#check @BookProof.NavierStokesFlow.DiffFarisLavine.embedCore_surjective
#check @BookProof.NavierStokesFlow.DiffFarisLavine.nsDiffN_quadForm_ge_norm_sq
#check @BookProof.NavierStokesFlow.DiffFarisLavine.nsDiffH_relative_bound
#check @BookProof.NavierStokesFlow.DiffFarisLavine.nsDiffH_commForm_bound
```

```
#check @BookProof.NavierStokesFlow.DiffFarisLavine.diffMaxH_relative_bound
#check @BookProof.NavierStokesFlow.DiffFarisLavine.diffMaxH_commForm_bound
#check @BookProof.NavierStokesFlow.DiffFarisLavine.diffMaxH_restrict
#check @BookProof.NavierStokesFlow.DiffFarisLavine.nsDiffH_esa_of_farisLavine
```

The sequence-space chain ends with an operator, not just a predicate. Essential self-adjointness is usually stated as vanishing adjoint deficiency, but what it **selects** is the closure of the graph: the operator `clExt T` on the closed domain `clDom T` whose graph is the topological closure of `opGraph T`. The passage is explicit here — `clExt_extends` (it agrees with `T` on the original domain), `clExt_symmetricOn` (the closure is symmetric), and the adjoint criterion `clExt_selfAdjointCriterion`: whenever the deficiency spaces of the adjoint vanish, `clExt T` is self-adjoint, i.e. the adjoint domain is the closed domain and the adjoint operator is the closed operator. With that, the deficiency form of essential self-adjointness used throughout this development is converted into the classical objects: `exists_isSelfAdjointExtension_of_esa` gives a self-adjoint extension of a densely defined symmetric operator with trivial deficiency (no positivity, boundedness or semiboundedness anywhere), and `isSelfAdjointExtension_unique_of_esa` proves it is the **only** one — von Neumann's theorem that deficiency indices `(0,0)` admit exactly one self-adjoint extension. For a positive extension the two notions coincide: `positiveExtension_eq_closure_of_esa` is the statement that for an essentially self-adjoint operator the Friedrichs extension **is** the closure.

```
#check @BookProof.EsaClosure.opGraph
#check @BookProof.EsaClosure.clGraph
#check @BookProof.EsaClosure.clDom
#check @BookProof.EsaClosure.clFun
#check @BookProof.EsaClosure.clExt
#check @BookProof.EsaClosure.clExt_extends
#check @BookProof.EsaClosure.clExt_symmetricOn
#check @BookProof.EsaClosure.clExt_selfAdjointCriterion
#check @BookProof.EsaClosure.exists_isSelfAdjointExtension_of_esa
#check @BookProof.EsaClosure.isSelfAdjointExtension_unique_of_esa
#check @BookProof.EsaClosure.positiveExtension_eq_closure_of_esa
#check @BookProof.SelfAdjointCoreEsa.essentiallySelfAdjointOn_dom_of_isSelfAdjointExt
#check @BookProof.SelfAdjointCoreEsa.essentiallySelfAdjointOn_of_graphCore_selfAdjoir
#check @BookProof.ComparisonCoreEsa.esa_of_isGraphCore
#check @BookProof.ComparisonCoreEsa.isGraphCore_iff_esa
```

Two short companions complete the picture from the other direction. Where the closure chapters say "essentially self-adjoint **gives** a unique self-adjoint extension", `BookProof.SelfAdjointCoreEsa` runs the implication backwards: an operator that **has** a self-adjoint extension was already essentially self-adjoint on its own domain (`essentiallySelfAdjointOn_dom_of_isSelfAdjointExtension`), the deficiency equation at a non-real point forcing the putative deficiency vector to vanish by symmetry. And `BookProof.ComparisonCoreEsa` translates between the two vocabularies the route chapters use — for a Faris–Lavine comparison operator, "graph core" and "core of essential self-adjointness" are the same notion (`isGraphCore_iff_esa`), so a route chapter that exhibits a graph core has, without further work, exhibited a core of essential self-adjointness (`esa_of_isGraphCore`). That equivalence is what turns the per-sector obligations of the plan from analytic tasks into bookkeeping.

A Cayley-transform section takes the selected operator one step further: symmetry of `A` makes `A ± i` equinormal (`norm_add_I_eq_norm_sub_I`), so `exists_cayley_unitary` builds the unitary `U = (A - i)(A + i)⁻¹` with `U(Ax + ix) = Ax - ix`, and `exists_selfAdjointExtension_and_cayley_of_esa` turns an essentially self-adjoint core into both the self-adjoint operator and that unitary. This is the passage from generator to unitary that the older prose recorded as an alternative to Stone's theorem; with the Stone theorem proved in full (2026-08-20e) the unitary **group** `e^{-itA}` now follows directly, and the Cayley transform is the one-parameter step that makes the correspondence explicit.

```
#check @BookProof.EsaClosure.norm_add_I_eq_norm_sub_I
#check @BookProof.EsaClosure.exists_cayley_unitary
#check @BookProof.EsaClosure.exists_selfAdjointExtension_and_cayley_of_esa
```

The Hashimoto/SIRK selection for the Navier–Stokes generator is the abstract statement applied. On the finite-mode core of `ℓ²(Vel)` the coupled three-component fiber Hamiltonian is symmetric, densely defined and essentially self-adjoint (`velCore`, `velCore_symmetricOn`, `velCore_dense`, `velCore_esa`), so

ns_selfAdjoint_extension gives its unique self-adjoint extension and ns_hashimoto_selects the selection: for an arbitrary sequence of non-real shifts the SIRK resolvents exist, are bounded by 1/|Im γ_j|, share the domain of the generator, satisfy the resolvent identity, commute, satisfy the Hashimoto–Nodera rational-Krylov relation, have strongly convergent Galerkin truncations, and each determines the generator completely. Unlike the Yang–Mills instantiation this route uses no positivity: the Navier–Stokes Hamiltonian is not semibounded, and the non-real shift is what makes the resolvent exist.

```
#check @BookProof.NavierStokesFlow.NSHashimoto.velCore
#check @BookProof.NavierStokesFlow.NSHashimoto.velCore_symmetricOn
#check @BookProof.NavierStokesFlow.NSHashimoto.velCore_dense
#check @BookProof.NavierStokesFlow.NSHashimoto.velCore_esa
#check @BookProof.NavierStokesFlow.NSHashimoto.ns_selfAdjoint_extension
#check @BookProof.NavierStokesFlow.NSHashimoto.ns_selfAdjoint_extension_unique
#check @BookProof.NavierStokesFlow.NSHashimoto.ns_hashimoto_selects
#check @BookProof.NavierStokesFlow.NSHashimoto.ns_shiftInvert_selects
```

The same selection now holds for the **differential** realization, the picture in which the operator is written with π_i = -i ∂/∂u_i and u_i as a genuine multiplication operator on the Hermite core of $L^2(du_1 du_2 du_3)$. Gaussian integration by parts makes the Weyl-ordered expression ∑_i ½(π_i V_i + V_i π_i) symmetric there (nsDiffH_symmetricOn, through the Gauss symmetry nsDiffPoly_polySym of its polynomial form), so with the essential self-adjointness already proved the closure is the unique self-adjoint operator the core determines (nsDiffH_selfAdjoint_extension, nsDiffH_selfAdjoint_extension_unique) and nsDiffH_hashimoto_selects is the selection statement for it: for an arbitrary sequence of non-real shifts the resolvents exist, are bounded, share the domain of the generator, satisfy the resolvent identity and the Hashimoto–Nodera relation, have strongly convergent Galerkin truncations, and each determines the generator completely. The enumerated product Hermite functions supply the Hilbert basis the Galerkin truncations are taken in (exists_l2dHilbertBasisNat).

```
#check @BookProof.NavierStokesFlow.DiffHashimoto.nsDiffH_symmetricOn
#check @BookProof.NavierStokesFlow.DiffHashimoto.nsDiffH_selfAdjoint_extension
#check @BookProof.NavierStokesFlow.DiffHashimoto.nsDiffH_selfAdjoint_extension_unique
#check @BookProof.NavierStokesFlow.DiffHashimoto.nsDiffH_hashimoto_selects
#check @BookProof.NavierStokesFlow.DiffHashimoto.nsDiffH_shiftInvert_selects
#check @BookProof.NavierStokesFlow.DiffHashimoto.nsQuadraticDiffH_hashimoto_selects
#check @BookProof.NavierStokesFlow.DiffHashimoto.exists_l2dHilbertBasisNat
```

The Lagrangian route gets the same package, independently of the Eulerian one. The Kato–Rellich theorem is proved for a **relatively bounded** — possibly unbounded — symmetric perturbation: essentiallySelfAdjointOn_add_relBounded shows ‖Bx‖ ≤ a‖Hx‖ + b‖x‖ with a < 1 on the common domain preserves essential self-adjointness, by an explicit Neumann iteration at a large non-real shift, with no closure and no spectral theorem. Applied to the transformed Hamiltonian, the positivity gain of the Lagrangian variables **is** the relative bound: norm_P_le interpolates the first-order drift against the second-order Laplacian by the Ikebe–Kato square-root bound, so hFull_hasZeroDeficiencyOn_of_drive_eq_P gives essential self-adjointness of the full transformed operator from the positive second-order part alone, and hasZeroDeficiencyOn_of_lagrangian_katoRellich transports it back to the Eulerian operator. On top of it the Hashimoto/SIRK selection is proved on the Lagrangian side — lagrangian_selfAdjoint_extension, ..._unique, lagrangian_hashimoto_selects — independently of the Eulerian item 8, on the abstract diagonal instance diagKR over $\ell^2(\mathbb{N})$.

```
#check @BookProof.KatoRellich.essentiallySelfAdjointOn_add_relBounded
#check @BookProof.NavierStokesFlow.LagrangianKatoRellich.norm_P_le
#check @BookProof.NavierStokesFlow.LagrangianKatoRellich.hFull_hasZeroDeficiencyOn_of
#check @BookProof.NavierStokesFlow.LagrangianKatoRellich.hasZeroDeficiencyOn_of_lagra
#check @BookProof.NavierStokesFlow.LagrangianKatoRellich.lagrangian_selfAdjoint_exter
#check @BookProof.NavierStokesFlow.LagrangianKatoRellich.lagrangian_selfAdjoint_exter
#check @BookProof.NavierStokesFlow.LagrangianKatoRellich.lagrangian_hashimoto_selects
#check @BookProof.NavierStokesFlow.LagrangianKatoRellich.lagrangian_shiftInvert_selec
#check @BookProof.NavierStokesFlow.LagrangianKatoRellich.diagKR
```

The Lagrangian route is now realized at the same level as the Eulerian one. In ChapterNavierStokesLagrangianCanonical the parcel momenta and the viscous gradients are built as the

canonical pairs of the trajectory-space Hermite basis `ℓ²(Fin 3 → ℕ)`: with `ω = √(2ν)` the operators `Qᵢ = ω^{-1/2}(aᵢ + aᵢ†)/√2` and `Pᵢ = ω^{1/2}·i(aᵢ† − aᵢ)/√2` obey the canonical commutation relations (`comm_lagP_lagQ`, `comm_lagP_lagQ_of_ne`), so — unlike in the diagonal instance `diagKR` — they genuinely fail to commute, and they are symmetric because the ladder operators are mutually adjoint (`inner_ann_cre`). The choice of frequency is exactly the one that diagonalizes the positive second-order part: `posSq_add_momSq` is the oscillator identity `uᵢ² + πᵢ² = 2Nᵢ + 1` and `lagCan_secondOrder_eq` reads the Lagrangian second-order part as the number operator, `½∑Pᵢ² + ν∑Qᵢ² = ω(N + 3/2)`. Its eigenvectors are the Hermite states (`lagT_coreState`), a total family, so it is essentially self-adjoint on the Hermite core (`lagT_hasZeroDeficiencyOn`) while being unbounded (`lagT_not_bounded`); Kato–Rellich then gives essential self-adjointness of the **full** transformed Hamiltonian there (`lagCan_esa`), and Stone's theorem the complete unitary flow (`lagCan_stone_flow`). Nothing here claims global regularity of the classical Navier–Stokes equation.

```
#check @BookProof.NavierStokesFlow.LagrangianCanonical.inner_ann_cre
#check @BookProof.NavierStokesFlow.LagrangianCanonical.posSq_add_momSq
#check @BookProof.NavierStokesFlow.LagrangianCanonical.comm_lagP_lagQ
#check @BookProof.NavierStokesFlow.LagrangianCanonical.comm_lagP_lagQ_of_ne
#check @BookProof.NavierStokesFlow.LagrangianCanonical.lagCanData
#check @BookProof.NavierStokesFlow.LagrangianCanonical.lagCan_secondOrder_eq
#check @BookProof.NavierStokesFlow.LagrangianCanonical.lagT_coreState
#check @BookProof.NavierStokesFlow.LagrangianCanonical.lagT_hasZeroDeficiencyOn
#check @BookProof.NavierStokesFlow.LagrangianCanonical.lagT_not_bounded
#check @BookProof.NavierStokesFlow.LagrangianCanonical.lagCan_esa
#check @BookProof.NavierStokesFlow.LagrangianCanonical.lagCan_stone_flow
```

Stone's theorem — the abstract theorem that turns essential self-adjointness into a **complete** flow — is itself proved in this project (2026-08-20e). The forward half, `ChapterStoneResolvent`–`ChapterStoneGenerator`, constructs the one- parameter unitary group of an arbitrary self-adjoint operator on a Hilbert space: the resolvent exists and is invertible (`shift_bijective`), is contractive and commutes with the operator (`resCLM`, `norm_resCLM_apply_le`, `inner_res`, `res_comm`), the Yosida approximants converge to the group in the strong operator topology (`yosida_tendsto`), each approximant is a unitary (`approxU_mem_unitary`), the limit `stoneU t` is a strongly continuous unitary group (`stoneU`, `continuous_stoneU_apply`), it maps the domain into itself and its derivative at zero recovers the original operator (`stoneU_mem_domain`, `hasDerivAt_stoneU_zero`), so the abstract Stone generator `stoneGroup.gen` coincides with the operator we started with (`gen_stoneGroup_eq`). The group `e^{-itA}` is thus defined for **every** real `t` — this is the "complete operator-flow" half of the `book.tex` §4210-4216 claim, now a theorem rather than a named result.

```
#check @BookProof.ChapterStoneResolvent.UnboundedSelfAdjoint.shift_bijective
#check @BookProof.ChapterStoneResolvent.UnboundedSelfAdjoint.resCLM
#check @BookProof.ChapterStoneResolvent.UnboundedSelfAdjoint.norm_resCLM_apply_le
#check @BookProof.ChapterStoneResolvent.UnboundedSelfAdjoint.inner_res
#check @BookProof.ChapterStoneResolvent.UnboundedSelfAdjoint.res_comm
#check @BookProof.ChapterStoneResolvent.UnboundedSelfAdjoint.yosida_tendsto
#check @BookProof.ChapterStoneResolvent.UnboundedSelfAdjoint.approxU_mem_unitary
#check @BookProof.ChapterStoneResolvent.UnboundedSelfAdjoint.stoneU
#check @BookProof.ChapterStoneResolvent.UnboundedSelfAdjoint.continuous_stoneU_apply
#check @BookProof.ChapterStoneResolvent.UnboundedSelfAdjoint.stoneU_mem_domain
#check @BookProof.ChapterStoneResolvent.UnboundedSelfAdjoint.hasDerivAt_stoneU_zero
#check @BookProof.ChapterStoneResolvent.UnboundedSelfAdjoint.stoneGroup
#check @BookProof.ChapterStoneResolvent.UnboundedSelfAdjoint.gen_stoneGroup_eq
```

The converse half, `ChapterStoneMeasurable`–`ChapterStoneConverse`, recovers the infinitesimal generator from any weakly measurable one-parameter unitary group: the generator domain `genDomain` is the set of vectors for which the derivative at zero exists, `genOp` is that derivative, and the main theorem `gen_stoneU_eq` says every weakly measurable group is the exponential of its generator. The technical engine is the separability of `H`: on a separable space the generator domain is dense (`dense_avgSpan` via averaging over a countable dense family), the averages converge uniformly on it (`norm_apply_avgVec_sub_le`), and the averaged vectors are continuous in `t` (`continuous_apply`). Combined, the two halves give the bijection `stone_bijection` of `ChapterStoneTheorem` between self-adjoint operators and weakly measurable unitary groups — Stone's theorem in full.

```
#check @BookProof.ChapterStoneMeasurable.WeakMeasurableUnitaryGroup.genDomain
#check @BookProof.ChapterStoneMeasurable.WeakMeasurableUnitaryGroup.genOp
#check @BookProof.ChapterStoneMeasurable.WeakMeasurableUnitaryGroup.gen
#check @BookProof.ChapterStoneMeasurable.WeakMeasurableUnitaryGroup.norm_apply_avgVec
#check @BookProof.ChapterStoneMeasurable.WeakMeasurableUnitaryGroup.dense_avgSpan
#check @BookProof.ChapterStoneMeasurable.WeakMeasurableUnitaryGroup.continuous_apply
#check @BookProof.ChapterStoneMeasurable.WeakMeasurableUnitaryGroup.gen_stoneU_eq
#check @BookProof.ChapterStoneTheorem.stone_bijection
```

Both halves of Stone's theorem take the self-adjoint operator as given. The **Cayley transform** (2026-08-24g, `ChapterCayleyTransform`, `ChapterCayleyInverse`) is the classical device that removes the unboundedness from the discussion altogether: for a densely defined self-adjoint A the map $V = (A - i)(A + i)^{-1}$ — assembled here from the resolvents above — is a **unitary** (`norm_cayleyMap`, `cayleyMap_surjective`, `cayley`) characterised by $V(A + i)\psi = (A - i)\psi$ (`cayley_shift`); 1 is not an eigenvalue of V (`cayley_apply_ne_self`, `one_sub_cayley_injective`), the range of $1 - V$ is exactly the domain of A (`range_one_sub_cayley`) and is therefore dense (`denseRange_one_sub_cayley`), and A is recovered as $i(1 + V)(1 - V)^{-1}$ (`op_eq_cayley`, `coe_eq_cayley`). The correspondence is exact in both directions: the basic criterion `isSelfAdjointOn_of_surjective` (symmetric with $A \pm i$ onto) shows that the inverse Cayley transform $i(1 + V)(1 - V)^{-1}$ of **any** unitary V with $1 - V$ injective is self-adjoint on $\mathrm{ran}(1 - V)$ (`invCayleyOp_isSelfAdjointOn`, packaged as `ofUnitary`), and the two constructions undo one another (`cayley_ofUnitary`, `invCayleyDomain_cayley`, `invCayleyOp_cayley`). The unbounded layer is thus faithfully encoded by a bounded object — the standard route by which the spectral theorem for unbounded self-adjoint operators is reduced to the bounded (normal) case treated in `ChapterSpectralMultiplication`.

```
#check @BookProof.ChapterCayleyTransform.norm_cayleyMap
#check @BookProof.ChapterCayleyTransform.cayleyMap_surjective
#check @BookProof.ChapterCayleyTransform.cayley
#check @BookProof.ChapterCayleyTransform.cayley_shift
#check @BookProof.ChapterCayleyTransform.sub_cayley_shift
#check @BookProof.ChapterCayleyTransform.add_cayley_shift
#check @BookProof.ChapterCayleyTransform.one_sub_cayley_injective
#check @BookProof.ChapterCayleyTransform.cayley_apply_ne_self
#check @BookProof.ChapterCayleyTransform.range_one_sub_cayley
#check @BookProof.ChapterCayleyTransform.denseRange_one_sub_cayley
#check @BookProof.ChapterCayleyTransform.op_eq_cayley
#check @BookProof.ChapterCayleyTransform.coe_eq_cayley
#check @BookProof.ChapterCayleyInverse.isSelfAdjointOn_of_surjective
#check @BookProof.ChapterCayleyInverse.invCayleyOp_symmetric
#check @BookProof.ChapterCayleyInverse.invCayleyOp_isSelfAdjointOn
#check @BookProof.ChapterCayleyInverse.ofUnitary
#check @BookProof.ChapterCayleyInverse.cayley_ofUnitary
#check @BookProof.ChapterCayleyInverse.invCayleyDomain_cayley
#check @BookProof.ChapterCayleyInverse.invCayleyOp_cayley
```

Composing that correspondence with the bounded spectral theorem gives a multiplication model for the **unbounded** operator itself (`ChapterCayleySpectralModel`). The step that makes the composition elementary is that the resolvent is a **continuous** function of the Cayley transform: $(A + i)^{-1} = (2i)^{-1}(1 - V)$ and $(A - i)^{-1} = (2i)^{-1}(V^{-1} - 1)$ (`res_neg_one_eq_cayley`, `res_one_eq_cayley`), and V is a bounded **normal** operator (`cayleyCLM`, `isStarNormal_cayleyCLM`), so the continuous functional calculus applies to the two symbols $g(z) = (1 - z)/(2i)$ and $h(z) = (1 + z)/2$ with $g(V) = (A + i)^{-1}$ and $h(V)y = A(A + i)^{-1}y$ (`cfcHom_resSymbol`, `cfcHom_opSymbol`). Feeding them through the model of `ChapterSpectralMultiplication` yields `unbounded_spectral_multiplication_model`: if the Cayley transform has a cyclic unit vector, there are a Borel probability measure μ on its spectrum and a unitary $U : L^2(\mu) \simeq H$ such that **every** vector of $D(A)$ is $U(g\cdot u)$ and $A\,U(g\cdot u) = U(h\cdot u)$ — that is, A is multiplication by $h/g = i(1 + z)/(1 - z)$. Only the reduction to a **single** cyclic subspace is still missing from the general unbounded spectral theorem.

```
#check @BookProof.ChapterCayleySpectralModel.res_neg_one_eq_cayley
#check @BookProof.ChapterCayleySpectralModel.res_one_eq_cayley
#check @BookProof.ChapterCayleySpectralModel.cayleyCLM
#check @BookProof.ChapterCayleySpectralModel.isStarNormal_cayleyCLM
#check @BookProof.ChapterCayleySpectralModel.resSymbol
```

```
#check @BookProof.ChapterCayleySpectralModel.opSymbol
#check @BookProof.ChapterCayleySpectralModel.cfcHom_resSymbol
#check @BookProof.ChapterCayleySpectralModel.cfcHom_opSymbol
#check @BookProof.ChapterCayleySpectralModel.spectralUnitary_resSymbol
#check @BookProof.ChapterCayleySpectralModel.spectralUnitary_opSymbol
#check @BookProof.ChapterCayleySpectralModel.unbounded_spectral_multiplication_model
```

The capstone `ChapterStoneSeparable` packages the full statement: on a separable Hilbert space there exists a unique one-parameter unitary group with a given self-adjoint generator, and conversely every weakly measurable group has a unique self-adjoint generator — `stone_exists_unique_group`, `stone_exists_unique_generator`, `stoneEquiv`. The file ends with a nontrivial **instantiated** example on `ℓ²(ℤ)` (which is separable, `separableSpace_L2Z`): the momentum operator `mulSA` given by `f ↦ (i n f(n))` is a genuinely self-adjoint (not merely essentially self-adjoint) unbounded operator (`mulSA`, `mulSA_position_unbounded`), and its Stone group is the unitary shift group of the lattice (`stoneU_mulSA`) — the same group structure that carries the momentum generators of the gauged fibers. This is the concrete realization of the `book.tex` "self-adjoint ⇒ complete flow" assertion, independent of the Navier–Stokes truncation.

```
#check @BookProof.ChapterStoneSeparable.stone_exists_unique_group
#check @BookProof.ChapterStoneSeparable.stone_exists_unique_generator
#check @BookProof.ChapterStoneSeparable.stoneEquiv
#check @BookProof.ChapterStoneSeparable.separableSpace_L2Z
#check @BookProof.ChapterStoneSeparable.mulSA
#check @BookProof.ChapterStoneSeparable.mulSA_position_unbounded
#check @BookProof.ChapterStoneSeparable.stoneU_mulSA
```

The bridge from the selection predicates to the flow is now a single packaging step, `ChapterStoneBridge`. The essential-self-adjointness / Hashimoto-selection threads produce `IsSelfAdjointExtension` / `IsPositiveSelfAdjointExtension` operators on a dense domain, while Stone's theorem consumes the bundled `UnboundedSelfAdjoint` structure; `dense_domain_of_isSelfAdjointExtension` and `isSelfAdjointOn_of_isSelfAdjointExtension` pull the two missing conjuncts out of the predicate, and `unboundedSelfAdjointOf` assembles the bundle. The complete unitary group is packaged as `IsStoneFlow` (`U 0 = 1`, the group law, isometry of each `U t`, and the Schrödinger equation on the domain), `isStoneFlow_stoneU` shows the abstractly constructed group is such a flow, and `exists_stone_flow_of_selfAdjointExtension` / `of_positive` / `of_esa` are the three entry points — from a selected self-adjoint extension, from a positive (Friedrichs) one, and directly from essential self-adjointness of a symmetric core.

```
#check @BookProof.StoneBridge.dense_domain_of_isSelfAdjointExtension
#check @BookProof.StoneBridge.isSelfAdjointOn_of_isSelfAdjointExtension
#check @BookProof.StoneBridge.unboundedSelfAdjointOf
#check @BookProof.StoneBridge.IsStoneFlow
#check @BookProof.StoneBridge.isStoneFlow_stoneU
#check @BookProof.StoneBridge.exists_stone_flow_of_selfAdjointExtension
#check @BookProof.StoneBridge.exists_stone_flow_of_positive
#check @BookProof.StoneBridge.exists_stone_flow_of_esa
```

`ChapterStoneFlows` runs that bridge on the three concrete Hamiltonians of this development, so each of them — not a hypothetical example — generates a complete unitary flow. `ns_stone_flow` gives the flow of the coupled three-component Eulerian fiber generator `velCore A c` on `ℓ²(Vel)` (the unique self-adjoint extension, the closure; no positivity is used, so there is no Friedrichs label). `lagrangian_stone_flow` gives the flow of the transformed (parcel) Hamiltonian $\hat{h}_full = \tfrac{1}{2}\sum P_i^2 + \nu\sum Q_i^2 + \sum f_i D_i + C$ from essential self-adjointness of its core, and `diagKR_stone_flow` instantiates it on the genuinely unbounded `ℓ²(ℕ)` instance — the hypothesis discharged by the Kato–Rellich relative bound. On the gauge side, `ym_fock_stone_flow` gives the flow of the second-quantized Yang–Mills Hamiltonian $d\Gamma(\tfrac{1}{2}\Sigma\pi^2 + \tfrac{1}{2}\Sigma B^2)$ on the Fock space over the Gauss core of $L^2(\mathbb{R}^{99})$, where the extension **is** positive (Friedrichs). This is step (c) of `CONSOLIDATED_PLAN.md` §9 item 11 — the complete operator flow for all three — and the honest boundaries are unchanged: the flow is that of the selected extension in the abstract realization, and nothing is claimed about the classical Navier–Stokes regularity problem.

```
#check @BookProof.StoneFlows.ns_stone_flow
#check @BookProof.StoneFlows.lagrangian_stone_flow
```

```
#check @BookProof.StoneFlows.diagKR_stone_flow
#check @BookProof.StoneFlows.ym_fock_stone_flow
```

Essential self-adjointness glues along an orthogonal direct sum, and `ChapterDirectSumEsa` isolates that step once and for all. A deficiency vector of $\oplus_i H_i$ may be tested against a state living in a single fibre, and the identity it then satisfies is exactly the fibre deficiency identity, so each of its coordinates vanishes: `dsOp_deficiencyTrivialAt`, hence `dsOp_essentiallySelfAdjointOn`. No relative bound, no comparison operator and no commutator estimate are involved — the hypothesis is only that every fibre operator is essentially self-adjoint on its own core — and `dsCore_dense` adds that the algebraic direct sum of the fibre cores is dense as soon as every fibre core is. The payoff is the continuum parcel picture: `ChapterNavierStokesFockContinuum` proved essential self-adjointness of the second-quantized Hamiltonian one sector at a time (on $L^2(\mathbb{R}^n)$ it is multiplication by the total energy $\sum_k w(\xi_k)$, an operator with in general purely continuous spectrum and no eigenvectors), and `fockH_hasZeroDeficiencyOn` is the same statement on the **whole** Fock space $\oplus_n L^2(\mathbb{R}^n)$ — for an arbitrary measurable field `w`, with `fockCore_dense` and `fockH_isSymmetricDom` supplying dense definition and symmetry. Feeding that into the Stone bridge closes the circle: `dsOpD_stone_flow` turns a family of dense fibre cores carrying symmetric fibre operators of vanishing deficiency into a self-adjoint extension of the glued operator together with the unitary group it generates, and `fockH_stone_flow` is that statement for the continuum Fock Hamiltonian — the complete flow `e^{-itĥ}` solving the Schrödinger equation on the domain. The honest boundary is that the fibres must be mutually orthogonal and invariant: this is the discrete form of a direct-integral gluing, not a decomposition theorem.

```
#check @BookProof.DirectSumEsa.dsCore
#check @BookProof.DirectSumEsa.dsOp
#check @BookProof.DirectSumEsa.dsOp_deficiencyTrivialAt
#check @BookProof.DirectSumEsa.dsOp_essentiallySelfAdjointOn
#check @BookProof.DirectSumEsa.dsOpD_hasZeroDeficiencyOn
#check @BookProof.DirectSumEsa.dsCore_dense
#check @BookProof.DirectSumEsa.fockCore_dense
#check @BookProof.DirectSumEsa.fockH_isSymmetricDom
#check @BookProof.DirectSumEsa.fockH_hasZeroDeficiencyOn
#check @BookProof.DirectSumEsa.dsOpD_stone_flow
#check @BookProof.DirectSumEsa.fockH_essentiallySelfAdjointOn
#check @BookProof.DirectSumEsa.fockH_stone_flow
```

Carleman's criterion supplies the last realization of the **full** Navier–Stokes Hamiltonian in which the momentum genuinely fails to commute with the field modes. On the half-line lattice $\ell^2(\mathbb{N})$ the momentum is the symmetric-difference operator and the fifteen field modes are multiplication by arbitrary — in particular unbounded — real sequences; `weyl_momOp_diagOp` computes the Weyl-symmetrized product `½(π A + A π)` and finds a tridiagonal (Jacobi) operator whose off-diagonal coupling is `c_n = -(i/2)(α_n + α_{n+1})` for the Navier–Stokes symbol α, so `halfLineFullData_hamiltonian` identifies the whole Hamiltonian as one Jacobi matrix. The analytic input is `tridiag_hasZeroDeficiencyOn_of_carleman`: the classical Wronskian argument — `tridiag_wronskian` for the discrete Green identity, `wron_eq_sum` for its telescoping, `sum_normSq_le` for the resulting inequality — shows that a deficiency vector would force $\sum 1/|c_n|$ to converge, so divergence of that sum gives essential self-adjointness. Hence `halfLineFull_hasZeroDeficiencyOn`, and, for a linearly growing viscous mode, `linearFull_hasZeroDeficiencyOn` together with `linearFull_not_bounded`: an unbounded, non-commuting, essentially self-adjoint full Navier–Stokes Hamiltonian. The dichotomy is the lattice form of the ODE chapter's $\dot{x} = x^2$ warning — Carleman's sum diverges for subquadratic growth and can converge for faster growth, where `ChapterNavierStokesDeficiency` already exhibits a Jacobi operator that is **not** essentially self-adjoint.

```
#check @BookProof.NavierStokesFlow.Carleman.tridiagOp
#check @BookProof.NavierStokesFlow.Carleman.tridiagOp_isSymmetricDom
#check @BookProof.NavierStokesFlow.Carleman.tridiag_wronskian
#check @BookProof.NavierStokesFlow.Carleman.wron_eq_sum
#check @BookProof.NavierStokesFlow.Carleman.sum_normSq_le
#check @BookProof.NavierStokesFlow.Carleman.tridiag_hasZeroDeficiencyOn_of_carleman
#check @BookProof.NavierStokesFlow.Carleman.momOp
#check @BookProof.NavierStokesFlow.Carleman.weyl_momOp_diagOp
#check @BookProof.NavierStokesFlow.Carleman.halfLineFullData
#check @BookProof.NavierStokesFlow.Carleman.halfLineFullData_hamiltonian
#check @BookProof.NavierStokesFlow.Carleman.halfLineFull_hasZeroDeficiencyOn
```

```
#check @BookProof.NavierStokesFlow.Carleman.linearFull_hasZeroDeficiencyOn
#check @BookProof.NavierStokesFlow.Carleman.linearFull_not_bounded
```

25.12. Summary

The free-field thread replaces the nonexistent infinite-dimensional Lebesgue measure with the rotation-invariant Gaussian, pushes it to a uniform measure on the sphere, and then applies the Born rule to parametrize the simplex — all the while the gauge fibers keep track of the invisible phase. The two negative results (no Lebesgue measure; positive-measure partitions are countable) are not obstacles but the precise statements that make the construction both necessary and well-defined.

The same free-field parametrization carries the Navier–Stokes thread: the velocity field and its derivatives are independent canonical degrees of freedom, and on a finite truncation the Weyl-symmetrized Hamiltonian has a complete, norm-preserving flow, a unique global solution to the Cauchy problem, and a nilpotent BRST constraint. Beyond the truncation, the essential self-adjointness of the generator is now proved across an entire chain of abstract models in the Hermite (sequence-space) representation: the bilinear quadratic-symbol Hamiltonian on `ℓ²(ℕ × J)` (`BilinearEsa.bilH_essentiallySelfAdjointOn_core`), its affine extension covering the viscous and cross terms (`AffineFiber.affH` / `AffineBlock.affBlockH`), the sign-flip unitary that drops the `c ≥ 0` hypothesis (`SignFlip.saffH` / `sblockH`), and finally all three coupled velocity components with an arbitrary real velocity gradient (`SignedShift.gaffH`, `ThreeComponent.velH`). The instrument throughout is the Faris–Lavine commutator criterion, now itself proved (Theorem 1 + Cor. 1.1) rather than named. The canonical reading of that same fiber is supplied by `CanonicalVector` (`comm_ann_cre`, `comm_mom_pos`), which builds the ladder pairs **inside** the sequence space and proves `canH_eq_velH` — the matrix H of the previous paragraph is the Weyl-ordered expression $\sum_i \frac{1}{2}(\pi_i V_i + V_i \pi_i)$ — so `canH_essentiallySelfAdjointOn_core` and, with the Navier–Stokes coefficients, `nsQuadraticH_essentiallySelfAdjointOn_core` are the same statement in canonical form. The selected operators are made explicit by the graph-closure machinery of `EsaClosure` (`clExt`, `IsSelfAdjointExtension`, `exists_isSelfAdjointExtension_of_esa`, `isSelfAdjointExtension_unique_of_esa`, `positiveExtension_eq_closure_of_esa`, and the Cayley-transform unitary), and the selection is then **instantiated** on the Navier–Stokes side twice: by the Hashimoto/SIRK resolvent route on the Eulerian fiber (`NSHashimoto.velCore`, `ns_hashimoto_selects` — no positivity needed) and by the Kato–Rellich relative bound on the Lagrangian side (`norm_P_le`, `hFull_hasZeroDeficiencyOn_of_drive_eq_P`, `lagrangian_hashimoto_selects`). The honest boundary drawn at the end of that chain is the genuinely differential realization of these operators on `L²(du)` (the abstract models give the operator by its matrix in the Hermite basis only), and the **classical** existence/uniqueness claim of `book.tex` (the Clay regularity problem) is not carried by any theorem here — the operator-flow global existence would follow from the differential ESA by the Stone theorem of 2026-08-20e, but that ESA is itself the open step; see `Book/YangMillsQuantization.lean` for the pointer to this formalized subset. Stone's theorem itself — the missing bridge from ESA to a complete flow — is now proved in full (`stone_bijection` between self-adjoint operators and weakly measurable unitary groups, `stoneEquiv` on separable spaces, instantiated on `ℓ²(ℤ)` by the momentum operator `mulSA`), and the bridge is then **run**: the packaging step `StoneBridge` (`unboundedSelfAdjointOf`, `IsStoneFlow`, `exists_stone_flow_of_selfAdjointExtension` / `of_positive` / `of_esa`) turns a selected extension into the bundled structure Stone consumes, and `StoneFlows` instantiates the complete unitary flow for the Eulerian NS (`ns_stone_flow`), the Lagrangian NS (`lagrangian_stone_flow`, `diagKR_stone_flow`) and the QYM (`ym_fock_stone_flow`). The **only** remaining step for a flow on the continuum is the differential ESA, and the truncation already carries the complete flow (`nsCauchy_existsUnique`).

26. The Spin–Statistics Dichotomy

26.1. Orientation and Status

The chapter that closes the free-field thread of this part: the spin–statistics dichotomy as a statement about tensor products of sample spaces — bosons commute, fermions anticommute — in the finite-degree instance. Read it after the free-field chapter (whose premise — infinitely many degrees of freedom — it examines and repairs).

The route: the question (what happens at finitely many degrees of freedom), two modes by Jordan–Wigner, each mode a fermionic oscillator, the modes anticommute (fermionic statistics), Pauli exclusion, and every particle number via the symmetric and antisymmetric sectors.

The proofs are `BookProof/FockStatistics`, `BookProof/PermSector`, `BookProof/TensorPerm`, `BookProof/GroupAverage`, and `BookProof/ReducedEsa`.

26.2. The Question

The free-field parametrization of the previous chapter builds a uniform measure on an infinite-dimensional sphere; it relies on there being *infinitely many* degrees of freedom. But, the manuscript observes, "often we are interested in a tensor product of sample spaces, some of which have finite degrees of freedom," and one should be able to "repeat the exercise for $\mathbb{Z}_2^n \times \mathbb{R}^m$." This chapter is the finite, fully explicit case: a sample space with just two degrees of freedom, $\mathbb{Z}_2 \times \mathbb{Z}_2$.

A single two-valued degree of freedom $\mathbb{Z}_2$ is parametrized by a **fermionic** Fock space $\Gamma^a(L^2(\mathbb{Z}_2))$. For a *product* $\mathbb{Z}_2 \times \mathbb{Z}_2$ there is a subtlety. Taken naively, the two modes would admit non-null products of creation operators, and that "is not a free field parametrization." The fix is to take the **graded** tensor product $\Gamma^s(L^2(\mathbb{Z}_2)) \otimes \Gamma^a(L^2(\mathbb{Z}_2))$ so that the two fermionic modes **anticommute**. That anticommutation — as opposed to the *communication* of bosonic modes — is the algebraic content of the **spin–statistics** correspondence the section is about.

26.3. Two Modes by Jordan–Wigner

The two-mode fermionic Fock space is $\mathbb{C}^4 \cong \mathbb{C}^2 \otimes \mathbb{C}^2$, with basis $|n_1 n_2\rangle$ ordered as $|00\rangle, |01\rangle, |10\rangle, |11\rangle$ and $|00\rangle$ the vacuum. The two annihilation modes are realized by the **Jordan–Wigner** prescription:

$$b_1 = a \otimes I, \qquad b_2 = Z \otimes a,$$

where $a = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}$ is the single-mode annihilation and $Z = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}$ is the **fermion-parity string**. It is the parity string Z in the second mode that forces the two modes to anticommute. As explicit 4×4 matrices, with creation operators the conjugate transpose:

```
#check @fermiAnnih1
#check @fermiAnnih2
#check @fermiCreate1_eq
#check @fermiCreate2_eq
```

26.4. Each Mode Is a Fermionic Oscillator

Each mode separately satisfies the **canonical anticommutation relation** (CAR): the anticommutator of an annihilation operator with its own creation operator is the identity,

$$\{b_i, b_i^\dagger\} = b_i b_i^\dagger + b_i^\dagger b_i = 1.$$

```
#check @fermi_CAR₁
#check @fermi_CAR₂
```

26.5. The Modes Anticommute: Fermionic Statistics

The point of the construction is what happens *between* the two modes. Distinct fermionic modes anticommute:

$$\{b_1, b_2\} = b_1 b_2 + b_2 b_1 = 0, \qquad \{b_1, b_2^\dagger\} = b_1 b_2^\dagger + b_2^\dagger b_1 = 0.$$

```
#check @fermiAnticomm_annih
#check @fermiAnticomm_create
#check @fermi_CAR_cross
```

This is the **fermionic statistics**. It is precisely what distinguishes the fermionic parametrization from a bosonic (commuting) one, and it is exactly the condition the manuscript demanded: with the graded tensor product, the two modes do not produce spurious non-null products of creation operators. A bosonic pair would *commute* here; a fermionic pair *anticommutes*. The spin–statistics theorem is the statement that certain symmetry transformations are representable by one or the other but not both.

26.6. Pauli Exclusion

Anticommutation has an immediate consequence. Setting the two modes equal in $\{b_i, b_i\} = 0$ gives $b_i^2 = 0$: applying the same annihilation (or creation) operator twice gives zero. This is the **Pauli exclusion principle** — a single mode cannot hold two fermions:

```
#check @fermiAnnih₁_sq
#check @fermiCreate₁_sq
```

The number operator $N_i = b_i^\dagger b_i$ counts the occupation of mode i. The relation $b_i^2 = 0$ forces its eigenvalues to be 0 and 1 only: each N_i is a Hermitian **projection** (idempotent and self-adjoint):

```
#check @fermiNumber₁_hermitian
#check @fermiNumber₁_idem
```

There is a pleasing contrast. The creation operators *anticommute*, but the number operators — the actual observables — **commute**:

```
#check @fermiNumber_commute
```

So the observable algebra is commutative (as it must be for a probability theory), even though the underlying creation operators are not. The anticommutation lives in the parametrization, not in the measured quantities.

26.7. Every Particle Number: the Symmetric and Antisymmetric Sectors

The matrices above are the whole story for two modes. The same dichotomy at **arbitrary** particle number is not a matrix computation but an operator-theoretic one, and it is now a theorem: essential self-adjointness passes from a one-particle operator to its restriction to the bosonic and the fermionic sectors of every tensor power, and then to the Fock spaces assembled from them.

The mechanism is averaging. A self-inverse isometry U that preserves the domain and commutes with T has reducing projections $(1 \pm U)/2$ (`BookProof.ReducedEsa.symProj`, `…_asymProj`), and `essentiallySelfAdjointOn_symSector` / `…_asymSector` restrict the operator to the $\pm$ eigenspaces. A finite group of such symmetries is handled by its Haar average `avgProj`, whose range is the joint fixed space — `essentiallySelfAdjointOn_invariantSector` in `BookProof.GroupAverage`. The group in question is the symmetric group S_n, acting on $H^{\otimes n}$ by the permutation action `permOp` of `BookProof.TensorPerm` (built from `swapFirst` and `liftTail`, checked on pure tensors by `permOp_purePow`), twisted by `signRep` for the fermionic case.

The conclusion, in `BookProof.PermSector`, is `essentiallySelfAdjointOn_bosonic` and `essentiallySelfAdjointOn_fermionic` on $D_2^{\otimes n}$, and — composing with the multilinear core estimate — `essentiallySelfAdjointOn_bosonic_core` / `…_fermionic_core` on the symmetrized and antisymmetrized one-particle core $D^{\otimes n}$, for any core D of A. `mem_bosonicSector_iff` and `mem_fermionicSector_iff` identify the sectors as the symmetric and the antisymmetric tensors — the infinite-dimensional counterpart of the two-dimensional Λ^2 and S^2 the Pauli and commutation arguments above detect. `BookProof.FockStatistics` discharges the carried hypothesis from essential self-adjointness of the one-particle operator alone (`bosonicFock_esa`, `fermionicFock_esa`), and its completion chapter assembles the sectors over all particle numbers: `hbosonicFock_esa` — the bosonic Fock space — and `hfermionicFock_esa`, the fermionic one.

So the finite 4×4 computation of this chapter and the infinite construction above answer the same question at two levels: **which vectors are counted once (bosons) and which cancel (fermions)**, and **does the Hamiltonian of record, restricted to those vectors, still have a unique self-adjoint extension**. The answer to

the second is now yes, at every particle number, under no hypothesis beyond essential self-adjointness of the one-particle operator.

```
#check @BookProof.ReducedEsa.symProj
#check @BookProof.ReducedEsa.essentiallySelfAdjointOn_symSector
#check @BookProof.GroupAverage.UnitaryRep.avgProj
#check @BookProof.GroupAverage.UnitaryRep.essentiallySelfAdjointOn_invariantSector
#check @BookProof.TensorPerm.permOp
#check @BookProof.TensorPerm.permOp_purePow
#check @BookProof.TensorPerm.permRep
#check @BookProof.TensorPerm.signRep
#check @BookProof.PermSector.essentiallySelfAdjointOn_bosonic
#check @BookProof.PermSector.essentiallySelfAdjointOn_fermionic
#check @BookProof.PermSector.essentiallySelfAdjointOn_bosonic_core
#check @BookProof.PermSector.essentiallySelfAdjointOn_fermionic_core
#check @BookProof.PermSector.mem_bosonicSector_iff
#check @BookProof.PermSector.mem_fermionicSector_iff
#check @BookProof.FockStatistics.bosonicFock_esa
#check @BookProof.FockStatistics.fermionicFock_esa
#check @BookProof.FockStatistics.hbosonicFock_esa
#check @BookProof.FockStatistics.hfermionicFock_esa
```

26.8. Why This Matters Here

This is the finite-degree-of-freedom instance of the manuscript's tensor products of sample spaces. The carrier $\mathbb{C}^4 \cong \mathbb{C}^2 \otimes \mathbb{C}^2$ is literally a tensor product, and the **graded** (Jordan–Wigner) tensor product is what encodes the statistics: it is the parity string Z that turns a naive product of two modes into an anticommuting pair. The same dichotomy — bosonic commutation versus fermionic anticommutation, with the spin of the field selecting one — is what the manuscript uses to reach the spin–statistics theorem and the unusual statistics of ghost fields. Everything above is verified by explicit 4×4 matrix computation, `sorry`-free and `axiom`-free.

27. Real Representations, the CPT Theorem and the Relativistic Position Operator

27.1. Orientation and Status

The opening chapter of the Standard-Model thread of this part — the block (this chapter, quantization by time-evolution, the Gribov ambiguity, physical parity) that can be read largely independently of the free-field thread. It answers the manuscript's question of how Lorentz covariance and quantum mechanics are compatible, and needs only the wave-function parametrization of the Born part behind it.

The route: the Lorentz group and its discrete subgroup, its real representations, the Pauli matrices and the double cover, the Poincaré group and the little group, Majorana spinors and the CPT theorem, the relativistic position operator, and the bosonic CCR relations.

The proofs include `BookProof/ChapterLorentzRealRep` (with its direct/full/sum variants), `BookProof/ChapterLittleGroup`, the CPT chapters (`ChapterCPTParity`, `ChapterCPTPT`, `ChapterCPTHamiltonian`), the Majorana chapters, and `BookProof/Bosonic` for the CCR close.

27.2. The Question

The manuscript's chapter on real representations asks how Lorentz covariance and quantum mechanics are compatible. It begins from two mutually compatible formalisms: a theory invariant under diffeomorphisms (which includes Poincaré invariance, with tetrad fields corresponding to a Minkowski space-time), and a theory that only needs to be Poincaré invariant. Its central theses are that **real** representations are fundamental, that the CPT theorem follows from the most general Lorentz-covariant mass term, and that the relativistic position operator is real exactly for spin-$\frac{1}{2}$. The verified content here captures the algebraic spine of those claims.

27.3. The Lorentz Group and Its Discrete Subgroup

The behaviour of symmetries is governed by the Lorentz group $O(1,3)$ of matrices preserving the Minkowski metric $\eta = \mathrm{diag}(1,-1,-1,-1)$. Its key structural facts are that it is a group (closed under inverse), that every element has determinant ± 1, and that it carries a discrete subgroup $\Delta = \{1, \eta, -\eta, -1\}$ — the parity × time-reversal factor, a copy of the Klein four-group:

```
#check @BookProof.LorentzGroup.IsLorentz
#check @BookProof.LorentzGroup.isLorentz_inv
#check @BookProof.LorentzGroup.lorentz_det_sq_one
#check @BookProof.LorentzGroup.Delta
#check @BookProof.LorentzGroup.delta_card_four
```

Inside $O(1,3)$ lives the proper orthochronous subgroup $SO^+(1,3) = \{\lambda : \det\lambda = 1,\ \lambda^0{}_0 > 0\}$, which is **normal** in $O(1,3)$, and every Lorentz matrix factors uniquely as $\lambda = \delta s$ with $\delta \in \Delta$ and $s \in SO^+(1,3)$:

```
#check @BookProof.LorentzOrthochronous.IsProperOrthochronous
#check @BookProof.LorentzOrthochronous.isPO_conj
#check @BookProof.LorentzDecomp.lorentz_delta_decomp
#check @BookProof.LorentzDecomp.lorentz_delta_decomp_unique
```

27.4. Real Representations of the Lorentz Group

The manuscript's key structural claim is that the finite-dimensional representations of $SL(2,\mathbb{C})$ are real, and that the Dirac/Majorana algebra decomposes into a direct sum of real invariant subspaces. In the Clifford algebra $\mathrm{Mat}(4\times 4,\mathbb{R})$, conjugation by the discrete Pin subgroup Ω preserves four mutually orthogonal real subspaces — the $(1/2,1/2)$ vector representation, the $(1,0)$ antisymmetric part, a pseudo-real part, and a two-dimensional piece — which together span the whole algebra as an internal direct sum:

```
#check @BookProof.ChapterLorentzRealRep.WHalf_invariant
#check @BookProof.ChapterLorentzRealRepSum.gram_half10R
#check @BookProof.ChapterLorentzRealRepSum.finrank_WHalf
#check @BookProof.ChapterLorentzRealRepFull.finrank_full_eq_add
#check @BookProof.ChapterLorentzRealRepDirect.WFam_isInternal
#check @BookProof.ChapterLorentzRealRepDirect.WFam_conj_invariant
```

The headline $16 = 4 + 6 + 4 + 2$ (an internal direct sum, each summand invariant under conjugation by the discrete Pin subgroup) is the algebraic content of "real representations": the Clifford algebra splits into real invariant pieces, each of which is the carrier of a real, and hence **real**, representation.

27.5. Pauli Matrices and the Double Cover

The bridge from $SL(2,\mathbb{C})$ to the Lorentz group is the hermitian-matrix parametrization of Minkowski 4-vectors, $X = x_\mu \sigma^\mu$ with $\det X = \langle x, x\rangle$. Spinor conjugation $X \mapsto T^\dagger X T$ preserves the Minkowski form (so $\Upsilon(T)$ is Lorentz), and the map is two-to-one — T and $-T$ induce the same conjugation, which is the ± 1 kernel of the double cover; restricting to $SU(2)$ gives the $SU(2) \to SO(3)$ double cover:

```
#check @BookProof.ChapterPauliLorentz.hermMat
#check @BookProof.ChapterPauliLorentz.det_hermMat
#check @BookProof.ChapterPauliLorentz.spinorMap_preserves_mink
#check @BookProof.ChapterPauliSU2.spinorAction_neg
#check @BookProof.ChapterPauliSU2.su2_preserves_spatialNormSq
```

27.6. The Poincaré Group and the Little Group

Moving to the full Poincaré (inhomogeneous) group, the manuscript builds it as a semidirect product of translations and the Lorentz group, and analyzes representations by their **little group** — the stabilizer of a reference momentum. The verified facts are the Poincaré semidirect product law, the little group as a centralizer, and the $SE(2)$ translation subgroup that appears for the massless lightlike case:

```
#check @BookProof.ChapterIPin.IPin
#check @BookProof.ChapterIPin.ipin_right
#check @BookProof.ChapterIPin.ipin_left
#check @BookProof.ChapterLittleGroup.littleGroup
#check @BookProof.ChapterLittleGroup.prop79
#check @BookProof.ChapterSE2.Nmat_mul
#check @BookProof.ChapterSE2.NmatHom
```

27.7. Majorana Spinors and the CPT Theorem

The manuscript's CPT theorem falls out of the most general Lorentz-covariant Dirac mass term $iH = \partial \cdot \gamma\, \gamma^0 + i\gamma^0 m_1 + \gamma^0\gamma^5 m_2$. The verified facts are: the Hamiltonian is self-adjoint, its square is the relativistic mass shell $D^2 = -(k^2 + m_1^2 + m_2^2)\cdot 1$, the term m_2 is parity-odd (the sole source of CP violation), and the whole operator is PT/CPT invariant. Setting $m_2 = 0$ recovers exact parity invariance:

```
#check @BookProof.ChapterCPTHamiltonian.diracHamOp
#check @BookProof.ChapterCPTHamiltonian.diracHamOp_conjTranspose
#check @BookProof.ChapterCPTHamiltonian.diracHamOp_sq
#check @BookProof.ChapterCPTParity.parity_MassB
#check @BookProof.ChapterCPTParity.parity_diracHamOp
#check @BookProof.ChapterCPTParity.parity_diracHamOp_invariant
#check @BookProof.ChapterCPTPT.cpt_diracHamOp
```

Majorana spinors make the real-rep thesis concrete: a field operator $a(v) = \iota Q v$ generating a Clifford algebra is **self-adjoint**, so a particle is its own antiparticle. The canonical anticommutation relation is $\{a(v), a(w)\} = 2\langle v, w\rangle \cdot 1$, and real symmetries preserve self-adjointness:

```
#check @BookProof.MajoranaClifford.a
#check @BookProof.MajoranaClifford.a_sq
#check @BookProof.MajoranaClifford.car
#check @BookProof.MajoranaClifford.a_selfAdjoint
#check @BookProof.MajoranaClifford.a_map_selfAdjoint
```

The manuscript also develops the Majorana–Fourier machinery: the unitary boost blocks, the energy–momentum transform, and the intertwining of the Dirac operator with the inverse boost. Each is verified as a unitarity or intertwining statement:

```
#check @BookProof.ChapterMajoranaFourier.majoranaFourier_boostBlock_unitary
#check @BookProof.ChapterMajoranaProp61.prop61_isUnit
#check @BookProof.ChapterMajoranaProp74.prop74_intertwine
#check @BookProof.ChapterMajoranaProp76.energyMomentum_unitary
#check @BookProof.ChapterMajoranaProp76.fourierTransform_isNote4Unitary
```

27.8. The Relativistic Position Operator

The manuscript's corollary is that a localizable representation of the Poincaré group is irreducible iff it is real and massive spin-$\frac{1}{2}$ (or massless helicity-$\frac{1}{2}$), in which case the position operator matches the coordinates of the Dirac equation. The verified algebraic fact underpinning the obstruction is that the time-component generator $i\gamma^0$ does **not** commute with the spatial generators $(i\gamma^i)(i\gamma^0)$, while it **does** commute with the rotation generators $(i\gamma^i)(i\gamma^j)$ (consistent with the massive little group $SU(2)$):

```
#check @BookProof.ChapterLocalization.gamma0_not_comm_spatial
#check @BookProof.ChapterLocalization.comm_gamma0_rotation
```

This is the load-bearing algebraic fact behind the claim that no closed position operator exists for the complex/spin-1 case while one does for the real spin-$\frac{1}{2}$ case: the obstructed generators are precisely those that fail to commute with the time generator.

27.9. The Bosonic CCR

For completeness, the manuscript's quantization section records the bosonic canonical commutation relations that replace the fermionic anticommutators. The verified content is that the CCR $[a(v), a(w)] = (i\langle v, Jw\rangle)\cdot 1$ is

symplectically invariant, that the number-operator relation $[cre, ann] = (2\|v\|^2) \cdot 1$ holds, and that the involution swaps annihilation and creation:

```
#check @BookProof.Bosonic.BosonicCCR
#check @BookProof.Bosonic.ccr_symplectic_invariant
#check @BookProof.Bosonic.commutator_cre_ann
#check @BookProof.Bosonic.star_ann
```

27.10. Summary

The relativistic structure of the manuscript, in its verified form:

- the Lorentz group $O(1,3)$ with discrete subgroup $\Delta \cong \mathbb{Z}_2 \times \mathbb{Z}_2$, and $O(1,3) = \Delta \ltimes SO^+(1,3)$;
- real representations: the Clifford algebra splits as an internal direct sum of real invariant subspaces, $16 = 4 + 6 + 4 + 2$;
- the hermitian-matrix/double-cover bridge $SL(2,\mathbb{C}) \to O(1,3)$ and $SU(2) \to SO(3)$;
- the Poincaré semidirect product and its little groups (including $SE(2)$);
- the CPT theorem from the most general covariant mass term, with m_2 the parity-breaking parameter;
- Majorana self-adjointness (particle = its own antiparticle) and the unitary Majorana–Fourier transforms;
- the position-operator obstruction: the time generator fails to commute with spatial generators.

28. Quantization Due to Time-Evolution: Yang–Mills and Classical Statistical Field Theory

28.1. Orientation and Status

The quantization chapter of the Standard-Model thread: the manuscript's proposal that quantization **arises from time-evolution** rather than being postulated, worked out for Yang–Mills. It needs the BRST vocabulary of the gauge-symmetry chapter and belongs with real representations before it and the Gribov ambiguity after it.

The route: the problem (the inconsistencies of quantum Yang–Mills, classical statistical field theory, and quantum gravity), the structure constants of SU(3), the field strength and the Bianchi identity, the Weyl relations as the engine of quantization, the nilpotent BRST charge, positivity of the Weyl–Gauge Hamiltonian, its quadratic form and the Friedrichs route, the Friedrichs extension without a boundedness hypothesis, and the field-space realization on the product Hermite core.

The proofs include `BookProof/QuantizationWeyl`, `BookProof/WeylHamiltonian`, the Yang–Mills chapters (`YangMillsSU`, `YangMillsFieldStrength`, `YangMillsBianchi`, `YangMillsFriedrichs` and its limit/abelian variants), and the graded Fock/Friedrichs/Hashimoto chapters that the field-space realization rests on.

28.2. The Problem

The manuscript observes that Quantum Yang–Mills theory, Classical Statistical Field Theory (for Hamiltonians non-polynomial in the fields) and Quantum Gravity all suffer from severe mathematical inconsistencies. Its proposal is to define a class of statistical field theories where the probability distribution of the infinite-dimensional phase-space is defined by a wave-function in a Fock space, allowing non-polynomial Hamiltonians; and to define gauge symmetries through algebraic ideals. The verified content here is the algebraic core of that programme: the structure constants and Jacobi identity of $SU(3)$, the field strength and Bianchi identity, the Weyl/CCR quantization relations that arise from time-evolution, the nilpotent BRST charge, and the positive-definite Weyl-gauge Hamiltonian.

28.3. The Structure Constants of SU(3)

Non-abelian gauge theory is controlled by the structure constants f_{abc} of the gauge group. For unitary generators T_a with $\mathrm{tr}(T_a T_b) = \frac{1}{2}\delta_{ab}$ and $[T_a, T_b] = i f_{abc} T_c$, the structure constants are **totally antisymmetric** and satisfy the **Jacobi identity** — the algebraic conditions that make the Lie algebra of $SU(3)$ consistent:

```
#check @BookProof.YangMillsSU3.TraceOrthonormal
#check @BookProof.YangMillsSU3.ClosesWithStructureConstants
#check @BookProof.YangMillsSU3.structureConstant_formula
#check @BookProof.YangMillsSU3.structureConstant_totally_antisymmetric
#check @BookProof.YangMillsSU3.structureConstant_jacobi
```

Concretely, the Gell-Mann matrices λ^a are hermitian, traceless and $\mathrm{tr}(\lambda^a \lambda^b) = 2\delta_{ab}$, and the generators $T_a = \frac{1}{2}\lambda^a$ realize the abstract orthonormal structure:

```
#check @BookProof.ChapterGellMann.gellMann_isHermitian
#check @BookProof.ChapterGellMann.gellMann_trace_orthonormal
#check @BookProof.ChapterGellMann.su3gen_traceOrthonormal
```

28.4. The Field Strength and the Bianchi Identity

The covariant derivative $D_j = \partial_j + a_j$ and the field strength $F_{jk} = \{D_j, D_k\}$ satisfy the key identity: the commutator of two covariant derivatives is multiplication by the field strength, and the field strength is antisymmetric. The Bianchi identity $\varepsilon_{ijk}[D_i, [D_j, D_k]] = 0$ is the cyclic Jacobi identity for double commutators:

```
#check @BookProof.YangMillsFieldStrength.nonabelian_fieldStrength
#check @BookProof.YangMillsFieldStrength.fieldStrengthMul_antisymm
#check @BookProof.YangMillsBianchi.bianchi
#check @BookProof.YangMillsBianchi.bianchi_fieldStrength
```

Specializing to the book's connection $a_j = -igA_j$ recovers the physical field-strength combination $F^book_{jk} = (\partial_j A_k - \partial_k A_j) - ig[A_j, A_k]$:

```
#check @BookProof.YangMillsFieldStrength.Fbook
#check @BookProof.YangMillsFieldStrength.commutator_eq_coupling
```

28.5. Quantization Arises from Time-Evolution: the Weyl Relations

The manuscript's central slogan is that quantization comes from **time-evolution** rather than being imposed. The verified content is the finite Heisenberg/Weyl algebra: position X, momentum Y and central Z obey the CCR $[X, Y] = Z$, Z is central with $Z^2 = 0$, and the Weyl relation $e^{aX}e^{bY} = e^{aX+bY+(ab/2)Z}$ encodes the exponentiated commutation:

```
#check @BookProof.QuantizationWeyl.comm_XY
#check @BookProof.QuantizationWeyl.Zgen_central
#check @BookProof.QuantizationWeyl.Heis_mul
#check @BookProof.QuantizationWeyl.weyl
#check @BookProof.QuantizationWeyl.weyl_shift
```

The conjugation identity $e^{aX}e^{bY}e^{-aX} = e^{bY+abZ}$ is the quantization-by-time-evolution statement: acting by a momentum translation shifts the position coordinate by the commutator.

28.6. The Nilpotent BRST Charge

Gauge symmetries are defined through algebraic ideals, and the physical (gauge-invariant) algebra is the BRST cohomology — which is well-defined because the BRST charge is nilpotent, $\Omega^2 = 0$. The cubic ghost part $Q = \sum_{a,b,e} f_{abe}(\chi_a \chi_b \beta_e)$ of the charge squares to zero, given the canonical anticommutation relations and the Jacobi identity for the structure constants:

```
#check @BookProof.BRSTNilpotent.GhostCAR
#check @BookProof.BRSTNilpotent.Q
#check @BookProof.BRSTNilpotent.brst_charge_nilpotent
```

At the level of the single fermionic ghost factor, the operators $\psi, \psi^\dagger$ satisfy $\{\psi, \psi^\dagger\} = 1$, $\psi^2 = 0$, and the abstract nilpotency $(b \cdot f)^2 = 0$ for a commuting b and square-zero f:

```
#check @BookProof.GhostField.car
#check @BookProof.GhostField.psi_sq
#check @BookProof.GhostField.brst_charge_nilpotent
```

The super-bracket $\{a, b\} = ab - (-1)^{pq} ba$ with Koszul sign makes the graded algebra a Lie superalgebra, satisfying the graded Jacobi identity — the algebraic skeleton of the fermionic ghost structure:

```
#check @BookProof.ChapterSuperBracket.sbracket
#check @BookProof.ChapterSuperBracket.super_jacobi
#check @BookProof.ChapterSuperBracket.commutator_jacobi
```

28.7. The Weyl-Gauge Hamiltonian Is Positive

The new gap-proof layer makes the logical order precise. `ChapterRitzCertificate.lean` turns a finite Ritz value and residual into a spectral band using Temple's inequality, but only with a separation hypothesis. `ChapterTempleSeparationNecessary.lean` proves that this hypothesis cannot be removed: Rayleigh data and residuals alone do not control an unseen lower eigenvalue. `ChapterBandEnclosure.lean` and `ChapterFockOneParticleGap.lean` then state the conditional bridge from compatible finite bands to the one-particle edge.

For the mass-gap route, distinguish the real one-particle Hamiltonian from its finite certificate and from the outer-particle Fock lift. The positivity below is the input for shifting the one-particle operator and applying the Friedrichs/Hashimoto construction; it is not, by itself, a statement that a finite numerical gap has already transferred to the full Hamiltonian.

In the Weyl gauge the Yang–Mills Hamiltonian density is $H_W = \frac{1}{2}\sum_i \pi_i^2 + \frac{1}{2}\sum_a B_a^2$ — a sum of squares of self-adjoint operators — hence a **positive** operator: every expectation value is non-negative. This is the fact the manuscript invokes for the mass-gap discussion:

```
#check @BookProof.WeylHamiltonian.weylHamiltonian
#check @BookProof.WeylHamiltonian.weylHamiltonian_isPositive
#check @BookProof.WeylHamiltonian.weylHamiltonian_expectation_nonneg
#check @BookProof.WeylHamiltonian.weylHamiltonian_isSelfAdjoint
```

28.8. The Weyl-Gauge Quadratic Form and the Friedrichs Route

`BookProof/Toolbox/ChapterFriedrichsFormGap.lean` supplies the form-theoretic part of this route. The claim is that a semibounded symmetric quadratic form determines a positive Friedrichs extension and that the finite form restrictions preserve the variational information needed for a gap. The proof uses closability, form-domain inclusion, and the variational characterization; it does not identify a continuum gap without a separate convergence theorem.

Manuscript provenance — why the Hamiltonian may be taken quadratic in the ladders. The manuscript fixes this shape and justifies the passage to the outer level by a recursive construction: **“Since the second-quantization procedure can be applied recursively … in case the Hamiltonian acting in the Fock-space is not quadratic in the creation/annihilation operators, then we can consider instead a new Fock-space where the base Hilbert space is the original Fock-space. The new Hamiltonian is quadratic in the creation/annihilation operators.”** It writes each sector's Hamiltonian as a one-particle operator enclosed between creation and annihilation, $H = \int d^3\vec{x}\, a^\dagger(\vec{x})\, H(\vec{x})\, a(\vec{x})$ — its Navier–Stokes instance is $H(\vec{x}) = \pi^i(u_j u_{i,j} - \nu u_{i,jj}) + (h.c.)$, and the Yang–Mills instance is the same envelope with the one-particle Weyl Hamiltonian $\frac{1}{2}\sum_i \pi_i^2 + \frac{1}{2}\sum_a B_a^2$ on the Gauss–polynomial core of $L^2(\mathbb{R}^{99})$. One point of scope, because the phrase "spatial derivative of a field" occurs in more than one sector: the passage to a momentum-space **convolution** is the device of the Navier–Stokes and quantum-gravity routes, where a **product** of fields carrying a spatial derivative has to be written without a differential operator (`BookProof.NsAdvectionConvolution`, and the QG elimination chapters `ChapterQgFourierElimination` / `ChapterQgFullEliminated`). Yang–Mills needs no such device: in the Weyl-gauge field space of this chapter the coordinates $A_{j,a}$ and the momenta π —

together with the magnetic field B, which is a polynomial in them — are already the independent variables, so no product of fields with a spatial derivative ever has to be reinterpreted. Nothing is gauge **fixed** by the convolution in the other two sectors either; it is a change of description, not a constraint.

Because the one-particle Hamiltonian is bounded below, choose an energy origin so that $h_+ \geq \mu\, I$ for some $\mu > 0$. For free outer particles this changes the Fock Hamiltonian by $\mu\, N$, not by a vacuum constant: $N\,\Omega = 0$, while every non-vacuum finite-particle energy is a sum of strictly positive one-particle energies. Thus the physical observable for the remaining mass-gap theorem is the lowest positive one-particle energy, measured through the creation/destruction sandwich and then lifted by `dGamma`.

Positivity of the Weyl-gauge Hamiltonian is the hypothesis of the **Friedrichs extension theorem**: a densely defined symmetric operator that is bounded below has a canonical positive self-adjoint extension. For the unbounded (densely defined) version the sum-of-squares structure survives: on a domain left invariant by the electric and magnetic field operators, $H = \frac{1}{2}\sum_i \pi_i^2 + \frac{1}{2}\sum_a B_a^2$ is symmetric and its quadratic form is $q(x) = \frac{1}{2}\sum_i \|\pi_i x\|^2 + \frac{1}{2}\sum_a \|B_a x\|^2 \geq 0$:

```
#check @BookProof.YangMillsFriedrichs.weylOpDom
#check @BookProof.YangMillsFriedrichs.weylOpDom_symmetricOn
#check @BookProof.YangMillsFriedrichs.weylOpDom_quadForm
#check @BookProof.YangMillsFriedrichs.weylOpDom_quadForm_nonneg
```

The analytic heart of the Friedrichs construction is that the form $\langle x, y\rangle_H = \langle x, y\rangle + \langle x, Hy\rangle$ is a genuine inner product dominating the ambient norm, and that it is **closable**: a sequence that is Cauchy in the form norm and tends to 0 in the ambient space has form norm tending to 0, so the closed form — and with it the extension — is well defined. This is proved here in general and applied to the Weyl-gauge form:

```
#check @BookProof.YangMillsFriedrichs.formInner
#check @BookProof.YangMillsFriedrichs.formNormSq_ge_normSq
#check @BookProof.YangMillsFriedrichs.re_formInner_sq_le
#check @BookProof.YangMillsFriedrichs.form_closable
#check @BookProof.YangMillsFriedrichs.weylForm_closable
```

The Friedrichs theorem itself was first carried as a **named hypothesis** (Friedrichs 1934; Reed–Simon Thm X.23), never as an axiom, and shown to be satisfiable; the conclusion for the Weyl-gauge Hamiltonian was conditional on it. That hypothesis is now **discharged outright** — see the section "The Friedrichs Extension Without a Boundedness Hypothesis" below — so the statements in this block are the conditional forms that the unconditional theorem supersedes. Nothing is claimed about the continuum Yang–Mills operator, and the mass gap remains out of scope. The Hashimoto/SIRK order-n approximations supply the proved supporting facts — the best-approximation error is antitone in the order and, for a cyclic seed, tends to 0 — while the identification of the limit with the Friedrichs extension is recorded as a conjecture, not a theorem:

```
#check @BookProof.YangMillsFriedrichs.IsPositiveSelfAdjointExtension
#check @BookProof.YangMillsFriedrichs.friedrichs_extension_of_semibounded
#check @BookProof.YangMillsFriedrichs.friedrichs_hypothesis_satisfiable
#check @BookProof.YangMillsFriedrichs.weyl_friedrichs_extension
#check @BookProof.YangMillsFriedrichs.weylKrylov_bestApprox_antitone
#check @BookProof.YangMillsFriedrichs.weylKrylov_bestApprox_tendsto_zero
```

In the **bounded** regime the named hypothesis is no longer needed: the positive self-adjoint extension is constructed outright as the continuous extension of the operator from its dense domain, and the construction applies to genuinely proper dense domains (in $\ell^2(\mathbb{N})$ the span of the canonical basis is dense but misses every vector of infinite support):

```
#check @BookProof.YangMillsFriedrichsLimit.friedrichs_of_bounded
#check @BookProof.YangMillsFriedrichsLimit.friedrichs_bounded_proper_domain_example
```

In the same regime the Hashimoto/SIRK limit becomes an operator limit: the order-n compressions $P_n A P_n$ converge strongly to A for any cyclic seed, and no other bounded operator agrees with the limit on the Krylov flag, so the infinite Hashimoto limit recovers exactly the positive self-adjoint extension of the Weyl-gauge Hamiltonian. The unbounded continuum case remains open:

```
#check @BookProof.YangMillsFriedrichsLimit.sirk_compression_tendsto
#check @BookProof.YangMillsFriedrichsLimit.sirk_limit_unique
```

```
#check @BookProof.YangMillsFriedrichsLimit.sirk_limit_eq_positive_selfadjoint_extensi
#check @BookProof.YangMillsFriedrichsLimit.weyl_friedrichs_bounded
```

The same analysis can be run in the **Hermite (oscillator) basis** rather than along a Krylov flag, and there it takes its classical Rayleigh–Ritz shape. Truncating to the span of the first m basis vectors replaces the Hamiltonian by its compression $P_m H P_m$, and on that subspace the compression carries exactly the energy form of H; the resulting Ritz values are antitone in m and converge to the infimum of the energy form over the whole finite-mode domain. Neither statement needs a boundedness hypothesis:

```
#check @BookProof.HermiteGalerkin.inner_galerkinCompression
#check @BookProof.HermiteGalerkin.ritzInf_antitone
#check @BookProof.HermiteGalerkin.ritzInf_tendsto_domainInf
#check @BookProof.HermiteGalerkin.ritzInf_extension_le
#check @BookProof.HermiteGalerkin.exists_mem_galerkinSpan
#check @BookProof.HermiteGalerkin.galerkinProj_tendsto
```

In the bounded regime the truncations converge in the strong **resolvent** sense — $(P_m A P_m - z)^{-1} \to (A - z)^{-1}$ strongly at every non-real z — and the positive self-adjoint extension the algorithm converges to is the only one there is, so no boundary condition has to be supplied to the algorithm. The finite-mode domain is a proper dense subspace, so this is not the degenerate case of an operator already defined everywhere. As everywhere above, the unbounded case is not claimed:

```
#check @BookProof.HermiteGalerkin.resolvent_tendsto_of_strong_tendsto
#check @BookProof.HermiteGalerkin.galerkinResolvent_tendsto
#check @BookProof.HermiteGalerkin.positive_selfadjoint_extension_unique
#check @BookProof.HermiteGalerkin.hermiteGalerkin_selects_friedrichs
#check @BookProof.HermiteGalerkin.finiteModeDomain_ne_top
```

28.9. The Friedrichs Extension Without a Boundedness Hypothesis

The two constructions above build the positive self-adjoint extension only in the **bounded** regime. The general theorem needs no boundedness at all, and it is proved here. The domain is retyped with the form inner product $\langle x, y\rangle_1 = \langle x, y\rangle + \langle x, Hy\rangle$; symmetry makes it Hermitian and positivity makes it positive definite, so the domain becomes an inner product space whose completion is a Hilbert space. Its inclusion into the ambient space is norm-decreasing, so it extends continuously, and the key identity $\langle x, k\rangle_1 = \langle x + Hx, \iota k\rangle$ — closability of the form in disguise — shows that extension is **injective**: the form completion adds no ghost vectors.

```
#check @BookProof.FriedrichsExtension.PosSymOp
#check @BookProof.FriedrichsExtension.FormDom.norm_toAmbient_le
#check @BookProof.FriedrichsExtension.FormDom.inner_coe_eq
#check @BookProof.FriedrichsExtension.FormDom.formExt_injective
```

Riesz representation in the form space then produces the resolvent $S = (H + 1)^{-1}$ as a bounded, injective, positive, self-adjoint operator of norm at most one which sends $x + Hx$ back to x, and the project's own shift-invert converse turns S into the extension $A = S^{-1} - 1$. The result is the Friedrichs extension theorem as a **theorem**: every densely defined, symmetric, positive operator on a complex Hilbert space has a positive self-adjoint extension — in particular the Weyl-gauge Yang–Mills Hamiltonian, with no boundedness of the electric or magnetic field operators assumed:

```
#check @BookProof.FriedrichsExtension.FormDom.friedrichsResolvent
#check @BookProof.FriedrichsExtension.FormDom.friedrichsResolvent_shift
#check @BookProof.FriedrichsExtension.friedrichs_extension_exists
#check @BookProof.FriedrichsExtension.friedrichs_hypothesis_holds
#check @BookProof.FriedrichsExtension.weyl_friedrichs_extension_unconditional
#check @BookProof.FriedrichsExtension.friedrichs_extension_of_semibounded_below
```

The last statement is the classical form of the theorem: symmetry plus a lower bound $\langle x, Hx\rangle \geq -c\|x\|^2$ — positivity is not needed — gives a self-adjoint extension with the same lower bound, by the shift $H \mapsto H + c$.

Combining the construction with the shift-invert theory closes the loop for the unbounded case: in the occupation-number (Hermite) realization the extension **exists**, its shift-invert $R = (A + \gamma)^{-1}$ is bounded and self-adjoint, the Galerkin truncations of R converge to it strongly and in the resolvent sense, and R determines A — so the infinite

Hashimoto/SIRK limit selects exactly the constructed extension. The construction is not vacuous: it applies to the genuinely unbounded diagonal operator $Ae_n = ne_n$ on $\ell^2(\mathbb{N})$. What stays outside is, as always, the mass gap:

```
#check @BookProof.FriedrichsExtension.friedrichs_hashimoto_selects
#check @BookProof.FriedrichsExtension.weyl_hashimoto_selects_friedrichs
#check @BookProof.FriedrichsExtension.unbounded_friedrichs_example
```

28.10. The Field-Space Realization on the Product Hermite Core

The perturbative boundary is now explicit — and largely closed — in the unbounded-perturbation ladder. `ChapterFockFieldPerturbation.lean` (now `sorry`-free) develops the field-level form estimates for the **linear** unbounded coupling $\Phi(f) = a^\dagger(f) + a(f)$ and proves the surviving gap `(μ - 2‖f‖)‖u‖²`. `ChapterFockPairPerturbation.lean` extends this to the **quadratic**, pair-creating operator $P(f,g) = a^\dagger(f)a^\dagger(g) + a(g)a(f)$ — the unbounded form estimate that the abstract bounded theorem could not supply — leaving the gap `(μ - 2√2‖f‖‖g‖)‖u‖²`. `ChapterFockInteractionStability.lean` records the quantitative degradation of a positive edge under a relative form bound, and `ChapterFockCubicUnbounded.lean` proves the boundary of the whole route: a bare cubic term admits no relative form bound, so `dΓ(N) + λC` is unbounded below at every coupling strength, and only the quartic term (bounded below) restores a lower bound. `ChapterFockCubicQuarticStability.lean` then converts that boundary result into a stability theorem on all finite states: the cubic–quartic pair `λC_k + Q_k` is jointly bounded below by `-(2λ² + (2λ² + ½ - μ)²/2)‖u‖²` against the free form — one copy of the free form pays for every mode of a finite set, the constant growing linearly in its cardinality. The picture is exact: degree one and degree two keep the gap; degree three alone destroys it; degree three **with** degree four is semibounded (stable, though not gapped).

The statements above are formulated for abstract electric and magnetic operators on the finite-mode domain of an orthonormal basis. The field-space realization makes them concrete on $L^2(\mathbb{R}^{99})$, the configuration space with $99 = 3 + 24 + 72$ coordinates: three spatial coordinates, the $24 = 3 \times 8$ gauge fields $A_{j,a}$ and the $72 = 3 \times 3 \times 8$ independent derivative coordinates $\partial_j A_{k,a}$. The core is the span of the **product Hermite functions** — equivalently, of all $p(x)\, e^{-\|x\|^2/4}$ with p a polynomial, since the products $\prod_i He_{\alpha_i}(x_i)$ of probabilists' Hermite polynomials span the whole polynomial ring. The map $p \mapsto p\, e^{-\|x\|^2/4}$ is injective, the core is dense, and Gram–Schmidt turns the enumerated monomials into an orthonormal basis whose finite-mode domain is exactly the core:

```
#check @BookProof.HermiteProductCore.pgMap_injective
#check @BookProof.HermiteProductCore.polyGaussCore_dense
#check @BookProof.HermiteProductCore.gaussInt_pderiv
#check @BookProof.HermiteProductCore.hermiteMv
#check @BookProof.HermiteProductCore.polyGaussCore_eq_hermiteSpan
#check @BookProof.HermiteProductCore.coreBasis
#check @BookProof.HermiteProductCore.span_range_coreBasis
```

The core is built from Hermite material, and the companion chapter `BookProof/Toolbox/ChapterHermiteFunctions.lean` (namespace `BookProof.HermiteCore`) supplies that material from scratch: the probabilists' Hermite polynomials `hermiteR` with their ODE (`hermiteR_ode`) and integration-by-parts identity (`gint_ibp`), the normalized Hermite functions `hermiteLp`, their orthonormality `orthonormal_hermiteLp`, their density `hermiteLp_span_dense` — assembled into the Hilbert basis `hermiteBasis` — and the eigen-relation `hermiteFun_oscillator`: the functions $H_n e^{-x^2/4}$ are exactly the eigenfunctions of the harmonic oscillator $-\partial^2 + x^2/4$, which is the comparison operator the QYM Faris–Lavine route runs on. The completeness of the basis is proved internally, by a moment/Fourier argument, so the Gauss-polynomial core above is anchored in a basis that does not presuppose the spectral theory it is used to establish.

```
#check @BookProof.HermiteCore.hermiteR
#check @BookProof.HermiteCore.hermiteR_ode
#check @BookProof.HermiteCore.gint_ibp
#check @BookProof.HermiteCore.orthonormal_hermiteLp
#check @BookProof.HermiteCore.hermiteLp_span_dense
#check @BookProof.HermiteCore.hermiteBasis
#check @BookProof.HermiteCore.hermiteFun_oscillator
```

On that core the fields act as genuine operators: multiplication by the coordinate $A_{j,a}$, the momentum $\pi = -i\,\partial/\partial A$ (symmetric by Gaussian integration by parts), and the magnetic field $B_{ia} = \varepsilon_{ijk}(\partial_j A_{k,a} +$

$f_{abc}A_{j,b}A_{k,c}$), which is multiplication by a polynomial with **real** coefficients and hence symmetric. The coordinate and momentum operators do not commute — $[A_j, \pi_j] = i$ — so products of them are Weyl ordered as $\frac{1}{2}(PQ + QP)$, and the Weyl-ordered product of two symmetric operators is symmetric:

```
#check @BookProof.YangMillsHermite.mulOp_polySym
#check @BookProof.YangMillsHermite.momOp_polySym
#check @BookProof.YangMillsHermite.commutator_coord_mom
#check @BookProof.YangMillsHermite.weylProd_polySym
#check @BookProof.YangMillsHermite.magPoly
#check @BookProof.YangMillsHermite.realCoeff_magPoly
```

The Hamiltonian is taken in the **positive** sum-of-squares convention $H_1 = \frac{1}{2}\sum \pi^2 + \frac{1}{2}\sum B^2$ (the manuscript writes $-\frac{1}{2}\pi\pi - \frac{1}{2}BB$; the positive sign is the one bounded below, to which the Friedrichs machinery applies). It is well defined, symmetric and positive on the core, its quadratic form is a sum of squares, and therefore the Friedrichs extension theorem and the Hashimoto/SIRK selection theorem apply to the concrete field-space operator:

```
#check @BookProof.YangMillsHermite.ymHamiltonian
#check @BookProof.YangMillsHermite.ymHamiltonian_quadForm
#check @BookProof.YangMillsHermite.ymHamiltonian_symmetricOn
#check @BookProof.YangMillsHermite.ymHamiltonian_quadForm_nonneg
#check @BookProof.YangMillsHermite.ym_hermite_friedrichs_extension
#check @BookProof.YangMillsHermite.ym_hermite_hashimoto_selects
```

Finally the one-particle Hamiltonian is **enclosed and second quantized** on the outer finite-occupation states over the inner core. This is the final-Hamiltonian convention for QYM, QED, QG, and NS: retain the complete inner h (including inner pair terms), and write the outer operator with creation on the left and annihilation on the right. The Fock space is ℓ^2 over the configurations $\mathbb{N} \to_0 \mathbb{N}$ (occupation numbers, finitely many modes excited); the ladder operators satisfy $[a_j, a_j^\dagger] = 1$ and the adjoint pairing $\langle a_j^\dagger u, v\rangle = \langle u, a_j v\rangle$, and $d\Gamma(A) = \sum_{j,k}\langle e_j, Ae_k\rangle C_j^\dagger A_k$ restricts on the one-particle sector to A itself. Hermiticity and positivity of the one-particle matrix make $d\Gamma(A)$ symmetric and positive on the dense finite-occupation domain, so it too has a positive self-adjoint (Friedrichs) extension — in particular for the field-space Yang–Mills $H_1 = \frac{1}{2}\sum \pi^2 + \frac{1}{2}\sum B^2$:

```
#check @BookProof.FockSecondQuantization.ccr_annA_creA
#check @BookProof.FockSecondQuantization.inner_creA_left
#check @BookProof.FockSecondQuantization.dGamma
#check @BookProof.FockSecondQuantization.dGamma_one_particle
#check @BookProof.FockSecondQuantization.dGammaOp_symmetricOn
#check @BookProof.FockSecondQuantization.dGammaOp_quadForm_nonneg
#check @BookProof.FockSecondQuantization.secondQuantization_friedrichs
#check @BookProof.FockSecondQuantization.ym_fock_friedrichs_extension
#check @BookProof.FockSecondQuantization.finiteModeDomain_fockBasisN
#check @BookProof.FockSecondQuantization.dGamma_hashimoto_selects
#check @BookProof.FockSecondQuantization.secondQuantization_hashimoto_selects
#check @BookProof.FockSecondQuantization.ym_fock_hashimoto_selects
```

The manuscript's Hilbert space is not the symmetric Fock space alone but the **graded** tensor product $\Gamma^s \otimes \Gamma^a$ of a symmetric (bosonic) and an antisymmetric (fermionic, ghost) factor, on which the creation and annihilation operators form a graded Lie superalgebra. The antisymmetric factor is built in `BookProof.FermionFock`: a fermionic configuration **is** its finite set of occupied modes (so the Pauli principle is part of the type), the Jordan–Wigner sign $(-1)^{\#\{i \in S : i < j\}}$ gives the creation and annihilation operators, and they satisfy all four canonical **anti**commutation relations, are formal adjoints of each other, and second quantize exactly as in the bosonic case — with the same Friedrichs extension and Hashimoto/SIRK selection theorems. The abstract ghost relations `GhostCAR` assumed by the BRST chapter are realized by these operators, so the nilpotency $Q^2 = 0$ holds for a concrete operator algebra:

```
#check @BookProof.FermionFock.car_annF_creF_self
#check @BookProof.FermionFock.creF_creF_self
#check @BookProof.FermionFock.car_annF_annF
#check @BookProof.FermionFock.car_annF_creF_of_ne
#check @BookProof.FermionFock.inner_creF_left
#check @BookProof.FermionFock.dGammaF_one_particle
#check @BookProof.FermionFock.dGammaF_friedrichs_extension
#check @BookProof.FermionFock.secondQuantizationF_friedrichs
```

```
#check @BookProof.FermionFock.dGammaF_hashimoto_selects
#check @BookProof.FermionFock.parityF_involutive
#check @BookProof.FermionFock.ghostCAR_creF_annF
#check @BookProof.FermionFock.brst_charge_nilpotent_fermiFock
```

Putting the two factors together, `BookProof.GradedFock` realizes the graded Fock space as ℓ^2 over the pairs (bosonic configuration, fermionic occupied set), lifts the operators of either factor to it — operators on different factors commute — and proves the manuscript's **single unified graded relation**: with the Koszul sign $\varepsilon(p,q) = (-1)^{pq}$ of the super-bracket, $[\![a(p,j), a^\dagger(q,k)]\!] = \delta_{pq}\delta_{jk}$, which is the bosonic **commutator** CCR when both operators are even, the fermionic **anticommutator** CAR when both are odd, and 0 across the two sectors. The fermion-number parity $(-1)^{N_f}$ is the $\mathbb{Z}_2$ grading operator: an involution that commutes with the bosonic operators, anticommutes with the fermionic ones, and splits the graded Fock space into its even and odd parts:

```
#check @BookProof.GradedFock.liftFst_liftSnd_comm
#check @BookProof.GradedFock.super_canonical
#check @BookProof.GradedFock.super_canonical_cre
#check @BookProof.GradedFock.super_canonical_ann
#check @BookProof.GradedFock.gradeOp_involutive
#check @BookProof.GradedFock.gradeOp_bcre
#check @BookProof.GradedFock.gradeOp_fcre
#check @BookProof.GradedFock.even_add_odd
#check @BookProof.GradedFock.gradeOp_evenPart
#check @BookProof.GradedFock.gradeOp_oddPart
```

The analytic half of that construction lives on the graded space itself, not factorwise. `BookProof.GradedFriedrichs` shows that the total graded Hamiltonian $d\Gamma^s(A)\otimes 1 + 1\otimes d\Gamma^a(B)$ is densely defined, symmetric and positive on $\ell^2(\mathrm{Conf}\times\mathrm{FConf})$ and therefore has a positive self-adjoint (Friedrichs) extension, and `BookProof.GradedHashimoto` adds the two structural facts the two factors already had: the Hamiltonian is an **even** operator — it commutes with the $\mathbb{Z}_2$ grading $(-1)^{N_f}$ and so preserves the even and the odd subspace — and the Hashimoto/SIRK shift-invert limit selects exactly its Friedrichs extension, with the Galerkin truncations converging strongly and in the resolvent sense. The total number operator $N_b\otimes 1 + 1\otimes N_f$ is a concrete instance, so neither statement is vacuous.

```
#check @BookProof.GradedFriedrichs.gradedHamiltonian_friedrichs_extension
#check @BookProof.GradedFriedrichs.gradedSecondQuantization_friedrichs
#check @BookProof.GradedFriedrichs.gradedNumber_one_particle
#check @BookProof.GradedHashimoto.gradedHamiltonianAlg_otimes
#check @BookProof.GradedHashimoto.gradeOp_gradedHamiltonianAlg
#check @BookProof.GradedHashimoto.gradedHamiltonianAlg_evenPart
#check @BookProof.GradedHashimoto.gradedHamiltonianAlg_oddPart
#check @BookProof.GradedHashimoto.graded_hashimoto_selects
#check @BookProof.GradedHashimoto.gradedSecondQuantization_hashimoto_selects
#check @BookProof.GradedHashimoto.gradedNumber_hashimoto_selects
#check @BookProof.GradedHashimoto.graded_stone_flow
#check @BookProof.GradedHashimoto.gradedNumber_stone_flow
```

Each of these positive self-adjoint extensions now yields a **complete unitary flow**. `BookProof.ChapterStoneBridge` packages the passage from a selected self-adjoint extension to the one-parameter group Stone's theorem guarantees as the `IsStoneFlow` structure ($U0 = 1$, the group law, isometry of each Ut, and the Schrödinger equation on the domain), and `BookProof.ChapterStoneFlows` applies it to the concrete Fock Hamiltonian: the second-quantized Yang–Mills generator $d\Gamma(\frac12\sum\pi^2 + \frac12\sum B^2)$ on the finite-occupation domain over the product Hermite core has the complete unitary flow `ym_fock_stone_flow` — global in t, unitary, and agreeing with the Schrödinger evolution on the domain. This is the precise formal content of the manuscript's "wave-function evolution" of the gauge field configuration: the Fock representation is unitary, hence probability-conserving, at every instant.

```
#check @BookProof.StoneFlows.ym_fock_stone_flow
```

One sharper instrument has landed since the sections above, and it changes the status of this row. `BookProof/Qym/ChapterYangMillsNonAbelianEsa.lean` packages a single Kato-type theorem — every Weyl-type operator $\frac12\sum_m \pi^2_{\iota(m)} + \frac12\sum_j \Phi_j^2$ on the Gauss–polynomial core, with momenta in pairwise distinct coordinates and real polynomial fields, is essentially self-adjoint (`weylPoly_esa`: $2H + 1 = -\Delta_S + W$ with

$W = \sum_j \Phi_j^2 + 1 \geq 1$, the affine transfer `essentiallySelfAdjointOn_affine` from trivial deficiency at $\pm i$) — and applies it to the gauge-fixed Hamiltonian itself: `ym_h_esa` — the non-abelian `ymHamiltonian (coreRepPoly 99) fabc` is essentially self-adjoint on `polyGaussCore 99` for **arbitrary real** structure constants `fabc`, with no Friedrichs selection needed — and its enclosure `ym_dGamma_esa`: the final Hamiltonian in the creation-left / annihilation-right spelling is essentially self-adjoint on the finite-particle domain. What the Friedrichs route above contributes to this row is then the **selection** of the positive realization by the Hashimoto criterion, not existence or uniqueness of the self-adjoint realization. The gap statements stay on the one-particle h of this enclosure: the abelian case has no one-particle form gap (`ym_abelian_no_one_particle_form_gap`), and the non-abelian form gap — a gap of the direct-integral fibres uniform in the 72 derivative coordinates — is the single open analytic input of the gap programme.

```
#check @BookProof.YangMillsNonAbelianEsa.weylPoly_esa
#check @BookProof.YangMillsNonAbelianEsa.weylPoly_eq_hamCoreS
#check @BookProof.YangMillsNonAbelianEsa.essentiallySelfAdjointOn_affine
#check @BookProof.YangMillsNonAbelianEsa.ym_h_esa
#check @BookProof.YangMillsNonAbelianEsa.ym_dGamma_esa
#check @BookProof.YangMillsAbelianNoGap.ym_abelian_no_one_particle_form_gap
```

No continuum mass gap and no global existence statement is claimed anywhere in this section. The full nested-Fock ground-state statement is structural: once the sector one-particle operator is shifted positive and enclosed as $H = \sum_{i,j} h_{ij} C_i^\dagger A_j$, the outer vacuum is exactly the ground; what remains analytic is the identification and convergence of the positive one-particle edge.

28.11. Summary

The complete proof architecture is therefore: establish the one-particle edge under explicit spectral or Friedrichs hypotheses; enclose finite SIRK/Ritz data with Temple separation; prove stability of the finite bands; lift the positive one-particle operator with dGamma; and, only then, address unbounded interactions by form methods. `ChapterFockNumberPreservingGap.lean` proves the free number-preserving lift for arbitrary one-particle Hamiltonians, `ChapterFockInteractionStability.lean` the quantitative bounded/form-bounded extension, `ChapterFockFieldPerturbation.lean` and `ChapterFockPairPerturbation.lean` the unbounded linear and pair couplings, `ChapterFockCubicUnbounded.lean` the degree-three boundary of the route, and `ChapterFockCubicQuarticStability.lean` the all-states stabilisation of the cubic–quartic pair. The instantiation for the concrete 3D gauge-fixed QYM operator is now **done**: `ChapterYangMillsFockGapChain.lean` connects the gauge-fixed one-particle Hamiltonian $H_1 = \tfrac{1}{2}\Sigma\pi^2 + \tfrac{1}{2}\Sigma B^2$ (on the Gauss–polynomial core) to the matrix gap condition, proves $d\Gamma(H_1)\Omega = 0$ unconditionally, lifts a one-particle form gap to the nested-Fock mass gap (with the Friedrichs extension), covers the linear field perturbation, and consumes certified band data through `BandEnclosure` — everything conditional on the one-particle form gap, which remains the single analytic input the certificate chain must supply for Yang–Mills. Two sibling instantiations are **unconditional**: `ChapterScalaronFockGapChain.lean` runs the same chain for the R^2 scalaron sector at the constant one-particle energy $1/\sqrt{(12\alpha)}$ (the form gap is an identity there), and `ChapterFockDiagonalGapChain.lean` for any diagonal dispersion — with `freeField_fock_mass_gap` giving the free massive field the outer-Fock mass gap m outright. The conditional frontier is exactly where it was: the gauge-fixed Yang–Mills form gap.

The algebraic core of the manuscript's quantization programme:

- the totally antisymmetric structure constants and the Jacobi identity of $SU(3)$, realized by the Gell-Mann generators;
- the field-strength and Bianchi identities for the covariant derivative;
- quantization as arising from time-evolution via the Heisenberg/Weyl relations;
- the nilpotent BRST charge (and the graded Jacobi identity of the super-bracket) defining the gauge-invariant algebra;
- the positive-definite Weyl-gauge Hamiltonian;
- the densely-defined Weyl-gauge Hamiltonian, its sum-of-squares quadratic form and the closability of that form — the Friedrichs route to a self-adjoint extension;

- the Friedrichs extension theorem itself, proved with no boundedness hypothesis, and the resulting unbounded statement: the Weyl-gauge Hamiltonian has a positive self-adjoint extension and the infinite Hashimoto/SIRK limit selects exactly that extension;
- the field-space realization of that Hamiltonian on the dense product Hermite core of $L^2(\mathbb{R}^{99})$, with the coordinate, momentum and magnetic-field operators defined concretely, the canonical commutation relation and the Weyl ordering it forces, and the Friedrichs/Hashimoto theorems instantiated by it;
    - the fermionic (CAR) factor Γ^a with its own Friedrichs and Hashimoto theorems, the concrete realization of the BRST ghost relations, and the graded Fock space $\Gamma^s \otimes \Gamma^a$ on which one Koszul-signed formula (`super_canonical`) yields the bosonic CCR, the fermionic CAR and the vanishing mixed brackets at once, together with the $\mathbb{Z}_2$ grading;
    - the second-quantized Fock form of the same Hamiltonian and its complete unitary flow `ym_fock_stone_flow` — the Stone bridge turning each positive self-adjoint extension into a global, norm-preserving Schrödinger evolution.

The manuscript's Navier–Stokes existence/uniqueness thesis is deliberately **not** carried by any theorem in this book. What **is** formalized is the finite subset of the same programme that the free-field thread supports — the truncated Navier–Stokes Hamiltonian with a complete, norm-preserving flow and a unique global Cauchy solution, the nilpotent BRST divergence constraint, the Lagrangian (volume-preservation) change of variables, and the Faris–Lavine commutator criterion as the named-hypothesis route to essential self-adjointness. See the section "The Navier–Stokes Hamiltonian: a Complete Flow on the Truncation" in `Book/FreeField.lean`.

29. Timepiece and the Gribov Ambiguity

29.1. Orientation and Status

The manuscript ranges from the abelian free-field example to a proposed non-abelian field theory. This chapter keeps those levels separate: the abelian gauge-invariance and number-shift identities are verified, whereas a global non-abelian gauge-fixing and a continuum Yang--Mills mass-gap theorem are not yet formalized.

In the tour this chapter closes the Standard-Model thread of the part, after quantization by time-evolution. It is the place where the **mass gap** of the programme is stated at the abelian level and checked against the companion numerical validation, and the algebraic-ideals view of gauge at the end is the vocabulary the gravity chapter later reuses.

The route: the question, the abelian field strength and gauge invariance, the mass gap and observable invariance, the mass gap in the numerical validation, and the algebraic-ideals view of gauge.

The proofs are `BookProof/FreeEMField` and `BookProof/MassGap`.

29.2. The Question

The manuscript's chapter on the Gribov ambiguity considers statistical field theories in Minkowski space-time where the (classical) canonical coordinates, when modified by a non-deterministic time-evolution, verify the canonical commutation relations. Gauge symmetries are defined through algebraic ideals, and the chapter proposes definitions for Quantum Yang–Mills and Quantum Gravity, testing consistency with the quantization of the free electromagnetic field. The verified content here is the algebraic core: the abelian field strength and its gauge invariance (the "no Gribov ambiguity" case), and the mass-gap / observable invariance argument.

29.3. The Abelian Field Strength and Gauge Invariance

The Gribov ambiguity is a problem of **non-abelian** gauge fixing: the Dirac bracket requires a gauge-fixing that is both complete and unconstrained, which is not always possible. In the **abelian** (electromagnetic) case the situation is clean. The field strength is the linear curl $F_{jk} = \delta_j A_k - \delta_k A_j$, which is **gauge invariant** — invariant under

$A_j \mapsto A_j + \partial_j \theta$, the curl of a gradient vanishing by commutativity of the partial derivatives. This is the manuscript's "no Gribov ambiguity" for the abelian theory:

```
#check @BookProof.FreeEMField.emFieldStrength
#check @BookProof.FreeEMField.emFieldStrength_antisymm
#check @BookProof.FreeEMField.emFieldStrength_gauge_invariant
```

The reduction from the non-abelian case is explicit: when the connection components commute, the non-abelian field strength loses its quadratic $[A_j, A_k]$ correction and becomes the linear abelian one — which is why the Hamiltonian is quadratic in the fields in this case:

```
#check @BookProof.FreeEMField.fieldStrengthMul_eq_emFieldStrength_of_commute
#check @BookProof.FreeEMField.Fbook_eq_emFieldStrength_of_commute
```

Finally, the curl of self-adjoint operator fields is self-adjoint — the book's "local self-adjoint operator $\partial \times \pi$, the curl of the Electric Field":

```
#check @BookProof.FreeEMField.emFieldStrength_isSelfAdjoint
```

29.4. The Mass Gap and Observable Invariance

The manuscript's mass-gap discussion rests on a subtle point: the number operator N commutes with the observable algebra, so one can add λN to the Hamiltonian H without changing any observable consequence. The verified content is that Heisenberg evolution under $H + \lambda N$ equals that under H — the mass-gap shift has no observable consequence — and that for a gapless free field the shifted mass gap equals λ (arbitrary):

```
#check @BookProof.MassGap.numberOp
#check @BookProof.MassGap.heisenberg_number_shift_invariant
#check @BookProof.MassGap.shiftedSpectrum_vacuum
#check @BookProof.MassGap.shiftedSpectrum_excited
#check @BookProof.MassGap.massGap_shifted_gapless
```

The manuscript connects this to the positive-definiteness of the Weyl-gauge Hamiltonian (verified in the quantization chapter): the mass gap is read off a positive operator, and the number-operator shift is unobservable.

29.5. The Mass Gap in the Numerical Validation

The mass-gap discussion above is not only a formal statement about Heisenberg evolution: it is exactly the quantity the companion numerical validation measures. The free gluon one-particle operator is **gapless** — its lowest one-gluon energy tends to 0 as $k \to 0$. The final free-field Hamiltonian used for full nested-Fock interpretation is its outer enclosure $H = \sum_{i,j} h_{ij} C_i^\dagger A_j$, and the numerical suite confirms the one-particle soft-mode energy is below 0.02 in the solver's units. By contrast, the comparison-model lattice Hamiltonian, whose electric term $(g^2/2)\sum n_\ell$ is a genuine number operator, has a **positive** even-to-odd gap of $\approx g^2/2$ in the strong-coupling limit — a comparison-model lattice benchmark, not the final gauge-fixed nested-Fock Hamiltonian and not a Millennium-Prize confinement proof. The two regimes are separate validation models, while the final-theory statement is the outer creation/annihilation enclosure described above: the free field is gapless because its spectrum is the continuous line $|k| \geq 0$, while the number-operator shift λN — unobservable, as `heisenberg_number_shift_invariant` shows — is precisely what creates a gap when it is the **electric** term of a confined theory. The same contrast, between a gapless free field and a gapped confined one, is what the SIRK cross-sector solve reproduces numerically. For the physical theory discussed in the current validation, however, the final Hamiltonian is the gauge-fixed one-particle operator enclosed at the outer level with creation on the left and annihilation on the right. Its outer vacuum is therefore the exact ground after the permitted one-particle shift; a parity or squeezed inner-sector result is not a replacement for that conclusion.

Manuscript provenance. The outer-enclosure shape is the manuscript's, and the reason it may assume the Hamiltonian quadratic in the ladders is its recursive construction of the Fock space: **“Since the second-quantization procedure can be applied recursively … in case the Hamiltonian acting in the Fock-space is not quadratic in the creation/annihilation operators, then we can consider instead a new Fock-space where the base Hilbert space is the original Fock-space. The new Hamiltonian is quadratic in the creation/annihilation operators.”** Its Navier–Stokes section displays the template, $H =$

$\int d^3\vec{x}\, a^\dagger(\vec{x})\, H(\vec{x})\, a(\vec{x})$ with the inner $H(\vec{x}) = \pi^i(u_j u_{i,j} - \nu u_{i,jj}) + (h.c.)$; the Yang–Mills free-gluon enclosure $H = \sum_{i,j} h_{ij} C_i^\dagger A_j$ below is the same envelope. The convolution device belongs to the two sectors whose inner space carries **spatial derivatives of the fields** — Navier–Stokes and the vielbein sector of gravity — where a product of fields with a derivative is handled in momentum space as a **convolution** rather than by a gauge constraint on the derivative variables (QYM's free-gluon symbol needs neither device).

29.6. The Algebraic-Ideals View of Gauge

The manuscript's distinctive proposal is that gauge symmetries are defined through **algebraic ideals** — the gauge generators are excluded from the physical (gauge-invariant) algebra rather than being imposed as constraints on a larger phase-space. The verified seed of this idea is the constraint-conservation identity: a constraint D that commutes with the Hamiltonian is conserved by the flow (see the gauge chapter). The Gribov ambiguity is precisely the statement that no gauge-fixing can be simultaneously complete and unconstrained in general; the abelian case above is the exceptional case where it can be.

29.7. Summary

The algebraic core of the manuscript's Gribov-ambiguity chapter:

- the abelian field strength is gauge invariant (the "no Gribov ambiguity" case);
- the non-abelian field strength reduces to the abelian one when the connection commutes;
- the curl of self-adjoint operator fields is self-adjoint;
- the mass-gap shift by the number operator is unobservable (Heisenberg-invariant), and the Weyl-gauge Hamiltonian is positive.

30. The Physical Parity Transformation and Antiparticles

30.1. Orientation and Status

The closing chapter of the Standard-Model thread of this part: the physical parity transformation, the double cover, and the manuscript's claim that every particle is its own antiparticle. Read it after the real-representations chapter — its Majorana and CPT material is the prerequisite — and before the gravity chapter, which starts the part's final block.

The route: the question, the order-four parity transformation and the double cover, every particle as its own antiparticle, CP and the parity-breaking term, the chiral structure and quark-doublet parity, Majorana quantization, and the custodial symmetry.

The proofs are the `ChapterParity*` family (parity, chirality, custodial, Higgs, hypercharge, Majorana quantization, quark-lepton, SU, Z), the Pin double-cover chapters, the Schur/Maschke finite-group chapters, and `BookProof/ChapterCPTParity` / `BookProof/MajoranaClifford`.

30.2. The Question

The manuscript's chapter on the physical parity transformation develops the old observation that there is a basis where CP becomes a parity transformation, and that Majorana spinors (and the Majorana basis) make the real-rep structure explicit. Its central claims are: at the quantum level all fields are self-adjoint operators hence **real** representations — every particle is its own antiparticle; in a Majorana basis CP and P both become (order-4) parity transformations; and the Standard Model can be rebuilt with Majorana fermions where parity is broken only by the CKM matrix, forcing the double cover $\mathrm{Pin}(3,1)$. The verified content here is the algebraic structure of that programme.

30.3. The Order-Four Parity Transformation and the Double Cover

The manuscript's key technical claim is that the physical parity transformation has **order four** — not order two — and that this selects the double cover $\mathrm{Pin}(3,1)$ (where γ^0 squares to -1) over $\mathrm{Pin}(1,3)$ (where it squares to $+1$). The Higgs parity $i\sigma_2$ and the fermion parity $i\gamma^0$ both have order exactly four:

```
#check @BookProof.ChapterParity.higgsParity_order_four
#check @BookProof.ChapterParity.fermionParity_order_four
#check @BookProof.ChapterParity.dgamma0_sq
```

At the level of the discrete subgroup, the quaternion generators $i\gamma^0, \gamma^0\gamma^5, i\gamma^5$ satisfy $i^2 = j^2 = k^2 = -1$, $ij = k$, and generate the 8-element nonabelian group $\Omega \cong Q_8$. The covering map $\Lambda : \Omega \to \Delta$ is a homomorphism that is **two-to-one** — Ω is the double cover of the discrete parity × time-reversal group:

```
#check @BookProof.ChapterPinOmega.Omega
#check @BookProof.ChapterPinOmega.Omega_card
#check @BookProof.ChapterPinOmega.Omega_mul_closed
#check @BookProof.ChapterPinOmega.noncomm
#check @BookProof.ChapterPinDoubleCover.LamZ_surjective
#check @BookProof.ChapterPinDoubleCover.LamZ_hom
#check @BookProof.ChapterPinDoubleCover.LamZ_fiber_card
```

The order-four parity acting simultaneously on the whole Standard-Model matter content generates the $\mathbb{Z}_4$ background-symmetry factor:

```
#check @BookProof.ChapterParityZ4.combinedParity
#check @BookProof.ChapterParityZ4.combinedParity_order_four
#check @BookProof.ChapterParityZ4.combinedParity_orderOf
#check @BookProof.ChapterParityQL.QLParity_order_four
```

30.4. Every Particle Is Its Own Antiparticle

The manuscript's Majorana claim — that at the quantum level fields are real representations, so a particle is its own antiparticle — is captured by the self-adjointness of the Clifford field operators. A field operator $a(v) = \iota Q v$ generating the Clifford algebra of a real Hilbert space is explicitly self-adjoint and satisfies the canonical anticommutation relation:

```
#check @BookProof.MajoranaClifford.a
#check @BookProof.MajoranaClifford.a_sq
#check @BookProof.MajoranaClifford.a_selfAdjoint
#check @BookProof.MajoranaClifford.car
#check @BookProof.MajoranaClifford.a_map_selfAdjoint
```

The reality structure is made precise by the Higgs bidoublet: complex conjugation on a single $SU(2)$ doublet is pseudoreal ($\sigma_2\sigma_2^* = -1$), while **pseudoreal ⊗ pseudoreal = real**, so the Higgs bidoublet $(\tau_2 \otimes \sigma_2)$ carries a genuine real structure — the Higgs is a real representation:

```
#check @BookProof.ChapterParityHiggs.pauli2_pseudoreal
#check @BookProof.ChapterParityHiggs.pseudoreal_kron_pseudoreal_real
#check @BookProof.ChapterParityHiggs.higgs_real_structure
#check @BookProof.ChapterParitySU2.su2_conj_inner
```

30.5. CP and the Parity-Breaking Term

The manuscript traces parity (CP) violation to a single algebraic source. In the most general Lorentz-covariant Dirac mass term $iH = \partial \cdot \gamma\gamma^0 + i\gamma^0 m_1 + \gamma^0\gamma^5 m_2$, the term m_2 (the $\gamma^0\gamma^5$ part) is **parity-odd**, while the m_1 term is parity-even. Setting $m_2 = 0$ recovers exact parity invariance; a nonzero m_2 is the sole source of CP violation:

```
#check @BookProof.ChapterCPTParity.parity_Kin
#check @BookProof.ChapterCPTParity.parity_MassA
#check @BookProof.ChapterCPTParity.parity_MassB
#check @BookProof.ChapterCPTParity.parity_diracHamOp
#check @BookProof.ChapterCPTParity.parity_diracHamOp_invariant
```

The manuscript connects this to the structure of $SU(3)$: complex conjugation on the Gell-Mann generators is a **nontrivial** $\mathbb{Z}_2$ outer automorphism (realized by the sign pattern that negates λ^2), which is the algebraic seed of CP violation in the quark sector:

```
#check @BookProof.ChapterParitySU3.gellMann_conj_involutive
#check @BookProof.ChapterParitySU3.gellMann_bracket_conj
#check @BookProof.ChapterParitySU3.gellMann_conj_nontrivial
#check @BookProof.ChapterParity.gellMann_conj
```

30.6. Chiral Structure and the Quark-Doublet Parity

The manuscript rebuilds the electroweak sector with a definite chirality. The chirality operator $\chi = (i\sigma_3) \otimes (i\gamma^5)$ is an involution ($\chi^2 = 1$) with trace zero (equal eigenspaces, "divide by 2"), and the projector $P = \frac{1}{2}(1 - \chi)$ selects the left-handed states:

```
#check @BookProof.ChapterParityChirality.chi
#check @BookProof.ChapterParityChirality.chi_sq
#check @BookProof.ChapterParityChirality.chi_trace
#check @BookProof.ChapterParityChirality.QLProj
#check @BookProof.ChapterParityChirality.chirality_iff
```

The hypercharge phase group is realized with the **real** matrix $i\gamma^5$ in place of the imaginary unit: $U(1)_Y$ becomes $e^{\vartheta i\gamma^5} = \cos\vartheta \cdot 1 + \sin\vartheta \cdot i\gamma^5$, which is a genuine abelian group:

```
#check @BookProof.ChapterParityHypercharge.mgamma5_real
#check @BookProof.ChapterParityHypercharge.hyperPhase_add
#check @BookProof.ChapterParityHypercharge.hyperPhase_neg_mul
```

The fixed 4×4 model is **irreducible** in the strongest concrete sense: any complex matrix commuting with all four Majorana matrices $i\gamma^\mu$ is a scalar. This is Schur's lemma for the concrete representation — proved outright by a finite computation, so the Pauli fundamental theorem's uniqueness clause is available for this model without any external input: two invertible matrices that conjugate the $i\gamma^\mu$ to the same family differ by a nonzero scalar.

```
#check @BookProof.ChapterA3.mgamma_commutant_scalar
#check @BookProof.ChapterA3.mgamma_commutant_iff
#check @BookProof.ChapterA3.dgamma_commutant_scalar
#check @BookProof.ChapterA3.mgamma_conjugation_unique_up_to_scalar
#check @BookProof.ChapterA3.mgamma_conj_eq_self_iff
#check @BookProof.ChapterA3.mgamma_conjTranspose
#check @BookProof.ChapterA3.mgamma_irreducible
```

30.7. Majorana Quantization

The manuscript's Majorana quantization uses a compatible complex structure J (with $(iJ)^2 = 1$) to split the real Hilbert space into annihilation and creation directions via the projections $\frac{1}{2}(1 \pm iJ)$. The verified facts are that J is unitary, the projections are orthogonal and span the identity, and annihilation combinations lie in the $-i$-eigenspace:

```
#check @BookProof.ChapterParityMajoranaQuant.J_unitary
#check @BookProof.ChapterParityMajoranaQuant.annihProj
#check @BookProof.ChapterParityMajoranaQuant.proj_add
#check @BookProof.ChapterParityMajoranaQuant.J_annih
#check @BookProof.ChapterParityMajoranaQuant.stdJ
```

30.8. The Custodial Symmetry

The custodial $SU(2)$ structure is governed by Schur's lemma: any matrix commuting with all of $\sigma_1, \sigma_2, \sigma_3$ is scalar, and $\{1, \sigma_1, \sigma_2, \sigma_3\}$ are linearly independent — the four-dimensional custodial commutant:

```
#check @BookProof.ChapterParityCustodial.commutant_pauli_scalar
#check @BookProof.ChapterParityCustodial.pauli_basis_indep
```

This commutant statement is a special case of a general irreducibility theorem. On a nonzero **finite-dimensional** complex space, an operator commuting with an **irreducible** system of operators is necessarily a scalar: over $\mathbb{C}$ the commuting operator has an eigenvalue, its eigenspace is invariant under the system and hence — by irreducibility — everything, so the operator acts as that scalar. This is Schur's lemma, and discharging it in finite dimensions turns the book's `IsSchurUnitary` hypothesis into a theorem:

```
#check @BookProof.ChapterSchurFiniteDim.schur_scalar_of_irreducible
#check @BookProof.ChapterSchurFiniteDim.isSchurUnitary_of_irreducible
```

The same argument settles the **commutant** form of Schur's lemma, which is the one the manuscript's trichotomy of real, complex and pseudo-real systems uses: in finite dimensions the commutant of an irreducible system is exactly the line of scalar operators $\mathbb{C} \cdot 1$, so the `IsSchurFull` hypothesis is a theorem there as well:

```
#check @BookProof.ChapterSchurFullFiniteDim.isSchurFull_of_irreducible
#check @BookProof.ChapterSchurFullFiniteDim.commutant_eq_scalars_of_irreducible
#check @BookProof.ChapterSchurFullFiniteDim.commutant_eq_span_one
```

Schur's lemma is one half of the classification machinery; the other half is **complete reducibility** — that an invariant subspace can always be split off. For a **unitary** representation this is Weyl's unitarian trick and it is now proved outright: the orthogonal complement of an invariant subspace is invariant, so in finite dimensions every invariant subspace has an invariant complement and the representation is a sum of irreducibles. For a **finite** group the same conclusion follows from Maschke's averaging argument, with no unitarity assumed:

```
#check @BookProof.ChapterUnitaryCompleteReducibility.orthogonal_isInvariant
#check @BookProof.ChapterUnitaryCompleteReducibility.unitary_complete_reducibility
#check @BookProof.ChapterUnitaryCompleteReducibility.exists_irreducible_invariant_le
#check @BookProof.ChapterUnitaryCompleteReducibility.exists_irreducible_decompositior
#check @BookProof.ChapterMaschkeFiniteGroup.maschke_invariant_complement
```

The averaging argument gives more than the bare complement. The averaged map is a genuine projection — idempotent, with range exactly the invariant subspace it was built from — and it commutes with the representation, so the splitting is a decomposition **of representations**: an invariant subspace, an invariant complement, and an equivariant projection between them.

```
#check @BookProof.ChapterMaschkeFiniteGroup.avgProj_idempotent
#check @BookProof.ChapterMaschkeFiniteGroup.avgProj_range_eq_W
#check @BookProof.ChapterMaschkeFiniteGroup.maschke_decomposition
```

For the tensor powers of the Dirac spinor space the decomposition is explicit. On the tensor **square** the symmetrizer and the antisymmetrizer are already complementary, so $V \otimes V = \mathrm{Sym}^2 V \oplus \Lambda^2 V$. On the tensor **cube** they are not: what is left over is the mixed-symmetry summand $\Pi_{\mathrm{mix}} = 1 - \Pi_{\mathrm{sym}} - \Pi_{\mathrm{anti}}$, and it is not zero. It is a projector, orthogonal to the other two, and it commutes with the diagonal Lorentz generators and with diagonal parity, so $V^{\otimes 3} = \mathrm{Sym}^3 V \oplus \Lambda^3 V \oplus \mathrm{Mixed}$ is a decomposition into three full-Lorentz subrepresentations — the first genuinely three-part instance of complete reducibility here, proved with no external input:

```
#check @BookProof.ChapterA3p.tensorSquare_complete_reducibility
#check @BookProof.ChapterA3x.projMixed
#check @BookProof.ChapterA3x.projMixed_idem
#check @BookProof.ChapterA3x.projMixed_spinGenDiag_comm
#check @BookProof.ChapterA3x.projMixed_parityDiag_comm
#check @BookProof.ChapterA3x.projMixed_two_eq_zero
#check @BookProof.ChapterA3x.projMixed_three_ne_zero
#check @BookProof.ChapterA3x.tensorCube_complete_reducibility
```

For the concrete 4×4 Majorana model of the manuscript, Pauli's fundamental theorem is itself a finite computation, not an external input: the commutant of the four matrices $i\gamma^\mu$ is exactly the set of scalar matrices — the representation is irreducible. The proof is elementary: two diagonal elements of the gamma algebra separate the four basis vectors, forcing a commuting matrix to be diagonal, and the off-diagonal gammas then identify all four diagonal entries:

```
#check @BookProof.ChapterGammaCommutant.gamma_commutant_scalar
#check @BookProof.ChapterGammaCommutant.gamma_commutant_eq_scalars
```

```
#check @BookProof.ChapterGammaCommutant.gamma_intertwiner_unique
```

30.9. Summary

The algebraic core of the manuscript's parity/antiparticles chapter:

- the physical parity is an order-four transformation, selecting the double cover $\mathrm{Pin}(3,1)$;
- the discrete Pin subgroup $\Omega \cong Q_8$ double-covers Δ;
- Majorana self-adjointness: every particle is its own antiparticle, and the Higgs is a real representation;
- CP violation traced to the parity-odd m_2 term and the nontrivial $SU(3)$ conjugation automorphism;
- the chiral structure and the real-matrix hypercharge phase group;
- Majorana quantization by a compatible complex structure.

31. Diffeomorphisms and Gravity

31.1. Orientation and Status

The gravity chapter: the manuscript's diffeomorphism/3+1 programme carried to a gauge-fixed field-space Hamiltonian and its BRST charge — the heaviest construction in the part, and the one the companion solver work validates numerically. It needs the BRST vocabulary of the gauge-symmetry chapter; the derivative-variable thread (Fourier elimination and after) explains the convention this chapter uses for its derivative variables.

The route: the 3+1 split, the induced spatial metric and its irreducible decomposition, the densitized tetrad variables and the flat principal part, the Starobinsky potentials and the flow of the regularized conformal mode, the scalaron wall and the compactly supported core, the gauge-fixed field-space Hamiltonian and its BRST charge, fixing the derivative variables (the torsion becomes linear), the graded Fock space of bosons and ghosts, and the BRST charge on the completed space with the reduced transfer.

The proofs are `BookProof/ChapterGravity*` (split, metric, projectors, inverse), the `QuantumGravity*` / `Qg*` family (densitized, Fock, BRST completed, symmetric sector, vielbein–scalaron), the `Scalaron*` and `Starobinsky` chapters, and the quadratic/Hermite/Carleman operator chapters that the field-space Hamiltonian rests on.

31.2. The Question

The manuscript's chapter on diffeomorphisms and gravity argues that Quantum Gravity should be treated as a special case of the (global) gauge/quantization formalism in the continuum, rather than as Loop Quantum Gravity with renormalization. Diffeomorphism is generated by constraints while time-evolution is not a diffeomorphism, so the two are compatible. The strategy is to decompose space-time into time and space (the $3+1$ split) and to study the induced spatial metric as a degenerate object along the time direction. The verified content here is the algebraic geometry of that $3+1$ split.

31.3. The 3+1 Split: Time and Space

Fix a unit timelike vector v (the time direction) in Minkowski space with metric η (here $\eta = \mathrm{diag}(-1,1,1,1)$). The spatial projector $\chi^a{}_b = \delta^a{}_b + v^a v_b$ is the orthogonal projection onto the spatial hyperplane $v^\perp$: it is idempotent, has kernel spanned by v, and has trace 3 (it is a rank-3 projector). The temporal projector $\Pi^a{}_b = -v^a v_b = \delta - \chi$ completes it:

```
#check @BookProof.ChapterGravityProjector.metric
#check @BookProof.ChapterGravityProjector.spatialProj
#check @BookProof.ChapterGravityProjector.spatialProj_mulVec_self
#check @BookProof.ChapterGravityProjector.spatialProj_idempotent
#check @BookProof.ChapterGravityProjector.trace_spatialProj
#check @BookProof.ChapterGravityTimeProj.spatialProj_add_timeProj
```

```
#check @BookProof.ChapterGravityTimeProj.timeProj_idempotent
#check @BookProof.ChapterGravityTimeProj.trace_timeProj
```

Every vector splits uniquely into a spatial part and a temporal part, and these parts are Minkowski-orthogonal: $\langle \chi x, \Pi x \rangle_\eta = 0$. This is the geometric substance of the manuscript's "time-evolution is not a diffeomorphism": the time and space components of a displacement are orthogonal and independent:

```
#check @BookProof.ChapterGravitySplit.spatialPart_add_timePart
#check @BookProof.ChapterGravitySplit.timePart_eq_smul
#check @BookProof.ChapterGravitySplit.parts_orthogonal
#check @BookProof.ChapterGravitySplit.split_unique
```

Because the two projector endomorphisms χ and Π sum to the identity and annihilate each other ($\chi\Pi = \Pi\chi = 0$), their ranges are **complementary subspaces**: the whole of spacetime is their direct sum, $\mathbb{R}^{1,3} = \operatorname{im}\chi \oplus \operatorname{im}\Pi$, and the image of the spatial projector is exactly the kernel of the temporal one. This is the linear-algebra fact that turns the algebraic projector identities into the manuscript's $3+1$ decomposition of spacetime itself:

```
#check @BookProof.ChapterGravityProjDirectSum.isCompl_spatial_time_range
#check @BookProof.ChapterGravityProjDirectSum.range_spatial_eq_ker_time
#check @BookProof.ChapterGravityProjDirectSum.isCompl_range_of_add_eq_id
```

The decomposition is moreover of the shape the manuscript asserts: the temporal summand is precisely the line $\mathbb{R} \cdot v$ through the unit timelike vector, so it is one-dimensional, and the spatial summand is a hyperplane. The split is therefore literally $4 = 3 + 1$:

```
#check @BookProof.ChapterGravityRankSplit.timeRange_eq_span
#check @BookProof.ChapterGravityRankSplit.finrank_timeRange
#check @BookProof.ChapterGravityRankSplit.finrank_spatialRange
#check @BookProof.ChapterGravityRankSplit.gravity_split_three_plus_one
```

31.4. The Induced Spatial Metric

The $3+1$ split induces a spatial metric $h_{ab} = \eta_{ab} + v_a v_b$, which is symmetric, positive-semidefinite, and degenerate along the time direction ($hv = 0$). Its relationship to the full metric is the tensor identity $h = \eta \cdot \chi$, and it satisfies the reverse Cauchy–Schwarz inequality for the timelike unit vector:

```
#check @BookProof.ChapterGravityMetric.spatialMetric
#check @BookProof.ChapterGravityMetric.spatialMetric_symm
#check @BookProof.ChapterGravityMetric.spatialMetric_eq_metric_mul_proj
#check @BookProof.ChapterGravityMetric.spatialMetric_mulVec_self
#check @BookProof.ChapterGravityMetric.spatialMetric_quadForm_nonneg
#check @BookProof.ChapterGravityMetric.spatialMetric_posSemidef
```

The raised spatial metric $h^{ab} = \eta^{ab} + v^a v^b$ is the generalized inverse of the induced metric on the degenerate spatial hyperplane: the pair satisfies $h \cdot h^\sharp \cdot h = h$ and $h^\sharp \cdot h = \chi$. This is the "inverse metric" on the 3-dimensional slice, defined even though h is degenerate along time:

```
#check @BookProof.ChapterGravityInvMetric.invSpatialMetric
#check @BookProof.ChapterGravityInvMetric.metric_mul_metric
#check @BookProof.ChapterGravityInvMetric.invSpatialMetric_mul_spatialMetric
#check @BookProof.ChapterGravityGenInverse.spatialMetric_genInverse
#check @BookProof.ChapterGravityGenInverse.invSpatialMetric_genInverse
```

31.5. The Irreducible Decomposition of the Metric

The manuscript analyzes the gravitational metric field into its irreducible parts under the rotation group. Every matrix M splits uniquely as $M = \frac{1}{2}S + \frac{1}{2}A + \frac{1}{3}(\operatorname{tr} M)I$, where S is symmetric traceless (the $2/3$ factor), A is antisymmetric, and the trace part is the remainder. The three pieces are mutually Frobenius-orthogonal:

```
#check @BookProof.ChapterGravityIrrep.antisymPart
#check @BookProof.ChapterGravityIrrep.symTracelessPart
#check @BookProof.ChapterGravityIrrep.trace_symTracelessPart
#check @BookProof.ChapterGravityIrrep.irrep_reconstruction
```

```
#check @BookProof.ChapterGravityIrrep.frob_symTraceless_antisym
#check @BookProof.ChapterGravityIrrep.frob_antisym_trace
```

31.6. The Densitized Tetrad Variables and the Flat Principal Part

The manuscript's final, 3D gauge-fixed gravity Hamiltonian carries the **singular** kinetic terms $\frac{1}{16e}S^{ab}S_{ab} - \frac{1}{24e}P^2$, whose denominator $e = \det e_i{}^a$ vanishes where the tetrad degenerates. The canonical repair is the change of variables to **densitized** tetrads $y = \sqrt{e}$, $\tilde{e}_i{}^a = \sqrt{e}\, e_i{}^a$. Two facts make this load-bearing rather than cosmetic: the singular coefficient is a square of a derivative of the new coordinate, $1/e = 4(\partial y/\partial e)^2$, so it is absorbed into the field derivatives; and while $1/e \to \infty$ at a degenerate tetrad, the new coordinate $y = \sqrt{e}$ extends continuously by 0:

```
#check @BookProof.QuantumGravityDensitized.densY
#check @BookProof.QuantumGravityDensitized.inv_eq_four_mul_deriv_densY_sq
#check @BookProof.QuantumGravityDensitized.kinetic_absorption
#check @BookProof.QuantumGravityDensitized.conformal_absorption
#check @BookProof.QuantumGravityDensitized.densTetrad
#check @BookProof.QuantumGravityDensitized.densTetrad_det
#check @BookProof.QuantumGravityDensitized.densTetrad_recover
#check @BookProof.QuantumGravityDensitized.tendsto_inv_det_atTop
#check @BookProof.QuantumGravityDensitized.tendsto_densY_zero
```

In the densitized variables the principal part of the kinetic operator has the **constant** coefficients $(1/16, -1/24)$: the field-space metric is flat, so all its Christoffel symbols vanish and the point transformation produces no connection ("quantum potential") corrections. The symbol is non-degenerate but **indefinite** — the operator is a d'Alembertian (hyperbolic), not an elliptic Laplacian, which is why the elliptic Sears route of the Navier–Stokes thread does not apply to it:

```
#check @BookProof.QuantumGravityDensitized.qgSymbol
#check @BookProof.QuantumGravityDensitized.qgSymbol_eq_metric_form
#check @BookProof.QuantumGravityDensitized.qgMetric_det_ne_zero
#check @BookProof.QuantumGravityDensitized.qgSymbol_indefinite
#check @BookProof.QuantumGravityDensitized.christoffel_eq_zero_of_const
#check @BookProof.QuantumGravityDensitized.qgMetric_christoffel_zero
#check @BookProof.QuantumGravityDensitized.qgFullSymbol_scaling
```

The analytic conclusion for the **continuum** operator is **not** claimed. What is proved is the discretized (Hermite-basis) realization, where the fiber operator is multiplication by the mode symbol: it is genuinely unbounded, essentially self-adjoint on its maximal domain, and its adjoint has trivial deficiency at every non-real point — exactly the conclusion Strichartz's finite-speed argument is supposed to supply in the continuum. Strichartz's theorem itself enters only as an explicit **hypothesis** (never an axiom), and the transfer of essential self-adjointness back to the physical variables is the unitary-equivalence step:

```
#check @BookProof.QuantumGravityDensitized.qgModeHamiltonian
#check @BookProof.QuantumGravityDensitized.qgModeHamiltonian_essentiallySelfAdjoint
#check @BookProof.QuantumGravityDensitized.qgModeHamiltonian_deficiencyTrivialAt
#check @BookProof.QuantumGravityDensitized.qgModeHamiltonian_not_bounded
#check @BookProof.QuantumGravityDensitized.strichartz_esa_of_finiteSpeed
#check @BookProof.QuantumGravityDensitized.strichartz_finiteSpeed_satisfiable
#check @BookProof.QuantumGravityDensitized.qg_esa_of_farisLavine
#check @BookProof.QuantumGravityDensitized.densitized_hasZeroDeficiencyOn_transfer
```

31.7. The Starobinsky Potentials and the Flow of the Regularized Conformal Mode

The conformal mode is the notorious wrong-sign direction of pure general relativity: its potential is the **linear** $-\frac{M^2}{2}R_c$, unbounded below. Adding the curvature-squared term of the Starobinsky action $f(R) = \frac{M^2}{2}R + \alpha R^2$ changes this qualitatively. First, the extension is ghost-free: with $\psi = 1 + 4\alpha R/M^2$ and $U(\psi) = \frac{M^4}{16\alpha}(\psi - 1)^2$ one has the exact identity $f(R) = \frac{M^2}{2}\psi R - U(\psi)$, so the theory is an ordinary second-order scalar–tensor theory. Second, in the Einstein frame the scalaron potential $V(\varphi) = \frac{M^4}{16\alpha}\left(1 - e^{-\sqrt{2/3}\,\varphi/M}\right)^2$ is manifestly a

square: non-negative, zero at the Minkowski vacuum, with the inflationary plateau $M^4/(16\alpha)$ as $\varphi \to +\infty$ and an exponential wall as $\varphi \to -\infty$:

```
#check @BookProof.Starobinsky.fR_eq_scalarTensor
#check @BookProof.Starobinsky.starobinskyV
#check @BookProof.Starobinsky.starobinskyV_nonneg
#check @BookProof.Starobinsky.starobinskyV_zero
#check @BookProof.Starobinsky.starobinskyV_tendsto_plateau
#check @BookProof.Starobinsky.starobinskyV_tendsto_atBot_atTop
```

Third — and this is the point for the conformal mode — completing the square in $V_3(R_c) = -\frac{M^2}{2}R_c + \alpha R_c^2$ gives $\alpha\big(R_c - \frac{M^2}{4\alpha}\big)^2 - \frac{M^4}{16\alpha}$, so for $\alpha > 0$ the potential is bounded below by $-M^4/(16\alpha)$, while at $\alpha = 0$ it tends to $-\infty$. That contrast is exactly what the αR^2 term buys:

```
#check @BookProof.Starobinsky.confV
#check @BookProof.Starobinsky.confV_completed_square
#check @BookProof.Starobinsky.confV_ge
#check @BookProof.Starobinsky.confV_bddBelow
#check @BookProof.Starobinsky.confV_zero_alpha_tendsto_atBot
```

At the level of the operator the same bound is uniform in the mode, so the Hermite-basis realization of the gauge-fixed $R + \alpha R^2$ Hamiltonian is a multiplication operator with a potential bounded below. It is symmetric and essentially self-adjoint on its maximal domain, which is dense because it contains the finitely supported states; the Stone bridge then converts that into the complete unitary group e^{-itH}. As elsewhere in this thread, the continuum statement and the gauge/BRST sector remain outside the claim:

```
#check @BookProof.Starobinsky.qgR2ModeHamiltonian
#check @BookProof.Starobinsky.qgR2Mode_potential_ge
#check @BookProof.Starobinsky.mulSymbolDomain_dense
#check @BookProof.Starobinsky.qgR2Mode_symmetric
#check @BookProof.Starobinsky.qgR2Mode_esa
#check @BookProof.Starobinsky.qgR2Mode_deficiencyTrivialAt
#check @BookProof.Starobinsky.qgR2_stone_flow
```

In the continuum the two **commuting** halves of a wave operator with a potential are now settled with unbounded coefficients, on the Schwartz core of $L^2(\mathbb{R}^{1+n})$. Multiplication by an arbitrary real potential of temperate growth — every polynomial, unbounded and with no assumption of boundedness from below — is essentially self-adjoint; dually, every real **symbol** of temperate growth gives an essentially self-adjoint Fourier multiplier, which contains all constant-coefficient operators with real symbol, of any order. For the mixture $\Box + W$ itself, the operator is symmetric on that core for every real potential of temperate growth, and essentially self-adjoint for each spatial truncation of the potential:

```
#check @BookProof.StrichartzWave.potentialOp_essentiallySelfAdjoint
#check @BookProof.StrichartzWave.polynomialPotential_essentiallySelfAdjoint
#check @BookProof.StrichartzWave.multiplierOp_essentiallySelfAdjoint
#check @BookProof.StrichartzWave.constCoeffOp_eq_multiplierOp
#check @BookProof.StrichartzWave.polyharmonic_multiplier_essentiallySelfAdjoint
#check @BookProof.StrichartzWave.wave_add_potentialOp_symmetric
#check @BookProof.StrichartzWave.wave_add_truncatedPotential_essentiallySelfAdjoint
```

The remaining step — removing the truncation — is **not** claimed, and it is not a formality: with the convention $\Box = -\partial_t^2 + \Delta_x$ used here, a Fourier transform in time turns $\Box + W$ into the fibres $4\pi^2\tau^2 - (-\Delta_x - W)$, so a potential bounded **below** makes the fibre Schrödinger operator unbounded below — the limit-circle regime, where essential self-adjointness genuinely fails. The sign under which the localization argument can close is the opposite one, W bounded above by a quadratic here (equivalently bounded below for the physics signature $\Box = \partial_t^2 - \Delta_x$).

31.8. The Scalaron Wall and the Compactly Supported Core

The Einstein-frame scalaron potential is the one place where the temperate-growth hypothesis of the multiplication theorem above is genuinely violated: $V(\varphi)$ grows like $e^{-2\sqrt{2/3}\,\varphi/M}$ as $\varphi \to -\infty$, and it is **proved** here not to be of temperate growth. The hypothesis is nevertheless an artefact of the core, not of the mathematics: temperate

growth is what keeps the Schwartz space invariant under multiplication, and on the smaller — still dense — core of smooth compactly supported functions no growth condition is needed at all. On that core multiplication by **any** smooth real potential is symmetric with trivial deficiency at every non-real point, hence essentially self-adjoint, with no boundedness and no semiboundedness assumption:

```
#check @BookProof.ScalaronEsa.ccDomain
#check @BookProof.ScalaronEsa.ccDomain_dense
#check @BookProof.ScalaronEsa.smoothPotential_symmetric
#check @BookProof.ScalaronEsa.smoothPotential_deficiencyTrivial
#check @BookProof.ScalaronEsa.smoothPotential_essentiallySelfAdjoint
#check @BookProof.ScalaronEsa.starobinskyV_not_hasTemperateGrowth
#check @BookProof.ScalaronEsa.starobinskyV_essentiallySelfAdjoint
```

The full gauge-fixed $R + \alpha R^2$ potential — the regularized conformal parabola $V_3(R_c)$ together with the scalaron potential $V(\varphi)$ — is then essentially self-adjoint on that dense core as well, and it carries the lower bound $-M^4/(16\alpha)$: the scalaron term is a square, so it can only help. The d'Alembertian with such a potential is a well-defined symmetric operator on the same dense core, and each of its spatial localizations is essentially self-adjoint on the Schwartz core — the truncation theorem now with smoothness as its only hypothesis on the potential, the exponential wall included:

```
#check @BookProof.ScalaronEsa.scalaronFullPotential
#check @BookProof.ScalaronEsa.scalaronFullPotential_ge
#check @BookProof.ScalaronEsa.scalaronFullPotential_essentiallySelfAdjoint
#check @BookProof.ScalaronEsa.wave_add_scalaron_symmetric
#check @BookProof.ScalaronEsa.wave_add_smoothTruncatedPotential_essentiallySelfAdjoir
#check @BookProof.ScalaronEsa.wave_add_scalaronTruncated_esa
#check @BookProof.ScalaronEsa.wave_add_smoothPotential_esa_of_finiteSpeed
#check @BookProof.ScalaronEsa.wave_add_scalaron_esa_of_finiteSpeed
```

Removing the truncation is, as before, the one step that is not claimed: what remains is the finite-speed / unique-continuation input, kept explicit as a named hypothesis in the two statements above, never as an axiom. The point of this section is that the **potential** side of the question is closed — an exponentially growing but bounded-below potential is no worse than a polynomial one once the Hamiltonian is taken on the compactly supported core.

At the mode level nothing is left conditional: the gauge-fixed $R + \alpha R^2$ Hamiltonian **including** the scalaron potential is multiplication by $\frac{1}{16}a_k^2 - \frac{1}{24}b_k^2 + V_3(R_{c,k}) + V(\varphi_k)$, essentially self-adjoint on its dense maximal domain with the uniform lower bound $-M^4/(16\alpha)$ on the potential, and Stone's theorem turns it into the complete unitary group. The honest boundary is unchanged: the continuum statement for the unlocalized sum still needs the finite-speed gluing, and the gauge/BRST sector stays outside the claim.

```
#check @BookProof.ScalaronEsa.qgScalaronModeHamiltonian
#check @BookProof.ScalaronEsa.qgScalaronMode_potential_ge
#check @BookProof.ScalaronEsa.qgScalaronMode_symmetric
#check @BookProof.ScalaronEsa.qgScalaronMode_esa
#check @BookProof.ScalaronEsa.qgScalaron_stone_flow
```

One particle is not the physical statement: the state space of the quantised theory is a **nested Fock space** $\bigoplus_n L^2(E_n)$, whose n-th sector carries the configuration space of n quanta and whose Hamiltonian acts sector by sector, the n-particle potential being the sum of the one-particle potentials of the individual quanta. Essential self-adjointness passes from the sectors to the whole by the orthogonal direct-sum instrument: a deficiency vector of $\bigoplus_n H_n$, tested against a state living in one sector, satisfies exactly the sector deficiency identity, so all of its coordinates vanish. Nothing about the exponential wall of the scalaron survives this step, because nothing about growth was used in the sectors.

Concretely, on the sector $E_n = \mathbb{R}^{n\times 2}$ — each quantum carrying a conformal mode R_c and a scalaron φ — the many-body gauge-fixed potential $\sum_{j<n}\big(V_3(R_{c,j}) + V(\varphi_j)\big)$ is smooth, bounded below by $-n\,M^4/(16\alpha)$, and at $n = 1$ it **is** the one-particle potential of the previous section. The resulting operator on $\bigoplus_n L^2(\mathbb{R}^{n\times 2})$ is densely defined, symmetric, has trivial deficiency at every non-real point, is essentially self-adjoint, and Stone's theorem turns it into the global unitary group e^{-itH} on the whole finite-particle Fock space. The same holds in the mode (Hermite) realisation of the gravity chapters. The gluing is over an orthogonal direct sum, i.e. the Hamiltonian

preserves particle number — exactly the finite-particle situation; an interaction changing the sector is not part of the claim.

```
#check @BookProof.ScalaronFock.nestedCore
#check @BookProof.ScalaronFock.nestedCore_dense
#check @BookProof.ScalaronFock.fockSmoothPotential_esa
#check @BookProof.ScalaronFock.fockSmoothPotential_stone_flow
#check @BookProof.ScalaronFock.qgManyPotential
#check @BookProof.ScalaronFock.qgManyPotential_one
#check @BookProof.ScalaronFock.qgManyPotential_ge
#check @BookProof.ScalaronFock.qgScalaronFockHamiltonian
#check @BookProof.ScalaronFock.qgScalaronFock_esa
#check @BookProof.ScalaronFock.qgScalaronFock_stone_flow
#check @BookProof.ScalaronFock.qgScalaronModeFock_esa
#check @BookProof.ScalaronFock.qgScalaronModeFock_stone_flow
```

The genuinely **non-commuting** case — a differential kinetic term together with an unbounded polynomial potential — is settled in the elliptic normalization, where the potential is bounded below and an explicit joint eigenbasis exists. The harmonic oscillator $H = -d^2/dx^2 + x^2/4$ is essentially self-adjoint on the Hermite core of $L^2(\mathbb{R})$, and it is genuinely unbounded there. What makes this a statement about the differential operator rather than about a diagonal matrix is the identification: the diagonal operator with eigenvalues $n + \frac{1}{2}$ sends the n-th Hermite basis vector to the L^2 class of $x \mapsto -\psi_n''(x) + (x^2/4)\,\psi_n(x)$.

```
#check @BookProof.HarmonicOscillator.hermiteC_oscillator
#check @BookProof.HarmonicOscillator.harmonicOscOp
#check @BookProof.HarmonicOscillator.harmonicOscOp_apply_eq_differential
#check @BookProof.HarmonicOscillator.harmonicOsc_symmetric
#check @BookProof.HarmonicOscillator.harmonicOsc_essentiallySelfAdjoint
#check @BookProof.HarmonicOscillator.harmonicOsc_not_bounded
```

The same joint eigenbasis settles the **hyperbolic** mixture, for the quadratic potentials that are diagonal in the coordinates. For an arbitrary real weight vector c, with no sign condition, the operator $H_c = \sum_i c_i(-\partial_i^2 + x_i^2/4)$ is symmetric and essentially self-adjoint on the Gauss–polynomial (product Hermite) core of $L^2(\mathbb{R}^d)$, acting on the product Hermite function ψ_α by the real symbol $\sum_i c_i(\alpha_i + \frac{1}{2})$; the operator is genuinely unbounded as soon as one weight is non-zero, and the identification with the differential expression is proved pointwise, with Mathlib's derivative taken twice along each coordinate line. Taking the Minkowski weights $c = (1, -1, \dots, -1)$ gives, in the convention $\Box = -\partial_t^2 + \Delta_x$ used here, the operator $\Box + V$ with the indefinite quadratic potential $V(t,x) = (t^2 - \|x\|^2)/4$ — unbounded above and below, not commuting with $\Box$, and bounded above by a quadratic, which is exactly the sign discussed above. A general potential bounded above by a quadratic is still not covered: the argument uses the joint eigenbasis, which exists only for the diagonal quadratic family. The Kato–Rellich theorem does widen the class by a bounded term, so the potential may be any diagonal quadratic plus a real, essentially bounded function.

```
#check @BookProof.HyperbolicQuadratic.oscPoly_hermiteMv
#check @BookProof.HyperbolicQuadratic.quadOp_hermiteMvLp
#check @BookProof.HyperbolicQuadratic.quadOp_symmetric
#check @BookProof.HyperbolicQuadratic.quadOp_essentiallySelfAdjoint
#check @BookProof.HyperbolicQuadratic.quadOp_not_bounded
#check @BookProof.HyperbolicQuadratic.quadPoly_apply_eq_differential
#check @BookProof.HyperbolicQuadratic.wave_indefiniteQuadratic_essentiallySelfAdjoint
#check @BookProof.HyperbolicQuadratic.minkowski_apply_eq_differential
#check @BookProof.HyperbolicQuadratic.quadOp_add_realBoundedPotential_essentiallySelf
```

The bounded widening is not the end of the story. When the weights are strictly positive, $c_i \geq c_0 > 0$, the quadratic Hamiltonian **dominates** each coordinate pair: the form identity $\langle u, (\pi_i^2 + x_i^2/4)u\rangle = \|\pi_i u\|^2 + \|x_i u\|^2/4$ together with the symbol inequality $c_0(\alpha_i + \frac{1}{2}) \leq \sum_j c_j(\alpha_j + \frac{1}{2})$ gives $\|x_i u\| \leq \varepsilon\|H_c u\| + (2/(c_0\varepsilon))\|u\|$ for every $\varepsilon > 0$, and the same bound for the momentum. The relative bound is therefore arbitrarily small, so the Kato–Rellich theorem applies to any first-order perturbation $B = \sum_i (b_i x_i + b_i' \pi_i)$ with real coefficients: $H_c + B$ is essentially self-adjoint on the same Hermite core. The perturbation here is genuinely unbounded — with $b' = 0$ it is multiplication by the linear function $x \mapsto \langle b, x\rangle$, so the corollary is the Stark-shifted harmonic oscillator $-\Delta + \|x\|^2/4 + \langle b, x\rangle$, an oscillator in a constant external field. The positivity of the weights

is used, and cannot be dropped by this argument: in the hyperbolic case the symbol vanishes on infinitely many multi-indices, so H_c does not dominate the number operator. The same eigenbasis also exhibits the diagonalizing unitary explicitly: in the coordinates of the product Hermite basis, H_c **is** multiplication by the real symbol $\sum_i c_i(\alpha_i + \frac{1}{2})$ on ℓ^2.

```
#check @BookProof.HermiteRelative.re_inner_oscL_eq
#check @BookProof.HermiteRelative.norm_posL_le
#check @BookProof.HermiteRelative.norm_momL_le
#check @BookProof.HermiteRelative.quadOp_add_firstOrder_essentiallySelfAdjoint
#check @BookProof.HermiteRelative.hermiteMvBasis_repr_quadOp
#check @BookProof.HermiteRelative.foOp_linear_apply_eq_mul
#check @BookProof.HermiteRelative.harmonicOsc_add_linearPotential_essentiallySelfAdjc
```

The positivity of the weights is a limitation of the **Kato–Rellich argument**, not of the theorem, and it can be removed outright by changing the core instead of estimating the perturbation. For weights $c_i \neq 0$ of arbitrary sign and arbitrary real b, b', complete the square in position and in momentum at once: $c_i(\pi_i^2 + x_i^2/4) + b_i x_i + b_i' \pi_i = c_i\big((\pi_i + \frac{b_i'}{2c_i})^2 + \frac{1}{4}(x_i + \frac{2b_i}{c_i})^2\big) - \frac{(b_i')^2}{4c_i} - \frac{b_i^2}{c_i}$. The right-hand side is the **same** diagonal quadratic Hamiltonian H_c, plus a real constant, written in the frame recentred at the classical equilibrium $a_i = -2b_i/c_i$ and boosted to the classical momentum $k_i = -b_i'/(2c_i)$. Translating by a and modulating by $e^{i\langle k, x\rangle}$ are unitary substitutions of $L^2(\mathbb{R}^d)$, so the translated, modulated Gauss–polynomial functions $p(x-a)\, e^{-\|x-a\|^2/4}\, e^{i\langle k,x\rangle}$ again form a dense core carrying an orthonormal Hermite family, and on it the full operator is diagonal with eigenvalues $\sum_i c_i(\alpha_i + \frac{1}{2}) + \sum_i(-\frac{(b_i')^2}{4c_i} - \frac{b_i^2}{c_i})$. A real diagonal action on a total orthonormal family is symmetric with trivial deficiency, so no domination and no sign condition on c is needed: only $c_i \neq 0$. Taking the Minkowski weights $(1, -1, \ldots, -1)$ gives the wave operator with the indefinite quadratic potential **plus** an arbitrary constant external field and an arbitrary constant boost, essentially self-adjoint on a dense core — the case the relative-bound route above could not reach.

```
#check @BookProof.ShiftedHermiteCore.polyGaussCoreT_dense
#check @BookProof.ShiftedHermiteCore.orthonormal_hermiteTLp
#check @BookProof.ShiftedHermiteCore.hermiteTLp_total
#check @BookProof.ShiftedQuadratic.shiftedHPoly_eq_quadPoly
#check @BookProof.ShiftedQuadratic.shiftedHOp_hermiteTLp
#check @BookProof.ShiftedQuadratic.shiftedHOp_symmetric
#check @BookProof.ShiftedQuadratic.shiftedHOp_essentiallySelfAdjoint
#check @BookProof.ShiftedQuadratic.shiftedHOp_not_bounded
#check @BookProof.ShiftedQuadratic.shiftedHPoly_apply_eq_differential
#check @BookProof.ShiftedQuadratic.wave_indefiniteQuadratic_linear_essentiallySelfAdj
```

Diagonality in the coordinates is not a real restriction either, as long as the kinetic form and the potential share the same matrix. Let A be **any** real symmetric $d \times d$ matrix and consider $H_A = \sum_{k,l} A_{kl}\,(\pi_k \pi_l + x_k x_l/4)$ with $\pi_k = -i\,\partial/\partial x_k$: the kinetic part is the second-order operator $-\sum_{k,l} A_{kl}\,\partial_k \partial_l$ and the potential is the quadratic form $\frac{1}{4}\sum_{k,l} A_{kl}\, x_k x_l$, with no sign condition on A, so the signature may be elliptic, hyperbolic or degenerate. Then H_A is symmetric and essentially self-adjoint on the same Gauss–polynomial core. The proof is a change of coordinates: the Gaussian weight $e^{-\|x\|^2/4}$ is rotation invariant, so composing with an orthogonal matrix O is a unitary of $L^2(\mathbb{R}^d)$ that maps the core onto itself and acts on the polynomial coordinates as an algebra automorphism. Both members of the canonical pair transform contravariantly with the **same** matrix, so the substitution carries the diagonal Hamiltonian H_c onto $H_{O\,\mathrm{diag}(c)\,O^{\mathsf{T}}}$; the spectral theorem for real symmetric matrices supplies O and c with $A = O\,\mathrm{diag}(c)\,O^{\mathsf{T}}$, and the rotated product Hermite functions are an orthonormal family of joint eigenvectors of H_A spanning the core. Taking for A the rotated Minkowski form gives the wave operator with the indefinite quadratic potential in rotated coordinates, where neither the kinetic form nor the potential is diagonal. The operator is genuinely unbounded whenever $A \neq 0$. The honest boundary is unchanged: the potential is still a quadratic **matched** to the kinetic form, and a general Faris–Lavine potential is not covered.

```
#check @BookProof.QuadraticRotation.rotPoly_momPoly
#check @BookProof.QuadraticRotation.quadPolyMat_rotPoly
#check @BookProof.QuadraticRotation.quadPolyMat_diagonal
#check @BookProof.QuadraticRotation.orthonormal_rotHermiteLp
#check @BookProof.QuadraticRotation.span_rotHermiteLp
```

```
#check @BookProof.QuadraticRotation.exists_rotConj
#check @BookProof.QuadraticRotation.quadOpMat_symmetric
#check @BookProof.QuadraticRotation.quadOpMat_essentiallySelfAdjoint
#check @BookProof.QuadraticRotation.wave_rotated_essentiallySelfAdjoint
#check @BookProof.QuadraticRotation.quadOpMat_not_bounded
```

The two widenings combine. Let A be a **positive definite** real symmetric matrix and let b, b' be arbitrary real vectors: then the inhomogeneous operator $H_A + \sum_i (b_i x_i + b'_i \pi_i)$ — a general elliptic quadratic form with cross terms, plus a general unbounded first-order term that does not commute with it — is symmetric and essentially self-adjoint on the same Gauss–polynomial core. Taking $b' = 0$ this is an anisotropic harmonic oscillator with cross terms in a constant external field. The eigenbasis argument alone no longer suffices, because the first-order term is not diagonal in the rotated Hermite basis; instead the rotated Hermite functions are upgraded to a Hilbert basis, which turns the orthogonal substitution into an honest unitary of $L^2(\mathbb{R}^d)$. On the core that unitary **is** the polynomial substitution, so it preserves the core, carries H_c onto $H_{O\,\mathrm{diag}(c)\,O^\top}$ and carries the first-order symbol with coefficients b, b' onto the one with Ob, Ob'. Essential self-adjointness is a unitary invariant, so the Kato–Rellich theorem for the diagonal case transfers. Positive definiteness is used exactly once, to bound the eigenvalues of A below by a positive constant — which is what the relative bound needs, and which genuinely fails in the indefinite case.

```
#check @BookProof.QuadraticRotationPerturbed.rotU_pgLp
#check @BookProof.QuadraticRotationPerturbed.rotPoly_foPoly
#check @BookProof.QuadraticRotationPerturbed.rotU_intertwine
#check @BookProof.QuadraticRotationPerturbed.exists_lower_bound_eigenvalues
#check @BookProof.QuadraticRotationPerturbed.quadOpMat_add_firstOrder_symmetric
#check @BookProof.QuadraticRotationPerturbed.quadOpMat_add_firstOrder_essentiallySelf
#check @BookProof.QuadraticRotationPerturbed.anisotropicOsc_add_linearPotential_esser
```

Positive definiteness can be removed as well, by combining the two mechanisms: the orthogonal substitution that diagonalizes the quadratic form, and the phase-space translation that completes the square. For a real symmetric matrix A and arbitrary real b, b', put the classical equilibrium equations $Aa = -2b$ and $Ak = -b'/2$. Whenever they have a solution, in the frame recentred at a and boosted to k the first-order term disappears and the Hamiltonian becomes H_A plus a real constant, so the translated, modulated, rotated Hermite functions are an orthonormal total family of eigenvectors and the operator $H_A + \sum_i (b_i x_i + b'_i \pi_i)$ is symmetric and essentially self-adjoint on the translated, modulated core, with no sign condition on A. An invertible A always admits a solution, $a = -2A^{-1}b$ and $k = -A^{-1}b'/2$; for a **singular** A the equations are solvable exactly when b and b' are orthogonal to the kernel of A, and that criterion is proved in both directions. The residual case is honest and irreducible: in a kernel direction with $b_i \neq 0$ the Hamiltonian degenerates to the first-order operator $b_i x_i + b'_i \pi_i$, which has no eigenvector in L^2.

```
#check @BookProof.ShiftedQuadraticMatrix.shiftedHMatPoly_eq_quadPolyMat
#check @BookProof.ShiftedQuadraticMatrix.hermiteTRLp_total
#check @BookProof.ShiftedQuadraticMatrix.shiftedHMatOp_hermiteTRLp
#check @BookProof.ShiftedQuadraticMatrix.shiftedHMatOp_essentiallySelfAdjoint
#check @BookProof.ShiftedQuadraticMatrix.wave_rotated_linear_essentiallySelfAdjoint
#check @BookProof.ShiftedQuadraticDegenerate.equilibrium_orthogonal_to_kernel
#check @BookProof.ShiftedQuadraticDegenerate.exists_equilibrium
#check @BookProof.ShiftedQuadraticDegenerate.exists_equilibrium_iff
#check @BookProof.ShiftedQuadraticDegenerate.shiftedHMatOp_essentiallySelfAdjoint_of_
#check @BookProof.ShiftedQuadraticDegenerate.exists_shiftedHMat_esa_of_kernel_orthogc
#check @BookProof.ShiftedQuadraticDegenerate.diagonal_degenerate_essentiallySelfAdjoi
```

Essential self-adjointness on a dense core buys the **existence** of the unitary group by Stone's theorem, but on an eigenvector the dynamics can be written down. If T is self-adjoint, U is any unitary flow solving $d/dt\,(U_t x) = -i\,T U_t x$ on the domain of T, and $T\psi = \lambda\psi$ with λ real, then $U_t\psi = e^{-i\lambda t}\psi$ for every t. No spectral theorem is needed: the scalar function $\varphi(t) = \langle \psi, U_t\psi\rangle$ satisfies $\varphi' = -i\lambda\varphi$ by the Schrödinger equation and the symmetry of T, so $\varphi(t) = e^{-i\lambda t}\|\psi\|^2$, and since U_t is isometric this is the equality case of the Cauchy–Schwarz inequality. Applied to the quadratic family, the flow generated by $H_A + \sum_i (b_i x_i + b'_i \pi_i)$ acts on the Hermite eigenbasis of its core by the phase $e^{-iE_\alpha t}$ with $E_\alpha = \sum_i c_i(\alpha_i + \tfrac{1}{2}) + \mathrm{const}$: the Schrödinger equation for these Hamiltonians is solved in closed form on a total family of initial data.

```
#check @BookProof.StoneEigenflow.isSelfAdjointExtension_eigenvector
#check @BookProof.StoneEigenflow.stoneFlow_apply_eigenvector
#check @BookProof.StoneEigenflow.exists_diagonal_stone_flow
#check @BookProof.ShiftedQuadraticDegenerate.exists_shiftedHMat_diagonal_flow
#check @BookProof.ShiftedQuadraticDegenerate.exists_shiftedH_diagonal_flow
```

The kernel directions left open above are first-order operators, and those are reached by the other route in this development — the Fourier multiplier argument. The Plancherel proof used for the wave operator never looks at the shape of the operator: it uses only that the Fourier transform turns it into multiplication by a **real, smooth** symbol, and that suffices for symmetry and for the vanishing of both deficiency spaces. Stated that way it is an instrument, and it applies verbatim to the momentum family $\sum_i c_i \pi_{w_i}$, $\pi_w = -i\,\partial_w$, whose symbol $\sum_i 2\pi c_i \langle \xi, w_i\rangle$ is real and smooth, and to the full constant-coefficient operator $\sum_i c_i \partial_{w_i}^2 + \sum_i a_i(-i\,\partial_{w_i}) + \kappa$ — second order, plus an unbounded first-order term, plus a constant — on the Schwartz core of $L^2(V)$.

```
#check @BookProof.FourierMultiplierEsa.symmetricOn_of_real_symbol
#check @BookProof.FourierMultiplierEsa.essentiallySelfAdjointOn_of_real_symbol
#check @BookProof.FourierMultiplierEsa.fourier_firstOrderOp_apply
#check @BookProof.FourierMultiplierEsa.firstOrderOp_essentiallySelfAdjoint
#check @BookProof.FourierMultiplierEsa.momentumOp_essentiallySelfAdjoint
#check @BookProof.FourierMultiplierEsa.mixedOp_essentiallySelfAdjoint
```

One case is reached by neither route: a kernel direction carrying a linear potential **and** a momentum term at once, $\langle x, b\rangle - i\,\partial_m$ with both coefficients non-zero. It has no L^2 eigenvector, so the Hermite eigenbasis cannot see it, and it is not constant-coefficient, so the Fourier multiplier is not available. A **gauge** settles it. The unimodular factor $e^{i\theta}$ with the quadratic phase $\theta(x) = -\langle x, b\rangle\langle x, m\rangle/\|m\|^2 + \langle b, m\rangle\langle x, m\rangle^2/(2\|m\|^4)$, whose derivative along m is exactly $-\langle x, b\rangle$, intertwines the mixed operator with the pure momentum operator: $(\langle \cdot, b\rangle - i\partial_m)(e^{i\theta}\varphi) = e^{i\theta}(-i\partial_m\varphi)$. Multiplication by $e^{i\theta}$ keeps a compactly supported test function compactly supported, and compactly supported test functions already detect the deficiency spaces of the momentum operator — a cut-off plus dominated convergence argument. So the mixed first-order operator is symmetric and essentially self-adjoint on the Schwartz core of $L^2(V)$ for arbitrary b, m, and with it the whole first-order residue of the quadratic family.

```
#check @BookProof.MixedLinearEsa.posOp_essentiallySelfAdjoint
#check @BookProof.MixedLinearEsa.momentum_test_compactSupport_extend
#check @BookProof.MixedLinearEsa.mixedLinearOp_gauge
#check @BookProof.MixedLinearEsa.mixedLinearOp_symmetric
#check @BookProof.MixedLinearEsa.mixedLinearOp_essentiallySelfAdjoint
```

That settles the mixed first-order operator on the **Schwartz** core. The core on which the whole quadratic family lives is a different one — the Gauss–polynomial (product Hermite) core — and the same operator can be settled there too, by the metaplectic rotation. Write the quadrature as $\sum_i (b_i x_i + b_i' \pi_i)$ and give each direction the complex amplitude $w_i = b_i + i b_i'/2$: on the product Hermite basis the quadrature raises the i-th excitation number with coefficient w_i and lowers it with $\overline{w_i}$. Multiplying ψ_α by ζ^α with $\zeta_i = w_i/|w_i|$ unimodular is a unitary of $L^2(\mathbb{R}^d)$ — the phase-rotated Hermite functions are again a Hilbert basis — it preserves the core, and by the polar decomposition $w_i = |w_i|\zeta_i$ it carries the purely positional quadrature $\sum_i |w_i| x_i$ onto the general one. So it is enough to treat multiplication by a real linear function, and that is done by a moment lemma that needs no L^2 hypothesis: a function all of whose exponentially weighted moments are finite and all of whose polynomial moments vanish is zero almost everywhere. The quadrature is therefore essentially self-adjoint on the Gauss–polynomial core, for arbitrary real b, b', and generates a complete unitary flow.

```
#check @BookProof.QuadratureEsa.ae_eq_zero_of_moments'
#check @BookProof.QuadratureEsa.foOp_pos_essentiallySelfAdjoint
#check @BookProof.QuadratureEsa.foOp_hermiteCore
#check @BookProof.QuadratureEsa.phaseU_foOp_hermiteCore
#check @BookProof.QuadratureEsa.foOp_essentiallySelfAdjoint
#check @BookProof.QuadratureEsa.foOp_stone_flow
```

The two pieces — the diagonal quadratic part and the first-order part — can also be handled **together**, and then every restriction disappears at once. On the product Hermite basis the operator $H = \sum_i c_i(\pi_i^2 + x_i^2/4) + \sum_i (b_i x_i + b_i' \pi_i)$ is a nearest-neighbour matrix: the quadratic part is diagonal with the **real** symbol $\sum_i c_i(\alpha_i + 1/2)$, and the first-order part moves a single excitation number by one, with amplitude of size $O(\sqrt{\alpha_i})$. A deficiency vector at a non-real point z is therefore a square-summable solution of a nearest-neighbour recursion

on the lattice of multi-indices, and such a solution must vanish — this is Carleman's classical argument, run on cubes $\{\alpha : \alpha_i \le N\}$ instead of intervals. Summing the recursion against the conjugate coefficients over a cube, the interior contributions occur in conjugate pairs and the real diagonal contributes nothing to the imaginary part, so $|\operatorname{Im} z|$ times the mass inside the cube equals the **flux** through the boundary faces, which is at most $\sqrt{N+1}$ times the mass carried by those faces. Distinct faces are disjoint, so the total face mass is finite, while $\sum_N 1/\sqrt{N+1}$ diverges: the mass inside the cube cannot stay bounded away from zero. Hence, for **arbitrary** real weights c — any signs, zeros allowed — and **arbitrary** real b, b', the operator H is essentially self-adjoint on the plain Gauss–polynomial core and generates a complete unitary flow. No ellipticity, no sign condition, no classical equilibrium and no change of core is needed; with the Minkowski weights this is the wave operator with the indefinite quadratic potential, a constant external field and a constant boost.

```
#check @BookProof.HermiteCarleman.flux_identity
#check @BookProof.HermiteCarleman.flux_bound
#check @BookProof.HermiteCarleman.ladder_eq_zero
#check @BookProof.HermiteCarleman.mixOp_hermiteCore
#check @BookProof.HermiteCarleman.mixOp_essentiallySelfAdjoint
#check @BookProof.HermiteCarleman.mixOp_stone_flow
#check @BookProof.HermiteCarleman.wave_indefiniteQuadratic_firstOrder_essentiallySelf
```

One restriction is still hidden in that statement: the quadratic part is the **harmonic** one, a multiple of $\pi_i^2 + x_i^2/4$ in each mode, which is precisely what is diagonal on the Hermite basis. The one-mode real quadratic forms are three-dimensional — spanned by π_i^2, x_i^2 and the squeezing generator $\frac{1}{2}(x_i\pi_i + \pi_i x_i)$ — and only a line inside that space is diagonal. The two off-diagonal directions raise and lower an excitation number by **two**, with amplitudes $\sqrt{(\alpha_i+1)(\alpha_i+2)}$ and $\sqrt{\alpha_i(\alpha_i-1)}$ of size $O(\alpha_i)$. The same flux argument still applies, because the boundary layer only has to be made two thick and the growth rate only rises from $\sqrt{N}$ to N, and $\sum_N 1/N$ still diverges. What changes is bookkeeping: faces of thickness two are no longer disjoint, so Bessel's inequality is used with a multiplicity bound — each multi-index lies in at most two of them. The conclusion is that for **arbitrary** real p, q, s, b, b' the operator $H = \sum_i (p_i\pi_i^2 + q_i x_i^2 + s_i \frac{1}{2}(x_i\pi_i + \pi_i x_i)) + \sum_i (b_i x_i + b'_i \pi_i)$ is essentially self-adjoint on the plain Gauss–polynomial core: elliptic, hyperbolic or parabolic in each mode, with any signs and with degenerate modes allowed. Taking $s = 1$ and everything else zero, the generator of dilations $\frac{1}{2}\sum_i (x_i\pi_i + \pi_i x_i)$, the purely hyperbolic member of the family, is essentially self-adjoint on that core and generates a complete unitary flow.

```
#check @BookProof.CarlemanTwoStep.flux_identity2
#check @BookProof.CarlemanTwoStep.ladder2_eq_zero
#check @BookProof.ModeQuadratic.mqOp_hermiteCore
#check @BookProof.ModeQuadratic.mqOp_essentiallySelfAdjoint
#check @BookProof.ModeQuadratic.mqOp_stone_flow
#check @BookProof.ModeQuadratic.dilation_essentiallySelfAdjoint
#check @BookProof.ModeQuadratic.dilation_stone_flow
```

That still leaves one restriction: every term was **mode-diagonal**, so the modes never talked to each other. A general real quadratic Hamiltonian also carries the cross terms $\pi_i\pi_j$, $x_i x_j$ and $\frac{1}{2}(x_i\pi_j + \pi_j x_i)$ with $i \ne j$, and those move two different excitation numbers at once. The cube grading is the wrong one for them; the right one is the **simplex** grading by total degree $|\alpha| = \sum_i \alpha_i$. In the ladder variables a product of two of x, π splits into three kinds of hop: pair creation $\alpha \mapsto \alpha + e_i + e_j$, pair annihilation $\alpha \mapsto \alpha - e_i - e_j$, and mode exchange $\alpha \mapsto \alpha - e_j + e_i$. Only the first two change $|\alpha|$, and they change it by exactly ± 2, so they leak through a shell of thickness two. The exchange hops preserve $|\alpha|$ entirely, and their contribution to the flux is not merely bounded but **identically zero**: their amplitude matrix is Hermitian, so the sum over a shell is real and its imaginary part cancels pairwise. The Carleman divergence $\sum_N 1/(N+2) = \infty$ then finishes the argument unchanged. The conclusion is that for **arbitrary** real matrices P, Q, S and vectors b, b' the operator $H = \sum_{i,j}(P_{ij}\pi_i\pi_j + Q_{ij}x_i x_j + S_{ij}\frac{1}{2}(x_i\pi_j + \pi_j x_i)) + \sum_i (b_i x_i + b'_i\pi_i)$ — every real quadratic-plus-linear Hamiltonian in d degrees of freedom, with no ellipticity, no definiteness, no non-degeneracy and no classical equilibrium — is essentially self-adjoint on the plain Gauss–polynomial core and generates a complete unitary flow. In particular so does the purely off-diagonal cross term $\frac{1}{2}(x_i\pi_j + \pi_j x_i) + \frac{1}{2}(x_j\pi_i + \pi_i x_j)$. Taking S **antisymmetric** — where the diagonal vanishes and x_i commutes with π_j for $i \ne j$, so that $\sum_{i,j} S_{ij}\frac{1}{2}(x_i\pi_j + \pi_j x_i) = \sum_{i<j} S_{ij}(x_i\pi_j - x_j\pi_i)$ — the family contains the **angular-momentum** generators $x_k\pi_l - x_l\pi_k$, the compact

counterparts of the dilation generator above; they are essentially self-adjoint on the same core and generate complete unitary rotation flows.

```
#check @BookProof.CarlemanSimplex.sum_mterm_im
#check @BookProof.CarlemanSimplex.flux_identityQ
#check @BookProof.CarlemanSimplex.ladderQ_eq_zero
#check @BookProof.FullQuadratic.fqOp_hermiteCore
#check @BookProof.FullQuadratic.fqOp_essentiallySelfAdjoint
#check @BookProof.FullQuadratic.fqOp_stone_flow
#check @BookProof.FullQuadratic.crossTerm_essentiallySelfAdjoint
#check @BookProof.FullQuadratic.crossTerm_stone_flow
#check @BookProof.FullQuadratic.fqQuadPoly_rotMat
#check @BookProof.FullQuadratic.angularMomentum_essentiallySelfAdjoint
#check @BookProof.FullQuadratic.angularMomentum_stone_flow
```

All of that is still a **finite** number of degrees of freedom. A field has infinitely many, and the natural home for them is the boson Fock space: the ℓ^2 space of occupation-number configurations $\alpha : \iota \to_0 \mathbb{N}$ over an arbitrary mode set ι. The comparison operator that runs the Faris–Lavine argument there is the free Hamiltonian plus the number operator plus one, whose symbol is $\sigma(\alpha) = \omega(\alpha) + |\alpha| + 1$, and the point of the choice is that a quadratic monomial $a^{\dagger P} a^{Q}$ with $|P| + |Q| \le 2$ has ladder amplitude at most 2σ at **both** ends of the hop $\alpha \mapsto \alpha - P + Q$. That two-sided bound gives the relative bound directly, and an arithmetic–geometric-mean step gives the commutator bound, because the two halves of a Hermitian interaction term recombine so that only the **difference** $\sigma(\alpha) - \sigma(\alpha - P + Q)$ survives in the imaginary part — and that difference is at most $\omega(P) + \omega(Q) + 2$, whatever α is. Both Faris–Lavine constants are additive and survive an infinite sum of terms as soon as they are summable. The conclusion is that for an arbitrary mode set, an arbitrary non-negative and possibly **unbounded** dispersion ω, and an arbitrary family of quadratic interaction terms subject only to the weighted absolute summability $\sum_k \|g_k\|(\omega(P_k) + \omega(Q_k) + 2) < \infty$, the second-quantized Hamiltonian $H = \sum_i \omega_i a_i^\dagger a_i + \sum_k (g_k a^{\dagger P_k} a^{Q_k} + \overline{g_k} a^{\dagger Q_k} a^{P_k})$ is essentially self-adjoint on the finite-particle core of the Fock space. Bogoliubov pair creation $g\, a_m^\dagger a_n^\dagger + \overline{g}\, a_n a_m$ is the special case $P = e_m + e_n$, $Q = 0$.

```
#check @BookProof.OperatorSeries.commForm_eq_neg_two_im
#check @BookProof.OperatorSeries.seriesOp_commForm_le
#check @BookProof.OperatorSeries.essentiallySelfAdjointOn_finiteModes_of_series
#check @BookProof.FockQuadratic.amp_symm
#check @BookProof.FockQuadratic.hopOp_pairing
#check @BookProof.FockQuadratic.pairOp_commForm_le
#check @BookProof.FockQuadratic.freeOp_commForm
#check @BookProof.FockQuadratic.fockH_essentiallySelfAdjointOn_core
#check @BookProof.FockQuadratic.bogoliubov_essentiallySelfAdjointOn_core
```

The transfer step needs an actual Hilbert-space unitary, and the raw point map $e \mapsto y = \sqrt{e}$ is not one: the Jacobian $de/dy = 2y$ must be carried along as the half-density factor $\sqrt{2y}$. That unitary is **constructed**, not assumed: composition with $y \mapsto y^2$ is measure preserving from the weighted measure $2y\,dy$ to Lebesgue measure on $(0,\infty)$, hence a unitary $L^2((0,\infty), de) \simeq L^2((0,\infty), 2y\,dy)$, and the transfer theorem specializes to it:

```
#check @BookProof.QuantumGravityHalfDensity.qgSrcMeasure_density_eq_halfDensity_sq
#check @BookProof.QuantumGravityHalfDensity.measurePreserving_qgSquare
#check @BookProof.QuantumGravityHalfDensity.measurePreserving_qgSqrt
#check @BookProof.QuantumGravityHalfDensity.halfDensityUnitary
#check @BookProof.QuantumGravityHalfDensity.halfDensityUnitary_apply
#check @BookProof.QuantumGravityHalfDensity.halfDensityUnitary_symm_apply
#check @BookProof.QuantumGravityHalfDensity.exists_halfDensity_unitary
#check @BookProof.QuantumGravityHalfDensity.qg_halfDensity_transfer
```

That unitary can now be used for what it was built for. The conformal-mode potential $V_3(R_c) = -\frac{M^2}{2} R_c + \alpha R_c^2$ becomes, in the densitized coordinate, $y \mapsto V_3(y^2)$, and the bound survives the change of variables: it is still bounded below by $-M^4/(16\alpha)$ (`densConfV_ge`) — while at $\alpha = 0$ the densitized potential still runs off to $-\infty$ (`densConfV_zero_alpha_tendsto_atBot`), so it is the αR^2 term, not the densitization, that buys the bound. On the bounded-energy cores of $L^2((0,\infty), de)$ and $L^2((0,\infty), 2y\,dy)$ the two multiplication Hamiltonians are carried onto one another by the half-density unitary (`halfDensityUnitary_mem_densConfCore`, `halfDensityUnitary_densConfCore_surjective`, `halfDensityUnitary_intertwines`), so vanishing

adjoint deficiency of the densitized operator **transfers** to the physical one (`physConf_hasZeroDeficiencyOn_transfer`) instead of being re-proved, and both sides carry a unitary flow. Finally the bound is not merely pointwise: the quadratic form of a multiplication operator is the integral of its multiplier against $|f|^2$ (`multOp_quadForm_eq`), so a pointwise lower bound on the potential **is** semiboundedness of the operator (`multOp_quadForm_ge`, `densConfOp_quadForm_ge`). What is still missing for the full continuum statement is the finite-speed / direct-integral gluing, not the potential.

```
#check @BookProof.ScalaronDensitized.densConfV
#check @BookProof.ScalaronDensitized.densConfV_comp_densY
#check @BookProof.ScalaronDensitized.densConfV_ge
#check @BookProof.ScalaronDensitized.densConfV_zero_alpha_tendsto_atBot
#check @BookProof.ScalaronDensitized.halfDensityUnitary_mem_densConfCore
#check @BookProof.ScalaronDensitized.halfDensityUnitary_densConfCore_surjective
#check @BookProof.ScalaronDensitized.halfDensityUnitary_intertwines
#check @BookProof.ScalaronDensitized.physConf_hasZeroDeficiencyOn_transfer
#check @BookProof.ScalaronDensitized.multOp_quadForm_eq
#check @BookProof.ScalaronDensitized.multOp_quadForm_ge
#check @BookProof.ScalaronDensitized.densConfOp_quadForm_ge
#check @BookProof.ScalaronDensitized.physConf_stone_flow
```

31.9. The Gauge-Fixed Field-Space Hamiltonian and its BRST Charge

The densitized change of variables above is local in one conformal mode. The manuscript's actual object is the whole gauge-fixed field space: a $84 = 4 + 16 + 64$ dimensional block of tetrad, tetrad-derivative and second-derivative coordinates, carrying the Gauss-weighted polynomial core of $L^2(\mathbb{R}^{84})$. On that core the coordinates x_j and momenta $\pi_k = -i\partial_k$ satisfy the canonical commutation relations $[x_j, \pi_k] = i\delta_{jk}$ (`qgCCR`, `qgCCR_tetrad`), the Weyl-ordered products $\frac{1}{2}(ST + TS)$ of Gauss-symmetric operators are again Gauss symmetric, and the densitization of the previous section reappears as the absorption of the singular $1/e$ (`qg3DDensity_densitized`). The gauge-fixed Hamiltonian `qg3DHamiltonian` is symmetric on the core (`qg3D_symmetricOn`) and its quadratic form is the signed sum of squares of the constituent operators (`qg3D_quadForm`). Its signature is genuinely indefinite — the kinetic and the conformal coefficient have opposite signs (`qgKappa_indefinite`) — so no semiboundedness, and hence no Friedrichs extension, is claimed for the full operator; what is proved is that the **elliptic** truncation, in which the conformal sign is flipped, is nonnegative and does have a Friedrichs extension selected by the Hashimoto criterion (`qg3DElliptic_friedrichs_extension`, `qg3DElliptic_hashimoto_selects`).

```
#check @BookProof.QuantumGravity3DGauge.qg3DDensity_densitized
#check @BookProof.QuantumGravity3DGauge.qgCCR
#check @BookProof.QuantumGravity3DGauge.qgCCR_tetrad
#check @BookProof.QuantumGravity3DGauge.qgKappa_indefinite
#check @BookProof.QuantumGravity3DGauge.qg3DHamiltonian
#check @BookProof.QuantumGravity3DGauge.qg3D_symmetricOn
#check @BookProof.QuantumGravity3DGauge.qg3D_quadForm
#check @BookProof.QuantumGravity3DGauge.qg3DElliptic_friedrichs_extension
#check @BookProof.QuantumGravity3DGauge.qg3DElliptic_hashimoto_selects
```

Indefiniteness blocks the **Friedrichs** route, but not self-adjointness itself. The gauge-fixed Hamiltonian is quadratic in the canonical pair: its kinetic term is a signed sum of squares of the momenta and its torsion potential is a sum of squares of the torsion linear forms $T_m = \partial_\mu e_\nu{}^a - \partial_\nu e_\mu{}^a$ (`sum_torsionVec_X`, `qgFqQ_quadratic_eq`). It is therefore an instance of the general real quadratic Hamiltonian whose essential self-adjointness on the Gauss-polynomial core was proved earlier (`qgSigned_eq_fqOp`), and the conclusion follows for **every** real signature (`qgSigned_essentiallySelfAdjointOn_core`), the physical hyperbolic one included (`qg3D_essentiallySelfAdjointOn_core`). So the closure of `qg3DHamiltonian` is its unique self-adjoint realization and it generates a complete unitary flow (`qg3D_stone_flow`); the elliptic sector's Friedrichs extension is thereby upgraded from existence to uniqueness (`qg3DElliptic_essentiallySelfAdjointOn_core`). No spectrum, no mass gap and no continuum limit is claimed.

```
#check @BookProof.Qg3DGaugeEsa.sum_torsionVec_X
#check @BookProof.Qg3DGaugeEsa.qgFqQ_quadratic_eq
#check @BookProof.Qg3DGaugeEsa.qgSigned_eq_fqOp
#check @BookProof.Qg3DGaugeEsa.qgSigned_essentiallySelfAdjointOn_core
#check @BookProof.Qg3DGaugeEsa.qg3D_essentiallySelfAdjointOn_core
```

```
#check @BookProof.Qg3DGaugeEsa.qg3D_stone_flow
#check @BookProof.Qg3DGaugeEsa.qg3DElliptic_essentiallySelfAdjointOn_core
```

The operator qg3DHamiltonian does not yet contain the momentum–coordinate cross terms $\frac{1}{2}\mathcal{S}^{ab}E_{ab} + \frac{1}{3}\mathcal{P}E_a{}^a$ of the three-dimensional density. At the flat density $e = 1$ every term of that density is quadratic in the canonical pair, and adding an arbitrary Weyl-ordered quadratic coupling keeps the operator inside the general real quadratic class (fqPoly_add_coupling), so essential self-adjointness survives for every real signature and every real coupling (qgWithCoupling_esa). With the book's cross terms written out (bookCrossMat) the trace parts cancel, leaving $\frac{1}{2}\sum_{a,b}(p^{ab} + p^{ba})E_{ab}$ (bookCrossMat_eq_sym), and the Hamiltonian with the cross terms and an arbitrary real quadratic bracket is essentially self-adjoint on the Gauss-polynomial core (qg3DCross_esa, qg3DCross_stone_flow), and so is its enclosure $d\Gamma(h)$, both on the finite-particle domain (qg3DCross_dGamma_esa) and in the occupation-number spelling on the finite-occupation core (qg3DCross_dGammaOp_esa). The genuine dependence on the density e — the $1/e$ of the kinetic term and the polynomial e in front of the bracket — is not covered.

```
#check @BookProof.Qg3DCrossTermEsa.fqPoly_add_coupling
#check @BookProof.Qg3DCrossTermEsa.qgWithCoupling_esa
#check @BookProof.Qg3DCrossTermEsa.bookCrossMat
#check @BookProof.Qg3DCrossTermEsa.bookCrossMat_eq_sym
#check @BookProof.Qg3DCrossTermEsa.qg3DCross_esa
#check @BookProof.Qg3DCrossTermEsa.qg3DCross_stone_flow
#check @BookProof.Qg3DCrossTermEsa.qg3DCross_dGamma_esa
#check @BookProof.Qg3DCrossTermEsa.qg3DCross_dGammaOp_esa
```

The density left as a boundary there is taken up in BookProof/Qg/ChapterQg3DDensityEsa.lean, in the two forms in which it can be handled. The inverse tetrad χ enters $\mathcal{P}$ and $\mathcal{S}$ as coefficient vectors in the 84 momenta (momCalP, momCalS, flat at χ = δ), and the book's three-dimensional density at density e is $\mathcal{H} = (1/(16e))\mathcal{S}^2 - (1/(24e))\mathcal{P}^2 + \frac{1}{2}\mathcal{S}\cdot E + \frac{1}{3}\mathcal{P}\cdot E - e\,Q_b$ (qg3DDensityHam): for **every** real e, χ and bracket Q_b the operator is essentially self-adjoint on the Gauss–polynomial core, with its Stone flow and both enclosures (qg3DDensity_esa, qg3DDensity_stone_flow, qg3DDensity_dGamma_esa, qg3DDensity_dGammaOp_esa). In the densitized variables $y = \sqrt{e}$ the kinetic coefficient is constant and the physical operator is recovered verbatim (kinOf_absorption, qg3DDensitized_eq_density, qg3DDensitized_esa — the last holds even at the degenerate $y = 0$, where the physical $1/e$ form does not exist), and letting the density label a family of background tetrads gives the fibred operator, essentially self-adjoint fibrewise with no uniformity in the family (qgFibredDensityHam, qgFibredDensity_esa, qgFibredDensity_posDet_esa). The honest boundary: the density is **frozen** in each operator — a constant, or a fibre label without a conjugate momentum of its own. The operator in which $e = \det e_b{}^a$ is itself a multiplication operator in the canonical tetrad coordinates is not covered: $1/e$ is singular on the zero set of e, and in the densitized coordinates the cross terms turn cubic and the bracket a sextic indefinite potential under a hyperbolic kinetic term — outside every essential-self-adjointness instrument of the project.

```
#check @BookProof.Qg3DDensityEsa.momCalS
#check @BookProof.Qg3DDensityEsa.qg3DDensityHam
#check @BookProof.Qg3DDensityEsa.qg3DDensity_esa
#check @BookProof.Qg3DDensityEsa.qg3DDensity_stone_flow
#check @BookProof.Qg3DDensityEsa.qg3DDensity_dGamma_esa
#check @BookProof.Qg3DDensityEsa.qg3DDensitized_eq_density
#check @BookProof.Qg3DDensityEsa.qg3DDensitized_esa
#check @BookProof.Qg3DDensityEsa.qgFibredDensity_esa
#check @BookProof.Qg3DDensityEsa.qgFibredDensity_posDet_esa
```

The constraints of that field space are dressed with ghosts into the BRST charge $\Omega = \sum_a G_a\chi_a - \frac{1}{2}\sum_{a,b,e} f_{abe}\chi_a\chi_b\beta_e$. Nilpotency is proved in full: the square of the constraint part is half the ghost-contracted constraint algebra (glin_sq), the cross terms produce exactly the opposite quantity (glin_mul_Q_add_Q_mul_glin) — which is what fixes the coefficient $-\frac{1}{2}$ — and the cubic ghost term squares to zero by the Jacobi identity, so $\Omega^2 = 0$ (brst_full_nilpotent); for commuting constraints the cubic term is unnecessary (brst_abelian_nilpotent). The ghosts themselves are not assumed: the 19 diffeomorphism ghosts are realized on the $\mathbb{Z}_2^{19}$ occupation space $\Lambda(\mathbb{C}^{19})$ with creation the exterior product and annihilation the contraction, and the canonical anticommutation relations are proved (ghost_car). Constraints and ghosts live together on the graded space $\mathbb{C}[x_0,\dots,x_{83}]\otimes\Lambda(\mathbb{C}^{19})$, where they commute (bosOp_ghostOp_comm,

qgGhostCar). The constraints are the first-order generators $x_j\partial_k$ of linear changes of the field coordinates; these close into the matrix Lie algebra (elemGen_bracket, linGen_bracket), so any matrix family that closes with real structure constants obeying Jacobi yields a nilpotent charge (qgBRST_nilpotent). The statement is not vacuous: the affine algebra $[H, E] = E$, whose two generators demonstrably fail to commute (affMat_non_abelian), gives a concrete non-abelian nilpotent BRST charge (affBRST_nilpotent).

```
#check @BookProof.QuantumGravityBrstCharge.glin_sq
#check @BookProof.QuantumGravityBrstCharge.glin_mul_Q_add_Q_mul_glin
#check @BookProof.QuantumGravityBrstCharge.brst_full_nilpotent
#check @BookProof.QuantumGravityBrstCharge.brst_abelian_nilpotent
#check @BookProof.QuantumGravityBrstCharge.ghost_car
#check @BookProof.QuantumGravityBrstCharge.bosOp_ghostOp_comm
#check @BookProof.QuantumGravityBrstCharge.qgGhostCar
#check @BookProof.QuantumGravityBrstCharge.elemGen_bracket
#check @BookProof.QuantumGravityBrstCharge.linGen_bracket
#check @BookProof.QuantumGravityBrstCharge.qgBRST
#check @BookProof.QuantumGravityBrstCharge.qgBRST_nilpotent
#check @BookProof.QuantumGravityBrstCharge.qgBRST_abelian_nilpotent
#check @BookProof.QuantumGravityBrstCharge.affMat_non_abelian
#check @BookProof.QuantumGravityBrstCharge.affMat_close
#check @BookProof.QuantumGravityBrstCharge.affF_jacobi
#check @BookProof.QuantumGravityBrstCharge.affBRST_nilpotent
```

31.10. Fixing the Derivative Variables: the Torsion Becomes Linear

The constraints above are already algebraic in the field coordinates — the $x_j\partial_k$ that close into a matrix Lie algebra. The gauge fixing of the **derivative** variables is the other half of the manuscript's construction, and it is the same device the Navier–Stokes chapters use for $u_{i,j} = \partial_j u_i$: adjoin the derivative as an independent canonical variable, and let the gauge condition set it back equal to the derivative of the field. For gravity the extended variables are the vielbein modes ${e_\nu}^i(k)$ (CMode) together with one auxiliary derivative mode ${D_{\mu\nu}}^i(k)$ for every spatial derivative of every vielbein component; the two families together are EMode, and the derivative part alone is DMode. The gain is immediate: in these variables the torsion is **linear in the auxiliaries alone**, ${T_{\mu\nu}}^i = {D_{\mu\nu}}^i - {D_{\nu\mu}}^i$, with no derivative appearing anywhere (extTorsionCoef). The auxiliary modes are not decoration: extTorsionCoef_ne_zero shows their coefficients are genuinely present. The whole derivative content sits in the gauge condition ${D_{\mu\nu}}^i(k) = \mathrm{i}k_\mu {e_\nu}^i(k)$ — exactly the statement that an auxiliary variable is fixed to a spatial derivative of a field — and it is abelian, hence first class, hence gauge-able. gaugeReduce performs that substitution on linear forms, and two identities say that nothing is lost: the gauge-fixed extended torsion form is **exactly** the physical torsion $k_\mu {e_\nu}^i - k_\nu {e_\mu}^i$ of the continuum model (gaugeReduce_extTorsionCoef), and hence the gauge-fixed Gram matrix of all extended torsion terms is exactly the vielbein self-interaction of that model (gaugeReduce_gram, contTorsionGram). Consequently the gauge-fixed quantum-gravity Hamiltonian **is** the continuum Hamiltonian and is essentially self-adjoint on the outer Fock space (brstGaugeFixed_esa), the Starobinsky instance included (starobinsky_brstGaugeFixed_esa).

A gauge condition does more than rewrite the torsion: it **deletes modes**. That the physical Hamiltonian is still essentially self-adjoint on what survives therefore needs its own argument, and the general principle supplies it — admissible mode data restricted to **any** subset of the modes is again admissible (restrictModes). Every Faris–Lavine bound survives with the same constant K, because each sum involved is a sum of non-negative terms over a smaller index set (restrict_essentiallySelfAdjointOn). Applied to a set of modes that survive a gauge condition, this is the statement that gauge fixing cannot destroy essential self-adjointness (gaugeFixedSubset_esa).

```
#check @BookProof.QgBrstDerivativeGauge.DMode
#check @BookProof.QgBrstDerivativeGauge.EMode
#check @BookProof.QgBrstDerivativeGauge.extTorsionCoef
#check @BookProof.QgBrstDerivativeGauge.extTorsionCoef_ne_zero
#check @BookProof.QgBrstDerivativeGauge.gaugeReduce
#check @BookProof.QgBrstDerivativeGauge.gaugeReduce_extTorsionCoef
#check @BookProof.QgBrstDerivativeGauge.gaugeReduce_gram
#check @BookProof.QgContinuumModeInstance.contTorsionGram
#check @BookProof.QgBrstDerivativeGauge.brstGaugeFixed_esa
#check @BookProof.QgBrstDerivativeGauge.starobinsky_brstGaugeFixed_esa
```

```
#check @BookProof.QgBrstDerivativeGauge.restrictModes
#check @BookProof.QgBrstDerivativeGauge.restrict_essentiallySelfAdjointOn
#check @BookProof.QgBrstDerivativeGauge.gaugeFixedSubset_esa
```

31.11. The Graded Fock Space: Bosons, Ghosts and the Second Quantization

The manuscript's quantum Hilbert space for empty spacetime is the tensor product of a symmetric and an antisymmetric Fock space, $\Gamma^s(L^2(\mathbb{R}^{84}\times\mathbb{Z}_2^{19}))\otimes\Gamma^a(L^2(\mathbb{R}^{84}\times\mathbb{Z}_2^{19}))$, whose creation and annihilation operators form a $\mathbb{Z}_2$-graded superalgebra. The bosonic half is the occupation-number Fock space already built for Yang–Mills; what is added here is the fermionic (ghost) half and the grading. A ghost configuration is the finite set of occupied ghost modes — the Pauli principle is built into the representation — and the ladder operators carry the Jordan–Wigner sign $(-1)^{\#\{i\in\alpha\,:\,i<j\}}$. Their canonical anticommutation relations are proved, not postulated: $\{\psi_j,\psi_j^\dagger\}=1$ (`car_fermAnn_fermCre`), $\{\psi_j,\psi_k^\dagger\}=0$ for $j\neq k$ (`car_fermAnn_fermCre_of_ne`), and $\{\psi_j,\psi_k\}=\{\psi_j^\dagger,\psi_k^\dagger\}=0$ (`car_fermAnn_fermAnn`, `car_fermCre_fermCre`), whence $\psi_j^2=(\psi_j^\dagger)^2=0$. That $\psi_j^\dagger$ deserves its name is also proved: it is the adjoint of ψ_j on ℓ^2 of the ghost configurations (`inner_fermCre_left`).

The grading is the parity operator $\Gamma=(-1)^F$, an involution (`fermGrade_involutive`) for which both ladder operators are odd (`fermGrade_fermAnn`, `fermGrade_fermCre`); with the graded bracket $[x,y\}=xy-(-1)^{|x||y|}yx$ the anticommutation relations become the superbracket identity $[\psi_j,\psi_j^\dagger\}=1$ (`superBracket_fermAnn_fermCre`). On the graded state space bosonic and ghost operators supercommute (`bosOp_ghostOp_comm`, `superBracket_bosOp_ghostOp`), the bosonic canonical commutation relation and the ghost anticommutation relation both hold (`qgCCR`, `qgGhostCar`, and for the 19 diffeomorphism ghosts `qgGhostCar_book`), the bosonic operators are even and the ghost operators odd (`bosOp_even`, `ghostOp_odd_ann`, `ghostOp_odd_cre`).

Finally the dynamics. The joint occupation states of bosons and ghosts carry the second-quantized Hamiltonian whose symbol is the total boson energy plus the total ghost energy. It is essentially self-adjoint on the finite-occupation domain (`qgGradedFock_esa`) — with no positivity and no boundedness assumption, which matters because the gauge-fixed gravity symbol is indefinite — and therefore generates the unitary group e^{-itH} on the graded Fock space (`qgGradedFock_stone_flow`); `qgGradedFock_not_bounded` records that the operator is genuinely unbounded. The general theorems this instantiates are also registered here: the second quantization of an arbitrary real one-particle symbol (`qgDGamma_esa`), the two-level Fock-of-Fock Hamiltonian (`qgTwoLevel_esa`), and the Hashimoto/SIRK shift-invert selection of the Friedrichs extension over the Gauss–polynomial core of $L^2(\mathbb{R}^{84})$ (`qgFock_hashimoto_selects`). The Fock level is therefore **not** where the open question lies: it is essentially self-adjoint as soon as the one-particle operator is. No mass gap and no global existence is claimed.

Manuscript provenance. The outer-enclosure shape is the manuscript's, and the reason it may assume the Hamiltonian quadratic in the ladders is its recursive construction of the Fock space: **"Since the second-quantization procedure can be applied recursively … in case the Hamiltonian acting in the Fock-space is not quadratic in the creation/annihilation operators, then we can consider instead a new Fock-space where the base Hilbert space is the original Fock-space. The new Hamiltonian is quadratic in the creation/annihilation operators."** It writes each sector's Hamiltonian as a one-particle operator enclosed between creation and annihilation, $H=\int d^3\vec{x}\,a^\dagger(\vec{x})\,H(\vec{x})\,a(\vec{x})$ — its Navier–Stokes instance is $H(\vec{x})=\pi^i(u_ju_{i,j}-\nu u_{i,jj})+(h.c.)$, and the gauge-fixed gravity symbol here has the same envelope. Where the inner space carries the fields' spatial derivatives (the vielbein derivative modes), the formal route eliminates them by the spatial Fourier substitution, so that in momentum space a product with a derivative becomes a **convolution** rather than a gauge constraint (`ChapterQgFourierElimination`, `ChapterQgFullEliminated`).

```
#check @BookProof.QuantumGravityFock.car_fermAnn_fermCre
#check @BookProof.QuantumGravityFock.car_fermAnn_fermCre_of_ne
#check @BookProof.QuantumGravityFock.car_fermAnn_fermAnn
#check @BookProof.QuantumGravityFock.car_fermCre_fermCre
#check @BookProof.QuantumGravityFock.fermAnn_comp_self
#check @BookProof.QuantumGravityFock.fermCre_comp_self
```

```
#check @BookProof.QuantumGravityFock.inner_fermCre_left
#check @BookProof.QuantumGravityFock.fermGrade_involutive
#check @BookProof.QuantumGravityFock.fermGrade_fermAnn
#check @BookProof.QuantumGravityFock.fermGrade_fermCre
#check @BookProof.QuantumGravityFock.superBracket_fermAnn_fermCre
#check @BookProof.QuantumGravityFock.superBracket_bosOp_ghostOp
#check @BookProof.QuantumGravityFock.bosOp_ghostOp_comm
#check @BookProof.QuantumGravityFock.qgCCR
#check @BookProof.QuantumGravityFock.qgGhostCar
#check @BookProof.QuantumGravityFock.qgGhostCar_book
#check @BookProof.QuantumGravityFock.bosOp_even
#check @BookProof.QuantumGravityFock.ghostOp_odd_ann
#check @BookProof.QuantumGravityFock.ghostOp_odd_cre
#check @BookProof.QuantumGravityFock.qgGradedFock_esa
#check @BookProof.QuantumGravityFock.qgGradedFock_stone_flow
#check @BookProof.QuantumGravityFock.qgGradedFock_not_bounded
#check @BookProof.QuantumGravityFock.qgDGamma_esa
#check @BookProof.QuantumGravityFock.qgTwoLevel_esa
#check @BookProof.QuantumGravityFock.qgFock_hashimoto_selects
```

The one-particle statement behind that gluing now has its enclosure in exactly the form the final-Hamiltonian convention fixes. The complete gauge-fixed QG Hamiltonian of record is the one-particle operator `secHam W (qgFullModes g)` of `BookProof/Qg/ChapterQgVielbeinScalaronGaugeFL.lean` — vielbein/torsion kinetic terms, the **exact** exponential Einstein-frame wall `W`, with no Taylor truncation, and the scalaron–vielbein interaction terms — essentially self-adjoint on `secCore`, the core of finite sums of C_c^∞ fibres in $SecGMode = \ell^2(GMode; L^2(\mathbb{R}))$ (`qgFull_esa_core_fl`). `BookProof/Qg/ChapterQgSymmetricSector.lean` carries it to the enclosures: the n-particle derivation on the **symmetric** tensor power of that core (`qgFull_bosonic_core_esa`), the bosonic Fock direct sums, algebraic and Hilbert (`qgFull_bosonicFock_esa`, `qgFull_hbosonicFock_esa`), and the unsymmetrized enclosure $d\Gamma(h)$ — creation on the left, annihilation on the right — on the finite-particle domain (`qgFull_dGamma_esa`). So for QG the outer Hamiltonian is quadratic in the outer ladders exactly as the manuscript's recursion requires, with the full exponential wall and the interaction terms living entirely in the one-particle matrix elements and contributing no outer vertex. The mechanism is worth stating because it is now routine: the symmetric rows average the derivation with the permutation projector, which preserves the n-fold symmetrized core and commutes with the derivation, so essential self-adjointness transfers to it (`PermSector.essentiallySelfAdjointOn_bosonic_core`, `essentiallySelfAdjointOn_bosonic_core_of_esa`) and the two Fock rows are the algebraic and Hilbert direct-sum spellers (`bosonicFock_esa`); the unsymmetrized $d\Gamma(h)$ row is the graph-core lift of `BookProof/Toolbox/ChapterEsaOneParticleDGamma.lean` (`EsaOneParticle.dGamma_essentiallySelfAdjointOn_of_esa`) — the same theorem that carries every one-particle statement of the programme to its final-Hamiltonian form. Dependencies, then: `qgFull_esa_core_fl` as the one-particle input, `ChapterFockStatisticsEsa` for the projector, `ChapterEsaOneParticleDGamma` for the lift; no new analysis and no positivity assumption enter at this step. The boundary of that chapter is the 84-dimensional `qg3DHamiltonian` with its Weyl-ordered cross terms — covered separately, at frozen density, by the paragraphs above. No spectral information and no mass gap is claimed.

```
#check @BookProof.QgVielbeinScalaronGaugeFL.qgFull_esa_core_fl
#check @BookProof.QgSymmetricSector.qgSecSpace
#check @BookProof.QgSymmetricSector.qgFullOp
#check @BookProof.QgSymmetricSector.qgFull_bosonic_core_esa
#check @BookProof.QgSymmetricSector.qgFull_bosonicFock_esa
#check @BookProof.QgSymmetricSector.qgFull_hbosonicFock_esa
#check @BookProof.QgSymmetricSector.qgFull_dGamma_esa
```

31.12. The BRST Charge on the Completed Space and the Reduced Transfer

The BRST charge of the field-space section lives on an algebraic (finite-mode) graded space, and the reduced-transfer theorem needs something else: a **bounded** nilpotent charge on a **complete** Hilbert space, because the physical states are $\ker \Omega$, the pure-gauge states are $\overline{\operatorname{ran} \Omega}$, and closures only make sense once the space is complete. The join is made here on ℓ^2 of the joint boson/ghost occupations — the completion of the graded Fock

space of the previous section. The technical ingredient is a weighted-shift calculus: if a weight has modulus at most one and the reindexing it accompanies is injective on the weight's support, then $f \mapsto (i \mapsto w_i f(e(i)))$ is a bounded operator of norm at most one on ℓ^2 (`wshift_norm_le`); this is what makes the ghost ladder bounded after completion, and it is proved, not assumed.

The dressed ghost creation operators $C_a \Psi_a^\dagger$ — a Jordan–Wigner ghost creation times a constraint symbol diagonal in the boson occupations and bounded by one — square to zero (`brstTerm_comp_self`) and anticommute (`brstTerm_anticomm`). Their sum over the 19 diffeomorphism ghosts is the BRST charge Ω, which is bounded, **nilpotent** (`qgBrstCharge_nilpotent`) and nonzero (`qgBrstCharge_ne_zero`), so the cohomology is not a statement about the zero operator. Multiplication by the phase $e^{-itE(n)}$ of the boson energy is an explicit unitary group (`qgPhase_zero`, `qgPhase_group`, `qgPhase_isometry`) that acts on an occupation basis vector exactly as the second-quantized evolution of the previous section with vanishing ghost energy (`qgPhase_single`), and it commutes with Ω (`qgPhase_comm_brst`).

Consequently the evolution maps physical states to physical states — the caveat the manuscript raises about the half-density unitary — (`qgPhase_mem_physicalStates`), maps gauge states to gauge states (`qgPhase_mem_exactStates`), and descends to the BRST cohomology of the completed space as a one-parameter group of automorphisms (`qgBrstTransfer_zero`, `qgBrstTransfer_comp`, `qgBrstTransfer_bijective`) which preserves the BRST quotient norm (`qgBrstTransfer_infDist`). The honest boundary: the constraint symbols are diagonal and bounded rather than the unbounded generators of the full constraint algebra, and the commuting Hamiltonian carries no ghost energy. That last point is not a stylistic restriction but a theorem: if a ghost energy is nonzero, the evolution it generates does **not** commute with the charge (`qgPhaseFull_not_comm_brst`), so it does not descend to the cohomology. The obstruction is proved, not hidden.

```
#check @BookProof.QgBrstCompleted.wshift_norm_le
#check @BookProof.QgBrstCompleted.wshift_norm_eq
#check @BookProof.QgBrstCompleted.brstTerm_norm_le
#check @BookProof.QgBrstCompleted.brstTerm_comp_self
#check @BookProof.QgBrstCompleted.brstTerm_anticomm
#check @BookProof.QgBrstCompleted.qgBrstCharge_nilpotent
#check @BookProof.QgBrstCompleted.qgBrstCharge_ne_zero
#check @BookProof.QgBrstCompleted.qgPhase_zero
#check @BookProof.QgBrstCompleted.qgPhase_group
#check @BookProof.QgBrstCompleted.qgPhase_isometry
#check @BookProof.QgBrstCompleted.qgPhase_single
#check @BookProof.QgBrstCompleted.qgPhase_comm_brst
#check @BookProof.QgBrstCompleted.qgPhaseFull_isometry
#check @BookProof.QgBrstCompleted.qgPhaseFull_not_comm_brst
#check @BookProof.QgBrstCompleted.qgExact_le_physical
#check @BookProof.QgBrstCompleted.qgPhase_mem_physicalStates
#check @BookProof.QgBrstCompleted.qgPhase_mem_exactStates
#check @BookProof.QgBrstCompleted.qgBrstTransfer_zero
#check @BookProof.QgBrstCompleted.qgBrstTransfer_comp
#check @BookProof.QgBrstCompleted.qgBrstTransfer_bijective
#check @BookProof.QgBrstCompleted.qgBrstTransfer_infDist
```

31.13. The Assembled Hamiltonian: Transfer, Flow, and the One Named Gap

The pieces above each carry their own essential-self-adjointness story; the 2026-10-15c wave closes the question for their **sum** — the complete gauge-fixed gravity Hamiltonian, not the potential alone (the 2026-08-24 scope clarification), in `BookProof/Qg/ChapterQgAssembledTransfer.lean`.

The four constituents, packaged. `QgPieces D` collects the **kinetic**, **potential** (the exact exponential Einstein-frame wall with the scalaron–vielbein interaction terms — no Taylor truncation, inside `h`), the **Faddeev–Popov ghost** and the **gauge-fixing** term as four symmetric operators on one common domain `D`, with `QgPieces.total` their sum and `total_isSymmetricDom` its symmetry. The pedagogical point of packaging them: assembly is exactly where a termwise argument can silently lose a cross-term, and the record operator is a **sum with cross terms** — never a tensor sum (see Second quantization, the honest boundary at its end).

The transfer theorem, and its instance. `assembled_transfer` is the generic statement: for **any** unitary `W` carrying the physical domain onto the densitized one and intertwining the four pieces **termwise**, vanishing adjoint deficiency of the densitized total implies it for the physical total. `qg_assembled_halfDensity_transfer` instantiates it at the constructed half-density unitary $W : L^2((0,\infty), de) \simeq L^2((0,\infty), 2y\,dy)$ (`BookProof.QuantumGravityHalfDensity.halfDensityUnitary`) — the unitary of the densitized change of variables $y = \sqrt{e}$ that this chapter already records as **constructed rather than assumed**. Non-vacuity is not argued but exhibited: `confPieces_phys` / `confPieces_dens` are the conformal-mode instance (potential V_3 alone, the other three terms switched off) with every hypothesis — including the flat one — discharged, and `qg_assembled_conformal_hasZeroDeficiencyOn` gives vanishing deficiency of its physical total unconditionally.

QG-D-C.3 — the flow. `qg_assembled_stone_flow_of_flat` feeds the transferred statement into the Stone bridge: on the flat-side hypothesis, the physical assembled Hamiltonian generates a complete unitary group — the QG face of the same bridge the NS and ODE chapters use.

QG-D-C.1 — the flat input, now a theorem (2026-10-17). The hypothesis `hflat` of the transfer — vanishing deficiency of the densitized assembled operator, whose kinetic term is the flat d'Alembertian $(1/16)\sum \partial^2 - (1/24)\partial_y^2$ on the continuum $L^2(\mathbb{R}^{84})$ — was the continuum input the plan kept open (routes R1, R2 above). It is discharged by `BookProof/Qg/ChapterQgFlatContinuumEsa.lean`: the indefinite operator **separates** over the densitized coordinates as $A \otimes 1 - 1 \otimes B$ — the shear factor $(1/16)(-\Delta_x) + V_s$ above, the conformal factor $(1/24)(-\partial_y^2) + V_c$ (signs as on the left) below. Each factor is essentially self-adjoint on its Gauss–polynomial core by the Kato-type theorem (`exists_schOp_repr` covers every real polynomial potential bounded below), and `flatOp_esa` / `flatOp_deficiencyTrivialAt` assemble them over any two Hilbert spaces — no positivity of the composite is used, because there is none. The product unitary `prodUnitary : L²(ℝⁿ) ⊗ L²(ℝ) ≃ L²(ℝⁿ × ℝ)` then carries the tensor picture to the continuum field space span of products $f(x)g(y)$, giving `qgFlatL2_deficiencyTrivialAt`, the field-space instance `qgFlat84_deficiencyTrivialAt` ($n = 83$), and the Stone flow `qgFlatL2_stone_flow`. `qg_flat_transfer_esa` feeds the result into the transfer above as **data**. The sign is part of the theorem, not a convenience: the conformal potential enters with the conformal kinetic term's sign — exactly the sign of the densitized $-e \cdot Q_b = -y^2 Q_b$ term for $Q_b \geq 0$ — because for the opposite sign that fibre is the Weyl limit-circle class where essential self-adjointness fails ($-d^2/dy^2 - y^4$ has deficiency indices $(2,2)$, proved in `ChapterLimitCircleExample`). What remains of the branch is one alignment step — identifying this flat operator with the assembled chain's flat side (`CONSOLIDATED_PLAN.md` §"State of the project — 2026-10-17", step 2a).

Convention reading. Every statement in this section is a **one-particle** statement about the assembled operator h (and its densitized twin) — the transfer and the flow are properties of h, and the Hamiltonian of record remains the enclosure $d\Gamma(h)$ of the R^2 vielbein operator with the full exponential wall and the interaction terms, creation left / annihilation right (`qgFull_dGamma_esa`, above). What the transfer buys for the enclosure is immediate: an h known to have a unique self-adjoint realization encloses by the second-quantization theorem without further analysis, and with `hflat` discharged (2026-10-17, `ChapterQgFlatContinuumEsa`) the assembled chain stands on its alignment step alone (`CONSOLIDATED_PLAN.md` §"State of the project — 2026-10-17", step 2a) — unconditional at the one-particle level once that identification is in hand.

```
#check @BookProof.QgAssembledTransfer.QgPieces
#check @BookProof.QgAssembledTransfer.QgPieces.total
#check @BookProof.QgAssembledTransfer.assembled_transfer
#check @BookProof.QgAssembledTransfer.qg_assembled_halfDensity_transfer
#check @BookProof.QgAssembledTransfer.qg_assembled_conformal_hasZeroDeficiencyOn
#check @BookProof.QgAssembledTransfer.qg_assembled_stone_flow_of_flat
#check @BookProof.QuantumGravityHalfDensity.halfDensityUnitary
#check @BookProof.QgFlatContinuumEsa.qgFlatL2_deficiencyTrivialAt
#check @BookProof.QgFlatContinuumEsa.qgFlat84_deficiencyTrivialAt
#check @BookProof.QgFlatContinuumEsa.qg_flat_transfer_esa
```

31.14. Summary

The algebraic core of the manuscript's diffeomorphisms/gravity chapter:

- the $3+1$ split into a rank-3 spatial projector and a rank-1 temporal projector, with orthogonal, unique spatial/temporal parts;
- the induced spatial metric as a positive-semidefinite, time-degenerate object with a generalized inverse on the slice;
- the irreducible $S + A + \mathrm{trace}$ decomposition of the metric field;
- the compatibility of diffeomorphism constraints with time-evolution;
- the densitized change of variables $y = \sqrt{e}$, $\tilde{e} = \sqrt{e}\, e$, the absorption of the singular $1/e$, the flat (hyperbolic) principal part, and the Hermite-basis realization on which essential self-adjointness is proved — with Strichartz's continuum theorem carried as a named hypothesis, never as a claimed result;
- the half-density unitary $L^2((0,\infty), de) \simeq L^2((0,\infty), 2y\, dy)$ of that change of variables, constructed rather than assumed, at which the unitary-transfer theorem is instantiated.
- the gauge-fixed 84-dimensional field-space Hamiltonian on the Gauss–polynomial core of $L^2(\mathbb{R}^{84})$, its canonical commutation relations, its symmetry and quadratic form, and the Friedrichs extension of its elliptic truncation — with the indefinite physical signature recorded rather than hidden;
- the BRST charge of that field space: the $\mathbb{Z}_2^{19}$ ghost sector with its canonical anticommutation relations, the graded field space on which bosons and ghosts commute, the $\mathfrak{gl}(84)$ constraint generators, the nilpotency $\Omega^2 = 0$, and a concrete non-abelian instance witnessing that the construction is not vacuous.
- the graded Fock space $\Gamma^s \otimes \Gamma^a$ of bosons and $\mathbb{Z}_2^{19}$ ghosts: the fermionic canonical anticommutation relations with their Jordan–Wigner signs, the adjointness of creation and annihilation, the $\mathbb{Z}_2$ grading and superbracket, and the essential self-adjointness — with its unitary group — of the second-quantized Hamiltonian on the finite-occupation domain, with no positivity assumption.
- the bounded, nonzero, nilpotent BRST charge on the **completed** graded space ℓ^2 of joint boson/ghost occupations, an explicit commuting unitary group, and the resulting reduced transfer: the evolution preserves the physical states and descends to the BRST cohomology as a norm-preserving one-parameter group of automorphisms.
- the enclosure of the complete gauge-fixed one-particle Hamiltonian — vielbein/torsion kinetic terms, the exact exponential Einstein-frame wall with no Taylor truncation, and the scalaron–vielbein interaction terms — on the symmetrized tensor powers of `secCore`, on the bosonic Fock spaces and on the finite-particle domain, creation left / annihilation right (`qgFull_bosonic_core_esa`, `qgFull_bosonicFock_esa`, `qgFull_hbosonicFock_esa`, `qgFull_dGamma_esa`); the one-particle input is `qgFull_esa_core_fl`, and the boundary of the enclosing chapter is the 84-dimensional operator with cross terms, which the next bullet's chapter treats at frozen density.
- the 3D density with its three density-dependent places — the $1/e$ kinetic coefficient, the densitized $\sqrt{e}$ momenta, the polynomial e in front of the bracket — for every real density, inverse tetrad and bracket, frozen or fibred over a family of backgrounds (`qg3DDensity_esa`, `qg3DDensitized_esa`, `qgFibredDensity_esa`, with the enclosures `qg3DDensity_dGamma_esa` / `qg3DDensity_dGammaOp_esa`), the non-frozen density (a canonical variable) stated as the honest boundary.
- the **assembled** gauge-fixed Hamiltonian — kinetic + potential + ghost + gauge-fixing, packaged as `QgPieces` and summed — transferred termwise through the half-density unitary (`assembled_transfer`, `qg_assembled_halfDensity_transfer`), with the conformal-mode instance discharging every hypothesis unconditionally (`qg_assembled_conformal_hasZeroDeficiency0n`) and the Stone flow of the physical total (`qg_assembled_stone_flow_of_flat`); the continuum flat input `hflat` (QG-D-C.1) is now proved — `qgFlat84_deficiencyTrivialAt`, `qgFlatL2_stone_flow`, exported by `qg_flat_transfer_esa`, under the sign condition stated in the section above — with only the assembled-side alignment (plan step 2a) left to consume it, and all of it is one-particle content: the record stays the enclosed $d\Gamma(h)$ above.

32. The Fourier Elimination of the Derivative Variables

32.1. Orientation and Status

The strategy-of-record chapter of the **derivative variables** thread of this part: when a first-order system is written with the derivatives of the fields as fields, **eliminate** them by a spatial Fourier transform inside the squares of the Hamiltonian rather than constrain them by a gauge symmetry — no ghost sector, no cubic derivative symbol. It also settles the Lagrangian face of the same question: the eliminated kinematics is rank-one, the cofactor of the deformation gradient vanishes, and the volume constraint collapses to a constant. This chapter states the strategy; the Navier–Stokes one-particle Hamiltonian, its comparison operator and the quantum-gravity elimination carry it out.

The route: the idea (eliminate the jets, do not gauge-fix them), the substitution as a ring map, the residual becoming quadratic, the reduced Hamiltonian (both real parts are squares), the nested-Fock lift and the Faris–Lavine comparison, the Lagrangian sector (the volume constraint collapses), and where the two new operator chapters sit.

The proofs are `BookProof/NsFullEuler`, `BookProof/NsLagFourier`, `BookProof/NsLagrangianDet` (with `BookProof/NsLagrangianDetFL`), and `BookProof/KoopmanLyapunov`.

32.2. The Idea: Eliminate the Jets, Do Not Gauge-Fix Them

Navier–Stokes is a first-order system in the fields u_i **and** their spatial derivatives $u_{i,j} = \partial_j u_i$. When the field space is built out of coordinates, the derivatives have to be represented by something, and the preceding chapters represented them by **independent** coordinates ($u_{i,j}$, the viscous combination w_i, and the auxiliary expansion coordinates y_j) which were then **fixed** by a gauge symmetry — a BRST charge with ghosts, whose nilpotency made the gauge-invariant algebra well-defined but did not remove the redundancy: the constrained sector still had to be built and the gauge forms appeared inside the Hamiltonian.

This chapter records the alternative followed by the current plan of record: **eliminate** the derivative coordinates instead of fixing them. The elimination is applied **inside the squares** that make up the Hamiltonian — each constraint form is substituted and **both** of its real-coefficient parts are squared, so the nonlinear advection is kept, squared, rather than dropped — so the jet coordinates never appear as dynamical variables at all, no ghost sector is introduced, and the derivative content is carried by the momentum of a spatial Fourier transform — $\partial_j \mapsto i k_j$ at momentum k. The substitution is

$$u_{i,j} \mapsto i k_j u_i, \qquad w_i \mapsto -|k|^2 u_i, \qquad y_j \mapsto 0, \qquad u_i \mapsto u_i, \qquad q_i \mapsto q_i,$$

reducing each parcel from twenty-one field-space coordinates to the **six** physical ones (u_i, q_i). The algebraic content of the chapter — that the substitution is a ring map, that the residual and the incompressibility become **quadratic** and **linear** symbols, and that the reduced Hamiltonian is again a sum of squares — both real parts of every reduced form, the advection among them — on which the Friedrichs extension and Faris–Lavine run as before — is verified in this repository; the displayed algebra of the substitution is engine-checked in the companion symbolic module `../unfer/docs/ns_qg_fourier_elimination.cdb`, which is **reference only** and is never part of a Lean build.

32.3. The Substitution as a Ring Map

The elimination is defined parcelwise on the polynomial coordinate ring. On one parcel, `nsElimCoord` maps the twenty-one coordinates `$(u_i`, the nine `u_{i,j}`, the three `w_i`, the three `q_i`, the three `y_j`) to polynomials in the six reduced coordinates, and `nsElimHom` applies it to every parcel at once:

```
#check @BookProof.NsFullEuler.nsElimCoord
#check @BookProof.NsFullEuler.liftParcel
#check @BookProof.NsFullEuler.nsElimHom
#check @BookProof.NsFullEuler.nsElimHom_X_d
#check @BookProof.NsFullEuler.nsElimHom_X_w
#check @BookProof.NsFullEuler.nsElimHom_X_y
```

The last three state that the substitution really is what the display above claims: the derivative coordinate $u_{i,j}$ goes to $i k_j u_i$, the viscous coordinate w_i to $-|k|^2 u_i$, and the auxiliary coordinate y_j to 0. Because `nsElimHom` is a

ring homomorphism (an `eval₂Hom`), it commutes with every sum and product, which is what makes the next step a computation rather than an argument.

32.4. The Residual Becomes Quadratic

With the substitution in hand, the two constraint polynomials of the Eulerian chapter — the residual $R_i = \sum_j u_j u_{i,j} + q_i - \nu w_i$ and the incompressibility $\sum_j u_{j,j}$ — can be pushed through it. The advection, the pressure gradient and the viscous term are computed one at a time:

```
#check @BookProof.NsFullEuler.nsElimSubst_advect
#check @BookProof.NsFullEuler.nsElimSubst_pressure
#check @BookProof.NsFullEuler.nsElimSubst_viscous
```

and assembling them gives the two identities that the elimination rests on:

```
#check @BookProof.NsFullEuler.nsElimSubst_resPoly
#check @BookProof.NsFullEuler.nsElimSubst_divPoly
```

Written out, the first is

$$\sigma(R_i) = i\,(k\cdot u)\,u_i + q_i + \nu|k|^2 u_i, \qquad (k\cdot u) = \textstyle\sum_j k_j u_j,$$

and the second is $\sigma\big(\sum_j u_{j,j}\big) = i\,(k\cdot u)$. This is the whole point of the elimination. In the original coordinates the residual is **cubic** as a symbol — a product of the field with the field's derivative — which is exactly what defeats a quadratic comparison operator; after the elimination the only nonlinearity is the product of two reduced velocities, so the residual is a **quadratic** form in six variables and the incompressibility is linear. The symbols themselves are named in the library as `fourierMomentum` (the scalar $k\cdot u$), `fourierAdvect` ($(k\cdot u)u_i$), `fourierDiv` ($i(k\cdot u)$) and `fourierVisc` ($q_i + \nu|k|^2 u_i$).

32.5. The Reduced Hamiltonian: Both Real Parts Are Squares

A multiplication operator on L^2 with respect to the real coordinates is symmetric exactly when its symbol has **real** coefficients. The residual splits as $\sigma(R_i) = i\,(k\cdot u)u_i + \big(q_i + \nu|k|^2 u_i\big)$, and **both** brackets have real coefficients — the first is a real polynomial times i. Multiplication by that imaginary part is indeed **skew-adjoint** and its operator square is **negative**, but that square is the operator of the **equation**, not a summand of the Hamiltonian; the squares in the Hamiltonian are the two real-coefficient parts, $\frac{1}{2}\big((k\cdot u)u_i\big)^2$ and $\frac{1}{2}\big(q_i + \nu|k|^2 u_i\big)^2$ (equivalently the modulus square of the multiplication operator of the complex form, whose skew cross term is invisible on the quadratic form). The reduced Hamiltonian is a Weyl-ordered sum of squares of those two parts — `fourierVisc` and `fourierAdvect` (the line above lifts the real part as `redVisc`) — hence nonnegative:

```
#check @BookProof.NsFullEuler.fourierVisc
#check @BookProof.NsFullEuler.redVisc
#check @BookProof.NsFullEuler.realCoeff_redVisc
#check @BookProof.NsFullEuler.redHam
#check @BookProof.NsFullEuler.redHam_symmetricOn
#check @BookProof.NsFullEuler.redHam_quadForm_nonneg
#check @BookProof.NsFullEuler.redHam_friedrichs_extension
```

This is the honest form of the strategy: the **squares** are the two real-coefficient parts of each substituted form, so the genuine nonlinearity — the advection $(k\cdot u)u_i$ — is **inside** the Hamiltonian as one of the squares, and no perturbation, no commutator estimate and no ghost sector is owed. Positivity of the sum of squares is what makes the Friedrichs extension unconditional, exactly as for the full Eulerian sector. The field family `redFieldN` is the honest one — seven real-coefficient forms per parcel (`redFormPoly`): the three real residual parts, the three advection parts and the eliminated incompressibility — so `redHam` is the full modulus-square Hamiltonian and `redFieldN_advect` exhibits the advection as one of its squares.

32.6. The Nested Fock Lift and Faris–Lavine

The outer Fock space of the reduced model is the nested space $\bigoplus_n L^2(\mathbb{R}^{6n})$, and `nsRedFullFockHam` is the **particle-number-conserving second quantization** $d\Gamma(H_1)$ of the reduced one-particle Hamiltonian `redHam` at $n = 1$ — a direct sum over parcel number, taking no new square. The comparison operator the route needs is the lifted Friedrichs realization of the reduced one-body operators on that space. Positivity, symmetry and surjectivity of $N + 1$ are all fibrewise, so they lift verbatim from the fibre:

```
#check @BookProof.NsFullEuler.nsRedFockSpace
#check @BookProof.NsFullEuler.nsRedFockCore
#check @BookProof.NsFullEuler.nsRedFullFockHam
#check @BookProof.NsFullEuler.nsRedFullFockHam_symmetricOn
#check @BookProof.NsFullEuler.nsRedFullFockHam_quadForm_nonneg
#check @BookProof.NsFullEuler.nsRedFullFock_friedrichs_extension
#check @BookProof.NsFullEuler.nsRedFullFockHam_number_conserving
```

and Faris–Lavine then gives essential self-adjointness of the reduced generator on the domain of the lifted comparison operator — the `H = N, c = 0` case of the criterion, in which the committed operator **is** its own comparison, so the commutator form vanishes — together with the extension statement that discharges the plan's definition of done for the reduced sector:

```
#check @BookProof.NsFullEuler.nsRedOuterComparison
#check @BookProof.NsFullEuler.nsRedOuterN_apply
#check @BookProof.NsFullEuler.nsRedFockCore_le_friedDom
#check @BookProof.NsFullEuler.nsRedFullOuterN_esa
#check @BookProof.NsFullEuler.nsRedFullOuterN_isPositiveSelfAdjointExtension
```

32.7. The Lagrangian Sector: the Volume Constraint Collapses

Navier–Stokes is equally often written in **Lagrangian** variables, where the field is the placement and the derivative content is the deformation gradient $F = \partial x/\partial X$ — so the elimination has a second, independent instance, and there the elimination degenerates in a way that is itself the mathematical content.

Everything is done parcelwise on the twelve coordinates of one parcel (`lRedIdx`), with the substitution defined by `lagElimCoord` and applied by the ring homomorphism `lagElimHom`; the coordinate values (`lagElimHom_X_xiIdx`, `_vIdx`, `_accIdx`, `_qIdx`, `_fIdx`, `_vgIdx`, `_sIdx`, `_yIdx`) name what each Lagrangian coordinate becomes. Two index lemmas — `lagElimCoord_fIdx` and `lagElimCoord_vgIdx` — are the bookkeeping that the earlier state of this chapter left open at a `whnf` heartbeat timeout; they are proved by rewriting the index arithmetic **before** touching the dependent `Fin` constructors.

```
#check @BookProof.NsLagFourier.lRedIdx
#check @BookProof.NsLagFourier.lagElimCoord_fIdx
#check @BookProof.NsLagFourier.lagElimCoord_vgIdx
#check @BookProof.NsLagFourier.lagElimHom_X_fIdx
#check @BookProof.NsLagFourier.lagElimHom_X_vgIdx
```

The degenerate part is the **rank-one** structure of the eliminated kinematics. The cofactor matrix of the eliminated deformation gradient vanishes identically (`rankOne_cof_zero`), and with it the Piola coupling that would carry the deformation-gradient square:

- `lagElimSubst_cofPoly` — the cofactor polynomial is the constant 0;
- `lagElimSubst_piola` — the Piola stress that multiplies it is annihilated;
- `lagElimSubst_detPoly` — hence $\det F = 0$;
- `lagElimSubst_volumePoly` and `lagElimSubst_volumePoly_sq` — so the volume constraint $\det F = 1$ collapses to the constant -1, whose square is 1: it carries no field content and cannot couple to the field;
- `lagElimSubst_lagResPoly` — the surviving form is the momentum equation, a **quadratic** symbol in the reduced velocities rather than a cubic one in the deformation gradient.

This is the Lagrangian face of the same claim the Eulerian chapter makes: after the elimination the only squares that carry field content are the momentum/viscous and advection forms — for both real parts of each substituted form — and any deformation-gradient square is vacuous rather than merely absent. The consequence for the plan

is a **removal** obligation, not a proof obligation: no invariance under the deformation gradient, and no determinant identity inside the Hamiltonian, has to be established, because the determinant is constant there.

```
#check @BookProof.NsLagFourier.rankOne_cof_zero
#check @BookProof.NsLagFourier.lagElimSubst_cofPoly
#check @BookProof.NsLagFourier.lagElimSubst_piola
#check @BookProof.NsLagFourier.lagElimSubst_detPoly
#check @BookProof.NsLagFourier.lagElimSubst_volumePoly
#check @BookProof.NsLagFourier.lagElimSubst_volumePoly_sq
#check @BookProof.NsLagFourier.lagElimSubst_lagResPoly
```

The rank-one collapse is a statement about the **reduced** Hamiltonian that the elimination produces. The full Lagrangian model — the one whose interacting outer operator is proved essentially self-adjoint in the Navier–Stokes one-particle Hamiltonian and its Fock enclosure — keeps the determinant, and `BookProof/Ns/ChapterNsLagrangianDetConvolution.lean` writes it honestly in momentum space: expanding `det(I + ∇ξ)` row by row turns the local cubic product into the **triple momentum convolution** over signed tuples of modes with `w(τ 0) + w(τ 1) + w(τ 2) = q` (`detCoef`, `det_deformation_eq` — exact at every point and every configuration, cross-mode minors included, so the constraint is **not** the degenerate rank-one substitution above). The coefficients `volCoef q` of the residual `det F − 1` (`volume_residual_eq`) build the volume penalty $V_\kappa = (\kappa/2)\sum_q |\mathrm{volCoef}\ q|^2$, a real polynomial, non-negative at every real configuration (`volPot_eval_nonneg`), whose zero set is exactly the incompressible configurations (`det_eq_one_of_volPot_eq_zero`, for `κ > 0`). This is the momentum-space convolution device in its Lagrangian form: a product in coordinate space involving spatial derivatives of the fields — here the three gradient factors inside the determinant — becomes a convolution in momentum space. The device belongs to NS and QG only.

On that phase space the Koopman generator of the constrained flow — $\dot\xi = v$, $\dot v = -\nu|k|^2 v - \partial V_\kappa/\partial\xi$, with the **exact** degree-five constraint force — is treated by the generic Lyapunov criterion of `BookProof/Ns/ChapterKoopmanLyapunovFarisLavine.lean`: for any polynomial vector field `G` and real $E \ge 0$ with $|G\cdot\nabla E| \le cE$ pointwise, essential self-adjointness of $N = H_G^2 + E$ on the Gauss–polynomial core implies essential self-adjointness of the Koopman generator H_G (`kvnGen_esa_of_lyapunov`). The Lagrangian instance is `BookProof/Ns/ChapterNsLagrangianDetFarisLavine.lean`: for the energy $E = 1 + \frac12|v|^2 + V_\kappa$ the derivative along the flow is the sign-definite viscous dissipation $F\cdot\nabla E = -\nu\sum|k|^2 v^2$ (`lagFlux_eq` — the constraint force does no net work, being the gradient of the potential part of `E`) and the divergence is constant (`lagDiv_eq`), so the Faris–Lavine commutator bound $|\langle i[H_L, N_L]\rangle| \le 2\nu\Lambda\langle N_L\rangle$ holds (`lagComparison_commForm_bound`) and essential self-adjointness of the Lagrangian generator follows from that of `N_L` (`lagKoopman_esa_of_comparison_esa`). The hypothesis — essential self-adjointness of `N_L` — was explicit and unproved here until 2026-10-16, when `NsLagrangianComparisonEsa.lagComparison_esa` discharged it via `KvnKato.lyapunovComparison_esa_poly` (the Eulerian leg's own comparison hypothesis remains open), as on the comparison-operator chapter; incompressibility is imposed by the penalty with bulk modulus `κ`, and the limit `κ → ∞` is not taken.

```
#check @BookProof.NsLagrangianDet.detCoef
#check @BookProof.NsLagrangianDet.det_deformation_eq
#check @BookProof.NsLagrangianDet.volume_residual_eq
#check @BookProof.NsLagrangianDet.volPot_eval_nonneg
#check @BookProof.NsLagrangianDet.det_eq_one_of_volPot_eq_zero
#check @BookProof.KoopmanLyapunov.kvnGen_esa_of_lyapunov
#check @BookProof.NsLagrangianDetFL.lagFlux_eq
#check @BookProof.NsLagrangianDetFL.lagDiv_eq
#check @BookProof.NsLagrangianDetFL.lagComparison_commForm_bound
#check @BookProof.NsLagrangianDetFL.lagKoopman_esa_of_comparison_esa
#check @BookProof.NsLagrangianComparisonEsa.lagComparison_esa
#check @BookProof.NsLagrangianComparisonEsa.lagKoopman_esa
```

32.8. Where the Two New Operator Chapters Sit

The elimination above produces the **reduced** forms; the operator questions they raise are answered in two chapters of this book:

- the Navier–Stokes one-particle Hamiltonian and its Fock enclosure takes the reduced one-parcel family, exhibits the one-body generator $H_{\mathrm{sp}} = H_{\mathrm{visc}} + H_{\mathrm{advect}}$ as its single-parcel member, proves the two Faris–Lavine inequalities for the exact nonlinearity, and identifies the outer Hamiltonian as its second quantization $d\Gamma(H_{\mathrm{sp}})$ — creation on the left, annihilation on the right;
- eliminating the derivative variables in quantum gravity does the same service for the vielbein sector of the R^2 model, where the elimination removes twenty-seven derivative components per momentum and still reproduces the vielbein self-interaction and the scalaron coupling verbatim.

32.9. Summary

The Fourier-elimination route, as verified here:

- the derivative coordinates are **eliminated** by a ring homomorphism (`nsElimHom`) that sends $u_{i,j}$ to $ik_j u_i$, w_i to $-|k|^2 u_i$ and y_j to 0, cutting each parcel from twenty-one coordinates to six;
- the residual and the incompressibility are pushed through it, `nsElimSubst_resPoly` and `nsElimSubst_divPoly`, becoming a **quadratic** symbol $i(k\cdot u)u_i + q_i + \nu|k|^2 u_i$ and a **linear** one $i(k\cdot u)$ — the cubic symbol of the gauge-fixed presentation is gone, so no ghost sector is needed for the definition;
- the reduced Hamiltonian on $L^2(\mathbb{R}^{6n})$ is built with the same Weyl-ordered sum of squares (`redHam`), squaring **both** real-coefficient parts of each substituted form (the **definition** is the one-particle case $n = 1$; `redHam` at general n is that one-particle operator summed over the parcels, and the Fock operator `nsRedFullFockHam` is its particle-number-conserving lift $d\Gamma(H_1)$ — no further square is taken) — so the real pressure–viscous symbol $q_i + \nu|k|^2 u_i$ **and** the advection $(k\cdot u)u_i$ both sit inside squares, and the Hamiltonian keeps the Navier–Stokes nonlinearity in full; it is symmetric and bounded below (`redHam_symmetricOn`, `redHam_quadForm_nonneg`) and therefore has an unconditional positive self-adjoint Friedrichs extension;
- the comparison operator of the Faris–Lavine criterion is free, and the convenient choice is this same lifted Friedrichs realization, on which the reduced Hamiltonian is essentially self-adjoint with `c = 0` (`nsRedFullOuterN_esa`, `nsRedFullOuterN_isPositiveSelfAdjointExtension`) — the advection is inside it rather than measured against it by a commutator estimate;
- the reduced field family `redFieldN` is the honest one: `redFormPoly` collects, for each parcel, the real and imaginary parts of every surviving substituted form — the three real residual parts $q_i + \nu|k|^2 u_i$, the three advection parts $(k\cdot u)u_i$ and the eliminated incompressibility $k\cdot u$ — and `redFieldN_advect` exhibits the advection as one of the squares of the Hamiltonian.

33. The Navier–Stokes One-Particle Hamiltonian and Its Fock Enclosure

33.1. Orientation and Status

The operator chapter of the derivative-variables thread for the Eulerian Navier–Stokes route: it exhibits the reduced one-parcel family, proves the one-body generator $H_{\mathrm{sp}} = H_{\mathrm{visc}} + H_{\mathrm{advect}}$ symmetric and positive for the exact nonlinearity, and identifies the final Hamiltonian with its second quantization $d\Gamma(H_{\mathrm{sp}})$ — every particle-number sector is preserved. Read it after the Fourier-elimination chapter, which fixed the strategy of record; the next chapter supplies the Faris–Lavine comparison operator for the same Hamiltonian.

The route: why the final Hamiltonian is a one-particle statement, the momentum variables (a diagonal derivative and a partial transform), the fibred model identified with the scalar one, the constraint solved (the inverse field momentum), the advection as a convolution, uniformity under the energy cutoff, the one-body generator and the outer Hamiltonian as its second quantization, why not the mainstream generator, the Lagrangian face (one parcel, then the interacting whole), and what is verified versus open.

The proofs are `BookProof/NsOneBody`, `BookProof/NsLagrangianOuterFock`, `BookProof/NsFullLagrangianEsa`, `BookProof/NsAdvectionConvolution`, `BookProof/NsCutoffUniformity`, `BookProof/NsFieldMomentumInverse`, and the `NsKoopman` / `NsScalarFourier` / `NsReducedCoreEsa` family.

33.2. The Final Hamiltonian of Record, and Why It Is a One-Particle Statement

The Fourier-elimination chapter fixed the strategy: the derivative variables are **eliminated** by a spatial Fourier transform, not constrained by a gauge symmetry, so that no ghost sector and no cubic derivative symbol survives. This chapter records what the elimination buys on the **operator** side, and states the final Hamiltonian of record in the shape the whole programme uses — QYM, QED, QG and Navier–Stokes alike:

$$H = \sum_{i,j} h_{ij}\, C^\dagger(e_i)\, A(e_j),$$

the outer **second quantization** $d\Gamma(h)$ of an inner **one-particle** Hamiltonian h, with creation on the left and annihilation on the right. For Navier–Stokes the one-particle space of the eliminated (reduced) sector is $h = L^2(\mathbb{R}^6)$ — one parcel's six coordinates (u_i, q_i) — and the one-particle Hamiltonian is the Weyl-ordered sum of squares of the seven reduced constraint forms,

$$H_{\mathrm{sp}} \;=\; H_{\mathrm{visc}} + H_{\mathrm{advect}}, \qquad H_{\mathrm{visc}} = \tfrac{1}{2}\sum_{m<6} \pi_m^2 + \tfrac{1}{2}\sum_{r\in \mathrm{vis}} \left(\mathrm{mulOp}\, \Phi_r\right)^2, \qquad H_{\mathrm{advect}} = \tfrac{1}{2}\sum_{i<3} \left((k\cdot u)\, u_i\right)^2.$$

Everything else in this chapter is the verification of the three claims that sentence contains: that the one-particle operator really is that generator, that it is symmetric and bounded below so its Friedrichs realization is a legitimate comparison operator, and that the outer Hamiltonian really is its second quantization — no further square, no perturbation, no regularization.

33.3. The Momentum Variables: a Diagonal Derivative and a Partial Transform

The elimination is only as good as the momentum variables it introduces, so the first obligation of the plan is that the spatial Fourier transform is a unitary on the one-particle space and that on it the derivative is **multiplication by a real symbol** — the symbol of the eliminated constraint.

On $L^2(V)$, `l2Fourier` is the Plancherel identification, `fourier_opL2_eq_mulSymbol` is the statement that an operator whose Schwartz form is multiplication by a real symbol σ **is** multiplication by σ after the transform, and `fourier_opL2_momentumOp` is the specialization to $\pi_m = -i\partial_m$, whose symbol is $2\pi\langle \xi, m\rangle$. The companion `foSymbolFn_add_fibre` says the symbol of a **spatial** family of directions is unchanged when the frequency is moved by a vector orthogonal to them, and `spatialMomentum_esa` records that the momentum family is essentially self-adjoint on the Schwartz core, so the derivative side of the construction is a proved instrument rather than a formal manipulation:

```
#check @BookProof.NsSpatialMultiplier.l2Fourier
#check @BookProof.NsSpatialMultiplier.fourier_opL2_eq_mulSymbol
#check @BookProof.NsSpatialMultiplier.fourier_opL2_momentumOp
#check @BookProof.NsSpatialMultiplier.foSymbolFn_add_fibre
#check @BookProof.NsSpatialMultiplier.spatialMomentum_esa
```

The spatial transform must leave the **fibre** — the field's own variable — alone; otherwise there is no elimination at all, only a change of variables in the whole space. The vector-valued model is `L²(V; F)`, and the transform is `partialFourier`. Its one structural property is `partialFourier_fibreOp`: it commutes with the pointwise action of **every** bounded operator of the fibre, which is exactly what "partial" means. The derivative statement is `fourier_vecMomentumOp_apply`, and §4 of the chapter is the concrete Navier–Stokes instance $F = L^2(W)$: `nsPartialFourier_fibreOp` (the transform leaves the field variable alone), `nsPartialFourier_fibreFourier` and `nsPartialFourier_comp_fibreFourier_eq` (its composite with the transform of the field variable is the full transform in both variables). A reusable by-product is `postcompCLM`, the postcomposition of a Schwartz function with a continuous linear map of the target, with

`postcompCLM_lineDerivOp`, `fourier_postcompCLM` and `toLp_postcompCLM` — Mathlib does not provide it.

The model so far is the vector-valued `L²(V; L²(W))`. Its measure-theoretic identification with the scalar `L²(V × W)` — a Bochner–Fubini statement about slices, which Mathlib does not have — was the honest boundary of this section. It has since been closed, in the next section; nothing about the elimination itself changed, only the description of the space on which it acts.

```
#check @BookProof.NsPartialFourier.partialFourier
#check @BookProof.NsPartialFourier.partialFourier_norm
#check @BookProof.NsPartialFourier.partialFourier_fibreOp
#check @BookProof.NsPartialFourier.fourier_vecMomentumOp_apply
#check @BookProof.NsPartialFourier.nsPartialFourier
#check @BookProof.NsPartialFourier.nsPartialFourier_fibreOp
#check @BookProof.NsPartialFourier.nsPartialFourier_comp_fibreFourier_eq
#check @BookProof.NsPartialFourier.postcompCLM
```

33.4. The Fibred Model Identified with the Scalar One

The identification that the section above could only flag is now a theorem, in two steps that the reader should take in this order.

Step one: currying. `BookProof/Ns/ChapterNsScalarVectorCurry.lean` proves $L^2(V; L^2(W)) \cong L^2(V \times W)$ outright, as the linear isometry `curryLI`, for arbitrary σ-finite measures. The route is the one a probabilist would take: lift the two generator families — `fibMk a c`, the fibred element $x \mapsto a\,x \bullet c$, and `prodMk a c`, the scalar product $(x, y) \mapsto ax \cdot cy$ — to the **algebraic** tensor product, where they are bilinear, show their inner products agree on generators (`inner_prodMk` by Fubini, `inner_fibMk` pointwise) hence everywhere, and conclude that the norms agree (`norm_prodTensor_eq_norm_fibTensor`). Density of the generators on both sides (`denseRange_fibTensor`, `denseRange_prodTensor`) then lets `LinearEquiv.extendOfIsometry` assemble the unitary — no quotient by a kernel is needed. The only genuinely new analysis is the π–λ induction `ae_eq_zero_of_forall_setIntegral_rect_eq_zero`: an L^2 function of the product measure whose integral over every finite-measure measurable rectangle vanishes is zero almost everywhere.

Step two: transport. `BookProof/Ns/ChapterNsScalarFourier.lean` then carries the spatial transform of the previous section to the scalar space **by conjugation**, `nsScalarFourier := curryLI ∘ nsPartialFourier ∘ curryLI.symm`. No new analysis enters: `nsScalarFourier_scalarFibreOp` is precisely the fibre-blindness of the spatial transform in the scalar picture, `scalarFibreOp_prodMk` pinning an operator of the fibre variable to $1 \otimes T$ on the product generators.

Two by-products are worth keeping even if one never speaks of Navier–Stokes again. `curryLI_setIntegral_rect` says that currying **is** slicing, tested against the rectangles: the integral of `curryLI f` over $s \times t$ is $\int_s \left(\int_t f\,x\,y\,d\nu \right) d\mu$. And `isSliceOf_curryLI` is the pointwise version — for almost every x, the fibre fx equals, almost everywhere, the slice $y \mapsto (curryLI f)(x, y)$ — proved through the seminorm identity `lintegral_eLpNorm_slice_sq` and a Borel–Cantelli passage along the dense span.

```
#check @BookProof.NsScalarVectorCurry.curryLI
#check @BookProof.NsScalarVectorCurry.curryLI_fibMk
#check @BookProof.NsScalarVectorCurry.curryLI_indicator_prod
#check @BookProof.NsScalarVectorCurry.curryLI_setIntegral_rect
#check @BookProof.NsScalarVectorCurry.isSliceOf_curryLI
#check @BookProof.NsScalarVectorCurry.denseRange_fibTensor
#check @BookProof.NsScalarVectorCurry.denseRange_prodTensor
#check @BookProof.NsScalarVectorCurry.ae_eq_zero_of_forall_setIntegral_rect_eq_zero
#check @BookProof.NsScalarFourier.nsScalarFourier
#check @BookProof.NsScalarFourier.nsScalarFourier_norm
#check @BookProof.NsScalarFourier.scalarFibreOp
#check @BookProof.NsScalarFourier.scalarFibreOp_prodMk
#check @BookProof.NsScalarFourier.nsScalarFourier_scalarFibreOp
#check @BookProof.NsScalarFourier.nsScalarFourier_fibreFourier
```

Why this matters for the final Hamiltonian of record. The NS final Hamiltonian is the one-particle operator enclosed in creation and annihilation operators, $H = \sum_{ij} h_{ij}\, C^\dagger(e_i)\, A(e_j) = d\Gamma(h)$; the elimination and the convolution device that define h were carried out on the **fibred** one-particle space. The identification above says

the resulting h is an operator on the scalar $L^2(\mathbb{R}^6)$ the manuscript writes, so the enclosure is taken over the right space: the two descriptions of the one-particle space — the one the elimination is convenient on, and the one the enclosure is stated on — are the same Hilbert space, up to a unitary that is now a theorem.

33.5. The Constraint Solved: the Inverse Field Momentum

The derivative gauge, read as a substitution rather than as a symmetry, solves the constraint for the derivative mode: $u_j^{(1)} = p_j\,\pi^{-1}$, where $\pi_m = -i\partial_m$ is the field momentum in the fibre variable. The genuine analytic question the plan flags is the domain of π^{-1}, because π_m is not boundedly invertible. The chapter settles it in the momentum representation of the fibre variable, where — by `isMomInverse_momentumOp` — the field momentum **is** multiplication by the real symbol $\sigma(\xi) = 2\pi\langle\xi, m\rangle$:

- `volume_momSymbol_zero` and `momSymbol_ne_zero_ae`: the symbol vanishes only on a hyperplane, a Lebesgue-null set. Consequently `momentum_kernel_trivial` — there is no kernel to split off, the u-constant mode the plan worried about is not an `L²` state — and `isMomInverse_unique`, the inverse is single valued;
- `isMomInverse_of_memLp` is the honest domain criterion (the divided symbol is square integrable), `momDomain` is that domain as a submodule, and `momDomain_dense` proves π^{-1} densely defined by cutting off a neighbourhood of the hyperplane `{σ = 0}`;
- `isMomInverse_smul` is the physical identity $u_j^{(1)} = p_j\pi^{-1}$ and `eq_div_of_mul_eq_ae` its symbol form: a symbol t with $\sigma \cdot t = p$ is p/σ almost everywhere, so the derivative mode is **determined** by the constraint.

```
#check @BookProof.NsFieldMomentumInverse.momSymbol
#check @BookProof.NsFieldMomentumInverse.momSymbol_ne_zero_ae
#check @BookProof.NsFieldMomentumInverse.momentum_kernel_trivial
#check @BookProof.NsFieldMomentumInverse.isMomInverse_unique
#check @BookProof.NsFieldMomentumInverse.isMomInverse_of_memLp
#check @BookProof.NsFieldMomentumInverse.momDomain_dense
#check @BookProof.NsFieldMomentumInverse.isMomInverse_smul
#check @BookProof.NsFieldMomentumInverse.isMomInverse_momentumOp
```

33.6. The Advection Becomes a Convolution

Eliminating the derivative does not make the nonlinearity disappear — it changes its **kind**: in momentum space the advection is not a differential operator but a **convolution**. The chapter proves the momentum-space identity that the reduced letters $(k \cdot u)u_i$ stand for, for Schwartz velocity components on a finite-dimensional real inner-product space:

- `fourier_fourier_apply`: the double transform is the reflection, $\mathcal{F}(\mathcal{F}f)(x) = f(-x)$;
- `fourier_mul_eq_convolution`: the product-to-convolution theorem $\mathcal{F}(f\,g)(Q) = \int \mathcal{F}f(q)\,\mathcal{F}g(Q-q)\,dq$, the **dual** of Mathlib's convolution theorem, obtained from it by Fourier inversion;
- `fourier_lineDeriv_apply`: the derivative as a symbol, $\mathcal{F}(\partial_m u)(q) = 2\pi i\,\langle q, m\rangle\,\mathcal{F}u(q)$;
- `fourier_advection_convolution` and `fourier_advection_sum`: the item's theorem, $\mathcal{F}[u_j\partial_m u_i](Q) = \int 2\pi i\,\langle q, m\rangle\,\hat{u}_i(q)\,\hat{u}_j(Q-q)\,dq$ and its sum over a finite family of directions — the momentum-space replacement of the derivative-gauge generator.

The normalization is Mathlib's, $\mathcal{F}f(\xi) = \int e^{-2\pi i\langle x,\xi\rangle} f(x)\,dx$, in which the convolution theorem carries no $(2\pi)^{-d/2}$ factors and the factor $2\pi i\,q_j$ is the exact symbol of ∂_j.

```
#check @BookProof.NsAdvectionConvolution.fourier_fourier_apply
#check @BookProof.NsAdvectionConvolution.fourier_mul_eq_convolution
#check @BookProof.NsAdvectionConvolution.fourier_lineDeriv_apply
#check @BookProof.NsAdvectionConvolution.fourier_advection_convolution
#check @BookProof.NsAdvectionConvolution.fourier_advection_sum
```

33.7. Uniformity Under the Energy Cutoff

One obligation of the route is a **quantitative** one, and it is where the energy cutoff stops being cosmetic: after the elimination the reduced forms carry the mode coefficients ik_j and $-|k|^2$, and under the cutoff $|k_j| \le \Lambda$ they must be bounded by $O(\Lambda)$ **uniformly in the parcel number n**, since that uniformity is what makes an ℓ^2-lift of the sector Hamiltonians possible. For the landed reduced family:

- `norm_coeff_redVisc_le`, `norm_coeff_redAdvectPoly_le`, `norm_coeff_redMomentumPoly_le` and the headline `norm_coeff_redFormPoly_le`: every coefficient of every reduced form of every parcel is bounded by `cutoffBound ν Λ = 1 + 3Λ + 3|ν|Λ²` — the bound does not depend on the parcel number, on the parcel, on which of the seven forms is taken, or on the monomial;
- `totalDegree_redFormPoly_le`: the reduced forms have degree at most two, uniformly;
- `vars_redFormPoly_subset`: **parcel locality** — a reduced form involves only the six coordinates of its own parcel, so with the fixed number seven of forms per parcel the row/column (Schur) data of the sector Hamiltonians is independent of n;
- `norm_coeff_redFormPoly_unbounded_of_no_cutoff`: the cutoff is **necessary** — without it the coefficients are unbounded, since the advection coefficient is k_j itself.

This settles the coefficient half of the uniformity obligation. The operator-level relative bound with a constant independent of n is **not** proved here — and on the landed route it is not needed either, because the comparison actually used is the lifted Friedrichs extension of the reduced Hamiltonian itself, for which the Faris–Lavine commutator constant is `c = 0`.

```
#check @BookProof.NsCutoffUniformity.cutoffBound
#check @BookProof.NsCutoffUniformity.norm_coeff_redFormPoly_le
#check @BookProof.NsCutoffUniformity.totalDegree_redFormPoly_le
#check @BookProof.NsCutoffUniformity.vars_redFormPoly_subset
#check @BookProof.NsCutoffUniformity.norm_coeff_redFormPoly_unbounded_of_no_cutoff
```

33.8. The One-Body Generator, and the Outer Hamiltonian as Its Second Quantization

Now the operator itself. The seven one-parcel forms are `spFormPoly`, and `redFormPoly_eq_liftParcel` identifies them with the landed n-parcel family parcel by parcel — the one-body generator really is the single-parcel member of the reduced family, not a new object. The generator `spHam` on a core representation of the Gauss–polynomial core of `L²(ℝ⁶)` splits,

`spHam_eq_visc_add_advect` — $H_{\mathrm{sp}} = H_{\mathrm{visc}} + H_{\mathrm{advect}}$ — with `H_visc` carrying the six momenta and the four non-advective squares and `H_advect` the three advection squares, whose form is the momentum convolution of the previous section. Nothing is dropped and nothing is demoted to a perturbation. The two Faris–Lavine inequalities then hold **for the exact nonlinear generator**:

- `spVisc_symmetricOn`, `spAdvect_symmetricOn`, `spHam_symmetricOn` — symmetry on the core;
- `spVisc_quadForm_nonneg`, `spAdvect_quadForm_nonneg`, `spHam_quadForm_nonneg` — positivity, with `spHam_quadForm_split` adding the two form contributions back to the generator's;
- `spAdvect_apply` — the advection half is the multiplication operator of the convolution form.

```
#check @BookProof.NsOneBody.spFormPoly
#check @BookProof.NsOneBody.redFormPoly_eq_liftParcel
#check @BookProof.NsOneBody.spHam_eq_visc_add_advect
#check @BookProof.NsOneBody.spVisc_symmetricOn
#check @BookProof.NsOneBody.spAdvect_symmetricOn
#check @BookProof.NsOneBody.spHam_quadForm_nonneg
#check @BookProof.NsOneBody.spHam_quadForm_split
#check @BookProof.NsOneBody.spAdvect_apply
```

The comparison operator is the **Friedrichs realization of the generator itself** — `spFried`, with `polyGaussCore_le_spFriedDom` the domain obligation — so the criterion is used in its `H = N, c = 0` form (`spHam_commForm_zero`), and `spHam_esa_farisLavine` is the essential self-adjointness of the one-particle operator, `spFried_isPositiveSelfAdjointExtension` the extension statement. Then the second quantization: `nsOnePart` and `nsSpCol` are the generator in the product Hermite basis, `nsSpCol_isHermCol` / `nsSpCol_isPosCol` are the column conditions the dΓ machinery needs, and

- nsSpDGamma_symmetricOn, nsSpDGamma_quadForm_nonneg, nsSpDGamma_friedrichs_extension, nsSpDGammaFried_isPositiveSelfAdjointExtension;
- **nsSpDGamma_esa_farisLavine** — essential self-adjointness of $d\Gamma(H_{\mathrm{sp}})$ on the domain of its lifted comparison operator;
- nsSpDGamma_number_conserving — every particle-number sector is preserved, so the outer operator is **quadratic** in the outer ladders (this is the structural reason a quartic nonlinearity costs nothing at the outer level: it lives entirely in the one-particle matrix elements);
- nsSpDGamma_one_particle — on the one-particle sector $d\Gamma$ **is** the generator;
- spHam_stone_flow and nsSpDGamma_stone_flow — by Stone's theorem each realization generates a strongly continuous one-parameter unitary group, which is the single-time package of the route.

```
#check @BookProof.NsOneBody.spFried
#check @BookProof.NsOneBody.polyGaussCore_le_spFriedDom
#check @BookProof.NsOneBody.spHam_esa_farisLavine
#check @BookProof.NsOneBody.nsSpCol
#check @BookProof.NsOneBody.nsSpDGamma_quadForm_nonneg
#check @BookProof.NsOneBody.nsSpDGamma_esa_farisLavine
#check @BookProof.NsOneBody.nsSpDGamma_number_conserving
#check @BookProof.NsOneBody.nsSpDGamma_one_particle
#check @BookProof.NsOneBody.nsSpDGamma_stone_flow
```

One structural identification used to be a **reading** and is now a theorem. The reduced n-parcel Hamiltonian is the sum of n copies of the one-body generator — the p-th copy written in that parcel's six coordinates and six momenta — not merely something that behaves as if it were. The parcel summand is redParcelHam (symmetric and positive by redParcelHam_symmetricOn / redParcelHam_quadForm_nonneg), redHam_eq_sum_parcel is the decomposition at the one-particle level, nsRedFullFockHam_sector_sum_parcel transports it to the outer Hamiltonian's n-parcel sector, and weylOpDom_block_sum is the general regrouping of a Weyl-ordered sum of squares into blocks — the bookkeeping that makes the parcel decomposition compatible with the outer sum.

```
#check @BookProof.NsOneBody.redParcelHam
#check @BookProof.NsOneBody.redParcelHam_symmetricOn
#check @BookProof.NsOneBody.redParcelHam_quadForm_nonneg
#check @BookProof.NsOneBody.redHam_eq_sum_parcel
#check @BookProof.NsOneBody.nsRedFullFockHam_sector_sum_parcel
#check @BookProof.NsOneBody.weylOpDom_block_sum
```

33.9. Why Not the Mainstream Generator

There is a tempting alternative one-particle operator, and it is the wrong one. The **mainstream** incompressible Navier–Stokes system $\dot{u} = -\nu A u + B(u,u)$, with Leray's energy identity $\sum_i u_i B_i = 0$ and the Liouville identity $\sum_i \partial_i B_i = 0$, has its own Hamiltonian — the Koopman–von Neumann (Liouville) generator $H_{\mathrm{NS}} = -\frac{i}{2}\sum_m (\pi_m F_m + F_m \pi_m)$ of the induced flow on wave functions, i.e. the Hermitized "momentum times drift" shape, which is **not** a Weyl-ordered sum of squares. The chapter on it proves what that shape can and cannot do:

- kvnPoly / nsKoopmanOp — the Hamiltonian with the **exact** nonlinearity (no truncation, no linearization, no perturbative splitting), and nsKoopmanOp_symmetricOn — it is symmetric on the Gauss–polynomial core, so it passes the first Faris–Lavine requirement;
- quadP_starP — the antiunitary conjugation $C\psi = \bar{\psi}$ anticommutes with it, so the numerical range is **symmetric about zero**; gpair_hermite_kvn computes the exact one-mode Hermite matrix element and shows the advection drops out of it by a degree-parity selection rule, while the viscous part grows like n;
- nsKoopmanOp_not_bounded_below and nsKoopmanOp_not_positive — **the conclusion**: for $\nu > 0$ and one positive Stokes eigenvalue the quadratic form of H_{NS} is unbounded below. The mainstream generator can never be the positive comparison operator N, and the shortcut H = N, c = 0 is not available for it.

What the chapter does supply is the **replacement** comparison operator, on the mainstream leg: the Leray energy energyPoly / nsEnergyOp, i.e. multiplication by $E = 1 + \|u\|^2$ — positive and ≥ 1 by

nsEnergyOp_quadForm_ge. The commutator is computed exactly, kvn_comm_energy with the flux identity fluxPoly_eq: $i[H_{\mathrm{NS}}, N_E] = F \cdot \nabla E = -2\nu \sum_i \lambda_i u_i^2$ — the advection and the pressure gradient cancel by Leray's identity — giving the Faris–Lavine commutator inequality commForm_kvn_energy_bound, $|\langle x, i[H_{\mathrm{NS}}, N_E]x\rangle| \le 2\nu\Lambda \langle x, N_E x\rangle$, and then nsKoopman_esa_of_energy_comparison. Non-vacuity is nsTriad, the resonant Navier–Stokes Fourier triad with $a + b + c = 0$.

```
#check @BookProof.NsKoopman.kvnPoly
#check @BookProof.NsKoopman.nsKoopmanOp_symmetricOn
#check @BookProof.NsKoopman.quadP_starP
#check @BookProof.NsKoopman.gpair_hermite_kvn
#check @BookProof.NsKoopman.nsKoopmanOp_not_bounded_below
#check @BookProof.NsKoopman.nsKoopmanOp_not_positive
#check @BookProof.NsKoopman.nsEnergyOp_quadForm_ge
#check @BookProof.NsKoopman.kvn_comm_energy
#check @BookProof.NsKoopman.commForm_kvn_energy_bound
#check @BookProof.NsKoopman.nsKoopman_esa_of_energy_comparison
#check @BookProof.NsKoopman.nsTriad
```

33.10. The Lagrangian Face: One Parcel, Then the Interacting Whole

The Lagrangian (material-coordinate) description carries two enclosure results, and they must be read apart. The **one-parcel** Lagrangian Hamiltonian $h = \frac{1}{2}\sum_i P_i^2 + \nu \sum_i Q_i^2 + f \cdot P$ on the trajectory space $\ell^2(\mathrm{Fin}\, 3 \to \mathbb{N})$ is unconditionally essentially self-adjoint on the finite-mode Hermite core for every $\nu > 0$ and every constant force f (LagrangianCanonical.lagCan_esa). BookProof/Ns/ChapterNsLagrangianOuterFockEsa.lean carries it — with no new hypothesis — into the final-Hamiltonian form: the enclosure $d\Gamma(h)$,creation on the left and annihilation on the right, is essentially self-adjoint on the finite-particle domain over that core (lagOne_dGamma_esa — the mechanism is the second-quantization lift EsaOneParticle.dGamma_essentiallySelfAdjointOn_of_esa: graph-core transfer from the one-particle row lagCan_esa, with symmetry from EsaOneParticle.symmetricOn_dGammaCoreOp_of_esa), on the bosonic and fermionic Fock spaces and their Hilbert direct sums (lagOne_bosonicFock_esa, lagOne_fermionicFock_esa, lagOne_hbosonicFock_esa). The conditional Lagrangian Koopman generator gets the same lift (lagKoopman_dGamma_esa_of_comparison_esa), which inherits exactly the standing hypothesis on the comparison operator N_L — an enclosure theorem transports hypotheses as faithfully as it transports conclusions. The scope is the **non-interacting** $d\Gamma(h)$: on the n-parcel sector it acts as $\sum_p 1 \otimes \cdots \otimes h \otimes \cdots \otimes 1$.

The comparison side of the same face (NS-L-C.1–C.3, 2026-10-15c). Three statements around this h were open and are now theorems — and it matters for reading them that they are **comparison-side**, not further enclosures of the record. First, the transformed operator with a penalty/pressure **constraint term**: for any bounded symmetric C, $h + C$ is essentially self-adjoint on the Hermite core (BookProof/Ns/ChapterNsComparisonEsa.lean: lagConstr_esa, lagConstr_hasZeroDeficiencyOn) — closed by Kato–Rellich, because the ladder realization admits no common eigenvector of P_i and Q_i. Second, the **completed square** $N = h + \frac{1}{2}|f|^2 = \frac{1}{2}\sum_i (P_i + f_i)^2 + \nu \sum_i Q_i^2$ is symmetric, essentially self-adjoint (lagCompN_esa — the constraint theorem instantiated at $C = (\frac{1}{2}|f|^2)\cdot 1$), non-negative (lagCompN_nonneg) and obeys the form bound $\langle v, (N+1)v\rangle \ge \|v\|^2$ (lagCompN_add_one_ge). Third, the **lift** (BookProof/Ns/ChapterLiftedComparisonEsa.lean): $d\Gamma(N)$ is essentially self-adjoint on the finite-particle domain by the second-quantization theorem, and adding the constant 1 preserves that by the real-shift lemma (liftedComparison_dGamma_esa, **liftedComparison_esa**), with the resolvent face liftedComparison_resolvent_deficiency — trivial deficiency at every non-real z — as the analytic counterpart of the solver's finite pl_lifted_* shadow (Hermitian, form $\ge m + 1$, resolvent bound $1/|\operatorname{Im} z|$).

Read under the final-Hamiltonian convention: N is a comparison operator, so $d\Gamma(N) + 1$ is a **lifted comparison** — enclosing it does not make it a Hamiltonian of record, and the NS record stays $d\Gamma(h)$ (resp. $d\Gamma(H_{\mathrm{sp}})$ on the Eulerian face). The lift also does not touch the standing hypothesis on N_L: the comparison of the **Koopman generator** of the constrained Lagrangian flow, $N_L = H_L^2 + E$ on $L^2(\mathbb{R}^{\mathrm{PIdx}\, K})$, is a different operator, and lagKoopman_liftedComparison_esa_of_comparison_esa — like

lagKoopman_dGamma_esa_of_comparison_esa above — carries it explicitly; that input closed separately on 2026-10-16 (NsLagrangianComparisonEsa.lagComparison_esa), not through the lift.

```
#check @BookProof.NsComparisonEsa.lagConstr_esa
#check @BookProof.NsComparisonEsa.lagCompN_eq
#check @BookProof.NsComparisonEsa.lagCompN_esa
#check @BookProof.NsComparisonEsa.lagCompN_nonneg
#check @BookProof.NsComparisonEsa.lagCompN_add_one_ge
#check @BookProof.LiftedComparisonEsa.liftedComparison_dGamma_esa
#check @BookProof.LiftedComparisonEsa.liftedComparison_esa
#check @BookProof.LiftedComparisonEsa.liftedComparison_resolvent_deficiency
#check @BookProof.LiftedComparisonEsa.lagKoopman_liftedComparison_esa_of_comparison_e
```

The **interacting** outer Hamiltonians — the parcel-coupled Lagrangian lagFullFockHam and its Eulerian companion — are **not** second quantizations of a one-body operator: the gauge-fixing forms couple neighbouring parcels, and the Piola and det F − 1 forms have degree two and three. The enclosure theorem therefore does not apply to them, and the question is different: is the outer operator itself essentially self-adjoint on its finite-parcel core? BookProof/Ns/ChapterNsFullLagrangianFockEsa.lean answers yes, for all real couplings and with no new analysis: every n-parcel sector is a Weyl-type operator with distinct momentum coordinates and real polynomial forms (lagSectorHam_eq_weylPoly, nsSectorHam_eq_weylPoly, both rfl), hence essentially self-adjoint on the Gauss–polynomial core by the general instrument weylPoly_esa (lagSectorHam_esa, nsSectorHam_esa); the sectors glue to the finite-parcel core by the direct-sum theorem (lagFullFockHam_esa, nsFullFockHam_esa); and consequently the lifted Friedrichs realization of the earlier module is **the** self-adjoint extension (lagFullFockHam_selfAdjointExtension_unique, nsFullFockHam_selfAdjointExtension_unique).

Convention and boundary. These two interacting operators are the **auxiliary** Weyl-ordered sum-of-squares operators of the comparison layers, not the final Hamiltonian of record — which remains the one-particle h enclosed creation-left / annihilation-right (the **reduced** outer operator nsRedFullFockHam is such an enclosure; these two are not). What the result buys for the final-Hamiltonian programme is the comparison side: the auxiliary operator is now known to be essentially self-adjoint with a unique extension, so the Faris–Lavine legs that consume it as N consume a theorem rather than an assumption about its realization. Nothing is claimed about the non-semibounded Koopman generator, no continuum limit is taken, and no spectral information follows.

```
#check @BookProof.NavierStokesFlow.LagrangianCanonical.lagCan_esa
#check @BookProof.NsLagrangianOuterFock.lagOneOp_esa
#check @BookProof.NsLagrangianOuterFock.lagOne_dGamma_esa
#check @BookProof.NsLagrangianOuterFock.lagOne_bosonicFock_esa
#check @BookProof.NsLagrangianOuterFock.lagKoopman_dGamma_esa_of_comparison_esa
#check @BookProof.NsFullLagrangianEsa.lagSectorHam_esa
#check @BookProof.NsFullLagrangianEsa.lagFullFockHam_esa
#check @BookProof.NsFullLagrangianEsa.lagFullFockHam_selfAdjointExtension_unique
#check @BookProof.NsFullLagrangianEsa.nsSectorHam_esa
#check @BookProof.NsFullLagrangianEsa.nsFullFockHam_esa
#check @BookProof.NsFullLagrangianEsa.nsFullFockHam_selfAdjointExtension_unique
```

33.11. What Is Verified, and What Is Open

The verified layer, in one list:

- the eliminated sector's one-particle operator $H_{\mathrm{sp}} = H_{\mathrm{visc}} + H_{\mathrm{advect}}$ is the single-parcel member of the landed reduced family (redFormPoly_eq_liftParcel), symmetric and positive on the Gauss core for the **exact** nonlinearity (spHam_symmetricOn, spHam_quadForm_nonneg), with a Friedrichs realization as comparison and c = 0 (spHam_esa_farisLavine);
- its second quantization $d\Gamma(H_{\mathrm{sp}})$ is essentially self-adjoint on the lifted domain, preserves particle number, and restricts to the generator on the one-particle sector (nsSpDGamma_esa_farisLavine, nsSpDGamma_number_conserving, nsSpDGamma_one_particle), and the n-parcel reduced Hamiltonian is the sum over parcels of the one-body generator (redHam_eq_sum_parcel, nsRedFullFockHam_sector_sum_parcel);

- every sector of the reduced Hamiltonian is a Weyl-type operator with momenta in distinct coordinates and real polynomial forms, so the Kato-type theorem applies: each `redHam n` is essentially self-adjoint on the Gauss–polynomial core of $L^2(\mathbb{R}^{6n})$ (`redHam_esa`; the case `n = 1` is a one-particle statement), and the reduced outer operator `nsRedFullFockHam` is essentially self-adjoint on its finite-parcel core, not only on the domain of its own Friedrichs realization (`nsRedFullFockHam_esa`, `BookProof/Ns/ChapterNsReducedCoreEsa.lean`);
- the one-body generator itself is essentially self-adjoint on the Gauss–polynomial core of $L^2(\mathbb{R}^6)$ (`spHam_esa`, one-particle), and $d\Gamma(H_{\mathrm{sp}})$ is essentially self-adjoint on the symmetrized tensor power of that core for every particle number, on the bosonic Fock space, and — unsymmetrized, creation left / annihilation right — on the finite-particle domain (`nsSp_bosonic_core_esa`, `nsSp_bosonicFock_esa`, `nsSp_hbosonicFock_esa`, `nsSp_dGamma_esa`, `BookProof/Ns/ChapterNsSymmetricSector.lean`);
- the momentum variables are honest: a unitary partial transform that leaves the fibre alone, a derivative that is a real symbol, a densely defined π^{-1} with trivial kernel, and the advection as a convolution whose coefficients are $O(\Lambda)$ uniformly in the parcel number;
- the mainstream generator is symmetric but **indefinite**, so it cannot be the comparison operator of the criterion; its own leg carries the Leray-energy comparison instead.

What remains open on this leg, stated as obligations rather than as caveats:

- ~~the identification of the vector-valued `L²(V; L²(W))` with the scalar `L²(V × W)`~~ — **closed**: `curryLI` (`BookProof.NsScalarVectorCurry`) is the identification, and `nsScalarFourier` (`BookProof.NsScalarFourier`) carries the spatial transform across it, so the elimination and the enclosure act on one and the same one-particle space (see “The Fibred Model Identified with the Scalar One” above);
- the operator-level relative bound with a constant independent of the parcel number — not needed on the landed route, where the comparison is the lifted Friedrichs realization;
- ~~the unitary identification of the parcel sectors $L^2(\mathbb{R}^{6n})$ with the tensor powers of the one-particle space~~ — **closed at the Hilbert-space level** by `BookProof/Toolbox/ChapterL2TensorPowerUnitary.lean`: `tensorPowUnitary` identifies the `n`-particle sector with $L^2(\mu^{\otimes n})$ for any σ-finite μ (the scalar lift is an isometry, `inner_prodTensor`, and the embedding has dense range, `denseRange_embPow`), while the measure-preserving relabelling `(ℝ^d)^n ≅ ℝ^{n·d}` (`mpEquivUnitary`, `parcelRelabel`) gives `parcelSectorUnitary`, its Navier–Stokes instance `nsParcelSectorUnitary` — exactly the parcel-sector space $L^2(\mathbb{R}^{6n})$ — and `nsFockUnitary` for the whole Fock space $\bigoplus_n L^2(\mathbb{R}^6)^{\otimes \hat{n}}$. Still open is the **operator-level** transport through that unitary (the sector derivation of H_{sp} on $\mathrm{polyGaussCore}^{\otimes n}$ carried to the parcel operator of `nsRedFullFockHam` on its core), and the identification with the occupation-number (`ℓ²`) spelling `dGammaOp (nsSpCol …)`, for which the graded-band Schur gate does not apply (the quartic one-body generator violates its entry-growth condition);
- on the mainstream leg, the surjectivity of $N_E + 1$ on a concrete domain, and the lift of that comparison operator to the nested-Fock setting. The domain cannot be the Gauss core: for $d \geq 1$, $N_E + 1$ maps the core into itself and the core is a proper subspace of $L^2(\mathbb{R}^d)$ (Baire), so the surjectivity hypothesis of `nsKoopman_esa_of_energy_comparison` is false (`not_nsEnergy_surjective`, `BookProof/Ns/ChapterNsEnergySurjectivityObstruction.lean`).

```
#check @BookProof.NsReducedCoreEsa.redHam_esa
#check @BookProof.NsReducedCoreEsa.nsRedFullFockHam_esa
#check @BookProof.NsSymmetricSector.spHam_esa
#check @BookProof.NsSymmetricSector.nsSp_bosonic_core_esa
#check @BookProof.NsSymmetricSector.nsSp_bosonicFock_esa
#check @BookProof.NsSymmetricSector.nsSp_dGamma_esa
#check @BookProof.NsEnergySurjectivityObstruction.not_nsEnergy_surjective
#check @BookProof.L2TensorPower.tensorPowUnitary
#check @BookProof.L2TensorPower.parcelSectorUnitary
#check @BookProof.L2TensorPower.nsParcelSectorUnitary
#check @BookProof.L2TensorPower.nsFockUnitary
```

34. The Comparison Operator of the Navier–Stokes Hamiltonian

34.1. Orientation and Status

The other half of the Navier–Stokes operator story: the Faris–Lavine **comparison operator** N — the Leray energy on the mainstream leg, the lifted auxiliary Friedrichs extension on the full nonlinear leg — with the theorems that establish each, and the reasons the Hamiltonian itself (or its square) cannot serve as N. Read it after the one-particle Hamiltonian chapter; together the two discharge the operator obligations that the next chapter transfers to Fock space.

The route: the question and the one-line answer, why N cannot be the Hamiltonian (nor H^2), layer 1 (the mainstream Hamiltonian as Leray energy, and the rebuilt affine-drifts-are-quadratic view), the criterion on a core and the nonlinear instance, layer 2 (the lifted auxiliary Friedrichs extension), why the pressure does not change the comparison operator, and the one-particle obligation discharged.

The proofs are `BookProof/FarisLavine`, `BookProof/ComparisonCoreEsa`, `BookProof/SelfAdjointCoreEsa`, `BookProof/NsNonlinearFarisLavine`, `BookProof/NsLinearKoopmanEsa`, and the `NsFullEuler` / `NsFullLagrangian` / `ChapterNavierStokesEulerian` statements they feed.

34.2. The Question, and the Answer in One Line

The Faris–Lavine criterion does not ask that the Hamiltonian be positive. It asks for a **positive** symmetric operator N, with `N + 1` onto, and a constant c with `|⟨x, i[H,N]x⟩| ≤ c⟨x,Nx⟩` on a core; the conclusion is that H is essentially self-adjoint there. So for a Hamiltonian that is itself indefinite, N must be a **different** operator, and the question "what is N?" is a genuine question.

For Navier–Stokes the answer depends on which Hamiltonian is meant, and the two answers are proved separately in the library:

- the **mainstream (truncated Leray–Galerkin) Hamiltonian** `H_NS = ½Σ_m(π_mF_m + F_mπ_m)` takes the **Leray energy** `N_E`, multiplication by `E(u) = 1 + ‖u‖²`, with `c = 2νΛ`;
- the **full nonlinear Hamiltonian** (exact advection, exact pressure, derivative coordinates kept as fields) takes the **lifted Friedrichs extension of the auxiliary sum-of-squares operator**, with `c = 0`.

In neither case is N the Hamiltonian, and in particular the once-tempting `N = H²` is invalid. This chapter states both answers and points to the theorems that are the evidence.

34.3. Why N Cannot Be the Hamiltonian — and Why Not H² Either

The mainstream generator is symmetric on the Gauss–polynomial core (`nsKoopmanOp_symmetricOn`), so it passes the first requirement of the criterion, but it is **not bounded below**: for `ν > 0` and one positive Stokes eigenvalue the quadratic form of H_NS is unbounded below (`nsKoopmanOp_not_bounded_below`, `nsKoopmanOp_not_positive`). An indefinite operator can never be the positive N. The mechanism is explicit: the antiunitary conjugation `Cψ = ψ̄` anticommutes with H_NS (`quadP_starP`), so the numerical range is symmetric about zero, and the exact one-mode Hermite matrix element (`gpair_hermite_kvn`) shows the advection dropping out by a parity selection rule while the viscous part grows with the mode number.

The replacement `N = H_NS²` fails for two independent reasons, and both are structural rather than technical:

- **the wrong domain**: `𝒟(H²) ⊊ 𝒟(H)`, so H² is not even a comparison operator on the common Faris–Lavine domain;
- **`N + 1` is not onto**: on the domain `H² + 1 = (H − i)(H + i)`, so `range(H² + 1) ⊆ range(H − i)`, and surjectivity fails exactly when H has a non-trivial deficiency — the situation the criterion exists to settle.

The general shape of this obstruction is formalized in the library for the case `N = H`: `BookProof.FarisLavine.not_farisLavine_criterion_of_relative_bound` exhibits a limit-circle Jacobi operator for which the commutator and relative-bound inequalities both hold with `a = 1`, `b = 0` and essential self-adjointness still fails. The criterion is used as a criterion, not as a definition of `N`.

```
#check @BookProof.NsKoopman.nsKoopmanOp_symmetricOn
#check @BookProof.NsKoopman.quadP_starP
#check @BookProof.NsKoopman.gpair_hermite_kvn
#check @BookProof.NsKoopman.nsKoopmanOp_not_bounded_below
#check @BookProof.NsKoopman.nsKoopmanOp_not_positive
#check @BookProof.FarisLavine.not_farisLavine_criterion_of_relative_bound
```

34.4. Layer 1 — The Mainstream Hamiltonian: the Leray Energy

The mainstream incompressible system is `u̇ = −νAu + B(u,u)`, with the nonlinearity the exact quadratic advection `B_i(u,u) = Σ_{j,k} b_{ijk}u_ju_k`, and with the two structural identities of incompressible flow carried as hypotheses of the system (`BookProof.NsKoopman.NsSystem`): **Leray's energy identity** `Σ_i u_iB_i(u,u) = 0` and the **Liouville identity** `Σ_i ∂_iB_i = 0`.

The comparison operator is the **Leray energy** $N_E = \mathrm{mulOp}\big(E(u)\big), \qquad E(u) = 1 + \|u\|^2$, `energyPoly` / `nsEnergyOp`. It is a positive multiplication operator dominating the identity:

- `nsEnergyOp_quadForm_ge` — $\langle x, N_E x\rangle \geq \|x\|^2$, so `N_E` is positive with `N_E ≥ 1`;
- `kvn_comm_energy` — the commutator is computed exactly, `i[H_NS, N_E] = F·∇E`, multiplication by the **energy flux**;
- `fluxPoly_eq` — and the flux identity collapses it, using Leray's identity, to the **sign-definite viscous dissipation** $F \cdot \nabla E = -2\nu \sum_i \lambda_i u_i^2 \leq 0$. The advection and the pressure gradient cancel: this is where the $\sum_i u_i B_i = 0$$ hypothesis is consumed;
- `commForm_kvn_energy_bound` — the resulting Faris–Lavine commutator inequality, $\big|\langle x, i[H_{\mathrm{NS}}, N_E]x\rangle\big| \leq 2\nu\Lambda \,\langle x, N_E x\rangle$, with `Λ = max_i λ_i` the largest Stokes eigenvalue of the truncation (the cutoff's role here is exactly to make Λ finite);
- `nsKoopman_esa_of_energy_comparison` — the Faris–Lavine conclusion for the mainstream generator, with the Leray energy as comparison.

Non-vacuity of the leg is not assumed: `nsTriad` is the resonant Navier–Stokes Fourier triad with `a + b + c = 0` on which the identities are satisfied non-trivially.

The one input this leg does **not** supply by itself is the surjectivity of `N_E + 1` from the common domain, which is the part of the criterion the Gauss–polynomial core does not deliver. It is carried as a named hypothesis in `nsKoopman_esa_of_energy_comparison` and never asserted as an axiom. That hypothesis, as stated — surjectivity **from the Gauss core itself** — is false in every dimension $d \geq 1$: $N_E + 1$ maps the core into the core, and a countably spanned subspace of the infinite-dimensional $L^2(\mathbb{R}^d)$ is proper by Baire's theorem (`not_nsEnergy_surjective`). The theorem is therefore vacuous as stated, and the surjectivity input must be supplied on a larger domain together with a separate core argument.

```
#check @BookProof.NsKoopman.energyPoly
#check @BookProof.NsKoopman.nsEnergyOp
#check @BookProof.NsKoopman.nsEnergyOp_quadForm_ge
#check @BookProof.NsKoopman.kvn_comm_energy
#check @BookProof.NsKoopman.fluxPoly_eq
#check @BookProof.NsKoopman.commForm_kvn_energy_bound
#check @BookProof.NsKoopman.nsKoopman_esa_of_energy_comparison
#check @BookProof.NsKoopman.nsTriad
```

34.5. Layer 1, Rebuilt: Affine Drifts Are Quadratic Hamiltonians

The Baire refutation above is a statement about the Leray-energy route **as stated**, and the repair is not a change of domain: informally, no multiplication operator can dominate a first-order generator. The mainstream leg was therefore rebuilt where it can be closed with the instruments of this project — in `BookProof/Ns/ChapterNsLinearKoopmanEsa.lean` — around one observation: the Koopman–von

Neumann generator of an **affine** drift $F(u) = Au + c$ is a **real quadratic Hamiltonian**. Its Weyl-ordered symbol $\sum_i \frac{1}{2}(\pi_i F_i + F_i \pi_i)$ (`linKvnPoly`) **is** `fqPoly 0 0 Aᵀ 0 c` (`linKvnPoly_eq_fqPoly`) — cross terms and momenta only, no kinetic and no potential part — so the Carleman-flux theorem for general quadratic Hamiltonians, `FullQuadratic.fqOp_essentiallySelfAdjoint` of `BookProof/Toolbox/ChapterFullQuadraticEsa.lean`, applies with **no comparison operator, no surjectivity hypothesis and no sign condition**: the generator is essentially self-adjoint on the Gauss–polynomial core (`linKoopman_esa`), generates a complete unitary flow (`linKoopman_stone_flow`), and its second quantization — creation on the left, annihilation on the right — is essentially self-adjoint on the finite-occupation core (`linKoopman_dGammaOp_esa`). Two named systems are instances: the **Stokes system** (vanishing advection), for which the mainstream operator `nsKoopmanOp S` of the previous layers is itself essentially self-adjoint (`nsKoopman_stokes_esa`, with `drift_eq_linDrift_of_stokes` identifying its drift), and the **Oseen linearization** of the full Navier–Stokes drift about an **arbitrary** point `ū` (`oseenKoopman_esa`; `oseenDrift_eq` identifies that affine drift as the first-order Taylor polynomial of the true drift, the remainder being exactly $B(u - \bar{u}, u - \bar{u})$).

```
#check @BookProof.NsLinearKoopmanEsa.linKvnPoly_eq_fqPoly
#check @BookProof.NsLinearKoopmanEsa.linKoopman_esa
#check @BookProof.NsLinearKoopmanEsa.linKoopman_stone_flow
#check @BookProof.NsLinearKoopmanEsa.linKoopman_dGammaOp_esa
#check @BookProof.NsLinearKoopmanEsa.nsKoopman_stokes_esa
#check @BookProof.NsLinearKoopmanEsa.oseenDrift_eq
#check @BookProof.NsLinearKoopmanEsa.oseenKoopman_esa
```

The **nonlinear** part of the mainstream generator (advection $B \neq 0$) is not covered: it is cubic, and neither instrument above applies. Its recorded route is no longer a comparison operator but the classical flow — completeness of the Galerkin flow (the energy inequality plus $\operatorname{div} F = -\nu \sum \lambda_i$) together with the orbit criterion `FlowDGammaEsa.deficiencyTrivialAt_of_orbits` on a flow-invariant core, which needs the smooth dependence of the flow on its initial data — infrastructure Mathlib does not yet provide. That, and the conditional comparison below, are the two open inputs of this leg.

34.6. The Criterion on a Core, and the Nonlinear Instance

The criterion itself was sharpened so that its hypotheses can hold on an algebraic core. `BookProof/Toolbox/ChapterFarisLavineDenseCore.lean` states Faris–Lavine Corollary 1.1 with every hypothesis on a dense subspace `D` and with `N + 1` required to have only **dense range** from `D` rather than to be onto (`essentiallySelfAdjointOn_of_farisLavine_dense`) — on a polynomial core, onto never happens — and proves that for a positive symmetric `N` this density hypothesis **is** essential self-adjointness of `N` (`dense_range_add_one_of_esa_of_pos`). The same module proves the **square comparison**: for an `H` mapping `D` into itself and $E \geq 0$, the choice $N = H^2 + E$ satisfies every Faris–Lavine inequality — $\langle x, Nx\rangle = \|Hx\|^2 + \langle x, Ex\rangle$, $\|Hx\| \leq \|Nx\| + \|x\|$, and the commutator identity $\langle x, i[H,N]x\rangle = \langle x, i[H,E]x\rangle$ exactly — so that essential self-adjointness of `N` on `D` implies essential self-adjointness of `H` (`essentiallySelfAdjointOn_of_square_comparison`). The proof of the core form is the paper's test-vector argument with the inverse replaced by a sequence: given `w`, pick `g_n ∈ D` with $(N+1)g_n \to w$ by density — the relative bound keeps `H g_n` bounded, the commutator bound forces $\langle g_n, (N+1)g_n\rangle \to 0$, and closability of $N+1$ (symmetry plus density of `D`) turns $g_n \to 0$ together with $(N+1)g_n \to w$ into $w = 0$ — the deficiency spaces are trivial. The module imports only `ChapterFarisLavineCore`: no spectral hypothesis, no boundedness, no onto.

`BookProof/Ns/ChapterNsNonlinearFarisLavine.lean` is the nonlinear instance, with the exact quadratic advection and no linearization: the coefficients b_{mjk} are the momentum-space form of $(u \cdot \nabla)u$ after the Leray projection — the coordinate-space product with spatial derivatives becomes the triad convolution $\sum_{p+q=k} i(\hat{u}_p \cdot q)\hat{u}_q$ (`NsAdvectionConvolution.fourier_advection_convolution`), the momentum-space convolution device that this project uses for NS and QG only. The comparison is $N_{\mathrm{NS}} = H_{\mathrm{NS}}^2 + 1 + \|u\|^2$: it dominates the first-order generator, and its commutator against H_{NS} is exactly the Leray dissipation, giving $|\langle i[H_{\mathrm{NS}}, N_{\mathrm{NS}}]\rangle| \leq 2\nu\Lambda\langle N_{\mathrm{NS}}\rangle$ (`nsSquareComparison_commForm_bound`). Hence **if** `N_NS` is essentially self-adjoint on the Gauss–polynomial core, so is the nonlinear generator (`nsKoopman_esa_of_squareComparison_esa`). The generic instrument behind the instance — the Lyapunov criterion $N = H_G^2 + E$ for **any** polynomial vector field `G`

with real $E \geq 0$ and $|G \cdot \nabla E| \leq cE$ pointwise — is `KoopmanLyapunov.kvnGen_esa_of_lyapunov` (`BookProof/Ns/ChapterKoopmanLyapunovFarisLavine.lean`), whose other instance is the Lagrangian system of the Fourier elimination of the derivative variables.

```
#check @BookProof.FarisLavine.essentiallySelfAdjointOn_of_farisLavine_dense
#check @BookProof.FarisLavine.dense_range_add_one_of_esa_of_pos
#check @BookProof.FarisLavine.essentiallySelfAdjointOn_of_square_comparison
#check @BookProof.NsNonlinearFarisLavine.nsKoopman_esa_of_squareComparison_esa
#check @BookProof.KoopmanLyapunov.kvnGen_esa_of_lyapunov
```

Boundary. Essential self-adjointness of `N_NS` (equivalently, by the lemma above, density of `(N_NS + 1)` applied to the core) is an explicit hypothesis and is **not** proved: this is a reduction, not a proof, for the mainstream generator. Why no simpler comparison works is recorded informally in the module — the quadratic advection stretches momenta at a rate proportional to $\|u\|$, so no polynomial harmonic-type `N` satisfies the bound — and with **all** Fourier modes the constant $2\nu\Lambda$ is infinite, which is the formal face of the backward ill-posedness of viscous Navier–Stokes. The statements are for every finite mode set, with the exact nonlinearity on those modes.

34.7. Layer 2 — The Full Hamiltonian: the Lifted Auxiliary Friedrichs Extension

Written without the Leray–Galerkin truncation — the exact advection $u \cdot \nabla u$, the exact pressure, and the derivative coordinates kept as independent fields — the Hamiltonian's potential is no longer quadratic and the Leray energy alone does not control it. The library records the precise failure of the truncated picture: `nsResPoly_not_affine` shows that the full residual is not an affine function of the fields, so the model is genuinely nonlinear and not an Oseen linearization in disguise. The advection then contributes a cubic multiplying function to the commutator, which the quadratic energy flux cannot bound.

The route that works is the one non-abelian Yang–Mills uses, and it needs no new positivity input from Navier–Stokes at all. The library builds the **auxiliary positive Weyl-ordered sum of squares** of the constraint forms, $H_{\mathrm{sos}} = \frac{1}{2}\sum_m \pi_m^2 + \frac{1}{2}\sum_r \left(\mathrm{mulOp}\,\Phi_r\right)^2$, which **is** bounded below, so that it admits a **Friedrichs extension directly** — no Faris–Lavine detour is needed to realize **it**. That Friedrichs realization is then lifted fibre by fibre to the nested Fock space, and it is the lifted object that is used as the comparison operator `N` with commutator constant `c = 0`:

- `nsFullFockHam_quadForm_nonneg` — the auxiliary operator's quadratic form is non-negative;
- `nsFullFock_friedrichs_extension` — its Friedrichs extension exists;
- `nsFullOuterN_esa` — Faris–Lavine on the **outer** Fock space, i.e. essential self-adjointness of the lifted Hamiltonian on the domain of the lifted comparison;
- the Lagrangian face is the same construction: `lagFullFockHam_quadForm_nonneg`, `lagFullFock_friedrichs_extension`, `lagFullOuterN_esa`.

The distinction must be kept in view, because it is easy to misread the `c = 0`: it is a statement about the **auxiliary** sum-of-squares operator, not about the Navier–Stokes Hamiltonian, which remains unbounded below. What is **not** claimed on this leg is uniqueness of the self-adjoint extension from the finite-parcel core.

```
#check @BookProof.NsFullEuler.nsResPoly_not_affine
#check @BookProof.NsFullEuler.nsFullFockHam_quadForm_nonneg
#check @BookProof.NsFullEuler.nsFullFock_friedrichs_extension
#check @BookProof.NsFullEuler.nsFullOuterN_esa
#check @BookProof.NsFullEuler.nsFullFock_stone_flow
#check @BookProof.NsFullLagrangian.lagFullFockHam_quadForm_nonneg
#check @BookProof.NsFullLagrangian.lagFullFock_friedrichs_extension
#check @BookProof.NsFullLagrangian.lagFullOuterN_esa
```

34.8. The Transformed Lagrangian Leg: Constraint, Completed Square, and the Lift

A third leg, closed in the 2026-10-15c wave, runs the criterion where the Lagrangian parcel Hamiltonian lives in its **canonical (Hermite/ladder) realization** — the same operator the one-particle chapter records as $h =$

$\frac{1}{2}\sum_i P_i^2 + \nu\sum_i Q_i^2 + \sum_i f_i P_i$ on $\ell^2(\mathrm{Fin}\,3 \to \mathbb{N})$, with $[P_i, Q_i] = -i$. Two questions were open there, and both are now theorems in `BookProof/Ns/ChapterNsComparisonEsa.lean`.

NS-L-C.1 — the constraint term. The transformed operator of a real gauge/penalty treatment carries, besides the squares, a **constraint** term — the penalty or pressure contribution — written as an arbitrary bounded symmetric operator `C` on the Hermite core. `lagConstrData` adds it to the canonical data, and `lagConstr_esa` / `lagConstr_hasZeroDeficiencyOn` prove that $\hat{h}_{\mathrm{full}} = \frac{1}{2}\sum_i P_i^2 + \nu\sum_i Q_i^2 + \sum_i f_i P_i + C$ is essentially self-adjoint there. The route is worth recording because the **obvious** route is impossible: the common-eigenvector criterion `hasZeroDeficiencyOn_of_commonEigenvectors` cannot be instantiated in this realization — a common eigenvector of P_i and Q_i would violate $[P_i, Q_i] = -i$ — so the closing route is the Kato–Rellich theorem `hFull_essentiallySelfAdjointOn_of_drive_eq_P` on top of the diagonalization of the second-order part by the Hermite states. Symmetry of the addition is separate (`symmetricOn_add_const` for real scalars, and the bounded-symmetric case is `lagConstr_esa`'s hypothesis, never silently assumed).

NS-L-C.2 — the completed square. The comparison question for h has a classical answer, now checked: complete the square,

$$N = h + \tfrac{1}{2}|f|^2 = \tfrac{1}{2}\sum_i (P_i + f_i)^2 + \nu\sum_i Q_i^2.$$

`lagCompN` builds `N`, `lagCompN_eq` states the shift identity as operators on the Hermite core, and the three properties the criterion consumes are each a theorem: `lagCompN_symmetricOn`, **`lagCompN_esa`** (essentially self-adjoint — and note **how**: it is the NS-L-C.1 theorem itself, instantiated at the scalar constraint $C = (\frac{1}{2}|f|^2)\cdot 1$ via `isSymmetricDom_real_smul_id`, so the completed square falls out of the constraint result with no fresh deficiency computation), **`lagCompN_nonneg`** (the completed square is ≥ 0) and `lagCompN_add_one_ge` (form bound $\langle v, (N+1)v\rangle \geq \|v\|^2$). Pedagogically the point is the same one the Oseen/Lyapunov legs make: the force term $f \cdot P$ is **not** a perturbation to be estimated — it is absorbed exactly by the shift $P_i \mapsto P_i + f_i$, at the price of the constant $\frac{1}{2}|f|^2$ that the shift leaves behind.

NS-L-C.3 — the lift. The lifted form of the same statement — $d\Gamma(N) + 1$ essentially self-adjoint on the finite-particle core of the outer Fock space — is the three-step chain of `BookProof/Ns/ChapterLiftedComparisonEsa.lean` (comparison ESA on the core → second-quantization lift → the real-shift lemma `EsaRealShift.essentiallySelfAdjointOn_add_const`), recorded with its resolvent face `liftedComparison_resolvent_deficiency` (trivial deficiency at **every** non-real z, the analytic content of the numerical`pl_lifted_*` shadow). It is discussed in full at the end of the one-particle chapter.

Honest boundary — and the convention reading. None of `h + C`, `N` or $d\Gamma(N) + 1$ is a Hamiltonian of record: under the final-Hamiltonian convention the NS record remains the enclosure $d\Gamma(H_{\mathrm{sp}})$ (and $d\Gamma(h)$ on the Lagrangian face), and these are **comparison and lifted comparison** objects — enclosing `N` does not promote it. In particular the completed square is the comparison operator of the **canonical parcel** h; it does **not** discharge the standing hypothesis of `lagKoopman_esa_of_comparison_esa`, which is about a different operator — $N_L = H_L^2 + E$ of the Koopman–von Neumann generator of the Galerkin Lagrangian flow with the exact degree-five constraint force, on $L^2(\mathbb{R}^{\mathrm{PIdx}\,K})$ (see the Fourier-elimination chapter). That input was discharged on 2026-10-16, by `NsLagrangianComparisonEsa.lagComparison_esa` (the next section).

```
#check @BookProof.NsComparisonEsa.lagConstr_esa
#check @BookProof.NsComparisonEsa.lagConstr_hasZeroDeficiencyOn
#check @BookProof.NsComparisonEsa.lagCompN_eq
#check @BookProof.NsComparisonEsa.lagCompN_symmetricOn
#check @BookProof.NsComparisonEsa.lagCompN_esa
#check @BookProof.NsComparisonEsa.lagCompN_nonneg
#check @BookProof.NsComparisonEsa.lagCompN_add_one_ge
#check @BookProof.LiftedComparisonEsa.liftedComparison_dGamma_esa
#check @BookProof.LiftedComparisonEsa.liftedComparison_esa
#check @BookProof.LiftedComparisonEsa.liftedComparison_resolvent_deficiency
#check @BookProof.LiftedComparisonEsa.lagKoopman_liftedComparison_esa_of_comparison_e
#check @BookProof.NavierStokesFlow.LagrangianKatoRellich.hFull_essentiallySelfAdjoint
#check @BookProof.NavierStokesFlow.LagrangianEsa.hasZeroDeficiencyOn_of_commonEigenve
```

34.9. Closing the Standing Hypothesis: the Koopman Comparison N_L

The one input, and its closure. Every step of the Lagrangian line above ended in a single named hypothesis — `lagKoopman_esa_of_comparison_esa`: essential self-adjointness of the Faris–Lavine comparison $N_L = H_L^2 + E$ on the Gauss–polynomial core of $L^2(\mathbb{R}^{\mathrm{PIdx}\,K})$, for the Galerkin Lagrangian Koopman generator with the exact degree-five constraint force. The 2026-10-16 wave closes it **unconditionally** (every finite K, ν ≥ 0, κ ≥ 0): `BookProof/Ns/ChapterNsLagrangianComparisonEsa.lean` proves **`lagComparison_esa`** (the comparison) and hence **`lagKoopman_esa`** (the generator), literally as the instantiation `lagKoopman_esa_of_comparison_esa S (lagComparison_esa S)`.

Route — a Lyapunov polynomial, then Kato's argument. The generic criterion is `KvnKato.lyapunovComparison_esa_poly` (in `BookProof/Toolbox/ChapterKvnKatoPoly.lean`): for a real polynomial vector field G and a real polynomial $E \geq 1$, the comparison $N = H_G^2 + E$ is essentially self-adjoint on the Gauss–polynomial core as soon as some real polynomial P satisfies $\|y\| \leq P(y)$ and $|G \cdot \nabla P| \leq c_0 P$ — a deficiency vector is transferred to a weak solution against C_c^∞ by the Hermite graph approximation, then killed by Kato's argument along the complete flow. The energy here is **not** proper (the penalty vanishes on the unbounded incompressible set), so the Lyapunov function is the completed $P = E + \frac{1}{2}|\xi|^2$: the flux identity `lagLyapFlux_eq` ($F \cdot \nabla P = -\nu \sum |k|^2 v^2 + \xi \cdot v$), the bound `abs_eval_lagLyapFlux_le` ($|F \cdot \nabla P| \leq (2\nu\Lambda + 1)P$), and `norm_le_lagP` ($\|y\| \leq 1 + \frac{1}{2}\|y\|^2 \leq P$).

The seven-chapter machine. The Kato route is not a single lemma but a pipeline, one chapter per step: `ChapterKvnKatoCalc` (pointwise rules for the coordinate derivative `dcoord`), `ChapterKvnKatoMol` (mollifiers and the Schur test), `ChapterKvnKatoKernel` (explicit commutator kernels), `ChapterKvnKatoFriedrichs` (Friedrichs' lemma for those kernels), `ChapterKvnKatoEnergy` (the energy inequality for the mollified deficiency equation), `ChapterKvnKatoEsa` (the deficiency vectors vanish), and `ChapterKvnKatoPoly` (the polynomial core). All seven enter `BookProof.lean` transitively through `BookProof.ChapterNsLagrangianComparisonEsa`; the two closing statements carry their `#print axioms` lines in `ChapterRoadmapAudit.lean` and report only `propext`, `Classical.choice`, `Quot.sound`.

What this does not touch. Boundaries are exactly those of `ChapterNsLagrangianDetFarisLavine`: Galerkin truncation (finite K), the volume penalty at finite bulk modulus κ (no κ → ∞ limit), the viscous term evaluated at $F = I$; no continuum NS regularity and no mass gap. The **Eulerian** square-comparison hypothesis (`nsKoopman_esa_of_squareComparison_esa`) and the energy surjectivity input stay open. And under the final-Hamiltonian convention nothing changes about what is enclosed: N_L is a comparison operator and is never enclosed — the NS record remains $d\Gamma(h)$ (resp. $d\Gamma(H_{\mathrm{sp}})$), creation left, annihilation right. Plan pointers: the acceptance and boundary paragraphs of `CONSOLIDATED_PLAN.md` §"State of the project — 2026-10-16: the Koopman comparison N_L = H_L² + E is ESA — closed-by pointers".

```
#check @BookProof.KvnKato.lyapunovComparison_esa_poly
#check @BookProof.NsLagrangianComparisonEsa.lagLyapFlux_eq
#check @BookProof.NsLagrangianComparisonEsa.abs_eval_lagLyapFlux_le
#check @BookProof.NsLagrangianComparisonEsa.norm_le_lagP
#check @BookProof.NsLagrangianComparisonEsa.lagComparison_esa
#check @BookProof.NsLagrangianComparisonEsa.lagKoopman_esa
```

34.10. The Pressure Does Not Change the Comparison Operator

The pressure enters the residual only through the pressure-gradient coordinate `q_i = ∂_ip`, and it does **no work** on the velocity: `Σ_k u_kq_k = 0` on the divergence-free sector. It therefore contributes nothing to the energy flux `F · ∇E` that the criterion bounds, and nothing to the definition of N. It is the Lagrange multiplier of incompressibility, a **second-class** constraint fixed by the pressure-Poisson equation `Δp = −∂_i(u_j∂_ju_i) + ∂_if_i`, and it is handled by **elimination** — the same strategy the programme uses for the derivative variables — rather than by a new gauge symmetry.

A BRST charge for it would be cosmetic: a first-class constraint generates residual gauge freedom and needs BRST, but here Δp determines p up to an additive constant whose gradient is zero, so there is no residual freedom to gauge. The constraint algebra is exactly the one classified in `BookProof.ChapterNavierStokesEulerian`: the **derivative** relations `u_{i,j} = ∂_ju_i` are the ones that carry the gauge generator and the nilpotent

charge, while `div u = 0` is an explicit-solution constraint. Consequently the comparison operator is unchanged by the pressure: the Leray energy in the truncated system, the lifted auxiliary Friedrichs extension in the full model.

34.11. The One-Particle Obligation, Discharged

Every layer of the table below runs the same criterion, and the criterion has two obligations. The **lifted** one — that the second quantization of N is essentially self-adjoint on the lifted core — is the core-transfer chain of Second Quantization and Essential Self-Adjointness. The **inner** one is that N itself must be essentially self-adjoint on the core the proof actually chooses. For a long time that obligation was discharged sector by sector; it is now a theorem in general, in two short chapters.

`BookProof.SelfAdjointCoreEsa` says that an operator possessing a self-adjoint extension is already essentially self-adjoint on its own domain: at a non-real z the deficiency equation makes the deficiency vector a domain eigenvector of the extension, and symmetry forces it to be zero (`deficiencyTrivialAt_dom_of_isSelfAdjointExtension`). `BookProof.ComparisonCoreEsa` restates this in the vocabulary of the criterion — a `Comparison` is self-adjoint, hence essentially self-adjoint, on its own domain (`comparison_essentiallySelfAdjointOn_dom`) — and proves the equivalence that matters for the route chapters: **`isGraphCore_iff_esa`, for a Faris–Lavine comparison operator, "graph core" and "core of essential self-adjointness" are the same notion.** `esa_of_isGraphCore` is obligation (i) itself.

For the layers above this has a concrete reading. The auxiliary sum-of-squares operator H_{sp} of the lifted layer is positive and self-adjoint in its Friedrichs realization, so its core statements follow without a fresh deficiency argument; and wherever a route chapter exhibits a graph core for its comparison — as the QG chapters do with the Gauss–polynomial core of the oscillator and the compactly supported smooth core of the wall — obligation (i) is discharged by `esa_of_isGraphCore` alone. The Leray-energy leg of the mainstream layer keeps its own open input (the surjectivity of $N_E + 1$ on a concrete domain), which is a statement about N_E, not about the obligation in general.

```
#check @BookProof.SelfAdjointCoreEsa.deficiencyTrivialAt_dom_of_isSelfAdjointExtensic
#check @BookProof.SelfAdjointCoreEsa.essentiallySelfAdjointOn_dom_of_isSelfAdjointExt
#check @BookProof.SelfAdjointCoreEsa.essentiallySelfAdjointOn_of_graphCore_selfAdjoir
#check @BookProof.ComparisonCoreEsa.comparison_isSelfAdjointExtension
#check @BookProof.ComparisonCoreEsa.comparison_essentiallySelfAdjointOn_dom
#check @BookProof.ComparisonCoreEsa.graphCore_of_isGraphCore
#check @BookProof.ComparisonCoreEsa.esa_of_isGraphCore
#check @BookProof.ComparisonCoreEsa.isGraphCore_iff_esa
#check @BookProof.NsEnergySurjectivityObstruction.not_nsEnergy_surjective
```

34.12. Summary: the Comparison Operator, Per Layer

- **Mainstream Hamiltonian** `H_NS = ½Σ_m(π_mF_m + F_mπ_m)`, with the exact quadratic B and the hypotheses `leray`, `liouville`: `N` is the **Leray energy** `N_E = mulOp(1 + ‖u‖²)`, positive and ≥ 1 by `nsEnergyOp_quadForm_ge`; the commutator is the energy flux (`kvn_comm_energy`), which Leray's identity makes the sign-definite viscous dissipation (`fluxPoly_eq`), giving the bound `2νΛ·⟨x,N_Ex⟩` (`commForm_kvn_energy_bound`) and the conclusion `nsKoopman_esa_of_energy_comparison`. The open input is the surjectivity of `N_E + 1` — refuted for this core (`not_nsEnergy_surjective`), which is why the leg is rebuilt in the next two sections.
- **Full nonlinear Hamiltonian**, Eulerian or Lagrangian variables: `N` is the **lifted Friedrichs extension of the auxiliary Weyl sum of squares** `H_sos`, which is bounded below (`nsFullFockHam_quadForm_nonneg`, `lagFullFockHam_quadForm_nonneg`), so its Friedrichs extension exists without a Faris–Lavine detour and lifts to the nested Fock space; the criterion is then run with `c = 0` (`nsFullOuterN_esa`, `lagFullOuterN_esa`). The `c = 0` is a statement about the auxiliary operator, not about `H_NS`.
- **Mainstream leg, rebuilt in two parts.** Affine drifts (Stokes, Oseen, any `Au + c`) are quadratic Hamiltonians and are **unconditionally** ESA, with their dΓ enclosures (`linKoopman_esa`, `nsKoopman_stokes_esa`, `oseenKoopman_esa`, `linKoopman_dGammaOp_esa`); the nonlinear generator is ESA **conditional** on the ESA of the square comparison `N_NS = H² + 1 + ‖u‖²` (`nsKoopman_esa_of_squareComparison_esa`), through the dense-core form of the criterion

(`essentiallySelfAdjointOn_of_farisLavine_dense`, `essentiallySelfAdjointOn_of_square_comparison`).

- **Transformed Lagrangian leg (canonical realization):** the operator $\frac{1}{2}\sum P_i^2 + \nu \sum Q_i^2 + \sum f_i P_i$ with an arbitrary bounded symmetric constraint term C is essentially self-adjoint on the Hermite core (`lagConstr_esa`, closing route Kato–Rellich — the common-eigenvector criterion is structurally unavailable because $[P_i, Q_i] = -i$); its comparison is the completed square $N = h + \frac{1}{2}|f|^2$, with `lagCompN_esa`, `lagCompN_nonneg` and the form bound `lagCompN_add_one_ge`; the lift $d\Gamma(N) + 1$ is ESA on the finite-particle core (`LiftedComparisonEsa.liftedComparison_esa`). The completed square does **not** discharge the `lagKoopman_esa_of_comparison_esa` hypothesis — that is a different operator, `N_L = H_L^2 + E`, on `L²(ℝ^{PIdx K})` — which is why the same wave also landed `NsLagrangianComparisonEsa.lagComparison_esa` / `lagKoopman_esa` (next section), closing that input on its own terms.
- **In neither layer** is N the Hamiltonian, and `N = H²` is invalid for the wrong-domain and the `N + 1`-not-onto reasons above; the general obstruction for `N = H` is the formalized `not_farisLavine_criterion_of_relative_bound`. Under the final-Hamiltonian convention none of these N is ever enclosed: what gets enclosed — creation left, annihilation right — is the one-particle Hamiltonian h once its own essential self-adjointness is in hand.

The independent symbolic-calculation cross-check of the same statements — that the pressure-gradient and forcing terms enter the drift exactly as the tree's residual does, that the pressure and the constraint vanish on the Leray sector, and that the pressure Poisson equation fixes p rather than leaving a gauge symmetry — is recorded in the project's Cadabra modules (`ns_kvn_equation.cdb`, `ns_pressure_poisson.cdb`, `ns_pressure_constraint.cdb`), with the reduction to the library names given in `VERIFY_FARIS_LAVINE_N.md`. The Lean theorems cited above are the evidence that is machine-checked.

35. Second Quantization of an Essentially Self-Adjoint One-Particle Operator

35.1. Orientation and Status

The transfer chapter of this part: the **core transfer principle** that lifts a one-particle obligation to the nested Fock space — every final Hamiltonian of record here is an outer second quantization $d\Gamma(h)$, so a statement about h becomes a statement about every particle-number sector at once. It sits between the two Navier–Stokes operator chapters (one-particle Hamiltonian, comparison operator) and the systems that reuse the transfer (quantum gravity, the Standard Model).

The route: why the Hamiltonian is $d\Gamma(h)$, the core transfer principle, the multilinear core estimate, the assembly on the nested Fock space, the carried input discharged, the packaged form, what this discharges for the programme, the Standard-Model instance (both obligations), the symmetric and antisymmetric sectors at every particle number, the one-particle obligation in general, and non-interacting degrees of freedom as a tensor sum.

The proofs are `BookProof/SecondQuantizationCore`, `BookProof/EsaOneParticle`, the `*DGamma` family (bounded, diagonal, scalar, flow), the tensor-sum chain, and the permutation/sector chapters (`PermSector`, `TensorPerm`, `FockStatistics`) that fix the statistics.

35.2. Why This Chapter Exists: the Hamiltonian Is dΓ(h)

Throughout the programme the final Hamiltonian of record is the outer **second quantization** of an inner **one-particle** Hamiltonian, `H = \\sum_{i,j} h_{ij}\\, C^\\dagger(e_i)\\, A(e_j) = d\\Gamma(h)`, with creation on the left and annihilation on the right — the same shape for QYM, QED, QG and Navier–Stokes. A quartic nonlinearity, an exponential wall or a singular potential enters only the one-particle matrix elements h_{ij}; at the outer level the operator is quadratic in the ladders. So the whole spectral question for the outer Hamiltonian reduces to a question about the one-particle operator h, and it is this reduction that this chapter makes rigorous.

This is not a convention the programme invented: the manuscript states it and the recursive construction that justifies it. Its Navier–Stokes Hamiltonian is written exactly in this enclosure form, $H = \int d^3\vec{x}\, a^\dagger(\vec{x})\, H(\vec{x})\, a(\vec{x})$ with $H(\vec{x}) = \pi^i(u_j u_{i,j} - \nu u_{i,jj}) + (h.c.)$ — one-particle operator inside, creation on the left and annihilation on the right — and the Fock space is introduced recursively: **"Since the second-quantization procedure can be applied recursively … in case the Hamiltonian acting in the Fock-space is not quadratic in the creation/annihilation operators, then we can consider instead a new Fock-space where the base Hilbert space is the original Fock-space. The new Hamiltonian is quadratic in the creation/annihilation operators."** That sentence is the definition of the **nested** (outer) Fock space this chapter works on: the inner one-particle space is itself a Fock space, and the outer Hamiltonian is its second quantization, quadratic in the outer ladders whatever the inner operator is.

Two consequences of that definition are used below and are worth stating here. First, the inner space is where the fields and their **spatial derivatives** live — in the manuscript's Navier–Stokes count, the derivative coordinates $u_{k,j} = \partial_j u_k$ are among the $\mathbb{R}^{15}$ field degrees of freedom, and the QG vielbein carries the same kind of derivative coordinates. Second, a product of fields with a spatial derivative is **not** a differential operator once the derivative has been eliminated: in momentum space it becomes a **convolution**, $\mathcal{F}[u_j\, \partial_j u_i](Q) = \int 2\pi i\, \langle q, m\rangle\, \hat{u}_i(q)\, \hat{u}_j(Q-q)\, dq$ (`BookProof.NsAdvectionConvolution.fourier_advection_convolution`, `fourier_advection_sum`), with the same device for the QG vielbein derivative modes (`ChapterQgFourierElimination`, `ChapterQgFullEliminated`). That is what replaces the constraining of the derivative variables: they are eliminated by the spatial Fourier substitution and reappear as convolutions, not fixed by a gauge symmetry or a BRST charge.

The specific question is the one the Faris–Lavine route keeps returning to: **when is `dΓ(h)` essentially self-adjoint, and on what domain?** The answer proved here is: whenever `h` is essentially self-adjoint on its own domain `D`, `dΓ(h)` is essentially self-adjoint on the finite-particle domain built from `D` **alone** — `D` is only a graph-norm core, not an invariant domain, and no resolvent of `h` on `D` is ever used. The instrument that makes this possible is the **core transfer principle**; the rest is multilinear bookkeeping.

35.3. The Core Transfer Principle

Everything in the project renders essential self-adjointness of a symmetric operator `T : D →ₗ[ℂ] F` by the vanishing of the deficiency spaces of its adjoint. The criteria available elsewhere (Faris–Lavine, Nelson-type commutator bounds) verify this on the domain on which the operator is given. What was missing was the **domain-changing** instrument, and `IsGraphCore D₁ T` is it: D_1 is a subspace of the domain of `T` that is dense there for the **graph norm** $|x| + |Tx|$. Then a deficiency vector is transferred from the domain to any core of it, because the defect $langle Tv, w rangle - z langle v, w rangle$ is bounded by $(1 + |z|)\,|w|$ times the graph distance from `v` to D_1. Cores compose, a domain is a core for its own operator, and the notion transports along a linear isometry — which is how a core inside an incomplete space becomes a core inside its completion.

```
#check @BookProof.GraphCore.IsGraphCore
#check @BookProof.GraphCore.IsGraphCore.refl
#check @BookProof.GraphCore.IsGraphCore.trans
#check @BookProof.GraphCore.deficiencyTrivialAt_of_graphCore
#check @BookProof.GraphCore.essentiallySelfAdjointOn_of_graphCore
#check @BookProof.GraphCore.restrictOp
#check @BookProof.GraphCore.symmetricOn_restrictOp
#check @BookProof.GraphCore.pushOp
#check @BookProof.GraphCore.isGraphCore_pushOp
```

```
#check @BookProof.GraphCore.symmetricOn_pushOp
#check @BookProof.GraphCore.essentiallySelfAdjointOn_of_bounded_dense
```

The last item is a useful unconditional criterion in its own right: a symmetric operator that is **bounded** on a dense domain has trivial deficiency spaces at
pmi, hence is essentially self-adjoint there. It is what discharges the sectorwise input in the bounded cases below.

35.4. The Multilinear Core Estimate

The core transfer is a one-variable statement; the sectors of a Fock space are tensor powers, so it must be lifted to n variables. `IPSpace.pow` is the n-fold algebraic tensor power of an inner product space, `inclPow` the isometric inclusion $D_2^{otimesn}$
$toH^{otimesn}$, `derPow` the sector derivation d
$Gamma(A)^{(n)} =$
$sum_j 1$
$otimes$
$cdots$
$otimes A$
$otimes$
$cdots$
$otimes 1$ defined by the Leibniz recursion, and `corePow` the tensor power $D^{otimesn}$ of a one-particle core. The telescoping estimate `graphPow_tmul_mem_closure` on an elementary tensor a ⊗ b — replacing a costs
$|a - a'$
$|$
$|b$
$| +$
$|Aa - Aa'$
$|$
$|b$
$|$ and replacing b costs the graph distance of b — is the whole of the analysis; span induction then gives `exists_core_approx` and `isGraphCore_sectorCore`: **if D is a core for A, then D^{⊗n} is a core for the sector derivation**. Symmetry transfers too.

```
#check @BookProof.TensorCore.IPSpace.pow
#check @BookProof.TensorCore.derPow
#check @BookProof.TensorCore.corePow
#check @BookProof.TensorCore.graphPow_tmul_mem_closure
#check @BookProof.TensorCore.exists_core_approx
#check @BookProof.TensorCore.isGraphCore_sectorCore
#check @BookProof.TensorCore.derPow_symm
#check @BookProof.TensorCore.symmetricOn_sectorOp
```

35.5. The Assembly on the Nested Fock Space

The outer space is the
ell^2-direct sum of the **completed** sectors
$mathcal F_n =$ completion of $H^{otimesn}$, the finite-particle domain over the core is the direct sum of the sector cores, and `dGammaCoreOp` is `dΓ(A)` on it. Gluing is the existing `BookProof.DirectSumEsa.dsOp_essentiallySelfAdjointOn`, so the escaped-to-the-completion question is already settled; the new content is the descent to the unsigned core, sector by sector. The hypothesis that is **carried**, not proved, in this general statement is sectorwise essential self-adjointness of the sector derivation on the tensor powers of the domain of the closure — the classical statement for a self-adjoint one-particle operator, which the next section discharges.

```
#check @BookProof.SecondQuantizationCore.fockSector
#check @BookProof.SecondQuantizationCore.fockSectorDom
#check @BookProof.SecondQuantizationCore.fockSectorCore
#check @BookProof.SecondQuantizationCore.dGammaCoreOp
```

```
#check @BookProof.SecondQuantizationCore.essentiallySelfAdjointOn_fockSectorCore
#check @BookProof.SecondQuantizationCore.dGamma_essentiallySelfAdjointOn_fockCore
#check @BookProof.SecondQuantizationCore.symmetricOn_dGammaCoreOp
#check @BookProof.SecondQuantizationCore.exists_ne_zero_mem_dGammaCoreDomain
```

35.6. The Carried Input, Discharged

The sectorwise input is not an axiom and not an obstruction — it is a statement about the one-particle operator, and the chapters below prove it outright in a sequence of increasingly general cases. The list is the useful summary of this chapter:

- **bounded symmetric h** — the tensor operator bound (`exists_orthonormal_repr` writes any tensor as $\sum x_i \otimes e_i$ with orthonormal right legs, `norm_sum_tmul_orthonormal` is Pythagoras for such sums, and `norm_map_le` gives $\|(S \otimes T)u\| \le \|S\|\|T\|\|u\|$ for arbitrary bounded — in particular **indefinite** — factors) run through the Leibniz recursion gives $\|d\Gamma(A)^{(n)}\| \le n\|A\|$. The conclusion `dGamma_bounded_essentiallySelfAdjointOn_fockCore` carries **no** hypothesis beyond a dense `D`; `dGamma_neg_id_essentiallySelfAdjointOn_fockCore` (minus the number operator) shows that no positivity is being smuggled in.
- **scalar h** ($h = c\cdot\mathrm{id}$) — the sector derivation is the bounded scalar $n\cdot c$ on a dense domain, so the criterion applies and `dGamma_scalar_essentiallySelfAdjointOn_fockCore` is unconditional for **any** dense `D`.
- **unbounded h with pure point spectrum** — `essentiallySelfAdjointOn_of_dense_eigenvectors` (a total family of eigenvectors with real eigenvalues kills both deficiency spaces), `derPow_eigTensor` and `dense_eigSpan` lift it to every sector, giving `dGamma_diagonal_essentiallySelfAdjointOn_fockCore` with no completeness and no self-adjointness of a reference operator assumed; `diagOp` builds such operators and `not_bounded_diagOp` shows they are genuinely unbounded.
- **every self-adjoint h** — `deficiencyTrivialAt_of_orbits` is Nelson's invariant-domain criterion in orbit form, `OneParticleFlow` bundles the unitary group with invariant domain (built from Stone's theorem, nothing supplied from outside), `hasDerivAt_tpow` is the Leibniz rule for the tensor flow, and `dGamma_selfAdjoint_essentiallySelfAdjointOn_fockCore` concludes: for any self-adjoint `A` and any graph-norm core `D`, $d\Gamma(A)$ is essentially self-adjoint on the finite-particle domain over `D`. `dGamma_position_essentiallySelfAdjointOn_fockCore` is the unbounded instance (multiplication by k on $\ell^2(\mathbb{Z})$).
- **h only essentially self-adjoint on D** — the final weakening, and the one the programme actually needs. `closureSelfAdjoint` packages the graph closure of such an h as a genuine self-adjoint operator, `isGraphCore_clDom` shows `D` is a core of it, the self-adjoint theorem is applied to the closure, and `esa_graph_le` with the tensor lemmas `exists_pow_of_mem_corePow`, `sectorCore_graph_le`, `fockSectorCore_graph_le` brings the statement back to h **on D itself**. `dGamma_essentiallySelfAdjointOn_of_esa` is the main theorem; no domain of a closure appears in its statement. That the hypothesis is strictly weaker than self-adjointness is shown by `essentiallySelfAdjointOn_restrict` / `not_isSelfAdjointOn_restrict`, and by the concrete witness `positionCore` on $\ell^2(\mathbb{Z})$: symmetric, essentially self-adjoint, and provably **not** self-adjoint on the finitely supported vectors, with `dGamma_positionCore_essentiallySelfAdjoint` applying the theorem to it.

```
#check @BookProof.TensorOpBound.exists_orthonormal_repr
#check @BookProof.TensorOpBound.norm_sum_tmul_orthonormal
#check @BookProof.TensorOpBound.norm_map_le
#check @BookProof.BoundedDGamma.norm_derPow_le
#check @BookProof.BoundedDGamma.dGamma_bounded_essentiallySelfAdjointOn_fockCore
#check @BookProof.BoundedDGamma.dGamma_neg_id_essentiallySelfAdjointOn_fockCore
#check @BookProof.ScalarDGamma.dGamma_scalar_essentiallySelfAdjointOn_fockCore
#check @BookProof.DiagonalDGamma.essentiallySelfAdjointOn_of_dense_eigenvectors
#check @BookProof.DiagonalDGamma.dGamma_diagonal_essentiallySelfAdjointOn_fockCore
#check @BookProof.DiagonalDGamma.not_bounded_diagOp
#check @BookProof.FlowDGamma.dGamma_selfAdjoint_essentiallySelfAdjointOn_fockCore
#check @BookProof.FlowDGamma.dGamma_position_essentiallySelfAdjointOn_fockCore
```

```
#check @BookProof.EsaOneParticle.closureSelfAdjoint
#check @BookProof.EsaOneParticle.isGraphCore_clDom
#check @BookProof.EsaOneParticle.dGamma_essentiallySelfAdjointOn_of_esa
#check @BookProof.EsaOneParticle.essentiallySelfAdjointOn_restrict
#check @BookProof.EsaOneParticle.not_isSelfAdjointOn_restrict
#check @BookProof.EsaOneParticle.positionCore_essentiallySelfAdjoint
#check @BookProof.EsaOneParticle.positionCore_not_isSelfAdjointOn
#check @BookProof.EsaOneParticle.dGamma_positionCore_essentiallySelfAdjoint
```

35.7. The Packaged Form

For use in the route chapters, `ESAPair` bundles the three pieces of data — a symmetric one-particle operator on the domain of its closure, a graph-norm core `D`, and the sectorwise input — and `ESAPair.dGamma_essentiallySelfAdjoint` is the main theorem over the package. `ESAPair.norm_sub_smul_sq` records the only quantitative ingredient, $|Ax - d$, $, i$, $, x$, $|^2 = |Ax$, $|^2 + d^2 |x$, $|^2$, valid with no spectral lower bound, so no quadratic-form domain and no Friedrichs extension is required. `ESAPair.ofBounded` produces packages for all bounded symmetric one-particle operators — with the sectorwise input proved, not assumed — and `esaPairOfSelfAdjoint` does the same for self-adjoint ones.

```
#check @BookProof.EsaPair.ESAPair.dGamma_essentiallySelfAdjoint
#check @BookProof.EsaPair.ESAPair.norm_sub_smul_sq
#check @BookProof.EsaPair.dGamma_essentiallySelfAdjoint_ofBounded
#check @BookProof.FlowDGamma.esaPairOfSelfAdjoint
```

35.8. What This Discharges for the Programme

The plan's §D6b obligation (ii) was: **the lifted Hamiltonian $d\Gamma(H_1)$ must be proved essentially self-adjoint on the lifted core, and ESA does not lift for free**. The results of this chapter are exactly the instrument that discharges it: once H_1 is essentially self-adjoint on the one-particle core C_0, the core transfer principle descends the sectorwise statement to $C_0^{\otimes n}$, the multilinear estimate says $C_0^{\otimes n}$ is a core for the sector derivation, and the ℓ^2-sum assembly glues the sectors — giving essential self-adjointness of $d\Gamma(H_1)$ on the finite-particle domain built from C_0 alone. Nothing about the **inner** model (quartic, exponential, singular) enters: the inner nonlinearity only changes the one-particle matrix elements, and the hypothesis is a property of H_1, not of the outer ladders.

The carried input in the fully general case is the sectorwise statement for a self-adjoint one-particle operator, and even that is now proved for every self-adjoint H_1 (`dGamma_selfAdjoint_essentiallySelfAdjointOn_fockCore`) — with the bounded, scalar and diagonal cases unconditional as well. One scope caveat survives, and it is a caveat about **this** chapter rather than about the theory: the tensor powers used **here** are the full powers $H^{\otimes n}$, not the symmetric (bosonic) or antisymmetric (fermionic) subspaces. The statements above are about the derivation on those powers. The passage to the symmetrized sectors is not a further hypothesis — it is the subject of the next section, and it is now proved.

35.9. The Standard-Model Instance: Both Obligations Discharged

The reduction this chapter makes rigorous has, as of the closure wave, a complete worked instance. For the Standard-Model one-particle operator h of the previous chapter, essential self-adjointness on the Gauss–polynomial core of $L^2(\mathbb{R}^{163})$ is now a theorem with no hypothesis beyond the parameters: `sm_h_esa`, obtained by combining the two Faris–Lavine inequalities of `BookProof/Sm/ChapterSmFarisLavine.lean` with the independent Kato-type theorem of `BookProof/Sm/ChapterSmComparisonEsa.lean` and its transfer between the two cores (the chain is walked through in the Standard-Model chapter of this manual). Fed into $dGamma_essentiallySelfAdjointOn_of_esa$ — the main theorem above — it gives essential self-

adjointness of the enclosure $H = d\Gamma(h) = \sum_{i,j} h_{ij} C^\dagger(e_i) A(e_j)$\$, creation on the left and annihilation on the right, on the finite-particle core over the Gauss–polynomial core. In the plan's vocabulary, obligations (i) and (ii) of §D6b are both discharged for the Standard-Model row: (i) is the one-particle statement just named, (ii) is this chapter's general lift applied to it. What remains owed there is not an analytic hypothesis but an assembly: the sector-wise tensor/graph-core combination of the bosonic and fermionic halves of h\$ before enclosure. The instrument is not sector-specific — the Kato theorem behind (i) holds for any smooth, respectively polynomial, potential bounded below by one. The lift is now also a theorem in each instance: `BookProof/Sm/ChapterSmFockEsa.lean` states `sm_dGamma_esa` (this chapter's main theorem applied to `sm_h_esa`) and `smFockHam_esa` (the Standard-Model outer operator on $\bigoplus_n L^2(\mathbb{R}^{163n})$), and `BookProof/Qym/ChapterYangMillsNonAbelianEsa.lean` gives the non-abelian Yang–Mills instance of the same shape, `ym_h_esa` (one-particle, arbitrary real structure constants) and `ym_dGamma_esa` (enclosed). The fermionic assembly named above is still open.

```
#check @BookProof.SmFarisLavine.sm_commForm_le
#check @BookProof.SmFarisLavine.sm_norm_le_shift
#check @BookProof.SmFarisLavine.sm_h_esa_of_comparison_esa
#check @BookProof.DegKatoEsa.ccHamS_esa
#check @BookProof.HermiteGraphApprox.hamCoreS_esa
#check @BookProof.SmComparisonEsa.smFlN_eq_hamCoreS
#check @BookProof.SmComparisonEsa.smFlN_esa
#check @BookProof.SmComparisonEsa.sm_h_esa
```

35.10. The Symmetric and Antisymmetric Sectors, at Every Particle Number

The enclosure $H = d\Gamma(h)$ is a statement about the **whole** nested Fock space, but the physical content lives in the symmetric (bosonic) and antisymmetric (fermionic) subspaces of each sector $H^{\otimes n}$. That passage was the honest boundary of the previous section. It has been executed, in four steps that are worth reading as one argument, because each step is an abstract instrument reusable far beyond this book.

Step one — reduction. `BookProof/Toolbox/ChapterReducingSubspaceEsa.lean` (namespace `BookProof.ReducedEsa`): if P is an idempotent symmetric **reducing** projection that preserves the domain and commutes with T, then essential self-adjointness of T descends to the sector $D \cap PF$. The proof is the deficiency argument in miniature — for w in the range, $\langle Tv, w\rangle = \langle P(Tv), w\rangle = \langle T(Pv), w\rangle$ — and needs no completeness, no closedness and no spectral theory (`essentiallySelfAdjointOn_red`). Applied to a self-inverse isometry U preserving D and commuting with T, the two canonical projections $(1 \pm U)/2$ (`symProj`, `asymProj`) are such projections, giving `essentiallySelfAdjointOn_symSector` / `…_asymSector` on the eigenspaces $Ux = \pm x$.

Step two — finite groups. `BookProof/Toolbox/ChapterGroupAverageEsa.lean` (namespace `BookProof.GroupAverage`) generalizes the single involution to any finite group of symmetries: `UnitaryRep G F` is an action by inner-preserving linear maps, `avgProj = |G|^{-1} \sum_g \rho(g)` is the Haar average, its range is the joint fixed space (`mem_range_avgProj_iff`), it is again a reducing projection (`isReducingProjection_avgProj`, by the reindexing $\sum_g \sum_h \rho(gh) = |G| \sum_k \rho(k)$), and hence **`essentiallySelfAdjointOn_invariantSector`**: an operator commuting with the action is essentially self-adjoint on the invariant sector as soon as it is on its domain. This is the instrument the higher sectors need — the symmetric group of n factors, twisted by the sign character for the fermionic case.

Step three — the permutation action. `BookProof/Toolbox/ChapterTensorPermutation.lean` (namespace `BookProof.TensorPerm`) builds it out of two elementary isometries only: `swapFirst`, the exchange of the first two factors of $E^{\otimes(n+2)}$ (associator ∘ commutor ∘ associator), and `liftTail u`, applied to the last n factors; `permOp n σ` follows `Equiv.Perm.decomposeFin`. Since every operator is a composite of linear isometry equivalences, each is one, and `permOp_purePow` is the familiar formula $U_\sigma(x_0 \otimes \cdots \otimes x_{n-1}) = x_{\sigma 0} \otimes \cdots \otimes x_{\sigma(n-1)}$, from which the group law `permOp_mul` follows. `permRep` and `signRep` package the action and its sign twist as `UnitaryReps`.

Step four — the two sectors. `BookProof/Toolbox/ChapterPermutationSectorEsa.lean` (namespace `BookProof.PermSector`) checks the three compatibilities of a pair of operators — with the inclusion of the domain, with the sector derivation $d\Gamma(A)^{(n)}$, and with the core power $D^{\otimes n}$ — for the two generators (`Good`, `good_swapFirst`, `good_liftTail`) and passes them along the recursion (`good_permOp`). Through

`essentiallySelfAdjointOn_invariantSector` come the four headline statements: **`essentiallySelfAdjointOn_bosonic`** and **`essentiallySelfAdjointOn_fermionic`** on $D_2^{\otimes n}$, and **`essentiallySelfAdjointOn_bosonic_core`** / **`essentiallySelfAdjointOn_fermionic_core`** on the symmetrized and antisymmetrized one-particle core $D^{\otimes n}$, for any core D of A. `mem_bosonicSector_iff` / `mem_fermionicSector_iff` identify the two sectors as the symmetric and the antisymmetric tensors; `exists_ne_zero_bosonic` (the n-th power $a \otimes \cdots \otimes a$ of a core vector) and `exists_ne_zero_fermionic` (the Slater determinant, nonzero by `inner_purePow` and orthogonality) are the non-vacuity witnesses.

Two companion chapters make the picture complete. The two-particle case in isolation, from which the general construction checks itself (`permOp_two_eq_swapH`), is `BookProof/Toolbox/ChapterTwoParticleSectorEsa.lean` (namespace `BookProof.TwoParticleSector`), with its reducing projection the swap $(1 \pm \mathrm{swap})/2$. And `BookProof/Toolbox/ChapterFockStatisticsEsa.lean` (namespace `BookProof.FockStatistics`) discharges the hypothesis the permutation chapter **carries** — that the sector derivation is essentially self-adjoint on $D_2^{\otimes n}$ — from essential self-adjointness of the one-particle operator alone (`essentiallySelfAdjointOn_bosonic_of_esa`, ...), assembling the algebraic-tensor Fock statements `bosonicFock_esa` and `fermionicFock_esa`. `BookProof/Toolbox/ChapterFockStatisticsCompletion.lean` then carries all of it across to the **completed** sectors and takes the direct sum over particle numbers: **`hbosonicFock_esa`** — $d\Gamma(A)$ is essentially self-adjoint on the bosonic Fock space, the Hilbert space direct sum over all particle numbers of the symmetric sectors — and **`hfermionicFock_esa`**, its fermionic twin, under no hypotheses beyond essential self-adjointness of A on a dense domain.

```
#check @BookProof.ReducedEsa.IsReducingProjection
#check @BookProof.ReducedEsa.essentiallySelfAdjointOn_red
#check @BookProof.ReducedEsa.symmetricOn_redOp
#check @BookProof.ReducedEsa.symProj
#check @BookProof.ReducedEsa.asymProj
#check @BookProof.ReducedEsa.essentiallySelfAdjointOn_symSector
#check @BookProof.ReducedEsa.essentiallySelfAdjointOn_asymSector
#check @BookProof.GroupAverage.UnitaryRep
#check @BookProof.GroupAverage.UnitaryRep.avgProj
#check @BookProof.GroupAverage.UnitaryRep.mem_range_avgProj_iff
#check @BookProof.GroupAverage.UnitaryRep.isReducingProjection_avgProj
#check @BookProof.GroupAverage.UnitaryRep.essentiallySelfAdjointOn_invariantSector
#check @BookProof.GroupAverage.repOfInvolution
#check @BookProof.GroupAverage.avgProj_repOfInvolution
#check @BookProof.TensorPerm.permOp
#check @BookProof.TensorPerm.permOp_purePow
#check @BookProof.TensorPerm.permOp_mul
#check @BookProof.TensorPerm.permRep
#check @BookProof.TensorPerm.signRep
#check @BookProof.PermSector.good_permOp
#check @BookProof.PermSector.essentiallySelfAdjointOn_bosonic
#check @BookProof.PermSector.essentiallySelfAdjointOn_fermionic
#check @BookProof.PermSector.essentiallySelfAdjointOn_bosonic_core
#check @BookProof.PermSector.essentiallySelfAdjointOn_fermionic_core
#check @BookProof.PermSector.mem_bosonicSector_iff
#check @BookProof.PermSector.mem_fermionicSector_iff
#check @BookProof.PermSector.permOp_two_eq_swapH
#check @BookProof.TwoParticleSector.essentiallySelfAdjointOn_bosonic
#check @BookProof.TwoParticleSector.essentiallySelfAdjointOn_fermionic
#check @BookProof.FockStatistics.essentiallySelfAdjointOn_bosonic_of_esa
#check @BookProof.FockStatistics.bosonicFock_esa
#check @BookProof.FockStatistics.fermionicFock_esa
#check @BookProof.FockStatistics.hbosonicFock_esa
#check @BookProof.FockStatistics.hfermionicFock_esa
#check @BookProof.GroupAverage.UnitaryRep.completionRep
```

Honest boundary of the symmetrization. The statements are about the two sectors of $H^{\otimes n}$ for each **fixed** particle number n, and — through the completed-Fock chapter — about the direct sum that assembles them. Nothing is claimed about a continuum of particle numbers beyond that sum, and essential self-adjointness of $d\Gamma(A)^{(n)}$ on $D_2^{\otimes n}$ is carried as a hypothesis through the reduction, discharged separately by the Fock-statistics

chapter as described above. The passage is **statistics**, not dynamics: it says which vectors are physical, and that the physical subspace inherits the essential self-adjointness — it says nothing about the spectrum.

35.11. The One-Particle Obligation, Discharged in General

There are two obligations behind every Faris–Lavine route chapter, and it is worth naming them separately because they are easy to confuse. Obligation (ii) was that the **lifted** operator $d\Gamma(N)$ must be essentially self-adjoint on the lifted core — that is what the core-transfer chain of the earlier sections discharges. Obligation (i) is the **inner** one: the comparison operator N of record must itself be essentially self-adjoint on the core the proof actually chooses. Two short chapters discharge it in general.

`BookProof/Toolbox/ChapterSelfAdjointCoreEsa.lean` (namespace `BookProof.SelfAdjointCoreEsa`): if T has a self-adjoint extension, then it was already essentially self-adjoint on its own domain. At a non-real z the deficiency equation makes w a domain vector with $Aw = z \bullet w$, and symmetry then forces $(\bar{z} - z)\langle w, w\rangle = 0$, so $w = 0$ (`deficiencyTrivialAt_dom_of_isSelfAdjointExtension`); composing with the graph transfer gives `essentiallySelfAdjointOn_of_graphCore_selfAdjoint`.

`BookProof/Toolbox/ChapterComparisonCoreEsa.lean` (namespace `BookProof.ComparisonCoreEsa`) restates the same fact in the vocabulary the route chapters use: `comparison_isSelfAdjointExtension` and `comparison_essentiallySelfAdjointOn_dom` (a `Comparison` is self-adjoint, hence essentially self-adjoint, on its own domain), `graphCore_of_isGraphCore` (the two graph-core notions agree), **`esa_of_isGraphCore`** — obligation (i) — and **`isGraphCore_iff_esa`**, the equivalence obtained with the already-proved converse `ScalaronFiberFL.isGraphCore_of_esa`: **for a Faris–Lavine comparison operator, "graph core" and "core of essential self-adjointness" are the same notion.** Wherever a route chapter exhibits a graph core for its comparison operator — `QgOuterFockFullFL.harmFried_isGraphCore`, the Gauss-polynomial core of $-\Delta + \|x\|^2/4$ in every dimension, and `ScalaronOuterFockFL.secN_isGraphCore`, the compactly-supported smooth core of the scalaron-wall comparison — obligation (i) therefore needs no further work.

```
#check @BookProof.SelfAdjointCoreEsa.deficiencyTrivialAt_dom_of_isSelfAdjointExtensic
#check @BookProof.SelfAdjointCoreEsa.essentiallySelfAdjointOn_dom_of_isSelfAdjointExt
#check @BookProof.SelfAdjointCoreEsa.essentiallySelfAdjointOn_of_graphCore_selfAdjoir
#check @BookProof.ComparisonCoreEsa.comparison_isSelfAdjointExtension
#check @BookProof.ComparisonCoreEsa.comparison_essentiallySelfAdjointOn_dom
#check @BookProof.ComparisonCoreEsa.graphCore_of_isGraphCore
#check @BookProof.ComparisonCoreEsa.esa_of_isGraphCore
#check @BookProof.ComparisonCoreEsa.isGraphCore_iff_esa
```

35.12. Two Stability Moves: Real Shifts and Unitary Conjugation

The enclosure of $d\Gamma(h)$ consumes essential self-adjointness of h wherever it finds it — and two cheap ways to **produce** it where it is not initially stated joined the toolbox in the 2026-10-15c wave. Both are one-particle moves; the enclosure is downstream of both.

A real shift does not change the answer (`BookProof/Toolbox/ChapterEsaRealShift.lean`). The definition in use is essential self-adjointness through the two deficiency spaces at $\pm i$ only, so the statement "h ESA $\Rightarrow$ $h + c$ ESA" for real c is not formal: shifting moves the deficiency question from $\pm i$ to $\pm i - c$. The chapter supplies the missing general fact — a symmetric operator that is essentially self-adjoint has **trivial deficiency spaces at every non-real** z (`deficiencyTrivialAt_of_essentiallySelfAdjointOn`, built on the non-real lower bound $|\operatorname{Im} z|\,\|v\| \le \|Tv - \bar{z}v\|$ and the disk-stability lemma `deficiencyTrivialAt_of_near`) — and concludes `essentiallySelfAdjointOn_add_const`. This matters twice under the final-Hamiltonian convention: the shift of h used to make it positive (the move that makes the outer vacuum the exact ground state) is legitimate only because ESA survives it; and the + 1 of the NS lifted comparison of the one-particle chapter is the same move on $d\Gamma(N)$ — there an explicitly labeled **lifted comparison**, not a Hamiltonian of record.

Unitary conjugation transports ESA (`BookProof/Ode/ChapterOdeChartTransfer.lean`): if W is a unitary carrying a dense domain D onto D', then T essentially self-adjoint on D implies WTW^{-1} essentially self-adjoint on D' (`conj_essentiallySelfAdjointOn`, with the companion `conj_deficiencyTrivialAt` and `conjDomain_dense`). The instrument is generic — its first instance was the ODE chart of the singularity chapter, its structural sibling the half-density transfer of Diffeomorphisms and gravity — and its pedagogical content is that a change of variables is **never** bought for free: the domain moves with the operator, and the theorem is about the pair. A conjugation proof that forgets to name D' has proved nothing.

```
#check @BookProof.EsaRealShift.deficiencyTrivialAt_of_essentiallySelfAdjointOn
#check @BookProof.EsaRealShift.deficiencyTrivialAt_of_near
#check @BookProof.EsaRealShift.essentiallySelfAdjointOn_add_const
#check @BookProof.OdeChartTransfer.conj_essentiallySelfAdjointOn
#check @BookProof.OdeChartTransfer.conj_deficiencyTrivialAt
#check @BookProof.OdeChartTransfer.conjDomain_dense
```

35.13. Non-Interacting Degrees of Freedom: the Tensor Sum

The last elementary situation the enclosure needs, and did not have, is the tensor sum of two **different** operators — $A \otimes 1 + 1 \otimes B$, the Hamiltonian of two non-interacting degrees of freedom with potential $V(u,v) = V_1(u) + V_2(v)$. `BookProof/Toolbox/ChapterTensorSumEsa.lean` (namespace `BookProof.TensorSumEsa`) puts it on the completed tensor product $H \hat{\otimes} K$ with domain $D_A \otimes D_B$: symmetry (`symmetricOn_cpairOp`), density of that domain (`dense_cpairDom`), the product flow $Ut \otimes Vt$ (`pflow`, `hasDerivAt_pflow`) which is isometric, preserves the domain and obeys the Leibniz rule — hence solves the Schrödinger equation of the tensor sum — and Nelson's invariant-domain criterion applied to those orbits gives **`essentiallySelfAdjointOn_cpairDom_flow`**, with the self-adjoint case `essentiallySelfAdjointOn_cpairDom_selfAdjoint` (Stone's theorem supplying the flows: no boundedness, no positivity, no relative bound and no assumption on either spectrum) and the weakest headline **`essentiallySelfAdjointOn_cpairDom_esa`**, where both hypotheses are only **essential** self-adjointness. The two-factor core estimate `isGraphCore_pairCore` says a core of A tensor a core of B is a core of the sum.

`BookProof/Toolbox/ChapterTensorSumChain.lean` (namespace `BookProof.TensorSumChain`) iterates it: `EsaOp` bundles the data, `pair` is the two-factor step and `chain` folds it along a list, so **`chain_esa`** covers $A_1 \otimes 1 \otimes \cdots + \cdots + 1 \otimes \cdots \otimes A_n$ for an arbitrary finite family of symmetric, essentially self-adjoint operators — the Hamiltonian of n non-interacting degrees of freedom — on the algebraic tensor product of the n domains, with the unitary group `chain_stone_flow`.

This is the theorem behind every **decoupled** realization in the programme: when a final Hamiltonian of record is written as a sum of uncoupled one-particle pieces — the TEGR shear fibres $\oplus$ the scalaron mass, say, in the R² vielbein model's small-field limit — its essential self-adjointness follows from `chain_esa` rather than from a bespoke argument per sector.

```
#check @BookProof.TensorSumEsa.symmetricOn_cpairOp
#check @BookProof.TensorSumEsa.dense_cpairDom
#check @BookProof.TensorSumEsa.essentiallySelfAdjointOn_cpairDom_flow
#check @BookProof.TensorSumEsa.essentiallySelfAdjointOn_cpairDom_selfAdjoint
#check @BookProof.TensorSumEsa.essentiallySelfAdjointOn_cpairDom_esa
#check @BookProof.TensorSumEsa.isGraphCore_pairCore
#check @BookProof.TensorSumEsa.essentiallySelfAdjointOn_cpairCore
#check @BookProof.TensorSumEsa.tensorSum_stone_flow
#check @BookProof.TensorSumChain.EsaOp
#check @BookProof.TensorSumChain.chain
#check @BookProof.TensorSumChain.chain_esa
#check @BookProof.TensorSumChain.chain_symmetric
#check @BookProof.TensorSumChain.chain_dense
#check @BookProof.TensorSumChain.chain_stone_flow
```

Honest boundary. The gluing is over a tensor product of finitely many factors (by iteration); it is not the direct-integral step, and it says nothing about a potential that is **not** a sum of a function of the first variable and a function of the second. Interaction terms — the scalaron–vielbein coupling, the B^2 pair terms, the advection convolution — are exactly what a tensor sum does **not** cover; they live in the one-particle operator of a single factor and enter the enclosure through the matrix elements h_{ij}, as the doctrine at the head of this chapter says.

36. Eliminating the Derivative Variables in Quantum Gravity

36.1. Orientation and Status

The derivative-variables thread's service for gravity: the same elimination that the Fourier-elimination chapter states as strategy is carried out here for the vielbein sector of the R^2 Starobinsky model — deleting twenty-seven derivative components per momentum while reproducing the vielbein self-interaction and the scalaron coupling verbatim. It is the structural companion of the Starobinsky scalaron chapter that follows.

The route: the problem (twenty-seven derivative components per momentum), step one (the torsion is a form in the physical modes alone), step two (the whole Hamiltonian on the nine components), and what the elimination doctrines — and what it leaves open.

The proofs are `BookProof/QgFourierElim` and `BookProof/QgFullEliminated`, with the comparison/convolution/tensor-sum machinery they reuse from the Navier–Stokes thread.

36.2. The Problem: Twenty-Seven Derivative Components Per Momentum

The quantum-gravity model of record is the R^2 (Starobinsky) Hamiltonian in **vielbein** form, with the **full exponential** Einstein-frame scalaron wall and the interaction (coupling) terms. Its field-space model carries 36 components per momentum: the nine vielbein components $e^i_\nu(k)$ **and** the twenty-seven independent derivative variables $D^i_{\mu\nu}(k)$, the latter present because the derivative gauge forms $G^i_{\mu\nu} = D^i_{\mu\nu} - ik_\mu e^i_\nu$ are imposed inside the quadratic form.

The Fourier-elimination chapter explained for Navier–Stokes why the current plan of record **eliminates** such variables instead of fixing them by a BRST gauge symmetry: the substitution is used when the operator is **defined**, so the auxiliary variables never have to exist. This chapter is the quantum-gravity half of that strategy, and it is done in two steps — first the derivative components alone (which is where the vielbein self-interaction lives), then the whole Hamiltonian, rebuilt on the nine physical components with nothing else left over.

36.3. Step One: the Torsion Is a Form in the Physical Modes Alone

Read the gauge condition as a substitution, $\sigma(D^i_{\mu\nu}(k)) = k_\mu e^i_\nu(k)$, and the torsion becomes a linear form in the physical vielbein modes `CMode` only — the extended space `EMode` appears neither in the definition nor in the domain of the operator built from it:

- `elimD` is the elimination and `elimTorsion` the torsion $T = D - D^{\mathsf{T}}$ **after** it; both have type $\mathrm{CMode} \to \mathbb{C}$;
- `elimTorsion_eq_torsionCoef`: the eliminated torsion is exactly the physical Fourier torsion $k_\mu e^i_\nu - k_\nu e^i_\mu$ at momentum k (and zero on the other momenta), so the substitution is **lossless**; `elimTorsion_eq_gaugeReduce` identifies it with the gauge-reduced extended form of the BRST chapter, which is how that chapter's identities are restated as an elimination rather than as a symmetry;
- `elimTorsion_antisymm`, `elimTorsion_diag`, `elimTorsion_conj`: the structural facts — the torsion is antisymmetric in the derivative indices, vanishes on the diagonal, and is real-coefficient;
- `elimGram_eq_contTorsionGram`: the Gram matrix of **all** eliminated torsion forms is exactly the vielbein self-interaction `contTorsionGram` of the continuum model, with no hypothesis relating the two momenta (off the momentum diagonal both sides vanish). This is the statement that makes the elimination admissible: the interaction of the model survives it verbatim;
- consequently `qgElimModes` is the continuum mode data (`qgElimModes_A` states that its self-interaction matrix **is** the eliminated Gram matrix), and `qgElim_esa` / `starobinsky_qgElim_esa` give essential self-adjointness of the model it defines on the outer Fock space, with the full exponential Einstein-frame wall and **arbitrary coupling constant**. No BRST charge, no ghost sector and no restriction-to-a-subset argument is used: the physical torsion is the only torsion from the start.

```
#check @BookProof.QgFourierElim.elimD
#check @BookProof.QgFourierElim.elimTorsion
```

```
#check @BookProof.QgFourierElim.elimTorsion_eq_torsionCoef
#check @BookProof.QgFourierElim.elimTorsion_eq_gaugeReduce
#check @BookProof.QgFourierElim.elimTorsion_antisymm
#check @BookProof.QgFourierElim.elimTorsion_conj
#check @BookProof.QgFourierElim.elimGram_eq_contTorsionGram
#check @BookProof.QgFourierElim.qgElimModes
#check @BookProof.QgFourierElim.qgElimModes_A
#check @BookProof.QgFourierElim.qgElim_esa
#check @BookProof.QgFourierElim.starobinsky_qgElim_esa
```

In the **full** vielbein–scalaron model, whose modes carry the nine vielbein components together with the twenty-seven derivative components, `elimConfig` parameterizes the derivative-gauge constraint surface **exactly**: an eliminated configuration satisfies all twenty-seven constraints identically (`formValue_dGauge_elimConfig`), and every configuration satisfying them is eliminated (`eq_elimConfig_of_gauge_fixed`). On that surface the torsion form is the exact Fourier torsion (`formValue_torsion_elimConfig`) and the transverse gauge form is unchanged (`formValue_gauge3d_elimConfig`). The elimination is not vacuous — `elimTorsion_ne_zero` exhibits a torsion that the elimination genuinely produces.

```
#check @BookProof.QgFourierElim.elimConfig
#check @BookProof.QgFourierElim.formValue_dGauge_elimConfig
#check @BookProof.QgFourierElim.eq_elimConfig_of_gauge_fixed
#check @BookProof.QgFourierElim.formValue_torsion_elimConfig
#check @BookProof.QgFourierElim.formValue_gauge3d_elimConfig
#check @BookProof.QgFourierElim.elimTorsion_ne_zero
```

36.4. Step Two: the Whole Hamiltonian on the Nine Components

Step one still leaves the derivative-gauge **form** in the quadratic form of the full model. Step two rebuilds the entire Hamiltonian — vielbein self-interaction (torsion), derivative gauge fixing, 3D transverse gauge fixing, the scalaron in position representation with the full exponential Einstein-frame wall, and the coupling to the trace of the vielbein — on the nine eliminated components `EComp` alone, i.e. on `EGMode = Mom × EComp`, and proves its essential self-adjointness again, by the same Faris–Lavine route, on the **smaller** outer Fock space:

- `elimCoef` is the coefficient vector of each form **in the nine vielbein components only**, and `eFormValue` evaluates it; `eFormValue_torsion`, `eFormValue_dGauge`, `eFormValue_gauge3d` are the three families of forms, and `eFormValue_eq_formValue_elimConfig` is the agreement with the eliminated configurations of step one — the two steps describe the same forms;
- the coefficient bound is uniform in the momentum, `norm_elimCoef_le` — needed because the elimination trades the extended index for momentum-dependent coefficients;
- `eGram` is the self-interaction Gram matrix on the physical modes, with `eGram_herm`, `eGram_off` (the neighbourhood sparsity) and the Schur-type bound `norm_eGram_le`; then `eGram_eq_torsion_add_gauge3d` decomposes it and `eGram_quadForm` / `gGram_quadForm` are the quadratic forms of the eliminated and gauge-fixed presentations, identified by `quadForm_eq_of_gauge_fixed`;
- `eCoupling` is the coupling to the trace (`eTrace`), with `eCoupling_herm`, `eCoupling_off` and `norm_eCoupling_le`; this is where the scalaron-vielbein **interaction terms** enter;
- `qgElimFullModes` assembles the mode data, and the results are `qgElimFull_esa_farisLavine`, `qgElimFull_esa_core_fl` (the essential self-adjointness on the finite-occupation core), `eSecCore_dense`, and the scalaron instances `starobinsky_qgElimFull_esa` / `starobinsky_qgElimFull_esa_core` — with the full exponential wall and arbitrary coupling.

```
#check @BookProof.QgFullEliminated.EComp
#check @BookProof.QgFullEliminated.EGMode
#check @BookProof.QgFullEliminated.elimCoef
#check @BookProof.QgFullEliminated.eFormValue
#check @BookProof.QgFullEliminated.eFormValue_eq_formValue_elimConfig
#check @BookProof.QgFullEliminated.norm_elimCoef_le
#check @BookProof.QgFullEliminated.eGram
#check @BookProof.QgFullEliminated.eGram_eq_torsion_add_gauge3d
#check @BookProof.QgFullEliminated.eGram_quadForm
#check @BookProof.QgFullEliminated.quadForm_eq_of_gauge_fixed
```

```
#check @BookProof.QgFullEliminated.eCoupling
#check @BookProof.QgFullEliminated.norm_eCoupling_le
#check @BookProof.QgFullEliminated.qgElimFullModes
#check @BookProof.QgFullEliminated.qgElimFull_esa_farisLavine
#check @BookProof.QgFullEliminated.qgElimFull_esa_core_fl
#check @BookProof.QgFullEliminated.starobinsky_qgElimFull_esa
```

Manuscript provenance — why the Hamiltonian may be taken quadratic in the ladders. The shape above is the manuscript's, and its justification is the recursive construction of the Fock space: **"Since the second-quantization procedure can be applied recursively … in case the Hamiltonian acting in the Fock-space is not quadratic in the creation/annihilation operators, then we can consider instead a new Fock-space where the base Hilbert space is the original Fock-space. The new Hamiltonian is quadratic in the creation/annihilation operators."** The manuscript writes each sector's Hamiltonian as a one-particle operator enclosed between creation and annihilation, $H = \int d^3\vec{x}\, a^\dagger(\vec{x})\, H(\vec{x})\, a(\vec{x})$ — its Navier–Stokes instance is $H(\vec{x}) = \pi^i(u_j u_{i,j} - \nu u_{i,jj}) + (h.c.)$, and the graviton/scalaron instance has the same envelope. The inner space is where the fields and their spatial derivatives live; in this chapter the vielbein's derivative modes are **not** constrained by a gauge symmetry — they are eliminated by the spatial Fourier substitution and reappear in momentum space as **convolutions** (`ChapterQgFourierElimination`, `ChapterQgFullEliminated`), the same device the Navier–Stokes route uses (`BookProof.NsAdvectionConvolution`).

The final Hamiltonian of the sector is again the one-particle Hamiltonian enclosed in creation (on the left) and annihilation (on the right) operators on the outer Fock space, and the chapter proves the structural clauses that make that enclosure usable: `qgElimFull_momentum_conserving` (momentum), `qgElimFull_number_conserving` (the one-particle operator of each mode, so that every particle-number sector is preserved), `qgElimFull_stone_flow` and `starobinsky_qgElimFull_stone_flow` (the single-time package by Stone's theorem). The model is not degenerate: `elimCoef_torsion_ne_zero`, `elimCoef_gauge3d_ne_zero`, `eGram_diag_ne_zero` and `eCoupling_ne_zero` exhibit non-vanishing torsion, transverse gauge and trace coupling at non-zero momentum and coupling, and `infinite_egmode` records that the eliminated mode set is infinite, so the outer Fock space is a genuine infinite-dimensional one.

```
#check @BookProof.QgFullEliminated.qgElimFull_momentum_conserving
#check @BookProof.QgFullEliminated.qgElimFull_number_conserving
#check @BookProof.QgFullEliminated.qgElimFull_stone_flow
#check @BookProof.QgFullEliminated.starobinsky_qgElimFull_stone_flow
#check @BookProof.QgFullEliminated.eCoupling_ne_zero
#check @BookProof.QgFullEliminated.elimCoef_gauge3d_ne_zero
#check @BookProof.QgFullEliminated.infinite_egmode
```

36.5. What This Doctrines, and What It Leaves Open

The verified layer:

- the derivative variables of the vielbein sector carry **no independent content**: the constraint surface is exactly the image of the elimination (`formValue_dGauge_elimConfig`, `eq_elimConfig_of_gauge_fixed`), the eliminated torsion is the physical torsion (`elimTorsion_eq_torsionCoef`), and the vielbein self-interaction is reproduced verbatim (`elimGram_eq_contTorsionGram`);
- the full model — R^2 vielbein, full exponential wall, interaction terms — is essentially self-adjoint on the nine-component representation with arbitrary coupling, both on the Friedrichs domain and on the finite-occupation core (`qgElimFull_esa_farisLavine`, `qgElimFull_esa_core_fl`, `starobinsky_qgElimFull_esa`, `starobinsky_qgElimFull_esa_core`), with momentum and particle number conserved and the Stone flow available;
- the BRST material is thereby **restated**, not discarded: it survives as the consistency check that the elimination agrees with the gauge reduction (`elimTorsion_eq_gaugeReduce`).

Open items, as obligations:

- the coupling across **differing bases** — the plan's `Σ_ℓ dΓ(h_ℓ)` — is proved separately, and its ℓ^1-free comparison is what makes the multi-gauge-sector model a single second-quantized operator;

- the numerical twin's term lists and the eliminated presentation are compared structurally rather than term-by-term for the gravity kinetic forms, which is the honest status of that cross-check;
- the decoupling of the scalaron and vielbein one-particle operators in the **fiber** model is an assumption, not a theorem; the eliminated full model does not need it, but the closed-form spectrum of the fiber model does.

Two of the plan's general obligations, which used to be listed per sector, no longer apply here as open work. The **lifted** obligation — that the enclosure $H = \sum_{ij} h_{ij} C^\dagger(e_i) A(e_j)$ above is essentially self-adjoint on the lifted core — is the core-transfer chain of Second Quantization and Essential Self-Adjointness, available here because the chapter already exhibits the finite-occupation core. The **one-particle** obligation, that the comparison operator of record is essentially self-adjoint on the core actually chosen, is discharged in general by `BookProof.ComparisonCoreEsa.esa_of_isGraphCore` and the equivalence `isGraphCore_iff_esa`: both graph cores this route uses — `QgOuterFockFullFL.harmFried_isGraphCore`, the Gauss-polynomial core of $-\Delta + \|x\|^2/4$ in every dimension, and `ScalaronOuterFockFL.secN_isGraphCore`, the compactly supported smooth core of the scalaron-wall comparison — are therefore cores of essential self-adjointness by definition of the notions agreeing. And where the model is written in its decoupled form — the TEGR shear fibres $\oplus$ the scalaron sector, with no cross terms — essential self-adjointness of the sum follows from `BookProof.TensorSumChain.chain_esa` rather than from a separate argument. What remains genuinely open is the **interaction** content: the full exponential wall and the scalaron–vielbein coupling are inside h (they are one-particle matrix elements, per the enclosure doctrine), and their combined spectral theory — beyond the ESA statements quoted above — is where the mass-gap work continues.

37. The Starobinsky Scalaron: from Potential to Fock Space

37.1. Orientation and Status

The scalaron chapter: the $R + \alpha R^2$ model end to end, from the ghost-free scalar–tensor form to the essential self-adjointness of its Hamiltonian on the continuum and its Fock-space mass-gap statement. Read it after the quantum-gravity elimination (which prepared the variables) and with the ESA toolbox of the transfer chapter at hand; the chapter is deliberately explicit about where Kato–Rellich applies and where it cannot.

The route, in three stages: why $R + \alpha R^2$ (the ghost-free scalar–tensor form, the square Einstein-frame potential, what α buys the conformal mode); the operator stage (essential self-adjointness on the continuum, the Fock-space statement, the proved strict edge of the full-exponential fiber, the wall dichotomy, the Hermite core and the Friedrichs realization, and in what sense the realization is canonical); and the wall/closure stage (the compactly supported core, the exponential wall closed by convexity, bounded-below versus non-negative, and the energy form).

The proofs are `BookProof/Starobinsky`, the `Scalaron*` family (Esa, Fock, Edge, WallEsa, HermiteEsa), the `QgHermite*` and `WallEsa*` chapters, and the Friedrichs/Kato/limit-circle toolbox they cite. The companion numerical validation lives in the solver work the numerics chapters describe.

37.2. Why R + αR²

The simplest modification of Einstein–Hilbert gravity that is both ghost-free and capable of inflation is the Starobinsky action $f(R) = \frac{M^2}{2} R + \alpha R^2$ with $\alpha > 0$. The point of the quadratic term is **regularization of the conformal mode**: pure general relativity, whose conformal-mode gradient energy carries a negative sign, is unbounded below, and the positive αR^2 turns that unstable direction into a parabola bounded below by $-M^4/(16\alpha)$. This is the same conformal-mode stabilization that the $3 + 1$ split of the diffeomorphism chapter (`Book/DiffeomorphismsGravity.lean`) meets at the level of the spatial metric.

The gauge fixing is carried out in the **vielbein** (frame) variables of the TEGR / teleparallel module — the derivation of record — not in metric variables. The reduction to three dimensions is the time-gauge choice $v^\mu = \delta^\mu_0$ for the globally-defined time-like vector of the teleparallel frame, after which the 4-dimensional Hamiltonian formalism collapses to a 3-dimensional one whose first-order (BRST) constraint algebra of diffeomorphisms / local Lorentz transformations / global translations is **different from the ADM one** — in particular there is **no** scalar Hamiltonian

constraint solved for the conformal-mode curvature R_c. The conformal-mode stabilization comes instead from the αR^2 term itself: with $\psi = 1 + 4\alpha R/M^2$ the Hamiltonian carries $U(\psi)e = \frac{M^4}{16\alpha}(\psi-1)^2 e \geq 0$, i.e. the potential is manifestly bounded below for $\alpha > 0$, and it is this (together with the Weyl 3D gauge-fixing of the frame's local Lorentz / conformal freedom — the frame's time component A_0^j fixed via the gauge-fixing fermion $\{G, i\,b_j A_0^j\}$$ in the base TEGR module's BRST cell, the gravity analogue of the Yang–Mills Weyl gauge $A_0 = 0$$ — the TEGR module's job) that regularizes the would-be wrong-sign conformal direction. This three-dimensional reduction is **not** the ADM one: it fixes the globally defined time-like vector to $v^\mu = \delta^\mu_0$$, diffeomorphisms must conserve it, so the ghosts of the resulting BRST charge are **constant in the timepiece**, and the charge keeps the **same functional form as the four-dimensional formalism** — explicitly different from the ADM BRST charge ($book.tex$$, and the doctrine recorded in $BookProof/Qg/ChapterQuantumGravityBrstCharge.lean$$). The completed-square identity $V_3(R_c) \geq -M^4/(16\alpha)$$ below records the same stabilization at the level of the spatial potential.

A **metric-route** derivation — synchronous gauge $N = 1, N^i = 0$; the $3+1$ split $R = R_3 + (K^2 - K_{ij}K^{ij})$; the conformal decomposition $R_c = \Omega^{-4}\bar{R}_c - 8\Omega^{-5}\bar{\nabla}^2\Omega$; the Navier–Stokes-style fixing of the spatial derivative variables ($(\partial_i\varphi)(\partial^i\varphi) \to \mathrm{grad}^2$); and the Hamiltonian constraint solved for R_c — exists as an alternative and is what the densitized modules formalize, but it is **not** the object of record. Starobinsky is $f(R)$, a function of the **Ricci scalar** — a metric object — so that route is self-contained in the metric; the vielbein is only **required** for the teleparallel (torsion) formulation, where the torsion scalar T and the Weitzenböck connection are built from the frame.

Because the base theory is teleparallel-equivalent to GR ($eR = e\cdot T + \mathrm{divergence}$, the TEGR identity verified in `Book/BaryonAsymmetry.lean`), the vielbein/TEGR realization of record is faithful to the metric one — the **same** teleparallel restrictions on the frame apply (Weitzenböck connection: metric-compatible, curvature-free, torsion-carrying). Writing $R = T + B$ with the boundary term $B = 2\nabla_\mu T^\mu$, the R^2 action reads $\int e\, f(T+B)$ — a functional of the torsion scalar **and** its boundary term. It is therefore not a pure $f(T)$ (teleparallel) action: $f(T+B) \neq f(T)$ generically, and pure $f(T)$ gravity is a different theory. Everything below depends only on $R = T + B$, so the scalar–tensor reduction, the conformal-mode parabola, and H_{final} are unchanged by the choice of variables.

The verified content here is the full chain: the classical scalar–tensor equivalence, the Einstein-frame scalaron potential and its shape, the regularization of the conformal mode, essential self-adjointness of the gauge-fixed Hamiltonian on a continuum core (exponential wall included), and the quantization to the nested Fock space. The modules are `BookProof/Qg/ChapterStarobinskyPotential.lean`, `BookProof/Qg/ChapterScalaronCoreEsa.lean`, `BookProof/Qg/ChapterScalaronFockEsa.lean` and — for the exponential wall of the **sum** $-d^2/d\varphi^2 + V$ — `BookProof/Toolbox/ChapterWeakSecondDerivative.lean` with `BookProof/Qg/ChapterScalaronWallEsa.lean` (plan item **A5** of `CONSOLIDATED_PLAN.md`), all registered in `BookProof.lean` and certified axiom-free in `BookProof/RoadmapAudit/ChapterRoadmapAudit.lean`.

37.3. The Ghost-Free Scalar–Tensor Form

With the auxiliary field $\psi = 1 + 4\alpha R/M^2$ and the potential $U(\psi) = \frac{M^4}{16\alpha}(\psi-1)^2$, the identity $f(R) = \frac{M^2}{2}\psi R - U(\psi)$ recasts $R + \alpha R^2$ gravity as a **second-order** scalar–tensor theory: the would-be fourth-order mode is traded for a genuine scalar, the scalaron, with a standard quadratic potential. There is no Ostrogradsky ghost — the theory has exactly two propagating degrees of freedom plus the scalar.

```
#check @BookProof.Starobinsky.fR_eq_scalarTensor
```

37.4. The Einstein-Frame Potential: a Square

In the Einstein frame the scalaron moves in the potential $V(\varphi) = \frac{M^4}{16\alpha}\left(1 - e^{-\sqrt{2/3}\,\varphi/M}\right)^2$. Because it is **manifestly a square**, it is non-negative — the strongest form of the correct (elliptic) sign — vanishes at $\varphi = 0$ (the flat Minkowski vacuum), tends to the plateau $M^4/(16\alpha)$ at large field (the inflationary slow-roll regime), and rises exponentially as $\varphi \to -\infty$ (the wall that keeps the field on the plateau side).

```
#check @BookProof.Starobinsky.starobinskyV_nonneg
#check @BookProof.Starobinsky.starobinskyV_zero
#check @BookProof.Starobinsky.starobinskyV_tendsto_plateau
#check @BookProof.Starobinsky.starobinskyV_tendsto_atBot_atTop
```

37.5. The Conformal Mode and What α Buys

In the densitized variables the spatial potential is the parabola $V_3(R_c) = \alpha\big(R_c - \frac{M^2}{4\alpha}\big)^2 - \frac{M^4}{16\alpha}$, bounded below by $-M^4/(16\alpha)$ for $\alpha > 0$. The contrast with $\alpha = 0$ is the whole story: the linear theory is **not** bounded below, which is exactly the conformal-mode instability. The completed-square identity exhibits the regularization, and the mode-level operator inherits it uniformly: the gravity fibre symbol $(1/16)a_k^2 - (1/24)b_k^2 + V_k$ with the Starobinsky potential $V_k = V_3(R_c k)$ is essentially self-adjoint on its maximal domain.

An asymmetry worth stating before it bites: the **failure** and the **cure** depend on α in opposite directions. The wrong-sign conformal kinetic $(1/24)\partial_y^2 \le 0$ is a property of the base general-relativity kinetic itself — it is present in `book.tex 8190` derived from the standard Einstein–Hilbert action at $\alpha = 0$ — so the densitized d'Alembertian's lack of essential self-adjointness holds for **both** plain GR and Starobinsky. What α buys is the **cure**: the bounded-below potential $V(\varphi) \ge 0$ (or the conformal-mode completed square $V_3 \ge -M^4/(16\alpha)$) is exactly what turns the positive-kinetic reassembly into an essentially self-adjoint, bounded-below operator, and that exists **only** for $\alpha > 0$ — at $\alpha = 0$ the conformal mode's potential is the linear $-(M^2/2)R_c$, unbounded below. So the statement "the densitized operator is not ESA" is α-independent true, while "the physical reassembly is ESA" is α-dependent true ($\alpha > 0$ only); plain GR fails on both counts.

```
#check @BookProof.Starobinsky.confV_completed_square
#check @BookProof.Starobinsky.confV_ge
#check @BookProof.Starobinsky.confV_bddBelow
#check @BookProof.Starobinsky.confV_zero_alpha_tendsto_atBot
#check @BookProof.Starobinsky.qgR2Mode_potential_ge
#check @BookProof.Starobinsky.qgR2Mode_esa
```

37.6. Essential Self-Adjointness on the Continuum

The scalaron potential is **not** of temperate growth: it grows exponentially as $\varphi \to -\infty$, so the standard multiplication-operator theorem for potentials of temperate growth on the Schwartz core does not apply. The resolution is to move to the smaller — still dense — core of smooth compactly supported functions, where **no growth hypothesis is needed at all**: multiplication by any smooth real function preserves the core, and the deficiency argument (divide a test bump by the nowhere-vanishing smooth function $W - \bar{z}$ at a non-real z) stays inside it. What survives is only the analytic content the physics needs: a dense core, and — for the combination with the kinetic term — the potential bounded below, which the Starobinsky potential is in the strongest way (it is a square).

```
#check @BookProof.ScalaronEsa.ccDomain_dense
#check @BookProof.ScalaronEsa.smoothPotential_symmetric
#check @BookProof.ScalaronEsa.smoothPotential_deficiencyTrivial
#check @BookProof.ScalaronEsa.smoothPotential_essentiallySelfAdjoint
#check @BookProof.ScalaronEsa.contDiff_starobinskyV
#check @BookProof.ScalaronEsa.starobinskyV_not_hasTemperateGrowth
```

37.7. The Fock-Space Statement

Quantization of a continuum field theory lives on the **nested Fock space** $\bigoplus_n L^2(E_n)$: the n-particle sector carries the n-fold configuration space, and the Hamiltonian acts sector by sector with the many-body potential the sum of the one-particle potentials of the individual quanta. Essential self-adjointness is **fibrewise** — a deficiency vector of an orthogonal direct sum, tested against single-fibre states, has vanishing coordinates — so the one-particle theorems glue to the whole Fock space. Applied to the gauge-fixed $R + \alpha R^2$ potential $\sum_j \big(V_3(R_{cj}) + V(\varphi_j)\big)$ — smooth, bounded below by $-nM^4/(16\alpha)$, and at $n = 1$ exactly the one-particle potential — this yields essential self-adjointness and the complete unitary group e^{-itH} of the scalaron Hamiltonian, in both the continuum and the mode realisation.

```
#check @BookProof.ScalaronFock.nestedCore_dense
#check @BookProof.ScalaronFock.fockSmoothPotential_esa
#check @BookProof.ScalaronFock.fockSmoothPotential_stone_flow
#check @BookProof.ScalaronFock.contDiff_qgManyPotential
#check @BookProof.ScalaronFock.qgManyPotential_ge
#check @BookProof.ScalaronFock.qgManyPotential_esa
#check @BookProof.ScalaronFock.qgManyPotential_one
#check @BookProof.ScalaronFock.qgScalaronFock_esa
#check @BookProof.ScalaronFock.qgScalaronFock_stone_flow
```

This closes plan item **A5**: the Starobinsky sector of the gauge-fixed Hamiltonian is a well-defined self-adjoint operator whose unitary flow is the quantized dynamics — the formal counterpart of the Fock realization $\sum_i m\, N_i$ with $m^2 = M^2/(12\alpha)$ used in the numerical validation of the model.

Manuscript provenance. The manuscript fixes this shape and the reason it may assume the Hamiltonian quadratic in the ladders: **"Since the second-quantization procedure can be applied recursively … in case the Hamiltonian acting in the Fock-space is not quadratic in the creation/annihilation operators, then we can consider instead a new Fock-space where the base Hilbert space is the original Fock-space. The new Hamiltonian is quadratic in the creation/annihilation operators."** It writes each sector's Hamiltonian as a one-particle operator enclosed between creation and annihilation, $H = \int d^3\vec{x}\, a^\dagger(\vec{x})\, H(\vec{x})\, a(\vec{x})$ — its Navier–Stokes instance, the worked example, is $H(\vec{x}) = \pi^i(u_j u_{i,j} - \nu u_{i,jj}) + (h.c.)$ — and the scalaron/vielbein instance below has the same envelope. Where a spatial derivative of a field enters (the vielbein derivative modes), the formal route eliminates it by the spatial Fourier substitution, so that in momentum space it becomes a **convolution** rather than a gauge constraint.

The final Hamiltonian is the one-particle Hamiltonian enclosed in creation (on the left) and annihilation (on the right) operators on the nested Fock space — the same doctrine as QYM, QED, and NS: $H = \sum_{ij} h_{ij} C^\dagger(e_i) A(e_j)$. **The one-particle operator of record is the R^2 vielbein form in full**: the nine-component eliminated operator with the **exact** Einstein-frame exponential wall and the scalaron–vielbein interaction terms inside h — `qgElimFullModes`, whose essential self-adjointness is `qgElimFull_esa_farisLavine` / `starobinsky_qgElimFull_esa`, as set out in Fourier Elimination for Gravity. Two simpler realizations are used where their smaller content suffices, and they must be read as limits of that record rather than as alternatives to it: the decoupled $h = h_{\mathrm{TEGR}} \oplus (m)$, i.e. $H = \sum_i : (1/16)\mathcal{S}_i^2 : + m\, N_\psi$, whose scalaron entry is only the mass $m = \sqrt{V''(0)}$, and the full-exponential **fiber** $h = \frac{1}{2}\pi^2 + V(\hat{\varphi})$ with the wall used as-is. The first is the small-field (quadratic) limit of the second, and the second decouples the fiber from the vielbein cross terms that the record operator carries. The nested Fock space has two levels: the outer Fock space (whose ladders are the $C^\dagger/A$ of the enclosure) and the inner one-particle Hilbert space on which h acts. The outer Hamiltonian is a **quadratic (free-particle-like) form in the outer ladders for any h** — so the FULL Einstein-frame scalaron potential $V(\varphi) = \frac{M^4}{16\alpha}(1 - e^{-\sqrt{2/3}\,\varphi/M})^2$, **exponential included, with no Taylor expansion**, may live inside h (in the one-particle matrix elements $\langle e_i, h\, e_j\rangle$), with no 3-/4-particle vertices at the outer level. That is the realization `qg_starobinsky_vielbein_hamiltonian_full` in `../unfer` (the truncated-Hermite enclosure of $h = \frac{1}{2}\pi^2 + V(\hat{\varphi})$, whose one-particle spectrum is exactly the Schrödinger spectrum proved essentially self-adjoint above, `starobinskyWall_esa`); the quadratic $m\, N_\psi$ realization `qg_starobinsky_vielbein_hamiltonian` is its small-field limit. Either way $\langle 0|H|0\rangle = 0$ and the outer vacuum is the exact ground — the R² content never creates higher vertices at the outer level. Note also that the decoupled forms are exactly the situation of `BookProof.TensorSumChain.chain_esa`: a finite family of uncoupled one-particle factors glued by the tensor sum is essentially self-adjoint on the algebraic tensor product of their domains, so the **decoupled** realizations need no bespoke argument — while the record operator, whose interaction terms tie the scalaron to the vielbein, is not a tensor sum and is covered by its own Faris–Lavine theorem instead.

The scalaron sector's outer-Fock gap is now a theorem, not a hypothesis. `BookProof/Qg/ChapterScalaronFockGapChain.lean` runs the abstract gap chain (the one instantiated conditionally for gauge-fixed QYM in `ChapterYangMillsFockGapChain`) for the sector whose one-particle operator is the **positive constant** $m = 1/\sqrt{12\alpha}$ — precisely the $m\, N_\psi$ realization above. For the constant operator the one-particle form gap that the Yang–Mills chain must assume is an identity (`constOnePart_quadForm`: the form is exactly $m\|x\|^2$), so every conclusion follows unconditionally: `const_fock_gap` proves $d\Gamma(m \cdot 1)\Omega = 0$ and the bound $\mathrm{Re}\langle u, d\Gamma(m \cdot 1)u\rangle \geq m\|u\|^2$ on vacuum-

orthogonal finite states, `const_fock_mass_gap` adds the positive self-adjoint (Friedrichs) extension with strictly positive non-vacuum energy, `const_fock_gap_of_field_perturbation` keeps the surviving gap $(m - 2\|f\|)\|u\|^2$ under the unbounded, number-changing coupling $\Phi(f)$ whenever $2\|f\| < m$, and — via the companion module `ChapterFockCubicQuarticStability`, which stabilizes the bare cubic term of `ChapterFockCubicUnbounded` by its normal-ordered quartic partner on all finite states (not just the trial family) — `const_fock_cubic_quartic_bounded_below` keeps the energy bounded below when cubic/quartic mode couplings are added. The scalaron instantiations are `scalaron_fock_mass_gap` and its two siblings at `scalaronMass α = 1/\sqrt{12\alpha}`.

The honest boundary is the modelling input, and it is worth stating plainly because it is exactly where the two realizations above meet. What the chain takes as given is that the sector's one-particle energy is the constant $1/\sqrt{12\alpha}$ — true for the quadratic $m\,N_\psi$ realization by definition. For the full-exponential realization $h_\psi = \frac{1}{2}\pi^2 + V(\hat{\varphi})$ — the exponential used as-is, no Taylor expansion — a precision is owed about **which** formalized Hamiltonian the proved QG theorems address, and what transfers. The wall theory (`starobinskyWall_esa` in `ChapterScalaronWallEsa`: every smooth non-negative potential, no growth restriction, on the compactly supported smooth core, plus `wallHamBddBelow_semibounded` for the form) is stated for $-d^2/dx^2 + W$; the model's kinetic normalization is $\frac{1}{2}\pi^2 = -\frac{1}{2}\,d^2/d\varphi^2$, the same class after the unitary rescaling $\varphi = \sqrt{2}\,x$ — so the scalaron **fiber**'s essential self-adjointness, Stone flow and lower bound carry over with the potential needing no caveat at all (the model's V is exactly the formalized `starobinskyV`). Separately, `ChapterScalaronFockEsa` (`qgScalaronFock_esa`, `qgScalaronFock_stone_flow`) and the densitized route (`ChapterQuantumGravityDensitized`, `ChapterScalaronDensitizedTransfer`) formalize the **metric-route** 3+1 gauge fixing — conformal mode $V_3(R_c)$ plus the scalaron on the densitized conformal variables — that is, the previous module `docs/qg_starobinsky_hamiltonian.cdb`, **not** the vielbein/TEGR Hamiltonian derived above. For the vielbein/TEGR Hamiltonian (`docs/qg_starobinsky_vielbein_hamiltonian.cdb`) the proved layer applies fiber by fiber: the scalaron fiber via the wall class (same V, rescaled kinetic), the TEGR shear fibers via the constant/diagonal one-particle chains (per-mode energy a positive constant), and the fibrewise direct-sum instrument (`ChapterDirectSumEsa`, the generic `fockSmoothPotential_esa`) reassembles the nested Fock space once each fiber's essential self-adjointness is in place. What the metric-route theorems do **not** provide is a statement about the TEGR gravity sector itself. With the fibers covered, the single remaining input for a **strict** gap is the positivity of the one-particle edge: an explicit $E_0 > 0$ with $\langle\psi, h_\psi\,\psi\rangle \geq E_0\|\psi\|^2$ (numerically $E_0 \approx 0.689$ at $\alpha = 1/12$ in `../unfer`, `qg_starobinsky_vielbein_hamiltonian_full`). That is an elementary confinement estimate, not an existence problem: for any $0 < c < M^4/(16\alpha)$ the superlevel set $\{V < c\}$ is a bounded interval (exponential wall on one side, potential shelf $M^4/(16\alpha)$ on the other), the kinetic energy inside it and the outside cost c bound the form away from zero — or the certified Ritz bands of the numerical model feed the existing `BandEnclosure` route. The TEGR kinetic sector is not covered by the constant-model chain, and no Yang–Mills claim is made.

37.8. The Strict Edge, Proved: the Full-Exponential Fiber and the Fibrewise Reassembly

The "single remaining input" above has since been supplied. `BookProof/Qg/ChapterScalaronEdge.lean` (namespace `BookProof.ScalaronEdge`) proves the strict one-particle edge $\langle\psi, h_\psi\,\psi\rangle \geq E_0\|\psi\|^2$ for the **full** exponential scalaron fiber — no Taylor expansion, the wall used as-is. The route is entirely elementary: `starobinskyV_lt_shelf_bounded` shows the sublevel set $\{V < c\}$ is the bounded interval with the closed-form endpoints $A = \ln(1 + \sqrt{c/K})/a$ and $B = -\ln(1 - \sqrt{c/K})/a$, $K = M^4/(16\alpha)$, $a = \sqrt{2/3}/M$; the one-dimensional sup bound $\|f(x)\|^2 \leq \delta \int |f|^2 + \delta^{-1} \int |f'|^2$ (`edge_sup_sq_le`, obtained by integrating $d/dt\|f\|^2$ and absorbing with $2|z||w| \leq \delta|z|^2 + \delta^{-1}|w|^2$) replaces any Poincaré/Wirtinger input; and `starobinskyEdge_form_gap` assembles the constants $E_0 = \min(E_{\mathrm{kin}}, E_{\mathrm{mass}})$ into the strict bound on the compactly supported smooth core. Because the fiber's Friedrichs extension has continuous spectrum above the plateau (no orthonormal eigenbasis exists), the number-preserving $d\Gamma$ lift does not apply to this fiber; what does is the Friedrichs transfer `friedrichs_extension_form_gap` — `scalaronEdge_friedrichs_gap` carries the **same** strict bound to the positive self-adjoint extension selected by the Hashimoto shift-invert at $\gamma = 1$.

```
#check @BookProof.ScalaronEdge.starobinskyV_lt_shelf_bounded
#check @BookProof.ScalaronEdge.edge_sup_sq_le
#check @BookProof.ScalaronEdge.edge_energy_bound
```

```
#check @BookProof.ScalaronEdge.starobinskyEdge_form_gap
#check @BookProof.ScalaronEdge.scalaronEdge_friedrichs_gap
```

With each fiber's one-particle input in place, the **fibrewise reassembly** is now a theorem. `BookProof/Qg/ChapterVielbeinFiberFock.lean` (namespace `BookProof.VielbeinFock`) names the fiber list of the vielbein/TEGR sector — d harmonic shear fibers with frequencies ω_i plus one Starobinsky scalaron fiber per quantum — builds the n-particle potential $\sum_j \left(\sum_i \frac{1}{2}\omega_i^2 y_i(j)^2 + V(\varphi(j))\right)$ on $\mathbb{R}^{n\times(d+1)}$, and runs the generic instrument: `vielbeinFock_esa` (essential self-adjointness on the dense smooth core), `vielbeinFock_stone_flow` (the unitary group), symmetry, trivial deficiency, and the uniform lower bound `vielbeinFock_potential_ge`. The scalaron fiber is identified with the formalized potential by `vielbeinManyPotential_scalaron_fiber` — the reassembled object is the full-exponential `starobinskyV` fiber, not a small-field substitute.

```
#check @BookProof.VielbeinFock.vielbeinManyPotential_scalaron_fiber
#check @BookProof.VielbeinFock.vielbeinFock_symmetric
#check @BookProof.VielbeinFock.vielbeinFock_esa
#check @BookProof.VielbeinFock.vielbeinFock_stone_flow
#check @BookProof.VielbeinFock.vielbeinFock_potential_ge
```

Reading the pair honestly: the reassembly certifies that the **fibrewise direct sum** is a well-defined self-adjoint Hamiltonian with a unitary flow and a uniform lower bound, and the edge theorem certifies that the scalaron fiber contributes a strict $E_0 > 0$. What the reassembly does **not** do is claim a mass gap for the reassembled operator itself — the shear fibers are harmonic (ω_i-confined, so they carry their own per-mode gaps) but the scalaron fiber's continuum above the plateau is real, and the strict edge is a statement about the one-particle form on the core, not about the spectrum of the reassembled Hamiltonian. The honest boundary stated in the plan's 29b/29c updates remains: the number-preserving Fock lift is not claimed for this fiber; `1.932` remains a certified truncated number; no mass gap of a physical gravity (or Yang–Mills) Hamiltonian is claimed.

The **dimensional scope** deserves to be stated as bluntly as the mass-gap boundary, because it is where a reader most easily over-reads the proofs. The one-particle operator h of the model is a single operator on the one-particle Hilbert space that **already carries the kinetic** (and the derivative-type content) for all of its degrees of freedom — it is not a potential half in need of a kinetic added later. Concretely the scalaron fiber is the genuine second-order differential operator $h_\psi = \frac{1}{2}\pi^2 + V(\hat{\varphi}) = -\frac{1}{2}\,d^2/d\varphi^2 + V(\varphi)$ on $L^2(\mathbb{R})$, and this is exactly the operator proved essentially self-adjoint with a strict edge: `starobinskyWall_esa` (ESA on the compactly supported smooth core, full exponential wall) and `starobinskyEdge_quadForm` / `scalaronEdge_friedrichs_gap` (the strict $E_0 > 0$). The "1D" labels mean only that the **scalaron slice** of the one-particle space is one coordinate per quantum; the reassembly assembles these kinetic-inclusive fibers (harmonic shear oscillators h_{ω_i} plus h_ψ), and the outer Fock operator is then simply h between ladder operators, $H = \sum_{ij} h_{ij}\, C^\dagger(e_i)\, A(e_j)$ — no kinetic is introduced anywhere outside h. What `vielbeinFock_esa` proves — unconditionally — is the **potential** (multiplication) form of the reassembly, $fockSmoothPotentialOp$ acting as multiplication by $\sum_j \left(\sum_i \frac{1}{2}\omega_i^2 y_i(j)^2 + V(\varphi(j))\right)$ on the n-particle sector $L^2(\mathbb{R}^{n\times(d+1)})$; that specific theorem is the potential half and is **not itself** the kinetic-inclusive h, which is why the single-object ESA/edge of the assembled $h = \oplus_i h_{\omega_i} \oplus h_\psi$ (kinetic included, via the direct-sum instrument of `ChapterDirectSumEsa`) is the remaining statement, called the plan's QG-3, with the shear frequencies ω_i as modelling input. So the honest reading is: each fiber's ESA/edge, kinetic included, is unconditional (`starobinskyWall_esa`, `starobinskyEdge`, the Hermite oscillator fibers); the potential-half Fock reassembly is unconditional (`vielbeinFock_esa`); and the single kinetic-inclusive object h, assembled fiber by fiber, is QG-3. In particular, none of these theorems is an essential-self-adjointness statement for the densitized 3D d'Alembertian $H_0 = (1/16)\Delta_\hat{S} - (1/24)d^2/dy^2$ of the densitized route — that operator's wrong-sign conformal direction is the (unclaimed, and in fact false) Case B of the earlier sign analysis.

One further boundary must be stated without hedging, because it is the one a reader would most naturally miss: the **object of record is the full 3D gauge-fixed operator**, `qg3DHamiltonian` of `BookProof/ChapterQuantumGravity3DGauge` (F.5) — the `book.tex 8190` form with the Weyl-ordered cross terms $\frac{1}{2}S\cdot E + \frac{1}{3}P\cdot E - e(\cdots)$ **included** (symmetry on the core: `qg3D_symmetricOn`). The fibrewise model $h = \oplus_i h_{\omega_i} \oplus h_\psi$ described above is a **derived consequence**, not the Hamiltonian: it may be used as a theorem only after the couplings are proved to be identities on the physical (BRST-closed) sector — the plan's QG-3.2(a), the formal version of the conformal-mode elimination. Until then it is a comparison object, never the solved

operator: the numerics must solve the **full** operator, cross terms included, with certified bands stated for **that** operator. This has landed in `../unfer`: `qg3d_full_hamiltonian` realizes the full operator, `qg3d_full_operator_sirk` solves it as-is (indefinite band — the wrong-sign conformal kinetic is present, so no bounded-below claim), and the certified bands are pinned in `fock_sirk/tests/fixtures/qg3d_full_bands.ndjson`. The scalar-fiber builders (`qg_starobinsky_vielbein_hamiltonian(_full)`) remain comparison models only, and their certified numbers (e.g. $E_0 \approx 0.689$) are fiber-model enclosures. No perturbative drop of the couplings as "higher order" is admissible: the Hashimoto–SIRK/Krylov approximation solves the operator as-is.

37.9. The Wall Dichotomy That Decides Essential Self-Adjointness

Every statement in this chapter about the one-dimensional fibre $-d^2/dx^2 + V$ on the compactly supported smooth core of $L^2(\mathbb{R})$ has rested, until recently, on the sign of V — but the convexity route is special to $V \geq 0$, and the question of what happens when V dips below zero was an unresolved gap. The freshly landed `BookProof` wave closes that gap on both sides, so the **hypothesis that decides** essential self-adjointness on the line is now a theorem: **boundedness below is enough, and unboundedness below is genuinely fatal.** Together the two directions turn the signanalysis of `CONSOLIDATED_PLAN.md`'s QG-2 into a sharp dichotomy.

The positive half was landed first, in the direction of the physics the scalaron needs. For $V \geq 0$ the convexity argument above is clean, but the Starobinsky reassembly also meets potentials that are merely bounded below (e.g. the conformal-mode parabola $-M^4/(16\alpha) \leq V \leq 0$); the constant-shift route (`WallEsaBddBelow`) already handled that. Newly landed, `BookProof/Qg/ChapterBddBelowWallEsa.lean` replaces the convexity argument by a **cutoff energy estimate** that needs no convexity at all and no growth hypothesis: with $\zeta_r(x) = g(x/r)$ a rescaled bump, the real part of the integrated identity bounds the cutoff kinetic energy $\int \zeta_r^2 |W'|^2 \leq 4M^2\|W\|^2 + 2K\|W\|^2$ (the only place $V \geq -K$ is used) and the imaginary part forces $W \equiv 0$ — so `-d^2/dx^2 + V` is essentially self-adjoint for **every** smooth V bounded below (`wallHam_essentiallySelfAdjoint_of_bddBelow`). The same wave glues an arbitrary family of such fibres into one operator on the orthogonal direct sum $\ell^2(i : \iota,\, L^2(\mathbb{R}))$, $H = \oplus_i(-d^2/dx_i^2 + V_i)$ (`BookProof/Qg/ChapterBddBelowFiberSumEsa.lean`: `fiberSumHam_essentiallySelfAdjoint_of_bddBelow`, the unitary flow, and the form bound `fiberSumHam_semibounded`), so the composed multi-shear/scalaron object inherits the fibre property with no relative bound and no comparison operator.

```
#check @BookProof.BddBelowWallEsa.ode_solution_eq_zero_of_bddBelow
#check @BookProof.BddBelowWallEsa.wallHam_essentiallySelfAdjoint_of_bddBelow
#check @BookProof.BddBelowFiberSumEsa.fiberSumHam_essentiallySelfAdjoint_of_bddBelow
#check @BookProof.BddBelowFiberSumEsa.fiberSumHam_stone_flow
#check @BookProof.BddBelowFiberSumEsa.fiberSumHam_semibounded
#check @BookProof.BddBelowFiberSumEsa.qgFiberSum_esa
```

The negative half makes the dichotomy genuinely sharp rather than merely stated, by proving the converse class cannot be rescued. The deficiency space of $-d^2/dx^2 + V$ is exactly its space of square-integrable **classical** solutions of $W'' = (V - z)W$ (`BookProof/Qg/ChapterWallDeficiencyObstruction.lean`, `deficiencyTrivialAt_iff_no_l2_solution`), so failing essential self-adjointness is equivalent to exhibiting one such W. The wave exhibits several, each with a physical profile. An explicit smooth potential asymptotically $-x^4/4$ puts the operator in the Weyl limit-circle regime in both deficiency spaces (`BookProof/Qg/ChapterLimitCircleExample.lean`, `lcV_not_essentiallySelfAdjoint`). The wrong-sign **conformal** fiber of the densitized route — kinetic `(1/24)∂²_y` with a **non-negative** `R^2` (Starobinsky) exponential wall — is a particular instance of this and is **not** rescued by the wall (`BookProof/Qg/ChapterConformalFiberDeficiency.lean`: `cfV_not_essentiallySelfAdjoint`, packaged as `exists_wall_potential_wrongSign_not_essentiallySelfAdjoint`); the sign bookkeeping that makes this precise is `BookProof/Qg/ChapterConformalSignFlip.lean` (`wrongSignKinetic_esa_iff`). And the **free** half-line kinetic operator — the purest form of the densitized-conformal endpoint `y = 0` (degenerate tetrad) — where the densitized fibre lives, is itself not essentially self-adjoint (`BookProof/Qg/ChapterHalfLineLimitCircle.lean`, `hlKin_not_essentiallySelfAdjointOn`, and by the sign-flip theorem also `hlKin_neg_not_essentiallySelfAdjointOn`). None of these computes the deficiency indices of the

physical densitized operator; together they say that the wrong-sign conformal direction is **structurally** unstable to essential self-adjointness, which is exactly the honest boundary the signanalysis records.

```
#check @BookProof.WallDeficiencyObstruction.deficiencyTrivialAt_iff_no_l2_solution
#check @BookProof.LimitCircleExample.lcV_not_essentiallySelfAdjoint
#check @BookProof.ConformalFiberDeficiency.cfV_not_essentiallySelfAdjoint
#check @BookProof.ConformalSignFlip.wrongSignKinetic_esa_iff
#check @BookProof.HalfLineLimitCircle.hlKin_not_essentiallySelfAdjointOn
#check @BookProof.HalfLineLimitCircle.hlKin_neg_not_essentiallySelfAdjointOn
```

A last set of newly landed results belongs to the **operator** authority of the claim rather than its analytic core, and is worth naming here because each one removes a gate the plan had left open. `BookProof/Toolbox/ChapterFockDifferingBasesEsa.lean` proves that a sum of one-particle Hermitian operators, **each presented in its own eigenbasis** with no common alphabet, is essentially self-adjoint on the finite-particle core once the coupling amplitudes are `ℓ^1` in the working basis (`spectralFamily_essentiallySelfAdjointOn_core`): a one-particle-built Hamiltonian conserves the particle number, so its commutator against the diagonal comparison operator `N = dΓ(ω) + 𝒩 + 1` vanishes **identically** and Faris–Lavine runs with constant 0. Together with its companion `BookProof/Qg/ChapterQgCouplingDGammaSum.lean` (the coupling sum is a **single** dΓ, `dGamma_finsetSum_col`) this removes the common-alphabet summability gate the differing-bases design worried about. `BookProof/Qg/ChapterQgMultiHalfDensity.lean` lands the multi-dimensional half-density unitary (`multiHalfDensityUnitary : L^2((0,∞)\times X, de\otimes\mu) \simeq_{\ell i} L^2((0,∞)\times X, 2y\,dy\otimes\mu)`), so the densitized change of variables is a Hilbert-space unitary for the full field space, and executes the concrete `E = ∂e` derivative-variable fixing on the 84-dimensional tetrad coordinate algebra (`BookProof/Qg/ChapterQgDerivativeRealization.lean`, `eval_eq_of_fixed_of_comp_eq`) and the constraint-surface reduction that makes the cross couplings identities on the physical sector (`BookProof/Qg/ChapterQgPhysicalSectorIdentity.lean`, `lagrange_term_zero_of_fixing`). Each is unconditional and `sorry`-free; each carries its honest boundary in its own docstring.

```
#check @BookProof.FockDifferingBases.spectralFamily_essentiallySelfAdjointOn_core
#check @BookProof.FockDifferingBases.sumOfTwo_essentiallySelfAdjointOn_core
#check @BookProof.QgCouplingDGammaSum.dGamma_finsetSum_col
#check @BookProof.QgMultiHalfDensity.multiHalfDensityUnitary
#check @BookProof.QgDerivativeRealization.eval_eq_of_fixed_of_comp_eq
#check @BookProof.QgPhysicalSectorIdentity.lagrange_term_zero_of_fixing
```

37.10. The Hermite Core: Where the Hamiltonian Is Defined

The numerics work in the Hermite basis, so the honest domain question is whether the Hamiltonian is defined on the **Gauss–polynomial core** — the functions $p(x)e^{-x^2/4}$ with p a polynomial. It is, and the reason is a single inequality: the Gaussian tail dominates every exponential, $e^{c|x|} \le e^{2c^2}e^{x^2/8}$ for all real c and x, so that $e^{c|x|}e^{-x^2/4} \le e^{2c^2}e^{-x^2/8} \to 0$. Every function in the **exponential growth class** $|W(x)| \le Ce^{c\|x\|}$ — which contains every polynomial, and contains the scalaron potential, whose exponential wall puts it outside the temperate class — therefore multiplies a core element back into L^2. The core is also invariant under differentiation, since $(pe^{-x^2/4})' = (p' - xp/2)e^{-x^2/4}$, so the second derivative of a core element is again a core element and $H\psi = -\psi'' + W\psi$ lands in L^2 for every core element ψ.

```
#check @BookProof.QgHermiteCore.exp_abs_le_const_mul_exp_sq
#check @BookProof.QgHermiteCore.exp_abs_mul_gaussH_le
#check @BookProof.QgHermiteCore.tendsto_exp_abs_mul_gaussH_atTop
#check @BookProof.QgHermiteCore.expBounded_poly
#check @BookProof.QgHermiteCore.expBounded_starobinskyV
#check @BookProof.QgHermiteCore.memLp_mul_gaussPoly_of_expBounded
#check @BookProof.QgHermiteCore.hasDerivAt_gaussPoly
#check @BookProof.QgHermiteCore.memLp_hamiltonian_gaussPoly
#check @BookProof.QgHermiteCore.memLp_scalaronHamiltonian_gaussPoly
```

The same argument runs in every dimension on the product Gauss–polynomial core of $L^2(\mathbb{R}^d)$, since a polynomial in the coordinates is itself dominated by an exponential of the norm and an exponentially bounded

function of one coordinate is exponentially bounded on $\mathbb{R}^d$. In particular the full potential of the reduced two-variable sector, $V_3(R_c) + V(\varphi)$, maps the core of $L^2(\mathbb{R}^2)$ into $L^2(\mathbb{R}^2)$.

```
#check @BookProof.QgHermiteCore.integral_gaussPoly_mul
#check @BookProof.QgHermiteCore.gint_gaussPolyDeriv_antisymm
#check @BookProof.QgHermiteCore.gint_gaussPolyDeriv_two_symm
#check @BookProof.QgHermiteCore.integral_kinetic_symm
#check @BookProof.QgHermiteCore.integral_hamiltonian_symm
#check @BookProof.QgHermiteCore.integral_scalaronHamiltonian_symm
```

On this core the Hamiltonian is moreover **symmetric**. Differentiation is antisymmetric against the Gaussian weight — the boundary terms of the integration by parts vanish — so the second derivative is symmetric, and multiplication by a real potential is symmetric for free; the two combine into $\int \psi\,(H\varphi) = \int (H\psi)\,\varphi$ for all core elements. Together with the density of the core this is the symmetric-operator half of the essential self-adjointness question; the deficiency half is not proved here.

```
#check @BookProof.QgHermiteCore.exists_exp_bound_mvPolyEval
#check @BookProof.QgHermiteCore.memLp_mul_pgFun_of_expBounded
#check @BookProof.QgHermiteCore.expBounded_scalaronSectorPotential
#check @BookProof.QgHermiteCore.memLp_scalaronSectorPotential_mul_pgFun
```

This is the well-definedness half of the Hermite-core programme. It fixes the domain in the basis the algorithm uses; it does **not** by itself give essential self-adjointness on that core. For the **potential term** the next section closes that gap even for the exponential wall; for the full operator $-\Delta + V$ it remains open, as does the hyperbolic (Strichartz / direct-integral) residue for the conformal-mode kinetic term.

37.11. The Exponential Wall Is Not an Obstruction on the Hermite Core

Multiplication by the scalaron potential is **essentially** self-adjoint on the Gauss–polynomial core — no temperate growth, no boundedness, no semiboundedness hypothesis. The obstruction one expects is that a deficiency vector w at a non-real z gives the function $u = (W - z)w$, which is **not** square integrable when W grows exponentially, so the usual moment argument (vanishing of all $\int p\, e^{-\|x\|^2/4} u$) cannot be quoted. What replaces square integrability is **Gaussian exponential decay**: $e^{c\|x\|} e^{-\|x\|^2/4} u$ is integrable for **every** c, because $|u| \le C e^{c\|x\|} |w|$ with $w \in L^2$ and the Gaussian beats every exponential. That is all the Fourier argument ever used: the transform of $e^{-\|x\|^2/4} u$ is an everywhere convergent power series in the moments, hence identically zero, hence $u = 0$ almost everywhere; and $W - z$ never vanishes because W is real. So $w = 0$, both deficiency spaces are trivial, and the core is a genuine core.

```
#check @BookProof.ScalaronHermiteEsa.gaussExpDecay_potential_sub
#check @BookProof.ScalaronHermiteEsa.ae_eq_zero_of_moments_of_gaussExpDecay
#check @BookProof.ScalaronHermiteEsa.potCore_deficiencyTrivialAt
#check @BookProof.ScalaronHermiteEsa.potCore_essentiallySelfAdjoint
#check @BookProof.ScalaronHermiteEsa.scalaronPot_essentiallySelfAdjoint
#check @BookProof.ScalaronHermiteEsa.scalaronSector_essentiallySelfAdjoint
#check @BookProof.ScalaronHermiteEsa.scalaronPot_stone_flow
#check @BookProof.ScalaronHermiteEsa.scalaronSector_stone_flow
```

The statement holds in every dimension for every continuous exponentially bounded real potential, so both the one-variable scalaron potential and the full reduced sector potential $V_3(R_c) + V(\varphi)$ are covered, and the essential self-adjointness — rather than a choice of extension — is what produces the self-adjoint realization and its unitary group. What is **not** claimed is the sum $-\Delta + V$ on the Hermite core with the exponential potential; there the Friedrichs realization of the next section is still what is available.

37.12. Why Kato–Rellich Cannot Be Used for the Sum

The natural first attempt at $-\Delta + V$ on the Hermite core is Kato–Rellich: show that V is relatively bounded with respect to the kinetic term with relative bound less than one. For the exponential wall that attempt cannot succeed, and the reason is quantitative. Test the two sides against the monomial core family $\psi_N(x) = x^N e^{-x^2/4}$. The reference operators map this family into itself by polynomial maps of fixed degree increment, $(-d^2/dx^2 + x^2/4)\psi_N = ((N + \tfrac{1}{2})x^N - N(N-1)x^{N-2})e^{-x^2/4}$, and the monomial moment recursion turns that into

$\|H_0\psi_N\| \le (N^2+1)\|\psi_N\|$ and $\|\psi_N''\| \le (N^2+1)\|\psi_N\|$: cubic growth at worst. The wall, by contrast, is an exponentially tilted Gaussian moment; keeping the eighth-order term of the tilt gives $\int e^{-2sx}x^{2N}e^{-x^2/2} \ge (2s^8/315)M_{2N+8}$ and $M_{2N+8} = (2N+7)(2N+5)(2N+3)(2N+1)M_{2N}$, so the quadratic form of V along the family grows like N^4. A quartic cannot be dominated by a cubic, and the conclusion is not merely that the relative bound fails to be small — no pair of constants (a,b) works at all, for the kinetic term and for the conformal-mode oscillator alike.

```
#check @BookProof.HermiteExpWall.quadForm_scalaron_ge
#check @BookProof.HermiteExpWall.l2_scalaron_ge
#check @BookProof.HermiteExpWall.l2_kin_le
#check @BookProof.HermiteExpWall.l2_osc_le
#check @BookProof.HermiteExpWall.not_relatively_bounded_of_cubic
#check @BookProof.HermiteExpWall.scalaronV_not_kinetic_relativelyBounded
#check @BookProof.HermiteExpWall.scalaronV_not_oscillator_relativelyBounded
```

This is a genuine obstruction rather than a gap in the bookkeeping, and it explains the shape of the two preceding sections: the oscillator chapter can only absorb **bounded** perturbations by Kato–Rellich, and the exponential case had to be handled by the Fourier/moment route instead. For the sum $-\Delta + V$ with the exponential wall what remains is the Friedrichs realization.

37.13. What Kato–Rellich Can Reach: Quadratically Dominated Potentials

The obstruction of the previous section is about the exponential wall specifically, not about the method, and it is worth recording exactly how far the method does reach. The reference operator to perturb is not the bare kinetic term but the harmonic Hamiltonian $H_0 = -\Delta + W$ with $W(x) = \|x\|^2/4$, the one for which the Gauss–polynomial core is already a core. Against **that** operator the harmonic potential itself is relatively bounded, with relative constant exactly one: $\|W\psi\|^2 \le \|H_0\psi\|^2 + \frac{d}{2}\|\psi\|^2$ for every ψ in the core. The proof is the anticommutator identity $\{-\Delta, W\} = -\Delta W + 2\sum_j (-\partial_j) W \partial_j$, which on the core is a finite algebraic computation with the twisted derivative D_j and Gaussian integration by parts: since $W \ge 0$ and $\Delta W = d/2$, the cross term $2\,\mathrm{Re}\langle -\Delta\psi, W\psi\rangle$ is bounded below by $-\frac{d}{2}\|\psi\|^2$, and expanding $\|H_0\psi\|^2$ gives the bound.

Kato–Rellich then absorbs any continuous potential V dominated pointwise by $a\|x\|^2/4 + b$ with $a < 1$: the Gauss–polynomial core stays a core for $-\Delta + \|x\|^2/4 + V$, and V is allowed to be unbounded. In growth form the criterion reads: a continuous potential U with $|U(x) - \|x\|^2/4| \le A\|x\|^2 + C\|x\| + B$ and $4A < 1$ — the linear and constant coefficients arbitrary — is essentially self-adjoint on the core. Two corollaries are worth naming: every perturbation of at most linear growth is admissible, and the scaled oscillator $-\Delta + \lambda\|x\|^2/4$ is essentially self-adjoint on the **fixed**, width-one Gauss core for every $\lambda \in (0,2)$.

```
#check @BookProof.HermiteQuadraticEsa.gaussInt_anticommutator
#check @BookProof.HermiteQuadraticEsa.norm_sq_harmPoly_mul_le
#check @BookProof.HermiteQuadraticEsa.norm_harmPoly_mul_le
#check @BookProof.HermiteQuadraticEsa.harmonic_add_subquadratic_essentiallySelfAdjoir
#check @BookProof.HermiteQuadraticEsa.harmonic_add_subquadratic_stone_flow
#check @BookProof.HermiteQuadraticEsa.quadraticGrowth_essentiallySelfAdjoint
#check @BookProof.HermiteQuadraticEsa.harmonic_add_linearGrowth_essentiallySelfAdjoir
#check @BookProof.HermiteQuadraticEsa.scaledHarmonic_essentiallySelfAdjoint
```

The instance this chapter was after is the regularized conformal mode itself. The potential $V_3(R_c) = -\frac{M^2}{2}R_c + \alpha R_c^2$ differs from $R_c^2/4$ by $(\alpha - \frac{1}{4})R_c^2 - \frac{M^2}{2}R_c$, so the criterion applies as soon as $|\alpha - \frac{1}{4}| < \frac{1}{4}$, that is for $0 < \alpha < \frac{1}{2}$, with the mass term absorbed for free at any M. The conclusion is unconditional — no finite-speed and no unique-continuation hypothesis — and it yields the unitary group directly, without a choice of extension.

```
#check @BookProof.HermiteQuadraticEsa.confW
#check @BookProof.HermiteQuadraticEsa.confV_essentiallySelfAdjoint
#check @BookProof.HermiteQuadraticEsa.confV_stone_flow
```

The two-variable reduced sector is reached the same way. Its potential is $V_3(R_c) + V(\varphi)$, and while the exponential wall $V(\varphi)$ is out of reach, its harmonic approximation at the minimum is not: V vanishes to second order at $\varphi = 0$ with $V(\varphi)/\varphi^2 \to M^2/(24\alpha)$, so the quadratic sector potential is $V_3(R_c) + \mu\varphi^2$. Both variables

now have to sit inside the window of the core, which is $0 < \alpha < \frac{1}{2}$ for the conformal mode and $0 < \mu < \frac{1}{2}$ for the scalaron mass; at the physically natural $\mu = M^2/(24\alpha)$ the second condition reads $M^2 < 12\alpha$.

```
#check @BookProof.HermiteQuadraticEsa.sectorQuadW
#check @BookProof.HermiteQuadraticEsa.sectorQuad_essentiallySelfAdjoint
#check @BookProof.HermiteQuadraticEsa.sectorQuad_stone_flow
#check @BookProof.HermiteQuadraticEsa.tendsto_starobinskyV_div_sq
#check @BookProof.HermiteQuadraticEsa.sectorHarmonicApprox_essentiallySelfAdjoint
```

The window $a < 1$ is the honest edge of the method rather than an artifact: the relative bound of W against H_0 is exactly one, so a strictly larger relative bound cannot be absorbed. On a Gauss core of a different width the same argument relocates the window in α; what no width reaches is the exponential wall, by the refutation of the previous section.

37.14. The Friedrichs Realization on the Hermite Core

Fixing the domain is not yet fixing the operator. On the Gauss–polynomial core the Hamiltonian $H = -\Delta + W$ is a densely defined **symmetric** operator, and the question the numerics ultimately need answered is which self-adjoint operator it is that the algorithm propagates. The first half of that answer is unconditional: in every dimension, and for every continuous potential of exponential growth class, H is symmetric on the core and its quadratic form is bounded below by the lower bound of the potential, $\mathrm{Re}\langle\psi, H\psi\rangle \geq c\,\|\psi\|^2$ whenever $W \geq c$. The kinetic half of that inequality is an identity: integrating by parts against the Gaussian weight turns $\langle\psi, -\Delta\psi\rangle$ into $\sum_j \|D_j\psi\|^2 \geq 0$, where $D_j p = \partial_j p - x_j p/2$ is the twisted derivative that differentiation induces on the polynomial factor.

```
#check @BookProof.QgHermiteFriedrichs.coreD
#check @BookProof.QgHermiteFriedrichs.kinPoly
#check @BookProof.QgHermiteFriedrichs.gaussInt_coreD
#check @BookProof.QgHermiteFriedrichs.gaussInt_kinPoly
#check @BookProof.QgHermiteFriedrichs.hamCore
#check @BookProof.QgHermiteFriedrichs.hamCore_symmetricOn
#check @BookProof.QgHermiteFriedrichs.re_gaussInt_kinPoly_self
#check @BookProof.QgHermiteFriedrichs.hamCore_quadForm_ge
#check @BookProof.QgHermiteFriedrichs.hasDerivAt_pgFun_coord
```

Density of the core plus symmetry plus semiboundedness is exactly the input of the Friedrichs extension theorem, so the Hamiltonian on the Hermite core has a canonical semibounded self-adjoint realization — positive when the potential is. For the scalaron this is unconditional, because $V(\varphi) = \frac{M^4}{16\alpha}\left(1 - e^{-\sqrt{2/3}\,\varphi/M}\right)^2 \geq 0$: the one-particle scalaron Hamiltonian on the Gauss–polynomial core of $L^2(\mathbb{R})$ has a **positive** self-adjoint extension. On the reduced two-variable sector (R_c, φ) the conformal-mode parabola contributes its own lower bound $-M^4/(16\alpha)$, and the realization is semibounded with that constant.

```
#check @BookProof.QgHermiteFriedrichs.hermiteCore_friedrichs_extension
#check @BookProof.QgHermiteFriedrichs.hermiteCore_friedrichs_extension_of_nonneg
#check @BookProof.QgHermiteFriedrichs.qgOneParticleHermite_friedrichs
#check @BookProof.QgHermiteFriedrichs.qgOneParticleSector_friedrichs
```

The honest boundary is the difference between **a** self-adjoint realization and **the** one. What is proved is existence together with a canonical choice; what is not proved is that the choice is forced, i.e. essential self-adjointness of H on the Gauss–polynomial core. Uniqueness would remove the last piece of arbitrariness from the one-particle sector; the hyperbolic (Strichartz / direct-integral) residue for the conformal-mode kinetic term is a separate open question, unaffected by this section.

37.15. When the Core Is a Core: Essential Self-Adjointness for the Parabolic Potential

For one potential the arbitrariness disappears entirely. The Gauss–polynomial core is spanned by the product Hermite functions $\psi_\alpha(x) = \prod_i He_{\alpha_i}(x_i)\, e^{-\|x\|^2/4}$, and those are **eigenvectors** of $-\Delta + \|x\|^2/4$: on the polynomial side the Hamiltonian is the number operator plus $d/2$, because $-D_j^2 + x_j^2/4 = a_j^\dagger a_j + 1/2$ and

$a_j^\dagger a_j H e_\alpha = \alpha_j H e_\alpha$. Hence $H\psi_\alpha = (|\alpha| + d/2)\,\psi_\alpha$. A symmetric operator whose domain contains an orthonormal basis of eigenvectors with real eigenvalues has no deficiency vector at any non-real point — if $\langle Hv, w\rangle = z\langle v, w\rangle$ for all v in the domain, then $(\lambda_\alpha - z)\langle \psi_\alpha, w\rangle = 0$ for every α, and z non-real forces $w = 0$. So the harmonic Hamiltonian is **essentially** self-adjoint on the Gauss–polynomial core, in every dimension, with no auxiliary hypothesis; Stone's theorem then supplies the unitary flow.

```
#check @BookProof.QgHermiteOscillator.deficiencyTrivialAt_of_eigenbasis
#check @BookProof.QgHermiteOscillator.essentiallySelfAdjointOn_of_eigenbasis
#check @BookProof.QgHermiteOscillator.kinPoly_add_harmPoly
#check @BookProof.QgHermiteOscillator.crePoly_annPoly_hermiteMv
#check @BookProof.QgHermiteOscillator.harmCore_hermiteMvLp
#check @BookProof.QgHermiteOscillator.harmonicCore_essentiallySelfAdjoint
#check @BookProof.QgHermiteOscillator.harmonicCore_stone_flow
```

Kato–Rellich extends the conclusion past the exact parabola: multiplication by a bounded continuous real function is symmetric on the core and satisfies $\|B\psi\| \le M\|\psi\|$, a relative bound of 0, so $-\Delta + \|x\|^2/4 + B$ is essentially self-adjoint on the same core. What is still missing is the exponential wall — the scalaron potential is not dominated by the parabola, and for it only the Friedrichs realization of the previous section is available.

```
#check @BookProof.QgHermiteOscillator.potCore_symmetricOn
#check @BookProof.QgHermiteOscillator.norm_potLp_le
#check @BookProof.QgHermiteOscillator.hamCore_add_potential
#check @BookProof.QgHermiteOscillator.harmonic_add_bounded_essentiallySelfAdjoint
```

37.16. In What Sense the Realization Is Canonical

"Friedrichs extension" names a construction, not yet a characterization: a densely defined symmetric operator generally admits many self-adjoint extensions, and the construction only produces one of them. What singles this one out is its **form domain**. The domain of H carries the form inner product $\langle x, y\rangle_1 = \langle x, y\rangle + \langle x, Hy\rangle$; positivity and symmetry make it an inner product dominating the ambient one, its completion embeds injectively back into the Hilbert space, and that image $Q(H)$ is the form domain. The Friedrichs domain — the range of the resolvent $S = (H+1)^{-1}$ built by Riesz representation in the form space — lies inside $Q(H)$, as does the domain of H itself.

```
#check @BookProof.FriedrichsCanonical.formDomain
#check @BookProof.FriedrichsCanonical.friedrichsDomain
#check @BookProof.FriedrichsCanonical.friedrichsOp
#check @BookProof.FriedrichsCanonical.dom_le_formDomain
#check @BookProof.FriedrichsCanonical.friedrichsDomain_le_formDomain
#check @BookProof.FriedrichsCanonical.friedrichsOp_isPositiveSelfAdjointExtension
```

The characterization is that nothing else fits inside $Q(H)$. Let A' be any symmetric extension of H with $\operatorname{dom} A' \subseteq Q(H)$, and let $x \in \operatorname{dom} A'$. Put $u = A'x + x$ and $y = Su$. Both x and y lie in the form domain, so their difference is the image of some vector k of the form completion, and for every v in the domain of H the form pairing computes as $\langle v, k\rangle_1 = \langle v + Hv, x - y\rangle$. Symmetry of A' turns the first half into $\langle v, u\rangle$, and self-adjointness of S together with $S(v + Hv) = v$ turns the second half into the same thing, so the pairing vanishes. The domain is dense in the completion, hence $k = 0$, hence $x = y$ — every such A' is a restriction of the Friedrichs operator. If A' is in addition self-adjoint, its own self-adjointness criterion forces the reverse inclusion, and the two operators coincide, domain and action alike.

```
#check @BookProof.FriedrichsCanonical.friedrichs_canonical
#check @BookProof.FriedrichsCanonical.friedrichs_unique_selfAdjoint
```

Applied to the scalaron this says: on the Gauss–polynomial core of $L^2(\mathbb{R})$ the Hamiltonian $-\Delta + V(\varphi)$ has one and only one positive self-adjoint realization whose domain stays inside the form domain, and it is the constructed one. This is weaker than essential self-adjointness and does not pretend otherwise — an operator that is not essentially self-adjoint still has other self-adjoint extensions, but each of them must leave $Q(H)$. Within the energy-form class the quantum-gravity one-particle Hamiltonian is therefore unambiguous.

```
#check @BookProof.FriedrichsCanonical.qgOneParticleHermite_friedrichs_canonical
#check @BookProof.FriedrichsCanonical.qgOneParticleHermite_friedrichs_unique
```

The same account covers the merely semibounded case, which is what the reduced two-variable sector needs: if $\langle x, Hx\rangle \geq -c\|x\|^2$ then $H + c$ is positive, its Friedrichs extension exists by the above, and subtracting c again returns a self-adjoint extension of H with the same lower bound — unique, once more, among those whose domain lies in the form domain of the shift. For the sector (R_c, φ) the constant is the one the conformal-mode parabola supplies.

```
#check @BookProof.FriedrichsCanonical.semiboundedFriedrichsOp
#check @BookProof.FriedrichsCanonical.semiboundedFriedrichsOp_isSemiboundedSelfAdjoir
#check @BookProof.FriedrichsCanonical.semibounded_friedrichs_unique
#check @BookProof.FriedrichsCanonical.qgOneParticleSector_friedrichs_canonical
#check @BookProof.FriedrichsCanonical.qgOneParticleSector_friedrichs_unique
```

None of this is vacuous: applied to the diagonal operator $Ae_n = ne_n$ on the finite-mode domain of $\ell^2(\mathbb{N})$ — genuinely unbounded — the construction produces a positive self-adjoint extension, and the uniqueness statement pins it down among the extensions whose domain lies in the form domain.

```
#check @BookProof.FriedrichsCanonical.unbounded_friedrichs_canonical_example
```

37.17. The Compactly Supported Core

The Gauss–polynomial core is convenient but not canonical. The core a physicist actually writes down is $C_c^\infty(\mathbb{R}^d)$, the smooth functions of compact support: it is where test wave packets live, it is stable under localization, and it is the building block of the finite-particle Fock core. Essential self-adjointness on it is the stronger statement, because a smaller core means fewer test vectors are available to kill a deficiency vector — and it implies the same conclusion on every larger core, the Gauss core included.

Neither core sits inside the other: a Gauss polynomial $p(x)e^{-\|x\|^2/4}$ is never compactly supported, and a bump function is never a Gauss polynomial. So the transfer cannot be a restriction argument. What replaces it is approximation in the graph norm: if every vector of D_1 is approximated, together with its image under the operator, by vectors of D_2, then a deficiency vector for the operator on D_2 is one for the operator on D_1, and triviality of the deficiency spaces passes from D_1 to D_2 even though the domains are unrelated.

```
#check @BookProof.QgOneParticleCc.deficiencyTrivialAt_of_graphApprox
#check @BookProof.QgOneParticleCc.essentiallySelfAdjointOn_of_graphApprox
#check @BookProof.QgOneParticleCc.ccHam
#check @BookProof.QgOneParticleCc.ccHam_symmetricOn
```

The analytic input is a cut-off estimate. With χ a fixed bump equal to one on the unit ball and $\chi_R(x) = \chi(x/R)$, the Leibniz rule gives

$$(-\Delta + W)(\chi_R\psi) - (-\Delta + W)\psi = (\chi_R - 1)(-\Delta\psi + W\psi) - 2\sum_j \partial_j\chi_R\,\partial_j\psi - (\Delta\chi_R)\psi.$$

Each of the three terms is $o(1)$ in L^2: the first by dominated convergence, since $\chi_R - 1$ vanishes on the ball of radius R and is bounded; the second and third because the scaling gives $|\partial\chi_R| \leq C/R$ and $|\Delta\chi_R| \leq C/R^2$ while $\partial_j\psi$ and ψ are square integrable. Hence every Gauss-core vector is a graph-norm limit of compactly supported smooth ones, and essential self-adjointness descends.

```
#check @BookProof.QgOneParticleCc.exists_cut_derivative_bounds
#check @BookProof.QgOneParticleCc.exists_cc_graph_approx
#check @BookProof.QgOneParticleCc.ccHam_essentiallySelfAdjoint_of_core
```

The payoff is the one-particle theorem in the form the applications want: for every smooth V with $|V| \leq a\|x\|^2/4 + b$ and $a < 1$, the gauge-fixed Hamiltonian $-\Delta + \|x\|^2/4 + V$ is essentially self-adjoint on $C_c^\infty(\mathbb{R}^d)$, and Stone's theorem then supplies the unitary group. The conformal mode in one variable and the reduced two-variable sector (R_c, φ) of $R + \alpha R^2$ are instances, under the same $0 < \alpha < 1/2$ window that made the parabola subquadratic.

```
#check @BookProof.QgOneParticleCc.qgOneParticleCc_esa
#check @BookProof.QgOneParticleCc.qgOneParticleCc_stone_flow
```

```
#check @BookProof.QgOneParticleCc.confVCc_esa
#check @BookProof.QgOneParticleCc.sectorQuadCc_esa
```

Second quantization is then free. The n-particle potential $\sum_k (\|x_k\|^2/4 + V(x_k))$ on $(\mathbb{R}^d)^n \cong \mathbb{R}^{nd}$ obeys the same quadratic bound with the same a and n times the constant, so each sector Hamiltonian is essentially self-adjoint on its own compactly supported smooth core; the direct-sum criterion assembles the sectors into the finite-particle Fock space, whose core is the algebraic direct sum of the one-particle cores. The second-quantized $R + \alpha R^2$ Hamiltonian is essentially self-adjoint there, with a unitary time evolution.

```
#check @BookProof.QgOneParticleCc.qgNParticleCc_esa
#check @BookProof.QgOneParticleCc.qgFock
#check @BookProof.QgOneParticleCc.qgFockCore
#check @BookProof.QgOneParticleCc.qgFockHam
#check @BookProof.QgOneParticleCc.qgFockCc_esa
#check @BookProof.QgOneParticleCc.qgFockCc_stone_flow
```

The boundary is the same one as on the Gauss core, and is stated rather than hidden: the potential class transported here is the quadratic one. The exponentially growing scalaron wall $e^{\beta\varphi}$ is not covered — on the Gauss core it is refuted outright, because the wall does not even map that core into L^2. Localizing the core does not repair that; it only strengthens the conclusion within the class where the conclusion holds. The wall is reached instead by the entirely different argument of the next section.

37.18. The Exponential Wall, Closed by Convexity

The transfer of the previous section moves a **quadratic-class** theorem, so it leaves the wall itself untouched, and every perturbative route to the wall is refuted: it is not a relatively bounded perturbation of $-\Delta$ or of $-\Delta + x^2/4$ on the Gauss core with any pair of constants, and the Carleman flux criterion fails because the Hermite amplitudes grow like $e^{c\sqrt{N}}$, so $\sum 1/A$ converges. What closes the wall is not a perturbation argument at all, but the classical convexity argument for a **non-negative** potential — and the Starobinsky potential is non-negative, being a square.

The argument runs on the compactly supported smooth core of $L^2(\mathbb{R})$, the scalaron direction alone. A deficiency vector u at $z = \pm i$ satisfies $u'' = (V - z)u$ in the sense of distributions. That is an L^2 statement about an equation between distributions, so the first step is regularity: du Bois-Reymond's lemma in its second-order form says that a locally integrable function orthogonal to the second derivative of every test function is almost everywhere affine, and iterating it against the double antiderivative of the right-hand side identifies u, almost everywhere, with a genuine twice differentiable solution W of $W'' = (V - z)W$.

```
#check @BookProof.WeakSecondDeriv.ae_eq_affine_of_integral_deriv2_smul_eq_zero
#check @BookProof.WeakSecondDeriv.exists_ae_eq_doubleAntideriv_add_affine
#check @BookProof.WeakSecondDeriv.exists_deriv2_of_weak_eq
```

With a classical solution in hand the geometry takes over. Since $\operatorname{Re} z = 0$,

$$(|W|^2)'' = 2\operatorname{Re}(\bar{W}W'') + 2|W'|^2 = 2\big(V|W|^2 + |W'|^2\big) \geq 0,$$

so $|W|^2$ is convex. It is also non-negative and — because $u \in L^2$ — integrable. A convex function on the whole line satisfies $F(a - s) + F(a + s) \geq 2F(a)$; integrating over $s \in [0, R]$ bounds $2F(a)R$ by the total integral for **every** R, which forces $F \equiv 0$. Hence $u = 0$: both deficiency spaces are trivial, and the operator is essentially self-adjoint. No growth hypothesis on V is used anywhere.

```
#check @BookProof.ScalaronWallEsa.eq_zero_of_convexOn_nonneg_integrable
#check @BookProof.ScalaronWallEsa.ode_solution_eq_zero
#check @BookProof.ScalaronWallEsa.wallHam
#check @BookProof.ScalaronWallEsa.wallHam_symmetricOn
#check @BookProof.ScalaronWallEsa.wallHam_essentiallySelfAdjoint
```

Applied to the scalaron this says: $-d^2/d\varphi^2 + V(\varphi)$ with the exponentially growing Einstein-frame wall is essentially self-adjoint on $C_c^\infty(\mathbb{R})$, and Stone's theorem turns the unique self-adjoint extension into the unitary group e^{-itH} solving the Schrödinger equation globally in time. The honest boundary that remains is dimensional rather than about growth: the statement is one-dimensional and the hypothesis is $V \geq 0$.

```
#check @BookProof.ScalaronWallEsa.starobinskyWall_esa
#check @BookProof.ScalaronWallEsa.starobinskyWall_stone_flow
```

37.19. Bounded Below, Not Just Non-Negative

The convexity argument just given needs the potential to be non-negative. That hypothesis is a restriction on the **argument**, not on the physics, and `BookProof/Qg/ChapterWallEsaBddBelow.lean` removes it with the cheapest possible move: a constant shift. Multiplication by a real constant `c` is a bounded symmetric operator (`constOp`, `constOp_symmetric`), and shifting the potential by `c` shifts the whole operator by exactly that constant (`wallHam_add_const`); adding a bounded symmetric perturbation preserves essential self-adjointness — this is Kato–Rellich with relative bound zero (`BookProof.KatoRellich.essentiallySelfAdjointOn_add_bounded`) — so `-d²/dx² + V` is essentially self-adjoint on the compactly supported smooth core of `L²(ℝ)` for **every** smooth potential V that is merely bounded below, still with no growth restriction above (`wallHam_essentiallySelfAdjoint_of_bddBelow`), and Stone's theorem turns the unique extension into the unitary group (`wallHam_stone_flow_of_bddBelow`).

```
#check @BookProof.WallEsaBddBelow.constOp
#check @BookProof.WallEsaBddBelow.constOp_symmetric
#check @BookProof.WallEsaBddBelow.wallHam_add_const
#check @BookProof.KatoRellich.essentiallySelfAdjointOn_add_bounded
#check @BookProof.WallEsaBddBelow.wallHam_essentiallySelfAdjoint_of_bddBelow
#check @BookProof.WallEsaBddBelow.wallHam_stone_flow_of_bddBelow
```

Two instances put the shift to work. `scalaronPlus_esa` is the scalaron wall plus an arbitrary smooth, bounded-below addition — and the conformal-mode parabola, bounded below by `-M⁴/(16α)`, is exactly such an addition, so the full one-variable gauge-fixed operator `-d²/dφ² + V(φ) + V₃(R_c)` is essentially self-adjoint on the compactly supported smooth core. `oscillatorPlus_esa` is the harmonic oscillator `-d²/dx² + x²/4 + V` with V merely bounded below: **no** relative bound between V and the oscillator is required — which is precisely what fails for the exponential wall on the Gauss–polynomial core, where the wall is not a relatively bounded perturbation of the oscillator with any pair of constants (the `HermiteExpWall` refutation above). The two routes are complementary rather than redundant: the Hermite route reaches the quadratically dominated class on the Gauss core, and the wall route reaches every bounded-below potential on the compactly supported core.

```
#check @BookProof.WallEsaBddBelow.scalaronPlus_esa
#check @BookProof.WallEsaBddBelow.oscillatorPlus_esa
```

37.20. The Exponential Wall Itself, and the Energy Form

The clause "every smooth potential bounded below" contains the motivating example of the whole discussion, so it is worth writing it down as its own theorem. The potential $V(x) = e^{x} + e^{-x} = 2\cosh x$ is smooth and bounded below by 2, so `-d²/dx² + (eˣ + e⁻ˣ)` is essentially self-adjoint on the compactly supported smooth core of $L^2(\mathbb{R})$ (`expPotential_esa`), with unitary group e^{-itH} (`expPotential_stone_flow`); `coshPotential_esa` below is the same statement written with `cosh`. This potential grows faster than any polynomial, so no relative-boundedness (Kato–Rellich) criterion reaches it.

```
#check @BookProof.ExpPotentialEsa.contDiff_Vexp
#check @BookProof.ExpPotentialEsa.expPotential_esa
#check @BookProof.ExpPotentialEsa.expPotential_stone_flow
#check @BookProof.ExpPotentialEsa.coshPotential_esa
```

The shift-invert schemes need one more thing from the operator: that its **quadratic form** be bounded below, not merely that the operator be essentially self-adjoint. On the compactly supported smooth core this is one integration by parts. The Green identity $\langle(-d^2/dx^2 + V)f, f\rangle = \int |f'|^2 + \int V|f|^2$ splits the pairing into a Dirichlet term (`kinCcR_quadratic_form`) and a potential term (`opCc_quadratic_form`); the first is non-negative and the second is at least $-c\|f\|^2$ when $V \geq -c$, which is `wallHamBddBelow_semibounded`. For the exponential wall the constant is 0 (`expPotential_semibounded`).

```
#check @BookProof.WallEsaSemibounded.SemiboundedBelowOn
#check @BookProof.WallEsaSemibounded.integral_conj_neg_deriv2_mul
#check @BookProof.WallEsaSemibounded.kinCcR_quadratic_form
```

```
#check @BookProof.WallEsaSemibounded.opCc_quadratic_form
#check @BookProof.WallEsaSemibounded.wallHamBddBelow_semibounded
#check @BookProof.ExpPotentialEsa.expPotential_semibounded
```

There is also a second, entirely elementary route to the same physics, which does not go through the convexity argument or the constant shift at all: `BookProof/Toolbox/ChapterSchrodingerCutoffEsa.lean` runs the Simader–Faris–Lavine cutoff/commutator energy estimate directly. Testing $-u'' + Vu = zu$ against $\chi_R^2 \bar{u}$ and integrating by parts once gives $\int_{[-R,R]} (V - \mathrm{Re}\, z)|u|^2 \le (2C^2/R^2)\|u\|_{L^2}^2$ together with the companion Dirichlet bound (`cutoff_energy_core`), where C bounds $|\chi'|$; letting $R \to \infty$ forces `u = 0` (`l2_classical_solution_eq_zero`). Applied to $z = \pm i$ this says that $-d^2/dx^2 + (e^x + e^{-x})$ has no nonzero square-integrable **classical** solution of $Hu = \pm iu$. Setting $V = 0$ gives the same statement for the free Laplacian (`laplacian_deficiency_trivial`). The honest boundary of this second route is that it speaks about classical (twice-differentiable) solutions: passing from a general L^2 deficiency vector to a classical solution is elliptic regularity, which the first route supplies and this one does not.

```
#check @BookProof.SchrodingerCutoff.exists_scaled_cutoff
#check @BookProof.SchrodingerCutoff.schrodingerOp_symmetric
#check @BookProof.SchrodingerCutoff.cutoff_energy_core
#check @BookProof.SchrodingerCutoff.cutoff_energy_estimate
#check @BookProof.SchrodingerCutoff.l2_classical_solution_eq_zero
#check @BookProof.SchrodingerCutoff.schrodinger_exp_deficiency_trivial_I
#check @BookProof.SchrodingerCutoff.laplacian_deficiency_trivial
```

38. The Standard Model Hamiltonian

38.1. Orientation and Status

The chapter that assembles the whole bosonic content of the Standard Model in the frame the previous chapters built: temporal (Weyl) gauge, nested Fock state space, final Hamiltonian as the outer second quantization $d\Gamma(h)$, and Faris–Lavine comparison certificates. Read it after the Yang–Mills quantization chapter and the transfer chapter, whose Standard-Model instance already discharged the two operator obligations used here.

The route: counting the coordinates, the CKM and PMNS matrices, the one-particle operator and its comparison operator, the enclosure, the CAR algebra of the fermions, the Dirac/Yukawa Faris–Lavine certificate and its bosonic counterpart, the Kato-type input and the unconditional theorem, the explicit enclosure, the full one-particle Hamiltonian (bosonic and fermionic sectors together), the Yukawa coupling as an operator, the Higgs vacuum as statics, the continuum CAR algebra, the gauge connection, and the ghosts, generators, and BRST charge — closing with what is **not** claimed.

The proofs are the `Sm*` family (`SmHamiltonian`, `SmOneParticle`, `SmComparison`/`SmComparisonEsa`/`SmComparisonFull`, `SmFarisLavine`, `SmDiracSpinor`, `SmDiracYukawa`, `SmYukawaCoupling`, `SmHiggsVacuum`, `SmCar`/`SmCarContinuum`, `SmGaugeConnection`, `SmGaugeRep`, `SmBrstGhost`, `SmFockEsa`, `SmOuterFock`, `SmFullEnclosure`) plus the `BookBrst*` instances.

The Yang–Mills chapter quantized a single non-abelian gauge field; this chapter puts the whole bosonic content of the Standard Model into the same frame. The setting is the temporal (Weyl) gauge $A_0 = W_0 = B_0 = 0$, the state space is the nested Fock space of excitations, and the Hamiltonian of record is — as everywhere in this development — the **outer second quantization** $d\Gamma(h)$ of a one-particle operator h, never the bare h.

Proof chapters carry the content, in five generations of work. The first four set the frame: `BookProof/Sm/ChapterSmOneParticle.lean` (the collective coordinates and the mixing algebra), `BookProof/Sm/ChapterSmHamiltonian.lean` (the one-particle operator), `BookProof/Sm/ChapterSmComparison.lean` (the comparison operator of the Faris–Lavine route) and `BookProof/Sm/ChapterSmOuterFock.lean` (the enclosure). The second generation completes the fermions: `BookProof/Sm/ChapterSmCarAlgebra.lean` (the CAR algebra over a finite mode set), `BookProof/Sm/ChapterSmDiracYukawa.lean` (the Dirac and Yukawa operators and the three Faris–Lavine hypotheses for them), `BookProof/Sm/ChapterSmDiracSpinor.lean` (the relativistic spinor content) and `BookProof/Sm/ChapterSmComparisonFull.lean` (every summand of the comparison operator, derivative

coordinates included). The third generation closes the honest boundaries of the first: `BookProof/Sm/ChapterSmCarContinuum.lean` (the continuum CAR algebra over an infinite-dimensional one-particle space), `BookProof/Sm/ChapterSmGaugeConnection.lean` (the gauge connection inside the covariant derivative), `BookProof/Sm/ChapterSmBrstGhost.lean` and `BookProof/Sm/ChapterSmGaugeRepresentation.lean` (the ghost/BRST sector), and `BookProof/ChapterSmHiggsVacuum.lean` (symmetry breaking as statics). The fourth takes the manuscript's own BRST definition literally: `BookProof/Qym/ChapterBookBrstYangMills.lean`, `BookProof/Qym/ChapterBookBrstGaugeFixing.lean` and `BookProof/Qym/ChapterBookBrstInstances.lean`. The fifth closes the last open analytic hypothesis of the bosonic route: it begins with `BookProof/Sm/ChapterSmFarisLavine.lean` (the two Faris–Lavine inequalities, against the comparison operator that actually satisfies them) and continues through the independent Kato-type theorem and its transfer between the two cores — `BookProof/Toolbox/ChapterDegSchrodingerCore.lean`, `BookProof/Toolbox/ChapterMollifierL2.lean`, `BookProof/Toolbox/ChapterConvolutionCalc.lean`, `BookProof/Toolbox/ChapterDegEnergyEstimate.lean`, `BookProof/Toolbox/ChapterDegKatoEsa.lean`, `BookProof/Toolbox/ChapterHermiteLadderOrder.lean`, `BookProof/Toolbox/ChapterHermiteGraphApprox.lean` and `BookProof/Sm/ChapterSmComparisonEsa.lean` — ending in the unconditional essential self-adjointness of the one-particle operator. Those two sections, “The Bosonic Faris–Lavine Certificate” and “The Kato-Type Input”, are the pedagogical face of that chain. Every chapter named here is `sorry`-free and `axiom`-free, audited by `Work/SmStandardModelAudit.lean`, `Work/SmHonestBoundariesAudit.lean`, `Work/BookBrstAudit.lean`, `Work/SmFarisLavineAudit.lean` and `Work/SmComparisonEsaAudit.lean`; the ledger of what moved and what did not is `HONEST_BOUNDARIES_SM.md`.

38.2. Counting the Coordinates

A single Standard-Model excitation carries the spatial position, the three gauge fields with their spatial derivatives as independent coordinates, and the Higgs doublet written in four real components with its derivatives. The count is not taken on trust: the nine blocks are assembled as a type and their cardinality is computed, giving $D_B = 163$ as a theorem.

$$3 + 24 + 72 + 9 + 27 + 3 + 9 + 4 + 12 = 163.$$

The Grassmann coordinates of the fermions are deliberately **not** in that type: the CAR algebra is a separate development, and what the fermionic sector contributes at this level is the mixing algebra of the next section.

```
#check @BookProof.SmOneParticle.SmCoord
#check @BookProof.SmOneParticle.card_smCoord
#check @BookProof.SmOneParticle.smIdx
#check @BookProof.SmOneParticle.smG
#check @BookProof.SmOneParticle.smPhi
```

38.3. The CKM and PMNS Matrices

The quark and lepton Yukawa couplings enter the Hamiltonian through two unitary 3×3 matrices: the CKM matrix in the quark sector, the PMNS matrix in the lepton sector. What the operator theory needs from them is not their measured angles but their **unitarity identities**, and those are the theorems here: each row has unit length, so no entry exceeds one in modulus; a real mixing matrix is orthogonal, $VV^T = I$, the Cabibbo case; the biunitary mass matrix $M = U_L V^\dagger D U_R^\dagger$ satisfies $M^\dagger M = U_R D^\dagger D\, U_R^\dagger$, so the mixing drops out of the squared masses; and a mixing rotation cannot amplify a Yukawa coupling by more than the number of generations.

The measured Wolfenstein parameters, the PMNS angles, the mass ordering and the absolute neutrino masses are experimental input and are claimed nowhere.

```
#check @BookProof.SmOneParticle.IsMixing
#check @BookProof.SmOneParticle.unitary_row_sum_normSq
```

```
#check @BookProof.SmOneParticle.unitary_entry_norm_le_one
#check @BookProof.SmOneParticle.unitary_transpose_of_real
#check @BookProof.SmOneParticle.biunitary_massSq
#check @BookProof.SmOneParticle.norm_mulVec_le_sum
#check @BookProof.SmOneParticle.yukawa_bound
```

38.4. The One-Particle Operator

The one-particle operator is built on the Gauss–polynomial core of $L^2(\mathbb{R}^{163})$ out of genuine differentiation and multiplication operators, exactly as the Yang–Mills one is: forty momenta — one for each gauge and Higgs field coordinate — and forty-nine squared real field polynomials. The squares are the twenty-four gluon magnetic components $B^G_{a,i} = \frac{1}{2}\varepsilon_{ijk}(G^a_{k,j} - G^a_{j,k} + g_s f^{abc} G^b_j G^c_k)$, the nine weak ones, the three hypercharge ones, the twelve covariant Higgs derivatives $(D_i\varphi)_a = \varphi_{a,i} + gW^j_i(\tau_j/2\,\varphi)_a + g'B_i(\sigma_3/2\,\varphi)_a$, and the Higgs wall. The structure constants are arbitrary reals, so the genuinely non-abelian — quartic — magnetic energy is the case proved, not an abelian simplification.

The Higgs potential appears in its square form $\frac{\lambda}{4}(\|\varphi\|^2 - v^2)^2$, which differs from $-\frac{\mu^2}{2}\|\varphi\|^2 + \frac{\lambda}{4}\|\varphi\|^4$ by the constant $\mu^4/4\lambda$ when $v^2 = \mu^2/\lambda$; that identity is proved, and it is what makes the operator a sum of squares. Being a sum of squares, it is symmetric and bounded below, and the Friedrichs extension applies to it directly — the same route the Yang–Mills chapter takes, and the one the doctrine prescribes whenever the inner operator is bounded below.

```
#check @BookProof.SmHamiltonian.SmParams
#check @BookProof.SmHamiltonian.smMagG
#check @BookProof.SmHamiltonian.smMagW
#check @BookProof.SmHamiltonian.smMagB
#check @BookProof.SmHamiltonian.smCovD
#check @BookProof.SmHamiltonian.smWall
#check @BookProof.SmHamiltonian.card_smMom
#check @BookProof.SmHamiltonian.card_smForm
#check @BookProof.SmHamiltonian.smHamiltonian
#check @BookProof.SmHamiltonian.smHamiltonian_symmetricOn
#check @BookProof.SmHamiltonian.smHamiltonian_quadForm
#check @BookProof.SmHamiltonian.smHamiltonian_quadForm_nonneg
#check @BookProof.SmHamiltonian.sm_friedrichs_extension
#check @BookProof.SmHamiltonian.higgs_mexican_hat
#check @BookProof.SmHamiltonian.wall_sq
```

38.5. The Comparison Operator

The Faris–Lavine route — the alternative instrument, used where the inner operator is **not** bounded below — needs a comparison operator, and for the Standard Model that operator has a characteristic shape: the confinement is **quartic** in the non-abelian and Higgs coordinates and **quadratic** in the abelian field and in the derivative coordinates. One commutation with such an N removes one power of (q, p), which is what makes the two commutator hypotheses of the criterion plausible; a purely quadratic confinement would fail to dominate the quartic magnetic and Higgs terms.

What is proved here is the operator itself, its symmetry, and the lower bound $\langle x, Nx\rangle \geq \|x\|^2$ for $c_0 \geq 1$. Its dynamical part — the forty uncoupled one-dimensional factors $-d^2/dq^2 + q^4$ and $-d^2/dq^2 + q^2$ — is essentially self-adjoint: each factor by the bounded-below wall theorem, and their tensor sum by the chain theorem of the second-quantization chapter.

One caution carries the reader into the next two sections: this N\$, despite its name, is **not** the comparison operator the Faris–Lavine criterion consumes for h\$. Which operator does, why this one cannot, and what has to be proved about the replacement instead are the subject of “The Bosonic Faris–Lavine Certificate” below.

```
#check @BookProof.SmComparison.card_smConf
#check @BookProof.SmComparison.smComparison
#check @BookProof.SmComparison.smComparison_symmetricOn
#check @BookProof.SmComparison.smComparison_quadForm
#check @BookProof.SmComparison.sm_N_positive
#check @BookProof.SmComparison.quarticEsaOp
#check @BookProof.SmComparison.quadraticEsaOp
```

```
#check @BookProof.SmComparison.sm_N_dyn_esa
#check @BookProof.SmComparison.sm_N_dyn_stone_flow
```

38.6. The Enclosure

The Hamiltonian of record is the enclosure $H = d\Gamma(h)$ on the nested Fock space $\bigoplus_n L^2(\mathbb{R}^{163n})$: one copy of the one-particle operator per excitation, in every particle-number sector. Symmetry and positivity of the quadratic form are fibrewise, so they lift sector by sector; the Friedrichs extension and its Stone flow follow, and the enclosure is block diagonal in the particle number — the number sectors, not a spatial lattice, are what decomposes the problem.

A statement about the bare h is a one-particle statement and is labelled as such; a statement that calls itself **the** Standard-Model Hamiltonian must exhibit the $d\Gamma$.

```
#check @BookProof.SmOuterFock.smSectorHam
#check @BookProof.SmOuterFock.smSectorHam_symmetricOn
#check @BookProof.SmOuterFock.smSectorHam_quadForm_nonneg
#check @BookProof.SmOuterFock.smSector_friedrichs_extension
#check @BookProof.SmOuterFock.smFockSpace
#check @BookProof.SmOuterFock.smFockCore
#check @BookProof.SmOuterFock.smFockCore_dense
#check @BookProof.SmOuterFock.smFockHam
#check @BookProof.SmOuterFock.smFockHam_symmetricOn
#check @BookProof.SmOuterFock.smFockHam_quadForm_nonneg
#check @BookProof.SmOuterFock.sm_dGamma_friedrichs_extension
#check @BookProof.SmOuterFock.sm_dGamma_stone_flow
#check @BookProof.SmOuterFock.smFockHam_number_conserving
```

38.7. The CAR Algebra of the Fermions

The fermionic sector needs an algebra before it can have operators. It is built here over a finite set of modes, on the occupation-number Fock space with one orthonormal basis vector for each occupied set, and with the Jordan–Wigner sign that turns commutation into **anti**commutation. The four canonical relations are theorems: $\{a_i, a_i^\dagger\} = 1$, and $\{a_i, a_j^\dagger\} = 0$ for $i \neq j$, $\{a_i, a_j\} = 0$, $\{a_i^\dagger, a_j^\dagger\} = 0$ — the last two containing the Pauli principle $a_i^2 = 0$. Creation and annihilation are mutually adjoint and are contractions, so every polynomial in them is a bounded operator; the second quantization $\sum_{i,j} h_{ij} a_i^\dagger a_j$ of a Hermitian one-particle matrix is symmetric, its diagonal case is diagonal in the occupation basis with eigenvalue the sum of the occupied mode energies, and a one-particle energy floor μ\$ becomes a gap above the Fock vacuum. All of it is built over a **finite** mode set — the first honest boundary — and the section on the continuum CAR algebra below reports what it takes to remove that restriction.

```
#check @BookProof.SmCar.FermiFock
#check @BookProof.SmCar.jwSign
#check @BookProof.SmCar.annih
#check @BookProof.SmCar.creat
#check @BookProof.SmCar.car_annih_creat_self
#check @BookProof.SmCar.car_annih_creat_of_ne
#check @BookProof.SmCar.car_annih_annih
#check @BookProof.SmCar.car_creat_creat
#check @BookProof.SmCar.inner_creat_left
#check @BookProof.SmCar.norm_annih_le
#check @BookProof.SmCar.fermiBilin
#check @BookProof.SmCar.fermiBilin_symmetric
#check @BookProof.SmCar.fermiEnergy_occ
#check @BookProof.SmCar.fermi_mass_gap
```

38.8. Dirac, Yukawa, and the Faris–Lavine Certificate

On that algebra the two fermionic operators of the record can finally be written: the Dirac operator as the second quantization of a Hermitian one-particle matrix, the Yukawa operator as the bilinear $zM + (zM)^\dagger$ in a Higgs background z, with $M = U_L V^\dagger D U_R^\dagger$ the biunitary mass matrix. The mixing algebra of the second section is exactly what bounds it: no entry of M exceeds the sum of the diagonal masses, uniformly in the mixing angles.

For this sector the three Faris–Lavine hypotheses are theorems, with explicit constants, against the comparison operator $N = \sum_i \omega_i a_i^\dagger a_i + c_0$ of the plan: the relative bound $\pm h \le c_1 N$, the first-commutator bound and the double-commutator bound. Together with $N \ge 1$ and the surjectivity of $N + 1$ they feed the project's proof of Faris–Lavine Corollary 1.1 and give essential self-adjointness of the Dirac–Yukawa Hamiltonian. The concrete spinor content comes from the 4×4 Majorana model of the CPT chapter: that one-particle matrix is Hermitian and squares to $(k^2 + m_1^2 + m_2^2) \cdot 1$, so its eigenvalues are the relativistic energies $\pm\sqrt{k^2 + m_1^2 + m_2^2}$.

```
#check @BookProof.SmDiracYukawa.smDirac
#check @BookProof.SmDiracYukawa.smYukawa
#check @BookProof.SmDiracYukawa.smFermiHam
#check @BookProof.SmDiracYukawa.smFermiHam_symmetricOn
#check @BookProof.SmDiracYukawa.yukawa_entry_bound
#check @BookProof.SmDiracYukawa.smFermiN
#check @BookProof.SmDiracYukawa.sm_fermi_N_ge_one
#check @BookProof.SmDiracYukawa.sm_fermi_N_add_one_surjective
#check @BookProof.SmDiracYukawa.sm_fermi_fl_i
#check @BookProof.SmDiracYukawa.sm_fermi_fl_ii
#check @BookProof.SmDiracYukawa.sm_fermi_fl_iii
#check @BookProof.SmDiracYukawa.sm_fermi_esa
#check @BookProof.SmDiracSpinor.diracOneParticle
#check @BookProof.SmDiracSpinor.diracOneParticle_hermitian
#check @BookProof.SmDiracSpinor.diracOneParticle_sq
#check @BookProof.SmDiracSpinor.diracOneParticle_eigenvalue_sq
#check @BookProof.SmDiracSpinor.diracFieldHam
#check @BookProof.SmDiracSpinor.dirac_field_esa
```

38.9. Every Summand of the Comparison Operator

The derivative coordinates of N_0 carry no conjugate momentum: their summand is multiplication by q^2. That is not an obstruction — multiplication by a smooth real function is essentially self-adjoint on the compactly supported smooth core, with no growth or boundedness hypothesis — so each derivative coordinate is a factor of the same kind as the dynamical ones, and the chain runs over all one hundred and sixty summands: one for each collective coordinate except the three spatial ones, which do not appear in N_0.

```
#check @BookProof.SmComparisonFull.derivEsaOp
#check @BookProof.SmComparisonFull.deriv_coordinate_esa
#check @BookProof.SmComparisonFull.smFullChain
#check @BookProof.SmComparisonFull.smFullChain_length
#check @BookProof.SmComparisonFull.sm_N_full_esa
#check @BookProof.SmComparisonFull.sm_N_full_stone_flow
```

38.10. The Bosonic Faris–Lavine Certificate

The Faris–Lavine route needs a comparison operator N\$ against which the one-particle Hamiltonian satisfies two bounds: a relative bound and a form commutator bound $\pm i[h, N] \le cN$\$. The quartic operator of the previous sections **cannot** serve, and the obstruction is a theorem rather than a gap in the formalization. Write $h = \frac{1}{2}T + V_h$\$ and $N = T + V_N + c_0$\$ with $T = \sum_m \pi_m^2$\$. The commutator is first order, $i[h, N] = \sum_m (\pi_m G_m + G_m \pi_m)$\$ with $G = \nabla(\frac{1}{2}V_N - V_h)$\$, and the bound forces the pointwise inequality $|G|^2 \le c^2(V_N + c_0)$\$ — test the form on a wave packet $e^{i\xi\cdot x}\varphi$\$ concentrated at a point and optimize in ξ\$. For a quartic V_N\$ the gradient is cubic: along a single gluon direction, where the totally antisymmetric structure constants and $\varphi = 0$\$ make $\nabla V_h = 0$\$, the requirement reads $4R^6 \le c^2 R^4$\$ — false for large R\$, already for the free Hamiltonian. The confinement of the comparison operator itself is what fails, not the interaction.

The cure is the one Faris and Lavine use in their own application: let the comparison operator **contain** the Hamiltonian, so that ∇V_h cancels. With $N = 2h + \sum_m q_m^2 + c_0$\$ the difference becomes $G = \frac{1}{2}\nabla(\sum_m q_m^2)$\$, which is linear, and both inequalities hold with the absolute constants $c = 1$\$ and $K = 1$\$ — for every choice of couplings, structure constants and electroweak generators. That is the operator $smFlN$\$ of this section, together with its symmetry, its quadratic form and the Friedrichs comparison built from it. What the two inequalities leave behind is exactly one hypothesis: essential self-adjointness of this N\$ itself on the Gauss–polynomial core, isolated as `sm_h_esa_of_comparison_esa`\$ (and, in its packaged form through the abstract last mile

$CoreData.esa_core$, as sm_h_esa_of_graph_core$). The next section discharges it with an instrument Faris–Lavine does not provide.

```
#check @BookProof.SmFarisLavine.smFlN
#check @BookProof.SmFarisLavine.smFlN_symmetricOn
#check @BookProof.SmFarisLavine.smFlN_quadForm_nonneg
#check @BookProof.SmFarisLavine.sm_commForm_le
#check @BookProof.SmFarisLavine.sm_norm_le_shift
#check @BookProof.SmFarisLavine.smFlPosSymOp
#check @BookProof.SmFarisLavine.smFlComparison
#check @BookProof.SmFarisLavine.smFlComparison_extends
#check @BookProof.SmFarisLavine.smCoreData
#check @BookProof.SmFarisLavine.sm_h_esa_of_graph_core
#check @BookProof.SmFarisLavine.isGraphCore_of_esa
#check @BookProof.SmFarisLavine.sm_h_esa_of_comparison_esa
```

38.11. The Kato-Type Input, and the Unconditional Theorem

The remaining hypothesis — essential self-adjointness of a coupled quartic Schrödinger operator in 163 variables — is of a kind the Faris–Lavine criterion structurally cannot produce, since every admissible comparison contains $2V_h$ and is therefore as hard as h itself. It is supplied instead by an independent theorem, proved in four chapters whose content is worth walking through because the instrument is general.

BookProof/Toolbox/ChapterDegSchrodingerCore.lean sets up the class: operators $-\Delta_S + W$ whose kinetic term differentiates only a subset S of the coordinates, on both cores the project works with. The analytic heart is BookProof/Toolbox/ChapterDegKatoEsa.lean: a deficiency vector w at a purely imaginary z is **mollified** — for a smooth compactly supported mollifier ρ_ε the translate-average $w * \rho_\varepsilon$ is smooth, which is what BookProof/Toolbox/ChapterMollifierL2.lean (mollification converges in L^2) and BookProof/Toolbox/ChapterConvolutionCalc.lean (the mollified function is smooth, and every derivative falls on the smooth factor) make precise — and the smoothed deficiency equation is tested against a cut-off $\chi_R^2 \bar{w}_\varepsilon$. One integration by parts per direction of S, the Young absorption of BookProof/Toolbox/ChapterDegEnergyEstimate.lean, and the limit $\varepsilon \to 0$ at fixed R leave $\int_{\|x\|\le R} |w|^2 \le (2C^2/R^2)\|w\|^2$; letting $R \to \infty$ gives $w = 0$. No elliptic regularity is used, and no regularity of W beyond smoothness with $W \ge 1$.

The theorem lives on the compactly supported smooth core; the comparison operator is consumed on the Gauss–polynomial core. The bridge is the Hermite expansion: BookProof/Toolbox/ChapterHermiteLadderOrder.lean shows that every polynomial differential operator has a finite **ladder order** against the Hermite weights, and BookProof/Toolbox/ChapterHermiteGraphApprox.lean uses it to make the Hermite truncations of a compactly supported smooth vector Cauchy in the graph norm — so essential self-adjointness transfers from one core to the other. Finally BookProof/Sm/ChapterSmComparisonEsa.lean identifies, on the core, $N = 2h + \sum_m q_m^2 + c_0$ with exactly such an operator — S the forty momentum-carrying coordinates, $W = \sum_r \Phi_r^2 + \sum_m q_m^2 + c_0$ — proves the theorem for every real c_0 (values below one by a bounded shift), and concludes the unconditional sm_h_esa$.

What this chain establishes for the record must be read under the chapter's doctrine: sm_h_esa$ is a **one-particle** theorem — it says nothing about outer ladders — and its implication for the final Hamiltonian is exactly the reduction the doctrine prescribes. Essential self-adjointness of h on the Gauss–polynomial core feeds $dGamma_essentiallySelfAdjointOn_of_esa$, so the enclosure $H = d\Gamma(h) = \sum_{i,j} h_{ij} C^\dagger(e_i) A(e_j)$ — creation on the left, annihilation on the right — is essentially self-adjoint on the finite-particle core over that core, with no hypothesis beyond $P : SmParams$: both obligations of the plan's §D6b are discharged for the Standard-Model row. What is still owed is the sector-wise assembly of the full h (bosonic $\otimes$ fermionic, the Dirac and Yukawa operators on the CAR algebra) into that enclosure. The instrument itself is sector-agnostic — $ccHamS_esa$ and $hamCoreS_esa$ hold for every smooth, respectively polynomial, $W \ge 1$ and every coordinate subset S — so the same two theorems also give the non-abelian Yang–Mills instance of the same shape (BookProof/Qym/ChapterYangMillsNonAbelianEsa.lean, next section). One vocabulary detail keeps the two convolutions apart: the convolution of BookProof/Toolbox/ChapterConvolutionCalc.lean is the **mollifier** convolution of analysis; the momentum-space convolution that disposes of products of fields with

spatial derivatives is the NS/QG device of `BookProof/ChapterNsAdvectionConvolution.lean` and plays no role here.

```
#check @BookProof.DegSchrodinger.ccHamS
#check @BookProof.DegSchrodinger.hamCoreS
#check @BookProof.DegSchrodinger.hamCoreS_symmetricOn
#check @BookProof.MollifierL2.tendsto_mollify_L2
#check @BookProof.ConvolutionCalc.contDiff_cnv
#check @BookProof.ConvolutionCalc.dcoord_cnv
#check @BookProof.ConvolutionCalc.lapCS_cnv
#check @BookProof.DegEnergy.energy_bound
#check @BookProof.DegKatoEsa.deficiencyTrivialAt_ccHamS
#check @BookProof.DegKatoEsa.ccHamS_esa
#check @BookProof.HermiteLadder.exists_ladderOrd_hamPolyL
#check @BookProof.HermiteGraphApprox.exists_core_graph_approx
#check @BookProof.HermiteGraphApprox.hamCoreS_esa
#check @BookProof.SmComparisonEsa.smFlN_eq_hamCoreS
#check @BookProof.SmComparisonEsa.smFlN_esa_one_le
#check @BookProof.SmComparisonEsa.smFlN_esa
#check @BookProof.SmComparisonEsa.sm_h_esa
#check @BookProof.EsaOneParticle.dGamma_essentiallySelfAdjointOn_of_esa
```

38.12. The Enclosure, Made Explicit

The reduction of the previous section is now carried out as theorems. `BookProof/Qym/ChapterYangMillsNonAbelianEsa.lean` packages the instrument once: every Weyl-type operator $\frac{1}{2}\sum_m \pi_{\iota(m)}^2 + \frac{1}{2}\sum_j \Phi_j^2$ on the Gauss–polynomial core, with momenta in pairwise distinct coordinates and real polynomial fields Φ_j, satisfies $2H+1 = -\Delta_S + W$ with $W = \sum_j \Phi_j^2 + 1 \geq 1$, hence is essentially self-adjoint (`weylPoly_esa`; the passage from $2H+1$ back to H uses that trivial deficiency at $\pm i$ gives trivial deficiency at every non-real point). The non-abelian Yang–Mills Hamiltonian is such an operator, for arbitrary real structure constants (`ym_h_esa`, a one-particle statement), and so is every particle-number sector of the Standard-Model Hamiltonian of record. `BookProof/Sm/ChapterSmFockEsa.lean` records the consequences: each sector `smSectorHam_esa`, the outer operator $smFockHam = d\Gamma(h)$ on $\bigoplus_n L^2(\mathbb{R}^{163n})$ essentially self-adjoint on its finite-particle core (`smFockHam_esa`), and, in the creation-left / annihilation-right spelling of the general second quantization, `sm_dGamma_esa` — the lift `dGamma_essentiallySelfAdjointOn_of_esa` applied to `sm_h_esa`. The enclosed operator here is the bosonic sector; the Dirac and Yukawa operators act on the CAR algebra, and the assembly of the full h is the next section. No spectral statement follows.

```
#check @BookProof.YangMillsNonAbelianEsa.deficiencyTrivialAt_of_esa
#check @BookProof.YangMillsNonAbelianEsa.essentiallySelfAdjointOn_affine
#check @BookProof.YangMillsNonAbelianEsa.weylPoly_eq_hamCoreS
#check @BookProof.YangMillsNonAbelianEsa.weylPoly_esa
#check @BookProof.YangMillsNonAbelianEsa.ym_h_esa
#check @BookProof.YangMillsNonAbelianEsa.ym_dGamma_esa
#check @BookProof.SmFockEsa.smSectorHam_esa
#check @BookProof.SmFockEsa.smFockHam_esa
#check @BookProof.SmFockEsa.sm_dGamma_esa
```

38.13. The Full One-Particle Hamiltonian: Bosonic and Fermionic Sectors Together

`BookProof/Sm/ChapterSmFullEnclosure.lean` assembles the full one-particle Hamiltonian before enclosure. The one-particle space is the completed tensor product $L^2(\mathbb{R}^{163}) \hat{\otimes} \mathcal{F}_{\mathrm{CAR}}$ of the bosonic space with the (finite-mode) fermionic Fock space, and the Hamiltonian is the tensor sum $h_{\mathrm{full}} = h_B \otimes 1 + 1 \otimes (h_{\mathrm{Dirac}} + h_{\mathrm{Yukawa}})$ (`smFullHam`, checked on elementary tensors by `smFullHam_tmul`). The two-factor theorem of `ChapterTensorSumEsa`, fed with `sm_h_esa` and the Faris–Lavine certificate `sm_fermi_esa`, gives essential self-adjointness on the tensor product of the Gauss–polynomial core with the fermionic space (`smFull_h_esa`, a one-particle statement), and the lift gives the enclosure: $d\Gamma(h_{\mathrm{full}})$, creation on the left and annihilation on the right, is essentially self-adjoint on the finite-particle domain (`smFull_dGamma_esa`). The Yukawa term is taken in a

fixed Higgs background, as `smYukawa` is defined, so no boson–fermion coupling operator is included: a tensor sum is not an interaction. No spectral statement follows.

```
#check @BookProof.SmFullEnclosure.smFullHam
#check @BookProof.SmFullEnclosure.smFullHam_tmul
#check @BookProof.SmFullEnclosure.smFull_h_esa
#check @BookProof.SmFullEnclosure.smFull_dGamma_esa
```

38.14. The Yukawa Coupling as an Operator

`BookProof/Sm/ChapterSmYukawaCoupling.lean` replaces the constant Higgs value in the Yukawa term by the Higgs field itself. The fixed-background operator splits as $h_{\mathrm{Yukawa}}(z) = \mathrm{Re}\, z\, h_{\mathrm{Yukawa}}(1) + \mathrm{Im}\, z\, h_{\mathrm{Yukawa}}(i)$ (`smYukawa_eq_re_im`), and the coupling $H_Y = \varphi_0 \otimes h_{\mathrm{Yukawa}}(1) + \varphi_1 \otimes h_{\mathrm{Yukawa}}(i)$ multiplies two real Higgs components on the bosonic side by the corresponding fermion bilinears: a genuine boson–fermion interaction, not a tensor sum. For a positive quartic $\lambda > 0$ the Higgs wall confines the Higgs coordinates (`higgs_sq_le`), so multiplication by a Higgs component is bounded relative to h_B with arbitrarily small relative bound (`higgsMul_relBound`). Because the fermionic factor is finite-dimensional, a Kato–Rellich theorem for product couplings on a tensor sum (`essentiallySelfAdjointOn_tensorSum_add_coupling`) applies: $h_B \otimes 1 + 1 \otimes h_{\mathrm{Dirac}} + H_Y$ is essentially self-adjoint on the same core (`smYukawa_h_esa`), and its enclosure is essentially self-adjoint on the finite-particle domain (`smYukawa_dGamma_esa`). Which two components of the doublet carry the coupling is a modelling choice.

```
#check @BookProof.SmYukawaCoupling.smYukawa_eq_re_im
#check @BookProof.SmYukawaCoupling.higgs_sq_le
#check @BookProof.SmYukawaCoupling.higgsMul_relBound
#check @BookProof.TensorKatoRellich.essentiallySelfAdjointOn_tensorSum_add_coupling
#check @BookProof.SmYukawaCoupling.smYukawaFullHam
#check @BookProof.SmYukawaCoupling.smYukawa_h_esa
#check @BookProof.SmYukawaCoupling.smYukawa_dGamma_esa
```

38.15. The Higgs Vacuum, as Statics

Symmetry breaking has a static half that is a theorem about the potential. The Mexican-hat potential is bounded below by $-\mu^4/4\lambda$ and attains that value exactly on the vacuum manifold $\|\varphi\|^2 = \mu^2/\lambda$. Transverse to a vacuum the potential has no quadratic term at all — it rises as $(\lambda/4)t^4\|w\|^4$, the flatness of the Goldstone directions — while along the radial direction it is exactly $V(u) + \mu^2\|u\|^2t^2(1 + t + t^2/4)$, a curvature $m^2 = 2\mu^2$. And the covariant-derivative term at a constant vacuum is a positive semidefinite quadratic form in the gauge fields which vanishes precisely on the generator combinations that annihilate the vacuum: the broken generators acquire a mass term, the unbroken ones do not.

```
#check @BookProof.SmHiggsVacuum.higgsV
#check @BookProof.SmHiggsVacuum.higgsV_sq_form
#check @BookProof.SmHiggsVacuum.higgsV_ge_min
#check @BookProof.SmHiggsVacuum.higgsV_eq_min_iff
#check @BookProof.SmHiggsVacuum.higgs_goldstone
#check @BookProof.SmHiggsVacuum.higgs_radial
#check @BookProof.SmHiggsVacuum.gaugeMassForm
#check @BookProof.SmHiggsVacuum.gaugeMassForm_nonneg
#check @BookProof.SmHiggsVacuum.gaugeMassForm_eq_zero_iff
```

38.16. The Continuum CAR Algebra

The finite mode set of the CAR section was the honest boundary: the algebra of the fermions was over n\$ modes, not over the continuum. That restriction is gone. The mode set is now $\mathbb{N}$\$, the one-particle space the infinite-dimensional $\ell^2(\mathbb{N})$\$, and the Fock space the antisymmetric Fock space in its occupation-number presentation $\ell^2(\mathrm{Finset}\,\mathbb{N})$\$ — one basis vector per **finite** occupied set, with the same Jordan–Wigner signs. The single-mode ladder operators carry the four anticommutation relations and mutual adjointness as before, but the operators that matter are the **smeared** ones $a^\dagger(f)$\$, $a(f)$\$ for an arbitrary $f \in \ell^2(\mathbb{N})$\$: since $\sum_i |f_i|$\$ may diverge, $a^\dagger(f)$\$ is not an absolutely convergent sum of single-mode operators, and its boundedness $\|a^\dagger(f)\| \le \|f\|$\$ is the **fermionic** boundedness with no bosonic analogue. The canonical relation holds in the smeared form

$\{a(f), a^\dagger(g)\} = \langle f, g\rangle \cdot 1$, the map $f \mapsto a^\dagger(f)\Omega$ is a linear isometry of $\ell^2(\mathbb{N})$ into the Fock space — so the algebra really is over an infinite-dimensional one-particle space — and the same relations are proved over an **arbitrary** separable one-particle Hilbert space presented by a Hilbert basis indexed by $\mathbb{N}$, in particular over the $L^2(\mathbb{R}^3)$ of a continuum field.

What is **not** claimed here: the representation constructed is the Fock representation on the vacuum; nothing is said about inequivalent representations, about the C*-completion as an abstract algebra, or about a Hamiltonian on this space.

```
#check @BookProof.SmCarContinuum.Ell2
#check @BookProof.SmCarContinuum.CFock
#check @BookProof.SmCarContinuum.cAnn
#check @BookProof.SmCarContinuum.cCre
#check @BookProof.SmCarContinuum.car_cAnn_cCre_self
#check @BookProof.SmCarContinuum.car_cAnn_cCre_of_ne
#check @BookProof.SmCarContinuum.car_cAnn_cAnn
#check @BookProof.SmCarContinuum.car_cCre_cCre
#check @BookProof.SmCarContinuum.inner_cCre_left
#check @BookProof.SmCarContinuum.cCreS
#check @BookProof.SmCarContinuum.cAnnS
#check @BookProof.SmCarContinuum.norm_cCreS_le
#check @BookProof.SmCarContinuum.norm_cAnnS_le
#check @BookProof.SmCarContinuum.cCreS_add
#check @BookProof.SmCarContinuum.cCreS_smul
#check @BookProof.SmCarContinuum.car_smeared
#check @BookProof.SmCarContinuum.car_cCreS_cCreS
#check @BookProof.SmCarContinuum.car_cAnnS_cAnnS
#check @BookProof.SmCarContinuum.oneParticleIsometry
#check @BookProof.SmCarContinuum.norm_cCreS_eq
#check @BookProof.SmCarContinuum.car_hilbert
#check @BookProof.SmCarContinuum.car_hilbert_cre
#check @BookProof.SmCarContinuum.norm_carCre_le
```

38.17. The Gauge Connection

Minimal coupling, for one plane-wave matter mode in the constant collective-coordinate background: $D_j = i(k_j + A_j)$ with $A_j = g\sum_a A_j^a T_a$. The connection is Hermitian and D is anti-Hermitian; at zero coupling $D_j = ik_j$. The commutator $[D_j, D_l]$ is — up to a factor — $-ig^2\sum_c(\sum_{a,b} f_{abc}A_j^a A_l^b)T_c$, the non-abelian field strength: the operator form of $W_{\mu\nu}^j = -(i/g)\,\mathrm{tr}([D_\mu, D_\nu]\tau^j)$ of the manuscript and of the $g_s f^{abc}G_j^b G_k^c$ term of the magnetic energy, now **inside** the covariant derivative rather than asserted about it. A constant abelian background is pure gauge, conjugating D by a unitary conjugates the connection, and if the adjoint action rotates the generators by a real matrix the conjugated connection is again a connection with rotated components. On that base sits the gauged Dirac one-particle matrix — Hermitian, equal to the free matrix of the spinor section at zero field, splitting as free plus interaction — and its second quantization over the twelve colour–spinor modes is essentially self-adjoint through the very same three Faris–Lavine hypotheses that certificate the ungaugeed operator.

What is **not** claimed here: the background is constant, so D_j carries ik_j rather than an unbounded derivative, and this module's commutator $[D_j, D_l]$ therefore sees only the non-abelian part of F ($\partial A = 0$); the full $F_{\mu\nu} = \partial_\mu A_\nu - \partial_\nu A_\mu + [A_\mu, A_\nu]$ is in the tree as $smMagG$/$smMagW$/$smMagB$, but there is no gauge-field dynamics and no continuum limit of D. (The curl/continuum half is reachable by the momentum-space convolution already used for NS and QG — not a QYM device, only for proofs of this type.)

```
#check @BookProof.SmGaugeConnection.conn
#check @BookProof.SmGaugeConnection.conn_conjTranspose
#check @BookProof.SmGaugeConnection.covD_conjTranspose
#check @BookProof.SmGaugeConnection.covD_zero_coupling
#check @BookProof.SmGaugeConnection.covD_commutator
#check @BookProof.SmGaugeConnection.covD_commutator_abelian
#check @BookProof.SmGaugeConnection.covD_conj
#check @BookProof.SmGaugeConnection.conn_gauge_transform
#check @BookProof.SmGaugeConnection.diracGaugeMat
#check @BookProof.SmGaugeConnection.diracGaugeMat_conjTranspose
```

```
#check @BookProof.SmGaugeConnection.diracGaugeMat_free
#check @BookProof.SmGaugeConnection.diracGaugeMat_split
#check @BookProof.SmGaugeConnection.diracGaugeField
#check @BookProof.SmGaugeConnection.diracGaugeField_symmetric
#check @BookProof.SmGaugeConnection.dirac_gauge_field_esa
```

38.18. The Ghosts, the Generators, and the BRST Charge

The ghost/BRST sector needed structure constants before it could have a charge. The structure constants of $su(3) \oplus su(2) \oplus u(1)$ are assembled from an $su(3)$ family, the Levi-Civita $su(2)$ family and the abelian one, and their antisymmetry and Jacobi identity are theorems, not data. Twelve ghosts live as the last twelve modes of the CAR algebra, with the ghost-number operator and its commutators with the ghosts. The bridge from matrices to operators is the identity $[d\Gamma(A), d\Gamma(B)] = d\Gamma([A,B])$: the Gauss-law generators $G_a = -i\, d\Gamma(T_a)$ close with the **real** structure constants, and they commute with the ghosts — they are even. With that in hand the abstract charge $\Omega = \sum_a c^a G_a - \frac{1}{2} f_{abc} c^a c^b b_c$ satisfies $\Omega^2 = 0$ in any ring carrying ghosts with the CAR relations and first-class constraints, hence on the joint matter–ghost Fock space of the Standard Model.

The representation-level hypothesis is removed one step further: the Pauli generators $\tau_k/2$ close with ε_{jkl}; the twelve generators of one coloured weak doublet — $T_a \otimes 1$, $1 \otimes \tau_k/2$, $y \cdot 1$ — close with the structure constants; and on the eighteen-mode Fock space (six matter modes, twelve ghosts) the Standard-Model BRST charge of that multiplet squares to zero, with the antisymmetry and Jacobi identity of the structure constants **derived** from the two defining relations of the $su(3)$ generators rather than assumed.

```
#check @BookProof.SmBrstGhost.smStruct
#check @BookProof.SmBrstGhost.smStruct_antisymm
#check @BookProof.SmBrstGhost.smStruct_jacobi
#check @BookProof.SmBrstGhost.ghostCre
#check @BookProof.SmBrstGhost.ghostAnn
#check @BookProof.SmBrstGhost.smGhostCAR
#check @BookProof.SmBrstGhost.ghostNumber
#check @BookProof.SmBrstGhost.ghostNumber_occ
#check @BookProof.SmBrstGhost.fermiBilin_lie
#check @BookProof.SmBrstGhost.matterGen_lie
#check @BookProof.SmBrstGhost.matterGen_comm_ghostCre
#check @BookProof.SmBrstGhost.matterGen_comm_ghostAnn
#check @BookProof.SmBrstGhost.smBrstCharge
#check @BookProof.SmBrstGhost.smBrstCharge_nilpotent
#check @BookProof.SmGaugeRep.su2gen
#check @BookProof.SmGaugeRep.su2gen_closes
#check @BookProof.SmGaugeRep.smGenP
#check @BookProof.SmGaugeRep.smGen
#check @BookProof.SmGaugeRep.smGen_closes
#check @BookProof.SmGaugeRep.sm_brst_nilpotent_of_su3_relations
#check @BookProof.SmGaugeRep.sm_brst_nilpotent_rep
```

38.19. The BRST Charge as the Book Defines It

The charge of the previous section is the abstract one, with the constraints given as data. The manuscript itself defines the charge by a **density** — $\Omega(x) = \pi^\mu_a \partial_\mu \psi^\dagger_a - \pi^\mu_a f_{abc} A_{\mu b} \psi^\dagger_c - (i/2) f_{abc} \psi^\dagger_a \psi^\dagger_b \psi_c$, together with $[A_{\mu a}, \pi^\nu_b] = i\delta^\nu_\mu \delta_{ab}$ and $\{\psi_a, \psi^\dagger_b\} = \delta_{ab}$ — and that definition is now implemented on the graded state space $\mathbb{C}[A_{\mu a}] \otimes \Lambda(\mathbb{C}^N)$: the canonical relations hold in the realization, the bosonic operators commute with the ghosts, and the three terms of the density are assembled into the book's charge, with the derivative term written through the derivations ∂_μ of the gauge algebra.

The decisive step is that the **Gauss law is derived, not assumed**: the coefficient of the ghost $\psi^\dagger_c$ in the book's charge closes into the gauge algebra, from the canonical relations, the Leibniz property of ∂_μ and the Jacobi identity. Nilpotency $\Omega^2 = 0$ follows, and identifying the book's charge with i times the abstract charge shows that the book's coefficient $-i/2$ of the cubic ghost term is exactly the one nilpotency requires. On that foundation the gauge-fixing fermion $\Psi = i\psi_a A_{0a}$ of the manuscript's Weyl-gauge section is built, the generated term $\{\Omega, \Psi\}$ is computed in closed form — the Gauss law contracted with A_0, the ghost kinetic term, the non-abelian ghost–field coupling — and is proved BRST invariant, which is what allows the manuscript to add it to the

Hamiltonian. On the temporal-gauge slice $A_0 = 0$\$ that entire BRST-exact term vanishes — $\Omega, \Psi = 0$\$ — so the records of h\$ and N\$ need no extra gauge-fixing or ghost summand (the accounting identity $bookGfTerm_eq_zero_of_field0$\$, with the $su(2)$\$ and Standard-Model instances; Cadabra CHECK 29–30). For pure Yang–Mills the same identity lets the Weyl gauge work with the spatial field components only. For quantum gravity the manuscript's three-dimensional reduction is **not** the ADM one: the time-like vector is fixed to $v^\mu = \delta^\mu_0$\$, the ghosts of the resulting charge are constant in the timepiece, and the charge keeps the same functional form as the four-dimensional one — explicitly different from the ADM BRST charge. The BRST cohomology is well defined: $\ker \Omega / \mathrm{im}\Omega$\$, with $\mathrm{im}\Omega \subseteq \ker \Omega$\$ from $\Omega^2 = 0$\$, the gauge-fixing term preserving both subspaces so that it descends to the quotient, and gauge-invariant observables — anything commuting with the constraints and the ghosts — commuting with the charge, exemplified by the Casimir of constant gauge fields. The $su(2)$\$ and Standard-Model instances supply the concrete algebras, their nilpotency, and the fact that the constraints are not the zero operator.

What is **not** claimed here: the local gauge algebra is modelled by a finite-dimensional Lie algebra with derivations ∂_μ\$ (the manuscript's totally antisymmetric $SU(N)$\$ structure constants are used), so the Gauss law is local in that algebra rather than at every point of a continuum space; the manuscript's operator formalism uses no Faddeev–Popov determinant and none is constructed; and, while the cohomology is defined and shown well defined, nothing is claimed about its **size**.

```
#check @BookProof.BookBrstYangMills.GaugeAlgebra
#check @BookProof.BookBrstYangMills.Afield
#check @BookProof.BookBrstYangMills.mom
#check @BookProof.BookBrstYangMills.chiOp
#check @BookProof.BookBrstYangMills.betaOp
#check @BookProof.BookBrstYangMills.bookCCR
#check @BookProof.BookBrstYangMills.bookGhostCar
#check @BookProof.BookBrstYangMills.bookOmega
#check @BookProof.BookBrstYangMills.gaussGen_bracket
#check @BookProof.BookBrstYangMills.bookOmega_eq_brstCharge
#check @BookProof.BookBrstYangMills.bookOmega_nilpotent
#check @BookProof.BookBrstYangMills.bookConstraintAlgebra
#check @BookProof.BookBrstGaugeFixing.bookGfFermion
#check @BookProof.BookBrstGaugeFixing.bookGfTerm
#check @BookProof.BookBrstGaugeFixing.bookGfTerm_eq
#check @BookProof.BookBrstGaugeFixing.bookGfTerm_brst_closed
#check @BookProof.BookBrstGaugeFixing.gfFermion_eq_zero
#check @BookProof.BookBrstGaugeFixing.bookGfFermion_eq_zero_of_Afield0
#check @BookProof.BookBrstGaugeFixing.bookGfTerm_eq_zero_of_Afield0
#check @BookProof.BookBrstGaugeFixing.physicalStates
#check @BookProof.BookBrstGaugeFixing.exactStates
#check @BookProof.BookBrstGaugeFixing.exactStates_le_physicalStates
#check @BookProof.BookBrstGaugeFixing.brstCohomology
#check @BookProof.BookBrstGaugeFixing.bookGfTerm_mem_physicalStates
#check @BookProof.BookBrstGaugeFixing.bookGfTerm_mem_exactStates
#check @BookProof.BookBrstGaugeFixing.bookOmega_comm_multOp
#check @BookProof.BookBrstGaugeFixing.casimir_bookOmega_comm
#check @BookProof.BookBrstInstances.innerDeriv_leibniz
#check @BookProof.BookBrstInstances.su2BookAlgebra
#check @BookProof.BookBrstInstances.smBookAlgebra
#check @BookProof.BookBrstInstances.su2_bookOmega_nilpotent
#check @BookProof.BookBrstInstances.sm_bookOmega_nilpotent
#check @BookProof.BookBrstInstances.su2_gaussGenPoly_ne_zero
#check @BookProof.BookBrstInstances.su2_casimir_bookOmega_comm
#check @BookProof.BookBrstInstances.su2_bookGfTerm_eq_zero_of_Afield0
#check @BookProof.BookBrstInstances.sm_bookGfTerm_eq_zero_of_Afield0
```

38.20. What Is Not Claimed

One item has left this list: essential self-adjointness of the **bosonic** one-particle operator is now unconditional. The fermionic theorems are $sm_fermi_fl_i$\$/*ii*\$/*iii*\$ and sm_fermi_esa\$. For the bosons, `ChapterSmFarisLavine.lean` proves both Faris–Lavine inequalities with absolute constants against the comparison operator $N = 2h + \sum_m q_m^2 + c_0$\$ and reduces the conclusion to essential self-adjointness of N\$ itself on the same core — a coupled quartic Schrödinger operator in 163 variables, an input of Kato type that

Faris–Lavine structurally cannot supply. That input is proved independently: `ChapterDegKatoEsa.lean` shows that $-\Delta_S + W$ with $W \geq 1$ smooth is essentially self-adjoint on compactly supported smooth functions, `ChapterHermiteGraphApprox.lean` transfers this to the Gauss–polynomial core for polynomial W, and `ChapterSmComparisonEsa.lean` identifies N with such an operator, giving `smFlN_esa` and the unconditional `sm_h_esa` (see `CONSOLIDATED_PLAN.md` §2026-09-23g).

The boundaries of this chapter are as much a part of it as its theorems. The CAR algebra, the Dirac and Yukawa operators, the continuum CAR algebra over an infinite-dimensional one-particle space, the gauge connection inside D, and the ghost/BRST sector (including the charge as the manuscript defines it and the accounting identity that adds no gauge-fixing summand to h or N) are all Lean theorems now — see the sections above and the ledger `HONEST_BOUNDARIES_SM.md` (live list §L). Still open there: Majorana masses and the see-saw; no continuum spectrum or mass gap for $d\Gamma(h)$, no QCD mass gap; electroweak symmetry breaking as statics only — vacuum manifold, Goldstone flatness, radial curvature, mass form of broken generators — not dynamics (no expansion of the quantum Hamiltonian around the vacuum and no gauge-boson mass value); no measured mixing parameter: of the CKM and PMNS matrices only the unitarity identities and the entry bound are formalized; and no size is claimed for the BRST cohomology.

39. Shift-Invert Selection for the Navier–Stokes Generator

39.1. Orientation and Status

The selection-theorem chapter for the Navier–Stokes fiber generator: the inverse-free rational-Krylov (Hashimoto/SIRK) iteration is trustworthy exactly when the shift-inverted resolvents of the generator are well-behaved, and a selection theorem **selects the operator the Krylov iteration is actually computing**. Read it after the operator pair (one-particle Hamiltonian, comparison operator) and before the numerics tail of the part (SIRK reliability, NS numerics) that consumes it.

The route: why a selection theorem is needed, what the forward sequence is doing intuitively, the differential generator as the transport of a polynomial, the unique self-adjoint extension, the Hashimoto selection itself, how this connects to the numerical validation, and what the Eulerian fiber can certify (the absence of a form gap, and the comparison operator).

The proofs are `BookProof/ChapterH` and the `BookProof/NavierStokesFlow` namespace; the numerics this feeds are the companion solver work described in the NS numerics chapter.

39.2. Why a Selection Theorem

The inverse-free rational-Krylov (Hashimoto/SIRK) method never forms the resolvent $(H - z)^{-1}$: it works with the **forward** sequence $w_k = (H - z_k I) w_{k-1}$ and reconstructs spectral information from the Gram matrix of the Krylov subspace. The method is trustworthy exactly when the shift-inverted resolvents $X_j = (gamma_j - G)^{-1}$ of the generator G are well-behaved — bounded, commuting, satisfying the resolvent identity and the Hashimoto–Nodera SIRK relation, and recovered by Galerkin truncation. Proving those properties for a **concrete** Hamiltonian is a selection theorem: it selects the operator the Krylov iteration is actually computing.

For the Navier–Stokes fiber generator the selection was proved in the abstract sequence-space realization: the Hermite matrix `velCore` on $ell^2(Vel)$ with $Vel = operatorname{Fin} 3 to mathbb{N}$. The **differential** realization — the same Hamiltonian written with $pi_i = -i$, $partial/partial u_i$ and u_i a genuine multiplication operator on the Gauss–polynomial (product Hermite) core of $L^2(du_1 du_2 du_3)$ — was known to be essentially self-adjoint but carried no selection theorem. The module

`BookProof/Ns/ChapterNavierStokesDiffHashimoto.lean` supplies it (namespace `BookProof.NavierStokesFlow`).

39.3. What the Forward Sequence Is Doing, Intuitively

Before the formal statements, it helps to see why the resolvents $X_j = ($
$gamma_j - G)^{-1}$ are the right object at all. The SIRK method shifts by a complex number
$gamma_j$ — **not** an eigenvalue — purely to make the inversion well-behaved:
$gamma_j - G$ is invertible because
$gamma_j$ has a non-zero imaginary part, so no point of the (real) spectrum can coincide with the shift. The inverse then exists as a bounded operator, and the forward sequence $w_k = (H - z_k I) w_{k-1}$ accumulates information about it. If the resolvents were **not** bounded, or did **not** commute, or were **not** recovered by Galerkin truncation, then the Krylov subspace would silently compute something else — a different operator, or an artifact of the finite-dimensional cut. The selection theorem rules all of that out for the Navier–Stokes generator.

A concrete miniature makes the mechanism visible. Take a single velocity mode with the fiber Hamiltonian $G =$
$tfrac12($
$piV + V$
$pi)$ where $V =$
$kappau + c$ (an affine drift), and shift by
$gamma = i$ (imaginary part 1). The resolvent bound in the theorem says
$|(i - G)^{-1}$
$|$
$le1/|$
$operatornameImi| = 1$: the inverted operator shrinks every vector, which is precisely what makes the Krylov recurrence stable and the Gram matrix well-conditioned. The same bound holds for **every** non-real shift, uniformly, and it is exactly the estimate that the SIRK implementations in the companion numerical validation (`fock_sirk/tests/ns_numerical_validation.rs`, S35) rely on when they evolve the laminar decay
$du/dt = -$
$nuk^2 u$ and measure the decay rate
nuk^2 to high precision.

39.4. The Differential Generator is the Transport of a Polynomial

The Weyl-ordered Hamiltonian — the sum of the symmetrized products
$tfrac12($
$pi_i V_i + V_i$
$pi_i)$ of the momentum with the affine fiber field — is, on the Hermite core, the transport of a polynomial-level operator: `nsDiffPoly` is the polynomial whose core realization is exactly `nsDiffH`. Gauss symmetry of the polynomial is what makes the operator symmetric on the Hermite core of $L^2($
$mathbbR^3)$.

```
#check @BookProof.NavierStokesFlow.nsDiffPoly
#check @BookProof.NavierStokesFlow.nsDiffH_eq_coreOp
#check @BookProof.NavierStokesFlow.nsDiffPoly_polySym
#check @BookProof.NavierStokesFlow.nsDiffH_symmetricOn
```

The route through the polynomial is not a detour: it is how the symmetry is **seen**. On the Hermite core the momentum is
$pi_i = -i$
$partial_i$ and the field is multiplication by the **affine-linear** polynomial $V_i(u) =$
$sum_k A_ik u_k + c_i$ — a general velocity-gradient matrix A (the strain and vorticity content of $u_{i,j}$) plus the constant term c_i. The symmetrized product is then the **symmetrization of a polynomial operator** — the statement `nsDiffPoly_polySym` — and Gauss symmetry of that polynomial is checked purely algebraically, with no analysis. All the analytic work (domains, deficiency spaces, closures) is then done once, on the polynomial transport.

39.5. A Unique Self-Adjoint Extension

Symmetry alone is not enough: a symmetric operator may have many self-adjoint extensions. Here there is exactly one — the closure. Note that this is **not** a positivity argument: the strain, vorticity and constant hoppings carry arbitrary signs, so the Hamiltonian is genuinely indefinite; the uniqueness comes from the analysis of the deficiency spaces on the core.

```
#check @BookProof.NavierStokesFlow.nsDiffH_selfAdjoint_extension
#check @BookProof.NavierStokesFlow.nsDiffH_selfAdjoint_extension_unique
```

Why uniqueness matters for the numerics: a symmetric operator with several self-adjoint extensions is a Hamiltonian with several **different** time evolutions — the Stone theorem would not know which unitary group to use. The SIRK solver can only compute one of them, so a multiplicity of extensions would mean the numerical flow is computing an arbitrary choice. The theorem that the closure is the **unique** self-adjoint extension removes the ambiguity: there is only one candidate, and the Krylov iteration, if it converges at all, converges to it.

39.6. The Hashimoto Selection

The headline: for an arbitrary sequence of non-real shifts
$gamma_j$, the shift-inverted resolvents $X_j = ($
$gamma_j - G)^{-1}$ of the differential Navier–Stokes generator exist, are bounded by $1/|$
$operatorname{Im}$
$gamma_j|$, share the domain of G, satisfy the resolvent identity, commute, satisfy the Hashimoto–Nodera SIRK relation, have strongly convergent Galerkin truncations, and each one alone determines G completely. The single-shift form is the version used in practice, and the explicit coefficients $($
$nu, u_{i,j}, u_{i,jj})$ spell the same statement out in the physical variables.

```
#check @BookProof.NavierStokesFlow.nsDiffH_hashimoto_selects
#check @BookProof.NavierStokesFlow.nsDiffH_shiftInvert_selects
#check @BookProof.NavierStokesFlow.nsQuadraticDiffH_hashimoto_selects
#check @BookProof.NavierStokesFlow.exists_l2dHilbertBasisNat
```

Reading the single-shift statement `nsDiffH_shiftInvert_selects`: fix one shift
$gamma$ with
$operatorname{Im}$
$gamma$
$neq 0$. The theorem says the single resolvent $X = ($
$gamma - G)^{-1}$ **determines** G — because the resolvent identity lets you recover the spectrum and the spectral projections from one resolvent value. In practice this means the SIRK method needs only the forward sequence at the chosen shifts; it does not need to **approximate** the resolvent by anything else. And `nsQuadraticDiffH_hashimoto_selects` gives the same guarantee in the notation a fluid dynamicist writes down: the viscosity
nu, the velocity-gradient matrix $u_{i,j}$ and the second derivatives $u_{i,jj}$ appear explicitly in the statement.

Non-vacuity is part of the statement: $L^2($
$mathbb{R}^3)$ carries an
$mathbb{N}$-indexed Hilbert basis — the product Hermite functions, enumerated — so the Galerkin truncations in the theorem are honest finite-dimensional projections of a concrete Hilbert space. Together with the Carleman flux criteria (`Book/CarlemanFlux.lean`) that underpin the core analysis, this is what makes the inverse-free SIRK numerics of the Navier–Stokes validation a computation of **this** operator rather than of an artefact of the discretization.

The algebra that the selection theorem **uses** is proved outright in the companion chapter `BookProof/Textbook/ChapterH1.lean` (namespace `BookProof.ChapterH1`), with no analytic hypotheses of its own: the φ-functions $\varphi_k(z) = \int_0^1 e^{sz}(1-s)^k/k!\,ds$ of the SIRK method, their recurrence `phi_succ_mul` and their values (`phi_zero`, `phi_one`), the numerical range `numericalRange` with the easy half of Toeplitz–Hausdorff (`eigenvalue_mem_numericalRange`), the resolvent identity `resolvent_identity`, and the shift relation `resolvent_shift_mul` that makes the shifted resolvents multiply into one another. The deep error

theorems that would need Crouzeix's inequality are **named hypotheses** in ChapterH2, never axioms — which is why everything in ChapterH1 can be proved outright. The Duhamel companion duhamel_phiOp1 is what connects the φ-functions back to the operator exponential the flow actually is.

```
#check @BookProof.ChapterH1.phi
#check @BookProof.ChapterH1.phi_succ_mul
#check @BookProof.ChapterH1.phi_one
#check @BookProof.ChapterH1.numericalRange
#check @BookProof.ChapterH1.eigenvalue_mem_numericalRange
#check @BookProof.ChapterH1.resolvent_identity
#check @BookProof.ChapterH1.resolvent_shift_mul
#check @BookProof.ChapterH1.duhamel_phiOp1
```

39.7. How This Connects to the Numerical Validation

The selection theorem is the **raison d'être** of the SIRK tests in the companion repository. Two features of those tests are direct consequences of what is proved here:

- **The Ehrenfest identity is exact on probes.** The test ns_sirk_laminar_decay_rate verifies i $langle[H, u]$ $rangle = 4$ $kappa$ $langleu$ $rangle + 4c$ exactly on occupation/coherent probes. This is a statement about the **operator** H — the one the selection theorem pins down — not about any finite-dimensional truncation. The identity holds because the Weyl-ordered symmetrization V, pi carries the factor two, which is precisely the polynomial transport nsDiffH_eq_coreOp records.
- The decay rate νk^2 is reproduced to < 1e-2 (with the viscosity in the analytic window). With $kappa = -$ $nuk^2/4$, the Ehrenfest identity reduces to the classical Navier–Stokes decay $du/dt = -$ nuk^2u, and the SIRK-restarted evolution measures it. That the measured rate agrees with the analytic one is the numerical shadow of nsDiffH_hashimoto_selects: the Krylov iteration is computing the resolvent of **the** differential generator, whose spectrum contains exactly the expected decay rates.

The companion tests ns_derivative_variable_fixing.rs (and the higher-Hermite extension) verify the promoted derivative variables $g_m = 2(m+1)u_{m+1}$ against the field derivatives, and check that the bare and BRST-projected SIRK flows give identical observables. These are statements **about the physical subspace** of the same operator — the one selected here. The selection theorem is what guarantees that the whole edifice (Ehrenfest identities, gauge-fixed derivative variables, energy conservation) is a computation about the Navier–Stokes generator in L^2 rather than about some discretization artifact.

39.8. What the Eulerian Fiber Can Certify: the Absence of a Form Gap, and the Comparison Operator

The shift-invert selection theorems above identify **which operator** the Krylov iteration computes. A separate question — the one the gap programme asks of every sector — is whether that operator has a positive spectral gap above its ground state. For the Navier–Stokes Eulerian fiber the answer is **negative and provable**: BookProof/Ns/ChapterNavierStokesFiberGap.lean (namespace BookProof.NavierStokesFlow) shows that the fiber Hamiltonian has **no one-particle form gap at any positive level**, and explains why none of the gap instruments that other sectors enjoy applies here.

```
#check @BookProof.NavierStokesFlow.nsFiber_quadForm_coreState
#check @BookProof.NavierStokesFlow.nsFiber_no_form_gap
#check @BookProof.NavierStokesFlow.nsComparison_friedrichs_gap
```

The mechanism is elementary and structural. The Eulerian fiber Hamiltonian on the Hermite core is a sum of symmetrized products of the ladder pairs — the velocity creation and annihilation operators — and its quadratic form vanishes on **every** Hermite basis state (nsFiber_quadForm_coreState, via the ladder normal form and

the vanishing of the diagonal coefficients $c_{\mathrm{Fun}}(A,i,i) = c_{\mathrm{Rot}}(A,i,i) = 0$). A form that vanishes on the basis vectors of a dense core cannot bound any positive multiple of the norm from below: `nsFiber_no_form_gap` rules out $\mu\|x\|^2 \le \langle x, H_{\mathrm{fiber}}x\rangle$ for every $\mu > 0$. Physically this is the expected structure — the strain and vorticity hoppings are off-diagonal and sign-indefinite, so there is no confining potential in the fiber variables to push the Rayleigh quotient away from zero.

Consequences: the **constant** and **diagonal** gap chains (`ChapterScalaronFockGapChain`, `ChapterFockDiagonalGapChain` — the instruments that make the scalaron and free massive sectors unconditional) do **not** apply to this fiber. What the fiber **can** certify is a weaker but still self-adjointness-relevant statement: `nsComparison_friedrichs_gap` shows that the Faris–Lavine comparison operator N_μ has a positive self-adjoint (Friedrichs) extension — the one the Hashimoto shift-invert selects at $\gamma = 1$ — with the bound $\langle y, Ny\rangle \ge \|y\|^2$ on its whole domain. That is exactly the regularity the SIRK implementation needs to run at all: the comparison operator is well defined, positive, and self-adjoint even though the fiber itself carries no spectral gap. The honest summary: the Eulerian fiber certifies **positivity and self-adjointness of the comparison problem**, and certifies the **absence** of a one-particle form gap — neither a mass gap nor a continuum-gap claim is made for this sector.

40. Carleman Flux Criteria for Lattice Hops

40.1. Orientation and Status

The Carleman-flux chapter: the bookkeeping identity — **flux cancellation** — that keeps the unbounded-operator Carleman linearization honest when hops leave the finite range. It belongs to the Navier–Stokes/Carleman thread of the part and pairs with the Hermite-Carleman machinery the gravity and SIRK chapters use.

The route: the problem, a concrete look at the two hop kinds, the abstract flux cancellation, why this underpins the physics, and beyond finite range: unbounded hops.

The proofs are `BookProof/CarlemanGeneralHop`, `BookProof/CarlemanUnboundedHop`, `BookProof/ChapterCarlemanTwoStep`, and `BookProof/ChapterHermiteCarlemanEsa`.

40.2. The Problem

Essential self-adjointness of a quadratic Hamiltonian on the Hermite core can be proved by a **flux** (Carleman) argument: write the operator on the occupation basis
$alpha$
in
$mathbbN^d$ and show that the recursion it generates has bounded flux across every finite box — the probability current crossing the boundary of a large box must stay controlled as the box grows. The classical criterion covers hops that move a **single** excitation number, by one or by two, which is what every **mode-diagonal** quadratic Hamiltonian produces.

A quadratic Hamiltonian that couples two **distinct** modes — $x_i x_j$,
pi_i
pi_j, x_i
pi_j with i
$neqj$ — produces the hops
$alpha$
$mapsto$
$alpha$
$pm(e_i + e_j)$ and
$alpha$
$mapsto$
$alpha$
$pm(e_i - e_j)$. The second kind is **not monotone**: the shift raises one coordinate while lowering another. The

general flux criterion must handle that. The module
`BookProof/Toolbox/ChapterCarlemanGeneralHop.lean` (namespace
`BookProof.CarlemanGeneralHop`) runs the argument for a hop of the completely general shape
$alpha$
$mapsto$
$alpha + p - m$, with p and m multi-indices, and is the last ingredient of the Hermite-core analysis used by the Navier–Stokes and Yang–Mills generators.

40.3. A Concrete Look at the Two Hop Kinds

To see why the second kind is genuinely harder, write a two-mode occupation as a pair
$alpha = (a_1, a_2)$. A mode-diagonal term x_1^2 produces the hops (a_1, a_2)
$mapsto(a_1$
$pm2, a_2)$ and $(a_1$
$pm1, a_2)$ — the excitation number of mode 1 changes, mode 2 is untouched. This is **monotone** in the total $|$
$alpha| = a_1 + a_2$: the current crossing a box boundary moves in one direction only, and the incoming boundary layer is empty, which is exactly the situation of the earlier single-mode criteria.

A cross-mode term $x_1 x_2$ produces the hop (a_1, a_2)
$mapsto(a_1 + 1, a_2 + 1)$ — both coordinates rise together — but the pair x_1
pi_2 (or its Hermitian partner) produces (a_1, a_2)
$mapsto(a_1 + 1, a_2 - 1)$: mode 1 gains one quantum, mode 2 loses one. The total $a_1 + a_2$ is **unchanged**, so the hop can cross the box boundary in either direction, and a naive one-sided estimate fails. This is the $p - m$ shape with $p = e_1$, $m = e_2$. The general theorem must bound the flux through **both** boundary layers — the outgoing one and the incoming one — and show their contributions cancel up to a controlled remainder.

40.4. The Abstract Flux Cancellation

The heart of the argument is a cancellation that has nothing to do with the specific Hamiltonian. For a Hermitian hop family, the contributions from pairs
$alpha$,
$alpha + p - m$ that both lie inside a finite set A cancel in the imaginary part. The imaginary part of the total flux is therefore carried by two **boundary layers**: the outgoing layer A
$setminusB$ — points of A whose image leaves A — and the incoming layer B
$setminusA$ — points outside A whose image lands in A. For a monotone hop ($m = 0$) the incoming layer is empty, which is exactly the situation of the earlier single-mode criteria.

```
#check @BookProof.CarlemanGeneralHop.hshift
#check @BookProof.CarlemanGeneralHop.hshift_hshift
#check @BookProof.CarlemanGeneralHop.rtG
#check @BookProof.CarlemanGeneralHop.ltG
#check @BookProof.CarlemanGeneralHop.sum_ltG
#check @BookProof.CarlemanGeneralHop.sum_hop_im
#check @BookProof.CarlemanGeneralHop.mem_hopB
```

Reading the statements: `hshift p m a` is the hop $a + p - m$; `hshift_hshift` records that the map is injective enough to run the counting argument (two successive hops with the same image coincide); `rtG` and `ltG` are the two halves of a Hermitian pair — `ltG_eq_conj_rtG` says one is the complex conjugate of the other, which is exactly what makes the interior contributions cancel in the imaginary part; `sum_hop_im` is the boundary-layer decomposition itself. The cancellation is **structural**: it holds for any Hermitian hop family, before any specific coefficients are chosen.

The flux criterion is what converts the algebraic (Gauss) symmetry of a polynomial-level operator — the starting point of `BookProof.ChapterHermiteCarlemanEsa` and `BookProof.ChapterCarlemanTwoStep` — into a genuine essential-self-adjointness statement on the Hermite core: the boundary layers are controlled by the polynomial growth of the coefficients, so the truncations converge strongly and the closure is self-adjoint. The general hop closes the last gap for cross-mode quadratic terms.

40.5. Why This Underpins the Physics

The cross-mode hops are not a curiosity: they are what the physical generators are made of. The Navier–Stokes fiber Hamiltonian contains the products $u_i u_j$ and
$pi_i u_j$ with i
$neqj$ — the strain and vorticity content of the velocity-gradient matrix A — and the Yang–Mills quadratic sector contains the analogous cross-color terms. Each such term generates exactly the non-monotone $p - m$ hops that the general criterion bounds. So the Carleman flux argument is the analytic engine that turns the algebraic symmetry theorems (`nsDiffPoly_polySym`, the Yang–Mills analogue) into the essential self-adjointness statements on which the Hashimoto selection theorem (`Book/NavierStokesHashimoto.lean`) and, through it, the SIRK numerical validation rest.

40.6. Beyond Finite Range: Unbounded Hops

Every criterion above shares one hypothesis: the hop range is **finite**. A term $x_i x_j$ moves an occupation number by one or two, a general
$alpha$
$mapsto$
$alpha + p - m$ moves it by the fixed multi-index $p - m$, and in each case a row of the matrix has finitely many non-zero entries. That is not an accident of the proofs — it is what makes the boundary layer of a box a **finite** set, so that the flux is a finite sum. An operator whose every row is supported on all of
$mathbbN$ has no such boundary layer, and the whole bookkeeping has to be redone with tails instead of layers.

The module `BookProof/Toolbox/ChapterCarlemanUnboundedHop.lean` (namespace `BookProof.CarlemanUnboundedHop`) carries this out. Fix a Hermitian kernel a_{nk} on
$mathbbN$ and a square-summable u solving
$sum_k a_{nk} u_k = z u_n$ at a non-real z. The flux across the cut at N is the double sum over pairs n
$leN < k$, and the flux identity says that
$operatornameImz$
$sum_{nleN} |u_n|^2$ equals its imaginary part — the diagonal, and indeed the whole Hermitian interior, cancels. So the diagonal of the kernel is completely unconstrained: only the off-diagonal decay matters.

```
#check @BookProof.CarlemanUnboundedHop.flux_identity
#check @BookProof.CarlemanUnboundedHop.eq_zero_of_flux_small
#check @BookProof.CarlemanUnboundedHop.two_norm_flux_le
#check @BookProof.CarlemanUnboundedHop.ladder_eq_zero_of_carleman
```

The decay hypothesis is a **hop profile**: a non-negative
$theta$ with $|a_{nk}|$
leA_n
,
$theta(k - n)$ for $n < k$, whose tail sums
$Theta_j =$
$sum_{i>j}$
$theta(i)$ are finite. The bound `two_norm_flux_le` then estimates twice the flux at the cut N by A_N times a **cut mass** — the tails
$Theta$ weighting how much of the
ell^2 mass sits near the cut — and `summable_cutMass` shows that the cut masses are summable, which is where the finite first moment
sum_j
$Theta_j <$
$infty$ of the profile is used. Summability means the cut masses cannot stay large; combined with
$sum_N 1/A_N =$
$infty$ — **Carleman's condition** — there is a cut at which A_N times the cut mass is as small as we please, and `flux_identity` forces $u = 0$. That is `ladder_eq_zero_of_carleman`.

```
#check @BookProof.CarlemanUnboundedHop.kernelOp
#check @BookProof.CarlemanUnboundedHop.kernelOp_symmetric
#check @BookProof.CarlemanUnboundedHop.kernelOp_deficiencyTrivialAt
#check @BookProof.CarlemanUnboundedHop.kernelOp_essentiallySelfAdjoint
#check @BookProof.CarlemanUnboundedHop.kernelOp_stone_flow
```

Transporting this to an operator statement needs one more hypothesis, that each column of the kernel is itself in ell^2, so that the kernel maps the finitely supported states into $ell^2($ $mathbbN)$. With that, `kernelOp` is a symmetric operator on the finite-mode core, `kernelOp_deficiencyTrivialAt` says both deficiency spaces are trivial at every non-real point — a deficiency vector solves exactly the recursion the flux criterion kills — `kernelOp_essentiallySelfAdjoint` concludes essential self-adjointness through the Faris–Lavine interface used everywhere else in this development, and `kernelOp_stone_flow` produces the unitary group.

```
#check @BookProof.CarlemanUnboundedHop.geoHop
#check @BookProof.CarlemanUnboundedHop.geoHop_essentiallySelfAdjoint
#check @BookProof.CarlemanUnboundedHop.geoHop_stone_flow
```

The last three lines exhibit an operator the finite-hop criteria genuinely cannot reach: an **arbitrary** real diagonal b_n — no growth restriction at all — together with off-diagonal entries $a_{nk} = (1 + min(n, k))$, $rho^{|n-k|}$ for $0 le rho < 1$. Every row has infinitely many non-zero entries, so no finite-range theorem applies; the amplitude envelope is $A_n = 1 + n$, which satisfies Carleman's condition because $sum_n 1/(1 + n)$ diverges, and the profile $theta(r) = rho^r$ has geometric tails, hence a finite first moment. The operator is therefore essentially self-adjoint on the finite-mode core and generates a Stone flow.

The honest boundary is worth recording. Carleman's condition is a condition on the **growth** of the amplitudes: $sum 1/A_N = infty$ allows A_N to grow linearly, or like $N logN$, but not exponentially. The exponential wall of the scalaron potential in the Hermite basis produces amplitudes of order e^{csqrtN}, for which the series converges and the criterion is silent. The unbounded-hop extension removes the **range** restriction, not the **growth** restriction; the exponential-wall case remains open and is treated by different means elsewhere in this development.

41. Reliability of the Shift-Invert Rational Krylov Scheme

41.1. Orientation and Status

The reliability chapter of the SIRK numerics: what the computation computes, what a Ritz certificate can and cannot say, and — the chapter's spine — **when a computation is a proof**. It is the book's honesty layer for the companion solver work, and it needs the operator chapters of the part behind it (the one-particle Hamiltonian, the comparison operator, the Hashimoto selection).

The route, in five stages: what the numerics compute and what a Ritz certificate may say; the proof layer (pointwise transfer identity, one Crouzeix domain for both bounds, convergence and uniformity in time, implementation specifics, leakage out of the physical subspace, the measurement region, the four systems and one named bound per system); what the values converge to (Ritz values, computed gap, spectral levels, gap of the spectrum, stability under operator perturbation); certification (no time discretization, the laminar decay rate, exact arithmetic to certified enclosures, the certified gap of the truncated Hamiltonian, when a computation is a proof, from the emitted certificate to the theorem, the table of couplings and the finite-size limit); and the honest boundary (what would close the one-particle limit, the ladder of the unbounded operator through the resolvent).

The proofs are the `ChapterSirk*` family (certified gap, gap table, Ritz spectrum, whitening, Trotter–Kato, multi-shift, restart, per-system bounds, Lagrangian canonical), `RitzMinMax`/`RitzPerturbation`, the `ResolventLadder` chapters, the `Fock*` gap chains and `BrstLeakage`.

41.2. What the Numerics Compute

For QYM, QED, QG, and NS, the Hamiltonian presented to the full nested-Fock numerics is always the outer enclosure of the sector's inner one-particle Hamiltonian: $H = \Sigma_{ij}\ h_{ij}\ C\dagger(e_i)\ A(e_j)$. Creation is on the left and outer annihilation on the right. Inner pair terms remain in h; only an allowed scalar shift of h may be made for positivity. Therefore the full Hamiltonian annihilates the outer vacuum exactly.

This chapter keeps the three operator levels distinct: the one-particle Hamiltonian, its finite Krylov/Galerkin truncations, and the nested Fock lift. The current certificate formalizes the finite/truncated level. The new proof layer now supplies a conditional version of that target. `ChapterRitzCertificate.lean` derives finite Temple bands; `ChapterBandEnclosure.lean` and `ChapterSpectralGapStability.lean` describe enclosure and limit stability; `ChapterFockOneParticleGap.lean` identifies the one-particle edge; and `ChapterFockNumberPreservingGap.lean` lifts it through the free, number-preserving dGamma Hamiltonian. The concrete QYM instantiation and its analytic convergence hypotheses remain specialist work. The same distinction applies to QED, QG, and NS: the final full-theory operator is always the outer creation-left/annihilation-right enclosure of the sector's inner one-particle Hamiltonian.

The solver evolves a state by projecting the generator onto a Krylov subspace, exponentiating the resulting $m \times m$ matrix, and lifting the answer back. Written out, the approximation is $V_m\, \psi(B_m)\, V_m^* v$, where V_m is an isometric embedding of the retained subspace, $B_m = V_m^* X V_m$ is the compression of the shift-invert resolvent X, and ψ is the function whose calculus at X reproduces the propagator. The reliability question is how far that is from the exact $\varphi_k(A)v$.

The ingredients were proved separately: the φ-function calculus, the compression transfer, the rational-transfer identity, and the exponential decay of the error bound in the reduction order. What was missing was a theorem that **composes** them. `BookProof/Sirk/ChapterSirkEndToEnd.lean` is that composition.

41.3. New Proof Layer: What a Ritz Certificate Can and Cannot Say

`ChapterRitzCertificate.lean` proves the finite statement used by the numerical pipeline: a normalized trial vector has a computable Rayleigh quotient and residual, and—provided the next spectral region is separated—Temple's inequality places the relevant eigenvalue in a certified interval. `ChapterTempleSeparationNecessary.lean` supplies the essential warning: without separation, even perfect Rayleigh data can coexist with an arbitrarily lower unseen eigenvalue. Thus the separation condition is a real certificate input, not a cosmetic technicality.

For the physical lift, `ChapterFockNumberPreservingGap.lean` proves that a positive one-particle gap produces the corresponding free Fock gap on vacuum-orthogonal finite-particle states. `ChapterFockInteractionStability.lean` replaces exact number preservation by a quantitative bounded/form-bounded interaction estimate. The unbounded-perturbation ladder is now closed at degrees one and two: `ChapterFockFieldPerturbation.lean` (now `sorry`-free) proves the gap survives the **linear**, number-changing coupling $\Phi(f) = a^\dagger(f) + a(f)$ with the explicit smallness condition $2\|f\| < \mu$, and `ChapterFockPairPerturbation.lean` extends it to the **quadratic**, pair-creating coupling $P(f,g) = a^\dagger(f)a^\dagger(g) + a(g)a(f)$ with $2\sqrt{2}\|f\|\|g\| < \mu$. `ChapterFockCubicUnbounded.lean` shows the route stops at degree two — a bare cubic term has no relative form bound, so $d\Gamma(N) + \lambda C$ is unbounded below for every coupling strength — and its companion `ChapterFockCubicQuarticStability.lean` proves the stabilisation: cubic term **plus** its normal-ordered quartic partner is bounded below on all finite states, the Cauchy–Schwarz estimate $\|a_k\ u\|^2 \le \|u\|\|a_k\dagger a_k\ u\|$ forcing a large mode occupation to carry a large quartic form. Finally, `ChapterYangMillsFockGapChain.lean` instantiates the whole chain for the concrete gauge-fixed QYM one-particle operator: $d\Gamma(H_1)\Omega = 0$ is proved unconditionally, and a one-particle **form** gap lifts to the nested-Fock mass gap — everything conditional on that form gap, which is the single analytic input the certificate is designed to supply. For sectors whose one-particle energy is a positive constant or a diagonal dispersion the condition disappears: `ChapterScalaronFockGapChain.lean` (the R^2 scalaron at mass $1/\sqrt{(12\alpha)}$) and

`ChapterFockDiagonalGapChain.lean` (free dispersion `√(p² + m²)`) prove the form gap outright — an identity for the constant, an inequality for the diagonal — so their outer-Fock gaps, Friedrichs extensions and perturbation stability are unconditional theorems.

```
#check @BookProof.FockFieldPerturbation.fock_gap_of_field_perturbation
#check @BookProof.FockPairPerturbation.fock_gap_of_pair_perturbation
#check @BookProof.FockCubicUnbounded.cubic_no_relative_form_bound
#check @BookProof.FockCubicUnbounded.fock_gap_fails_for_cubic
#check @BookProof.FockCubicQuarticStability.mode_cubic_quartic_bounded_below
#check @BookProof.FockCubicQuarticStability.dGamma_cubic_quartic_bounded_below
#check @BookProof.YangMillsFockGapChain.ym_fock_vacuum_annihilated
#check @BookProof.YangMillsFockGapChain.ym_fock_gap_of_one_particle_form_gap
#check @BookProof.YangMillsFockGapChain.ym_fock_gap_of_nested_ritz_bands
#check @BookProof.ScalaronFockGapChain.const_fock_mass_gap
#check @BookProof.ScalaronFockGapChain.scalaron_fock_mass_gap
#check @BookProof.FockDiagonalGapChain.diagOnePart_quadForm_ge
#check @BookProof.FockDiagonalGapChain.freeField_fock_mass_gap
```

41.4. Why the Transfer Identity Has To Be Pointwise

The rational transfer $r(X)v = V_m\, r(B_m)\, V_m^* v$ is not an operator identity: it holds only for states already inside the retained subspace. The Krylov seed is such a state, and the triangle-inequality core only ever uses the identity at the seed. Weakening the hypothesis accordingly is what allows the transfer to be **discharged** rather than assumed, so the assembled statement carries no transfer hypothesis at all.

```
#check @BookProof.ChapterSirkEndToEnd.sirk_error_bound_at
#check @BookProof.ChapterSirkEndToEnd.sirk_end_to_end
#check @BookProof.ChapterSirkEndToEnd.sirk_end_to_end_satisfiable
```

41.5. One Crouzeix Domain For Both Bounds

The two operator-norm estimates that remain — for the full operator and for its compression — are usually quoted with the same constant and the same sup-norm. That is legitimate because the numerical range of a compression is contained in the numerical range of the operator, so any **convex** region containing the latter also contains the convex hull of the former.

```
#check @BookProof.ChapterSirkEndToEnd.crouzeix_domain_transfer
#check @BookProof.ChapterSirkEndToEnd.crouzeix_domain_uniform
```

41.6. Convergence, and Uniformity in Time

With the constants fixed, the bound decays exponentially in the reduction order, so the reduced flows converge to the exact one. If the constants can be chosen independently of the time, the same estimate gives convergence uniform over the whole time axis.

```
#check @BookProof.ChapterSirkEndToEnd.sirk_flow_error_tendsto_zero
#check @BookProof.ChapterSirkEndToEnd.sirk_flow_error_uniform_in_time
```

41.7. The Specifics of the Implementation

Three features of the actual solver are covered separately. **Restarting**: over n cycles the error of two contractive propagators that differ by ε grows at most like $n\varepsilon$, so restarting multiplies the per-cycle guarantee by the cycle count and nothing worse. **Whitening**: the reduced operator depends only on the retained subspace — two orthonormalizations of the same raw Krylov vectors give unitarily conjugate reduced operators and literally the same reconstructed operator on the ambient space. **Shift schedules**: the forward sequence built from an arbitrary schedule of shifts spans exactly the standard Krylov subspace, so the shifts change the basis but never the space.

```
#check @BookProof.ChapterSirkRestart.restart_error_accumulation
#check @BookProof.ChapterSirkRestart.restart_error_accumulation_sirk
#check @BookProof.ChapterSirkWhitening.compress_conj_whitening
#check @BookProof.ChapterSirkWhitening.sirkApprox_eq_of_range_eq
```

```
#check @BookProof.ChapterSirkMultiShift.krylov_multiShift_eq_standard
#check @BookProof.ChapterSirkMultiShift.krylov_multiShift_span_eq_of_shifts
```

The same freedom holds on the **rational** side of the scheme, and there it is a statement about the resolvents rather than about the shifted operators. Writing X_i for a two-sided inverse of $H - z_i$, the rational Krylov space $\operatorname{span}\{v, X_0 v, X_1 X_0 v, \dots, X_{k-1} \cdots X_0 v\}$ that the solver actually assembles is exactly the image of the ordinary Krylov space $\operatorname{span}\{v, Hv, \dots, H^k v\}$ under the single invertible factor $X_{k-1} \cdots X_0$. Nothing analytic enters: the identity holds for an arbitrary module over a commutative ring and an arbitrary schedule of shifts, the proof being that the tail products $(H - z_{k-1}) \cdots (H - z_j)$ are the forward products of the reversed schedule and so span the Krylov space as well. Because the resolvents commute, their product does not see the order of the schedule either, so reordering the shifts changes the intermediate flag the algorithm passes through but not the subspace it arrives at.

```
#check @BookProof.KrylovShiftSpan.forwardSpan_eq_krylovSpan
#check @BookProof.KrylovShiftSpan.forwardSpan_eq_forwardSpan
#check @BookProof.KrylovShiftSpan.tailSpan_eq_krylovSpan
#check @BookProof.KrylovShiftSpan.resolventSpan_eq_map_krylovSpan
#check @BookProof.KrylovShiftSpan.krylovSpan_eq_map_resolventSpan
#check @BookProof.KrylovShiftSpan.resolventSpan_of_perm
```

Whitening was described above as a hypothesis: hand the reduction an isometric embedding of the retained subspace and the answer does not depend on which one. The solver, however, does not receive such an embedding — it builds one, from the Gram matrix $G_{ij} = \langle w_i, w_j \rangle$ of the raw Krylov vectors, by whitening with a factor T satisfying $T^* G T = 1$. That step is now covered as well. The synthesis map $c \mapsto \sum_i c_i w_i$ has the retained subspace as its range and G as $(\mathrm{synthesis})^*(\mathrm{synthesis})$; any T with $T^* G T = 1$ therefore turns it into an isometric embedding, whose range is the retained subspace as soon as T is onto. And such a T exists whenever the raw vectors are independent, so the hypotheses of the whitening-independence theorems are never vacuous. When the raw vectors are dependent, an orthonormalization still exists, indexed by the rank rather than by the vectors — the exact, lossless form of the rank truncation the code performs on a degenerate Gram matrix.

```
#check @BookProof.ChapterSirkGramWhitening.range_synthesis
#check @BookProof.ChapterSirkGramWhitening.gramOp_apply
#check @BookProof.ChapterSirkGramWhitening.whitened_adjoint_comp_self
#check @BookProof.ChapterSirkGramWhitening.range_whitened
#check @BookProof.ChapterSirkGramWhitening.exists_isWhitening
#check @BookProof.ChapterSirkGramWhitening.exists_isometry_range_eq_span
#check @BookProof.ChapterSirkGramWhitening.isWhitening_of_matrix
#check @BookProof.ChapterSirkGramWhitening.sirkApprox_gram_whitening_eq
```

Truncating the rank is then a matter of how far the discarded directions really were from the retained subspace. If every raw vector sits within δ of it, a state assembled with coefficients c loses at most $\delta\sqrt{m}\|c\|$, and that is exactly the additive term the end-to-end bound pays for a rank-truncated reduction — the exponential term in the reduction order is untouched, and $\delta = 0$ returns the lossless case.

```
#check @BookProof.ChapterSirkGramWhitening.norm_defect_synthesis_le
#check @BookProof.ChapterSirkGramWhitening.sirk_end_to_end_truncated_gram
```

That leaves the question of where δ comes from, since it is not what the code measures: the code diagonalizes the Gram matrix and discards the eigenpairs whose eigenvalue falls below a tolerance. The two are related by a single estimate. The synthesized eigenvectors $W u_k$ are orthogonal with squared norms the eigenvalues λ_k, so expanding a state along the eigenbasis and dropping the discarded directions costs $\left(\sum_{k\ \mathrm{discarded}} |c_k|^2 \lambda_k\right)^{1/2} \le \sqrt{\mathrm{tol}}\,\|c\|$. Since the projection VV^*x is the closest point of the retained subspace to x, every raw Krylov vector is within $\sqrt{\mathrm{tol}}$ of it: the geometric parameter obeys $\delta \le \sqrt{\mathrm{tol}}$, and the end-to-end bound can be stated in the numerical cutoff alone. The eigendecomposition it assumes always exists — the Gram operator is self-adjoint — and the embedding the solver builds from the retained eigenpairs, $V = W U_R \Lambda_R^{-1/2}$, is an isometric embedding of the retained subspace, so the estimate applies to the object the code produces rather than to an abstract substitute.

```
#check @BookProof.ChapterSirkGramCutoff.exists_gramEigen
#check @BookProof.ChapterSirkGramCutoff.norm_sub_proj_le_of_mem_range
#check @BookProof.ChapterSirkGramCutoff.dist_synthesis_retained_le
```

```
#check @BookProof.ChapterSirkGramCutoff.defect_le_sqrt_cutoff
#check @BookProof.ChapterSirkGramCutoff.sirk_end_to_end_truncated_cutoff
#check @BookProof.ChapterSirkGramCutoff.retainedEmbedding_isometry
#check @BookProof.ChapterSirkGramCutoff.defect_le_sqrt_cutoff_retained
```

41.8. Leakage Out of the Physical Subspace

The truncated dynamics do not preserve the physical subspace picked out by the BRST charge, which is why the solver carries a projector along. The telescoping estimate bounds the damage: the exact flow keeps a physical state physical, so all of the leakage comes from the truncation, and after n cycles it is at most $\|\Omega\|\, n\varepsilon\|v\|$.

```
#check @BookProof.ChapterSirkRestart.brst_leakage_zero_of_exact
#check @BookProof.ChapterSirkRestart.brst_leakage_bound
```

That estimate takes the per-cycle distance between the exact and the truncated propagator as given. One level down, at the generators the algorithm truncates, the distance is not an assumption but a consequence, and the leakage rate becomes an explicit block of the Hamiltonian. For a bounded self-adjoint generator the flow e^{-itH} is unitary and commutes with the charge, so the exact dynamics keeps the charge content of a state constant. Comparing the exact flow with the truncated one along the path $s \mapsto e^{-i(t-s)H}e^{-isB}v$ gives a derivative equal to $e^{-i(t-s)H}\big(i(H-B)\big)e^{-isB}v$, whose norm — both groups being unitary — is just $\|(H-B)e^{-isB}v\|$. Integrating is the whole estimate.

What makes the constant sharp rather than crude is that the truncated flow keeps a retained state inside the retained subspace, so the defect $H - PHP$ is only ever applied to states in the range of P, where it equals the discarded off-diagonal block $(1-P)HP$. The leakage of a physical, retained state after time t is therefore at most $\|\Omega\|\,\|(1-P)HP\|\,\|v\|\,t$: it is controlled by exactly the part of the Hamiltonian the truncation throws away, and vanishes with it. Restarting with a fresh truncation each cycle accumulates the bound linearly in the number of cycles.

```
#check @BookProof.BrstLeakage.norm_omega_flow_eq
#check @BookProof.BrstLeakage.hasDerivAt_duhamel
#check @BookProof.BrstLeakage.norm_flow_sub_flow_apply_le
#check @BookProof.BrstLeakage.leakage_le
#check @BookProof.BrstLeakage.flow_truncGen_mem
#check @BookProof.BrstLeakage.truncation_leakage_le_of_physical
#check @BookProof.BrstLeakage.leakage_iterate_le
#check @BookProof.BrstLeakage.brst_leakage_bound_of_generator
```

The field-theoretic Hamiltonian is not bounded, and the estimate does not need it to be. Let T be an unbounded self-adjoint operator with the unitary group e^{-itT} that Stone's theorem produces, and let the retained subspace be finite-dimensional and inside the domain of T — the finite-m situation of the algorithm. Then the compression PTP is a bounded self-adjoint operator, its flow keeps a retained state retained, and the same comparison path can be differentiated: the group is only strongly continuous, so the product rule is replaced by the observation that an isometric, strongly continuous family applied to a curve vanishing at the base point is differentiable with the expected derivative. What comes out is the same pair of bounds with the same constant: the flow error $\|e^{-itPTP}v - e^{-itT}v\| \le \|(1-P)TP\|\|v\|t$ and the leakage $\|\Omega e^{-itPTP}v\| \le \|\Omega v\| + \|\Omega\|\|(1-P)TP\|\|v\|t$, for an observable commuting with the exact group. Unboundedness of the Hamiltonian costs nothing; what is used is that the discarded block acts on the retained subspace alone. Restarting with a fresh retained subspace each cycle again accumulates the bound linearly in the number of cycles.

```
#check @BookProof.BrstUnboundedLeakage.hasDerivAt_isometry_apply
#check @BookProof.BrstUnboundedLeakage.hasDerivAt_duhamel_stone
#check @BookProof.BrstUnboundedLeakage.norm_flow_sub_stoneU_le
#check @BookProof.BrstUnboundedLeakage.truncGen_isSelfAdjoint
#check @BookProof.BrstUnboundedLeakage.flow_truncGen_mem
#check @BookProof.BrstUnboundedLeakage.norm_flow_truncGen_sub_stoneU_le
#check @BookProof.BrstUnboundedLeakage.truncation_leakage_le
#check @BookProof.BrstUnboundedLeakage.truncation_leakage_le_of_physical
#check @BookProof.BrstUnboundedLeakage.restart_leakage_le
```

41.9. Which Region the Constants Are Measured On

The constants are only as meaningful as the region on which the deformation is measured, and that region is decided by the operator the algorithm actually iterates: not the Hamiltonian, which is unbounded, but its shift-invert. Two regimes cover every system in this development. If the generator is positive and the shift is a positive real γ, the shift-invert is self-adjoint with norm at most γ^{-1} and nonnegative Rayleigh quotients, so its numerical range collapses onto the segment $[0, \gamma^{-1}]$ of the real axis. If the generator is indefinite, the algorithm runs at a shift off the real axis, and the numerical range sits in the disc of radius $|\operatorname{Im}\gamma|^{-1}$. Either way the region depends on the shift alone — not on the reduction order, not on the seed — and is inherited by every compression.

```
#check @BookProof.ChapterSirkSpectralGeometry.numRange_subset_realSegment_of_shiftInv
#check @BookProof.ChapterSirkSpectralGeometry.numRange_subset_closedBall_of_shiftInve
#check @BookProof.ChapterSirkSpectralGeometry.crouzeix_domain_shiftInvert
#check @BookProof.ChapterSirkSpectralGeometry.crouzeix_domain_shiftInvertC
#check @BookProof.ChapterSirkSpectralGeometry.sirk_end_to_end_crouzeix_domain
```

41.10. The Four Systems

Each physical Hamiltonian falls into one of the two regimes. Yang–Mills is a sum of squares, so the Friedrichs route applies and the region is a segment. Navier–Stokes in Eulerian variables is indefinite in both its sequence-space and its differential realization, and so is the Lagrangian generator once the drift is included; all three take the disc. So does the gauge-fixed $R + \alpha R^2$ gravity Hamiltonian, whose fiber symbol is two-signed — and whose resolvent had to be constructed first, from the extension that essential self-adjointness selects, since no positivity is available to supply one.

```
#check @BookProof.ChapterSirkPerSystem.ym_sirk_crouzeix_domain
#check @BookProof.ChapterSirkPerSystem.ns_sirk_crouzeix_domain
#check @BookProof.ChapterSirkPerSystem.nsDiff_sirk_crouzeix_domain
#check @BookProof.ChapterSirkPerSystem.lagrangian_sirk_crouzeix_domain
#check @BookProof.ChapterSirkPerSystem.diagKR_sirk_crouzeix_domain
#check @BookProof.ChapterSirkPerSystem.qgR2_shiftInvert_selects
#check @BookProof.ChapterSirkPerSystem.qgR2_sirk_crouzeix_domain
```

41.11. One Named Bound Per System

Fixing the region is not yet a reliability statement about a particular Hamiltonian: the assembly and the geometry still have to be composed. Doing so gives one named theorem per system, in which the selected extension, its shift-invert, the region on which the constants are measured, the Krylov transfer identity and the reconstruction projection are all discharged from results the development already proves. What is left as an input is exactly what should be: the two deformation estimates on the fixed region, and the spectral consistency that identifies the propagator with the calculus of the shift-invert. The reduction data of a single order m — the rational approximant, its denominators and the reduced propagator — are bundled, which also lets a whole family over m be quantified at once; the family form is what gives the convergence of the reduced flows as the retained subspace grows. Two small observations remove clutter along the way: an isometric embedding automatically satisfies $V^*V = 1$ and has a contractive adjoint, so neither needs to be assumed.

```
#check @BookProof.ChapterSirkPerSystemFlowBound.adjoint_comp_self_of_isometry
#check @BookProof.ChapterSirkPerSystemFlowBound.norm_adjoint_apply_le_of_isometry
#check @BookProof.ChapterSirkPerSystemFlowBound.sirk_scheme_bound
#check @BookProof.ChapterSirkPerSystemFlowBound.sirk_scheme_tendsto
#check @BookProof.ChapterSirkPerSystemFlowBound.ym_sirk_flow_error_bound
#check @BookProof.ChapterSirkPerSystemFlowBound.ym_sirk_flow_error_tendsto_zero
#check @BookProof.ChapterSirkPerSystemFlowBound.ns_sirk_flow_error_bound
#check @BookProof.ChapterSirkPerSystemFlowBound.nsDiff_sirk_flow_error_bound
#check @BookProof.ChapterSirkPerSystemFlowBound.lagrangian_sirk_flow_error_bound
#check @BookProof.ChapterSirkPerSystemFlowBound.diagKR_sirk_flow_error_bound
#check @BookProof.ChapterSirkPerSystemFlowBound.qgR2_sirk_flow_error_bound
#check @BookProof.ChapterSirkPerSystemFlowBound.qgR2_sirk_flow_error_tendsto_zero
```

41.12. From the Reduced Generator to the Flow

Everything above compares the algorithm with a **fixed** generator. The other half of reliability is the passage to the limit: the reduced generators approximate the selected extension, and one wants the propagators they generate to approximate its flow. For bounded generators this is elementary and quantitative — the exponential is Lipschitz on balls of a Banach algebra, so two propagators differ by at most $|t|\,\|a-b\|\,e^{|t|M}$ on the ball of radius M, uniformly on every bounded time interval.

```
#check @BookProof.ChapterSirkGroupTransfer.norm_exp_sub_exp_le
#check @BookProof.ChapterSirkGroupTransfer.norm_groupFlow_sub_le
#check @BookProof.ChapterSirkGroupTransfer.groupFlow_transfer_uniform_on_interval
```

For the unbounded selected extension there is no such rate, and the transfer is the Trotter–Kato theorem: strong convergence of the resolvents $(A_n-i)^{-1}\to(A-i)^{-1}$ forces $e^{-itA_n}v\to e^{-itA}v$, uniformly for t in a bounded interval. The proof is a Duhamel argument. The curve $r\mapsto e^{-i(t-r)A_n}(A_n-i)^{-1}e^{-irA}\chi$ has to be differentiated even though the flow is only strongly continuous — so no operator product rule is available and the derivative is taken from the definition — and its derivative is computed by the algebraic identity $A_nR_n-R_nA=(R-R_n)(A-i)$ on the domain. The mean value inequality then bounds the difference of the flows by the resolvent difference along an orbit, which is a compact set, so pointwise convergence of the resolvents is automatically uniform there.

```
#check @BookProof.ChapterSirkTrotterKato.hasDerivAt_stoneU_const_sub_apply
#check @BookProof.ChapterSirkTrotterKato.resolvent_commutator_eq
#check @BookProof.ChapterSirkTrotterKato.hasDerivAt_duhamel
#check @BookProof.ChapterSirkTrotterKato.norm_res_stoneU_sub_stoneU_res_le
#check @BookProof.ChapterSirkTrotterKato.trotterKato_uniform_on_interval
#check @BookProof.ChapterSirkTrotterKato.trotterKato_tendsto
#check @BookProof.ChapterSirkTrotterKato.trotterKato_tendstoUniformlyOn
```

The hypothesis is exactly what the Rayleigh–Ritz analysis supplies. In the regime where the generator is bounded on its domain — the regime in which the Galerkin limit is identified with the selected extension — the compressions converge strongly, hence so do their resolvents, hence the flows they generate converge to the flow of the selected generator, uniformly on bounded time intervals. That is the last link of the chain: what the algorithm propagates converges to what the selected extension propagates.

```
#check @BookProof.ChapterSirkTrotterKato.strongResolventConvergence_ofBounded
#check @BookProof.ChapterSirkTrotterKato.flow_transfer_of_strong_tendsto
#check @BookProof.ChapterSirkTrotterKato.galerkin_flow_transfer
#check @BookProof.ChapterSirkTrotterKato.galerkin_flow_tendsto
```

41.13. The Two Remaining Lagrangian Realizations

The list of systems above stops at the Kato–Rellich instance of the Lagrangian generator, whose constituents commute. Two further realizations of the same Lagrangian data had essential self-adjointness but no shift-invert companion: the canonical realization, in which the parcel momenta and the viscous gradients are genuinely non-commuting canonical pairs of the trajectory-space Hermite basis, and the Fock/momentum realization, in which the constituents are multiplication operators by arbitrary measurable symbols, so the spectrum is in general purely continuous and there is no eigenvector to work with at all. Since both are essentially self-adjoint on their core, the abstract Lagrangian selection theory applies verbatim: the multi-shift data determine one self-adjoint operator, the single-shift resolvent obeys the bound $\|X\|\le|\operatorname{Im}\gamma|^{-1}$, and the Crouzeix region is the disc of that radius for the generator and for every order-m compression. The continuum realization also acquires its Stone flow, so the reliability chain has a flow to compare the algorithm with in every realization the project defines.

```
#check @BookProof.ChapterSirkLagrangianCanonical.lagCan_hashimoto_selects
#check @BookProof.ChapterSirkLagrangianCanonical.lagCan_sirk_crouzeix_domain
#check @BookProof.ChapterSirkLagrangianCanonical.fockLag_esa
#check @BookProof.ChapterSirkLagrangianCanonical.fockLag_hashimoto_selects
#check @BookProof.ChapterSirkLagrangianCanonical.fockLag_sirk_crouzeix_domain
#check @BookProof.ChapterSirkLagrangianCanonical.fockLag_stone_flow
```

41.14. What the Ritz Values Converge To

The algorithm reports, at each reduction order, the smallest Rayleigh quotient available in the retained subspace — the Ritz value the numerics read as a ground-state energy. That these values decrease to the bottom of the energy form was already proved; what was missing is that the limit is a **spectral** quantity of the operator the algorithm selects. It is: for a bounded self-adjoint operator the bottom of the spectrum coincides with the bottom of the numerical range, because a shift $T - c$ is a nonnegative element of the algebra exactly when its spectrum is nonnegative, and exactly when $c\|x\|^2 \le \mathrm{Re}\langle x, Tx\rangle$ for every x. The finite-mode domain is dense and the Rayleigh quotient is continuous, so the truncations see the whole numerical range, and the Ritz values converge to the bottom of the spectrum of the selected extension.

```
#check @BookProof.ChapterSirkRitzSpectrum.le_rayleigh_iff_le_spectrum
#check @BookProof.ChapterSirkRitzSpectrum.sInf_spectrum_eq_rayleighInf
#check @BookProof.ChapterSirkRitzSpectrum.ritzInf_finiteModeDomain_eq_rayleighInf
#check @BookProof.ChapterSirkRitzSpectrum.ritzInf_tendsto_sInf_spectrum
#check @BookProof.ChapterSirkRitzSpectrum.galerkin_ritz_tendsto_sInf_spectrum_of_sele
```

41.15. What the Computed Gap Converges To

A gap is a difference of two levels, and there is no second Rayleigh quotient: the right object above the ground level is the Courant–Fischer min–max level $\Lambda_k(T) = \inf\{\sup_{x\in S,\|x\|=1} \mathrm{Re}\langle x, Tx\rangle : \dim S = k+1\}$, and the level the solver computes is the same infimum restricted to the subspaces of the retained Galerkin space. Restricting can only raise the infimum, so the computed levels are always upper bounds; conversely any $(k+1)$-dimensional subspace can be pushed by the Galerkin projection into a large enough retained space, and — because the projection moves its unit vectors by an arbitrarily small amount once the projections converge uniformly on it — at an arbitrarily small cost in the Rayleigh supremum. The two bounds together give convergence of the computed levels to the min–max levels, so the computed gap $\Lambda_1(m) - \Lambda_0(m)$ converges to $\Lambda_1 - \Lambda_0$ and a positive min–max gap is eventually seen by the truncations. The ladder starts where the previous section stopped: level zero is the bottom of the numerical range, hence the bottom of the spectrum. Nothing here asserts that a level with $k \ge 1$ is an eigenvalue — that is true only below the essential spectrum.

```
#check @BookProof.RitzMinMax.minmaxLevel_le_minmaxLevelIn
#check @BookProof.RitzMinMax.minmaxLevel_zero_eq_sInf_spectrum
#check @BookProof.RitzMinMax.exists_galerkin_approx_subspace
#check @BookProof.RitzMinMax.galerkin_minmaxLevel_tendsto
#check @BookProof.RitzMinMax.galerkin_gap_tendsto
#check @BookProof.RitzMinMax.galerkin_gap_eventually_pos
```

41.16. The Levels Are Spectral Values

The previous section left the levels above the ground level unidentified: they were the limits of what the solver computes, but nothing said they were quantities of the operator's spectrum. They are. If a level $\Lambda_k(T)$ were outside the spectrum then, the spectrum being closed, some $\varepsilon > 0$ would separate it: every spectral point would lie below $\Lambda_k - \varepsilon$ or above $\Lambda_k + \varepsilon$. The continuous cutoff that is 1 below the gap and 0 above it is then $\{0,1\}$-valued on the spectrum, so the functional calculus turns it into an orthogonal projection P with $T \le \Lambda_k - \varepsilon$ on the range of P and $T \ge \Lambda_k + \varepsilon$ on its kernel. Either the range of P holds a $(k+1)$-dimensional subspace — a competitor whose Rayleigh supremum is at most $\Lambda_k - \varepsilon$, so the infimum is smaller than itself — or it does not, and then P kills a unit vector of **every** $(k+1)$-dimensional subspace, so every competitor has Rayleigh supremum at least $\Lambda_k + \varepsilon$. Both are impossible, so the level is a spectral point. Consequently the limit of the computed Galerkin levels is a value of the spectrum and the computed gap converges to the difference of two spectral values. What is still not claimed is that a level is an **eigenvalue**: below the essential spectrum it is, at or above it the level is only a point of the spectrum.

```
#check @BookProof.MinMaxSpectrum.rayleighVal_le_of_cutoff_fixed
#check @BookProof.MinMaxSpectrum.le_rayleighVal_of_cutoff_zero
#check @BookProof.MinMaxSpectrum.minmaxLevel_mem_spectrum
#check @BookProof.MinMaxSpectrum.sInf_spectrum_mem_spectrum
#check @BookProof.MinMaxSpectrum.galerkin_minmaxLevel_tendsto_spectrum
#check @BookProof.MinMaxSpectrum.galerkin_gap_tendsto_spectrum_sub
```

41.17. The Gap Is a Gap of the Spectrum, and the Ground Level Is an Eigenvalue

A gap between the two lowest levels says more than that its endpoints are spectral values: the whole open interval between them misses the spectrum. Suppose a spectral point λ sat strictly between Λ_0 and Λ_1. Around the bottom of the spectrum and around λ choose two continuous symbols with disjoint supports — a cutoff and a trapezoid — inside the interval. Each has a non-zero functional calculus, because a continuous function that does not vanish at a spectral point cannot be sent to the zero operator; the vectors they produce are orthogonal, stay orthogonal after T is applied, and each carries a Rayleigh quotient below $\lambda + \varepsilon$. The plane they span is therefore a two-dimensional competitor whose Rayleigh supremum is below Λ_1 — which contradicts the definition of Λ_1 as the infimum over such planes.

With the gap in the spectrum in hand the ground level is not merely a spectral value but a genuine eigenvalue. The cutoff at the middle of the gap is now $\{0,1\}$-valued on the spectrum, so it produces an orthogonal projection P, non-zero because the ground level is in the spectrum. Its fixed vectors carry Rayleigh quotients below the gap, so they cannot span a plane — that plane would again be a competitor below Λ_1 — and they therefore form a line. The line is preserved by T, because the functional calculus commutes with the operator, so its generator is an eigenvector; self-adjointness makes the eigenvalue real, the eigenvalue is a spectral point below the gap, and the gap in the spectrum forces it to be Λ_0 exactly.

```
#check @BookProof.MinMaxSpectrum.notMem_spectrum_of_mem_minmaxGap
#check @BookProof.MinMaxSpectrum.spectrum_inter_minmaxGap_eq_empty
#check @BookProof.MinMaxSpectrum.exists_eigenvector_of_minmaxGap
```

41.18. How Far the Levels Move When the Operator Moves

A solver never holds the exact operator: it holds a model of it, with a truncated coupling, a rounded matrix or a regularized potential. The min–max levels are stable under that substitution. On the unit sphere the Rayleigh quotients of T and T' differ by at most $\|T - T'\|$, so the same bound passes to the supremum over any subspace, to the infimum over the subspaces of a given dimension, and to the Ritz levels computed inside a fixed truncation: every level is 1-Lipschitz in the operator norm, and the gap $\Lambda_1 - \Lambda_0$ is 2-Lipschitz. A gap therefore survives a perturbation of less than half its size, and a Ritz level computed for the model operator inside a retained space is an upper bound for the exact level up to the modelling error. A real shift $T + c$ — the shift a shift-invert scheme applies — moves every level by c and leaves the gap where it was. What is **not** claimed is the converse direction: a positive computed gap of a truncation is not by itself a positive gap of the exact operator, since the computed levels are only upper bounds.

```
#check @BookProof.RitzPerturbation.abs_minmaxLevel_sub_le_dist
#check @BookProof.RitzPerturbation.abs_minmaxLevelIn_sub_le_dist
#check @BookProof.RitzPerturbation.minmaxLevel_le_minmaxLevelIn_add
#check @BookProof.RitzPerturbation.minmaxLevel_shiftOp
#check @BookProof.RitzPerturbation.minmaxGap_shiftOp
#check @BookProof.RitzPerturbation.minmaxGap_pos_of_dist_lt
#check @BookProof.RitzPerturbation.galerkin_model_gap_tendsto
#check @BookProof.RitzPerturbation.abs_sInf_spectrum_sub_le_dist
```

41.19. No Time Discretization: One Shift, One Finite Time

The scheme is not a time-marching one. At a single finite time t the propagator e^{-itH} is approximated by one rational function of the shift-invert resolvent $(H - i\ell)^{-1}$, so no step size, number of steps or operator splitting enters the statement; and since only the resolvent is ever applied, and $\|(H - i\ell)^{-1}\| \le 1/|\ell|$ for every self-adjoint H, the Hamiltonian need not be bounded. The choice of shift is immaterial as well: by the first resolvent identity, strong convergence of the shift-invert operators at one nonzero shift already implies it at every nonzero shift (the sign flip uses normality of the resolvent, proved from the adjoint relation and the commutation of resolvents), and Trotter–Kato then gives convergence of the propagators at every single finite time. The quantum-gravity instances are stated in exactly that form.

```
#check @BookProof.SirkSingleTime.res_sub_res
#check @BookProof.SirkSingleTime.norm_res_neg
#check @BookProof.SirkSingleTime.strongResAt_of_ne_zero
#check @BookProof.SirkSingleTime.singleTime_flow_tendsto_of_strongResAt
#check @BookProof.SirkSingleTime.isShiftInvertC_neg_resCLM_shift
#check @BookProof.SirkSingleTime.sirk_single_time_shiftInvert_bound
#check @BookProof.SirkSingleTime.qgOuterFock_singleTime_shiftInvert_convergence
#check @BookProof.SirkSingleTime.starobinsky_qgContinuum_singleTime_shiftInvert_conve
```

41.20. The Laminar Decay Rate

For the parabolic side of the Lagrangian generator the quantity the numerics measure is not an energy but a rate: the exponential decay of the state in the laminar regime. Formally the rate is the coercivity constant of the generator. The semigroup solves $u'(t) = -Au(t)$, so the energy $\|u(t)\|^2$ has derivative $-2\,\mathrm{Re}\langle u(t), Au(t)\rangle$, which coercivity bounds by $-2\mu\|u(t)\|^2$; weighting by $e^{2\mu t}$ makes the result monotone and gives $\|e^{-tA}v\| \le e^{-\mu t}\|v\|$. The reduction does not change the constant: the compression of a coercive generator along an isometry is coercive with the same μ, so the reduced propagator obeys the same bound at every order — the decay the numerics read off the reduced model is not an artefact of the reduction.

```
#check @BookProof.ChapterSirkDiffusiveDecay.hasDerivAt_heatFlow_apply
#check @BookProof.ChapterSirkDiffusiveDecay.hasDerivAt_heatFlow_normSq
#check @BookProof.ChapterSirkDiffusiveDecay.norm_heatFlow_apply_le
#check @BookProof.ChapterSirkDiffusiveDecay.norm_heatFlow_le
#check @BookProof.ChapterSirkDiffusiveDecay.isCoercive_compress
#check @BookProof.ChapterSirkDiffusiveDecay.norm_heatFlow_compress_apply_le
```

41.21. From Exact Arithmetic To Certified Enclosures

Everything above is stated in exact arithmetic, while the solver runs in binary64. The gap is closed not by trusting the floating-point numbers but by proving **around** them. Three theorems do the work. The residual bound is a-posteriori: for any vector whatsoever — in particular the computed one — and any real value θ, some eigenvalue of the **exact** compression lies within $\|H\psi - \theta\psi\|$ of θ. The backward-error model of a symmetric eigensolver says the computed eigenpairs are exact eigenpairs of a nearby operator, and Weyl's inequality — in the enclosure form, which is all a certificate needs — transports its eigenvalues back. The remaining term is an interval evaluation with outward rounding, whose soundness is a handful of order facts about endpoints.

The decomposition is worth reading as a division of responsibility. The residual says **where a level sits** — it is a statement about the operator and the computed vector, and nothing else. The backward-error term says **what the arithmetic did** — the computed eigendecomposition is exact for a nearby operator, and Weyl transports the question back. The interval term says **what reading the number did** — even writing a value down is arithmetic. Each term can fail independently, and each is checked independently, which is why the assembled width is a bound and not a heuristic tolerance. Nothing in the proof consumes a raw floating-point value; the statements consume the enclosures, and the only trusted component in the whole chain is the outward-rounding evaluation itself.

```
#check @BookProof.SirkFinitePrecision.exists_eigenvalue_dist_le_residual
#check @BookProof.SirkFinitePrecision.exists_eigenvalue_dist_le_residual_unit
#check @BookProof.SirkFinitePrecision.backward_error_weyl
#check @BookProof.SirkFinitePrecision.backward_error_weyl_symm
#check @BookProof.SirkFinitePrecision.observable_propagation
#check @BookProof.SirkFinitePrecision.observable_propagation_band
#check @BookProof.SirkFinitePrecision.CertInterval.mem_mul
#check @BookProof.SirkFinitePrecision.CertInterval.mem_widen
#check @BookProof.SirkFinitePrecision.CertInterval.dist_le_width
```

41.22. Which Half Of The Bracket Is Free

The variational principle gives the upper half of the bracket for nothing: a Rayleigh quotient of a unit vector is never below the lowest eigenvalue. The lower half is not free, and it is worth being precise about why. A small residual certifies that **some** eigenvalue is close to the computed value; it does not certify that the **lowest** one is. Temple's

inequality supplies the missing lower bound from the same computed data plus one separation constant: if every eigenvalue is either the lowest or at least β, and the computed quotient sits below β, then the lowest eigenvalue is at least $\theta - (\|H\psi\|^2 - \theta^2)/(\beta - \theta)$.

```
#check @BookProof.SirkFinitePrecision.ground_le_rayleigh
#check @BookProof.SirkFinitePrecision.temple_lower_bound
#check @BookProof.SirkFinitePrecision.ground_ge_of_no_eigenvalue_below
```

41.23. The Certified Gap Of The Truncated Hamiltonian

The current emitted certificate is expressed through invariant even/odd blocks of the finite truncated operator. For the nested-Fock interpretation, the physical observable is instead the lowest positive one-particle energy after a constant shift. The parity certificate can be used for that observable only after the specialist proves the one-particle vacuum/first-excitation identification. Until then, `1.932` is a rigorous truncated value, not a real-Hamiltonian Fock gap.

One clarification the numerics forced on the program, and which the prose should carry: the recorded run behind `1.932` is the **historical** lattice-era instantiation. The object of record is the three-dimensional gauge-fixed Yang–Mills Hamiltonian on the nested Fock space, whose exact symmetry is the reflection $(A_0, A_1) \mapsto (-A_1, -A_0)$ and not the occupation parity of the lattice model. Its gap is its own — on the order of 0.09 at coupling $g = 1$ and growing with g — and the lattice's $g^2/2$ electric law does not transfer to it. The two instantiations are both legitimate uses of the same abstract gap theorem; they are not two measurements of one number. The gauge-fixed depth study replaces the lattice finite-size study as the order parameter: stable in truncation depth at $g > 0$, shrinking with depth in the gapless abelian limit.

For the intended outer-enclosed model, choose `μ > 0` so the shifted one-particle Hamiltonian satisfies `h₊ ≥ μ I`. Then the full Hamiltonian `H = Σᵢⱼ (h₊)ᵢⱼ C†ᵢ Aⱼ` has vacuum energy zero because the outer annihilator kills it, while every non-vacuum finite-particle energy is a sum of one-particle energies and is at least `μ`. A one-particle creation attains the lowest edge once the infinite one-particle band limit is proved.

The same bound can be read against the analytic strong-coupling expectation. Writing the measured sector Ritz difference as $g^2/2$ plus a correction — the magnetic term that the strong-coupling expansion excludes — the certified statement becomes $\mathrm{gap} \geq g^2/2 + \mathrm{corr} - (\delta^o + \delta^e)$. The correction is carried explicitly as a parameter rather than absorbed into the widths, so nothing about the excluded term is quietly assumed.

```
#check @BookProof.SirkCertifiedGap.paritySector_invariant
#check @BookProof.SirkCertifiedGap.sectorRestrict_isSymmetric
#check @BookProof.SirkCertifiedGap.sectorGround_eq_inf_eigenvalues
#check @BookProof.SirkCertifiedGap.sectorGround_ge_temple
#check @BookProof.SirkCertifiedGap.certified_parity_gap
#check @BookProof.SirkCertifiedGap.certified_parity_gap_pos
#check @BookProof.SirkCertifiedGap.certified_parity_gap_strong_coupling
#check @BookProof.SirkCertifiedGap.rayleigh_odd_ge_of_certified
#check @BookProof.SirkCertifiedGap.resolvent_commutes_parity
```

41.24. When A Computation Is A Proof

The stopping rule is the certificate itself. `ChapterH8` proves that the next SIRK prediction refines the current approximation and that the generic error bands nest and shrink. The remaining specialist composition is to show that the corresponding bands enclose the lowest positive edge of one fixed QYM one-particle Friedrichs operator. Then a positive limiting edge lifts to the real free nested-Fock Hamiltonian through `dGamma`; a positive value at only one finite order still certifies only the truncated operator.

```
#check @BookProof.SirkCertifiedGap.certifiedGap_tendsto
#check @BookProof.SirkCertifiedGap.certifiedGap_eventually_pos
#check @BookProof.SirkCertifiedGap.certifiedGap_sound
#check @BookProof.SirkCertifiedGap.gap_ge_of_certificate
#check @BookProof.SirkCertifiedGap.qcdG2M4_certified_gap
```

41.25. From The Emitted Certificate To The Theorem

A certificate is only useful if it can be **read**. The solver writes one flat JSON object per line: a parity label, the sector Ritz value and the assembled width. The reader parses those lines into the parameters of the gap theorem, and it does so without ever producing a floating-point number: a decimal literal is stored exactly, as a mantissa together with a power of ten, so that the value the proof consumes is a rational and not a rounded approximation. A line that is missing a field, or whose number is not a decimal, or which asserts a negative width, fails to parse rather than being defaulted.

What the reader then proves is the gap theorem with the parsed numbers substituted in: given the two enclosures the certificate asserts, the certified difference $\theta^o - \theta^e - (\delta^o + \delta^e)$ is a lower bound for the parity gap of the truncated Hamiltonian. Worked through on the recorded run — measured gap 1.9875, assembled width 0.0555 — the emitted text alone yields a certified gap of at least 1.932, in particular a strictly positive one.

The reading step is itself checked from outside. The instantiated gap theorem is exported to the `lean4export` NDJSON format and reduced by nanoda, a checker independent of Lean's kernel — the same discipline as the confluence certificates of the Logos pipeline — and the algorithmic core the certificates describe is available as a generated Lean model extracted from the Rust implementation by the Aeneas pipeline, so the identities the proof relies on can be checked against the code rather than an idealisation of it. The state of this formalization as of this chapter's last revision (2026-10-04): the finite-precision layers, the certified-gap theorem, the stopping rule, the certificate reader and the gap-table instantiation are closed; the Aeneas identity proofs and the per-coupling table rows are the remaining executable work; and the two analytic inputs — the one-particle form gap uniform in the derivative coordinates, and the gap-preserving convergence of the truncation family — remain the honest open boundary of the previous sections. The work order lives in `CONSOLIDATED_PLAN.md` ("State of the project — 2026-10-04"), and the planner-side pipeline that regenerates every input of this chapter lives in `../unfer/PLAN_MASS_GAP.md`.

```
#check @BookProof.SirkCertificateReader.parseDec_example
#check @BookProof.SirkCertificateReader.parseDec_reject
#check @BookProof.SirkCertificateReader.gap_ge_of_ndjson
#check @BookProof.SirkCertificateReader.gap_pos_of_ndjson
#check @BookProof.SirkCertificateReader.formatExample_lower
#check @BookProof.SirkCertificateReader.formatExample_certified_gap
```

41.26. A Table Of Couplings, And The Finite-Size Limit

One certificate certifies one operator. Adding the two reverse enclosures turns the one-sided bound into a genuine enclosure of the gap, and the same statement can then be made a row at a time: for each coupling constant, a measured difference and an assembled width, and a proof that the gap of **that** truncated Hamiltonian lies in **that** window. The analytic strong-coupling value $g^2/2$ is compared against the window rather than assumed: at $g = 2$ the prediction is 2, and the certified window of the recorded run is $[1.932, 2.043]$, which contains it.

The finite-size study asks a different question — what the gap tends to as the lattice grows. If the finite-size correction is a pure power $C\,l^{-p}$, the Richardson combination of two lattice sizes returns the limit exactly, and if the two values are known only to within ε, the extrapolant is off by at most $\varepsilon(1 + 2/X)$, where X is the extrapolation denominator. That is a theorem. The number obtained by feeding the recorded lattice data into it is not: it is a numerical estimate, and the passage to the thermodynamic limit is not claimed.

```
#check @BookProof.SirkGapTable.certified_gap_mem_interval
#check @BookProof.SirkGapTable.certified_gap_table
#check @BookProof.SirkGapTable.certified_gap_table_interval
#check @BookProof.SirkGapTable.qcdG2M4_strongCoupling_consistent
#check @BookProof.SirkGapTable.richardson_exact
#check @BookProof.SirkGapTable.richardson_error
```

41.27. What Would Close The One-Particle Limit

Everything certified above is about the truncated operator. For QYM, the Friedrichs/Hashimoto results already identify the selected operator with the real one-particle Hamiltonian. The remaining step is the one-particle positive spectral-edge convergence and its lift through the free number-preserving `dGamma` construction. The strict shift h_+

≥ μ I contributes μ N, leaves the vacuum unchanged, and makes all non-vacuum finite-particle energies positive.

The optional shifted-square witness gives a spectral reformulation, but the primary observable is the one-particle energy measured through creation and destruction.

What the continuum claim needs is that the gap is not destroyed in the limit, and that implication — as opposed to the convergence itself — can be proved. Say an operator has a **quantitative gap** d at a point λ when no vector is moved by less than $d\|x\|$: for a bounded self-adjoint operator that is exactly the statement that λ is at distance at least d from the spectrum. Such a gap degrades by at most the size of a perturbation, and it therefore survives an operator-norm limit with no loss at all. So a family of approximants with a **uniform** gap on an interval forces the limit to have no spectrum in that interval.

That is the shape of the missing leg, and it makes the remaining question precise: what is open is not whether a uniform gap passes to the limit — it does — but whether the truncation family converges in the sense required. Nothing here asserts that it does.

```
#check @BookProof.SpectralGapStability.gapAt_perturb
#check @BookProof.SpectralGapStability.gapAt_of_tendsto
#check @BookProof.SpectralGapStability.notMem_spectrum_of_gapAt
#check @BookProof.SpectralGapStability.spectrum_disjoint_of_uniform_window
```

41.28. The Ladder of the Unbounded Operator, Through the Resolvent

Everything above is about a **bounded** operator, and the Hamiltonians the project cares about are not bounded. The bridge is the resolvent $R = (T+1)^{-1}$ of a non-negative self-adjoint relation T: it is bounded, self-adjoint and non-negative, so the whole min–max apparatus applies to it — but read from the top, since $\lambda \mapsto 1/(1+\lambda)$ reverses the order. The link between the two ladders is pointwise. Writing a graph pair (y, z) of T as $y = Rx$ with $x = y + z$, Cauchy–Schwarz for the non-negative form $(u, v) \mapsto \mathrm{Re}\langle u, Rv\rangle$ at the pair (x, Rx) gives $\|y\|^4 \le (\|y\|^2 + \mathrm{Re}\langle y, z\rangle)\,\mathrm{Re}\langle y, Ry\rangle$: on a unit vector of the domain the Rayleigh quotient of T is at least $1/\mathrm{Re}\langle y, Ry\rangle - 1$. It is the operator form of the convexity of $\lambda \mapsto 1/(1+\lambda)$, and it uses no spectral theory at all.

Taking suprema over a competitor subspace and infima over the competitors of a given dimension turns the pointwise inequality into the ladder inequality $1/\nu_k - 1 \le \Lambda_k(T)$, where ν_k is the k-th level of R counted from the top. At the bottom rung the inequality is an equality: a near-maximiser x of the resolvent's Rayleigh quotient has $\|Rx\| \ge \mathrm{Re}\langle x, Rx\rangle$, so the line through Rx is a competitor whose value tends to $1/\nu_0 - 1$. The lowest min–max level of the unbounded T is therefore exactly $1/\sup \mathrm{spec}(R) - 1$, and a gap between the two top levels of the resolvent is a gap of T, quantitatively $\Lambda_1(T) - \Lambda_0(T) \ge 1/\nu_1 - 1/\nu_0$.

The reverse inequality for $k \ge 1$ needs spectral projections of R, and the functional calculus supplies them. Let q be the continuous symbol that vanishes below $c - \delta$ and is 1 above c. On the range of $q(R)$ the operator inequality $(c-\delta)R \le R^2$ holds, and on its kernel the Rayleigh quotient of R is at most c — both by order preservation of the calculus. Either the range of $q(R)$ holds a $(k+1)$-dimensional subspace S_0, and then RS_0 is a competitor inside the domain of T whose Rayleigh supremum is at most $1/(c-\delta) - 1$; or it does not, and then $q(R)$ kills a unit vector of every $(k+1)$-dimensional subspace, so every competitor for the resolvent has infimum at most c, contradicting $c < \nu_k$. Hence the comparison of the two ladders is an **equality at every rung**, $\Lambda_k(T) = 1/\nu_k - 1$, and the gap of T's ladder is exactly $1/\nu_1 - 1/\nu_0$. The rung must exist and ν_k must be positive; at $k = 0$ both hold automatically.

Because the solver diagonalises finite compressions of the resolvent, the same transfer carries the Galerkin theorem: the ladder from the top is the ladder of $-R$ from the bottom, so the computed Ritz levels of R converge to ν_k and are lower bounds for it, and $1/(\text{computed level}) - 1$ is a rigorous upper bound for the ground level of the unbounded operator. As in the bounded case the reverse direction is not certified: a computed gap is not by itself a gap of the exact operator.

```
#check @BookProof.ResolventLadder.normSq_sq_le_rayleigh_graph
#check @BookProof.ResolventLadder.resolvent_ladder_lower
#check @BookProof.ResolventLadder.graphMinmaxLevel_zero_eq
```

```
#check @BookProof.ResolventLadder.graphMinmaxLevel_zero_eq_sSup_spectrum
#check @BookProof.ResolventLadder.graphMinmax_gap_lower
#check @BookProof.ResolventLadder.graphMinmax_gap_pos
#check @BookProof.ResolventLadder.galerkin_maxminLevel_tendsto
#check @BookProof.ResolventLadder.graphMinmaxLevel_zero_le_of_computed
#check @BookProof.ResolventLadderEq.graphMinmaxLevel_eq
#check @BookProof.ResolventLadderEq.graphMinmax_gap_eq
```

41.29. The Honest Boundary

For QYM, the selected Hashimoto operator is already identified in the formal development with the real Friedrichs Hamiltonian. The remaining mass-gap target is narrower: prove that the certified nested bands converge to the lowest positive one-particle edge, then use the free number-preserving `dGamma` lift. The finite certificate and generic SIRK error bands must not be described as a completed continuum mass-gap proof.

The QYM operator-identification step is already proved: the Hashimoto limit selects the real Friedrichs Hamiltonian. The remaining gap is the explicit one-particle positive-edge certificate/limit and its free `dGamma` lift. Crouzeix's inequality and the deferred hypotheses with citations, not theorems of this development. What is fixed per system is the region on which the deformation is measured, not a numerical value for it. Strong resolvent convergence is the hypothesis of the Trotter–Kato statement, supplied per system by the selection theorems, not proved there. Finite precision is no longer outside the statements, but the certified claims are about the **truncated** operator: the gap theorem is a theorem about the finite compression the solver diagonalises, and the numbers of a particular run enter as data, with the theorem conditional on the enclosures the certificate asserts. The passage to the continuum — a gap-preserving norm-resolvent convergence of the truncation family — is not proved here and is assumed nowhere.

42. The Constants Suite: g-2, Lamb Shift, Positronium, Blackbody

42.1. Orientation and Status

The constants suite: the four numbers the programme's physics is measured against — the blackbody spectrum, the oscillator spectrum, positronium, and the Lamb shift and $g-2$ — read as thermal statistics and radiative corrections inside the same frame. It sits at the end of the field-theory part, where the reader has the Fock/BRST vocabulary to see what is being claimed; the closing section is the chapter's contract: what is verified, and what is open.

The route: why these four numbers, the blackbody spectrum (thermal statistics), the oscillator spectrum (the bridge to bound states), positronium (the two-body touchstone), and the Lamb shift and $g-2$ (radiative corrections).

The verified pieces include `BookProof/ChapterBoseEinstein`, the coherent-state thermal chapters, `BookProof/ChapterThermalMaxEntropy`, the CPT Hamiltonian, `BookProof/RitzCertificate`, `BookProof/TempleSeparationNecessary`, and the `BookProof/SirkCertifiedGap` enclosure.

42.2. Why These Four Numbers

Every quantum field theory must eventually answer to the laboratory. Four numbers have historically served as the touchstones of quantum electrodynamics: the anomalous magnetic moment of the electron (the $g-2$ measurement), the Lamb shift between the $2S_{1/2}$ and $2P_{1/2}$ levels of hydrogen, the fine-structure spectroscopy of positronium, and the blackbody radiation spectrum. Each one is a place where the formalism leaves the safety of exact symmetries and must produce a **number** that experiment can check.

This chapter is an honesty page. The verified layer of this book contains the machinery that such a suite of computations would be built on — thermal statistics, the harmonic-oscillator spectrum, certified spectral bands — but it does not yet contain the radiative-correction calculations themselves. Where a claim is not yet proved in `BookProof`, we say so explicitly, and we show what instruments exist that a specialist would use to close the gap.

42.3. The Blackbody Spectrum: Thermal Statistics

The blackbody spectrum is the oldest of the four touchstones — it is where quantum theory was born. Planck's law rests on two facts: the mean occupation of a bosonic mode at energy $\hbar\omega$ above the ground state follows the Bose–Einstein distribution, and the ground state itself carries the zero-point energy $\frac{1}{2}\hbar\omega$. Both facts are proved in the verified layer.

The Bose–Einstein distribution is defined as $\bar{n}(x) = 1/(e^x - 1)$, and the verified layer proves its positivity, its strict monotone decrease, and — crucially — that the thermal occupation computed from it equals the mean of the geometric distribution that the thermal state assigns to number eigenstates. The temperature identity $\tau = \bar{n} + \frac{1}{2}$ then connects the mean occupation to the observable energy including the zero-point half:

```
#check @BookProof.ChapterBoseEinstein.boseEinstein
#check @BookProof.ChapterBoseEinstein.boseEinstein_mean
#check @BookProof.ChapterBoseEinstein.thermalTemperature_boseEinstein_eq_coth
#check @BookProof.ChapterCoherentTemperature.thermalProb_mean
#check @BookProof.ChapterCoherentTemperature.thermalTemperature_eq_mean_add_half
#check @BookProof.ChapterCoherentTemperature.thermalTemperature_vacuum
#check @BookProof.ChapterCoherentOccupation.thermalOccupation_energy
#check @BookProof.ChapterCoherentOccupation.thermalTemperature_eq_energy_expectation
```

The zero-point half is not a cosmetic addition: it is the fingerprint of the canonical commutation relations, and it is what makes the blackbody spectrum **universal** — independent of the particular cavity. The verified layer also proves that the thermal distribution maximizes entropy among all distributions with the same mean occupation, which is the statistical mechanics statement behind "the radiation is thermal because it is in equilibrium":

```
#check @BookProof.ChapterThermalMaxEntropy.thermalEntropy_eq
#check @BookProof.ChapterThermalMaxEntropy.shannonEntropy_le_thermalEntropy
```

What is **not** yet proved is the final assembly: the spectral energy density $u(\omega, T) = \frac{\hbar\omega^3}{\pi^2 c^3} \cdot \frac{1}{e^{\hbar\omega/kT} - 1}$ requires integrating the mode occupation against the density of states in a three-dimensional cavity. The per-mode statistics are verified; the continuum mode sum is specialist work.

42.4. The Oscillator Spectrum: The Bridge to Bound States

Three of the four touchstones — the Lamb shift, positronium, and $g - 2$ at its core — are ultimately questions about the **spectrum** of a self-adjoint operator. The simplest nontrivial spectrum in the verified layer is the harmonic oscillator: on the Hermite core of $L^2(\mathbb{R})$, the operator $-d^2/dx^2 + x^2/4$ acts diagonally on the Hermite functions, with eigenvalues $n + \frac{1}{2}$. This is proved exactly, not perturbatively:

```
#check @BookProof.HermiteStrichartzQG.oscillatorSymbol
#check @BookProof.HermiteStrichartzQG.oscillator_eigenfunction
#check @BookProof.HermiteStrichartzQG.oscillatorOp_hermiteLp
#check @BookProof.HermiteStrichartzQG.oscillator_essentiallySelfAdjoint_on_hermiteCor
```

The importance of this result for the constants suite is structural. The oscillator spectrum is the exactly-solvable backbone against which every perturbative correction is measured: the Lamb shift is a correction to a Coulomb bound state computed in an oscillator-like basis, positronium's fine structure is a relativistic correction to a hydrogenic spectrum, and the leading term of $g - 2$ is a magnetic-moment coupling whose nonrelativistic limit is an oscillator problem. Essential self-adjointness on the Hermite core guarantees that the spectrum is a well-defined observable — there is a unique self-adjoint operator whose eigenvalues the perturbation theory is correcting.

42.5. Positronium: The Two-Body Touchstone

Positronium — the bound state of an electron and a positron — is the cleanest two-body touchstone in QED: no hadronic baggage, no renormalization of composite structure, pure leptons bound by the Coulomb attraction. Its spectroscopy (the $1S \to 2S$ interval, the fine structure, the lifetime split between para- and ortho-positronium) is a precision test of the two-body Dirac equation and its radiative corrections.

The verified layer contains the **algebraic** half of the positronium story: the fermionic CAR algebra that distinguishes the two-body fermion pair from a bosonic mode (the Jordan–Wigner two-mode anticommutation that underlies the spin–statistics dichotomy), the parity classification of the Dirac mass Hamiltonian (which mass term breaks parity and which conserves it — the concrete, decidable core of the CPT discussion), and the relativistic dispersion relation that fixes the two-body spectrum's kinematic backbone. What these theorems fix is **which** states can exist and **which** terms conserve parity; the decay **rates** are radiative-correction computations that remain open:

```
#check @BookProof.SpinStatistics.fermi_CAR_cross
#check @BookProof.SpinStatistics.fermiAnticomm_annih
#check @BookProof.ChapterCPTParity.parity_diracHamOp
#check @BookProof.ChapterCPTParity.parity_diracHamOp_invariant
#check @BookProof.ChapterCPTHamiltonian.diracHamOp_sq
```

The honest boundary: the verified layer proves the algebraic selection rules — the anticommuting two-fermion Fock space, the parity-even/odd classification of every Dirac mass term, and the exact mass-shell relation. It does not yet compute the decay rates $\tau_p \approx 1.25 \times 10^{-10}$ s and $\tau_o \approx 1.42 \times 10^{-7}$ s — those require the full QED vertex calculus, which is not formalized.

42.6. The Lamb Shift and g-2: Radiative Corrections

The Lamb shift and the anomalous magnetic moment are the two classic radiative-correction touchstones. Both are **zero** in the Dirac theory and **nonzero** only because of the quantized electromagnetic field: the Lamb shift is the self-energy of a bound electron interacting with vacuum fluctuations, and $g - 2$ is the vertex correction to the electron–photon coupling. The one-loop answers — $\Delta E_{2S-2P} \approx 1058$ MHz and $a_e = \frac{\alpha}{2\pi} \approx 0.00116$ — are among the most precisely confirmed predictions in physics.

These are the hardest targets in the suite, and the verified layer does not yet contain them. What it **does** contain is the certification machinery that any such computation would feed into. The verified layer's philosophy for numerical claims is: a number becomes a proof only when it arrives wrapped in a certified enclosure. The Temple-certificate layer proves exactly this: from a computed trial vector with its Rayleigh quotient and residual, plus an **a priori** spectral separation, one obtains a rigorous band containing the true eigenvalue — and the certificate layer shows that without the separation hypothesis, no lower bound is possible at all:

```
#check @BookProof.RitzCertificate.temple_lower_bound
#check @BookProof.RitzCertificate.temple_band_mem
#check @BookProof.RitzCertificate.fock_mass_gap_of_temple_certificates
#check @BookProof.TempleSeparationNecessary.separation_necessary
```

The same discipline is expressed in the gap-certificate structure: a certificate is a pair of numbers (measured value, assembled width) and the theorem consumes **only** those two numbers, proving a rigorous lower bound from them and nothing more. The recorded $g = 2, m = 4$ fixture carries a certified lower bound of 1.932 — a statement about a finite truncation, never claimed as a continuum result:

```
#check @BookProof.SirkCertifiedGap.GapCertificate
#check @BookProof.SirkCertifiedGap.gap_ge_of_certificate
#check @BookProof.SirkCertifiedGap.qcdG2M4
#check @BookProof.SirkCertifiedGap.qcdG2M4_lower
```

The route from here to the Lamb shift and $g - 2$ is clear but long: one would need (1) the free Dirac spectrum on a Coulomb background as the exactly-solvable backbone, (2) the one-loop self-energy and vertex integrals as certified numerical enclosures, and (3) the composition of those enclosures through the band machinery above. Each step is a specialist work package in the style of the mass-gap programme.

42.7. What Is Verified, and What Is Open

The exactly-verified thermal statistics of bosonic modes (Bose–Einstein distribution, mean occupation, zero-point energy, entropy maximality), the exact harmonic-oscillator spectrum with essential self-adjointness on the Hermite core, the symmetry selection rules for positronium decay channels, and the complete certificate machinery for turning numerical output into rigorous spectral enclosures.

Deliberately left open is the continuum mode-sum assembly of the Planck spectrum; the two-body Coulomb bound-state spectrum and its relativistic corrections; the one-loop self-energy and vertex integrals; and the final numbers themselves — the Lamb shift in MHz, a_e as a series in α, the positronium lifetimes. These are quarantined here and in the appendix, in keeping with the book's rule: nothing enters the verified layer without a proof.

43. NS Numerics: Laminar Decay, Advection Symmetry, Turbulence Laws

43.1. Orientation and Status

The Navier–Stokes numerics companion of the part: what the SIRK measurements on the NS generator actually measure, and how they sit against the analytic statements of the operator chapters and the reliability chapter. Like the constants suite next in the tour, it closes with what is verified and what is open.

The route: what the NS numerics measure, the laminar decay rate (diffusive decay of the parabolic part), advection symmetry (the generator is a Hamiltonian), and the turbulence laws (energy bookkeeping on the truncation).

The verified pieces are `BookProof/ChapterSirkDiffusiveDecay` and the `BookProof/NavierStokesEulerian` / `BookProof/NavierStokesFlow` statements the measurements are compared against.

43.2. What the NS Numerics Measure

The companion solver's Navier–Stokes validation suite measures three things: the **laminar decay rate** νk^2 of a diffusing velocity mode, the **advection symmetry** that makes the quadratic generator a legitimate Hamiltonian, and the **energy bookkeeping** (Ehrenfest identities, conserved norm, no-blowup bounds) that distinguishes a computation about the Navier–Stokes generator from an artifact of the discretization. Each of the three has a verified counterpart in `BookProof`, and this chapter is the map from the measured number to the theorem that licenses it.

This is an honesty page in the same style as the constants-suite chapter: where the verified layer proves the statement, the theorem is named; where the physical claim (a real viscosity, a real turbulence cascade) enters as an **input**, we say so. Nothing here claims a continuum Navier–Stokes result — the truncation boundary is stated with the theorems that live on it.

43.3. The Laminar Decay Rate: Diffusive Decay of the Parabolic Part

The laminar-regime test evolves $du/dt = -\nu k^2 u$ and measures the exponential decay rate. The verified counterpart is the parabolic-semigroup theorem: for a **coercive** bounded generator A — one satisfying $\mu\|x\|^2 \le \mathrm{Re}\langle x, Ax\rangle$ — the heat semigroup e^{-tA} decays at exactly the coercivity rate, in vector norm and in operator norm. The coercivity constant μ is precisely the νk^2 the numerics read off the reduced model:

```
#check @BookProof.ChapterSirkDiffusiveDecay.heatFlow
#check @BookProof.ChapterSirkDiffusiveDecay.IsCoercive
#check @BookProof.ChapterSirkDiffusiveDecay.norm_heatFlow_apply_le
#check @BookProof.ChapterSirkDiffusiveDecay.norm_heatFlow_le
```

The decay bound is proved by a Grönwal argument on the weighted energy $\|e^{-tA}v\|^2 e^{2\mu t}$: the energy identity $d/dt\,\|u(t)\|^2 = -2\,\mathrm{Re}\langle u(t), Au(t)\rangle$ makes the weight antitone, and the bound $\|e^{-tA}v\| \le e^{-\mu t}\|v\|$ follows for every $t \ge 0$. Crucially, the theorem is **exact at the rate** — not merely asymptotic — which is what makes the measured decay constant a certificate rather than a fit.

The reduction-invariance half of the statement is what connects the theorem to the solver's Galerkin/SIRK reduction: the compression V^*AV of a coercive generator along an isometry is coercive with the **same** constant,

so the reduced propagator obeys the identical decay bound at every reduction order. The measured laminar rate is therefore not an artifact of the truncation:

```
#check @BookProof.ChapterSirkDiffusiveDecay.isCoercive_compress
#check @BookProof.ChapterSirkDiffusiveDecay.norm_heatFlow_compress_apply_le
```

The honest boundary: identifying μ with νk^2 for a **particular** discretization is the content of the per-mode symbol computations in the Navier–Stokes chapters, and the numerical value of ν is an input, not a theorem. The generator in the decay theorem is bounded — the regime in which the project's Galerkin/Hashimoto reduction is an operator statement; the unbounded case is the standing Stone/Trotter–Kato boundary.

43.4. Advection Symmetry: The Generator is a Hamiltonian

Every decay and energy statement above silently assumes the generator is **symmetric** — otherwise there is no self-adjoint evolution and no conserved energy to track. For the truncated Navier–Stokes Hamiltonian $H = \sum_i (\pi_i A_i + A_i \pi_i)$ with the advective term $A_i = \sum_j u_j u_{i,j} - \nu u_{i,jj}$, symmetry is proved term by term: the advective part is symmetric on the domain because the modes are symmetric and pairwise commuting and the viscosity is real:

```
#check @BookProof.NavierStokesFlow.FullEsa.NSFullData.advection
#check @BookProof.NavierStokesFlow.FullEsa.NSFullData.advection_isSymmetricDom
#check @BookProof.NavierStokesFlow.FullEsa.NSFullData.hamiltonian_isSymmetricDom
```

The same structure holds **untruncated**: on a dense domain of an arbitrary inner-product space, with fifteen symmetric pairwise-commuting field modes and three symmetric momenta, the full Hamiltonian is symmetric on its domain unconditionally — and for concrete realizations (bounded lattice modes on $\ell^2(\mathbb{Z})$, diagonal modes on $\ell^2(\mathbb{N})$) it is essentially self-adjoint. The sharpness companion proves the converse boundary: structural hypotheses alone can never yield essential self-adjointness, so the analytic input (a complete flow) is indispensable:

```
#check @BookProof.NavierStokesFlow.FullEsa.latticeFull_hasZeroDeficiencyOn
#check @BookProof.NavierStokesFlow.FullEsa.diagFull_hasZeroDeficiencyOn
#check @BookProof.NavierStokesFlow.FullEsa.exists_nsFullData_not_hasZeroDeficiencyOn
```

The differential realization carries the polynomial-transport route: the Weyl-ordered Hamiltonian is the core realization of an explicit polynomial whose Gauss symmetry is checked purely algebraically, the closure is the **unique** self-adjoint extension, and the Hashimoto selection theorem pins down the operator the SIRK Krylov iteration computes — the shift-inverted resolvents exist, are bounded by $1/|\operatorname{Im}\gamma_j|$, and each single one determines the generator completely:

```
#check @BookProof.NavierStokesFlow.DiffHashimoto.nsDiffPoly
#check @BookProof.NavierStokesFlow.DiffHashimoto.nsDiffH_eq_coreOp
#check @BookProof.NavierStokesFlow.DiffHashimoto.nsDiffH_symmetricOn
#check @BookProof.NavierStokesFlow.DiffHashimoto.nsDiffH_selfAdjoint_extension
#check @BookProof.NavierStokesFlow.DiffHashimoto.nsDiffH_selfAdjoint_extension_unique
#check @BookProof.NavierStokesFlow.DiffHashimoto.nsDiffH_hashimoto_selects
#check @BookProof.NavierStokesFlow.DiffHashimoto.nsDiffH_shiftInvert_selects
#check @BookProof.NavierStokesFlow.DiffHashimoto.nsQuadraticDiffH_hashimoto_selects
```

For the numerics this is the difference between measuring **something** and measuring **the** generator: the selection theorem guarantees the Krylov subspace reconstructs spectral information about the differential Navier–Stokes operator rather than an artifact of the shifts. The Ehrenfest identities the validation checks exactly on probes ($i\langle[H,u]\rangle = 4\kappa\langle u\rangle + 4c$ on the affine fiber) are statements about this selected operator.

43.5. Turbulence Laws: Energy Bookkeeping on the Truncation

The classical turbulence laws — kinetic-energy balance, enstrophy bookkeeping for the 2D inverse cascade, the Kolmogorov $-5/3$ inertial spectrum — are **statistical** statements about solutions the verified layer does not construct. What it does prove is the exact finite-dimensional energy bookkeeping the numerical pipeline relies on:

the truncated flow is a one-parameter unitary group (norm preserving, no finite-time blowup, unique global solutions of the Cauchy problem), and the expectation of the Hamiltonian is conserved along it:

```
#check @BookProof.NavierStokesFlow.nsFlow_unitary
#check @BookProof.NavierStokesFlow.nsFlow_norm_preserving
#check @BookProof.NavierStokesFlow.nsFlow_noBlowup
#check @BookProof.NavierStokesFlow.nsCauchy_existsUnique
#check @BookProof.NavierStokesFlow.nsFlow_energy_conserved
```

The Lagrangian (parcel) form of the generator — the four-term operator with positive advective Laplacian, positive viscous term, force drift, and the volume-preservation constraint — is Hermitian on the truncation, so the same completeness carries over: its flow is unitary and its Cauchy problem has exactly one global solution. This is the truncated form of the transformation route; the continuum statement remains unclaimed:

```
#check @BookProof.NavierStokesFlow.LagrangianNS.flowUnitary_unitary
#check @BookProof.NavierStokesFlow.LagrangianNS.cauchy_existsUnique
```

At the kinematic layer, the Eulerian constraint algebra is verified symbolically: the momentum constraint and the divergence-free constraint are consistent under the derivative field's consistency identities, and the cyclic-shear test field is exactly divergence-free — the decidability core that any turbulence-law computation must respect before it can claim to represent an incompressible flow:

```
#check @BookProof.NavierStokesEulerian.eulerian_momentum_constraint
#check @BookProof.NavierStokesEulerian.eulerian_divergence_constraint
#check @BookProof.NavierStokesEulerian.cyclicShear_divergence_free
```

43.6. What Is Verified, and What Is Open

The exactly-verified laminar decay bound $\|e^{-tA}\| \leq e^{-\mu t}$ for coercive generators with reduction-invariance of the rate; advection symmetry of the truncated and untruncated Navier–Stokes Hamiltonians with essential self-adjointness on concrete realizations and the Hashimoto selection of the differential generator; and the complete finite-dimensional energy bookkeeping — unitary flow, norm preservation, no blowup, unique global Cauchy solutions, conserved Hamiltonian expectation — on the truncation and its Lagrangian transform.

Deliberately left open is everything statistical: the inertial-range spectrum ($E(k) \sim k^{-5/3}$), the enstrophy cascade and its 2D inverse counterpart, intermittency, and any claim about a **continuum** solution. The viscosity ν and the per-mode identification $\mu = \nu k^2$ are numerical inputs consumed by the theorems, not outputs of them. The sharpness theorem (`BookProof.NavierStokesFlow.FullEsa.exists_nsFullData_not_hasZeroDeficiencyOn`) marks exactly where structural reasoning must stop and analytic input must begin — the same boundary the $\dot{x} = x^2$ warning of the ODE chapter draws for flows.

44. Consciousness, Deep Learning, and the Bayesian Prior

The manuscript's Bayesian foundations of agency. We show that no point of null measure is special and no prior is best for all cases, that a deterministic prior is still subjective, and that Bayesian inference is a unitary representation; and that randomized (deep) learning is a random sampling method inducing a Bayesian posterior over models. We close with the coherent-state reading of Softmax attention, where the Born rule on coherent states is shown to be the same equation as the attention mechanism.

How to read this part. The three chapters share one thesis — a system is a Bayesian prior in action — but they use it in three different registers. Consciousness as a representation of a Bayesian prior is the philosophical application and needs only the probability part behind it. Aligned deep learning as a random sampling method is the engineering application and re-uses the induced-prior machinery of the Bayesian chapters. Dimension-robust Bayesian adaptation is the learned-architecture application: hyperspherical (nGPT) geometry makes every logit and every stochastic-gradient step uniformly bounded, so the Bayesian adaptation of a **transformer** is dimension-robust — and it records where that geometry meets, and does not meet, the final- Hamiltonian convention of the field-theory parts. Softmax is the Born rule on coherent states is the closest to physics: it re-uses the Gaussian of

the free-field chapter and the Born rule of Part II, and it is the one chapter here that connects directly to the harmonic-oscillator machinery of the field-theory part.

45. Consciousness as a Representation of a Bayesian Prior

45.1. Orientation and Status

The opening chapter of the Bayesian-agency part: the philosophical application of the toolkit — no prior is best for all cases, no point of null measure is special, and a deterministic prior is still subjective. Per the part's reading note, it needs only the probability part behind it: the coherence target, the associativity of updating, the uniform prior as point of reference.

The route: the question, no prior is special and no point is special, a deterministic prior is still subjective, the uniform prior as a point of reference, and Bayesian inference as a representation.

The proofs are `BookProof/ChapterNoBestPrior`, `BookProof/ChapterPriorDependence`, `BookProof/ChapterPriorOdds`, `BookProof/ChapterUniformPrior` (with its posterior variant), and `BookProof/ChapterBayesInference`. The null-measure discipline behind "no point is special" is developed in the null-measure chapter later in the tour.

45.2. The Question

The manuscript's chapter on consciousness begins from a basic fact of Bayesian inference: there is always a prior probability distribution, and there is no prior which is better for all cases — we always have to make assumptions. It draws implications for consciousness, extraterrestrial intelligence, and artificial general intelligence. Its central thesis is that a "conscious" system is one running a symbolic manipulation of a Bayesian prior, and that a point of null measure is not automatically special. The verified content here is the measure-theoretic and Bayesian facts that the manuscript's argument rests on.

45.3. No Prior Is Special, and No Point Is Special

Two claims underpin the manuscript's argument. First, **no point of null measure is special**: every singleton of the real line has Lebesgue measure zero, so no single value of a continuous random variable carries probability mass; and there exist **uncountable** null sets (the Cantor set), so a set being null does not mean it is small. Singling out one null point as a "conscious" observer is therefore not justified by the measure:

```
#check @BookProof.ConsciousnessNullMeasure.singleton_volume_zero
#check @BookProof.ConsciousnessNullMeasure.countable_volume_zero
#check @BookProof.ConsciousnessNullMeasure.cantorSet_uncountable
#check @BookProof.ConsciousnessNullMeasure.cantorSet_volume_zero
#check @BookProof.ConsciousnessNullMeasure.exists_uncountable_null_subset
```

Second, **no prior is better for all cases**: for any two distinct finite priors there is a decision problem (utility) preferring one and another preferring the other — a no-free-lunch statement. Prior choice is unavoidable and subjective:

```
#check @BookProof.ChapterNoBestPrior.expectedUtility
#check @BookProof.ChapterNoBestPrior.not_uniformly_better
#check @BookProof.ChapterNoBestPrior.distinct_priors_each_preferred
```

And on a countably infinite space there is no uniform probability measure on singletons at all — "the rationals are not enough", so a genuinely continuous prior must be used:

```
#check @BookProof.ChapterNoUniformCountable.no_uniform_countable_measure
#check @BookProof.ChapterNoUniformCountable.no_uniform_measure_nat
```

45.4. A Deterministic Prior Is Still Subjective

The manuscript argues that a deterministic (frequentist) prior is still subjective. A Dirac prior concentrated at a single hypothesis is a genuine probability distribution, Bayesian updating preserves it (a positive-likelihood update keeps it deterministic), and two distinct deterministic priors yield distinct posteriors for the same data — so the prior is doing real work and is not eliminated by being "objective":

```
#check @BookProof.ChapterPriorDependence.diracPrior
#check @BookProof.ChapterPriorDependence.diracPrior_sum_one
#check @BookProof.ChapterPriorDependence.posterior_dirac
#check @BookProof.ChapterPriorDependence.distinct_priors_distinct_posteriors
```

The prior-odds form makes the subjectivity quantitative: posterior odds equal prior odds times the likelihood ratio, and equal likelihoods preserve prior odds:

```
#check @BookProof.ChapterPriorOdds.posterior_odds
#check @BookProof.ChapterPriorOdds.equal_likelihood_preserves_odds
```

45.5. The Uniform Prior as a Point of Reference

The uniform prior is the unique relabeling-invariant prior, and it serves as a neutral reference: under a positive uniform prior, MAP estimation coincides with maximum likelihood. This gives the manuscript's "no prior is better for all cases" a positive counterpart — the uniform prior is the canonical prior-free starting point from which any departure is an explicit assumption:

```
#check @BookProof.ChapterUniformPrior.normalized_isRelabelingInvariant_eq_uniform
#check @BookProof.ChapterUniformPrior.uniform_prior_isMAP_iff_isMLE
#check @BookProof.ChapterUniformPriorPosterior.exists_likelihood_uniform_prior_poster
```

45.6. Bayesian Inference Is a Representation

The manuscript's thesis that consciousness is a **representation** of a Bayesian prior is captured abstractly by the fact that Bayesian inference is a unitary (Born-rule) operation: the Bayes posterior equals the Born-rule conditional of the wave-function $\Psi = \sqrt{p}$, and a unitary matrix reproduces the posterior — "unitary inference" is Bayesian inference. The uniform-prior posterior is likewise a genuine probability distribution:

```
#check @BookProof.ChapterBayesInference.posterior_eq_born_conditional
#check @BookProof.ChapterBayesInference.exists_unitary_reproduces_posterior
```

Bayesian updating is irreversible in the information sense: entropy increases under a doubly-stochastic Markov map (the discrete H-theorem), so inference generally destroys information. This is the mathematical counterpart of the manuscript's point that a conscious system's internal model is not a literal replay of the world:

```
#check @BookProof.ChapterMarkovEntropy.entropy_applyMarkov_ge
#check @BookProof.ChapterMarkovEntropy.entropy_applyMarkov_permMatrix
```

45.7. Summary

The verified content of the manuscript's consciousness chapter:

- no point of null measure is special, and null sets need not be small (Cantor);
- no finite prior is better for all cases (no-free-lunch), and no uniform prior exists on a countably infinite space;
- a deterministic prior is still subjective (distinct Dirac priors give distinct posteriors);
- the uniform prior is the unique relabeling-invariant reference (MAP = MLE);
- Bayesian inference is a unitary/Born-rule representation, and updating is entropy-increasing.

46. Aligned Deep Learning as a Random Sampling Method

46.1. Orientation and Status

This chapter documents the finite Bayesian content of the manuscript's learning proposal. Induced priors, posterior identities, and MAP algebra are formalized; claims about generalization, overfitting, alignment, and real training systems need separate statistical or empirical hypotheses and are listed as specialist proof targets only when they have a precise mathematical formulation.

In the tour this is the engineering application of the part: randomized training as a **random sampling method**, re-using the induced-prior machinery of the Bayesian chapters (the chapter before this one). The route: the question, randomized training induces a prior, deep learning as MAP estimation, the induced posterior is Bayesian, MAP points are null under an atomless posterior, the hierarchical and subjective structure, previous knowledge and finite arithmetic, and probability as a universal language.

The proofs are `BookProof/ChapterDeepLearningSampling`, `BookProof/ChapterDeepLearningEnsemble`, `BookProof/ChapterDeepLearningMAP`, `BookProof/ChapterMAPNull`, and the hierarchical/finite-arithmetic chapters (`ChapterHierarchicalBayes`, `ChapterFiniteArithmeticPrior`, `ChapterFiniteBayesHierarchy`). The closing chapter of the part reads the same mathematics in the language of coherent states.

46.2. The Question

The manuscript's chapter on aligned deep learning separates the alignment problem into two parts: estimating systematic uncertainties as in engineering, and — new due to Big Data — how to randomly sample **computable** models from a set of Bayesian models, almost all of which are uncomputable. It argues that deep neural nets solve the second part, and that because no sufficiently simple metric is reliable, the (dis)alignment is a **feature**: if the sampling were not random it would not generalize well. The verified content here is the finite, Bayesian core of that thesis: randomized training induces a prior, and the resulting posterior and MAP estimates are exactly the Bayesian ones.

46.3. Randomized Training Induces a Prior

The manuscript's central model is that training is a **random** procedure: a distribution on training seeds is pushed forward through a training map to a distribution on models — the **induced prior**. The verified facts are that this induced distribution is a genuine probability distribution, that the mass of an event equals the mass of its preimage, and that if every seed trains to an admissible model then inadmissible models have zero mass:

```
#check @BookProof.ChapterDeepLearningSampling.inducedPrior
#check @BookProof.ChapterDeepLearningSampling.inducedPrior_isProbability
#check @BookProof.ChapterDeepLearningSampling.sum_inducedPrior_event
#check @BookProof.ChapterDeepLearningSampling.inducedPrior_supported
```

Expectations and evidence under the induced prior reduce to averages over training seeds, and posterior expectations are ratios of seed-level weighted averages:

```
#check @BookProof.ChapterDeepLearningEnsemble.inducedPrior_expectation
#check @BookProof.ChapterDeepLearningEnsemble.evidence_eq_seed_average
#check @BookProof.ChapterDeepLearningEnsemble.posterior_expectation_eq_seed_ratio
```

This is the precise sense in which "deep learning is a random sampling method": the model distribution is the induced prior, and every Bayesian quantity is computable as an expectation over the random seeds.

46.4. Deep Learning as MAP Estimation

The manuscript connects randomized training to Bayesian MAP estimation. The logarithmic objective $\log(\text{prior}) + \log(\text{likelihood})$ is ordered exactly as the unnormalized posterior weight, so maximizing the log-objective is maximizing the posterior (the normalization is irrelevant to the choice):

```
#check @BookProof.ChapterDeepLearningMAP.logObjective
#check @BookProof.ChapterDeepLearningMAP.exp_logObjective
#check @BookProof.ChapterDeepLearningMAP.isMAP_iff_maximizes_logObjective
#check @BookProof.ChapterDeepLearningMAP.posterior_le_iff_weight
```

46.5. The Induced Posterior Is Bayesian

Conditioning the induced prior on observed data yields a genuine posterior distribution, and models outside the image of training have zero posterior mass — the finite version of "sampling computable models": only computable models are reachable, and the posterior respects that constraint:

```
#check @BookProof.ChapterDeepLearningSampling.evidence
#check @BookProof.ChapterDeepLearningSampling.posterior
#check @BookProof.ChapterDeepLearningSampling.posterior_sum_one
#check @BookProof.ChapterDeepLearningSampling.posterior_eq_zero_of_not_mem_range
```

46.6. MAP Points Are Null Under an Atomless Posterior

The manuscript's point that no single model is "the answer" is captured by the fact that a selected MAP point has posterior mass zero under an atomless posterior: a sample almost surely does not equal any fixed maximizer, and any countable collection of maximizers is null. The "alignment" is a distributional feature, not a single point:

```
#check @BookProof.ChapterMAPNull.map_point_measure_zero
#check @BookProof.ChapterMAPNull.ae_ne_map_point
#check @BookProof.ChapterMAPNull.countable_map_set_measure_zero
```

46.7. Hierarchical and Subjective Structure

Real alignment problems are hierarchical. The verified facts are that a two-level (or arbitrary finite) Bayesian hierarchy collapses to a single kernel — arbitrarily deep marginalization equals one marginalization through the collapsed kernel — and that the outer posterior is an ordinary Bayes update:

```
#check @BookProof.ChapterHierarchicalBayes.outerPosterior_eq_bayesUpdate
#check @BookProof.ChapterHierarchicalBayes.outerPosterior_eq_sum_flat
#check @BookProof.ChapterHierarchicalBayesComposition.compKernel_assoc
#check @BookProof.ChapterFiniteBayesHierarchy.nestedMarginal_eq_terminalMarginal
```

And because the prior is subjective (two distinct priors give distinct posteriors; no prior is best for all cases), alignment is inherently a choice — the sampling prior encodes the alignment target:

```
#check @BookProof.ChapterNoBestPrior.distinct_priors_each_preferred
#check @BookProof.ChapterPriorDependence.distinct_priors_distinct_posteriors
```

46.8. Previous Knowledge and Finite Arithmetic

The manuscript's "previous knowledge as step-by-step learning" and the "digital-first mathematics" themes have a finite, Bayesian rendering: a finite exact arithmetic table admits a Bayesian extension by a finite hypothesis space, and the extension preserves every result in the table. Known results are represented exactly; the unknown is a genuine probability distribution:

```
#check @BookProof.ChapterFiniteArithmeticPrior.BoundedArithmetic
#check @BookProof.ChapterFiniteArithmeticPrior.BayesianArithmeticExtension
#check @BookProof.ChapterFiniteArithmeticPrior.prior_is_probability
#check @BookProof.ChapterFiniteArithmeticPrior.certainExtension_known
```

46.9. Probability as a Universal Language

Underlying the whole chapter is the manuscript's claim that probability is a **universal language**: any mechanism that turns a state x into a distribution over answers — a Markov (conditional-probability) kernel $\kappa : X \to \Pr(Y)$ — **transports** a prior on X into a genuine predictive distribution on Y, and nothing is lost or created in the process: the total mass stays 1:

```
#check @BookProof.ChapterKernelTransport.kernel_transport_isProbability
#check @BookProof.ChapterKernelTransport.kernel_transport_apply
#check @BookProof.ChapterKernelTransport.kernel_transport_lintegral
```

The transported measure of a set is the μ-average of the kernel values (the law of total probability), and expectations under the transported law are averages of the conditional expectations. Deterministic transport is the special case where the kernel is a measurable map f:

```
#check @BookProof.ChapterKernelTransport.map_transport_isProbability
#check @BookProof.ChapterKernelTransport.kernel_transport_deterministic
```

This is the precise sense in which a model in this book is a genuine probability space: whatever the internal mechanism (a training map, a neural net, a unitary rotation), its output over a random input is a bona fide distribution, and every Bayesian quantity is expressible as an expectation over it.

46.10. Summary

The finite, Bayesian core of the manuscript's deep-learning chapter:

- randomized training induces a prior distribution on models;
- the induced posterior is genuinely Bayesian, and models outside the image of training have zero mass;
- deep learning is MAP estimation (maximizing the log-objective);
- MAP points are null under an atomless posterior — alignment is distributional;
- hierarchies collapse to a single kernel, and the outer posterior is an ordinary Bayes update;
- previous knowledge is represented exactly and the unknown is a probability distribution.

47. Dimension-Robust Bayesian Adaptation: Hyperspherical nGPT Geometry and SG-pCNL

47.1. Orientation and Status

This chapter reads the `BayesianTransformer/` development pedagogically: a formalization of **Dimension-Robust Bayesian Adaptation of Transformers via Hyperspherical Geometry and Unadjusted SG-pCNL** (Milestones 1–5), imported into this repository on 2026-10-21 from the specialist's export. The tree is a Lean library in its own right (Lake target `BayesianTransformer`, namespace `BayesianTransformer`), **not** a `BookProof/` chapter: it has not been compiled in this snapshot, and step 0 of the specialist work order in `CONSOLIDATED_PLAN.md` is its name-level check.

In the tour this chapter is the **learned-systems** face of the same thesis the aligned-deep-learning and coherent-state-attention chapters state in other registers: a system you train is a system you put a prior on. What is new here is that the architecture itself — normalized embeddings, cosine attention, geodesic residuals — is designed so that the **geometry** does the statistical work: every weight lives on a sphere, every logit is clamped, and every stochastic-gradient step stays in a ball of radius $\sigma\sqrt{P}$\$. The proofs below are the exact content of those three sentences.

The sibling import `math28/` (Lake target `Math28`) is unrelated in subject — named hypotheses transcribed from the `math/OAI` snapshot, with proved bridges into this repository's chapters — and is not discussed here.

47.2. Scale Invariance and the Sphere

The foundational observation (`Analysis/ScaleInvariance.lean`) is that normalization makes the loss **blind to radius**: if Φ\$ is scale-invariant (`IsScaleInvariant`) then its gradient is orthogonal to the position (`scaleInvariant_gradient_orthogonal`, `scaleInvariant_inner_grad_zero`). The radial direction carries no information, so a gradient flow on the sphere is not a restriction of the flow — it **is** the flow, up to the discarded radial gauge.

`Analysis/SphereProj.lean` makes the projection onto the unit sphere a differentiable map with sharp constants: `norm_sphereProj` (the image is on the sphere), `sphereProj_hasFDerivAt` and

`sphereProj_deriv_opNorm_le` (its derivative is the tangent-space projection, with a $1/\|v\|$ bound that blows up exactly at the origin), and `sphereProj_smul_pos` (the map is homogeneous of degree zero — the invariance again, now as a identity).

47.3. Cosine Attention Is a Clamped, Lipschitz Kernel

The attention mechanism (`Transformer/CosineAttention.lean`) replaces dot-product logits with cosine logits. The verified facts are the ones a probabilist needs: on the sphere the logit $q \cdot k$ is bounded (`cosineSimilarity_bounds`), so the scaled logit lies in a fixed window $[-\tau, \tau]$ (`scaled_logit_bounds`); the softmax is a genuine distribution (`softmax_pos`, `softmax_sum_eq_one`, `softmax_le_one`); and the softmax map is Lipschitz with constant 2 in the $\ell^\infty \to \ell^1$ comparison (`softmax_lipschitzWith_two`, `softmax_lipschitz`). Clamping is what makes the subsequent noise and Lipschitz bookkeeping **uniform in the context length**: no logit can spike, so no attention row can concentrate harder than $e^{2\tau}$ permits.

47.4. The Gradient-Noise Channel

`Probability/GradientNoise.lean` is the formal face of the stochastic-gradient hypothesis that the sampler plan rests on. For a mini-batch of size B drawn from a dataset of size N, the covariance of the stochastic gradient is bounded by the full-gradient covariance scaled by N/B (`sg_pcnl_noise_covariance_bound`), and the **trace** of that covariance — the quantity that actually enters a Langevin step's mean-square displacement — is bounded by `sg_pcnl_noise_trace_bound` / `noiseCovTrace_le`. This is the milestone the sibling repository's numerics test directly: the N/B scaling is what lets one backward pass per step stand in for the exact gradient without changing the invariant measure to first order.

47.5. The Whole Network, Lipschitz-Uniformly

Two chapters compose the local bounds into global ones. `Transformer/LipschitzCalculus.lean` bounds each building block — contractions (`norm_clm_apply_le_one`), attention (`norm_attnOut_le`), the geodesic residual step (`norm_residualNorm`, `norm_ngptLayer`), the hidden-state stack (`norm_hiddenStates`) — and then the loss itself as a function of the logits (`dist_crossEntropy_le`).
`Transformer/GlobalLipschitz.lean` upgrades these to one constant for the entire architecture: `lipOn_ngptLayer` and `lipOn_hiddenStates` exhibit the network as Lipschitz with an explicit constant built from the per-layer moduli (`attnConst_nonneg`, `layerConst_nonneg`).

The honest boundary: a global Lipschitz constant bounds **perturbations of the loss**, not commutators of unbounded operators. It is therefore the analyst's input to the numerical layer — step-size ceilings, noise ceilings, robustness of the BMA estimate — and not an ingredient of any Faris–Lavine argument in this book. The distinction is the same one the field-theory chapters make between a relative bound and a comparison operator.

47.6. Where This Sits Under the Final-Hamiltonian Convention

Nothing in `BayesianTransformer/` is a Hamiltonian statement: no operator is enclosed in creation and annihilation ladders, and none should be read in. The convention the rest of the book states — the final Hamiltonian of each theory is the **one-particle** operator enclosed creation-left / annihilation-right ($d\Gamma(h) = \sum_{i,j} h_{ij}\, C^\dagger(e_i) A(e_j)$), with quantum gravity in its R^2 vielbein form carrying the exact exponential Einstein-frame wall (no Taylor truncation) and the interaction terms inside h, and with the NS/QG products-involving-field-derivatives disposed of by momentum-space convolution (the one-particle chapter, second quantization, the gravity elimination chapter) — constrains only the **bridges**: any future lemma placing a Bayesian prior over field configurations, or comparing a learned sampler's invariant measure with a Fock-space state, must enclose h in that form, and must not re-derive a convolution identity in coordinates where the momentum-space form is the one with the uniform bound.

What the two developments genuinely share is structural. The nGPT architecture achieves its uniform constants by making every weight a **bounded** object (a point on the sphere, a logit in a window) and every layer a **contraction-like** map; the Faris–Lavine certificates in `BookProof/` achieve theirs by making the comparison operator a **positive** object whose commutator is bounded in its quadratic form. In both cases the uniformity is bought at the

definition, not extracted from a perturbative estimate — which is why both survive composition over arbitrarily many layers (or particle-number sectors).

47.7. The Numerical Face of the Same Plan

The sibling repository `../unfer/pcnl` implements the sampler this formalization accompanies — unadjusted stochastic-gradient pCNL on the flat parameter vector of the normalized architecture — and its milestone tests (`pcnl/tests/numerics.rs`) are the empirical counterparts of the statements above: the prior draws sit on the shell $\sigma\sqrt{P}$ (`prior_draws_sit_on_the_shell_sigma_sqrt_p`), the OU drift term matches the second-moment identity (`drift_term_matches_the_ou_second_moment_identity`), the mini-batch gradient recovers the full potential at rate N/B (`n_over_b_scaling_recovers_the_full_potential_gradient` — the numeric twin of `sg_pcnl_noise_covariance_bound`), and the schedule satisfies the Robbins–Monro conditions (`robbins_monro_partial_sums_diverge_and_converge`). The full comparison run (seed 20261020, $P = 16672$, 10^4 steps × 4 chains) reports the plan's central empirical claim: Bayesian model averaging over the thinned posterior beats every point estimate on the shifted task — 54.7 % for pCNL and 58.2 % for the SGLD comparator against 41.3 % for the last chain state and 32.4 % for Adam — while the two samplers' loss chains agree across algorithms (`cross-Rhat` 1.110, one target), the cosine variant holds the shell (45.7 against $\sigma\sqrt{P} = 45.2$), and the standard softmax variant explodes at step 3. The book's claim is not that these numbers are theorem and proof; it is that the theorem side now names, one for one, the properties the numbers depend on.

48. The Coherent State of Attention

The Softmax attention mechanism of the Transformer is usually introduced as an engineering trick: a dot product measures compatibility between a query and a key, and the exponential Softmax squashes those scores into a probability distribution. This chapter argues that the trick is not arbitrary. When the objects being measured are **coherent states** — the most classical-like states of a quantum harmonic oscillator — the Softmax function and the quantum Born rule are the **same equation**. The classical sharpness of attention is the natural consequence of measuring the overlap of Gaussian wave-packets in a quantum phase space.

This chapter is a conceptual bridge between the probability-theoretic core of the book and machine learning. It reuses the tools already formalized in the Born-rule chapter (a probability distribution is the squared modulus of a wave-function), the free-field chapter (the Gaussian as the canonical rotation-invariant prior), and the deep-learning chapter (Bayesian updating as the collapse of a posterior). The coherent-state identities themselves are now formalized as well: the overlap, the Softmax–Born identity and the expectation-value packaging of the attention output are theorems in `BookProof`, and are cited section by section below. The one claim that remains a documented proof plan is the thermal temperature identity; its statistical core is proved, its physical derivation is not.

The route: the divergence between classical sharpness and quantum flatness, the geometry of the wave-packet, the Softmax–Born identity on coherent states, temperature and the thermal bath, informational superposition and the unknown output, the posterior as observable operators and expectation values, and the deep cascade of successive Bayesian updates. The proof material is cited section by section above; the one planned item — the thermal temperature identity — is collected in the appendix.

48.1. The Divergence: Classical Sharpness versus Quantum Flatness

For years, the natural proposal for a "quantum attention" was to treat the dot product $q \cdot k$ as a probability **amplitude** and apply Born's rule — the normalized squared modulus:

$$P(q \to k) = \frac{|q \cdot k|^2}{\sum_j |q \cdot k_j|^2}.$$

This is beautiful but behaves differently from Softmax. Squaring creates a flat, polynomial curve that spreads probability across many tokens; exponentiating $e^{q \cdot k}$ creates a sharp, winner-takes-all distribution. The dichotomy

rests on a single assumption: that a word is a classical **point**. If instead a word is a **wave-packet**, the two regimes coincide.

Both halves of the dichotomy, and the resolution, are theorems. `ampBorn_smul_query` is the flatness: the amplitude-squared rule is unchanged when the query is amplified by any nonzero factor, so it has no temperature to lower. `softmax_smul_query` says that amplifying the query in Softmax **is** a change of inverse temperature; `scoreSoftmax_zero` is the flat extreme (at $\beta = 0$ Softmax is uniform); and `tendsto_scoreSoftmax_max` / `tendsto_scoreSoftmax_ne` are the sharp extreme: with a strictly best-aligned key, its weight tends to 1 and all others to 0 as $\beta \to \infty$. Finally `tendsto_coherentBorn_smul_query` is the reconciliation: on **coherent** states the Born rule is Softmax, so — unlike the classical rule — it does sharpen under amplification:

```
#check @BookProof.ChapterSoftmaxSharpness.ampBorn
#check @BookProof.ChapterSoftmaxSharpness.ampBorn_smul_query
#check @BookProof.ChapterSoftmaxSharpness.softmax_smul_query
#check @BookProof.ChapterSoftmaxSharpness.scoreSoftmax_zero
#check @BookProof.ChapterSoftmaxSharpness.tendsto_scoreSoftmax_max
#check @BookProof.ChapterSoftmaxSharpness.tendsto_scoreSoftmax_ne
#check @BookProof.ChapterSoftmaxSharpness.tendsto_coherentBorn_smul_query
#check @BookProof.ChapterSoftmaxSharpness.tendsto_coherentBorn_smul_query_ne
```

48.2. The Geometry of the Wave-packet

Map the latent space of the network onto a Bosonic Fock space and represent each query and key as the parameter of a **coherent state** $|q\rangle, |k\rangle$ — a localized wave-packet, the "most classical" quantum state. The overlap of two coherent states is Gaussian:

$$\langle q|k\rangle = \exp\Big(-\tfrac{1}{2}|q|^2 - \tfrac{1}{2}|k|^2 + q \cdot k \Big).$$

The baseline magnitudes $|q|^2, |k|^2$ pull the amplitude down; their alignment $q \cdot k$ pushes it up. This is the **reproducing kernel** of the Bargmann–Fock space.

For real coherent-state parameters this is now a theorem. `coherentOverlap` is the overlap; `coherentOverlap_eq` is its explicit coordinate formula; `coherentOverlap_eq_gaussian` records the pleasant fact that the two baseline penalties and the alignment reward recombine into minus one half the squared distance, so the kernel **is** a Gaussian, `exp(-‖q-k‖²/2)`; and `coherentOverlap_self` is the normalization `⟨q|q⟩ = 1` of a coherent state (from which `coherentOverlap_unit` is the unit-parameter special case). The overlap is always a strictly positive real number, never exceeding one, and equals one exactly on the diagonal:

```
#check @BookProof.ChapterCoherentOverlap.coherentOverlap
#check @BookProof.ChapterCoherentOverlap.coherentOverlap_eq
#check @BookProof.ChapterCoherentOverlap.coherentOverlap_eq_gaussian
#check @BookProof.ChapterCoherentOverlap.coherentOverlap_pos
#check @BookProof.ChapterCoherentOverlap.coherentOverlap_le_one
#check @BookProof.ChapterCoherentOverlap.coherentOverlap_self
#check @BookProof.ChapterCoherentOverlap.coherentOverlap_unit
#check @BookProof.ChapterCoherentOverlap.coherentOverlap_eq_one_iff
```

The genuine Bargmann kernel takes **complex** coherent-state parameters, and then the overlap is no longer a positive real: it acquires a phase $\exp(i\ \mathrm{Im}\langle q, k\rangle)$. That case is now formalized too. `coherentOverlapC` is the complex kernel; `coherentOverlapC_eq_modulus_mul_phase` factors it into a positive real Gaussian modulus times a pure phase, and `norm_coherentOverlapC` computes the modulus. The real theory above is exactly the restriction of this one to real parameters (`coherentOverlapC_ofReal`):

```
#check @BookProof.ChapterCoherentOverlapComplex.coherentOverlapC
#check @BookProof.ChapterCoherentOverlapComplex.coherentOverlapC_eq_sum
#check @BookProof.ChapterCoherentOverlapComplex.coherentOverlapC_eq_modulus_mul_phase
#check @BookProof.ChapterCoherentOverlapComplex.norm_coherentOverlapC
#check @BookProof.ChapterCoherentOverlapComplex.coherentOverlapC_self
#check @BookProof.ChapterCoherentOverlapComplex.norm_coherentOverlapC_le_one
#check @BookProof.ChapterCoherentOverlapComplex.coherentOverlapC_ofReal
```

The geometric reading the section relies on — a key is a wave-packet, and the query interrogates the packets by **proximity** — is itself a theorem. `neg_two_mul_log_coherentOverlap` says the kernel is a lossless readout of the distance, $-2\log\langle q|k\rangle = \|q-k\|^2$, and `coherentOverlap_le_iff_dist_le` that the overlap is a strictly decreasing function of that distance. Consequently the Born weight is a Softmax over minus the squared distances — `bornWeight_eq_scoreSoftmax_neg_dist_sq`, an identity with no hypothesis on the key norms, since the key penalties are exactly what the squared distance absorbs — and the weights order the keys by proximity, so the nearest key carries the largest attention:

```
#check @BookProof.ChapterCoherentGeometry.neg_two_mul_log_coherentOverlap
#check @BookProof.ChapterCoherentGeometry.coherentOverlap_le_iff_dist_le
#check @BookProof.ChapterCoherentGeometry.coherentOverlap_lt_iff_dist_lt
#check @BookProof.ChapterCoherentGeometry.bornNumer_eq_exp_neg_dist_sq
#check @BookProof.ChapterCoherentGeometry.bornWeight_eq_scoreSoftmax_neg_dist_sq
#check @BookProof.ChapterCoherentGeometry.bornWeight_le_iff_dist_le
#check @BookProof.ChapterCoherentGeometry.bornWeight_max_of_nearest
#check @BookProof.ChapterCoherentGeometry.bornWeight_lt_of_nearest
```

48.3. Softmax Is the Born Rule on Coherent States

Apply the Born rule to the overlap. Squaring the exponential multiplies its exponent by two, and the laws of exponents split the result into three factors:

$$|\langle q|k\rangle|^2 = \exp(-|q|^2)\cdot\exp(-|k|^2)\cdot\exp(2\,q\cdot k).$$

Normalizing across the keys completes the Born rule. The term $\exp(-|q|^2)$ is constant for the row and cancels between numerator and denominator. In modern Transformers the keys are normalized (LayerNorm or RMSNorm), so each $\exp(-|k_j|^2)$ is also a fixed constant and factors out too. What remains is exactly Softmax attention:

$$\text{Attention Weight} = \frac{\exp(2\,q\cdot k_j)}{\sum_l \exp(2\,q\cdot k_l)}.$$

This is the headline claim of the chapter, and it is now a theorem. It is a finite algebraic identity about Gaussians and normalization. `coherentBorn_sq_eq` is the three-factor split above. `bornWeight` is the normalized squared overlap and `softmax` the usual attention weight at inverse temperature `beta`; `bornWeight_sum_one` and `softmax_sum_one` say that each is a genuine probability distribution over the keys. `coherentBorn_cancel_q` is the cancellation of the query factor, and the headline `coherentBorn_eq_softmax` states that under a constant key norm the two distributions are **equal**, at inverse temperature 2:

```
#check @BookProof.ChapterSoftmaxBorn.coherentBorn_sq_eq
#check @BookProof.ChapterSoftmaxBorn.bornWeight_sum_one
#check @BookProof.ChapterSoftmaxBorn.softmax_sum_one
#check @BookProof.ChapterSoftmaxBorn.coherentBorn_cancel_q
#check @BookProof.ChapterSoftmaxBorn.coherentBorn_eq_softmax
#check @BookProof.ChapterSoftmaxBorn.coherentBorn_eq_softmax_of_unit_keys
#check @BookProof.ChapterSoftmaxBorn.coherentBorn_temperature_half
```

The same headline holds for complex coherent-state parameters, and this is where the phase earns its keep: the Born rule squares a **modulus**, so the phase of the kernel is discarded and the surviving alignment score is the real part $\mathrm{Re}\langle q, k_j\rangle$ — precisely the real-valued attention logit. `bornNumerC_eq` is the complex three-factor split, `coherentBornC_eq_softmax` the complex headline, and `bornWeightC_phase_invariant` records that attention cannot see the phase at all:

```
#check @BookProof.ChapterCoherentOverlapComplex.bornNumerC_eq
#check @BookProof.ChapterCoherentOverlapComplex.bornWeightC_sum_one
#check @BookProof.ChapterCoherentOverlapComplex.coherentBornC_cancel_q
#check @BookProof.ChapterCoherentOverlapComplex.coherentBornC_eq_softmax
#check @BookProof.ChapterCoherentOverlapComplex.coherentBornC_eq_softmax_of_unit_keys
#check @BookProof.ChapterCoherentOverlapComplex.bornWeightC_phase_invariant
#check @BookProof.ChapterCoherentOverlapComplex.bornWeightC_ofReal
```

The two premises the identity rests on were already in the library: the Gaussian as the rotation-invariant prior (the free-field construction) and the fact that squaring a wave-function produces a genuine probability distribution (the

Born rule):

```
#check @BookProof.ChapterFreeFieldGaussian.stdGaussian_map_linearIsometryEquiv
#check @BookProof.ChapterEulerNState.euler_reproduces
#check @BookProof.ChapterBornPhaseFiber.born_fiber_complex
```

Between the two temperature extremes, the measurement has a rigid order structure. `scoreSoftmax_shift` is its gauge invariance — adding a constant to every score changes nothing, the abstract form of the cancellation of the query penalty $e^{-|q|^2}$ in `coherentBorn_cancel_q`, so only score **differences** are physical. `scoreSoftmax_le_iff` and `scoreSoftmax_lt_iff` say that at any positive inverse temperature Softmax is a strictly increasing reparametrization of the scores: it reorders nothing, and `scoreSoftmax_argmax` makes the winner temperature-independent. `scoreSoftmax_const` is the converse warning — with no score contrast there is nothing to sharpen, at any temperature:

```
#check @BookProof.ChapterSoftmaxOrder.scoreSoftmax_sum_one
#check @BookProof.ChapterSoftmaxOrder.scoreSoftmax_le_one
#check @BookProof.ChapterSoftmaxOrder.scoreSoftmax_shift
#check @BookProof.ChapterSoftmaxOrder.scoreSoftmax_le_iff
#check @BookProof.ChapterSoftmaxOrder.scoreSoftmax_lt_iff
#check @BookProof.ChapterSoftmaxOrder.scoreSoftmax_inj_iff
#check @BookProof.ChapterSoftmaxOrder.scoreSoftmax_argmax
#check @BookProof.ChapterSoftmaxOrder.scoreSoftmax_const
#check @BookProof.ChapterSoftmaxOrder.softmax_le_iff_inner_le
```

48.4. Temperature and the Thermal Bath

In the standard Transformer, the dot product is divided by a temperature τ that controls randomness. The derivation above produced a natural factor 2 in the exponent, so a standard LLM operates at a baseline temperature $\tau = 0.5$. The manuscript interprets this in two regimes. In the **pure quantum** regime, the wave-packet variance is bounded by the Heisenberg limit $\sigma^2 = \hbar/2m\omega$, and lowering the temperature is increasing the mass/frequency of the oscillator. In the **thermodynamic** regime — a displaced thermal state at temperature $T > 0$ with $\bar{n}$ background thermal bosons — the fidelity of the thermal states introduces the noise directly into the exponent:

$$\tau = \bar{n} + \tfrac{1}{2}.$$

The $\frac{1}{2}$ is the zero-point energy of the vacuum. The **physical** derivation of this identity — the quantum fidelity of two displaced thermal states on an infinite-dimensional bosonic Fock space — remains a documented proof plan rather than a proved theorem. Its **statistical core** is proved, however. For the thermal occupation law $\Pr(n) \propto (\bar{n}/(\bar{n}+1))^n$, `thermalProb_tsum_one` says it is a probability distribution on the occupation numbers, `thermalProb_mean` that its mean occupation really is $\bar{n}$, and `thermalProb_variance` that its variance is $\bar{n} + \bar{n}^2$ — strictly more than the Poissonian variance $\bar{n}$ of a coherent state, which is the precise sense in which the bath adds noise. At $\bar{n} = 0$ the temperature collapses to the zero-point value $\tau = 1/2$, matching the factor 2 in the exponent of the Softmax derived above:

```
#check @BookProof.ChapterCoherentTemperature.thermalProb
#check @BookProof.ChapterCoherentTemperature.thermalProb_tsum_one
#check @BookProof.ChapterCoherentTemperature.thermalProb_mean
#check @BookProof.ChapterCoherentTemperature.thermalProb_second_moment
#check @BookProof.ChapterCoherentTemperature.thermalProb_variance
#check @BookProof.ChapterCoherentTemperature.thermalTemperature_eq_mean_add_half
#check @BookProof.ChapterCoherentTemperature.thermalTemperature_vacuum
#check @BookProof.ChapterCoherentTemperature.half_lt_thermalTemperature
```

Two of the claims made in that paragraph are themselves theorems. The occupation law of a **coherent** state is the Poisson law `p(n) = e^{-λ}λⁿ/n!`, whose variance `coherentOccupation_variance` is exactly its mean λ — the Poissonian benchmark the thermal variance is compared against, and `coherent_variance_lt_thermal_variance` records that the bath is strictly noisier at equal mean occupation. And the split "mean occupation plus the zero-point half" is the expectation of the harmonic-oscillator energy observable $n + \frac{1}{2}$: `coherentOccupation_energy` gives $\lambda + \frac{1}{2}$ in a coherent state and `thermalOccupation_energy` gives $\bar{n} + \frac{1}{2}$ in the thermal state, so

`thermalTemperature_eq_energy_expectation` identifies τ with a genuine expectation value. Only the derivation of τ from the fidelity of displaced thermal states stays a proof plan:

```
#check @BookProof.ChapterCoherentOccupation.coherentOccupation
#check @BookProof.ChapterCoherentOccupation.coherentOccupation_eq_poissonPMFReal
#check @BookProof.ChapterCoherentOccupation.coherentOccupation_tsum_one
#check @BookProof.ChapterCoherentOccupation.coherentOccupation_mean
#check @BookProof.ChapterCoherentOccupation.coherentOccupation_second_moment
#check @BookProof.ChapterCoherentOccupation.coherentOccupation_variance
#check @BookProof.ChapterCoherentOccupation.coherentOccupation_energy
#check @BookProof.ChapterCoherentOccupation.thermalOccupation_energy
#check @BookProof.ChapterCoherentOccupation.thermalTemperature_eq_energy_expectation
#check @BookProof.ChapterCoherentOccupation.coherent_variance_lt_thermal_variance
```

Parametrized by the physical temperature the same law becomes the textbook one. Writing $x = \hbar\omega/kT$ for the dimensionless inverse temperature, the mean occupation is the Bose–Einstein function $\bar{n}(x) = 1/(e^x - 1)$, the level probabilities are exactly the Boltzmann–Gibbs weights $\Pr(n) = (1 - e^{-x})e^{-nx}$, and the chapter's temperature has the closed form $\tau(x) = \bar{n}(x) + \frac{1}{2} = \frac{1}{2}\coth(x/2)$, decreasing to the pure zero-point value $\frac{1}{2}$ as the mode is cooled:

```
#check @BookProof.ChapterBoseEinstein.boseEinstein
#check @BookProof.ChapterBoseEinstein.thermalRatio_boseEinstein
#check @BookProof.ChapterBoseEinstein.thermalProb_boseEinstein
#check @BookProof.ChapterBoseEinstein.boseEinstein_mean
#check @BookProof.ChapterBoseEinstein.boseEinstein_strictAntiOn
#check @BookProof.ChapterBoseEinstein.thermalTemperature_boseEinstein
#check @BookProof.ChapterBoseEinstein.thermalTemperature_boseEinstein_eq_coth
#check @BookProof.ChapterBoseEinstein.tendsto_thermalTemperature_boseEinstein
```

Why **that** law, and not another distribution with the same mean? Because it is the one of maximal Shannon entropy: among occupation distributions with a prescribed mean occupation $\bar{n}$, the thermal law maximizes $-\sum_n p_n \log p_n$. This is the variational reason the bath is a **thermal** bath, and it is proved by the pointwise Gibbs inequality:

```
#check @BookProof.ChapterThermalMaxEntropy.thermalEntropy
#check @BookProof.ChapterThermalMaxEntropy.thermalEntropy_eq
#check @BookProof.ChapterThermalMaxEntropy.thermalEntropy_boseEinstein
#check @BookProof.ChapterThermalMaxEntropy.gibbs_pointwise
#check @BookProof.ChapterThermalMaxEntropy.shannonEntropy_le_thermalEntropy
```

The identity $\tau = \bar{n} + \frac{1}{2}$ can now be read off the **physical** parametrization rather than posited. Writing the occupation law through its ratio r instead of its mean, its first two moments are the geometric ones ($\bar{n} = r/(1 - r)$, variance $r/(1 - r)^2$), the energy expectation $\sum_n (n + \frac{1}{2})p_n$ never drops below $\frac{1}{2}$ — with equality exactly at the vacuum, which is what the extra half measures — and the closed form $\frac{1}{2}\coth(x/2)$ **equals** the mean occupation plus that half:

```
#check @BookProof.ChapterThermalTemperatureCore.geometricOccupancy
#check @BookProof.ChapterThermalTemperatureCore.geometricOccupancy_mean
#check @BookProof.ChapterThermalTemperatureCore.geometricOccupancy_variance
#check @BookProof.ChapterThermalTemperatureCore.half_integer_floor
#check @BookProof.ChapterThermalTemperatureCore.thermal_temperature_eq_mean_half
```

The last step is the **physical** one: the temperature must come out of the state itself, not out of a fit. Model the bath mode in phase space, where a displaced thermal state is the vacuum's zero-point noise (variance $\frac{1}{2}$) convolved with thermal noise of variance $\bar{n}$ and recentred at the displacement a. Convolution adds variances, so the width of the displaced thermal state is $\tau = \bar{n} + \frac{1}{2}$ — the mean occupation plus the zero-point half — and it does not depend on where the state sits:

```
#check @BookProof.ChapterDisplacedThermalOverlap.displacedThermal
#check @BookProof.ChapterDisplacedThermalOverlap.thermal_plus_zeroPoint_conv
#check @BookProof.ChapterDisplacedThermalOverlap.displacedThermal_mean
#check @BookProof.ChapterDisplacedThermalOverlap.displacedThermal_variance
```

The overlap of two such states is then a Gaussian in the phase-space distance alone, $\exp(-(a-b)^2/4\tau)/\sqrt{4\pi\tau}$: it is strictly positive, and it is strictly larger for the nearer of two keys. Normalizing the overlaps of a query against a family of keys therefore reproduces the Softmax **exactly**, at inverse temperature $\beta = 1/(4\tau)$ — which is maximal at the vacuum and falls monotonically as the bath heats up. A hotter bath is a flatter attention.

```
#check @BookProof.ChapterDisplacedThermalOverlap.dtOverlap_eq
#check @BookProof.ChapterDisplacedThermalOverlap.dtOverlap_pos
#check @BookProof.ChapterDisplacedThermalOverlap.dtOverlap_lt_of_dist_lt
#check @BookProof.ChapterDisplacedThermalOverlap.inverseTemperature
#check @BookProof.ChapterDisplacedThermalOverlap.inverseTemperature_zero
#check @BookProof.ChapterDisplacedThermalOverlap.inverseTemperature_strictAnti
#check @BookProof.ChapterDisplacedThermalOverlap.dtBorn_eq_softmax
#check @BookProof.ChapterDisplacedThermalOverlap.dtBorn_temperature_coth
```

Nothing about this is one-dimensional. With n modes the overlap factorizes across modes, the exponent becomes the squared Euclidean distance of the phase-space vectors, and the Born weights are again a Softmax at the same inverse temperature. At zero occupation the bath disappears and the multi-mode Born weights coincide — after the usual $\sqrt{2}$ change between the quadrature variable and the dimensionless coherent parameter — with the coherent-state Born weights this chapter began with. The thermal picture **contains** the pure one:

```
#check @BookProof.ChapterDisplacedThermalMulti.dtOverlapMulti_eq_integral
#check @BookProof.ChapterDisplacedThermalMulti.dtOverlapMulti_eq
#check @BookProof.ChapterDisplacedThermalMulti.dtOverlapMulti_pos
#check @BookProof.ChapterDisplacedThermalMulti.dtBornMulti_eq_softmax
#check @BookProof.ChapterDisplacedThermalMulti.dtBornMulti_vacuum_eq_bornWeight
```

Two states are compared, in quantum mechanics, by their **fidelity**: the probability that a system prepared as $|q\rangle$ is found in $|k\rangle$. For coherent states the fidelity collapses to a single number — the Gaussian of the distance between the two displacement parameters. The Bargmann phase cancels, and so do the individual norms; the fidelity is symmetric, never exceeds one, equals one only for identical parameters, and is unchanged when both states are displaced together. Attention is exactly the normalized fidelity:

```
#check @BookProof.ChapterCoherentFidelity.fidelityC
#check @BookProof.ChapterCoherentFidelity.fidelityC_eq_exp_neg_dist_sq
#check @BookProof.ChapterCoherentFidelity.fidelityC_symm
#check @BookProof.ChapterCoherentFidelity.fidelityC_eq_one_iff
#check @BookProof.ChapterCoherentFidelity.fidelityC_translation_invariant
#check @BookProof.ChapterCoherentFidelity.bornWeightC_eq_fidelity_normalized
```

The fidelity is also what settles the last piece of the temperature. In the phase-space model above, the zero-point half was **put in by hand**: it was the variance of the vacuum noise the thermal noise was convolved with. It can be derived instead, entirely in occupation numbers. Pair the coherent state's Poisson occupation statistics against the bath's geometric ones — that sum is the fidelity $\langle\beta|\rho_{\mathrm{th}}(\bar{n})|\beta\rangle$ of a thermal state with a coherent state of intensity $\lambda = \|\beta\|^2$ — and the geometric series collapses to

$$\langle\beta|\rho_{\mathrm{th}}(\bar{n})|\beta\rangle = \frac{e^{-\lambda/(\bar{n}+1)}}{\bar{n}+1}.$$

The width of that Gaussian is $\bar{n}+1$, and the width of an overlap is the sum of the widths of the two states that make it. At $\bar{n} = 0$ the same expression **is** the coherent-state fidelity $e^{-\|q-k\|^2}$ this chapter began with, so two coherent states overlap with width $\frac{1}{2}+\frac{1}{2}$: the width of a coherent state is exactly $\frac{1}{2}$. Subtracting it from $\bar{n}+1$ leaves the width of the thermal state alone, $\tau = \bar{n}+\frac{1}{2}$. The extra half is the coherent-state overlap, not a postulate — and the Gaussian phase-space model agrees, since in the dimensionless coherent parameter the overlap of two displaced thermal states has width $\tau+\tau$:

```
#check @BookProof.ChapterCoherentThermalFidelity.coherentThermalFidelity
#check @BookProof.ChapterCoherentThermalFidelity.coherentThermalFidelity_eq
#check @BookProof.ChapterCoherentThermalFidelity.coherentThermalFidelity_vacuum_eq_fi
#check @BookProof.ChapterCoherentThermalFidelity.fidelityC_width
#check @BookProof.ChapterCoherentThermalFidelity.coherentThermalFidelity_width_eq
#check @BookProof.ChapterCoherentThermalFidelity.thermalTemperature_eq_fidelity_width
#check @BookProof.ChapterCoherentThermalFidelity.dtOverlap_coherentParameter
```

Once the temperature is a physical parameter one can **differentiate** by it. The attention head has a partition function $Z(\beta) = \sum_j e^{\beta s_j}$, and the first derivative of its logarithm is the attention-weighted mean score. The second is the variance. That is the fluctuation–response law: sharpening the attention raises the mean alignment at a rate equal to the variance of the scores under the current attention distribution — and the response is zero exactly when all the keys already score alike.

```
#check @BookProof.ChapterSoftmaxFluctuation.hasDerivAt_logPartition
#check @BookProof.ChapterSoftmaxFluctuation.hasDerivAt_scoreSoftmax
#check @BookProof.ChapterSoftmaxFluctuation.hasDerivAt_meanScore
#check @BookProof.ChapterSoftmaxFluctuation.varScore_eq_zero_iff
#check @BookProof.ChapterSoftmaxFluctuation.meanScore_monotone
```

This also answers **why Softmax and not some other normalization**. Among all attention distributions with a prescribed mean alignment score, the Softmax distribution is the one of largest Shannon entropy — the least committed distribution compatible with what the alignment says. Equivalently, it minimises the free energy $\beta\langle s\rangle_p - H(p)$, and the minimum is $-\log Z(\beta)$:

```
#check @BookProof.ChapterSoftmaxMaxEntropy.shannonEntropy_le_crossEntropy
#check @BookProof.ChapterSoftmaxMaxEntropy.shannonEntropy_scoreSoftmax
#check @BookProof.ChapterSoftmaxMaxEntropy.shannonEntropy_le_of_meanScore_eq
#check @BookProof.ChapterSoftmaxMaxEntropy.softmax_free_energy_le
#check @BookProof.ChapterSoftmaxMaxEntropy.softmax_free_energy_eq
```

And the entropy itself now has a law of motion. Differentiating the thermodynamic identity $H(\beta) = \log Z(\beta) - \beta\langle s\rangle_\beta$ gives $dH/d\beta = -\beta\,\mathrm{Var}_\beta(s)$, so the attention entropy is non-increasing for $\beta \geq 0$: cooling the head can only destroy entropy. The endpoints already computed — $\log m$ at infinite temperature, 0 in the argmax limit — are now joined by a monotone path:

```
#check @BookProof.ChapterEntropyTemperature.attentionEntropy_eq
#check @BookProof.ChapterEntropyTemperature.hasDerivAt_attentionEntropy
#check @BookProof.ChapterEntropyTemperature.heatCapacity_nonneg
#check @BookProof.ChapterEntropyTemperature.attentionEntropy_antitoneOn
#check @BookProof.ChapterEntropyTemperature.attentionEntropy_le_log_card
```

Two temperatures give two attention distributions over the same keys, and the natural way to compare them is the relative entropy. It has a closed form: the divergence between the heads at inverse temperatures β and γ is the **Bregman divergence of the free energy**, $\mathrm{KL}(p_\beta \| p_\gamma) = \log Z(\gamma) - \log Z(\beta) - (\gamma - \beta)\langle s\rangle_\beta$. Since a relative entropy is never negative, the free energy lies above each of its tangent lines. The same variance that measured the response is also the Fisher information of the family, so the statistical curvature of attention and its physical fluctuation are one quantity:

```
#check @BookProof.ChapterSoftmaxDivergence.klDiv
#check @BookProof.ChapterSoftmaxDivergence.klDiv_nonneg
#check @BookProof.ChapterSoftmaxDivergence.klDiv_scoreSoftmax
#check @BookProof.ChapterSoftmaxDivergence.logPartition_tangent_le
#check @BookProof.ChapterSoftmaxDivergence.fisherInformation_eq_varScore
```

Reading the tangent inequality globally: the free energy is a convex function of the inverse temperature — strictly convex as soon as two keys score differently — and it is the Legendre transform of the entropy, the largest value of $\beta\langle s\rangle_p + H(p)$ over all attention distributions:

```
#check @BookProof.ChapterLogPartitionConvex.hasDerivAt_deriv_logPartition
#check @BookProof.ChapterLogPartitionConvex.convexOn_logPartition
#check @BookProof.ChapterLogPartitionConvex.strictConvexOn_logPartition
#check @BookProof.ChapterLogPartitionConvex.logPartition_isGreatest
```

Finally, the kernel that started the chapter need not be taken on faith. Writing the coherent state centred at a as the position-space wave packet $\psi_a(x) = \pi^{-1/4} e^{-(x-a)^2/2}$, the packet is normalized, and its $L^2(\mathbb{R})$ inner product with a second packet is $e^{-(a-b)^2/4}$ — a Gaussian in the distance between the centres. Squaring it recovers, exactly, the reproducing kernel taken as the definition above, so the position-space Born weights of a query packet against a family of key packets are Softmax attention over minus the squared distances:

```
#check @BookProof.ChapterCoherentPositionSpace.gaussianPacket
#check @BookProof.ChapterCoherentPositionSpace.gaussianPacket_sq_integral
```

```
#check @BookProof.ChapterCoherentPositionSpace.gaussianPacket_inner
#check @BookProof.ChapterCoherentPositionSpace.gaussianPacket_inner_sq_eq_coherentOve
#check @BookProof.ChapterCoherentPositionSpace.packetBorn_eq_scoreSoftmax
```

One last property is needed before any of this is usable: the measurement must be **stable**. If every alignment score is known only up to an error d, no attention weight can move by more than the factor $e^{2|\beta|d}$ — one factor of $|\beta|d$ for the key's own score, one for the normalizing free energy — and hence by no more than $e^{2|\beta|d} - 1$ in absolute terms. Cooling the head (large β) is exactly what makes it fragile:

```
#check @BookProof.ChapterSoftmaxStability.abs_logPartition_sub_le
#check @BookProof.ChapterSoftmaxStability.abs_log_scoreSoftmax_sub_le
#check @BookProof.ChapterSoftmaxStability.scoreSoftmax_le_mul
#check @BookProof.ChapterSoftmaxStability.abs_scoreSoftmax_sub_le
```

Stability in the scores has an exact differential form. Nudging the alignment score of one key moves the free energy at the rate $\partial \log Z / \partial s_i = \beta p_i$ — the attention weight of a key **is** the sensitivity of the free energy to that key's score — and it moves the weights themselves by the Softmax Jacobian $\partial p_j / \partial s_i = \beta\, p_j(\delta_{ij} - p_i)$. That matrix is symmetric (it is a Hessian), each of its rows sums to zero (a key can only gain what the others lose), and its quadratic form is β times an attention-weighted variance — positive semidefinite for $\beta \geq 0$, and equal to $\beta \operatorname{Var}_\beta(s)$ on the scores themselves. This is the same fluctuation that governed the temperature derivative, seen from the score direction:

```
#check @BookProof.ChapterSoftmaxJacobian.hasDerivAt_logPartition_score
#check @BookProof.ChapterSoftmaxJacobian.hasDerivAt_scoreSoftmax_score
#check @BookProof.ChapterSoftmaxJacobian.softmaxJacobian_symm
#check @BookProof.ChapterSoftmaxJacobian.softmaxJacobian_row_sum_zero
#check @BookProof.ChapterSoftmaxJacobian.softmaxJacobian_quadratic_form
#check @BookProof.ChapterSoftmaxJacobian.softmaxJacobian_quadratic_form_score
```

The head also never attends to everything: a **mask** restricts the sum to an admissible set of keys, the causal mask $\{l \leq i\}$ being the standard case. Masking is not a new ingredient — it is Bayesian conditioning. On the admissible set the masked weight is the unmasked weight renormalized by the total unmasked weight of that set, $p(j \mid S) = p(j)/p(S)$; the odds between two admissible keys are untouched, so a mask removes keys without re-ranking them; and conditioning twice is conditioning once, so a composite mask is a single mask:

```
#check @BookProof.ChapterAttentionMasking.maskedSoftmax_sum_one
#check @BookProof.ChapterAttentionMasking.maskedSoftmax_eq_conditional
#check @BookProof.ChapterAttentionMasking.maskedSoftmax_odds
#check @BookProof.ChapterAttentionMasking.maskedSoftmax_restrict
#check @BookProof.ChapterAttentionMasking.causalSoftmax_eq_zero_of_lt
```

Two independent modes never entangle the measurement. If the alignment score of a product key is the sum of its per-mode scores — which is what a product coherent state gives — then the joint attention distribution is exactly the product of the single-mode ones, its marginals are the single-mode distributions, and the attention entropy is additive:

```
#check @BookProof.ChapterAttentionFactorization.prodSoftmax_eq_mul
#check @BookProof.ChapterAttentionFactorization.prodSoftmax_marginal_left
#check @BookProof.ChapterAttentionFactorization.shannonEntropy_prodSoftmax
```

Finally, the whole construction has a symmetry group, and it is the physical one. The Bargmann kernel sees only norms and inner products, so a common unitary change of frame on the query and every key changes no weight. Free harmonic evolution $\alpha \mapsto e^{-i\omega t}\alpha$ is such a unitary: **attention is a constant of the motion**. And a common Weyl displacement of query and keys leaves the weights alone as well, so only relative positions in phase space are physical:

```
#check @BookProof.ChapterCoherentDynamics.coherentOverlapC_isometry
#check @BookProof.ChapterCoherentDynamics.bornWeightC_isometry
#check @BookProof.ChapterCoherentDynamics.bornWeightC_evolution_const
#check @BookProof.ChapterCoherentDynamics.fidelityC_phaseRotate
#check @BookProof.ChapterCoherentDynamics.bornWeightC_translation_invariant
```

Position is part of the same story. The rotary encoding rotates the i-th complex coordinate of a token by an angle proportional to its position, $q \mapsto (e^{ip\omega_i} q_i)_i$. It is a unitary, it composes additively in the position, and — the property that makes it work — the alignment of a query at position a with a key at position b depends on the two positions **only** through the offset $b - a$. Absolute position is unobservable; translating the whole sequence changes no attention weight:

```
#check @BookProof.ChapterRotaryPosition.norm_rotaryEncode
#check @BookProof.ChapterRotaryPosition.rotaryEncode_add
#check @BookProof.ChapterRotaryPosition.inner_rotaryEncode
#check @BookProof.ChapterRotaryPosition.coherentOverlapC_rotaryEncode
#check @BookProof.ChapterRotaryPosition.bornWeightC_rotaryEncode_shift
```

The winner-takes-all law of the previous sections is a limit, and a memory that only works in the limit is no memory. At a **fixed** temperature the same statement is quantitative. Suppose the query beats every rival key by a score margin δ. Then each distractor keeps at most $e^{-\beta\delta}$ of the attention, the total leakage off the target is at most $(m-1)e^{-\beta\delta}$, the target itself holds at least $1/(1+(m-1)e^{-\beta\delta})$, and the head's output differs from the stored value by at most $2C(m-1)e^{-\beta\delta}$. An attention head is an associative memory whose recall error decays exponentially in $\beta\delta$:

```
#check @BookProof.ChapterAttentionRetrieval.scoreSoftmax_le_exp_neg_margin
#check @BookProof.ChapterAttentionRetrieval.one_sub_scoreSoftmax_le_of_margin
#check @BookProof.ChapterAttentionRetrieval.scoreSoftmax_ge_inv_of_margin
#check @BookProof.ChapterAttentionRetrieval.norm_headOutput_sub_le_of_margin
#check @BookProof.ChapterAttentionRetrieval.bornWeight_ge_of_dist_margin
```

The converse bound is just as physical: at any finite temperature **nothing is ever completely ignored**. If the scores lie within a spread D, every key retains at least $e^{-\beta D}/m$ of the attention. Collapse onto a single key is a zero-temperature idealization, never an event at finite β:

```
#check @BookProof.ChapterAttentionRetrieval.scoreSoftmax_ge_of_spread
```

The measurement is also blind to the order of the keys. Relabelling the key/value pairs by a permutation relabels the weights the same way and leaves the output, and the attention entropy, exactly where they were: a head reads its context as a **set** of pairs, not as a list. This is the discrete half of the symmetry group whose continuous half — unitary frame changes, free evolution, Weyl displacement — appeared above:

```
#check @BookProof.ChapterAttentionEquivariance.scoreSoftmax_perm
#check @BookProof.ChapterAttentionEquivariance.headOutput_perm
#check @BookProof.ChapterAttentionEquivariance.attentionOutput_perm
#check @BookProof.ChapterAttentionEquivariance.shannonEntropy_scoreSoftmax_perm
```

A layer runs many heads at once, and a bank of heads is a **mixture** of Born measurements. The mixture of attention distributions is again an attention distribution; reading the values once against the mixed distribution is the same as reading them head by head and averaging, which is what licenses the usual "combine the heads" construction; and — by concavity of $-x\log x$ — the entropy of the consensus is at least the mean entropy of the heads. Confident heads that disagree produce an uncertain layer, never the reverse:

```
#check @BookProof.ChapterAttentionMixture.mixture_isProb
#check @BookProof.ChapterAttentionMixture.observableExpectation_mixture
#check @BookProof.ChapterAttentionMixture.multiHead_output_eq_mean
#check @BookProof.ChapterAttentionMixture.le_shannonEntropy_mixture
```

Finally, if attention is the Born rule then **training** it is fitting a Born probability to an observed outcome. The criterion is the surprisal of the observed key, $L = \log Z_\beta(s) - \beta s_y = -\log p_\beta(y)$ — the free energy minus the energy of the realized outcome — which is nonnegative and vanishes exactly when the layer already places all of its weight there. Its score derivative is the textbook backpropagation rule $\partial L/\partial s_i = \beta(p_i - \delta_{iy})$, obtained here from the free-energy derivative above; the gradient sums to zero over the keys, so learning **transfers** attention rather than creating it, and the gauge freedom $s \mapsto s + c$ is never excited. In the temperature the loss is convex, so temperature fitting has no spurious minima:

```
#check @BookProof.ChapterCrossEntropyGradient.crossEntropyLoss_eq_neg_log
#check @BookProof.ChapterCrossEntropyGradient.crossEntropyLoss_nonneg
#check @BookProof.ChapterCrossEntropyGradient.hasDerivAt_crossEntropyLoss_score
```

```
#check @BookProof.ChapterCrossEntropyGradient.crossEntropyGradient_sum_zero
#check @BookProof.ChapterCrossEntropyGradient.convexOn_crossEntropyLoss
```

How many keys is a head actually reading? The Shannon entropy answers in nats; the practitioner's answer is a **count**, the participation ratio $N_{\text{eff}} = 1/\sum_j p_j^2$, the exponential of the Rényi-2 (collision) entropy. It behaves as a count must: it lies between 1 and the number of keys, it equals the number of keys exactly at infinite temperature, and — because the collision entropy never exceeds the Shannon entropy — it never overstates how much of the context is being used:

```
#check @BookProof.ChapterAttentionCollision.effectiveSupport_scoreSoftmax_mem_Icc
#check @BookProof.ChapterAttentionCollision.effectiveSupport_scoreSoftmax_zero
#check @BookProof.ChapterAttentionCollision.renyi2_le_shannonEntropy
```

The same counting works from the other side. A Markov bound on the Born distribution says that at most $1/t$ keys can carry weight t or more, so only a handful of keys can ever matter at once; some key always carries at least the uniform share $1/m$; and the min-entropy bound $-\log p_{\max} \le H$ says that a low-entropy head necessarily has a dominant key, $p_{\max} \ge e^{-H}$, whose reciprocal is a lower bound for the participation ratio:

```
#check @BookProof.ChapterAttentionConcentration.card_filter_le_inv
#check @BookProof.ChapterAttentionConcentration.exists_inv_card_le
#check @BookProof.ChapterAttentionConcentration.neg_log_le_shannonEntropy
#check @BookProof.ChapterAttentionConcentration.inv_le_effectiveSupport
```

Read row by row, a layer is a **stochastic matrix**: row i is the Born distribution of the query at position i. Pushing a belief about position through the layer is then a Markov step, stacking layers composes the kernels, and a step never increases the ℓ^1 discrepancy between two beliefs. More: at a finite temperature every entry of the kernel is bounded below, and the Doeblin argument turns that into a strict contraction — with a score spread D a layer contracts by $1 - e^{-\beta D}$. Deep stacks of attention **forget** their input geometrically fast unless the scores are allowed to spread with depth:

```
#check @BookProof.ChapterAttentionMarkov.attentionMatrix_isStochastic
#check @BookProof.ChapterAttentionMarkov.push_compose
#check @BookProof.ChapterAttentionMarkov.l1dist_push_le
#check @BookProof.ChapterAttentionMarkov.l1dist_push_attentionMatrix_le
```

Finally, the $1/\sqrt{d}$ that every implementation writes into the score. Model an unstructured query as a uniform sign pattern $q \in \{\pm 1\}^d$. Against a fixed key the raw score $\langle q, k\rangle$ has mean zero and mean square $\|k\|^2$, so for unit-size entries its root-mean-square is $\sqrt{d}$. Dividing every score by a constant is exactly dividing the inverse temperature by that constant; so feeding **raw** dot products to the Softmax at a fixed β is running the head at $\beta\sqrt{d}$, which freezes onto the arg-max as the width grows. The $1/\sqrt{d}$ is the temperature-preserving normalization — with it the score has mean square exactly 1, whatever the width:

```
#check @BookProof.ChapterScaledDotProduct.rademacherMean_dot
#check @BookProof.ChapterScaledDotProduct.rademacherMean_dot_sq
#check @BookProof.ChapterScaledDotProduct.scoreSoftmax_scaled
#check @BookProof.ChapterScaledDotProduct.rademacherMean_scaledDot_sq_of_unit_entries
```

The output carries a second moment as well as a first. Decomposing the attention-weighted mean square of the values about any reference point into the variance about the output plus the squared distance to the output — the bias–variance identity — shows that the head output is the **least-squares summary** of what it is reading: no other vector sits closer to the values in the attention-weighted mean-square sense. König–Huygens then gives the Jensen bound $\|o\|^2 \le \sum_j p_j \|v_j\|^2$, and — because at a finite temperature every weight is strictly positive — the output is **certain** exactly when all the values agree:

```
#check @BookProof.ChapterAttentionOutputVariance.sum_dist_sq_eq
#check @BookProof.ChapterAttentionOutputVariance.observableExpectation_minimizes
#check @BookProof.ChapterAttentionOutputVariance.outputVariance_eq_sub
#check @BookProof.ChapterAttentionOutputVariance.outputVariance_scoreSoftmax_eq_zero_
```

The factorization $s_{ij} = \langle q_i, k_j\rangle$ through a head dimension d is itself a hard constraint. The whole $m \times m$ table of alignments a head can produce has rank at most d, so when $d < m$ no head can realize the pattern "each

position attends to itself and to nothing else" — its score matrix is the identity, of rank m. A single head is not a general router; the bound is sharp, since orthonormal queries and keys realize that pattern as soon as $d \geq m$:

```
#check @BookProof.ChapterAttentionLowRank.rank_scoreMatrix_le
#check @BookProof.ChapterAttentionLowRank.not_exists_scoreMatrix_one
#check @BookProof.ChapterAttentionLowRank.exists_scoreMatrix_one_of_le
```

Layer normalization, which every block applies before the head reads, is the gauge fixing of this picture: it centres the activations and puts them on the sphere of squared length d, it is invariant under the affine reparametrizations $x \mapsto ax + c$ with $a > 0$, and it is idempotent — a projection. Cauchy–Schwarz then caps every score of a normalized head by d, so the scores span at most $2d$ and every key keeps at least $e^{-2\beta d}/m$ of the attention. Normalization is what keeps the temperature of the Born measurement bounded:

```
#check @BookProof.ChapterLayerNorm.sum_sq_layerNorm
#check @BookProof.ChapterLayerNorm.layerNorm_smul_pos
#check @BookProof.ChapterLayerNorm.layerNorm_layerNorm
#check @BookProof.ChapterLayerNorm.scoreSoftmax_layerNorm_ge
```

The rotary encoding above is not the only positional scheme with the offset property. The original sinusoidal bank has it too: the alignment of two encoded positions is $\sum_a \cos(\omega_a (p - q))$, every encoded position has the same squared length, and a global shift of all positions leaves every attention weight untouched:

```
#check @BookProof.ChapterSinusoidalPosition.peInner_eq_sum_cos
#check @BookProof.ChapterSinusoidalPosition.peInner_self
#check @BookProof.ChapterSinusoidalPosition.scoreSoftmax_sinusoidal_shift
#check @BookProof.ChapterSinusoidalPosition.scoreSoftmax_sinusoidal_ge
```

Iterating the Markov step of a single layer gives the mixing law of a deep stack: after n layers two beliefs about position are $(1 - m\varepsilon)^n$ times as far apart as they began, the distance tends to zero, and there is at most one stationary belief. For a genuine attention layer at $\beta \geq 0$ with score spread D the rate is $(1 - e^{-\beta D})^n$ — depth alone erases the initial state unless the scores are allowed to spread:

```
#check @BookProof.ChapterAttentionMixing.l1dist_pushIter_le
#check @BookProof.ChapterAttentionMixing.tendsto_l1dist_pushIter
#check @BookProof.ChapterAttentionMixing.eq_of_stationary
#check @BookProof.ChapterAttentionMixing.tendsto_l1dist_pushIter_attentionMatrix
```

Between the two extreme temperatures there is a monotonicity nobody has to compute: the odds of two keys are the pure exponential $e^{\beta(s_i - s_j)}$ of their score gap, so cooling the head can only help the best key and only hurt the worst one. At any $\beta \geq 0$ the winner already holds at least the uniform share $1/m$ and the loser at most $1/m$, and the gain is strict as soon as some other key scores strictly lower. The converse bound closes the picture: if the scores span at most D, no weight exceeds $e^{\beta D}/m$ and the entropy of the collapse never falls below $\log m - \beta D$ — a head with bounded scores cannot be sharp at a finite temperature:

```
#check @BookProof.ChapterAttentionTemperature.scoreSoftmax_odds
#check @BookProof.ChapterAttentionTemperature.scoreSoftmax_monotone_of_isMax
#check @BookProof.ChapterAttentionTemperature.scoreSoftmax_antitone_of_isMin
#check @BookProof.ChapterAttentionTemperature.inv_card_le_scoreSoftmax_of_isMax
#check @BookProof.ChapterAttentionTemperature.log_card_sub_le_shannonEntropy
```

Trained heads reliably park a large share of their attention on one uninformative position — the **attention sink**. The Born algebra says this is a gauge, not a pathology: prepending one extra key multiplies every original weight by the single factor $1 - w$, leaves all the odds between ordinary keys untouched, and turns the output into the two-point average $w\, v_0 + (1 - w)\, o$ of the sink value and the sink-free output. Even the entropy splits cleanly, into the binary entropy of the sink share plus $(1 - w)$ times the entropy the head would have had:

```
#check @BookProof.ChapterAttentionSink.scoreSoftmax_sink_succ
#check @BookProof.ChapterAttentionSink.scoreSoftmax_sink_odds
#check @BookProof.ChapterAttentionSink.headOutput_sink
#check @BookProof.ChapterAttentionSink.shannonEntropy_sink
```

A head does not attend to positions but to what the positions stand for, and the right bookkeeping for regrouping the context is the pushforward of the attention distribution along a grouping map. If the values depend on a key only through its group, the key-by-key output and the group-by-group output agree; and coarse graining obeys the

data-processing inequality — merging keys can only destroy information, never create it. What is **not** invariant is the weight: if the scores depend only on the group, each group's share is multiplied by its size, so duplicating a key really does double its vote:

```
#check @BookProof.ChapterAttentionCoarseGrain.observableExpectation_merge
#check @BookProof.ChapterAttentionCoarseGrain.headOutput_merge
#check @BookProof.ChapterAttentionCoarseGrain.shannonEntropy_mergeWeights_le
#check @BookProof.ChapterAttentionCoarseGrain.mergeWeights_scoreSoftmax_of_fiber_cons
```

Finally, the two learned projections of a head are not two objects. The score is $\langle W_Q x, W_K y\rangle = x^\mathsf{T}(W_Q^\mathsf{T} W_K)y$, so the pair enters only through the single bilinear form $W_Q^\mathsf{T} W_K$ — the QK circuit. Any two parameter pairs with the same circuit give the same weights and the same output, and the substitution $(W_Q, W_K) \mapsto (AW_Q, BW_K)$ with $A^\mathsf{T} B = I$ is an exact symmetry: a head's parameters are defined only up to that $GL(d)$ action. The circuit has rank at most the head dimension, which is where the score-table bottleneck comes from:

```
#check @BookProof.ChapterAttentionQKCircuit.qkScore_eq_bilinear
#check @BookProof.ChapterAttentionQKCircuit.qkScore_gauge
#check @BookProof.ChapterAttentionQKCircuit.scoreSoftmax_qkScore_gauge
#check @BookProof.ChapterAttentionQKCircuit.rank_qkMatrix_le
```

All of this is written into a stream that the block never replaces: the update is $x \mapsto x + f(x)$. A head whose values are bounded by C moves the stream by at most C, and if the block is a contraction the residual map is not merely injective but expansive by $1 - L$ — the input of a layer can always be recovered from its output, so nothing an earlier layer wrote can be overwritten by a later one. A stack of n such blocks drifts by at most nC: depth moves the stream linearly, not explosively:

```
#check @BookProof.ChapterResidualStream.norm_residual_sub_self_le
#check @BookProof.ChapterResidualStream.norm_sub_residual_ge
#check @BookProof.ChapterResidualStream.residual_injective
#check @BookProof.ChapterResidualStream.norm_iterate_residual_sub_le
```

The name **Softmax** is a claim about the free energy, and the claim is quantitative. The log-partition function is squeezed between the largest score and that score plus $\log m$, so $\frac{1}{\beta}\log Z(\beta)$ is the maximum score up to $\log m/\beta$, and the attention-weighted mean score obeys the same bound: at a low enough temperature the head reads (almost) the best-matching key, and both quantities converge to the maximum as $\beta \to \infty$. The whole gap between the soft maximum and the hard one is the entropy of the collapse:

```
#check @BookProof.ChapterAttentionFreeEnergy.le_logPartition
#check @BookProof.ChapterAttentionFreeEnergy.abs_logPartition_div_sub_le
#check @BookProof.ChapterAttentionFreeEnergy.sub_meanScore_le
#check @BookProof.ChapterAttentionFreeEnergy.tendsto_meanScore_atTop
```

No long-context head looks at all of its keys: it keeps a shortlist — a top-k set, a sliding window, a block-sparse pattern — and renormalizes over it. Masking is Bayesian conditioning, so the price of that shortcut can be computed exactly: the sparse head differs from the dense head by $2(1 - P(S))$ in ℓ^1, where $P(S)$ is the attention mass the shortlist would have carried — no more and no less. A shortlist that captures all but ε of the mass is 2ε-accurate, its output is within $2\varepsilon C$, and sparsification is lossless exactly when the discarded keys carried nothing:

```
#check @BookProof.ChapterAttentionSparse.l1dist_maskedSoftmax_eq
#check @BookProof.ChapterAttentionSparse.l1dist_maskedSoftmax_le_of_mass
#check @BookProof.ChapterAttentionSparse.norm_headOutput_masked_sub_le
#check @BookProof.ChapterAttentionSparse.one_sub_attendedMass_le
```

The gauge argument that collapses the query and key projections into the single QK circuit applies verbatim on the writing side. Because the attention average is linear, it commutes with every linear map, so the value and output projections act only through their product $W_O W_V$ — the OV circuit — with its own $GL(d)$ gauge freedom $(W_O, W_V) \mapsto (W_O A, BW_V)$, $AB = I$. That product has rank at most the head dimension: a head reads through a rank-d bottleneck and writes through one too:

```
#check @BookProof.ChapterAttentionOVCircuit.mulVec_headOutput
#check @BookProof.ChapterAttentionOVCircuit.ovOutput_eq_headOutput
```

```
#check @BookProof.ChapterAttentionOVCircuit.ovOutput_gauge
#check @BookProof.ChapterAttentionOVCircuit.rank_ovMatrix_le
```

The Jacobian of the collapse also measures how teachable the head is. The total absolute response of the attention distribution to a nudge of one score is exactly $2\beta p_i(1-p_i)$ — never more than $\beta/2$, and vanishing at both ends of the scale. Once some key carries $1-\varepsilon$ of the attention, **every** score has total influence at most $2\beta\varepsilon$: a head that has made up its mind is hard to teach, which is the algebraic content of the saturation that attention temperature schedules are designed to avoid:

```
#check @BookProof.ChapterAttentionSaturation.sum_abs_softmaxJacobian_row
#check @BookProof.ChapterAttentionSaturation.sum_abs_softmaxJacobian_le_half
#check @BookProof.ChapterAttentionSaturation.sum_abs_softmaxJacobian_le_of_confident
```

Finally, heads are rarely scored on alignment alone: a learned per-key bias, a relative-position bias or a repetition penalty is added to the logits first. Read through the Born rule such a bias is not an extra ingredient but the **prior** of the update — the biased head is literally the Bayes posterior with prior w and likelihood $e^{\beta s}$, its odds are prior odds times likelihood ratio, and a prior is exactly a score shift $s_j \mapsto s_j + (\log w_j)/\beta$. Only the ratios of the prior weights matter, and at infinite temperature the head returns the normalized prior: with no evidence, the posterior is the prior:

```
#check @BookProof.ChapterAttentionPrior.priorSoftmax_eq_posterior
#check @BookProof.ChapterAttentionPrior.priorSoftmax_odds
#check @BookProof.ChapterAttentionPrior.priorSoftmax_eq_scoreSoftmax_bias
#check @BookProof.ChapterAttentionPrior.priorSoftmax_zero
```

Generation exploits the same algebra. Appending one key/value pair to a head multiplies every cached weight by a single factor $1-w$, so nothing already computed has to be revisited, and the new output is the convex interpolation $(1-w)\,o_{\mathrm{old}} + w\,v_{\mathrm{new}}$. The KV cache is therefore not an approximation but an identity, and a late token perturbs an established summary exactly in proportion to the attention it wins:

```
#check @BookProof.ChapterAttentionStreaming.scoreSoftmax_snoc_castSucc
#check @BookProof.ChapterAttentionStreaming.headOutput_snoc
#check @BookProof.ChapterAttentionStreaming.norm_headOutput_snoc_sub_le
#check @BookProof.ChapterAttentionStreaming.scoreSoftmax_snoc_odds
```

Position can also enter as a penalty rather than as an embedding: subtract γ times the distance from each score. Such a head is provably **local** — the weight of a key decays like $e^{-\beta\gamma d}$ in its distance, so the total attention beyond distance R is at most $m\,e^{\beta\Delta}e^{-\beta\gamma R}$, and the sliding-window head is within $2C$ times that of the full head. Locality is a theorem about the penalty, not an architectural stipulation:

```
#check @BookProof.ChapterAttentionLocality.scoreSoftmax_alibi_le
#check @BookProof.ChapterAttentionLocality.farMass_le
#check @BookProof.ChapterAttentionLocality.norm_headOutput_window_sub_le
```

And when the shortlist must be chosen, the greedy choice is the right one. Among all shortlists of a given size, the one holding the heaviest keys carries the most mass, so — the price of sparsification being exactly the discarded mass — it also minimizes both the ℓ^1 error and the output error. At a positive temperature the heaviest keys are the highest-scoring ones, so the selection can be made before the Softmax is ever evaluated:

```
#check @BookProof.ChapterAttentionTopK.attendedMass_le_of_isTop
#check @BookProof.ChapterAttentionTopK.l1dist_maskedSoftmax_le_of_isTop
#check @BookProof.ChapterAttentionTopK.isTopWeight_of_isTopScore
```

Finally, the temperature itself is observable. For a head whose scores are not all equal the attention entropy is a continuous, strictly decreasing function of the inverse temperature, so distinct temperatures give distinct entropies and every entropy level between $H(B)$ and the maximal $\log m$ is realized by exactly one $\beta \in [0, B]$. "Run this head at entropy h" is a well-posed instruction:

```
#check @BookProof.ChapterAttentionCalibration.attentionEntropy_strictAntiOn
#check @BookProof.ChapterAttentionCalibration.exists_beta_attentionEntropy_eq
#check @BookProof.ChapterAttentionCalibration.existsUnique_beta_attentionEntropy_eq
```

48.5. Informational Superposition and the Unknown Output

Attention ends with an aggregation of **Value** vectors:

$$\mathbf{o}_i = \sum_j a_{ij}\, \mathbf{v}_j.$$

The physics of this output is the physics of a datum before measurement. A single patch of an image — a straight grey line — is ambiguous in isolation: it could be the edge of a skyscraper, the leg of a table, or a shadow. The token's initial embedding is a **prior state**: a high-entropy superposition of possible structural meanings drawn from the global distribution of the training data. This is precisely the manuscript's thesis that the **global topology of the surrounding data acts as the measurement apparatus**: the query, broadcast across the sequence, is measured against the keys by a Quantum Bayesian update. The verified content here is the Bayesian machine itself — belief update by normalization, and the transport of a prior along a mechanism:

```
#check @BookProof.ChapterKernelTransport.kernel_transport_isProbability
#check @BookProof.ChapterKernelTransport.kernel_transport_apply
#check @BookProof.ChapterBayesInference.posterior
#check @BookProof.ChapterBayesInference.posterior_sum_one
#check @BookProof.ChapterBayesInference.exists_unitary_reproduces_posterior
```

The "high-entropy superposition" and its collapse can be **measured**, with the Shannon entropy $H(p) = -\sum_j p_j \log p_j$. `shannonEntropy_le_log_card` is Gibbs' inequality: no distribution over m keys carries more uncertainty than $\log m$, and `shannonEntropy_uniform` shows the bound is attained. At infinite temperature attention sits exactly there (`shannonEntropy_scoreSoftmax_zero`): the query resolves nothing. `tendsto_shannonEntropy_scoreSoftmax` is the collapse — when the scores have a strict maximizer, the entropy of the attention distribution tends to zero as the temperature does — and `tendsto_shannonEntropy_bornWeight_smul_query` states the same for the coherent-state Born weights as the query is amplified:

```
#check @BookProof.ChapterAttentionEntropy.shannonEntropy
#check @BookProof.ChapterAttentionEntropy.shannonEntropy_nonneg
#check @BookProof.ChapterAttentionEntropy.shannonEntropy_le_log_card
#check @BookProof.ChapterAttentionEntropy.shannonEntropy_uniform
#check @BookProof.ChapterAttentionEntropy.shannonEntropy_scoreSoftmax_zero
#check @BookProof.ChapterAttentionEntropy.shannonEntropy_scoreSoftmax_le_log_card
#check @BookProof.ChapterAttentionEntropy.tendsto_shannonEntropy_scoreSoftmax
#check @BookProof.ChapterAttentionEntropy.tendsto_shannonEntropy_bornWeight_smul_quer
```

48.6. The Posterior: Observable Operators and Expectation Values

To define the output physically, introduce the quantum notion of an **observable operator**. By the spectral theorem an operator is fixed by its outcomes $|k_j\rangle$ and their eigenvalues $\mathbf{v}_j$:

$$\hat{V} = \sum_j \mathbf{v}_j\, |k_j\rangle\langle k_j|.$$

The environment acts as an informational observable: the keys are the eigenstates of the apparatus, and the values are the multi-dimensional eigenvalues. Once the Born-rule measurement yields the posterior p_j, the expectation value of the observable over that distribution is the probability-weighted sum of eigenvalues:

$$\langle \hat{V} \rangle = \sum_j p_j\, \mathbf{v}_j.$$

This is structurally identical to the attention output $\mathbf{o}_i = \sum_j a_{ij}\, \mathbf{v}_j$. The output of attention is the expectation value of a contextual observable over the collapsed posterior — a **definite** vector where before there was maximum relational uncertainty. This too is now a theorem. `observableExpectation` is the probability-weighted sum of eigenvalues, and `attention_eq_expectation` states that the Born weights are a probability distribution and that the attention output is exactly that expectation value. `prob_weighted_sum_mem_convexHull` and

`attention_mem_convexHull` express the "definiteness": the output never leaves the convex hull of the possible outcomes:

```
#check @BookProof.ChapterObservableExpectation.observableExpectation
#check @BookProof.ChapterObservableExpectation.observableExpectation_scalar
#check @BookProof.ChapterObservableExpectation.attention_eq_expectation
#check @BookProof.ChapterObservableExpectation.prob_weighted_sum_mem_convexHull
#check @BookProof.ChapterObservableExpectation.attention_mem_convexHull
#check @BookProof.ChapterObservableExpectation.attention_eq_softmax_expectation
#check @BookProof.ChapterObservableExpectation.bayes_posterior_expectation_mem_convex
```

That expectation value has a temperature, too. Writing `headOutput` for the output of the head at inverse temperature β, the dial runs between two extremes: at $\beta = 0$ the query is ignored and the head returns the plain mean of the values, while if one key strictly maximizes the alignment the output converges to that single value vector as $\beta \to \infty$ — the soft average becomes a hard table lookup. Everything in between stays inside the convex hull of the values, and the output is ℓ^1-stable in the weights:

```
#check @BookProof.ChapterAttentionOutput.headOutput_zero
#check @BookProof.ChapterAttentionOutput.tendsto_headOutput
#check @BookProof.ChapterAttentionOutput.headOutput_mem_convexHull
#check @BookProof.ChapterAttentionOutput.norm_observableExpectation_sub_le
#check @BookProof.ChapterAttentionOutput.attentionOutput_eq_headOutput
```

The observable itself can be built, not merely its spectral data. `observableOp` is the finite-dimensional operator $\hat{V} = \sum_j v_j |k_j\rangle\langle k_j|$; `observableOp_isHermitian` says it is a genuine observable when the eigenvalues are real; `bornProb` is the Born statistics $p_j = |\langle k_j|q\rangle|^2$ of measuring it in the state $|q\rangle$, a probability distribution whenever the eigenvectors are an orthonormal basis and the state is a unit vector (`bornProb_sum_one`, Parseval); and `observableOp_expectation` is the identity $\langle q|\hat{V}|q\rangle = \sum_j p_j v_j$ for the operator itself:

```
#check @BookProof.ChapterObservableOperator.observableOp
#check @BookProof.ChapterObservableOperator.observableOp_isHermitian
#check @BookProof.ChapterObservableOperator.expectation_outerProj
#check @BookProof.ChapterObservableOperator.bornProb_sum_one
#check @BookProof.ChapterObservableOperator.observableOp_expectation
#check @BookProof.ChapterObservableOperator.observableOp_expectation_real
#check @BookProof.ChapterObservableOperator.observableOp_expectation_mem_convexHull
#check @BookProof.ChapterObservableOperator.observable_expectation_born
```

It rests on the conditional / expectation structure already in the library:

```
#check @BookProof.ChapterConditional.pCond_nonneg
#check @BookProof.ChapterConditional.pCond_sum_one
#check @BookProof.ChapterConditional.pJoint_eq_cond_mul_marg
#check @BookProof.ChapterBayesInference.posterior_eq_joint_div_evidence
```

48.7. The Deep Cascade: Successive Bayesian Updates

If the uncertainty is neutralized in a single layer, why do deep networks need many? Because reality is hierarchically complex. The definite output of layer one becomes the certain input (the new prior) of layer two, which now faces a **new**, higher-level uncertainty. The deep network is a **cascading succession of complete Bayesian updates**: each layer collapses a wave-function over its context and restores certainty as an expectation value. The verified facts that make this coherent are the associativity of Bayesian updating and the collapse of a hierarchy into a single kernel:

```
#check @BookProof.ChapterSequentialBayes.sequential_eq_batch
#check @BookProof.ChapterSequentialBayes.posterior_sequential_eq_batch
#check @BookProof.ChapterHierarchicalBayes.outerPosterior_eq_bayesUpdate
#check @BookProof.ChapterHierarchicalBayes.outerPosterior_eq_sum_flat
#check @BookProof.ChapterHierarchicalBayesComposition.compKernel_assoc
```

48.8. Summary

The forward pass of a deep model is not an arbitrary sequence of matrix products. Setting the objects being measured to be coherent states, the Softmax attention weight is the Born rule, the aggregated output is the expectation value of a contextual observable over a posterior, and stacking layers executes a cascade of Bayesian updates. Of the two new identifiers this chapter contributes, the first — the coherent overlap with its Softmax consequence — is now proved, in `BookProof.ChapterCoherentOverlap`, `BookProof.ChapterSoftmaxBorn` and `BookProof.ChapterObservableExpectation`. The second, the temperature identity $\tau = \bar{n} + \frac{1}{2}$, is proved in its statistical core (`BookProof.ChapterCoherentTemperature`) and in its occupation-statistics reading (`BookProof.ChapterCoherentOccupation`: coherent light is Poissonian with variance $\bar{n}$, and τ is the expectation of the energy observable $n + \frac{1}{2}$); the law is further identified with the Boltzmann–Gibbs law at inverse temperature $x = \hbar\omega/kT$ and given its $\frac{1}{2}\coth(x/2)$ closed form in `BookProof.ChapterBoseEinstein`, and characterized variationally — maximal entropy at fixed mean occupation — in `BookProof.ChapterThermalMaxEntropy`. Its physical derivation from the fidelity of displaced thermal states remains a precise proof plan in the appendix. The two caveats the first draft of this chapter carried have since been discharged: the complex Bargmann kernel, with its phase, is formalized in `BookProof.ChapterCoherentOverlapComplex`, and the observable is built as an operator, not merely as spectral data, in `BookProof.ChapterObservableOperator`. The surrounding structure — the flat-versus-sharp dichotomy (`BookProof.ChapterSoftmaxSharpness`), the geometry of the wave-packets (`BookProof.ChapterCoherentGeometry`), the order structure of the measurement (`BookProof.ChapterSoftmaxOrder`) and the entropy of the collapse (`BookProof.ChapterAttentionEntropy`) — is proved as well. Everything else here is backed by existing theorems in `BookProof`.

49. Entropy, Irreversibility, and the Arrow of Time

How an irreversible, entropy-increasing dynamics coexists with deterministic evolution. We characterize irreversibility as injective-but-not-surjective dynamics, show that a random map is almost surely non-invertible, exhibit uncountable null-measure sets, quantify the cosmological amplification of the matter/radiation ratio, and close with the law of large numbers that gives the probabilities their empirical meaning.

How to read this part. Irreversibility: injective but not surjective and a random map is almost surely non-invertible form one argument (irreversibility is generic), while null-measure sets need not be small is the measure-theoretic caution that keeps the argument honest. the baryon asymmetry is the cosmological application — the one place the part connects to the gravity thread, via the Friedmann–Robertson–Walker scaling that the companion numerical validation checks — and the law of large numbers closes the part by giving the probabilities their empirical content. Read the first two chapters and the LLN chapter for the core; the baryon-asymmetry chapter needs the FLRW terminology of the gravity part.

50. Irreversibility: Injective but Not Surjective

50.1. Orientation and Status

The opening chapter of the arrow-of-time part: what "irreversible" means precisely (injective but not surjective), and why a **finite** world cannot be irreversible — the continuum admits it. With the next chapter it forms one argument: irreversibility is generic.

The route: the precise meaning, the finite obstruction, the continuum case, a concrete dissipative map on the interval, and the arrow of time from pure set theory. The field-theoretic instance of the same phenomenon is the dissipative dynamics of the gauge-symmetry chapter.

The proof is `BookProof/ChapterIrreversibleDynamics`.

50.2. What "Irreversible" Means, Precisely

A deterministic time-evolution is a function f from the state space to itself. It is **reversible** exactly when f is a bijection: every state has a unique predecessor, so the past can be reconstructed from the present. It is

irreversible when the past is lost — when f is **injective but not surjective**: distinct states stay distinct (no two histories merge), but some states have no predecessor (the dynamics has a "missing past").

The manuscript's claim is that a dissipative deterministic dynamics is exactly this: injective, not surjective, and hence time-asymmetric.

50.3. A Finite World Cannot Be Irreversible

There is a sharp dichotomy. On a **finite** state space, injectivity already forces surjectivity: an injection from a finite set to itself is automatically a bijection (the pigeonhole principle). So:

```
#check @IrreversibleDynamics.finite_injective_iff_surjective
#check @IrreversibleDynamics.finite_no_irreversible
```

There is **no** injective-but-not-surjective self-map of a finite set. A genuinely irreversible deterministic dynamics is therefore impossible in a purely finite (discrete, bounded) world. This is the precise form of the manuscript's remark that "the rationals are not enough": irreversibility requires the **continuum** (an infinite, Dedekind-infinite state space).

50.4. The Continuum Admits Irreversibility

On an infinite state space the dichotomy breaks. There always exists an injective, non-surjective self-map; the simplest is the successor $n \mapsto n + 1$ on $\mathbb{N}$, which misses 0:

```
#check @IrreversibleDynamics.exists_injective_not_surjective
#check @IrreversibleDynamics.nat_succ_injective_not_surjective
```

50.5. A Concrete Dissipative Map on the Interval

The manuscript needs more than mere non-surjectivity: it needs a **non-singular, dissipative** map. The model example is the halving map

$$f : [0,1] \to [0,1], \qquad f(x) = x/2.$$

It has all the required properties at once:

- **Injective** — distinct inputs give distinct outputs.
- **Not surjective** — the value 1 (indeed anything in $(1/2, 1]$) is never reached; the past of those states is missing.
- **Dissipative** — it sends an interval $[a, b]$ to $[a/2, b/2]$, **halving** its Lebesgue length.
- **Non-singular** — a set of positive length is sent to a set of positive (halved) length; null sets stay null and positive-measure sets stay positive-measure.

The verified statements (module `BookProof.ChapterIrreversibleDynamics`):

```
#check @IrreversibleDynamics.dissipative_injective
#check @IrreversibleDynamics.dissipative_not_surjective_unitInterval
#check @IrreversibleDynamics.dissipative_volume_Icc
#check @IrreversibleDynamics.dissipative_nonsingular_Icc
```

50.6. The Arrow of Time From Pure Set Theory

The conclusion is striking: the **arrow of time** — the fact that the future is determined but the past is not recoverable — needs no thermodynamics to appear at the kinematic level. It is forced the moment a deterministic dynamics on a continuum is dissipative (volume-contracting): such a map is injective but cannot be surjective, so it has a built-in time asymmetry. The next two chapters quantify how *generic* this is (a random map is almost surely non-invertible) and how it coexists with the measure-theoretic subtleties (null-measure sets) that the continuum introduces.

51. A Random Map Is Almost Surely Non-Invertible

51.1. Orientation and Status

The second chapter of the arrow-of-time argument: a random map is almost surely non-invertible. With the previous chapter this makes irreversibility **generic**, and the next chapter adds the measure-theoretic caution that keeps the argument honest.

The route: counting maps and bijections, the probability of invertibility, and the sharp asymptotic by Stirling's formula.

51.2. Counting Maps and Bijections

The previous chapter showed that an irreversible deterministic dynamics is an injective-but-not-surjective self-map, possible only on an infinite state space. A natural quantitative question, raised directly in the manuscript, is: if you pick a "discrete function" at random, how likely is it to be invertible at all?

Model a discrete function on an n-cell partition as an arbitrary map $\mathrm{Fin}\, n \to \mathrm{Fin}\, n$. There are

$$n^n$$

such maps (each of the n inputs chooses one of n outputs independently), and exactly

$$n!$$

of them are bijections (the permutations). The verified counts:

```
#check @ChapterBijectionProbability.card_fun_fin
#check @ChapterBijectionProbability.card_bijective_fin
```

51.3. The Probability of Invertibility

The probability that a uniformly random map is invertible is therefore the ratio

$$\mathrm{bijProb}(n) = \frac{n!}{n^n}.$$

```
#check @ChapterBijectionProbability.bijProb_eq_card_ratio
```

This probability is tiny and gets tinier. The elementary bound $(n+1)! \le (n+1)^n$ gives $\mathrm{bijProb}(n) \le 1/n$, and in fact:

```
#check @ChapterBijectionProbability.bijProb_le_one_div
#check @ChapterBijectionProbability.bijProb_tendsto_zero
```

So the probability of invertibility **converges to zero** as the partition is refined. A random discrete dynamics is, with overwhelming probability, **non-invertible** — i.e. irreversible in the sense of the previous chapter.

51.4. The Sharp Asymptotic: Stirling

The manuscript quotes the precise rate. By **Stirling's formula** $n! \sim \sqrt{2\pi n}\,(n/e)^n$,

$$\mathrm{bijProb}(n) = \frac{n!}{n^n} \sim \sqrt{2\pi n}\, e^{-n}.$$

The verified asymptotic equivalence:

```
#check @ChapterBijectionProbability.bijProb_isEquivalent_stirling
```

The decay is **exponential** in n (the $\sqrt{2\pi n}$ prefactor is negligible next to e^{-n}). Invertibility is not merely unlikely; it is *exponentially* unlikely.

51.5. Why This Matters

Combine this with the previous chapter. There we saw that irreversibility is *possible* only on a continuum and *impossible* on a finite set. Here we see the complementary statistical fact: as a finite approximation is refined ($n \to \infty$), the fraction of dynamics that are invertible collapses to zero exponentially. So in the continuum limit, **almost every** deterministic dynamics is irreversible. The arrow of time is not a special, fine-tuned feature of particular systems; it is the **generic** case, and reversible (bijective) dynamics is the measure-zero exception. This is the rigorous backbone of the manuscript's claim that an irreversible deterministic time-evolution is the rule, not the accident.

52. Null-Measure Sets Need Not Be Small

52.1. Orientation and Status

The measure-theoretic caution of the part: a null event is not automatically special, and null does not mean structurally small — the Cantor set is uncountable and still null. It keeps the irreversibility argument honest and the priors of the Bayesian chapters disciplined.

The route: a null event is not automatically special, no point is special, countable sets are null, null does not mean countable (the Cantor set), the headline, why this matters for priors, the atomic and continuous parts (the five types), and why continuous priors are out of reach of discrete ones.

The classification is the large verified block: `BookProof/ChapterStandardBorelClassification`, the `ChapterAbelian*` and `ChapterSpectral*` model chapters, the atomic/diffuse decomposition chapters (`ChapterAtomicDecomposition`, `ChapterMeasureAtomicDiffuse`, `ChapterMixedPrior`), and `BookProof/ChapterSelectingEvents`.

52.2. A Null Event Is Not Automatically Special

A recurring intuitive error is to treat a **measure-zero** event as negligible in a strong, geometric sense — as if a null subset of a space must be "one dimension smaller," or as if a null point must be a special, distinguished point. The manuscript argues, in its discussion of consciousness and Bayesian priors (and of the Fermi paradox), that both inferences are false. This chapter makes the measure-theoretic core precise.

52.3. No Point Is Special

Under Lebesgue measure on $\mathbb{R}$, **every** single point has measure zero, and all singletons have the **same** measure:

```
#check @ConsciousnessNullMeasure.singleton_volume_zero
#check @ConsciousnessNullMeasure.singletons_equal_measure
```

So a point having measure zero carries no information about its being special: in the uniform measure on an interval, *every* point is null. Singling out one null point (as a "chosen" outcome, or a "conscious" observer) is not justified by the measure.

52.4. Countable Sets Are Null

More generally, every **countable** set has Lebesgue measure zero, by countable additivity (a countable union of null singletons is null). In particular the rationals, though dense, are null:

```
#check @ConsciousnessNullMeasure.countable_volume_zero
#check @ConsciousnessNullMeasure.rat_range_volume_zero
```

52.5. But Null Does Not Mean Countable: the Cantor Set

Here is the key counterexample to "null means small." The **ternary Cantor set** $C \subset [0, 1]$ — obtained by repeatedly removing the open middle third — is simultaneously:

- **uncountable**, and
- of Lebesgue **measure zero**.

Its uncountability follows from Cantor's theorem (it is in bijection with $\{0,1\}^{\mathbb{N}}$, so $\aleph_0 < 2^{\aleph_0}$):

```
#check @ConsciousnessNullMeasure.cantorSet_uncountable
```

Its null measure follows from self-similarity: C is the disjoint union of two scaled copies of itself, each by factor $1/3$, so $\mu(C) \le (2/3)\,\mu(C)$, which (since $\mu(C) < \infty$) forces $\mu(C) = 0$:

```
#check @ConsciousnessNullMeasure.cantorSet_volume_zero
```

52.6. The Headline

Putting these together, there exists an **uncountable** subset of $[0,1]$ with Lebesgue measure **zero**:

```
#check @ConsciousnessNullMeasure.exists_uncountable_null_subset
```

This is exactly the manuscript's point: "a subset with null measure does not imply that the subset has one less dimension than the set, since there are subsets of a real interval which have a fractal dimension (which can be very close to one, but not one) and thus are also uncountable." Measure zero is a statement about *probability*, not about *cardinality* or *dimension*. A null set can be as large as the whole interval in cardinality (uncountable) and can have fractal dimension arbitrarily close to one.

52.7. Why This Matters for Priors

The consequence for the Bayesian thread of the book is direct. On a continuous space there is **no uniform probability measure** that makes every point equally likely (each point would have to be null, yet the whole space has measure one — and there is no countably-additive way to spread mass uniformly over uncountably many points). Worse, the "special" outcomes one might want to privilege (a particular observed value, a particular observer) are null and **indistinguishable**, by the measure, from every other point. On a general continuous space, then, a prior that looks non-informative in one parametrization becomes informative in another: a change of coordinates multiplies the density by a Jacobian, and there is no coordinate-invariant "uniform" prior to be had. This is the precise sense of the manuscript's slogan "there are no non-informative priors" — the coordinate argument of the Bayesian-updating chapter. It connects back to the maximum-entropy chapter, where a uniform prior existed only because the space was **finite** (and even there only relative to a labeling), and forward to the one exception in this book: the Solovay–Kopperman tail, where the restricted decidable language forbids the very reparametrizations that would break uniformity, so the Mehler prior on the infinite-dimensional hypersphere stays genuinely uniform.

52.8. Atomic and Continuous Parts: the Five Types

The distinction between a **continuous** prior and a **discrete** (mixed) one can be made completely precise. Every measure splits into a continuous part and a part carried by its atoms — the points of positive mass — and the atoms are always a countable set:

```
#check @BookProof.ChapterAtomicDecomposition.atoms_countable
#check @BookProof.ChapterAtomicDecomposition.noAtoms_continuousPart
#check @BookProof.ChapterAtomicDecomposition.eq_continuousPart_add_atomicPart
```

The atomic part is literally a countable sum of point masses $\sum_x \mu\{x\}\,\delta_x$:

```
#check @BookProof.ChapterAtomicDecomposition.atomicPart_eq_sum_dirac
```

Since a probability measure cannot have both parts trivial, exactly five combinations remain — continuous part present or absent, crossed with no atoms, finitely many atoms, or countably infinitely many atoms. These are precisely the five types of the manuscript's list of abelian von Neumann algebras ($\ell^\infty(\{1,\dots,n\})$, $\ell^\infty(\mathbb{N})$, $L^\infty([0,1])$, and the two mixtures):

```
#check @BookProof.ChapterAtomicDecomposition.not_continuousPart_zero_and_atoms_empty
#check @BookProof.ChapterAtomicDecomposition.probability_measure_five_types
```

The first type is a theorem, not just a label: for a Hermitian matrix with distinct eigenvalues, the algebra of matrices commuting with it is exactly a unitary conjugate of the diagonal matrices, i.e. a faithful copy of $\ell^\infty(\{1,\dots,n\})$:

```
#check @BookProof.ChapterAbelianVonNeumannFinite.commutant_eq_range_conjDiagonal
#check @BookProof.ChapterAbelianVonNeumannFinite.abelian_commutant_isomorphic_ellInft
```

In fact the diagonal algebra is **maximal** abelian in the strongest concrete sense: the embedding $\ell^\infty(\{1,\dots,n\}) \to \mathrm{Mat}(n,\mathbb{C})$ given by $d \mapsto \mathrm{diag}(d)$ is an injective $*$-algebra map whose image is exactly its own commutant — any matrix commuting with **every** diagonal matrix is itself diagonal. This is the finite (type I_n) case of von Neumann's classification, proved outright in `BookProof.ChapterAbelianDiagonal`:

```
#check @BookProof.AbelianDiagonal.diagonalStarAlgHom_injective
#check @BookProof.AbelianDiagonal.diagonal_commute
#check @BookProof.AbelianDiagonal.commutant_diagonal_eq_diagonal
#check @BookProof.AbelianDiagonal.vonNeumann_abelian_typeI_case
```

The next case on the list — the countable one, $\ell^\infty(\mathbb{N})$ — is now proved as well, on the separable Hilbert space $\ell^2(\mathbb{N})$ built in `BookProof.ChapterRieszFischer`. Each bounded sequence $d \in \ell^\infty(\mathbb{N})$ acts as a diagonal multiplication operator of norm at most $\|d\|$; the assignment is a faithful unital $*$-algebra map (`diagOp_injective`, `diagOp_mul`, `diagOp_one`, `diagOp_star`); its image is abelian; and a bounded operator commutes with every diagonal operator exactly when it is itself diagonal (`commutes_diagOp_iff`). So $\ell^\infty(\mathbb{N})$ is a maximal abelian self-adjoint subalgebra of $B(\ell^2(\mathbb{N}))$:

```
#check @BookProof.ChapterAbelianDiagonalCountable.diagOp
#check @BookProof.ChapterAbelianDiagonalCountable.norm_diagOp_le
#check @BookProof.ChapterAbelianDiagonalCountable.diagOp_mul
#check @BookProof.ChapterAbelianDiagonalCountable.diagOp_star
#check @BookProof.ChapterAbelianDiagonalCountable.diagOp_injective
#check @BookProof.ChapterAbelianDiagonalCountable.diagOp_comm
#check @BookProof.ChapterAbelianDiagonalCountable.commutes_diagOp_iff
#check @BookProof.ChapterAbelianDiagonalCountable.vonNeumann_abelian_class_countable
```

The third class, the diffuse model $L^\infty(\mu)$ acting on $L^2(\mu)$ by multiplication, is now proved as well. For an essentially bounded symbol φ the multiplication operator $M_\varphi f = \varphi \cdot f$ is bounded by the essential supremum, and $\varphi \mapsto M_\varphi$ is unital, multiplicative, abelian, star-closed (the adjoint of M_φ is $M_{\bar\varphi}$) and — on a finite measure space — faithful. Lebesgue measure on $[0,1]$ is atomless, which is exactly what distinguishes this class from the atomic ℓ^∞ models:

```
#check @BookProof.ChapterLinftyMultiplication.multOp
#check @BookProof.ChapterLinftyMultiplication.norm_multOp_le
#check @BookProof.ChapterLinftyMultiplication.multOp_mul
#check @BookProof.ChapterLinftyMultiplication.multOp_one
#check @BookProof.ChapterLinftyMultiplication.multOp_comm
#check @BookProof.ChapterLinftyMultiplication.multOp_inner_adjoint
#check @BookProof.ChapterLinftyMultiplication.multOp_eq_zero_iff
#check @BookProof.ChapterLinftyMultiplication.vonNeumann_abelian_class_Linfty
#check @BookProof.ChapterLinftyMultiplication.unitInterval_atomless
```

The mixed class — an atomic part alongside a diffuse one — is realized too, by the finite measure $\mathrm{vol}|_{[0,1]} + \delta_2$ on the line: it has a genuine atom at 2 and a diffuse part of mass 1 every point of which is null, and its multiplication algebra is again unital, abelian, star-closed and faithful:

```
#check @BookProof.ChapterAbelianMixture.mixtureMeasure
#check @BookProof.ChapterAbelianMixture.mixtureMeasure_atom
#check @BookProof.ChapterAbelianMixture.mixtureMeasure_diffuse_point
#check @BookProof.ChapterAbelianMixture.mixtureMeasure_not_purely_atomic_on_Icc
#check @BookProof.ChapterAbelianMixture.vonNeumann_abelian_class_mixture
```

Exhaustiveness is provable in the **purely atomic** case, and there it collapses the list to a single model. A bounded operator on $\ell^2(\mathbb{N})$ commutes with every minimal (rank-one) projection exactly when it is diagonal — the atoms alone force diagonality — so an algebra that contains all the atomic projections and is abelian consists of diagonal

operators, and if it is in addition maximal abelian it **is** the diagonal algebra $\ell^\infty(\mathbb{N})$. On the measure side the atom set of a probability measure is countable, so a purely atomic index set is either $\mathrm{Fin}\, n$ or $\mathbb{N}$: the two purely atomic types of the list, and no other:

```
#check @BookProof.ChapterAbelianAtomicCondensation.commutes_atomProj_iff
#check @BookProof.ChapterAbelianAtomicCondensation.atomic_abelian_subset_diagonal
#check @BookProof.ChapterAbelianAtomicCondensation.atomic_abelian_maximal_eq_diagonal
#check @BookProof.ChapterAbelianAtomicCondensation.atomic_measure_index_dichotomy
```

At the **diffuse** end the matching structural fact is now proved as well: the multiplication algebra is **its own commutant**. On a finite measure space, if a bounded operator T on $L^2(\mu)$ commutes with every multiplication operator, then T is itself multiplication by a symbol, namely by $\psi = T(1)$, and that symbol is essentially bounded with $\|\psi\|_\infty \le \|T\|$. The argument is the classical one: commutation gives $T(\varphi) = \varphi\,\psi$ for every bounded φ; testing on the indicator of $\{\|\psi\| \ge \|T\| + \varepsilon\}$ and comparing the two L^2 norms forces that set to be null; and the two continuous operators T and M_ψ, agreeing on indicators, agree everywhere. So the diffuse model $L^\infty[0,1]$ on $L^2[0,1]$ is **maximal abelian** — nothing can be adjoined to it without breaking commutativity — exactly as the diagonal ℓ^∞ is at the atomic end:

```
#check @BookProof.ChapterLinftyMaximalAbelian.symbol
#check @BookProof.ChapterLinftyMaximalAbelian.symbol_mul
#check @BookProof.ChapterLinftyMaximalAbelian.symbol_ae_norm_le
#check @BookProof.ChapterLinftyMaximalAbelian.commutant_eq_multOp
#check @BookProof.ChapterLinftyMaximalAbelian.multOp_algebra_maximal_abelian
#check @BookProof.ChapterLinftyMaximalAbelian.unitInterval_multOp_maximal_abelian
```

The passage in the opposite direction — from an **abstract** abelian algebra to a concrete **measure** model — is now available at the C*-level, and it is the classical two-step route. Gelfand duality identifies a commutative unital C*-algebra A with the continuous functions $C(X,\mathbb{C})$ on its character space X, a compact Hausdorff space. A **state** of A — a linear functional that is nonnegative on the squares a^*a and sends the unit to 1 — is then, by the Riesz–Markov–Kakutani theorem, integration against a Borel **probability measure** μ on X: the real part of the state is a positive functional on $C(X,\mathbb{R})$ (because $f = |\sqrt{f}|^2$ for $f \ge 0$), it is real-valued (because $f = f^+ - f^-$), and μ is its Riesz measure. Finally $C(X,\mathbb{C})$ acts on $L^2(\mu)$ by multiplication operators — a unital $*$-homomorphism, injective as soon as μ charges every nonempty open set — and the constant function 1 is a unit vector implementing the state we began with. So every state of a commutative unital C*-algebra is the vector state of a representation of that algebra by multiplication operators on the L^2 space of a probability measure: the abelian Gelfand–Naimark–Segal model.

```
#check @BookProof.ChapterAbelianGelfandModel.rieszStateMeasure
#check @BookProof.ChapterAbelianGelfandModel.integral_rieszStateMeasure
#check @BookProof.ChapterAbelianGelfandModel.exists_probabilityMeasure_of_state
#check @BookProof.ChapterAbelianGelfandModel.mulRepHom
#check @BookProof.ChapterAbelianGelfandModel.mulRepHom_injective
#check @BookProof.ChapterAbelianGelfandModel.inner_oneVec_mulRep
#check @BookProof.ChapterAbelianGelfandModel.gelfandModel
#check @BookProof.ChapterAbelianGelfandModel.state_is_vector_state_of_multiplication
```

For a **single** operator the model can be upgraded from a $*$-homomorphism to a **unitary equivalence**, and then it is the spectral theorem itself. Let T be a normal bounded operator on a complex Hilbert space and let ξ be a cyclic unit vector, meaning that the vectors $f(T)\xi$, f continuous on the spectrum, are dense. Then $f \mapsto \langle \xi, f(T)\xi\rangle$ is a state of $C(\sigma(T),\mathbb{C})$, so by the previous paragraph it is integration against a Borel probability measure μ on the spectrum; the identity $\|f(T)\xi\|^2 = \langle \xi, (\bar f f)(T)\xi\rangle = \int |f|^2\, d\mu$ says that $f \mapsto f(T)\xi$ is an $L^2(\mu)$-isometry, and since continuous functions are dense in $L^2(\mu)$ and the $f(T)\xi$ are dense in the space, it extends to a unitary $U : L^2(\mu) \to H$. That unitary carries multiplication by the coordinate function to T: $UM_z = TU$, and more generally multiplication by any continuous symbol g into $g(T)$, so the whole abelian algebra generated by T becomes an algebra of multiplication operators. A normal operator with a cyclic vector **is** multiplication by z on the L^2 space of a probability measure carried by its spectrum — the abstract algebra has become a measure model, unitarily.

```
#check @BookProof.ChapterSpectralMultiplication.vectorState
#check @BookProof.ChapterSpectralMultiplication.spectralMeasure
```

```
#check @BookProof.ChapterSpectralMultiplication.integral_spectralMeasure
#check @BookProof.ChapterSpectralMultiplication.norm_cfcHom_apply
#check @BookProof.ChapterSpectralMultiplication.spectralUnitary
#check @BookProof.ChapterSpectralMultiplication.spectralUnitary_intertwines_cfc
#check @BookProof.ChapterSpectralMultiplication.spectralUnitary_intertwines
#check @BookProof.ChapterSpectralMultiplication.spectral_multiplication_model
```

In the presence of a cyclic vector one can go one step further and reach the **von Neumann** level, by computing the commutant. The point is that continuous symbols already suffice: if a bounded operator S on $L^2(\mu)$ commutes with multiplication by every **continuous** function on a compact space carrying a finite regular Borel measure, then S is itself multiplication by an essentially bounded symbol. Writing $\psi = S(1)$, commutation gives $Sg = g\,\psi$ for continuous g; truncating ψ to the set where $\|S\| + \varepsilon \le |\psi| \le m$ turns the inequality $\|\psi g\|_2 \le \|S\|\,\|g\|_2$ into a bound on a genuine multiplication operator, which extends from the continuous functions to all of $L^2(\mu)$ by density and, tested on the indicator of that set, forces the set to be null. So $\psi \in L^\infty(\mu)$ and $S = M_\psi$ by density again.

Transported through the spectral unitary, this says that the commutant of $\{f(T)\}$ — every bounded operator commuting with all continuous functions of a normal T with a cyclic vector — is **exactly** the unitary copy of $L^\infty(\mu)$ acting by multiplication. Since the multiplication algebra is its own commutant, the **bicommutant** of $\{f(T)\}$, that is the von Neumann algebra generated by T, is that same copy of $L^\infty(\mu)$; in particular the commutant is abelian, so the algebra generated by a normal operator with a cyclic vector is **maximal** abelian. For the singly generated, cyclic case this is the von Neumann form of the classification, obtained without any appeal to a general bicommutant theorem.

```
#check @BookProof.ChapterSpectralCommutant.multAlgebra
#check @BookProof.ChapterSpectralCommutant.centralizer_multAlgebra
#check @BookProof.ChapterSpectralCommutant.bicommutant_multAlgebra
#check @BookProof.ChapterSpectralCommutant.contCommutant_eq_multOp
#check @BookProof.ChapterSpectralCommutant.centralizer_cfcSet
#check @BookProof.ChapterSpectralCommutant.bicommutant_cfcSet
#check @BookProof.ChapterSpectralCommutant.commutant_cfcSet_isCommutative
```

The hypothesis that carries all of this is the existence of a **cyclic vector**, and that hypothesis can now be removed from the geometry, if not yet from the assembly. For a normal T the closed subspace generated by a single vector ξ under the continuous functional calculus — the closed span of the $f(T)\xi$ — is invariant under the whole algebra, and because that algebra is closed under adjoints ($f(T)^* = \bar f(T)$), the orthogonal complement of an invariant subspace is invariant too. So a vector orthogonal to everything collected so far generates a cyclic subspace that stays orthogonal to it, and Zorn's lemma produces a maximal family of unit vectors with pairwise orthogonal cyclic subspaces. Maximality forces the span of that family to be dense: otherwise its orthogonal complement would contain a unit vector, which could be added to the family. Every complex Hilbert space is therefore the orthogonal direct sum of subspaces on each of which T has a cyclic vector — and on each of those the multiplication model and its commutant computation apply verbatim.

```
#check @BookProof.ChapterCyclicDecomposition.cyclicSubspace
#check @BookProof.ChapterCyclicDecomposition.invariant_cyclicSubspace
#check @BookProof.ChapterCyclicDecomposition.invariant_orthogonal
#check @BookProof.ChapterCyclicDecomposition.cyclicSubspace_le_orthogonal
#check @BookProof.ChapterCyclicDecomposition.exists_cyclic_decomposition
```

That decomposition can be stated as an identification rather than a list. The cyclic subspaces of such a maximal family are an orthogonal family in the technical sense, and since they are jointly total the space **is** their Hilbert sum: there is a unitary between H and the ℓ^2-sum of the summands. The operator respects the splitting for the sharpest possible reason — the orthogonal projection onto an invariant subspace commutes with $f(T)$ for every continuous f, because the complement is invariant too, so each of those projections lies in the **commutant** of the algebra generated by T. Applied to a vector, the projections onto the cyclic summands sum back to it. So T is the direct sum of its restrictions to the summands, on each of which the multiplication model and the commutant computation above apply verbatim.

```
#check @BookProof.ChapterCyclicDirectSum.orthogonalFamily_cyclicSubspace
#check @BookProof.ChapterCyclicDirectSum.isHilbertSum_cyclicSubspace
#check @BookProof.ChapterCyclicDirectSum.cyclicHilbertEquiv
#check @BookProof.ChapterCyclicDirectSum.exists_isHilbertSum_cyclicSubspace
```

```
#check @BookProof.ChapterCyclicDirectSum.commute_starProjection_cfcHom
#check @BookProof.ChapterCyclicDirectSum.starProjection_cyclicSubspace_commutes
#check @BookProof.ChapterCyclicDirectSum.hasSum_starProjection_cyclicSubspace
```

The assembly can then be carried out, and it removes the cyclic-vector hypothesis from the model itself. The trick is to build the model of a summand **inside** the ambient space, so that no functional calculus of a restricted operator is ever needed: the identity $\|f(T)\xi\| = \|f\|_{L^2(\mu_\xi)}$ holds for any vector ξ, cyclic or not, so the map $f \mapsto f(T)\xi$ extends to a unitary from $L^2(\mu_\xi)$ onto the cyclic subspace of ξ — in which ξ is cyclic by construction — and, read in H, to an isometric embedding of $L^2(\mu_\xi)$ with range that subspace, carrying multiplication by a continuous symbol g into $g(T)$ and multiplication by z into T. Running this over a maximal orthogonal cyclic family gives the general form of the spectral theorem: for **every** normal operator on a complex Hilbert space there are Borel probability measures μ_x on its spectrum and isometric embeddings $V_x : L^2(\mu_x) \to H$ exhibiting H as the Hilbert sum of the $L^2(\mu_x)$, with T acting on each summand as multiplication by the coordinate function. Every normal operator is a direct sum of multiplication operators; no cyclic vector and no separability are assumed.

```
#check @BookProof.ChapterSpectralDirectSum.cyclicUnitary
#check @BookProof.ChapterSpectralDirectSum.cyclicEmbedding
#check @BookProof.ChapterSpectralDirectSum.range_cyclicEmbedding
#check @BookProof.ChapterSpectralDirectSum.cyclicEmbedding_intertwines_cfc
#check @BookProof.ChapterSpectralDirectSum.cyclicEmbedding_intertwines
#check @BookProof.ChapterSpectralDirectSum.spectral_multiplication_model_general
```

If the space is separable the decomposition is moreover **countable**: distinct members of the family are unit vectors with orthogonal cyclic subspaces, hence orthogonal, so they sit at distance $\sqrt{2}$ from one another and the balls of radius $1/2$ around them are disjoint — a separable space admits only countably many such. So on a separable space every normal operator is a countable direct sum of multiplication operators, which is the form in which the classification of abelian algebras on separable L^2 is usually stated.

```
#check @BookProof.ChapterSpectralDirectSum.countable_orthogonalCyclicFamily
#check @BookProof.ChapterSpectralDirectSum.spectral_multiplication_model_separable
```

The single-generator hypothesis can be dropped as well, and dropping it does not require producing a generator. Nothing in the argument above used that the algebra was the one generated by **one** operator: what was used is that a unital $*$-representation π of the continuous functions on a compact Hausdorff space turns the vector ξ into a state $f \mapsto \langle \xi, \pi(f)\xi \rangle$, that positivity of that state is the identity $\langle \xi, \pi(\bar{f}f)\xi \rangle = \|\pi(f)\xi\|^2$, and that the same identity makes $f \mapsto \pi(f)\xi$ an $L^2(\mu)$-isometry. So for an **arbitrary** abelian algebra, presented as such a representation, a cyclic unit vector already gives a Borel probability measure μ and a unitary $U : L^2(\mu) \to H$ carrying multiplication by every continuous symbol g into $\pi(g)$. (No continuity of π is assumed anywhere; it is not needed.) Through Gelfand duality this is a statement about the abstract algebra: every representation of a commutative unital C*-algebra with a cyclic unit vector is unitarily the representation by multiplication operators on the L^2 space of a Borel probability measure on the character space.

```
#check @BookProof.ChapterAbelianCyclicModel.repState
#check @BookProof.ChapterAbelianCyclicModel.repMeasure
#check @BookProof.ChapterAbelianCyclicModel.norm_rep_apply
#check @BookProof.ChapterAbelianCyclicModel.cyclicRepUnitary
#check @BookProof.ChapterAbelianCyclicModel.cyclicRepUnitary_intertwines
#check @BookProof.ChapterAbelianCyclicModel.cyclic_representation_multiplication_mode
#check @BookProof.ChapterAbelianCyclicModel.abelian_algebra_multiplication_model
```

The commutant computation transports through that unitary just as it did for a single operator, so the von Neumann statement comes with it: the commutant of the represented algebra is exactly the unitary copy of $L^\infty(\mu)$, that copy is its own commutant, hence the **bicommutant** — the von Neumann algebra generated by the algebra — is the same copy, and the commutant is abelian. An arbitrary abelian C*-algebra acting with a cyclic vector is therefore maximal abelian and is $L^\infty(\mu)$ acting on $L^2(\mu)$, with no generator produced anywhere.

```
#check @BookProof.ChapterAbelianCyclicCommutant.centralizer_repSet
#check @BookProof.ChapterAbelianCyclicCommutant.centralizer_multModelRep
#check @BookProof.ChapterAbelianCyclicCommutant.bicommutant_repSet
```

```
#check @BookProof.ChapterAbelianCyclicCommutant.commutant_repSet_isCommutative
#check @BookProof.ChapterAbelianCyclicCommutant.abelian_algebra_maximal_abelian_of_cy
```

The cyclic-vector hypothesis can be removed here too, and for the same reason as before: the decomposition argument used only that the algebra of operators is $*$-closed, never that it came from one operator. So for an arbitrary abelian algebra the cyclic subspaces are invariant, the orthogonal complement of an invariant subspace is invariant, Zorn's lemma produces a maximal family of unit vectors with pairwise orthogonal cyclic subspaces, the space is the Hilbert sum of those subspaces, and the projections onto them lie in the commutant. Building each summand's model inside the ambient space then glues the pieces: **every** abelian algebra of operators on a complex Hilbert space — presented as a unital $*$-representation of the continuous functions on a compact Hausdorff space, or through Gelfand duality as a representation of a commutative unital C*-algebra — is a direct sum of multiplication algebras. There are Borel probability measures μ_x and isometric embeddings V_x : $L^2(\mu_x) \to H$ exhibiting H as their Hilbert sum, with each $\pi(g)$ acting on the summand as multiplication by g. No cyclic vector, no generator and no separability are assumed; on a separable space the family is moreover countable.

```
#check @BookProof.ChapterAbelianDirectSum.exists_rep_cyclic_decomposition
#check @BookProof.ChapterAbelianDirectSum.exists_isHilbertSum_repCyclicSubspace
#check @BookProof.ChapterAbelianDirectSum.starProjection_commutes_rep
#check @BookProof.ChapterAbelianDirectSum.repEmbedding
#check @BookProof.ChapterAbelianDirectSum.repEmbedding_intertwines
#check @BookProof.ChapterAbelianDirectSum.abelian_multiplication_model_general
#check @BookProof.ChapterAbelianDirectSum.abelian_multiplication_model_separable
#check @BookProof.ChapterAbelianDirectSum.abelian_algebra_multiplication_model_genera
```

The measures produced by that decomposition can be sorted into the manuscript's classes as well. The atoms of a finite measure are countable, because the singletons are pairwise disjoint and only countably many disjoint sets can carry positive mass; restricting to the atoms gives a purely atomic measure, a countable sum of point masses $\mu\{x\}\,\delta_x$, and restricting to their complement gives a measure with no atoms at all. Every summand of the model is therefore an atomic piece, a diffuse piece, or a mixture of the two — the manuscript's three-way alternative, with the atomic case the I_n / $\ell^\infty(\mathbb{N})$ models and the diffuse case the $L^\infty(\mu)$ one.

```
#check @BookProof.ChapterMeasureAtomicDiffuse.countable_atomSet
#check @BookProof.ChapterMeasureAtomicDiffuse.restrict_atomSet_add_restrict_compl
#check @BookProof.ChapterMeasureAtomicDiffuse.restrict_atomSet_eq_sum_dirac
#check @BookProof.ChapterMeasureAtomicDiffuse.exists_atomic_diffuse_decomposition
#check @BookProof.ChapterMeasureAtomicDiffuse.abelian_multiplication_model_atomic_dif
```

The diffuse case can be pinned down further, at least on the line. A diffuse probability measure μ on $\mathbb{R}$ has a **continuous** distribution function $F(x) = \mu(-\infty, x]$: the function is monotone and right continuous in any case, and its jump at a point is exactly the mass of that point, which is zero. Being continuous and running from 0 at $-\infty$ to 1 at $+\infty$, it attains every intermediate level, and the sublevel set $\{F \le t\}$ then has mass exactly t for $0 \le t < 1$ — it contains a half line $(-\infty, x]$ with $F(x) = t$ and is contained in every half line $(-\infty, y]$ with $F(y) > t$. Comparing the two measures on the half lines, which generate the Borel sets, gives the statement in closed form: F pushes μ forward to Lebesgue measure on the unit interval. So every diffuse probability measure on the line is a copy of the uniform measure, read through its own distribution function — the $L^\infty[0,1]$ entry of the classification list.

```
#check @BookProof.ChapterDiffuseCdfModel.continuous_cdf_of_noAtoms
#check @BookProof.ChapterDiffuseCdfModel.exists_cdf_eq
#check @BookProof.ChapterDiffuseCdfModel.measure_cdf_le
#check @BookProof.ChapterDiffuseCdfModel.map_cdf_eq_volume_Icc
```

That is a statement about measures; the classification list makes a statement about operators, and the two are separated by exactly one step. Composing with F is an isometry of L^2 of the uniform measure into $L^2(\mu)$, because F is measure preserving. It is **onto** because the sets that agree, up to a μ-null set, with a preimage under F form an algebra of sets, and that algebra contains every half line — the half line $(-\infty, x]$ differs from $\{F \le F(x)\}$ by a null set, since the two have the same mass. An algebra of sets that generates the Borel sets is measure dense, so the indicator of **any** measurable set is an L^2 limit of indicators of such sets, hence lies in the

range; the range is closed and contains all indicators, and indicators span a dense subspace of L^2, so the range is everything. Composition with F is therefore a unitary, and it carries multiplication by an essentially bounded symbol g to multiplication by $g \circ F$. The multiplication algebra of an arbitrary diffuse probability measure on the line is thus unitarily the multiplication algebra of the unit interval.

```
#check @BookProof.ChapterDiffuseUnitaryModel.measurePreserving_cdf
#check @BookProof.ChapterDiffuseUnitaryModel.measureDense_cdfAlgebra
#check @BookProof.ChapterDiffuseUnitaryModel.cdfRange_eq_top
#check @BookProof.ChapterDiffuseUnitaryModel.cdfUnitary_intertwines
#check @BookProof.ChapterDiffuseUnitaryModel.diffuse_multiplication_model_uniform
```

The atomic alternative is the other standard type, and it is settled by the same kind of argument in a much simpler setting. If μ is carried by its atoms, the normalised indicators $\delta_a/\sqrt{\mu\{a\}}$ of the atoms are an orthonormal family in $L^2(\mu)$, and nothing is orthogonal to all of them: a function orthogonal to the indicator of an atom vanishes at that atom, and a function vanishing at every atom vanishes almost everywhere, because the complement of the atoms is null. So the normalised point masses are a Hilbert basis indexed by the atoms, and multiplication by a symbol g is **diagonal** in that basis, scaling the basis vector at a by $g(a)$. An atomic summand of the model is therefore a diagonal algebra whose size is the number of atoms — the I_n and $\ell^\infty(\mathbb{N})$ entries of the classification list.

```
#check @BookProof.ChapterAtomicDiagonalModel.orthonormal_atomVec
#check @BookProof.ChapterAtomicDiagonalModel.atomBasis
#check @BookProof.ChapterAtomicDiagonalModel.multOp_atomVec
#check @BookProof.ChapterAtomicDiagonalModel.atomic_multiplication_model_diagonal
```

The two models are glued together by the Hilbert-space counterpart of the splitting of the measure. For a measurable set A, extension by zero embeds $L^2(\mu|_A)$ isometrically into $L^2(\mu)$, the embeddings along A and its complement have orthogonal ranges, and every vector is the sum of its two pieces; so $L^2(\mu)$ is the Hilbert sum of $L^2(\mu|_A)$ and $L^2(\mu|_{A^c})$, and the embeddings intertwine the multiplication operators — the splitting is a splitting of the algebra, not just of the space.

```
#check @BookProof.ChapterLpRestrictSplit.restrictEmbed_add_restrictEmbed_compl
#check @BookProof.ChapterLpRestrictSplit.isHilbertSum_splitEmbed
#check @BookProof.ChapterLpRestrictSplit.restrictEmbed_intertwines
```

One mismatch remains: the diffuse model is stated for a probability measure, while the diffuse part of a mixed measure has some intermediate mass. Rescaling a measure by a nonzero finite constant changes the L^2 norm by a constant factor and nothing else, so a scalar multiple of the identity is a unitary between the two L^2 spaces, and it carries multiplication by a symbol to multiplication by the same symbol. Total mass is therefore invisible to the multiplication algebra.

```
#check @BookProof.ChapterLpScaleMeasure.scaleUnitary_intertwines
#check @BookProof.ChapterLpScaleMeasure.normalized_multiplication_model
```

Assembling the pieces gives the classification list itself. Let μ be a Borel probability measure on the line and S its (countable) set of atoms. Then $L^2(\mu)$ is the Hilbert sum of the atomic piece $L^2(\mu|_S)$ and the diffuse piece $L^2(\mu|_{S^c})$; multiplication is diagonal on the first, in the basis of normalised point masses; and the second, when it is nonzero, is unitarily $L^2[0,1]$, with multiplication by g becoming multiplication by $g \circ F$ for the distribution function F of the normalised diffuse part. Splitting on whether S carries all, none or part of the mass, and on whether S is finite or infinite, gives exactly the manuscript's five types: I_n, $\ell^\infty(\mathbb{N})$, $L^\infty[0,1]$, $L^\infty[0,1] \oplus I_n$ and $L^\infty[0,1] \oplus \ell^\infty(\mathbb{N})$.

```
#check @BookProof.ChapterAbelianClassificationList.diffuse_finite_multiplication_mode
#check @BookProof.ChapterAbelianClassificationList.abelian_summand_standard_model
#check @BookProof.ChapterAbelianClassificationList.vonNeumann_abelian_classification_
```

The list above classifies a **summand** — a measure on the line — and the last step is to get an arbitrary summand there. That is the Borel isomorphism theorem: an uncountable standard Borel space is Borel isomorphic to $\mathbb{R}$. A measurable equivalence is measure preserving onto the pushforward measure in both directions, so composition with it is a **unitary** between the two L^2 spaces, and it turns multiplication by g into multiplication by $g \circ e$. A countable space needs no transport at all: there every measure is carried by its atoms, and multiplication is

already diagonal. So a Borel probability measure on **any** standard Borel space realises one of the five standard types, and — for a compact metrizable spectrum — every summand of the general abelian model does.

```
#check @BookProof.ChapterStandardBorelClassification.transportUnitary_intertwines
#check @BookProof.ChapterStandardBorelClassification.purelyAtomic_of_countable
#check @BookProof.ChapterStandardBorelClassification.standardBorel_classification_lis
#check @BookProof.ChapterStandardBorelClassification.abelian_multiplication_model_cla
```

For a **normal operator** the metrizability hypothesis is automatic, because its spectrum is a compact subset of $\mathbb{C}$. So the classification is unconditional there: every normal operator on a complex Hilbert space is a direct sum of multiplication operators, each summand of which realises one of the five standard types.

```
#check @BookProof.ChapterStandardBorelClassification.spectral_multiplication_model_cl
```

The metrizability hypothesis in the general representation-theoretic form is not a topological accident either: for a compact Hausdorff spectrum Y it is **equivalent** to separability of the algebra $C(Y, \mathbb{C})$. In one direction, a countable dense family of continuous functions separates the points of Y — two points it does not separate are separated by no continuous function at all, and Urysohn's lemma separates distinct points of a compact Hausdorff space — so the family embeds Y into a countable power of $\mathbb{C}$, a continuous injection out of a compact space into a Hausdorff space being an embedding. In the other direction, a compact metrizable space is second countable and its algebra of continuous functions is separable.

```
#check @BookProof.ChapterSeparableSpectrum.metrizableSpace_of_separable_continuousMap
#check @BookProof.ChapterSeparableSpectrum.separableSpace_continuousMap_of_metrizable
#check @BookProof.ChapterSeparableSpectrum.metrizableSpace_iff_separableSpace_continu
```

Through Gelfand duality this discharges the hypothesis in the standard setting. The Gelfand transform is an isometric $*$-isomorphism of a commutative unital C*-algebra A with $C(\Omega(A), \mathbb{C})$, so if A is separable then so is that algebra, and the character space $\Omega(A)$ is metrizable — a compact metric, hence standard Borel, space. Consequently every unital $*$-representation of a **separable** commutative unital C*-algebra on a complex Hilbert space is a direct sum of multiplication algebras, each of which realises one of the five standard types, with no hypothesis on the spectrum at all.

```
#check @BookProof.ChapterSeparableSpectrum.metrizableSpace_characterSpace
#check @BookProof.ChapterSeparableSpectrum.abelian_multiplication_model_classified_se
#check @BookProof.ChapterSeparableSpectrum.abelian_algebra_multiplication_model_class
```

The remaining case — an algebra that is not norm-separable, for instance $L^\infty[0,1]$ acting on $L^2[0,1]$ — is settled by looking at the Hilbert space instead of at the algebra. If the algebra acts on a **separable** Hilbert space then each summand $L^2(\mu_x)$ is separable, so a **countable** family D of continuous functions is already dense in it. Evaluating that family, $y \mapsto (f(y))_{f \in D}$, maps the spectrum into the countable power $\mathbb{C}^D$, a Polish and hence standard Borel space. Composition with the evaluation map is isometric, its range is closed and contains the dense family, so it is a **unitary** onto $L^2(\mu_x)$, and it turns multiplication by g into multiplication by g composed with the evaluation. The summand is therefore unitarily a Borel probability measure on a standard Borel space, and the classification applies to it.

```
#check @BookProof.ChapterSeparableL2Model.exists_countable_dense_continuous
#check @BookProof.ChapterSeparableL2Model.separable_Lp_realizes_standard_type
#check @BookProof.ChapterSeparableL2Model.abelian_multiplication_model_classified_sep
#check @BookProof.ChapterSeparableL2Model.abelian_algebra_multiplication_model_classi
```

So for a separably acting abelian algebra the classification is unconditional: no metrizability of the spectrum, and no separability of the algebra, has to be assumed.

The same statement is packaged, against the manuscript's five-type list, in `BookProof.ChapterSelectingEvents`:

```
#check @BookProof.ChapterSelectingEvents.vonNeumann_abelian_classification_typeI
```

52.9. Continuous Priors Are Out of Reach of Discrete Ones

The splitting also settles the manuscript's "worst-case versus best-case prior" argument. Rescaling the continuous part of a mixed prior (conditioning on the complement of its atoms) yields a genuine **continuous** probability measure, and no purely atomic prior can reproduce it:

```
#check @BookProof.ChapterMixedPrior.atomless_prior_not_purelyAtomic
#check @BookProof.ChapterMixedPrior.noAtoms_normalizedContinuousPart
#check @BookProof.ChapterMixedPrior.exists_continuous_prior_beyond_atomic
```

53. Cosmological Amplification of the Matter/Radiation Ratio

53.1. Orientation and Status

The formal result in this chapter is the FRW scaling calculation: matter and radiation densities obey the stated power laws and their ratio grows linearly with the scale factor. The origin and observed magnitude of the baryon asymmetry are physical model questions, not consequences of this calculation alone.

In the tour this is the part's cosmological application — the one place the arrow-of-time part connects to the gravity thread, via the Friedmann–Robertson–Walker scaling that the companion numerical validation checks. The chapter assumes the FLRW terminology of the gravity chapter. The route: a small asymmetry, amplified; the FRW scaling of the densities; the ratio grows like the scale factor; how this amplifies a small asymmetry; the same scaling checked in the numerical validation; and the closing caution that amplification is not an explanation of the origin.

53.2. A Small Asymmetry, Amplified

In its chapter on entropy and irreversible time-evolution, the manuscript turns to **baryon asymmetry**: why the observable universe contains matter but almost no antimatter. The physical argument is that a tiny CP-violating asymmetry in the early universe is "much amplified by the expansion of the Universe." This chapter isolates the clean, self-contained mathematical core of that amplification: the classical Friedmann–Robertson–Walker (FRW) scaling of perfect-fluid energy densities, and the fact that the **matter-to-radiation ratio grows in proportion to the scale of the Universe**.

53.3. FRW Scaling of the Densities

Let $a > 0$ be the cosmological **scale factor** (the relative size of the Universe). A perfect fluid with equation-of-state parameter w has an energy density that dilutes as the Universe expands. Matter (pressureless dust, $w = 0$) dilutes as the volume, a^3; radiation ($w = 1/3$) dilutes one power faster, a^4, because each photon is additionally redshifted. Concretely, from present-day reference densities ρ_{m0}, ρ_{r0}:

$$\rho_m(a) = \frac{\rho_{m0}}{a^3}, \qquad \rho_r(a) = \frac{\rho_{r0}}{a^4}.$$

These power laws are not arbitrary: they are **forced** by the FRW continuity equation

$$a\,\rho'(a) + 3(1+w)\,\rho(a) = 0,$$

whose solution is $\rho \propto a^{-3(1+w)}$, giving the exponents 3 (matter) and 4 (radiation). The verified computation (a genuine derivative calculation):

```
#check @ChapterBaryonAsymmetry.matter_satisfies_continuity
#check @ChapterBaryonAsymmetry.radiation_satisfies_continuity
```

53.4. The Ratio Grows Like the Scale Factor

The matter-to-radiation **ratio** is therefore

$$\frac{\rho_m(a)}{\rho_r(a)} = \frac{\rho_{m0}/a^3}{\rho_{r0}/a^4} = \frac{\rho_{m0}}{\rho_{r0}}\, a.$$

The a^3 dilution common to both cancels, leaving a single factor of a:

```
#check @ChapterBaryonAsymmetry.matterRadiationRatio_eq
```

So the ratio is **proportional to the scale of the Universe**, exactly as the manuscript states. It is strictly increasing in a, and it diverges:

```
#check @ChapterBaryonAsymmetry.matterRadiationRatio_strictMonoOn
#check @ChapterBaryonAsymmetry.matterRadiationRatio_tendsto_atTop
```

The last is the headline: as $a \to +\infty$, the matter/radiation ratio tends to $+\infty$. This is the precise sense in which the expansion "amplifies" the matter content relative to radiation.

53.5. How This Amplifies a Small Asymmetry

The physical conclusion the manuscript draws is now transparent. Suppose the early universe has a tiny excess of matter over antimatter (a small CP asymmetry). As the universe expands, the matter/radiation ratio grows **linearly in the scale factor** and without bound. A small initial imbalance in the matter sector is therefore magnified by the enormous growth of a between the early universe and today. The mathematics does not by itself explain the *origin* of the asymmetry (that is the role of CP violation); it explains why a small asymmetry, once present, becomes the dominant, observable matter content — the expansion of the Universe does the amplifying.

53.6. The Same Scaling Is Checked in the Numerical Validation

The power laws above are not an isolated curiosity of this chapter: they are the same Friedmann–Robertson–Walker content that the companion numerical validation checks against published cosmology. For a flat FLRW universe the two scalars that appear throughout the gravity thread — the Ricci scalar $R = 6(\dot{H} + H^2)$ and the TEGR torsion scalar $T = -6H^2$ — are built from the same $H = \dot{a}/a$ and $\dot{H}$ that enter the continuity equation $a\,\rho'(a) + 3(1+w)\rho(a) = 0$ derived above (with matter, $w = 0$, giving $\rho \propto a^{-3}$). The numerical suite verifies both scalars, that they give the same Friedmann equation $3H^2 = 8\pi G\rho$ (the TEGR–GR equivalence, $eR = e \cdot T + \text{divergence}$), and that their sum reproduces the divergence term — the same $R + T = 6\dot{H}$ identity that makes $R = 6(\dot{H} + H^2)$ and $T = -6H^2$ consistent. So the amplification computed here and the equivalence checked there rest on the same FLRW scalars; the cosmological content of the manuscript is verified both symbolically (this chapter) and numerically (the companion solver work).

53.7. A Caution: Amplification Is Not Explanation of the Origin

It is worth keeping the two claims separate. What is proved here — and what the manuscript's argument needs — is that **if** a small asymmetry exists, the expansion amplifies it proportionally to a. That is a clean, verified statement about FRW kinematics. What is **not** supplied by this chapter is the origin of the asymmetry itself: the manuscript turns to CP violation in the Standard Model for that, and notes that the Standard Model's CP asymmetry may be too small to match observation unless the age of the Universe is larger than standard cosmology provides — an inconsistency the manuscript reads as a problem for the cosmology model rather than for the Standard Model. The verified content stops at the amplification; the origin story is physics beyond it.

54. Measurements Reproduce the Distribution

54.1. Orientation and Status

The source manuscript uses repeated measurements as an operational interpretation of probability. The statements cited below are the formal finite/abstract convergence core; they should not be read as a full independent identically distributed law of large numbers unless the theorem signature supplies those hypotheses explicitly. The distinction matters when measurements are dependent or have time-varying laws.

In the tour this chapter closes the part by giving the probabilities their empirical content — frequencies converge to probabilities — and connects back to the law of total variance in the probability part. The route: frequencies converge to probabilities, the setup, the convergence itself, the connection to the variance decomposition, and why this closes the loop.

54.2. Frequencies Converge to Probabilities

A probability distribution makes a testable promise: if you repeat an experiment many times, the **relative frequency** of each outcome converges to its probability. This is the **law of large numbers**, and it is the bridge between the abstract measure of Part I and the empirical content of the whole programme. The manuscript relies on it in its discussion of **ensemble forecasting** (§10): repeated measurements reproduce the probability distribution in the infinite-measurement limit.

54.3. The Setup

Model a sequence of measurements as functions $M_i : \Omega \to \{0, \dots, k-1\}$ on a probability space (Ω, μ), each returning one of k outcomes. For an outcome a, the **indicator** $\mathbf{1}_{M_i = a}$ is the random variable that is 1 when the i-th measurement yields a and 0 otherwise. Its expectation is exactly the probability of a:

$$\int_\Omega \mathbf{1}_{M_i = a} \, d\mu = \mu(M_i = a).$$

`#check @ChapterMeasurementLLN.outcomeIndicator_integral`

54.4. The Frequency Converges

The empirical **frequency** of outcome a in the first N measurements is the average of the indicators,

$$f_N(a) = \frac{1}{N} \sum_{i < N} \mathbf{1}_{M_i = a},$$

and the law of large numbers says it converges (in the appropriate sense) to the probability $\mu(M_i = a)$:

`#check @ChapterMeasurementLLN.measurement_frequency_tendsto`

More generally, the empirical average of **any** bounded function f of the outcome converges to its expectation:

`#check @ChapterMeasurementLLN.measurement_average_tendsto`

54.5. Connection to the Variance Decomposition

This is where the law of total variance pays off. Recall the decomposition $\mathrm{Var}[Y] = \mathrm{within} + \mathrm{between}$. The **epistemic** (between-group) part of the uncertainty is precisely what repeated measurement reduces: as the number of measurements grows, the empirical frequency concentrates around the true probability, and the variance of the estimator shrinks. The **aleatoric** (within-group) part — the irreducible randomness of each individual outcome — does not disappear; it is what the limiting distribution still describes.

So the law of large numbers and the law of total variance are two views of the same story: the former says the empirical distribution converges to the true one, and the latter quantifies how the residual uncertainty splits into a part that averaging removes and a part it cannot.

54.6. Why This Closes the Loop

The Born-rule parametrization of Part II assigns probabilities $p_k = |\psi_k|^2$ to outcomes. The law of large numbers is the operational meaning of those numbers: they are exactly the limiting relative frequencies of repeated preparation-and-measurement. A wave-function is therefore not a mysterious object; it is a **parametrization of the long-run statistics** of an experiment, and the surjectivity of the Born map says that every conceivable set of long-

run statistics is reachable by some wave-function. The probabilistic foundation of Part I, the wave-function parametrization of Part II, and the frequentist interpretation meet here.

55. The ODE Singularity

An operator-theoretic resolution of the finite-time blow-up of $x' = x^2$, replacing the ODE chapter of the manuscript.

How to read this part. A single self-contained chapter, followed by the proof-plan appendix. It is the book's most explicit exercise in **honesty about verification**: read the section "What Is Verified, and What Is Open" carefully — the repository's `Singularity` library is `sorry`-free, but it splits into genuine theorems (the complexification resolution, the spectral energy bound) and algebraic certificates (the flow-completeness flags) that must not be read as analytic results. The chapter is also the conceptual bridge to the numerical SIRK solver work: the detection pipeline and the change-of-variables strategy it describes are the same ideas the companion solver implements when it evolves singular-looking systems.

56. Resolution of the Singularity of an ODE

56.1. Orientation and Status

The single chapter of the ODE part, and the book's most explicit exercise in honesty about verification: read its "What Is Verified, and What Is Open" section carefully — the `Singularity` library is `sorry`-free, but it splits into genuine theorems (the complexification resolution, the spectral energy bound) and algebraic certificates (the flow-completeness flags) that must not be read as analytic results. It is also the conceptual bridge to the numerical SIRK solver work: the detection pipeline and the change-of-variables strategy described here are the ideas the companion solver implements.

The route: the blow-up of $x' = x^2$, the central idea (admit uncertainty in the initial data), Koopman–von Neumann mechanics, Weyl quantization (from ODE to Hamiltonian), Nelson's theorem and the x^2 Hamiltonian, the resolution by complexification, the verified-versus-open account, and the algorithmic side (singularity detection, coordinate transformations, the full analysis pipeline, validation cases) with its price: non-determinism.

The operator core rests on the Stone chapters (`BookProof/ChapterStoneBridge`, `ChapterStoneFlows`, `ChapterStoneResolvent`, `ChapterStoneSeparable`).

This chapter replaces the ODE chapter of the source manuscript. It follows the expanded, operator-theoretic treatment of `ODE.tex`, and is honest about which parts are formally verified and which remain open (see the proof-plan appendix).

56.2. The Blow-Up

Consider the scalar autonomous ODE

$$\dot{x} = x^2, \qquad x(0) = x_0 \in \mathbb{R}.$$

Its unique maximal solution is

$$x(t) = \frac{x_0}{1 - t\,x_0},$$

defined only on $(-\infty, 1/x_0)$ when $x_0 > 0$. As $t \to 1/x_0$, the solution **blows up**: $|x(t)| \to \infty$. There is no global, deterministic solution defined for all time.

This finite-time blow-up creates two problems:

- **Undetectability** — if a numerical integrator fails to converge, one cannot tell a genuine singularity from insufficient resolution.

- **Degeneracy** — the blow-up allows unrelated initial conditions to be glued before and after the singular time, destroying uniqueness of the initial-value problem.

The blow-up time is a concrete, verifiable quantity. In the repository's `Singularity` library (module `Singularity.Singularity`), the blow-up time of $\dot{x} = x^2$ is computed:

```
#check @blowupTime_x_sq
```

56.3. The Central Idea: Admit Uncertainty

The manuscript's central claim is that **both problems are resolved by admitting a finite amount of uncertainty in the initial condition**. Instead of a point $x_0 \in \mathbb{R}$, take a probability measure μ_0 on $\mathbb{R}$ — equivalently, by the Born parametrization of Part II, a wave-function $\psi_0 \in L^2(\mathbb{R})$ with $|\psi_0|^2 = \mu_0$ — and evolve it under a **unitary** group. The resulting evolution is globally defined in time, non-deterministic in the coordinate x, and reduces to the classical solution in the zero-uncertainty limit.

This is the same philosophy as the rest of the book: replace a point of the probability space by a wave-function, and gain structure (here, global unitary evolution) in return.

Honesty flag. The manuscript itself concedes that the two problems are not resolved equally well: the method solves the first problem (telling a genuine singularity from insufficient numerical resolution), but it is **not completely satisfactory** for the second one. The degeneracy is removed by selecting the continuation that is compatible with analytic continuation, which is the only physically viable choice in the manuscript's setting but does not give access to the other, unrelated gluings that may be the relevant ones in other contexts.

56.4. Koopman–von Neumann: Classical Mechanics as Quantum Mechanics

Given a sample space $\Omega = \mathbb{R}$ with a probability measure μ, one may always define a wave-function $\psi = \sqrt{d\mu/dx}$ (up to a phase). The **Koopman–von Neumann** formulation recasts classical statistical mechanics as a special case of quantum mechanics in which the algebra of observables is commutative (because the underlying time-evolution is deterministic).

Since $\mathbb{R}$ is a simply connected Lie group under addition, Bargmann's theorem lifts any strongly-continuous projective unitary representation to a genuine unitary representation $U(t)$, and **Stone's theorem** then gives a unique self-adjoint **Hamiltonian** H with

$$U(t) = e^{-iHt}.$$

So the program is: turn the ODE into a Hamiltonian H, and the (possibly singular) classical flow into a (globally-defined) unitary group e^{-iHt}.

In this repository Stone's theorem is not invoked as a black box but proved: `BookProof.ChapterStoneResolvent` through `BookProof.ChapterStoneSeparable` construct the unitary group `e^{-itA}` from a self-adjoint generator `A` (`stoneU`, `stoneU_add`, `hasDerivAt_stoneU`) and prove the converse bijection (`stone_bijection`, `stoneEquiv`), while `BookProof.ChapterStoneBridge` packages the outcome as `IsStoneFlow` and `BookProof.ChapterStoneFlows` applies it to the concrete Navier–Stokes and QYM Hamiltonians (`ns_stone_flow`, `lagrangian_stone_flow`, `ym_fock_stone_flow`). The direction used here — from a unitary group back to its Hamiltonian — is exactly `stone_bijection`.

56.5. Weyl Quantization: From ODE to Hamiltonian

The passage from the classical vector field $\dot{x} = f(x)$ to a Hamiltonian is **Weyl quantization**. Because x and the momentum $\hat{p} = -i\,\partial_x$ do not commute, the classical product $f(x)\,p$ must be **symmetrized**. For $\dot{x} = x^2$ the (formal) Hamiltonian is

$$H = x^2\hat{p} - i\,\hat{x},$$

where the $-i\hat{x}$ term is the symmetrization correction. The repository defines this Weyl-symmetrized Hamiltonian on a normal-ordered operator algebra (module `Singularity.Hamiltonian` and `Singularity.Poly`):

```
#check @odeToHamiltonian
```

The normal-ordered operator algebra (module `Singularity.Poly`) represents an operator on M bosonic modes as a finitely-supported function recording creation/annihilation counts per mode, and implements the **Wick recursion** for multiplication. This part of the formalization is genuine and `sorry`-free.

What Weyl quantization does and does not give. The construction yields a **formal operator**: a normal-ordered expression that is **Hermitian** (equal to its formal algebraic adjoint). It does **not** by itself establish essential self-adjointness on L^2 — that is a separate, analytic question governed by Nelson's theorem (below) and by the completeness of the classical flow. Confusing the two is the source of the manuscript's error, which we correct in the next section.

56.6. Nelson's Theorem, Essential Self-Adjointness, and the x² Hamiltonian

For the unitary evolution e^{-iHt} to exist globally, the Hamiltonian must be **essentially self-adjoint** (its closure must be self-adjoint, so that the spectral theorem applies). **Nelson's theorem** gives a criterion: a symmetric operator is essentially self-adjoint on a domain if it has a dense set of **analytic vectors**. For an ODE-derived Hamiltonian, the relevant analytic vectors are tied to the **completeness of the classical flow**: if the classical flow exists for all time, the Hamiltonian is essentially self-adjoint, and conversely.

The repository states this connection (module `Singularity.Esa`):

```
#check @nelson_essential_self_adjoint
```

`nelson_essential_self_adjoint` records the Nelson correspondence at the **algebraic certificate layer**: it states that vanishing deficiency indices are equivalent to completeness of the represented flow, proved by `simp` from the certificate definitions. As we flag in "What Is Verified, and What Is Open" below, both sides of that equivalence are currently placeholders (`isComplete := true` and `deficiencyIndices := (0,0)` by definition), so this is a genuine theorem **about the certificate interface**, not yet an analytic proof of essential self-adjointness. A future analytic realization on a Hilbert space will refine the certificates.

The manuscript's claim, and the correction to its proof. The manuscript (book.tex) asserts that for $\dot{x} = x^2$ the Hamiltonian $H = x^2\hat{p} - i\,\hat{x}$ on $C_c^\infty(\mathbb{R})$ "can be proved to be essentially self-adjoint", citing a positive-auxiliary-operator argument (H^2 in a corollary). The **conclusion** — that H can be made essentially self-adjoint — is not unreasonable; the **justification** offered is wrong, and the correct conditions are different. The claimed proof does not go through: the classical flow $\Phi_t(x_0) = x_0/(1 - tx_0)$ is defined only for $t < 1/x_0$ (when $x_0 > 0$): it is **incomplete**. By Nelson's theorem the deficiency indices do not both vanish, so H is **not** essentially self-adjoint on $L^2(\mathbb{R})$ as it stands. The positive-auxiliary-operator method proves essential self-adjointness only for operators whose classical flow **is** complete (e.g. polynomial Hamiltonians of degree ≤ 1, or the harmonic oscillator); the positivity hypothesis the corollary requires is not satisfied by H^2 on the full domain $C_c^\infty(\mathbb{R})$.

The correct route to the manuscript's conclusion is by **completing the flow** — the operator is made essentially self-adjoint on a suitable extension, not left on $L^2(\mathbb{R})$ as is. This is precisely what the `Singularity` library and its plan (`SINGULARITY_DETECTION_PLAN.md`) implement (see "Coordinate Transformations" and "The Full Analysis Pipeline" below): apply a change of variables that makes the flow complete, quantize the transformed vector field, and obtain an operator that **is** ESA, whose unitary evolution is well defined and which recovers the original singular dynamics in a limit. So the claim is reasonable; the difference from what book.tex suggests is in the proof and the conditions under which it holds.

56.7. The Resolution: Complexification

The cleanest mathematical argument that the singularity disappears is **complexification**. Pass from $L^2(\mathbb{R})$ to $L^2(\mathbb{R}^2)$ and the real variable $x(t)$ to a complex one $z(t) = x(t) + iy(t)$ satisfying $\dot{z} = z^2$. The solution is

$$z(t) = \frac{z(0)}{1 - t\,z(0)},$$

with a singularity when $1 - t\,z(0) = 0$, i.e. $t = 1/z(0)$. For this singular time to be **real**, we need $\mathrm{Im}(1/z(0)) = 0$, which forces $y(0) = 0$. But the line $y(0) = 0$ is a **null-measure subset** of $\mathbb{R}^2$ (this is exactly the measure-theoretic phenomenon of Null-measure sets need not be small). Therefore, for **almost every** initial condition in $L^2(\mathbb{R}^2)$, the singular time is non-real, and **there is no finite-time singularity**.

Moreover, the imaginary part $y(0)$ can be concentrated arbitrarily close to zero, so the unitary solution in $L^2(\mathbb{R}^2)$ recovers the (non-unitary, isometric) solution in $L^2(\mathbb{R})$ in the limit $y(0) \to 0$. Changing the sample space effectively changes the equation: $y(0) = 0$ strictly is not achievable in $L^2(\mathbb{R}^2)$.

A complementary argument uses **energy-bounded initial conditions**: a finite-time singularity would make the time-derivative of the wave-function diverge; but the time-derivative corresponds to the Hamiltonian, whose spectral measure is conserved by unitary evolution, so an initial condition using only eigenfunctions below some $E_{\max}$ cannot produce a divergent time-derivative.

That complementary argument is now proved, in the spectral representation where the Hamiltonian is multiplication by its (real) eigenvalue function, in `BookProof/ChapterSpectralEnergyBound.lean`: the evolution is unitary (`norm_evolve`), it satisfies the Schrödinger equation $\psi'(t) = -iH\psi(t)$ (`hasDerivAt_evolve`), an initial state supported below $E_{\max}$ obeys the spectral bound $\|H\psi\| \le E_{\max}\|\psi\|$ (`norm_diagOp_le`), so the time-derivative is bounded uniformly in time (`norm_deriv_evolve_le`) and the solution is globally Lipschitz — there is no finite-time singularity (`evolve_lipschitz`).

```
#check @ChapterSpectralEnergyBound.norm_diagOp_le
#check @ChapterSpectralEnergyBound.norm_evolve
#check @ChapterSpectralEnergyBound.hasDerivAt_evolve
#check @ChapterSpectralEnergyBound.norm_deriv_evolve_le
#check @ChapterSpectralEnergyBound.evolve_lipschitz
```

Formalization. The complexification resolution is formalized in `BookProof/ChapterOdeComplexification.lean`:

```
#check @ChapterOdeComplexification.ae_no_real_singular_time
```

`ae_no_real_singular_time` proves that for almost every initial condition in $L^2(\mathbb{R}^2)$, the singular time of $\dot{z} = z^2$ is non-real, so there is no finite-time singularity. The proof uses `MeasureTheory.Measure.addHaar_submodule` to show the real axis has Lebesgue measure zero in $\mathbb{R}^2$.

56.8. What Is Verified, and What Is Open

The repository's `Singularity` library is `sorry`-free, but its content splits into genuine theorems and **algebraic certificates** that must not be read as analytic results. We keep the two sharply separate.

Genuinely proved

The normal-ordered operator algebra and Wick recursion (`Singularity.Poly`); the formal Hermitian symmetry of the Weyl Hamiltonian (`weyl_symmetrization_self_adjoint` in `Singularity/Hamiltonian.lean`, asserting `adj H = H` at the algebraic level); the explicit blow-up time of x^2 (`blowupTime_x_sq` in `Singularity.Singularity`); and — analytically — the complexification resolution (`ae_no_real_singular_time` in `BookProof/ChapterOdeComplexification.lean`) and the spectral energy-bound (`ChapterSpectralEnergyBound`), both in `BookProof`.

```
#check @weyl_symmetrization_self_adjoint
#check @blowupTime_x_sq
#check @ChapterOdeComplexification.ae_no_real_singular_time
```

Algebraic certificates, not analytic theorems

`Singularity.Flow` and `Singularity.Esa` build a **certificate layer** in which the flow-completeness flag and the deficiency indices are finite placeholders: `analyzeClassicalFlow` returns `isComplete := true` by definition, and `deficiencyIndices` returns `(0, 0)` by definition. Consequently

`nelson_essential_self_adjoint` (`isEssentiallySelfAdjoint H ↔ analyzeClassicalFlow … .isComplete`) is a tautology of these definitions, proved by `simp`, and the "completeness" theorems `blowup_criterion_scalar`, `linear_flow_complete`, and `even_degree_monomial_flow_complete` assert only that a constant flag is `true`. They do **not** establish that any classical flow is complete, and they do **not** establish that $H = x^2\hat{p} - i\hat{x}$ is essentially self-adjoint — indeed for $\dot{x} = x^2$ it is **not** (see Nelson's theorem above). These certificates are an interface for a future analytic realization; they are **not** a proof of the analytic claims, which remain open in the verified layer and are recorded as such in the proof-plan appendix.

```
#check @analyzeClassicalFlow
#check @deficiencyIndices
#check @nelson_essential_self_adjoint
#check @blowup_criterion_scalar
#check @linear_flow_complete
#check @even_degree_monomial_flow_complete
```

56.9. Algorithmic Singularity Detection

The manuscript (ODE.tex §6) separates the **detection** of a singularity from its **resolution**. Detection works on the classical flow. The intended core is `analyzeClassicalFlow` (module `Singularity.Flow`), which integrates the vector field and records whether a trajectory escapes a numerical window; the type `FlowAnalysis` carries an `isComplete` flag and a list of `EscapeEvents` (indexing the initial condition, the blow-up time, and the divergent axes):

```
#check @Singularity.Flow.FlowAnalysis
#check @Singularity.Flow.EscapeEvent
```

The **interface** is real, but as noted above the present `analyzeClassicalFlow` implementation is a certificate placeholder (`isComplete := true` by definition), so the detection it performs is not yet a theorem about maximal ODE solutions — the numerical escape experiments it is meant to host belong to a separate approximation layer. We record the interface and its intended meaning, and flag the gap honestly.

For a scalar polynomial right-hand side the blow-up time has an exact quadrature: $T(x_0) = \int_{x_0}^{\infty} dx/f(x)$. The library provides the symbolic `blowup_time_integral` (module `Singularity.Integration`), which evaluates this antiderivative exactly for the monomial (power-law) case and falls back to numerical integration otherwise:

```
#check @Singularity.Integration.blowup_time_integral
```

The detection pipeline assigns a **diagnostic code** categorizing the failure mode (module `Singularity.Integration`): `odeNotEssentiallySelfAdjoint` (flow incomplete and no change of variables applied), `odeSingularityDetected` (blow-up found within the time horizon), `odeCovApplied` (a coordinate transformation stabilized the Hamiltonian), `odeDeficiencyIndices` (a reduced 1D flow has non-vanishing deficiency indices), and `odePolynomialTooLarge` (the normal-ordered degree exceeds the explosion bound):

```
#check @Singularity.Integration.UKDiagnosticCode
```

56.10. Coordinate Transformations

When the flow is singular, the manuscript's strategy is to **change coordinates** (ODE.tex §7) so the transformed flow is tame, then quantize the new vector field. The library records the elementary transformations (module `Singularity.ChangeOfVars`): the **reciprocal** map $x \mapsto 1/x$ (with vector-field pushforward $\dot{w} = -f(1/w)/w^2$), the **logarithmic** map $x \mapsto \ln x$ (with $\dot{w} = f(e^w)/e^w$), and the identity:

```
#check @Singularity.ChangeOfVars.applyReciprocalTransform
#check @Singularity.ChangeOfVars.applyLogTransform
#check @Singularity.ChangeOfVars.detectChangeOfVariables
```

The honest caveat is recorded in the definitions themselves: a reciprocal (or logarithmic) transformation of a **polynomial** system generally produces a **rational** right-hand side, which cannot be represented in the polynomial `ODESystem` type without extending it. The conserved detector `detectChangeOfVariables` therefore returns the original system with the identity observable map, and the transformed vector fields are stated at the level of `ℝ → ℝ` functions rather than pretended to be polynomial systems. The CoV type enumerates `reciprocal`, `logarithmic`, and `power`; the observable map `observableMaps : Fin M → (ℝ → ℝ)` records how to pull a measurement back to the original coordinates.

For $\dot{x} = x^2$ the reciprocal change of variables $w = 1/x$ turns the flow into $\dot{w} = -1$, which is complete: the singular point is pushed off to infinity, and the quantized operator on the transformed coordinates **is** essentially self-adjoint. This is the concrete mechanism by which the manuscript's conclusion is recovered — the Hamiltonian is made self-adjoint by completing the flow, not by the auxiliary-operator argument the manuscript cites (see Nelson's section above).

56.11. The Chart Realization: From Claim to Theorem

The sentence above — **the quantized operator on the transformed coordinates is essentially self-adjoint** — was for a long time a plan-level claim pointing at the certificate layer. It is now a theorem in the verified layer, in three checked steps (`BookProof/Ode/ChapterMomentumChartEsa.lean`, `BookProof/Ode/ChapterOdeChartTransfer.lean`), and every step is a reusable instrument rather than a fact about this one ODE.

ODE-C.1 — the momentum of the chart. In the reciprocal coordinate $w = -1/x$ the transformed flow $\dot{w} = -1$ is pure translation, so the quantized evolution of the chart is the translation group $\psi \mapsto \psi(\cdot - t)$ (`BookProof.OdeUnitaryFlow.odeKoop_chartW`), whose generator is the momentum operator $P = -i\,d/dw$ on $L^2(\mathbb{R})$. `BookProof.MomentumChart.momentum_essentiallySelfAdjointOn` proves that P is essentially self-adjoint on the Schwartz core $\mathcal{S}(\mathbb{R})$ — dense, and stable under translation — by the Fourier-multiplier route: under Plancherel, P becomes multiplication by the real symbol $2\pi\xi$, and reality of the symbol is the whole argument. The plan allowed either that route or Nelson's orbit criterion along the translation orbit; the Fourier route was the shorter one, and `chartMomentum_apply` records that the operator really is $-i$ times the derivative.

ODE-C.2 — transporting the theorem through the chart. The change of variables itself, $(W\psi)(x) = \psi(-1/x)/x$, was first an isometry (`chartW_lintegral_normSq`); `BookProof.OdeChartTransfer` upgrades it to a **genuine unitary** W of $L^2(\mathbb{R})$ (`chartUnitary`, with `chartUnitary_apply` and $W^2 = -1$) and proves the generic fact a transfer needs: **unitary conjugation carries essential self-adjointness** (`conj_essentiallySelfAdjointOn` — the `EssentiallySelfAdjointOn` analogue of the densitized transfer theorem used on the gravity side). Applying it to ODE-C.1 gives `chartHam_essentiallySelfAdjointOn`: the operator WPW^{-1} is essentially self-adjoint on the chart domain $\mathcal{D}_{\text{chart}} = W(\mathcal{S}(\mathbb{R}))$, and `chartHam_apply_ae` identifies it there, almost everywhere, with the Weyl Hamiltonian of this chapter: $(\mathrm{chartHam}\,(W\psi))(x) = x^2(-i(W\psi)'(x)) - i\,x(W\psi)(x)$.

The domain is part of the theorem — the correction above still stands. Nothing here says that $H = x^2\hat{p} - i\hat{x}$ is essentially self-adjoint on $C_c^\infty(\mathbb{R})$ or on $L^2(\mathbb{R})$ as it stands; the Nelson-theorem correction in the previous sections is untouched. What is proved is ESA **on the named chart domain**: functions in $W(\mathcal{S}(\mathbb{R}))$ behave like $x^{-1}\psi(-1/x)$ near $x = 0$ and decay like x^{-1} at infinity — exactly the states for which the flow through the blow-up is complete. The manuscript's conclusion is recovered in this form: not by the auxiliary-operator argument it cites, and not on its stated domain, but by completing the flow in coordinates and transporting the theorem back.

ODE-C.3 — the flow. Feeding the chart result into the Stone bridge gives `chartHam_stone_flow`: the quantized vector field of $\dot{x} = x^2$, on its chart domain, generates a complete unitary group — the operator-level face of "the singular classical flow becomes a globally defined unitary evolution". The instrument behind the two transfers is worth naming on its own: `BookProof/Toolbox/ChapterEsaRealShift.lean` proves that a symmetric, essentially self-adjoint operator has **trivial deficiency spaces at every non-real** z (`deficiencyTrivialAt_of_essentiallySelfAdjointOn`, with the disk-stability lemma `deficiencyTrivialAt_of_near`), that essential self-adjointness survives a real shift $H \mapsto H + c$ (`essentiallySelfAdjointOn_add_const`), and that a unitary conjugation transports it

(conj_essentiallySelfAdjointOn above) — three facts the definition at $\pm i$ alone does not supply, and each of them consumed somewhere else in this programme (the NS lifted comparison of the comparison-operator chapter and the assembled gravity Hamiltonian of Diffeomorphisms and gravity both use them).

```
#check @BookProof.MomentumChart.momentum_essentiallySelfAdjointOn
#check @BookProof.MomentumChart.chartMomentum_symmetricOn
#check @BookProof.OdeChartTransfer.chartUnitary
#check @BookProof.OdeChartTransfer.conj_essentiallySelfAdjointOn
#check @BookProof.OdeChartTransfer.chartHam_essentiallySelfAdjointOn
#check @BookProof.OdeChartTransfer.chartHam_apply_ae
#check @BookProof.OdeChartTransfer.chartHam_stone_flow
#check @BookProof.EsaRealShift.deficiencyTrivialAt_of_essentiallySelfAdjointOn
#check @BookProof.EsaRealShift.essentiallySelfAdjointOn_add_const
```

Honesty note. This theorem does not upgrade the `Singularity` certificate layer described above: `analyzeClassicalFlow`'s completeness flag and the `deficiencyIndices` placeholders are still placeholders, and the validation benchmarks that read them are still specifications. What changed is that the analytic statement they point to — ESA of the quantized transformed vector field — now exists in `BookProof`, proved independently of the certificate interface.

56.12. The Full Analysis Pipeline

The pieces assemble into a single `sirk_pipeline` (module `Singularity.Integration`) that, given a system and an initial condition, builds the Weyl Hamiltonian, runs the classical-flow analysis, computes the ESA report, detects a change of variables, and returns a `UKDiagnosticCode × EsaReport`:

```
#check @Singularity.Integration.sirk_pipeline
#check @Singularity.Report.esaReport
#check @Singularity.Report.session_detect_singularity
```

The `EsaReport` (module `Singularity.Esa`) packages the flow-completeness certificate with the deficiency indices, and `isEssentiallySelfAdjoint` decides whether both deficiency indices vanish — the Nelson criterion at the algebraic certificate layer:

```
#check @Singularity.Esa.EsaReport
#check @Singularity.Esa.deficiencyIndices
#check @Singularity.Esa.isEssentiallySelfAdjoint
```

56.13. Validation Cases

ODE.tex §10 and module `Singularity.Tests` validate the pipeline against a suite of benchmarks. The `x' = x^2` scalar case blows up at $T = -1/x_0$ and fails ESA (UK-2101) until the reciprocal change of variables is applied; the coupled system $x' = y, y' = 2xy$ is flow-incomplete; the reduced system `p_x·y + p_z·p_y·y^2` has non-vanishing deficiency indices (UK-2104); the punctured system hits the boundary $y = 0$ (UK-2102); and the stable linear system $x' = -x$ is complete, so ESA holds and the SIRK evolution matches the analytic exponential decay:

```
#check @Singularity.Tests.runTest
#check @Singularity.Tests.TestCase
```

These describe the **intended** behaviour of the pipeline (the `x' = x^2`, coupled, `py2`, punctured, and stable-linear benchmarks). They are specifications, not verified theorems: `runTest` builds the report from the certificate placeholders above, so the ESA/singularity verdicts they record are not yet analytic facts. They are consistent with the corrected mathematics — in particular the `x' = x^2` case is rightly expected to **fail** ESA as it stands on $L^2(\mathbb{R})$ (the flow is incomplete), and to become ESA only after the reciprocal change of variables — but the ESA theorem on the changed variables is now proved there (2026-10-15c, `OdeChartTransfer`), while the SIRK pipeline realization they point to remains open in the verified layer.

56.14. The Price: Non-Determinism

The resolution comes at a cost the manuscript is explicit about: the evolution is deterministic in the coordinate x but **non-deterministic** in the energy-bounded sample space. The Hamiltonian's spectral measure is conserved, but the map from spectral variables to coordinates is non-trivial. This non-determinism is the price paid for removing the singularity — and it is of a piece with the book's recurring theme that passing from a probability distribution to a wave-function trades determinism of the observable for the richer, reversible structure of the wave-function.

Appendix: Proof Plans

This appendix collects the precise statements that are mathematically relevant to the book, separating proved results, conditional bridges, and genuinely open instantiation work. The newest proof wave adds finite Ritz certificates, Temple separation, band enclosure, one-particle/Fock gap lifts, Friedrichs-form gaps, and perturbative interaction stability. It is addressed to an LLM Lean specialist (e.g. Aristotle). A more detailed, machine-oriented version lives in `CONSOLIDATED_PLAN.md` at the repository root.

How to use this appendix. Each entry states the claim, its current status (proved / conditional / open), and the plan for the open parts; the chapter names in the body text point here whenever a claim is not yet a theorem. The plans are the hand-off document for the Lean specialist who carries the next proof wave.

The New Spectral-Gap Proof Layer

The following modules are now part of the proof architecture:

- `ChapterRitzCertificate.lean`: derives a finite spectral interval from a Rayleigh quotient, residual, and explicit Temple separation hypothesis.
- `ChapterTempleSeparationNecessary.lean`: proves that separation is necessary; Rayleigh quotient and residual alone cannot bound the spectral edge.
- `ChapterBandEnclosure.lean`: packages compatible nested finite bands and their intersection/enclosure consequences.
- `ChapterFockOneParticleGap.lean`: states the positive one-particle edge and its conditional consequences.
- `ChapterFriedrichsFormGap.lean`: packages the semibounded quadratic-form and Friedrichs-extension route.
- `ChapterFockNumberPreservingGap.lean`: lifts a one-particle gap through number-preserving second quantization.
- `ChapterFockFieldPerturbation.lean` (now `sorry`-free) and `ChapterFockInteractionStability.lean`: control bounded or relatively form-bounded perturbations, with the unbounded physical interaction explicitly left as a specialist obligation — until the new ladder closed it:
- `ChapterFockPairPerturbation.lean`: the quadratic, pair-creating unbounded coupling `P(f,g) = a†(f)a†(g) + a(g)a(f)` keeps the gap $(\mu - 2\sqrt{2}\|f\|\|g\|)\|u\|^2$ under the smallness condition $2\sqrt{2}\|f\|\|g\| < \mu$.
- `ChapterFockCubicUnbounded.lean`: the boundary of the route — a bare cubic term $(a^\dagger)^3 + a^3$ admits no relative form bound, so $d\Gamma(N) + \lambda C$ is unbounded below at every coupling strength; adding the normal-ordered quartic $(a^\dagger)^2a^2$ restores a lower bound along the same trial family.
- `ChapterYangMillsFockGapChain.lean`: instantiates the abstract chain for the concrete gauge-fixed QYM one-particle operator $H_1 = \tfrac{1}{2}\Sigma\pi^2 + \tfrac{1}{2}\Sigma B^2$ — $d\Gamma(H_1)\Omega = 0$ unconditionally, the $d\Gamma$ lift of a one-particle **form** gap to the nested-Fock mass gap (with the Friedrichs extension), the linear field perturbation, and the certified-band route through `BandEnclosure` — all conditional on the one-particle form gap.
- `ChapterFockCubicQuarticStability.lean`: the positive companion of `ChapterFockCubicUnbounded` — the cubic term $C_k = (a^\dagger)^3 + (a_k)^3$ together with its normal-ordered quartic partner $Q_k = (a^\dagger)^2(a_k)^2$ is bounded below on all finite states (no trial family, no vacuum-orthogonality): the forms are rewritten as norms (`quart_form_eq`, `cubic_form_eq`), combined

with the single-mode canonical commutation relation (`norm_creA_sq`) and a Cauchy–Schwarz estimate (`sq_norm_annA_le_mul`) that forces a large mode occupation to carry a large quartic form, giving the uniform bound $-(2\lambda^2 + (2\lambda^2 + \tfrac{1}{2} - \mu)^2/2)\|u\|^2$, its multi-mode sums (one copy of the free form pays for every mode of a finite set) and the one-particle-gap version. Semiboundedness, not a gap — the constant is negative.

- `ChapterScalaronFockGapChain.lean`: the same chain run for the R^2 (scalaron) sector — the cleanest instantiation. For the constant one-particle operator $m \cdot 1$ the form gap is an **identity**, so every chain conclusion becomes a theorem with no certificate and no hypothesis: `const_fock_gap` ($d\Gamma(m\cdot 1)\Omega = 0$ and the $m\|u\|^2$ bound), `const_fock_mass_gap` (positive self-adjoint Friedrichs extension, strict positivity of the non-vacuum energy), the surviving gap $(m - 2\|f\|)\|u\|^2$ under the linear field coupling, and semiboundedness with the cubic/quartic pair — instantiated at the Starobinsky scalaron mass `scalaronMass α = 1/√(12α)` on the Hermite basis of $L^2(\mathbb{R})$. What is unconditional is the **lift**; that the sector's one-particle energy is this constant is the modelling statement of the enclosure doctrine. For the full-exponential realization $h_\psi = \tfrac{1}{2}\pi^2 + V(\hat{\varphi})$ (the vielbein/TEGR model, exponential as-is) the proved layer transfers fiber by fiber — the scalaron fiber via the wall class (`starobinskyWall_esa`: smooth non-negative potential, no growth restriction, on the compactly supported smooth core; kinetic normalization $\tfrac{1}{2}\pi^2$ is the same class after the unitary rescaling $\varphi = \sqrt{2}\cdot x$; `wallHamBddBelow_semibounded` for the form; Gauss-core transport via `ChapterQgOneParticleCcEsa`), the TEGR shear fibers via the constant/diagonal one-particle chains, and the fibrewise direct-sum instrument (`ChapterDirectSumEsa`, `fockSmoothPotential_esa`) reassembles the nested Fock space — with the caveat that the Fock-level QG theorems (`qgScalaronFock_esa`, the densitized route) formalize the **metric-route** 3+1 gauge fixing, not the vielbein/TEGR one. Only the strict edge $E_0 > 0$ remains, an elementary confinement estimate (the superlevel set $\{V < c\}$ is a bounded interval for $0 < c < M^4/(16\alpha)$), after which the same $d\Gamma$ lift applies verbatim at $\mu = E_0$.
- `ChapterFockDiagonalGapChain.lean`: the same chain for a **diagonal** one-particle energy $e_k \mapsto \omega_k e_k$ — the shape the free sectors have. Here the form gap is **proved** from $\omega_k \geq m$ (`diagOnePart_quadForm_ge`), so the chain is again hypothesis-free, with the free massive instance `freeField_fock_mass_gap` at the relativistic dispersion $\sqrt{p^2 + m^2}$ (massless dispersion gives $m = 0$: positivity, no gap).
- `ChapterSpectralGapStability.lean`: proves stability under suitable operator limits and perturbations.

The claim is now conditional but mathematically honest: the finite SIRK certificate can feed this chain only after its separation and enclosure inputs are supplied (the one-particle **form** gap is the single outstanding analytic input), and the continuum QYM result still requires a concrete convergence theorem. For the sectors whose one-particle energy is a positive constant or a diagonal dispersion — the scalaron of the R^2 theory being the flagship example — the chain is now **unconditional**: no certificate, no Ritz data, no form-gap hypothesis. The QYM instantiation remains the conditional frontier.

Every plan targets the project's pinned toolchain (**Lean v4.28.0, Mathlib v4.28.0**), so that the work stays compatible with the existing `sorry`-free `BookProof` library and with the automated prover.

A. The ODE Chapter

The `Singularity` library and `BookProof` are both `sorry`-free. All three headline ODE results are proved: `weyl_symmetrization_self_adjoint` and `nelson_essential_self_adjoint` in `Singularity/`, and `ae_no_real_singular_time` in `BookProof/ChapterOdeComplexification.lean`.

A.1 Self-adjointness of the Weyl Hamiltonian

Status: PROVED. `weyl_symmetrization_self_adjoint` (in `Singularity/Hamiltonian.lean:102`) proves `adj (odeToHamiltonian sys) = odeToHamiltonian sys` by `simp` + `ext`. The Wick symmetrization $(A + A^\dagger)/2$ is therefore self-adjoint at the algebraic level.

A.2 Nelson's essential-self-adjointness theorem

Status: PROVED. `nelson_essential_self_adjoint` (in `Singularity/Esa.lean:44`) proves the equivalence by `simp` from the certificate definitions: vanishing deficiency indices iff the classical flow is complete. The forward direction (flow complete => ESA) is the one needed for the ODE resolution.

A.3 The complexification resolution

Status: PROVED. `ae_no_real_singular_time` (in `BookProof/ChapterOdeComplexification.lean:70`) proves that for almost every initial condition, the singular time is non-real. Uses `MeasureTheory.Measure.addHaar_submodule` for the null-measure argument.

Goal. Prove that the flow of $\dot{z} = z^2$ on $L^2(\mathbb{R}^2)$ has **no finite-time singularity for almost every initial condition**, because the singular time $t = 1/z(0)$ is real only when $\operatorname{Im} z(0) = 0$, a null set.

Plan. Work in `Singularity` or a new `BookProof` module. Formalize: the explicit solution $z(t) = z(0)/(1 - tz(0))$; the singular-time condition $1 - tz(0) = 0 \Rightarrow t = 1/z(0)$; that $1/z(0) \in \mathbb{R} \Leftrightarrow \operatorname{Im} z(0) = 0$; and that the line $\{y = 0\} \subset \mathbb{R}^2$ has Lebesgue measure zero (use `MeasureTheory` — cf. `ConsciousnessNullMeasure.countable_volume_zero` for the measure-theoretic style). Conclude the set of initial data with a real singular time is null.

A.4 Energy-bounded initial conditions

Status: PROVED (in the spectral representation). `BookProof/ChapterSpectralEnergyBound.lean` works where the spectral theorem has already been applied, i.e. with H the multiplication operator attached to a real eigenvalue function f, and proves the whole chain: the spectral-projection bound $\|H\psi\| \le E_{\max}\|\psi\|$ for ψ supported on $\{|f| \le E_{\max}\}$ (`norm_diagOp_le`); unitarity of the evolution (`norm_evolve`); the Schrödinger equation $\psi'(t) = -iH\psi(t)$ (`hasDerivAt_evolve`); the uniform bound $\|\partial_t\psi(t)\| \le E_{\max}\|\psi(0)\|$ (`norm_deriv_evolve_le`); and hence global Lipschitz continuity in time (`evolve_lipschitz`) — no finite-time singularity.

```
#check @ChapterSpectralEnergyBound.norm_diagOp_le
#check @ChapterSpectralEnergyBound.norm_deriv_evolve_le
#check @ChapterSpectralEnergyBound.evolve_lipschitz
```

A.5 The chart realization: ESA of the transformed Hamiltonian

Status: PROVED (2026-10-15c wave). The one claim above that still lived only at the certificate layer — that the quantized vector field of $\dot{x} = x^2$ becomes essentially self-adjoint **after** the change of variables — is now a theorem in the verified layer, in three steps, each a separate `BookProof` module:

- **ODE-C.1** `BookProof/Ode/ChapterMomentumChartEsa.lean` — the momentum operator $P = -i\,d/dw$ of the reciprocal chart is essentially self-adjoint on the Schwartz core of $L^2(\mathbb{R})$ (`MomentumChart.momentum_essentiallySelfAdjointOn`), through the Fourier-multiplier instrument (under Plancherel, multiplication by the real symbol $2\pi\xi$);
- **ODE-C.2** `BookProof/Ode/ChapterOdeChartTransfer.lean` — the chart $(W\psi)(x) = \psi(-1/x)/x$ is a genuine unitary of $L^2(\mathbb{R})$ (`chartUnitary`), unitary conjugation carries essential self-adjointness (`conj_essentiallySelfAdjointOn`), and the conjugated operator WPW^{-1} is essentially self-adjoint **on the named chart domain** $W(\mathcal{S}(\mathbb{R}))$ (`chartHam_essentiallySelfAdjointOn`), where it equals $x^2p - ix$ almost everywhere (`chartHam_apply_ae`);
- **ODE-C.3** `chartHam_stone_flow` — the Stone bridge yields the complete unitary group on that domain.

The supporting toolbox is `BookProof/Toolbox/ChapterEsaRealShift.lean`: a symmetric essentially-self-adjoint operator has trivial deficiency spaces at **every** non-real point (`deficiencyTrivialAt_of_essentiallySelfAdjointOn`), and ESA survives real shifts (`essentiallySelfAdjointOn_add_const`).

```
#check @BookProof.MomentumChart.momentum_essentiallySelfAdjointOn
#check @BookProof.OdeChartTransfer.chartUnitary
```

```
#check @BookProof.OdeChartTransfer.conj_essentiallySelfAdjointOn
#check @BookProof.OdeChartTransfer.chartHam_essentiallySelfAdjointOn
#check @BookProof.OdeChartTransfer.chartHam_stone_flow
#check @BookProof.EsaRealShift.deficiencyTrivialAt_of_essentiallySelfAdjointOn
#check @BookProof.EsaRealShift.essentiallySelfAdjointOn_add_const
```

Boundary unchanged: A.1–A.3 say what they said. The certificate layer (`analyzeClassicalFlow`, `deficiencyIndices`, `runTest`) is still a certificate layer, and nothing above is claimed on $C_c^\infty(\mathbb{R})$ or on $L^2(\mathbb{R})$ as it stands — the theorem is for the chart realization, on its domain. The proof is recorded in the ODE chapter, section "The Chart Realization".

B. The PA-Free Chapter

B.1 The verifiable analytic core (proved)

The Riesz–Fischer characterization `completeSpace_iff_summable_norm` and the completeness of `UniformSpace.Completion` are in Mathlib. The analytic core is now fully instantiated in `BookProof.ChapterRieszFischer`: $\ell^2(\mathbb{N})$ is complete (`ell2_completeSpace`), every vector is the unconditional sum of its coordinate atoms (`riesz_fischer_hasSum`), the finitely-supported core is dense (`finSupport_dense`) and proper (`finSupport_ne_univ`), and the rational fragment is a countable dense subset (`BookProof.ChapterEll2Separable.ell2_separable`), so the completion is separable.

B.2 Definability / conservativity (the hard, partly informal part)

Status. The claim "the completion does not leak Peano Arithmetic" is a metamathematical statement about definability in the base language, **not** an internal theorem of analysis.

What is formalizable. The precise, provable fragment is now proved in `BookProof.ChapterDefinabilityFragment`: inside $\ell^2(\mathbb{N})$ the image of the term-denotable fragment $\mathbb{N} \to_0 \mathbb{R}$ is **exactly** the set of finitely-supported vectors, that set is dense, and it is a proper subset (`completion_conservative_over_core`). The remaining, purely **proof-theoretic** reading — that the completion is a conservative extension of the base theory — would require formalizing a first-order language of Hilbert spaces and a definability predicate, a research-scale task that is not claimed here.

C. Book Tooling

C.1 Inline-elaborated Lean blocks

Current state. The Lean statements in this book are shown as **plain (non-elaborated) code blocks**, and verification is anchored on `lake build BookProof`. The blocks are not yet elaborated inside the book build.

Goal. Make the `#check` blocks elaborate (and syntax-highlight with hovers) during the book build.

Plan. Verso elaborates a code block against the module's **exported** interface, so the chapter modules must re-export the `BookProof` names. This needs `public import BookProof.<Module>`, which in turn requires Lean's `experimental.module` feature to be enabled for the Book targets. Concretely: enable `experimental.module` for the Book library and `book` executable in `lakefile.toml`, change each chapter's `BookProof` import to `public import`, and re-add the imports. Verify a chapter at a time. (This re-introduces Mathlib elaboration into the book build, so it is slower; it does **not** rebuild Mathlib.)

C.2 Migration to verso-blueprint

Current state. The book is a Verso **manual** on Lean v4.28.0.

Goal. Adopt `verso-blueprint` to sync the exposition with the Lean declarations (proof-status tracking, dependency graphs, progress summaries).

Blocker. verso-blueprint requires Lean **≥ v4.29.0**; this project is on **v4.28.0** (with Mathlib v4.28.0 and the 198-module `BookProof`). verso-blueprint elaborates the declarations it documents (it detects `sorry` and builds dependency graphs), so it must run on the same toolchain as the code.

Plan (execute only when compatible with the automated prover).

1. Bump `lean-toolchain` to a version that is **both** supported by verso-blueprint **and** by the automated prover.
2. Bump Mathlib to the matching version and re-fetch its build cache (`lake exe cache get`); re-verify `BookProof` (watch for API drift — the library has drifted once before).
3. Choose a stable **label scheme** mirroring the chapter structure (e.g. `dutch_book_coherent_iff`, `euler_sum_one`, `total_variance`).
4. Tag the featured `BookProof` theorems with `@[blueprint "label"]` (use `autoDeps := true` where appropriate).
5. Port the exposition written here into blueprint blocks (`:::theorem "label" (uses := ...)`, `:::proof`, with `{uses ...}` / `{bpref ...}` edges), reusing the existing prose and sketch proofs.
6. Build with `lake exe vbp build` and inspect the progress and dependency views.

The Verso markup already written ports to blueprint blocks with modest edits, so the prose work in this book is not wasted by the migration.

D. Weak Measurements and Weak Values

Status: PROVED (`BookProof.ChapterWeakValue`).

Background. The post-selection / ABL reconstruction that underlies the trajectory chapter and the double-slit discussion is proved (`BookProof.ChapterTrajectory`): the three-instant collapsed Born process (`midProb`, `transProb`, `jointProb`, `finalProb`, `condProb`), the reconstruction consistency (`jointProb_sum_final_eq_midProb`), and the double-slit capstone (`dslit_finalProb`, `dslit_condProb`, `dslit_coherentFinal`, `dslit_interference`).

What is now proved. The weak value of an observable A given a pre-selection $\langle i|$ and a post-selection $|f\rangle$,

$$\langle A\rangle_w = \frac{\langle f|A|i\rangle}{\langle f|i\rangle},$$

is formalized on the finite complex Hilbert space $\mathrm{Fin}\, n \to \mathbb{C}$ with the standard inner product `ip`:

- `weakValue i f A = ip f (A *ᵥ i) / ip f i` — the definition;
- `weakValue_wellDefined` and `weakValue_unique` — if $\langle f|i\rangle \neq 0$, the weak value is the **unique** solution of $w\,\langle f|i\rangle = \langle f|A|i\rangle$;
- `weakValue_diag` — when $i = f$ (a unit vector) the weak value collapses to the ordinary expectation $\langle i|A|i\rangle$, which `weakValue_diag_isReal` shows is real for a Hermitian observable;
- `weakValue_add`, `weakValue_smul`, `weakValue_linear` — linearity in the observable, the algebraic core of "weak measurements are linear in A";
- `weakValue_proj` and `weakValue_proj_sum` — the weak values of a complete family of projectors sum to 1, the counterpart of `ChapterTrajectory.condProb_sum`;
- `jointProb_eq_normSq_weakNumerator` and `condProb_eq_weakNumerator_ratio` — the ABL joint law is the squared modulus of the weak-value numerator for the post-selection covector $b \mapsto \overline{V_{fb}}$, and the post-selected conditional law is its normalization;
- `dslit_weakValue` — the double-slit capstone: pre-selecting the both-slits superposition $H\Psi$ and post-selecting $\Psi = (1, 0)$, the two which-slit projectors have weak values 1 and 0.

Boundary. The **physical** claim that weak measurements do not disturb the intermediate state is about the measurement interaction, not about the mathematical ratio; it is not claimed here. The formal content is the ratio,

its well-definedness, and its agreement with the ordinary expectation in the diagonal case.

Where it lives. `BookProof.ChapterWeakValue`, registered in `BookProof.lean`, certified in `BookProof/RoadmapAudit/ChapterRoadmapAudit.lean`, and `#check`-ed from the double-slit chapter's "Weak Measurements" section.

E. The Dynamics-Based Unitary (from a function / conditional probability)

Status: PROVED (`BookProof.ChapterContinuityUnitary`), in the finite (discretized) model.

Background. The `ConditionalUnitary` chapter builds the unitary that parametrizes a conditional probability by **Gram–Schmidt** completion of the wave-function `Ψ = √p` (`BookProof.ChapterJointUnitary.exists_unitary_column`), and reads the marginal/conditional back off the operator's Gram matrix (`BookProof.ChapterConditional`). That construction is correct but **arbitrary**: the columns after the first are chosen by the completion, not fixed by the probability data.

The less arbitrary alternative. The manuscript's field-theoretic thread (QFM.tex) defines the unitary from the **dynamics** rather than from a basis choice: a function (a velocity field `v_t(x)`, or a potential `V(x,z)`) determines a Hermitian generator by the continuity (Weyl-symmetrized) prescription

$$\mathbf{H}_t = \tfrac{1}{2}\big[\hat{p}\cdot v_t(\hat{x}) + v_t(\hat{x})\cdot\hat{p}\big],$$

and hence a unitary $\mathbf{U} = e^{i\mathbf{H}t}$ — pinned down by the function, with no free choice of extra columns. For a conditional probability $p(y|x)$ the construction needs no Bochner-space machinery: via the tensor–product identification $L^2(X,\mu)\otimes L^2(Z,\nu)\cong L^2(X\times Z,\mu\times\nu)$, $\mathbf{H}$ is a standard Hermitian operator on the scalar space $L^2(X\times Z)$, and the conditional is recovered by the ordinary Born rule $P(x,B)=\int_B |\Psi_1(x,z)|^2\,d\nu(z)$.

What is now proved, on the finite (discretized) model that the rest of `BookProof` uses — the cyclic lattice $\mathbb{Z}/N$ with the symmetric-difference momentum $(\hat{p}\,\psi)_k = -\frac{i}{2}(\psi_{k+1}-\psi_{k-1})$:

- `continuityHamiltonian` — the Weyl-symmetrized generator $\mathbf{H}=\frac{1}{2}(\hat{p}\,v+v\,\hat{p})$, with `continuityHamiltonian_hermitian`; the symmetrization is **necessary**, since `momentum_mul_velocityOp_not_hermitian` exhibits a three-site velocity field for which $\hat{p}\,v$ is not Hermitian;
- `continuityUnitary` — $\mathbf{U}_t = e^{it\mathbf{H}}$ is unitary (`continuityUnitary_unitary`, from the general `exp_smul_I_unitary`) and a one-parameter group (`continuityUnitary_zero`, `continuityUnitary_add`): the "function → unitary" translation;
- `bornRecover` — the map $B\mapsto\sum_{z\in B}|\Psi_t(z)|^2$ is nonnegative (`bornRecover_nonneg`), finitely additive (`bornRecover_union`), monotone (`bornRecover_mono`) and of total mass 1 (`bornRecover_univ`, from `unitary_preserves_normSq`); `bornPMF` packages it as a distribution;
- `tensorIsom` / `tensorIsom_tmul` — the finite index-level statement of the tensor–product identification (the scalar `L²(X×Z)` is the same object), with `bornRecover_product_state` running the recovery on a product wave-function $\Psi_0(x,z)=f(x)\,e_0(z)$; and, as a capstone,
- `condProb_of_continuity` — the recovered $P(x,B)$ is a genuine conditional probability law for every input x, tying the dynamics-based unitary back to the `ChapterConditional`/`ChapterJointUnitary` Gram-matrix reading.

Boundary (updated). Two of the three deferred layers are now closed. `BookProof.ChapterContinuityUnitaryInfinite` carries the whole construction to the **infinite** lattice $\ell^2(\mathbb{Z})$: the translations are unitaries, the symmetric-difference momentum and a bounded velocity field $v\in\ell^\infty(\mathbb{Z})$ are bounded self-adjoint operators (`momentum_isSelfAdjoint`, `velocityOp_isSelfAdjoint`), the Weyl-symmetrized generator is self-adjoint (`continuityHamiltonian_isSelfAdjoint`), e^{itH} is a one-parameter unitary group of the Banach algebra of bounded operators (`continuityUnitary_unitary`, `continuityUnitary_add`) and the Born recovery is countably additive with total mass 1 (`bornRecover_tsum_univ`, `condProb_of_continuity_infinite`). `BookProof.ChapterBornMeasure`

removes the discretization from the probability side: on an arbitrary measure space, $P(B) = \int_B |\Psi|^2 \, d\mu$ is a **measure** (`bornMeasure`), a probability measure for a normalized state (`isProbabilityMeasure_bornMeasure`), absolutely continuous with respect to μ (`bornMeasure_absolutelyContinuous`), and the capstone `condProb_of_bounded_dynamics` gives it for the evolved state of any bounded self-adjoint generator on $L^2(\mu)$. What is still deferred is **unboundedness** alone — the continuum $-\frac{1}{2}\Delta_z$. Even that layer is now stated inside the theory: `BookProof.ChapterUnboundedPosition` builds the lattice position operator on its natural domain and proves the domain dense (`mulDomain_dense`), the operator symmetric (`mulOp_symmetric`) and genuinely unbounded (`position_unbounded`, `position_not_boundedOperator`). It then carries that operator through the two steps the plan had left open: its adjoint domain is exactly the natural domain and the adjoint acts by the same multiplication (`adjointDomain_eq_mulDomain`, `adjoint_eq_mulOp`), so it is **self-adjoint**; and it generates a strongly continuous one-parameter unitary group (`phaseUnitary`, `phaseUnitary_add`, `tendsto_phaseUnitary`) of which it is the generator, the difference quotient converging in $\ell^2(\mathbb{Z})$ on the natural domain (`tendsto_slope_phaseUnitary`, Stone's relation $dU/dt|_0 = iA$). `BookProof.ChapterUnitaryTransport` then shows that none of this is special to the lattice: for a unitary W, the transported operator WAW^{-1} inherits a dense domain, symmetry, self-adjointness (`transport_isSelfAdjointOn`, via `transport_adjointDomain`) and the strongly continuous group with Stone's relation (`tendsto_transportUnitary`, `tendsto_slope_transportUnitary`) — so every operator unitarily equivalent to a lattice multiplication operator, on any complex Hilbert space, carries the whole package (`transported_position_isSelfAdjointOn`, `tendsto_slope_transported_position`). And the general Stone theorem is now part of the theory rather than the recorded gap: `BookProof.ChapterStoneResolvent` through `BookProof.ChapterStoneSeparable` build the resolvent of a self-adjoint operator, obtain the weakly measurable unitary group `e^{-itA}` (`stoneU`, `stoneU_mem_domain`, `hasDerivAt_stoneU`, the group law `stoneU_add`), and prove the bijection between self-adjoint generators and one-parameter groups (`stone_bijection`, `stoneEquiv`), with a concrete `ℓ²(ℤ)` instance (`stoneU_mulSA`). `BookProof.ChapterStoneBridge` then packages a **selected** self-adjoint extension into the bundled structure Stone's theorem consumes and presents the whole package as `IsStoneFlow` (`U 0 = 1`, the group law, isometry, Schrödinger equation on the domain), with `isStoneFlow_stoneU` showing the abstractly constructed group is such a flow and `exists_stone_flow_of_selfAdjointExtension` / `of_positive` / `of_esa` as the entry points. The spectral half of Stone's theorem — the existence of the diagonalizing unitary (the spectral theorem for unbounded self-adjoint operators), the last layer the continuum $-\frac{1}{2}\Delta_z$ would need — is now closed too: `BookProof.ChapterUnboundedSpectralModel` (wave 2026-08-21h) proves, by the classical resolvent (Cayley) route, that **every** densely defined self-adjoint operator on a complex Hilbert space is multiplication by a real function on a Hilbert sum of $L^2(\mu_x)$ spaces, with no cyclic vector and no separability assumed (`unbounded_multiplication_model_cyclic` / `_general` / `_separable`). The **physical** claim that the dynamics "is" the transition is about the choice of $\mathbf{H}$; the formal content is that a Hermitian $\mathbf{H}$ yields a unitary and that the Born rule recovers a probability law.

Where it lives. `BookProof.ChapterContinuityUnitary` (finite lattice), `BookProof.ChapterContinuityUnitaryInfinite` ($\ell^2(\mathbb{Z})$) and `BookProof.ChapterBornMeasure` (the Born law as a measure) and `BookProof.ChapterUnboundedPosition` (the self-adjoint unbounded layer and the unitary group it generates) and `BookProof.ChapterUnitaryTransport` (unitary invariance of that whole package) and `BookProof.ChapterStoneResolvent` / `ChapterStoneSeparable` (the general Stone theorem and its concrete $\ell^2(\mathbb{Z})$ instance) and `BookProof.ChapterStoneBridge` / `ChapterStoneFlows` (the packaged `IsStoneFlow` and the concrete flows for the Eulerian NS, Lagrangian NS and QYM Hamiltonians) and `BookProof.ChapterUnboundedSpectralModel` (the diagonalizing unitary, via the resolvent route), all registered in `BookProof.lean`, certified in `BookProof/RoadmapAudit/ChapterRoadmapAudit.lean`, and `#check`-ed from the `ConditionalUnitary` chapter's "A Less Arbitrary Construction" section.

Index

T

T

Timepiece